

Notas del curso de Métodos Numéricos

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2.1. Ejercicios generales

2.1.1. 7,1 Conversiones en base 2

2.1.1.1. Ejercicios a mano

```
# Asignación de funciones

# Parte entera

binario <- function(x) {
  if (x == 0) return("0")
  nueva_x <- abs(x)
  iteracion <- ""
  while (nueva_x > 0) {
    iteracion <- paste0(nueva_x %% base, iteracion)
    nueva_x <- nueva_x %/% base
  }
  if (x < 0) iteracion <- paste0("-", iteracion)
  return(iteracion)
}

# Parte decimal

decimal <- function(x2) {
  if (x2 == 0) return("0")
  nueva_x <- x2
  iteracion <- "."
  for (i in 1:10) {
    nueva_x <- nueva_x * base
    if (nueva_x >= 1)
    {
      iteracion <- paste0(iteracion, "1")
      nueva_x <- nueva_x - 1
    }
  }
}
```

EJERCICIO 7.1 Convertir los siguientes números de base 10 a base 2.

1. $324 = 162 \times 2 + 0 \rightarrow 0$ $(324)_{10} = (101000100)_2$
 $162 = 81 \times 2 + 0 \rightarrow 0$
 $81 = 40 \times 2 + 1 \rightarrow 1$
 $40 = 20 \times 2 + 0 \rightarrow 0$
 $20 = 10 \times 2 + 0 \rightarrow 0$
 $10 = 5 \times 2 + 0 \rightarrow 0$
 $5 = 2 \times 2 + 1 \rightarrow 1$
 $2 = 1 \times 2 + 0 \rightarrow 0$
 $1 = 0 \times 2 + 1 \rightarrow 1$

2. $27 = 13 \times 2 + 1 \rightarrow 1$ $(27)_{10} = (11011)_2$
 $13 = 6 \times 2 + 1 \rightarrow 1$
 $6 = 3 \times 2 + 0 \rightarrow 0$
 $3 = 1 \times 2 + 1 \rightarrow 1$
 $1 = 0 \times 2 + 1 \rightarrow 1$

3. $1423 = 711 \times 2 + 1 \rightarrow 1$ $(1423)_{10} = (10110001111)_2$
 $711 = 355 \times 2 + 1 \rightarrow 1$
 $355 = 177 \times 2 + 1 \rightarrow 1$
 $177 = 88 \times 2 + 1 \rightarrow 1$
 $88 = 44 \times 2 + 0 \rightarrow 0$
 $44 = 22 \times 2 + 0 \rightarrow 0$
 $22 = 11 \times 2 + 0 \rightarrow 0$
 $11 = 5 \times 2 + 1 \rightarrow 1$
 $5 = 2 \times 2 + 1 \rightarrow 1$
 $2 = 1 \times 2 + 0 \rightarrow 0$
 $1 = 0 \times 2 + 1 \rightarrow 1$

Figura 2.1: Ejercicios 1, 2 y 3

4. 235.25

Parte entera	Parte decimal
$235 = 117 \times 2 + 1 \rightarrow 1$	$Q = 0.25$
$117 = 58 \times 2 + 1 \rightarrow 1$	$2Q = 0.5$
$58 = 29 \times 2 + 0 \rightarrow 0$	$b_1 = \lceil 2Q \rceil = 0 \rightarrow 0$
$29 = 14 \times 2 + 1 \rightarrow 1$	$F_1 = \text{fracc}(2Q) = 0.5$
$14 = 7 \times 2 + 0 \rightarrow 0$	$2F_1 = 1 \rightarrow 1$
$7 = 3 \times 2 + 1 \rightarrow 1$	$b_2 = \lceil 2F_1 \rceil = 1$
$3 = 1 \times 2 + 1 \rightarrow 1$	$F_2 = \text{fracc}(2F_1) = 0$
$1 = 0 \times 2 + 1 \rightarrow 1$	

$(235.25)_{10} = (11101011.01)_2$

5. 41.596

Parte entera	Parte decimal
$41 = 20 \times 2 + 1 \rightarrow 1$	$Q = 0.596$
$20 = 10 \times 2 + 0 \rightarrow 0$	$2Q = 1.192$
$10 = 5 \times 2 + 0 \rightarrow 0$	$b_1 = \lceil 2Q \rceil = 1 \rightarrow 1$
$5 = 2 \times 2 + 1 \rightarrow 1$	$F_1 = 0.192$
$2 = 1 \times 2 + 0 \rightarrow 0$	$2F_1 = 0.384$
$1 = 0 \times 2 + 1 \rightarrow 1$	$b_2 = 0 \rightarrow 0$
	$F_2 = 0.384$
	$2F_2 = 0.768$
	$b_3 = 0 \rightarrow 0$
	$F_3 = 0.768$
	$2F_3 = 1.536$
	$b_4 = 1 \rightarrow 1$
	$F_4 = 0.536$
	$2F_4 = 1.072$
	$b_5 = 1 \rightarrow 1$
	$F_5 = 0.072$
	$2F_5 = 0.144$
	$b_6 = 0 \rightarrow 0$
	$F_6 = 0.144$
	$2F_6 = 0.288$
	$b_7 = 0 \rightarrow 0$
	$F_7 = 0.288$
	$2F_7 = 0.576$
	$b_8 = 0 \rightarrow 0$
	$F_8 = 0.576$
	$2F_8 = 1.152$
	$b_9 = 1 \rightarrow 1$
	$F_9 = 0.152$
	$2F_9 = 0.304$
	$b_{10} = 0 \rightarrow 0$
	$F_{10} = 0.304$
	$2F_{10} = 0.608$
	$b_{11} = 0 \rightarrow 0$
	$F_{11} = 0.608$
	$2F_{11} = 1.216$
	$b_{12} = 1 \rightarrow 1$
	$F_{12} = 0.216$
	$2F_{12} = 0.432$
	$b_{13} = 0 \rightarrow 0$
	$F_{13} = 0.432$
	$2F_{13} = 0.864$
	$b_{14} = 0 \rightarrow 0$
	$F_{14} = 0.864$
	$2F_{14} = 1.728$
	$b_{15} = 1 \rightarrow 1$
	$F_{15} = 0.728$
	$2F_{15} = 1.456$
	$b_{16} = 1 \rightarrow 1$
	$F_{16} = 0.456$
	$2F_{16} = 0.912$
	$b_{17} = 0 \rightarrow 0$
	$F_{17} = 0.912$
	$2F_{17} = 1.824$
	$b_{18} = 1 \rightarrow 1$
	$F_{18} = 0.824$
	$2F_{18} = 1.648$
	$b_{19} = 1 \rightarrow 1$
	$F_{19} = 0.648$
	$2F_{19} = 1.296$
	$b_{20} = 0 \rightarrow 0$
	$F_{20} = 0.296$
	$2F_{20} = 0.592$
	$b_{21} = 0 \rightarrow 0$
	$F_{21} = 0.592$
	$2F_{21} = 1.184$
	$b_{22} = 1 \rightarrow 1$
	$F_{22} = 0.184$
	$2F_{22} = 0.368$
	$b_{23} = 0 \rightarrow 0$
	$F_{23} = 0.368$
	$2F_{23} = 0.736$
	$b_{24} = 0 \rightarrow 0$
	$F_{24} = 0.736$
	$2F_{24} = 1.472$
	$b_{25} = 1 \rightarrow 1$
	$F_{25} = 0.472$
	$2F_{25} = 0.944$
	$b_{26} = 0 \rightarrow 0$
	$F_{26} = 0.944$
	$2F_{26} = 1.888$
	$b_{27} = 1 \rightarrow 1$
	$F_{27} = 0.888$
	$2F_{27} = 1.776$
	$b_{28} = 1 \rightarrow 1$
	$F_{28} = 0.776$
	$2F_{28} = 1.552$
	$b_{29} = 1 \rightarrow 1$
	$F_{29} = 0.552$
	$2F_{29} = 1.104$
	$b_{30} = 0 \rightarrow 0$
	$F_{30} = 0.552$
	$2F_{30} = 1.104$
	$b_{31} = 1 \rightarrow 1$
	$F_{31} = 0.552$
	$2F_{31} = 1.104$
	$b_{32} = 0 \rightarrow 0$
	$F_{32} = 0.552$
	$2F_{32} = 1.104$
	$b_{33} = 1 \rightarrow 1$
	$F_{33} = 0.552$
	$2F_{33} = 1.104$
	$b_{34} = 0 \rightarrow 0$
	$F_{34} = 0.552$
	$2F_{34} = 1.104$
	$b_{35} = 1 \rightarrow 1$
	$F_{35} = 0.552$
	$2F_{35} = 1.104$
	$b_{36} = 0 \rightarrow 0$
	$F_{36} = 0.552$
	$2F_{36} = 1.104$
	$b_{37} = 1 \rightarrow 1$
	$F_{37} = 0.552$
	$2F_{37} = 1.104$
	$b_{38} = 0 \rightarrow 0$
	$F_{38} = 0.552$
	$2F_{38} = 1.104$
	$b_{39} = 1 \rightarrow 1$
	$F_{39} = 0.552$
	$2F_{39} = 1.104$
	$b_{40} = 0 \rightarrow 0$
	$F_{40} = 0.552$
	$2F_{40} = 1.104$
	$b_{41} = 1 \rightarrow 1$
	$F_{41} = 0.552$
	$2F_{41} = 1.104$
	$b_{42} = 0 \rightarrow 0$
	$F_{42} = 0.552$
	$2F_{42} = 1.104$
	$b_{43} = 1 \rightarrow 1$
	$F_{43} = 0.552$
	$2F_{43} = 1.104$
	$b_{44} = 0 \rightarrow 0$
	$F_{44} = 0.552$
	$2F_{44} = 1.104$
	$b_{45} = 1 \rightarrow 1$
	$F_{45} = 0.552$
	$2F_{45} = 1.104$
	$b_{46} = 0 \rightarrow 0$
	$F_{46} = 0.552$
	$2F_{46} = 1.104$
	$b_{47} = 1 \rightarrow 1$
	$F_{47} = 0.552$
	$2F_{47} = 1.104$
	$b_{48} = 0 \rightarrow 0$
	$F_{48} = 0.552$
	$2F_{48} = 1.104$
	$b_{49} = 1 \rightarrow 1$
	$F_{49} = 0.552$
	$2F_{49} = 1.104$
	$b_{50} = 0 \rightarrow 0$
	$F_{50} = 0.552$
	$2F_{50} = 1.104$
	$b_{51} = 1 \rightarrow 1$
	$F_{51} = 0.552$
	$2F_{51} = 1.104$
	$b_{52} = 0 \rightarrow 0$
	$F_{52} = 0.552$
	$2F_{52} = 1.104$
	$b_{53} = 1 \rightarrow 1$
	$F_{53} = 0.552$
	$2F_{53} = 1.104$
	$b_{54} = 0 \rightarrow 0$
	$F_{54} = 0.552$
	$2F_{54} = 1.104$
	$b_{55} = 1 \rightarrow 1$
	$F_{55} = 0.552$
	$2F_{55} = 1.104$
	$b_{56} = 0 \rightarrow 0$
	$F_{56} = 0.552$
	$2F_{56} = 1.104$
	$b_{57} = 1 \rightarrow 1$
	$F_{57} = 0.552$
	$2F_{57} = 1.104$
	$b_{58} = 0 \rightarrow 0$
	$F_{58} = 0.552$
	$2F_{58} = 1.104$
	$b_{59} = 1 \rightarrow 1$
	$F_{59} = 0.552$
	$2F_{59} = 1.104$
	$b_{60} = 0 \rightarrow 0$
	$F_{60} = 0.552$
	$2F_{60} = 1.104$
	$b_{61} = 1 \rightarrow 1$
	$F_{61} = 0.552$
	$2F_{61} = 1.104$
	$b_{62} = 0 \rightarrow 0$
	$F_{62} = 0.552$
	$2F_{62} = 1.104$
	$b_{63} = 1 \rightarrow 1$
	$F_{63} = 0.552$
	$2F_{63} = 1.104$
	$b_{64} = 0 \rightarrow 0$
	$F_{64} = 0.552$
	$2F_{64} = 1.104$
	$b_{65} = 1 \rightarrow 1$
	$F_{65} = 0.552$
	$2F_{65} = 1.104$
	$b_{66} = 0 \rightarrow 0$
	$F_{66} = 0.552$
	$2F_{66} = 1.104$
	$b_{67} = 1 \rightarrow 1$
	$F_{67} = 0.552$
	$2F_{67} = 1.104$
	$b_{68} = 0 \rightarrow 0$
	$F_{68} = 0.552$
	$2F_{68} = 1.104$
	$b_{69} = 1 \rightarrow 1$
	$F_{69} = 0.552$
	$2F_{69} = 1.104$
	$b_{70} = 0 \rightarrow 0$
	$F_{70} = 0.552$
	$2F_{70} = 1.104$
	$b_{71} = 1 \rightarrow 1$
	$F_{71} = 0.552$
	$2F_{71} = 1.104$
	$b_{72} = 0 \rightarrow 0$
	$F_{72} = 0.552$
	$2F_{72} = 1.104$
	$b_{73} = 1 \rightarrow 1$
	$F_{73} = 0.552$
	$2F_{73} = 1.104$
	$b_{74} = 0 \rightarrow 0$
	$F_{74} = 0.552$
	$2F_{74} = 1.104$
	$b_{75} = 1 \rightarrow 1$
	$F_{75} = 0.552$
	$2F_{75} = 1.104$
	$b_{76} = 0 \rightarrow 0$
	$F_{76} = 0.552$
	$2F_{76} = 1.104$
	$b_{77} = 1 \rightarrow 1$
	$F_{77} = 0.552$
	$2F_{77} = 1.104$
	$b_{78} = 0 \rightarrow 0$
	$F_{78} = 0.552$
	$2F_{78} = 1.104$
	$b_{79} = 1 \rightarrow 1$
	$F_{79} = 0.552$
	$2F_{79} = 1.104$
	$b_{80} = 0 \rightarrow 0$
	$F_{80} = 0.552$
	$2F_{80} = 1.104$
	$b_{81} = 1 \rightarrow 1$
	$F_{81} = 0.552$
	$2F_{81} = 1.104$
	$b_{82} = 0 \rightarrow 0$
	$F_{82} = 0.552$
	$2F_{82} = 1.104$
	$b_{83} = 1 \rightarrow 1$
	$F_{83} = 0.552$
	$2F_{83} = 1.104$
	$b_{84} = 0 \rightarrow 0$
	$F_{84} = 0.552$
	$2F_{84} = 1.104$
	$b_{85} = 1 \rightarrow 1$
	$F_{85} = 0.552$
	$2F_{85} = 1.104$
	$b_{86} = 0 \rightarrow 0$
	$F_{86} = 0.552$
	$2F_{86} = 1.104$
	$b_{87} = 1 \rightarrow 1$
	$F_{87} = 0.552$
	$2F_{87} = 1.104$
	$b_{88} = 0 \rightarrow 0$
	$F_{88} = 0.552$
	$2F_{88} = 1.104$
	$b_{89} = 1 \rightarrow 1$
	$F_{89} = 0.552$
	$2F_{89} = 1.104$
	$b_{90} = 0 \rightarrow 0$
	$F_{90} = 0.552$
	$2F_{90} = 1.104$
	$b_{91} = 1 \rightarrow 1$
	$F_{91} = 0.552$
	$2F_{91} = 1.104$
	$b_{92} = 0 \rightarrow 0$
	$F_{92} = 0.552$
	$2F_{92} = 1.104$
	$b_{93} = 1 \rightarrow 1$
	$F_{93} = 0.552$
	$2F_{93} = 1.104$
	$b_{94} = 0 \rightarrow 0$
	$F_{94} = 0.552$
	$2F_{94} = 1.104$
	$b_{95} = 1 \rightarrow 1$
	$F_{95} = 0.552$
	$2F_{95} = 1.104$
	$b_{96} = 0 \rightarrow 0$
	$F_{96} = 0.552$
	$2F_{96} = 1.104$
	$b_{97} = 1 \rightarrow 1$
	$F_{97} = 0.552$
	$2F_{97} = 1.104$
	$b_{98} = 0 \rightarrow 0$
	$F_{98} = 0.552$
	$2F_{98} = 1.104$
	$b_{99} = 1 \rightarrow 1$
	$F_{99} = 0.552$
	$2F_{99} = 1.104$
	$b_{100} = 0 \rightarrow 0$
	$F_{100} = 0.552$
	$2F_{100} = 1.104$
	$b_{101} = 1 \rightarrow 1$
	$F_{101} = 0.552$
	$2F_{101} = 1.104$
	$b_{102} = 0 \rightarrow 0$
	$F_{102} = 0.552$
	$2F_{102} = 1.104$
	$b_{103} = 1 \rightarrow 1$
	$F_{103} = 0.552$
	$2F_{103} = 1.104$
	$b_{104} = 0 \rightarrow 0$
	$F_{104} = 0.552$
	$2F_{104} = 1.104$
	$b_{105} = 1 \rightarrow 1$
	$F_{105} = 0.552$
	$2F_{105} = 1.104$
	$b_{106} = 0 \rightarrow 0$
	$F_{106} = 0.552$
	$2F_{106} = 1.104$
	$b_{107} = 1 \rightarrow 1$
	$F_{107} = 0.552$
	$2F_{107} = 1.104$
	$b_{108} = 0 \rightarrow 0$
	$F_{108} = 0.552$
	$2F_{108} = 1.104$
	$b_{109} = 1 \rightarrow 1$
	$F_{109} = 0.552$
	$2F_{109} = 1.104$
	$b_{110} = 0 \rightarrow 0$
	$F_{110} = 0.552$
	$2F_{110} = 1.104$
	$b_{111} = 1 \rightarrow 1$
	$F_{111} = 0.552$
	$2F_{111} = 1.104$
	$b_{112} = 0 \rightarrow 0$
	$F_{112} = 0.552$
	$2F_{112} = 1.104$
	$b_{113} = 1 \rightarrow 1$
	$F_{113} = 0.552$
	$2F_{113} = 1.104$
	$b_{114} = 0 \rightarrow 0$
	$F_{114} = 0.552$
	$2F_{114} = 1.104$

```
    else {  
      iteracion <- paste0(iteracion, "0")  
    }  
    if (nueva_x == 0) break  
  }  
  return(iteracion)  
}
```

2.1.1.2. 1. 324

```
x <- 324 #Número a convertir  
base <- 2 #Base a la que se convierte  
  
resultado <- binario(x)
```

2.1.1.2.1. Procedimiento

```
paste0("El número ", x, " convertido a base ", base, " es: ", resultado)
```

2.1.1.2.2. Resultado

```
## [1] "El número 324 convertido a base 2 es: 101000100"
```

2.1.1.3. 2. 27

```
x <- 27 #Número a convertir  
base <- 2 #Base a la que se convierte  
  
resultado <- binario(x)
```

2.1.1.3.1. Procedimiento

```
paste0("El número ", x, " convertido a base ", base, " es: ", resultado)
```

2.1.1.3.2. Resultado

```
## [1] "El número 27 convertido a base 2 es: 11011"
```

2.1.1.4. 3. 1423

```
x <- 1423 #Número a convertir  
base <- 2 #Base a la que se convierte  
  
resultado <- binario(x)
```

2.1.1.4.1. Procedimiento

```
paste0("El número ", x, " convertido a base ", base, " es: ", resultado)
```

2.1.1.4.2. Resultado

```
## [1] "El número 1423 convertido a base 2 es: 10110001111"
```

2.1.1.5. 4. 235,25

```
# Parte entera

x1 <- 235 #Número a convertir
base <- 2 #Base a la que se convierte

resultado1 <- binario(x1)

paste0("La parte entera es: ", resultado1)
```

2.1.1.5.1. Procedimiento

```
## [1] "La parte entera es: 11101011"

# Parte decimal

x2 <- 0.25 #Número a convertir
base <- 2 #Base a la que se convierte

resultado2 <- decimal(x2)

paste0("La parte decimal es: ", resultado2)

## [1] "La parte decimal es: .01"

numero <- x1 + x2
```

```
paste0("Juntando las dos partes, el número ", numero, " es: ", resultado1, resultado2)
```

2.1.1.5.2. Resultado

```
## [1] "Juntando las dos partes, el número 235.25 es: 11101011.01"
```

2.1.1.6. 5. 41,596

```
# Parte entera

x1 <- 41 #Número a convertir
base <- 2 #Base a la que se convierte

resultado1 <- binario(x1)

paste0("La parte entera es: ", resultado1)
```

2.1.1.6.1. Procedimiento

```
## [1] "La parte entera es: 101001"
```

```
# Parte decimal

x2 <- 0.596 #Número a convertir
base <- 2 #Base a la que se convierte

resultado2 <- decimal(x2)

paste0("La parte decimal es: ", resultado2)

## [1] "La parte decimal es: .1001100010"

numero <- x1 + x2
```

```
paste0("Juntando las dos partes, el número ", numero, " es: ", resultado1, resultado2)
```

2.1.1.6.2. Resultado

```
## [1] "Juntando las dos partes, el número 41.596 es: 101001.1001100010"
```

2.1.2. 7,2 Breve historia de métodos numéricos y base 2

2.1.2.1. 1. Breve historia

Historia y evolución de los métodos numéricos

El análisis numérico representa una disciplina fundamental dentro de las matemáticas aplicadas, dedicada al diseño y estudio de algoritmos para obtener soluciones aproximadas a problemas matemáticos complejos. Estos procedimientos sistemáticos, conocidos como métodos numéricos, han evolucionado a la par del desarrollo tecnológico y científico de la humanidad, constituyendo un puente esencial entre la teoría matemática y las aplicaciones prácticas en ingeniería, física, economía y muchas otras áreas del conocimiento.

Los primeros indicios de haberse utilizado los métodos numéricos se remontan a las civilizaciones antiguas, donde ya existía la necesidad de resolver problemas de cálculo en la vida cotidiana. Los babilonios que se conocen por haber utilizado una de las primeras herramientas para el cálculo conocido como el ábaco, alrededor del 2000 a.C., utilizaban métodos iterativos para calcular raíces cuadradas, como lo demuestran las tablillas de barro que se encontraron y se tienen conservadas hoy en día. No obstante, fueron los griegos quienes establecieron un precedente conceptual fundamental con Arquímedes (287-212 a.C.), cuyo “método de exhaustión” permitió acotar el valor de (π) y calcular el área del círculo mediante polígonos regulares, encapsulando la esencia de la aproximación sucesiva que caracteriza a los métodos numéricos modernos.

El desarrollo histórico de estas técnicas experimentó un impulso significativo durante los siglos XVII y XVIII con el surgimiento del cálculo infinitesimal y las contribuciones de genios matemáticos como Isaac Newton, Leonhard Euler y Carl Friedrich Gauss. Newton no solo desarrolló los fundamentos del cálculo, sino que también aportó algoritmos prácticos como el método que lleva su nombre para encontrar raíces de ecuaciones. Euler estableció las bases para resolver ecuaciones diferenciales con su propuesta numérica, mientras que Gauss desarrolló procedimientos fundamentales para sistemas de ecuaciones lineales y técnicas de integración avanzada. Sin embargo, la aplicación de estos métodos permaneció limitada a la tecnología de su época.

La verdadera revolución en el campo ocurrió durante el siglo XX con la aparición de las computadoras electrónicas. Como señala Vargas et al. (s. f.) de la UNAM, esta innovación tecnológica transformó radicalmente la disciplina, permitiendo implementar algoritmos numéricos para resolver problemas a una escala antes inconcebible. Este periodo presenció el surgimiento formal del análisis numérico como una disciplina dedicada no solo a crear métodos, sino también a analizar su estabilidad, convergencia y el impacto de los errores de redondeo. La creación de librerías

estandarizadas como EISPACK y LINPACK consolidó las mejores prácticas y garantizó la confiabilidad de los cálculos, estableciendo estándares de calidad en la implementación computacional.

Su importancia radica en su capacidad para resolver problemas que carecen de solución analítica exacta o que son demasiado complejos para abordarse mediante métodos tradicionales. Desde la industria aeroespacial y la predicción meteorológica, hasta en la economía y el entrenamiento de sistemas de inteligencia artificial, estos algoritmos permiten modelar y analizar sistemas del mundo real con un nivel de precisión sin precedentes. La evolución de los métodos numéricos ha estado marcada por la interacción constante entre las necesidades prácticas y el desarrollo teórico. Cada avance tecnológico, desde el ábaco hasta las supercomputadoras modernas, ha expandido los horizontes de lo que es posible calcular. El Renacimiento trajo consigo el desarrollo de métodos para ecuaciones algebraicas, mientras que la era industrial impulsó la creación de técnicas para ecuaciones diferenciales que modelaban sistemas físicos complejos. El siglo XX, con la computación digital, permitió implementar estos métodos de manera sistemática y a gran escala además de rápida, dando lugar a nuevas áreas como el análisis de error y el estudio de la estabilidad numérica.

Entornos de programación como MATLAB, Python con sus librerías especializadas han puesto estas herramientas al alcance de todos, permitiendo a estudiantes, investigadores y profesionales implementar soluciones numéricas sin requerir conocimientos profundos en computación. Esta accesibilidad ha acelerado la innovación y ha permitido que los métodos numéricos encuentren aplicación en campos tan diversos como la bioinformática, la ciencia de datos y la inteligencia artificial.

En perspectiva, la historia de los métodos numéricos representa un viaje continuo desde la intuición hasta el rigor algorítmico, desde la aproximación manual hasta la simulación computacional de alta fidelidad. Como bien señala Vargas et al. (s. f.), estos métodos constituyen hoy una herramienta indispensable en el arsenal de cualquier científico o ingeniero, permitiendo traducir problemas del mundo real a formulaciones matemáticas y encontrar soluciones útiles incluso cuando los métodos analíticos tradicionales resultan insuficientes. Esto solo afirma que la humanidad siempre busca cuantificar la información que posee y este avanza al mismo tiempo que la humanidad misma.

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2.1.2.2. 2. Conversiones en base 2 a mano

2.1.2.3. 3. Conversiones en base 2 en R

2.1.2.3.1. a) 246

```
x <- 246 #Número a convertir
base <- 2 #Base a la que se convierte

resultado <- binario(x)
```

2.1.2.3.1.1. Procedimiento

```
paste0("El número ", x, " convertido a base ", base, " es: ", resultado)
```

2.1.2.3.1.2. Resultado

EJERCICIO 7.2-2 Realiza las siguientes conversiones de base 10 a base 2.

a) $246 = 123 \times 2 + 0 \rightarrow 0$ $(246)_{10} = (11110110)_2$
 $123 = 61 \times 2 + 1 \rightarrow 1$
 $61 = 30 \times 2 + 1 \rightarrow 1$
 $30 = 15 \times 2 + 0 \rightarrow 0$
 $15 = 7 \times 2 + 1 \rightarrow 1$
 $7 = 3 \times 2 + 1 \rightarrow 1$
 $3 = 1 \times 2 + 1 \rightarrow 1$
 $1 = 0 \times 2 + 1 \rightarrow 1$

b) 345.68

Parte entera

$$\begin{aligned} 345 &= 172 \times 2 + 1 \rightarrow 1 \\ 172 &= 86 \times 2 + 0 \rightarrow 0 \\ 86 &= 43 \times 2 + 0 \rightarrow 0 \\ 43 &= 21 \times 2 + 1 \rightarrow 1 \\ 21 &= 10 \times 2 + 1 \rightarrow 1 \\ 10 &= 5 \times 2 + 0 \rightarrow 0 \\ 5 &= 2 \times 2 + 1 \rightarrow 1 \\ 2 &= 1 \times 2 + 0 \rightarrow 0 \\ 1 &= 0 \times 2 + 1 \rightarrow 1 \end{aligned}$$

Parte decimal

$$\begin{aligned} Q &= 0.68 & 2F &= 1.36 \\ 2Q &= 1.36 & b_6 &= 1 \rightarrow 1 \\ b_1 &= [12Q] = 1 \rightarrow 1 & F_6 &= 0.52 \\ F_1 &= 0.36 & 2F_6 &= 1.04 \\ 2F_1 &= 0.72 & b_7 &= 1 \rightarrow 1 \\ b_2 &= 0 & F_7 &= 0.04 \\ F_2 &= 0.72 & 2F_7 &= 0.08 \\ 2F_2 &= 1.44 & b_8 &= 0 \rightarrow 0 \\ b_3 &= 1 & F_8 &= 0.08 \\ F_3 &= 0.44 & 2F_8 &= 0.16 \\ 2F_3 &= 0.88 & b_9 &= 0 \rightarrow 0 \\ b_4 &= 0 & F_9 &= 0.16 \\ F_4 &= 0.88 & 2F_9 &= 0.32 \\ 2F_4 &= 1.76 & b_{10} &= 0 \rightarrow 0 \\ b_5 &= 1 & F_{10} &= 0.32 \\ F_5 &= 0.76 \end{aligned}$$

$$(345.68)_{10} = (101011001.1010111000)_2$$

Figura 2.3: Ejercicios a y b

c) 4586632.2846

Parte entera

$4586632 = 2293316 \times 2 + 0 \rightarrow 0$	$1119 = 559 \times 2 + 1 \rightarrow 1$
$2293316 = 1146658 \times 2 + 0 \rightarrow 0$	$559 = 279 \times 2 + 1 \rightarrow 1$
$1146658 = 573329 \times 2 + 0 \rightarrow 0$	$279 = 139 \times 2 + 1 \rightarrow 1$
$573329 = 286664 \times 2 + 1 \rightarrow 1$	$139 = 69 \times 2 + 1 \rightarrow 1$
$286664 = 143332 \times 2 + 0 \rightarrow 0$	$69 = 34 \times 2 + 1 \rightarrow 1$
$143332 = 71666 \times 2 + 0 \rightarrow 0$	$34 = 17 \times 2 + 0 \rightarrow 0$
$71666 = 35833 \times 2 + 0 \rightarrow 0$	$17 = 8 \times 2 + 1 \rightarrow 1$
$35833 = 17916 \times 2 + 1 \rightarrow 1$	$8 = 4 \times 2 + 0 \rightarrow 0$
$17916 = 8958 \times 2 + 0 \rightarrow 0$	$4 = 2 \times 2 + 0 \rightarrow 0$
$8958 = 4479 \times 2 + 0 \rightarrow 0$	$2 = 1 \times 2 + 0 \rightarrow 0$
$4479 = 2239 \times 2 + 1 \rightarrow 1$	$1 = 0 \times 2 + 1 \rightarrow 1$
$2239 = 1119 \times 2 + 1 \rightarrow 1$	

Parte decimal

$Q = 0.2846$	$b_5 = 1$	$2F_9 = 1.4304$
$2Q = 0.5692$	$F_5 = 0.11072$	$b_{10} = 1$
$b_1 = \lfloor 2Q \rfloor = 0 \rightarrow 0$	$2F_5 = 0.2144$	$F_{10} = 0.4304$
$F_1 = 0.5692$	$b_6 = 0 \rightarrow 0$	
$2F_1 = 1.1384$	$F_6 = 0.2144$	
$b_2 = 1 \rightarrow 1$	$2F_6 = 0.4288$	
$F_2 = 0.384$	$b_7 = 0 \rightarrow 0$	
$2F_2 = 0.768$	$F_7 = 0.4288$	
$b_3 = 0 \rightarrow 0$	$2F_7 = 0.8576$	
$F_3 = 0.2768$	$b_8 = 0 \rightarrow 0$	
$2F_3 = 0.5536$	$F_8 = 0.8576$	
$b_4 = 0 \rightarrow 0$	$2F_8 = 1.7152$	
$F_4 = 0.5536$	$b_9 = 1 \rightarrow 1$	
$2F_4 = 1.1072$	$F_9 = 0.7152$	

$(4586632.2846)_{10} = (1000101111110010001000.0100100011...)_{2}$

d) 984365.27463	
Parte entera	Parte decimal
$984365 = 492182 \times 2 + 1 \rightarrow 1$	$Q = 0.27463$
$492182 = 246091 \times 2 + 0 \rightarrow 0$	$2Q = 0.54926$
$246091 = 123045 \times 2 + 1 \rightarrow 1$	$b_1 = \lfloor 1.201 \rfloor = 0 \rightarrow 0$
$123045 = 61522 \times 2 + 1 \rightarrow 1$	$F_1 = 0.54926$
$61522 = 30761 \times 2 + 0 \rightarrow 0$	$2F_1 = 1.09852$
$30761 = 15380 \times 2 + 1 \rightarrow 1$	$b_2 = 1 \rightarrow 1$
$15380 = 7690 \times 2 + 0 \rightarrow 0$	$F_2 = 0.9852$
$7690 = 3845 \times 2 + 0 \rightarrow 0$	$2F_2 = 0.9704$
$3845 = 1922 \times 2 + 1 \rightarrow 1$	$b_3 = 0 \rightarrow 0$
$1922 = 961 \times 2 + 0 \rightarrow 0$	$F_3 = 0.4704$
$961 = 480 \times 2 + 1 \rightarrow 1$	$2F_3 = 0.9408$
$480 = 240 \times 2 + 0 \rightarrow 0$	$b_4 = 0 \rightarrow 0$
$240 = 120 \times 2 + 0 \rightarrow 0$	$F_4 = 0.4704$
$120 = 60 \times 2 + 0 \rightarrow 0$	$2F_4 = 0.9408$
$60 = 30 \times 2 + 0 \rightarrow 0$	$b_5 = 0 \rightarrow 0$
$30 = 15 \times 2 + 0 \rightarrow 0$	$F_5 = 0.4704$
$15 = 7 \times 2 + 1 \rightarrow 1$	$2F_5 = 0.9408$
$7 = 3 \times 2 + 1 \rightarrow 1$	$b_6 = 1 \rightarrow 1$
$3 = 1 \times 2 + 1 \rightarrow 1$	$F_6 = 0.4704$
$1 = 0 \times 2 + 1 \rightarrow 1$	$2F_6 = 0.9408$
$2F_9 = 1.22112$	$b_7 = 1 \rightarrow 1$
$b_{10} = 1 \rightarrow 1$	$F_7 = 0.4704$
$F_{10} = 0.22112$	$2F_7 = 0.9408$
	$b_8 = 0 \rightarrow 0$
	$F_8 = 0.4704$
	$2F_8 = 0.9408$
	$b_9 = 0 \rightarrow 0$
	$F_9 = 0.4704$
	$2F_9 = 0.9408$
	$b_{10} = 1 \rightarrow 1$
	$F_{10} = 0.4704$
	$2F_{10} = 0.9408$
	$b_{11} = 0 \rightarrow 0$
	$F_{11} = 0.4704$
	$2F_{11} = 0.9408$
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	$2F_{121} = 0.9408$
	$b_{122} = 0 \rightarrow 0$
	$F_{122} = 0.4704$
	$2F_{122} = 0.9408$
	$b_{123} = 0 \rightarrow 0$
	$F_{123} = 0.4704$
	$2F_{123} = 0.9408$
	b_{124

e) $79905523 = 39952761 \times 2 + 1 \rightarrow 1$

$39952761 = 19976380 \times 2 + 1 \rightarrow 1$

$19976380 = 9988190 \times 2 + 0 \rightarrow 0$

$9988190 = 4994095 \times 2 + 0 \rightarrow 0$

$4994095 = 2497047 \times 2 + 1 \rightarrow 1$

$2497047 = 1248523 \times 2 + 1 \rightarrow 1$

$1248523 = 624261 \times 2 + 1 \rightarrow 1$

$624261 = 312130 \times 2 + 1 \rightarrow 1$

$312130 = 156065 \times 2 + 0 \rightarrow 0$

$156065 = 78032 \times 2 + 1 \rightarrow 1$

$78032 = 39016 \times 2 + 0 \rightarrow 0$

$39016 = 19508 \times 2 + 0 \rightarrow 0$

$19508 = 9754 \times 2 + 0 \rightarrow 0$

$9754 = 4877 \times 2 + 0 \rightarrow 0$

$4877 = 2438 \times 2 + 1 \rightarrow 1$

$2438 = 1219 \times 2 + 0 \rightarrow 0$

$1219 = 609 \times 2 + 1 \rightarrow 1$

$609 = 304 \times 2 + 1 \rightarrow 1$

$304 = 152 \times 2 + 0 \rightarrow 0$

$152 = 76 \times 2 + 0 \rightarrow 0$

$76 = 38 \times 2 + 0 \rightarrow 0$

$38 = 19 \times 2 + 0 \rightarrow 0$

$19 = 9 \times 2 + 1 \rightarrow 1$

$9 = 4 \times 2 + 1 \rightarrow 1$

$4 = 2 \times 2 + 0 \rightarrow 0$

$2 = 1 \times 2 + 0 \rightarrow 0$

$1 = 0 \times 2 + 1 \rightarrow 1$

$(79905523)_{10} = (10011000011010000101110011)_2$

```
## [1] "El número 246 convertido a base 2 es: 11110110"
```

2.1.2.3.2. b) 345,68

```
# Parte entera

x1 <- 345 #Número a convertir
base <- 2 #Base a la que se convierte

resultado1 <- binario(x1)

paste0("La parte entera es: ", resultado1)
```

2.1.2.3.2.1. Procedimiento

```
## [1] "La parte entera es: 101011001"

# Parte decimal

x2 <- 0.68 #Número a convertir
base <- 2 #Base a la que se convierte

resultado2 <- decimal(x2)

paste0("La parte decimal es: ", resultado2)
```

```
## [1] "La parte decimal es: .1010111000"

numero <- x1 + x2
```

```
paste0("Juntando las dos partes, el número ", numero, " es: ", resultado1, resultado2)
```

2.1.2.3.2.2. Resultado

```
## [1] "Juntando las dos partes, el número 345.68 es: 101011001.1010111000"
```

2.1.2.3.3. c) 4586632,2846

```
# Parte entera

x1 <- 4586632 #Número a convertir
base <- 2 #Base a la que se convierte

resultado1 <- binario(x1)

paste0("La parte entera es: ", resultado1)
```

2.1.2.3.3.1. Procedimiento

```
## [1] "La parte entera es: 10001011111110010001000"
```

```
# Parte decimal

x2 <- 0.2846 #Número a convertir
base <- 2 #Base a la que se convierte

resultado2 <- decimal(x2)

paste0("La parte decimal es: ", resultado2)

## [1] "La parte decimal es: .0100100011"

numero <- x1 + x2
```

```
paste0("Juntando las dos partes, el número ", numero, " es: ", resultado1, resultado2)
```

2.1.2.3.3.2. Resultado

```
## [1] "Juntando las dos partes, el número 4586632.2846 es: 10001011111110010001000.0100100011"
```

2.1.2.3.4. d) 984365,27463

```
# Parte entera

x1 <- 984365 #Número a convertir
base <- 2 #Base a la que se convierte

resultado1 <- binario(x1)

paste0("La parte entera es: ", resultado1)
```

2.1.2.3.4.1. Procedimiento

```
## [1] "La parte entera es: 11110000010100101101"
```

```
# Parte decimal

x2 <- 0.27463 #Número a convertir
base <- 2 #Base a la que se convierte
```

```
resultado2 <- decimal(x2)

paste0("La parte decimal es: ", resultado2)
```

```
## [1] "La parte decimal es: .0100011001"
```

```
numero <- x1 + x2
```

```
paste0("Juntando las dos partes, el número ", numero, " es: ", resultado1, resultado2)
```

2.1.2.3.4.2. Resultado

```
## [1] "Juntando las dos partes, el número 984365.27463 es: 11110000010100101101.0100011001"
```

2.1.2.3.5. e) 79905523

```
x <- 79905523 #Número a convertir
base <- 2 #Base a la que se convierte

resultado <- binario(x)
```

2.1.2.3.5.1. Procedimiento

```
paste0("El número ", x, " convertido a base ", base, " es: ", resultado)
```

2.1.2.3.5.2. Resultado

```
## [1] "El número 79905523 convertido a base 2 es: 100110000110100001011110011"
```

2.1.2.4. 4. Tipos de Errores

Descripción Exhaustiva de los Tipos de Errores en Métodos Numéricos

En el ámbito de los métodos numéricos, la confiabilidad de un resultado depende críticamente de la comprensión y el control de los errores inherentes a todo proceso de cálculo. Dado que las soluciones rara vez son exactas, es fundamental cuantificar y gestionar las discrepancias entre el valor calculado y el valor real o teórico. Para expresar numéricamente este nivel de confianza, se emplea el concepto de “cifras significativas”, que permite comunicar la precisión de una medición o cálculo de manera estandarizada.

1. Clasificación General de los Errores

Los errores en los cálculos numéricos se pueden clasificar en dos categorías principales según su origen:

1,1. Errores Inherentes o del Modelo

Estos errores son intrínsecos al problema mismo y surgen antes de iniciar cualquier cálculo numérico. Sus fuentes principales son:

Incertidumbres en los datos de entrada: Ninguna magnitud física puede medirse con exactitud absoluta debido a limitaciones de los instrumentos y condiciones ambientales.

Equivocaciones: Errores humanos groseros, como una lectura incorrecta de un instrumento, una mala transcripción de datos o un error de cálculo simple.

Simplificaciones del modelo matemático: Las suposiciones y simplificaciones que se realizan para modelar un fenómeno físico o matemático, ignorando ciertos detalles que pueden afectar la precisión.

1,2. Errores del Método Numérico

Estos errores son introducidos por las aproximaciones necesarias para resolver el problema de manera computacional. Se subdividen en:

Error de Truncamiento: Surge cuando un procedimiento matemático exacto, que requiere un número infinito de operaciones, es reemplazado por uno aproximado y finito. Un ejemplo clásico es el uso de un número limitado de términos en una serie de Taylor para aproximar una función, o la discretización de un proceso continuo usando un tamaño de paso (h). La notación $O(h^n)$ se utiliza para describir el orden de magnitud de este error.

Error de Redondeo: Se produce debido a que las computadoras tienen una precisión finita y deben representar los números reales con una cantidad limitada de bits.

2. Cuantificación de los Errores

Para evaluar la magnitud de un error, se utilizan las siguientes definiciones:

Error Absoluto (E): Representa la diferencia directa entre el valor real (V_{real}) y el valor aproximado (V_{aprox}).

$$E = |V_{real} - V_{aprox}|$$

Error Relativo (e): Proporciona una medida del error en relación con la magnitud del valor real, lo que lo hace más significativo para comparar precisiones en diferentes escalas. Suele expresarse como porcentaje.

$$e = \frac{|V_{real} - V_{aprox}|}{|V_{real}|} \times 100$$

En la práctica del análisis numérico, donde el valor real exacto a menudo se desconoce, es común aproximar el error absoluto en un proceso iterativo como la diferencia entre dos iteraciones consecutivas: $E \approx |V_i - V_{i-1}|$.

3. Errores en Punto Flotante y Precisión de Máquina

Dado que las computadoras almacenan números en formato binario de punto flotante, la representación exacta de muchos números reales es imposible. Un número real (x) se almacena como una aproximación, $fl(x)$, tal que:

$$x = fl(x) \times (1 + \varepsilon), \text{ donde } |\varepsilon| \leq \varepsilon_{mach}$$

El valor ε_{mach} se conoce como “precisión de máquina” y representa el error de redondeo relativo máximo posible para un sistema computacional dado. Su valor depende directamente del número de bits asignados a la mantisa.

4. Aproximación Numérica: Convergencia, Estabilidad y Robustez

Una “aproximación numérica” es un valor que se acerca a la solución real, y la diferencia entre ambos constituye el error. En este contexto, se evalúan propiedades clave de los métodos numéricos:

Convergencia: Un método es convergente si, a medida que avanza el proceso iterativo (por ejemplo, reduciendo el tamaño de paso h), la solución aproximada se acerca progresivamente a la solución exacta. Un método converge de forma monótona si $|x_n - x_{n-1}| \leq |x_{n-1} - x_{n-2}|$.

Estabilidad: Un método es estable si los pequeños errores iniciales (como los de redondeo) no se amplifican.

Robustez: Un método robusto combina la convergencia y la estabilidad, siendo capaz de proporcionar resultados precisos.

La búsqueda de métodos que sean a la vez convergentes, estables y robustos es uno de los objetivos centrales en el diseño y la selección de algoritmos numéricos efectivos.

2.2. Solucion de Sistemas de Ecuaciones Lineales por métodos directos

2.2.1. 7.3. Eliminación Gaussiana Simple (Método directo)

2.2.1.1. Ejercicios a mano

2.2.1.2. Sistema 1

$$x_1 - 2x_2 + 0,5x_3 = -5$$

$$-2x_1 + 5x_2 - 1,5x_3 = 0$$

$$-0,2x_1 + 1,75x_2 - x_3 = 10$$

```
A <- matrix(c(1, -2, 0.5,
              -2, 5, -1.5,
              -0.2, 1.75, -1),
            nrow = 3, byrow = TRUE)
b <- matrix(c(-5, 0, 10), nrow = 3)

Ab <- cbind(A,b); (Ab)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]  1.0 -2.00  0.5  -5
```

Ejercicio 7.3 Gaussian Simple

6. $8x_1 + 2x_2 - 2x_3 = -2$ $\begin{vmatrix} 8 & 2 & -2 & -2 \end{vmatrix}$ R_1
 $10x_1 + 2x_2 + 4x_3 = 4$ $\Rightarrow \begin{vmatrix} 10 & 2 & 4 & 4 \end{vmatrix}$ R_2
 $12x_1 + 2x_2 + 2x_3 = 6$ $\begin{vmatrix} 12 & 2 & 2 & 6 \end{vmatrix}$ R_3

Operaciones

$m_1 = -(10/8)$
 $m_1 R_1 + R_2$
 $[(-10/8)(8) + 10] = 0$
 $[(-10/8)(2) + 2] = -0.5$
 $[(-10/8)(-2) + 4] = 6.5$
 $[(-10/8)(-2) + 4] = 6.5$

$m_2 = -(12/8) = -12/8$
 $m_2 R_1 + R_3$
 $[(-12/8)(8) + 12] = 0$
 $[(-12/8)(2) + 2] = -1$
 $[(-12/8)(-2) + 2] = 5$
 $[(-12/8)(-2) + 6] = 9$

$m_3 = -(-1/-0.5) = -2$
 $m_3 R_2 + R_3$
 $[(-2)(0) + 0] = 0$
 $[(-2)(-0.5) + (-1)] = 0$
 $[(-2)(6.5) + 5] = -8$
 $[(-2)(6.5) + 9] = -4$

Sustitución Regresiva

$x_3 = -4/-8 = 0.5$
 $x_2 = \frac{6.5 - 6.5(0.5)}{-0.5} = \frac{3.25}{-0.5} = -6.5$
 $x_1 = \frac{-2 - 2(-6.5) + 2(0.5)}{8} = \frac{12}{8} = 1.5$

Figura 2.7: Ejercicio 6

Comprobación

$$8(1.5) + 2(-6.5) - 2(0.5) = -2 \Rightarrow 12 - 13 - 1 = -2 \Rightarrow -2 = -2$$

$$10(1.5) + 2(-6.5) + 4(0.5) = 4 \Rightarrow 15 - 13 + 2 = 4 \Rightarrow 4 = 4$$

$$12(1.5) + 2(-6.5) + 2(0.5) = 6 \Rightarrow 18 - 13 + 1 = 6 \Rightarrow 6 = 6$$

Solución: $x_1 = 1.5, x_2 = -6.5, x_3 = 0.5$

7. $5x_1 + 2x_2 + x_3 - x_4 = 1$
 $2x_1 - x_2 + 3x_3 + 2x_4 = 12$
 $4x_1 + x_2 - 2x_3 + 3x_4 = 5$
 $-2x_1 + 2x_2 + x_3 + x_4 = 2$

$$\Rightarrow \begin{bmatrix} 5 & 2 & 1 & -1 & 1 \\ 2 & -1 & 3 & 2 & 12 \\ 4 & 1 & -2 & 3 & 5 \\ -2 & 2 & 1 & 1 & 2 \end{bmatrix}$$

Operaciones

$m_1 = -(2/5) = -2/5$

$m_1 R_1 + R_2 = [(-2/5)(5) + 2] = 0$
 $[(-2/5)(2) - 1] = -1.8$
 $[(-2/5)(1) + 3] = 2.6$
 $[(-2/5)(-1) + 2] = 2.4$
 $[(-2/5)(1) + 12] = 11.6$

$$\begin{bmatrix} 5 & 2 & 1 & -1 & 1 \\ 0 & -1.8 & 2.6 & 2.4 & 11.6 \\ 4 & 1 & -2 & 3 & 5 \\ -2 & 2 & 1 & 1 & 2 \end{bmatrix}$$

$m_2 = -(4/5) = -4/5$

$m_2 R_1 + R_3 = [(-4/5)(5) + 4] = 0$
 $[(-4/5)(2) + 1] = -0.6$
 $[(-4/5)(1) - 2] = -2.8$
 $[(-4/5)(-1) + 3] = 3.8$
 $[(-4/5)(1) + 5] = 4.2$

$$\begin{bmatrix} 5 & 2 & 1 & -1 & 1 \\ 0 & -1.8 & 2.6 & 2.4 & 11.6 \\ 0 & -0.6 & -2.8 & 3.8 & 4.2 \\ -2 & 2 & 1 & 1 & 2 \end{bmatrix}$$

Figura 2.8: Ejercicio 7

$m_3 = -(-2/5) = 2/5$
 $m_3 R_1 + R_4 = [(2/5)(5) - 2] = 0$
 $[(2/5)(2) + 2] = 2.8$
 $[(2/5)(11) + 1] = 1.4$
 $[(2/5)(-1) + 1] = 0.6$
 $[(2/5)(1) + 2] = 2.4$

5	2	1	-1	1
0	-1.8	2.6	2.4	11.6
0	-0.6	-2.8	3.8	4.2
0	2.8	1.4	0.6	2.4

$m_4 = -(-0.6/-1.8) = -1/3$
 $m_4 R_2 + R_3 = [(-1/3)(-1.8) - 0.6] = 0$
 $[(-1/3)(2.6) - 2.8] = -11/3$
 $[(-1/3)(2.4) + 3.8] = 3$
 $[(-1/3)(11.6) + 4.2] = 1/3$

5	2	1	-1	1
0	-1.8	2.6	2.4	11.6
0	0	-11/3	3	1/3
0	2.8	1.4	0.6	2.4

$m_5 = -(2.8/-1.8) = 14/9$
 $m_5 R_2 + R_4 = [(14/9)(-1.8) + 2.8] = 0$
 $[(14/9)(2.6) + 1.4] = 49/9$
 $[(14/9)(2.4) + 0.6] = 13/3$
 $[(14/9)(11.6) + 2.4] = 181/9$

5	2	1	-1	1
0	-1.8	2.6	2.4	11.6
0	0	-11/3	3	1/3
0	0	49/9	13/3	181/9

$m_6 = -(49/9)/(-11/3) = 49/33$
 $m_6 R_3 + R_4 = [(49/33)(-11/3) + 49/9] = 0$
 $[(49/33)(3) + 13/3] = 290/33$
 $[(49/33)(13) + 181/9] = 691/33$

5	2	1	-1	1
0	-1.8	2.6	2.4	11.6
0	0	-11/3	3	1/3
0	0	0	290/33	691/33

Sustitución regresiva

$x_4 = (691/33) / (290/33) = 2.382759$
 $x_3 = [(1/3) - 3(2.382759)] / (-11/3) = 1.858621$
 $x_2 = [11.6 - 2.6(1.858621) - 2.4(2.382759)] / -1.8 = -0.5827586$
 $x_1 = [1 - 2(-0.5827586) - (1.858621) + 2.382759] / 5 = 0.534931$

Figura 2.9: Ejercicio 7-2

Comprobación

$$5(0.537931) + 2(-0.5827586) + (1.858621) - 2.382759 = 1$$

$$2.689655 - 1.1655172 + 1.858621 - 2.382759 = 1 \rightarrow 0.999999 \approx 1$$

$$2(0.537931) - (-0.5827586) + 3(1.858621) + 2(2.382759) = 12$$

$$1.075862 + 0.5827586 + 5.575863 + 4.765518 = 12 \rightarrow 12.000001 \approx 12$$

$$4(0.537931) - 0.5827586 - 2(1.858621) + 3(2.382759) = 5$$

$$2.151724 - 0.5827586 - 3.717242 + 7.148277 = 5 \rightarrow 5.000000 \approx 5$$

$$-2(0.537931) + 2(-0.5827586) + (1.858621) + (2.382759) = 2$$

$$-1.075862 - 1.1655172 + 1.858621 + 2.382759 = 2 \rightarrow 1.999999 \approx 2$$

Solución: $X_1 = 0.537931, X_2 = -0.5827586, X_3 = 1.858621, X_4 = 2.382759$

Figura 2.10: Ejercicio 7-3

```
## [2,] -2.0  5.00 -1.5    0
## [3,] -0.2  1.75 -1.0   10
```

```
# Primera eliminación
```

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

2.2.1.2.1. Procedimiento

```
## [1] 2
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] 0.2
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1]  1.0 -2.0  0.5 -5.0
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1]  0.0  1.0 -0.5 -10.0
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.00 1.35 -0.90 9.00
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1 -2.00 0.5   -5
## A2      0 1.00 -0.5  -10
## A3      0 1.35 -0.9   9
```

```
# Segunda eliminación
```

```
Ab <- NuevaA
m3 <- -Ab[3,2] / Ab[2,2]
(m3)
```

```
##      A3
## -1.35
```

```
A3 <- m3 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0.000 0.000 -0.225 22.500
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1  -2  0.500 -5.0
## A2      0   1 -0.500 -10.0
## A3      0   0 -0.225 22.5
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x3 <- Ab[3,4] / Ab[3,3]; (x3)
```

```
##      A3
## -100
```

```
x2 <- (Ab[2,4] - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
##      A2
## -60
```

```
x1 <- (Ab[1,4] - (Ab[1,2] * x2) - (Ab[1,3] * x3)) / Ab[1,1]; (x1)
```

```
##      A1
## -75
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] -5
II <- b[2,]; (II)

## [1] 0
III <- b[3,]; (III)

## [1] 10
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: -4.999999999999999"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: -2.8421709430404e-14"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 10"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: -4.999999999999999 tecnicamente es: -5"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: -2.8421709430404e-14 tecnicamente es: 0"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: 10 tecnicamente es: 10"

cat("Por lo tanto, los valores son:")
```

2.2.1.2.2. Resultados

Por lo tanto, los valores son:

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = -75"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = -60"
```

```
paste0("x3 = ", x3)
```

```
## [1] "x3 = -100"
```

2.2.1.3. Sistema 2

$$3x_1 - x_2 + 6x_4 = 2,3$$

$$4x_1 + 2x_2 - x_3 - 5x_4 = 6,9$$

$$-5x_1 + x_2 - 3x_3 = -36$$

$$10x_2 - 4x_3 + 7x_4 = -36$$

```
A <- matrix(c(3, -1, 0, 6,
              4, 2, -1, -5,
              -5, 1, -3, 0,
              0, 10, -4, 7),
            nrow = 4, byrow = TRUE)
b <- matrix(c(2.3, 6.9, -36, -36), nrow = 4)

Ab<- cbind(A,b); (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    3  -1    0    6  2.3
## [2,]    4   2  -1   -5  6.9
## [3,]   -5   1  -3    0 -36.0
## [4,]    0  10  -4    7 -36.0
```

```
# Primera eliminación
```

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

2.2.1.3.1. Procedimiento

```
## [1] -1.333333
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] 1.666667
```

```
m3 <- -Ab[4,1] / Ab[1,1]
(m3)
```

```
## [1] 0
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 3.0 -1.0 0.0 6.0 2.3
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.000000 3.333333 -1.000000 -13.000000 3.833333
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.0000000 -0.6666667 -3.0000000 10.0000000 -32.1666667
```

```
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)
```

```
## [1] 0 10 -4 7 -36
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```



```
##      [,1]      [,2] [,3] [,4]      [,5]
## A1      3 -1.0000000  0      6  2.300000
## A2      0  3.3333333 -1     -13  3.833333
## A3      0 -0.6666667 -3      10 -32.166667
## A4      0 10.0000000 -4      7 -36.000000
```

Segunda eliminación

```
Ab <- NuevaA
m4 <- -Ab[3,2] / Ab[2,2]
(m4)
```

```
## A3
## 0.2
```

```
m5 <- -Ab[4,2] / Ab[2,2]
(m5)
```

```
## A4
## -3
```

```
A3<- m4 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1]  0.0  0.0 -3.2  7.4 -31.4
```

```
A4<- m5 * Ab[2,] + Ab[4,]
(A4)
```

```
## [1]  0.0  0.0 -1.0 46.0 -47.5
```

```
NuevaA<- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2] [,3] [,4]      [,5]
## A1      3 -1.000000  0.0  6.0  2.300000
## A2      0  3.333333 -1.0 -13.0  3.833333
## A3      0  0.000000 -3.2  7.4 -31.400000
## A4      0  0.000000 -1.0 46.0 -47.500000
```

Tercera eliminación

```
Ab <- NuevaA
m6 <- -Ab[4,3] / Ab[3,3]
(m5)
```

```
## A4
## -3
```

```
A4<- m6 * Ab[3,] + Ab[4,]
(A4)
```

```
## [1]  0.0000  0.0000  0.0000 43.6875 -37.6875
```

```
NuevaA<- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]
## A1      3 -1.000000  0.0  6.0000  2.300000
```

```
## A2    0  3.333333 -1.0 -13.0000  3.833333
## A3    0  0.000000 -3.2   7.4000 -31.400000
## A4    0  0.000000  0.0  43.6875 -37.687500
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x4 <- Ab[4,5] / Ab[4,4]; (x4)
```

```
##          A4
## -0.8626609
x3 <- (Ab[3,5] - (Ab[3,4] * x4)) / Ab[3,3]; (x3)
```

```
##          A3
##  7.817597
x2 <- (Ab[2,5] - (Ab[2,3] * x3) - (Ab[2,4] * x4)) / Ab[2,2]; (x2)
```

```
##          A2
##  0.1309013
x1 <- (Ab[1,5] - (Ab[1,2] * x2) - (Ab[1,4] * x4)) / Ab[1,1]; (x1)
```

```
##          A1
##  2.535622
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 2.3
II <- b[2,]; (II)
```

```
## [1] 6.9
III <- b[3,]; (III)
```

```
## [1] -36
IV <- b[4,]; (IV)
```

```
## [1] -36
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 2.3"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ", c2)
```

```
## [1] "El resultado es: 6.9"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ", c3)
```

```
## [1] "El resultado es: -36"
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ", c4)
```

```
## [1] "El resultado es: -36"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es ", I)
```

```
## [1] "En la ecuación 1: 2.3 técnicamente es 2.3"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es", II)
```

```
## [1] "En la ecuación 2: 6.9 técnicamente es6.9"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es", III)
```

```
## [1] "En la ecuación 3: -36 técnicamente es-36"
```

```
paste0("En la ecuación 4: ", c4, " técnicamente es", IV)
```

```
## [1] "En la ecuación 4: -36 técnicamente es-36"
```

```
cat("Por lo tanto, los valores son:")
```

2.2.1.3.2. Resultados

```
## Por lo tanto, los valores son:
```

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = 2.53562231759657"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = 0.130901287553648"
```

```
paste0("x3 = ", x3)
```

```
## [1] "x3 = 7.8175965665236"
```

```
paste0("x4 = ", x4)
```

```
## [1] "x4 = -0.862660944206009"
```

2.2.1.4. Sistema 3

$$0,003000x_1 + 59,14x_2 = 59,17$$

$$5,291x_1 - 6,130x_2 = 46,78$$

Para este ejercicio se utiliza redondeo a 4 cifras significativas.

```
A <- matrix(c(0.003000, 59.14,
              5.291, -6.130),
            nrow = 2, byrow = TRUE)
b <- matrix(c(59.17, 46.78), nrow = 2)

Ab <- cbind(A,b); (Ab)
```

2.2.1.4.1. Procedimiento

```
##      [,1] [,2] [,3]
## [1,] 0.003 59.14 59.17
## [2,] 5.291 -6.13 46.78
```

Primera eliminación

```
m1 <- round(-Ab[2,1] / Ab[1,1], 4)
(m1)
```

```
## [1] -1763.667
```

```
A1 <- round(Ab[1,], 4)
(A1)
```

```
## [1] 0.003 59.140 59.170
```

```
A2 <- round((m1 * Ab[1,] + Ab[2,]), 4)
(A2)
```

```
## [1] 0.0 -104309.4 -104309.4
```

```
NuevaA <- rbind(A1, A2)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]
## A1 0.003    59.14    59.17
## A2 0.000 -104309.38 -104309.38
```

Con esto sabemos los valores de x_n

Sustitución regresiva

```
Ab <- NuevaA
x2 <- Ab[2,3] / Ab[2,2]; (x2)
```

```
## A2
```

```
## 1
```

```
x1 <- (Ab[1,3] - (Ab[1,2] * x2)) / Ab[1,1]; (x1)
```

```
## A1
```

```
## 10
```

Comprobación

```
I <- b[1,]; (I)
```

```
## [1] 59.17
```

```
II <- b[2,]; (II)
```

```
## [1] 46.78
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 59.17"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2); paste0("El resultado es: ", c2)
```

```
## [1] "El resultado es: 46.7800000000002"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 59.17 técnicamente es: 59.17"
```



```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 46.7800000000002 técnicamente es: 46.78"
```

```
cat("Por lo tanto, los valores son:")
```

2.2.1.4.2. Resultados

```
## Por lo tanto, los valores son:
```

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = 10.00000000000004"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = 1"
```

2.2.1.5. Sistema 4

$$4x_1 + 2x_2 = 2$$

$$2x_1 + 3x_2 + x_3 = -1$$

$$x_2 + \frac{5}{2}x_3 = 3$$

```
A <- matrix(c(4, 2, 0,
              2, 3, 1,
              0, 1, 5/2),
            nrow = 3, byrow = TRUE)
b <- matrix(c(2, -1, 3), nrow = 3)

Ab <- cbind(A,b); (Ab)
```

2.2.1.5.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]   4    2  0.0    2
## [2,]   2    3  1.0   -1
## [3,]   0    1  2.5    3
```

```
# Primera eliminación
```

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

```
## [1] -0.5
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] 0
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 4 2 0 2
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0 2 1 -2
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.0 1.0 2.5 3.0
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1     4     2  0.0     2
## A2     0     2  1.0    -2
## A3     0     1  2.5     3
```

```
# Segunda eliminación
```

```
Ab <- NuevaA
m3 <- -Ab[3,2] / Ab[2,2]
(m3)
```

```
## A3
```

```
## -0.5
```

```
A3 <- m3 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0 0 2 4
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1     4     2     0     2
## A2     0     2     1    -2
## A3     0     0     2     4
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x3 <- Ab[3,4] / Ab[3,3]; (x3)
```

```
## A3
```

```
## 2
```

```
x2 <- (Ab[2,4] - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
## A2
```

```
## -2
```

```
x1 <- (Ab[1,4] - (Ab[1,2] * x2) - (Ab[1,3] * x3)) / Ab[1,1]; (x1)
```

```
## A1
```

```
## 1.5
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 2
```

```
II <- b[2,]; (II)
```

```
## [1] -1
```

```
III <- b[3,]; (III)
```

```
## [1] 3
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 2"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ", c2)
```

```
## [1] "El resultado es: -1"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ", c3)
```

```
## [1] "El resultado es: 3"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 2 técnicamente es: 2"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -1 técnicamente es: -1"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 3 técnicamente es: 3"
```

```
paste0("Por lo tanto, los valores son:")
```

2.2.1.5.2. Resultados

```
## [1] "Por lo tanto, los valores son:"
```

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = 1.5"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = -2"
```

```
paste0("x3 = ", x3)
```

```
## [1] "x3 = 2"
```

2.2.1.6. Sistema 5

$$3x_1 - 0,1x_2 - 0,2x_3 = 7,85$$

$$0,1x_1 + 7x_2 - 0,3x_3 = -19,3$$

$$0,3x_1 - 0,2x_2 + 10x_3 = 71,4$$

```
A <- matrix(c(3, -0.1, -0.2,
              0.1, 7, -0.3,
              0.3, -0.2, 10),
            nrow = 3, byrow = TRUE)
b <- matrix(c(7.85, -19.3, 71.4), nrow = 3)

Ab <- cbind(A,b); (Ab)
```

2.2.1.6.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  3.0 -0.1 -0.2  7.85
## [2,]  0.1  7.0 -0.3 -19.30
## [3,]  0.3 -0.2 10.0  71.40
```

Primera eliminación

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

```
## [1] -0.03333333
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] -0.1
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1]  3.00 -0.10 -0.20  7.85
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1]  0.0000000  7.0033333 -0.2933333 -19.5616667
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1]  0.000 -0.190 10.020 70.615
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## A1      3 -0.100000 -0.200000  7.85000
## A2      0  7.003333 -0.293333 -19.56167
## A3      0 -0.190000 10.020000  70.61500
```

Segunda eliminación

```
Ab <- NuevaA
```



```
m3 <--Ab[3,2] / Ab[2,2]
(m3)
```

```
##          A3
## 0.02712994
```

```
A3 <- m3 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0.00000 0.00000 10.01204 70.08429
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## A1      3 -0.100000 -0.2000000  7.85000
## A2      0  7.003333 -0.2933333 -19.56167
## A3      0  0.000000 10.0120419  70.08429
```

Con esto sabemos los valores de x_n

Sustitución regresiva

```
Ab <- NuevaA
x3 <- Ab[3,4] / Ab[3,3]; (x3)
```

```
## A3
## 7
```

```
x2 <- (Ab[2,4] - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
## A2
## -2.5
```

```
x1 <- (Ab[1,4] - (Ab[1,2] * x2) - (Ab[1,3] * x3)) / Ab[1,1]; (x1)
```

```
## A1
## 3
```

Comprobación

```
I <- b[1,]; (I)
```

```
## [1] 7.85
```

```
II <- b[2,]; (II)
```

```
## [1] -19.3
```

```
III <- b[3,]; (III)
```

```
## [1] 71.4
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 7.85"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -19.3"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 71.4"
```

```
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 7.85 tecnicamente es: 7.85"
```

```
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -19.3 tecnicamente es: -19.3"
```

```
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 71.4 tecnicamente es: 71.4"
```

```
paste0("Por lo tanto, los valores son:")
```

2.2.1.6.2. Resultados

```
## [1] "Por lo tanto, los valores son:"
```

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = 3"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = -2.5"
```

```
paste0("x3 = ", x3)
```

```
## [1] "x3 = 7"
```

2.2.1.7. Sistema 6

$$8x_1 + 2x_2 - 2x_3 = -2$$

$$10x_1 + 2x_2 + 4x_3 = 4$$

$$12x_1 - 2x_2 + 2x_3 = 6$$

```
A <- matrix(c(8, 2, -2,
              10, 2, 4,
              12, 2, 2),
            nrow = 3, byrow = TRUE)
b <- matrix(c(-2, 4, 6), nrow = 3)

Ab <- cbind(A,b); (Ab)
```

2.2.1.7.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]    8    2   -2   -2
## [2,]   10    2    4    4
## [3,]   12    2    2    6
```

Primera eliminación

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

```
## [1] -1.25
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] -1.5
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 8 2 -2 -2
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.0 -0.5 6.5 6.5
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0 -1 5 9
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
```

```
## A1      8  2.0 -2.0 -2.0
```

```
## A2      0 -0.5  6.5  6.5
```

```
## A3      0 -1.0  5.0  9.0
```

Segunda eliminación

```
Ab <- NuevaA
```

```
m3 <- -Ab[3,2] / Ab[2,2]
(m3)
```

```
## A3
```

```
## -2
```

```
A3 <- m3 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0 0 -8 -4
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
```

```
## A1      8  2.0 -2.0 -2.0
```

```
## A2      0 -0.5  6.5  6.5
```

```
## A3      0  0.0 -8.0 -4.0
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x3 <- Ab[3,4] / Ab[3,3]; (x3)
```

```
## A3
## 0.5
```

```
x2 <- (Ab[2,4] - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
## A2
## -6.5
```

```
x1 <- (Ab[1,4] - (Ab[1,2] * x2) - (Ab[1,3] * x3)) / Ab[1,1]; (x1)
```

```
## A1
## 1.5
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] -2
```

```
II <- b[2,]; (II)
```

```
## [1] 4
```

```
III <- b[3,]; (III)
```

```
## [1] 6
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: -2"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 4"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 6"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: -2 técnicamente es: -2"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 4 técnicamente es 4"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 6 técnicamente es 6"
```

```
paste0("Por lo tanto, los valores son:")
```

2.2.1.7.2. Resultados

```
## [1] "Por lo tanto, los valores son:"
```

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = 1.5"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = -6.5"
```

```
paste0("x3 = ", x3)
```

```
## [1] "x3 = 0.5"
```

2.2.1.8. Sistema 7

$$5x_1 + 2x_2 + x_3 - x_4 = 1$$

$$2x_1 - x_2 + 3x_3 + 2x_4 = 12$$

$$4x_1 + x_2 - 2x_3 + 3x_4 = 5$$

$$-2x_1 + 2x_2 + x_3 + x_4 = 2$$

```
A <- matrix(c(5, 2, 1, -1,
              2, -1, 3, 2,
              4, 1, -2, 3,
              -2, 2, 1, 1),
            nrow = 4, byrow = TRUE)
b <- matrix(c(1, 12, 5, 2), nrow = 4)

Ab <- cbind(A,b); (Ab)
```

2.2.1.8.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    5    2    1   -1    1
## [2,]    2   -1    3    2   12
## [3,]    4    1   -2    3    5
## [4,]   -2    2    1    1    2
```

Primera eliminación

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

```
## [1] -0.4
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] -0.8
```

```
m3 <- -Ab[4,1] / Ab[1,1]
A1 <- Ab[1,]
(A1)
```

```
## [1]  5  2  1 -1  1
```



```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.0 -1.8 2.6 2.4 11.6
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.0 -0.6 -2.8 3.8 4.2
```

```
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)
```

```
## [1] 0.0 2.8 1.4 0.6 2.4
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      5  2.0  1.0 -1.0  1.0
## A2      0 -1.8  2.6  2.4 11.6
## A3      0 -0.6 -2.8  3.8  4.2
## A4      0  2.8  1.4  0.6  2.4
```

```
# Segunda eliminación
```

```
Ab <- NuevaA
m4 <- Ab[3,2] / Ab[2,2]
(m4)
```

```
##          A3
## -0.3333333
```

```
m5 <- Ab[4,2] / Ab[2,2]
(m5)
```

```
##          A4
## 1.5555556
```

```
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0.0000000 0.0000000 -3.6666667 3.0000000 0.3333333
```

```
A4 <- m5 * Ab[2,] + Ab[4,]
(A4)
```

```
## [1] 0.0000000 0.0000000 5.4444444 4.3333333 20.4444444
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      5  2.0  1.000000 -1.000000  1.000000
## A2      0 -1.8  2.600000  2.400000 11.600000
## A3      0  0.0 -3.666667  3.000000  0.333333
## A4      0  0.0  5.444444  4.333333 20.444444
```

```
# Tercera eliminación
```

```
Ab <- NuevaA
m6 <- Ab[4,3] / Ab[3,3]
(m6)
```

```
##      A4
## 1.484848
```

```
A4 <- m6 * Ab[3,] + Ab[4,]
(A4)
```

```
## [1] 0.000000 0.000000 0.000000 8.787879 20.939394
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      5  2.0  1.000000 -1.000000  1.000000
## A2      0 -1.8  2.600000  2.400000 11.600000
## A3      0  0.0 -3.666667  3.000000  0.333333
## A4      0  0.0  0.000000  8.787879 20.939393
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x4 <- Ab[4,5] / Ab[4,4]; (x4)
```

```
##      A4
## 2.382759
```

```
x3 <- (Ab[3,5] - (Ab[3,4] * x4)) / Ab[3,3]; (x3)
```

```
##      A3
## 1.858621
```

```
x2 <- (Ab[2,5] - (Ab[2,4] * x4) - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
##      A2
## -0.5827586
```

```
x1 <- (Ab[1,5] - (Ab[1,4] * x4) - (Ab[1,3] * x3) - (Ab[1,2] * x2)) / Ab[1,1]; (x1)
```

```
##      A1
## 0.537931
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 1
```

```
II <- b[2,]; (II)
```

```
## [1] 12
```

```

III <- b[3,]; (III)

## [1] 5
IV <- b[4,]; (IV)

## [1] 2
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)

## [1] "El resultado es: 1"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)

## [1] "El resultado es: 12"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)

## [1] "El resultado es: 5"
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)

## [1] "El resultado es: 2"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 1 técnicamente es: 1"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 12 técnicamente es: 12"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 5 técnicamente es: 5"
paste0("En la ecuación 3: ", c4, " técnicamente es: ", IV)

## [1] "En la ecuación 3: 2 técnicamente es: 2"

paste0("Por lo tanto, los valores son:")

2.2.1.8.2. Resultados

## [1] "Por lo tanto, los valores son:"
paste0("x1 = ", x1)

## [1] "x1 = 0.537931034482759"
paste0("x2 = ", x2)

## [1] "x2 = -0.582758620689656"
paste0("x3 = ", x3)

## [1] "x3 = 1.85862068965517"
paste0("x4 = ", x4)

## [1] "x4 = 2.38275862068965"

```

2.2.1.9. Sistema 8

$$2x_1 + 3x_2 + 2x_3 + 4x_4 = 4$$

$$4x_1 + 10x_2 - 4x_3 = -8$$

$$-3x_1 - 2x_2 - 5x_3 - 2x_4 = -4$$

$$-2x_1 + 4x_2 + 4x_3 - 7x_4 = -1$$

```
A <- matrix(c(2, 3, 2, 4,
              4, 10, -4, 0,
              -3, -2, -5, -2,
              -2, 4, 4, -7),
            nrow = 4, byrow = TRUE)
b <- matrix(c(4, -8, -4, -1), nrow = 4)

Ab <- cbind(A,b); (Ab)
```

2.2.1.9.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2    3    2    4    4
## [2,]    4   10   -4    0   -8
## [3,]   -3    -2   -5   -2   -4
## [4,]   -2    4    4   -7   -1
```

Primera eliminación

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

```
## [1] -2
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] 1.5
```

```
m3 <- -Ab[4,1] / Ab[1,1]
A1 <- Ab[1,]
(A1)
```

```
## [1] 2 3 2 4 4
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1]    0    4   -8   -8  -16
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1]  0.0  2.5 -2.0  4.0  2.0
```

```
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)
```

```
## [1]    0    7    6   -3    3
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      2 3.0   2    4    4
## A2      0 4.0  -8   -8  -16
## A3      0 2.5  -2    4    2
## A4      0 7.0   6   -3    3
```

```
# Segunda eliminación
```

```
Ab <- NuevaA
m4 <-- Ab[3,2] / Ab[2,2]
(m4)
```

```
##      A3
## -0.625
```

```
m5 <-- Ab[4,2] / Ab[2,2]
(m5)
```

```
##      A4
## -1.75
```

```
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0 0 3 9 12
```

```
A4 <- m5 * Ab[2,] + Ab[4,]
(A4)
```

```
## [1] 0 0 20 11 31
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      2  3   2   4   4
## A2      0  4  -8  -8  -16
## A3      0  0   3   9  12
## A4      0  0  20  11  31
```

```
# Tercera eliminación
```

```
Ab <- NuevaA
m6 <-- Ab[4,3] / Ab[3,3]
(m6)
```

```
##      A4
## -6.666667
```

```
A4 <- m6 * Ab[3,] + Ab[4,]
(A4)
```

```
## [1] 0 0 0 -49 -49
```



```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      2      3      2      4      4
## A2      0      4     -8     -8    -16
## A3      0      0      3      9     12
## A4      0      0      0    -49    -49
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x4 <- Ab[4,5] / Ab[4,4]; (x4)
```

```
## A4
## 1
x3 <- (Ab[3,5] - (Ab[3,4] * x4)) / Ab[3,3]; (x3)
```

```
## A3
## 1
x2 <- (Ab[2,5] - (Ab[2,4] * x4) - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
## A2
## 0
x1 <- (Ab[1,5] - (Ab[1,4] * x4) - (Ab[1,3] * x3) - (Ab[1,2] * x2)) / Ab[1,1]; (x1)
```

```
## A1
## -1
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 4
II <- b[2,]; (II)
```

```
## [1] -8
III <- b[3,]; (III)
```

```
## [1] -4
IV <- b[4,]; (IV)
```

```
## [1] -1
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 4"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -8"
```

```

c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)

## [1] "El resultado es: -4"
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)

## [1] "El resultado es: -1"
paste0("En la ecuación 1: ", c1, " debería ser ", I)

## [1] "En la ecuación 1: 4 debería ser 4"
paste0("En la ecuación 2: ", c2, " debería ser ", II)

## [1] "En la ecuación 2: -8 debería ser -8"
paste0("En la ecuación 3: ", c3, " debería ser ", III)

## [1] "En la ecuación 3: -4 debería ser -4"
paste0("En la ecuación 3: ", c4, " debería ser ", IV)

## [1] "En la ecuación 3: -1 debería ser -1"

paste0("Por lo tanto, los valores son:")

```

2.2.1.9.2. Resultados

```

## [1] "Por lo tanto, los valores son:"
paste0("x1 = ", x1)

## [1] "x1 = -1"
paste0("x2 = ", x2)

## [1] "x2 = 0"
paste0("x3 = ", x3)

## [1] "x3 = 1"
paste0("x4 = ", x4)

## [1] "x4 = 1"

```

2.2.1.10. Sistema 9

$$1,133x_1 + 5,281x_2 - 2,454x_3 = 6,414$$

$$24,14x_1 - 1,21x_2 + 5,281x_3 = 113,8$$

$$-10,123x_1 + 6,387x_2 - x_3 = 1$$

```

A <- matrix(c(1.133, 5.281, -2.454,
              24.14, -1.21, 5.281,
              -10.123, 6.387, -1),
            nrow = 3, byrow = TRUE)

```

```
b <- matrix(c(6.414, 113.8, 1), nrow = 3)
```

```
Ab <- cbind(A,b); (Ab)
```

2.2.1.10.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  1.133  5.281 -2.454  6.414
## [2,] 24.140 -1.210  5.281 113.800
## [3,] -10.123  6.387 -1.000  1.000
```

Primera eliminación

```
m1 <- -Ab[2,1] / Ab[1,1]
(m1)
```

```
## [1] -21.30627
```

```
m2 <- -Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] 8.934687
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1]  1.133  5.281 -2.454  6.414
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1]  0.00000 -113.72839  57.56658 -22.85839
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1]  0.00000  53.57108 -22.92572  58.30708
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1  1.133  5.28100 -2.45400  6.41400
## A2  0.000 -113.72839  57.56658 -22.85839
## A3  0.000  53.57108 -22.92572  58.30708
```

Segunda eliminación

```
Ab <- NuevaA
m3 <- -Ab[3,2] / Ab[2,2]
(m3)
```

```
##      A3
## 0.471044
```

```
A3 <- m3 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1]  0.000000  0.000000  4.190672 47.539771
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## A1 1.133    5.2810 -2.454000   6.41400
## A2 0.000 -113.7284 57.566578 -22.85839
## A3 0.000    0.0000  4.190672  47.53977
```

Con esto sabemos los valores de x_n

Sustitución regresiva

```
Ab <- NuevaA
x3 <- Ab[3,4] / Ab[3,3]; (x3)
```

```
##      A3
## 11.34419
x2 <- (Ab[2,4] - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
##      A2
## 5.943147
x1 <- (Ab[1,4] - (Ab[1,2] * x2) - (Ab[1,3] * x3)) / Ab[1,1]; (x1)
```

```
##      A1
## 2.530346
```

Comprobación

```
I <- b[1,]; (I)
```

```
## [1] 6.414
II <- b[2,]; (II)
```

```
## [1] 113.8
III <- b[3,]; (III)
```

```
## [1] 1
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 6.414"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 113.8"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 0.9999999999999961"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 6.414 técnicamente es: 6.414"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 113.8 técnicamente es: 113.8"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 0.999999999999961 técnicamente es: 1"
```

```
paste0("Por lo tanto, los valores son:")
```

2.2.1.10.2. Resultados

```
## [1] "Por lo tanto, los valores son:"
```

```
paste0("x1 = ", x1)
```

```
## [1] "x1 = 2.53034570295305"
```

```
paste0("x2 = ", x2)
```

```
## [1] "x2 = 5.94314679783675"
```

```
paste0("x3 = ", x3)
```

```
## [1] "x3 = 11.3441890467896"
```

2.2.1.11. Sistema 10

$$A = \begin{bmatrix} 2 & 3 & 2 & 4 \\ 4 & 10 & -4 & 0 \\ -3 & -2 & -5 & -2 \\ -2 & 4 & 4 & -7 \end{bmatrix}$$

$$b = \begin{bmatrix} 9 \\ -15 \\ 6 \\ 2 \end{bmatrix}$$

```
A <- matrix(c(2, 3, 2, 4,
              4, 10, -4, 0,
              -3, -2, -5, -2,
              -2, 4, 4, -7),
            nrow = 4, byrow = TRUE)
b <- matrix(c(9, -15, 6, 2), nrow = 4)
Ab <- cbind(A,b); (Ab)
```

2.2.1.11.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2    3    2    4    9
## [2,]    4   10   -4    0  -15
## [3,]   -3   -2   -5   -2    6
## [4,]   -2    4    4   -7    2
```

```
# Primera eliminación
```

```

m1 <- -Ab[2,1] / Ab[1,1]
(m1)

## [1] -2
m2 <- -Ab[3,1] / Ab[1,1]
(m2)

## [1] 1.5
m3 <- -Ab[4,1] / Ab[1,1]
A1 <- Ab[1,]
(A1)

## [1] 2 3 2 4 9
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)

## [1] 0 4 -8 -8 -33
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)

## [1] 0.0 2.5 -2.0 4.0 19.5
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)

## [1] 0 7 6 -3 11
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

##      [,1] [,2] [,3] [,4] [,5]
## A1      2 3.0  2   4   9.0
## A2      0 4.0 -8  -8 -33.0
## A3      0 2.5 -2   4  19.5
## A4      0 7.0  6  -3  11.0

# Segunda eliminación

Ab <- NuevaA
m4 <- -Ab[3,2] / Ab[2,2]
(m4)

##      A3
## -0.625
m5 <- -Ab[4,2] / Ab[2,2]
(m5)

##      A4
## -1.75
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)

## [1] 0.000 0.000 3.000 9.000 40.125

```



```
A4 <- m5 * Ab[2,] + Ab[4,]
(A4)
```

```
## [1] 0.00 0.00 20.00 11.00 68.75
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      2      3      2      4      9.000
## A2      0      4     -8     -8    -33.000
## A3      0      0      3      9     40.125
## A4      0      0     20     11     68.750
```

```
# Tercera eliminación
```

```
Ab <- NuevaA
m6 <- Ab[4,3] / Ab[3,3]
(m6)
```

```
##      A4
## -6.666667
```

```
A4 <- m6 * Ab[3,] + Ab[4,]
(A4)
```

```
## [1] 0.00 0.00 0.00 -49.00 -198.75
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      2      3      2      4      9.000
## A2      0      4     -8     -8    -33.000
## A3      0      0      3      9     40.125
## A4      0      0      0    -49   -198.750
```

Con esto sabemos los valores de x_n

```
# Sustitución regresiva
```

```
Ab <- NuevaA
x4 <- Ab[4,5] / Ab[4,4]; (x4)
```

```
##      A4
## 4.056122
x3 <- (Ab[3,5] - (Ab[3,4] * x4)) / Ab[3,3]; (x3)
```

```
##      A3
## 1.206633
x2 <- (Ab[2,5] - (Ab[2,4] * x4) - (Ab[2,3] * x3)) / Ab[2,2]; (x2)
```

```
##      A2
## 2.27551
```

```

x1 <- (Ab[1,5] - (Ab[1,4] * x4) - (Ab[1,3] * x3) - (Ab[1,2] * x2)) / Ab[1,1]; (x1)

##          A1
## -8.232143
# Comprobación

I <- b[1,]; (I)

## [1] 9
II <- b[2,]; (II)

## [1] -15
III <- b[3,]; (III)

## [1] 6
IV <- b[4,]; (IV)

## [1] 2
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)

## [1] "El resultado es: 9"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)

## [1] "El resultado es: -15"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)

## [1] "El resultado es: 6"
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)

## [1] "El resultado es: 2"
paste0("En la ecuación 1: ", c1, " debería ser ", I)

## [1] "En la ecuación 1: 9 debería ser 9"
paste0("En la ecuación 2: ", c2, " debería ser ", II)

## [1] "En la ecuación 2: -15 debería ser -15"
paste0("En la ecuación 3: ", c3, " debería ser ", III)

## [1] "En la ecuación 3: 6 debería ser 6"
paste0("En la ecuación 3: ", c4, " debería ser ", IV)

## [1] "En la ecuación 3: 2 debería ser 2"

paste0("Por lo tanto, los valores son:")

```

2.2.1.11.2. Resultados

```
## [1] "Por lo tanto, los valores son:"
```

```

paste0("x1 = ", x1)

## [1] "x1 = -8.23214285714286"
paste0("x2 = ", x2)

## [1] "x2 = 2.27551020408163"
paste0("x3 = ", x3)

## [1] "x3 = 1.20663265306122"
paste0("x4 = ", x4)

## [1] "x4 = 4.05612244897959"

```

2.2.2. 7.4. Eliminación Gaussiana con Pivoteo (Método directo)

2.2.2.1. Ejercicios a mano

EJERCICIO 7.4. Gaussiana con pivoteo parcial.

2. $5x_1 - 8x_2 + x_3 = -71$
 $-2x_1 + 6x_2 - 9x_3 = 134$
 $3x_1 - 5x_2 + 2x_3 = -58$

$\rightarrow$

$$\begin{bmatrix} 5 & -8 & 1 & -71 \\ -2 & 6 & -9 & 134 \\ 3 & -5 & 2 & -58 \end{bmatrix}$$

Operaciones

$m_1 = -(-2/5) = 2/5$

$m_1 R_1 + R_2 =$

$$\begin{aligned} [(2/5)(5) - 2] &= 0 \\ [(2/5)(-8) + 6] &= 2.8 \\ [(2/5)(1) - 9] &= -8.6 \\ [(2/5)(-71) + 134] &= 105.6 \end{aligned}$$

$m_2 = -3/5$

$m_2 R_1 + R_3 =$

$$\begin{aligned} [(-3/5)(5) + 3] &= 0 \\ [(-3/5)(-8) - 5] &= -0.2 \\ [(-3/5)(1) + 2] &= 1.4 \\ [(-3/5)(-71) - 58] &= -15.4 \end{aligned}$$

Final augmented matrix:

$$\begin{bmatrix} 5 & -8 & 1 & -71 \\ 0 & 2.8 & -8.6 & 105.6 \\ 0 & -0.2 & 1.4 & -15.4 \end{bmatrix}$$

Figura 2.11: Ejercicio 2

2.2.2.2. Sistema 1

$$0,4x_1 - 1,5x_2 + 0,75x_3 = -20$$

$$-0,5x_1 - 15x_2 + 10x_3 = -10$$

$m_2 = -(-0.2)/(2.8) = \frac{1}{14}$
 $m_3 R_2 + R_3 = \begin{bmatrix} (1/14)(2.8) + 0.2 & 0 & 0 & 0 \\ (1/14)(-8.6) + 1.4 & 0 & 2.8 & -8.6 \\ (1/14)(105.6) - 13.4 & 0 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 2.8 & -8.6 & 10.6 \\ 0 & 0 & 11 & -55 \end{bmatrix}$

Volver ceros
 $m_4 = -(8.6)(1/14) = \frac{602}{55}$
 $m_4 R_3 + R_2 = \begin{bmatrix} (602/55)(-55) + 105.6 & 0 & 0 & 0 \\ (602/55)(1/14) - 8.6 & 0 & 2.8 & 0 \\ (602/55)(0) + 2.8 & 0 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 19.6 & 0 & 0 & -55 \\ 0 & 2.8 & 0 & 19.6 \\ 0 & 0 & 11 & -55 \end{bmatrix}$

$m_5 = -(1/14) = -\frac{14}{14}$
 $m_5 R_3 + R_1 = \begin{bmatrix} (-14)(-55) - 1 & 5 & -8 & 0 \\ (-14)(1/14) + 1 & 0 & 2.8 & 0 \\ (-14)(0) - 8 & 0 & 0 & 11 \end{bmatrix} = \begin{bmatrix} -61 & 5 & -8 & 0 \\ 0 & 2.8 & 0 & 19.6 \\ 0 & 0 & 11 & -55 \end{bmatrix}$

$m_6 = -(-8/2.8) = \frac{20}{7}$
 $m_6 R_2 + R_1 = \begin{bmatrix} (20/7)(19.6) - 61 & 5 & 0 & 0 \\ (20/7)(2.8) - 8 & 0 & 2.8 & 0 \\ 0 & 0 & 11 & -55 \end{bmatrix} = \begin{bmatrix} -5 & 5 & 0 & 0 \\ 0 & 2.8 & 0 & 19.6 \\ 0 & 0 & 11 & -55 \end{bmatrix}$

Soluciones
 $5x_1 = -5 \Rightarrow x_1 = -\frac{5}{5} = -1$
 $2.8x_2 = 19.6 \Rightarrow x_2 = 19.6/2.8 = 7$
 $\frac{11}{14}x_3 = -\frac{55}{14} \Rightarrow x_3 = -\frac{55}{14}(\frac{14}{11}) = -10$

Solución
 $x_1 = -1$
 $x_2 = 7$
 $x_3 = -10$

Comprobación
 $5(-1) - 8(7) + (-10) = -71 \Rightarrow -5 - 56 - 10 = -71 \Rightarrow -71 = -71$
 $-2(-1) + 6(7) - 9(-10) = 134 \Rightarrow 2 + 42 + 90 = 134 \Rightarrow 134 = 134$
 $3(-1) - 5(7) + 2(-10) = -58 \Rightarrow -3 - 35 - 20 = -58 \Rightarrow -58 = -58$

Figura 2.12: Ejercicio 2-2

$$\begin{aligned} 4. \quad & -x_2 + 4x_3 - x_4 = -1 \\ & -x_1 + 4x_2 - x_3 = 2 \\ & -x_1 - x_3 + 4x_4 = 4 \\ & 4x_1 - x_2 = 10 \end{aligned} \Rightarrow \begin{bmatrix} 0 & -1 & 4 & -1 & -1 \\ -1 & 4 & -1 & 0 & 2 \\ -1 & 0 & -1 & 4 & 4 \\ 4 & -1 & 0 & 0 & 10 \end{bmatrix}$$

Operaciones:

Pivotes $R_4 \leftrightarrow R_1$

$m_1 = -(-1/4) = 1/4$

$m_1 R_1 + R_2 = [(1/4)(4) - 1, (1/4)(-1) + 4, (1/4)(0) - 1, (1/4)(0) + 0, (1/4)(10) + 2]$

$m_1 R_1 + R_3 = [(1/4)(4) - 1, (1/4)(-1) + 0, (1/4)(0) - 1, (1/4)(0) + 4, (1/4)(10) + 4]$

$m_1 R_1 + R_4 = [0, 15/4, -1, 0, 9/2]$

$m_1 R_1 + R_5 = [0, -1/4, -1, 4, 13/2]$

$$\begin{bmatrix} 4 & -1 & 0 & 0 & 10 \\ 0 & 15/4 & -1 & 0 & 9/2 \\ 0 & -1/4 & -1 & 4 & 13/2 \\ 0 & -1 & 4 & -1 & -1 \end{bmatrix}$$

$m_2 = -(-1/4)/(15/4) = +1/15$

$m_3 = -(-1)/(15/4) = 4/15$

$m_2 R_2 + R_3 = [(1/15)(15/4) - 1/4, (1/15)(-1) - 1, (1/15)(0) + 4, (1/15)(9/2) + 13/2]$

$m_3 R_2 + R_4 = [0, (4/15)(15/4) - 1, (4/15)(-1) + 4, (4/15)(0) - 1, (4/15)(9/2) - 1]$

$m_2 R_2 + R_5 = [0, 0, -15/15, 4, 34/5]$

$m_3 R_2 + R_6 = [0, 0, 56/15, -1, 5]$

$$\begin{bmatrix} 4 & -1 & 0 & 0 & 10 \\ 0 & 15/4 & -1 & 0 & 9/2 \\ 0 & 0 & -15/15 & 4 & 34/5 \\ 0 & 0 & 56/15 & -1 & 5 \end{bmatrix}$$

Figura 2.13: Ejercicio 4

Pivoteo $R_3 \leftrightarrow R_4$

$$m_4 = -(-16/5)/(56/5) = 16/56 = \frac{2}{7}$$

$$m_4 R_3 + R_4 = [0, 0, (2/7)(56/5) - 16/5, (2/7)(-1) + 1, (2/7)(1/3) + 6.8]$$

$$m_4 R_3 + R_4 = [0, 0, 0, \frac{26}{7}, \frac{43}{7}]$$

$$\begin{bmatrix} 4 & -1 & 0 & 0 & 10 \\ 0 & \frac{15}{4} & -1 & 0 & \frac{9}{2} \\ 0 & 0 & \frac{56}{13} & -1 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{26}{7} & \frac{43}{7} \end{bmatrix}$$

Volver ceros

$$m_5 = -(-1)/(\frac{26}{7}) = \frac{7}{26}$$

$$m_5 R_4 + R_3 = [0, 0, \frac{56}{13}, (\frac{7}{26})(\frac{26}{7}) - 1, (\frac{7}{26})(\frac{43}{7}) + \frac{1}{3}] = [0, 0, \frac{56}{13}, 0, \frac{133}{65}]$$

$$\begin{bmatrix} 4 & -1 & 0 & 0 & 10 \\ 0 & \frac{15}{4} & -1 & 0 & \frac{9}{2} \\ 0 & 0 & \frac{56}{13} & 0 & \frac{133}{65} \\ 0 & 0 & 0 & \frac{26}{7} & \frac{43}{7} \end{bmatrix}$$

$$m_6 = -(-1)/(\frac{56}{13}) = \frac{13}{56}$$

$$m_6 R_3 + R_2 = [0, \frac{15}{4}, (\frac{13}{56})(\frac{56}{13}) - 1, 0, (\frac{13}{56})(\frac{133}{65}) + \frac{9}{2}] = [0, \frac{15}{4}, 0, 0, \frac{525}{104}]$$

$$\begin{bmatrix} 4 & -1 & 0 & 0 & 10 \\ 0 & \frac{15}{4} & 0 & 0 & \frac{525}{104} \\ 0 & 0 & \frac{56}{13} & 0 & \frac{133}{65} \\ 0 & 0 & 0 & \frac{26}{7} & \frac{43}{7} \end{bmatrix}$$

$$m_7 = -(-1)/(\frac{15}{4}) = \frac{4}{15}$$

$$m_7 R_2 + R_1 = [4, (\frac{4}{15})(\frac{15}{4}) - 1, 0, 0, (\frac{4}{15})(\frac{525}{104}) + 10] = [4, 0, 0, 0, \frac{245}{13}]$$

$$\begin{bmatrix} 4 & 0 & 0 & 0 & \frac{245}{13} \\ 0 & \frac{15}{4} & 0 & 0 & \frac{525}{104} \\ 0 & 0 & \frac{56}{13} & 0 & \frac{133}{65} \\ 0 & 0 & 0 & \frac{26}{7} & \frac{43}{7} \end{bmatrix}$$

Figura 2.14: Ejercicio 4-2

Soluciones

$$\frac{26}{7}x_4 = \frac{48}{7} \Rightarrow x_4 = \left(\frac{48}{7}\right)\left(\frac{7}{26}\right) = \frac{48}{26} \approx 1.846154$$

$$(56/15)x_3 = 133/65 \Rightarrow x_3 = \left(\frac{133}{65}\right)\left(\frac{15}{56}\right) = \frac{57}{104} \approx 0.5480769$$

$$(15/4)x_2 = (525/104) \Rightarrow x_2 = (525/104)\left(\frac{4}{15}\right) = \frac{35}{26} \approx 1.346154$$

$$4x_1 = \frac{295}{26} \Rightarrow x_1 = \left(\frac{295}{26}\right)\left(\frac{1}{4}\right) = \frac{295}{104} \approx 2.836538$$

Comprobación

$$-\left(\frac{35}{26}\right) + 4\left(\frac{57}{104}\right) - \left(\frac{48}{26}\right) = -1 \Rightarrow \frac{57}{26} - \frac{83}{26} = -1 \Rightarrow -\frac{26}{26} = -1 \Rightarrow -1 = -1$$

$$-\left(\frac{295}{104}\right) + 4\left(\frac{35}{26}\right) - \frac{57}{104} = 2 \Rightarrow \frac{140}{13} - \frac{94}{13} = 2 \Rightarrow \frac{26}{13} = 2 \Rightarrow 2 = 2$$

$$-\left(\frac{295}{104}\right) - \left(\frac{57}{104}\right) + 4\left(\frac{48}{26}\right) = 4 \Rightarrow \frac{96}{13} - \frac{44}{13} = 4 \Rightarrow \frac{52}{13} = 4 \Rightarrow 4 = 4$$

$$4\left(\frac{295}{104}\right) - \left(\frac{35}{26}\right) = 10 \Rightarrow \frac{295}{26} - \frac{35}{26} = 10 \Rightarrow \frac{260}{26} = 10 \Rightarrow 10 = 10$$

Soluciones:

$$x_1 = \frac{295}{104} \approx 2.836538$$

$$x_2 = \frac{35}{26} \approx 1.346154$$

$$x_3 = \frac{57}{104} \approx 0.5480769$$

$$x_4 = \frac{48}{26} \approx 1.846154$$

Figura 2.15: Ejercicio 4-3

$$-10x_1 - 9x_2 + 2,5x_3 = 30$$

```
A <- matrix(c(0.4, -1.5, 0.75,
              -0.5, -15, 10,
              -10, -9, 2.5),
            nrow = 3, byrow = TRUE)
b <- matrix(c(-20, -10, 30), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.2.2.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  0.4 -1.5  0.75 -20
## [2,] -0.5 -15.0 10.00 -10
## [3,] -10.0 -9.0  2.50  30
```

Primer pivoteo

```
NuevaA <- Ab[c(3, 2, 1), ]
(NuevaA)
```

2.2.2.2.1.1. Primera eliminación

```
##      [,1] [,2] [,3] [,4]
## [1,] -10.0 -9.0  2.50  30
## [2,] -0.5 -15.0 10.00 -10
## [3,]  0.4 -1.5  0.75 -20
```

Volver ceros

```
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
R1 <- NuevaA[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## R1  -10 -9.000 2.5000  30
## R2   0 -16.875 10.9375 -35
## R3   0 -46.500 21.2500 -470
```

Segundo pivoteo

```
Ab <- NuevaA[c(1, 3, 2), ]
(Ab)
```

2.2.2.2.1.2. Segunda eliminación

```
##      [,1] [,2] [,3] [,4]
```

```
## R1  -10  -9.000  2.5000  30
## R3   0 -46.500 21.2500 -470
## R2   0 -16.875 10.9375 -35
```

```
# Volver ceros
```

```
m3 <- - Ab[3,2] / Ab[2,2]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m3 * Ab[2,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1  -10  -9.0  2.500000  30.0000
## R2   0 -46.5 21.250000 -470.0000
## R3   0   0.0  3.225806 135.5645
```

Volver ceros con eliminación hacia atrás

```
# Tercera columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1  -10  -9.0  2.500000  30.0000
## R2   0 -46.5 21.250000 -470.0000
## R3   0   0.0  3.225806 135.5645
```

```
m4 <- - Ab[2,3] / Ab[3,3]
m5 <- - Ab[1,3] / Ab[3,3]
R1 <- m5 * Ab[3,] + Ab[1,]
R2 <- m4 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1  -10  -9.0  0.000000  -75.0625
## R2   0 -46.5  0.000000 -1363.0312
## R3   0   0.0  3.225806 135.5645
```

```
# Segunda columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1  -10  -9.0  0.000000  -75.0625
## R2   0 -46.5  0.000000 -1363.0312
## R3   0   0.0  3.225806 135.5645
```

```
m6 <- - Ab[1,2] / Ab[2,2]
```

```
R1 <- m6 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
```

```

R3 <- Ab[3,]

NuevaA <- rbind(R1, R2, R3)
(NuevaA)

##      [,1] [,2]      [,3]      [,4]
## R1  -10   0.0 0.000000   188.7500
## R2    0 -46.5 0.000000 -1363.0312
## R3    0   0.0 3.225806   135.5645

# Soluciones

Ab <- NuevaA; (Ab)

##      [,1] [,2]      [,3]      [,4]
## R1  -10   0.0 0.000000   188.7500
## R2    0 -46.5 0.000000 -1363.0312
## R3    0   0.0 3.225806   135.5645

x1 <- Ab[1,4] / Ab[1,1]
x2 <- Ab[2,4] / Ab[2,2]
x3 <- Ab[3,4] / Ab[3,3]

# Comprobación

I <- b[1,]; (I)

## [1] -20

II <- b[2,]; (II)

## [1] -10

III <- b[3,]; (III)

## [1] 30

c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: -20"

c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: -10"

c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 30"

paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: -20 técnicamente es: -20"

paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: -10 técnicamente es: -10"

paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 30 técnicamente es: 30"

```

```
cat("Soluciones del sistema:")
```

2.2.2.2.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = -18.875
```

```
cat("x2 =", x2)
```

```
## x2 = 29.3125
```

```
cat("x3 =", x3)
```

```
## x3 = 42.025
```

2.2.2.3. Sistema 2

$$5x_1 - 8x_2 + x_3 = -71$$

$$-2x_1 + 6x_2 - 9x_3 = 134$$

$$3x_1 - 5x_2 + 2x_3 = -58$$

```
A <- matrix(c(5, -8, 1,
              -2, 6, -9,
              3, -5, 2),
            nrow = 3, byrow = TRUE)
b <- matrix(c(-71, 134, -58), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.2.3.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]    5   -8    1  -71
## [2,]   -2    6   -9  134
## [3,]    3   -5    2  -58
```

2.2.2.3.1.1. Primera eliminación No se necesita hacer el pivoteo ya que se encuentra acomodado

```
# Volver ceros
```

```
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## R1     5  -8.0  1.0 -71.0
```



```
## R2    0  2.8 -8.6 105.6
## R3    0 -0.2  1.4 -15.4
```

2.2.2.3.1.2. Segunda eliminación

Tampoco se necesita pivoteo en esta ocasión

```
# Volver ceros
```

```
Ab <- NuevaA
m3 <- - Ab[3,2] / Ab[2,2]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m3 * Ab[2,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1     5 -8.0  1.0000000 -71.000000
## R2     0  2.8 -8.6000000 105.600000
## R3     0  0.0  0.7857143 -7.857143
```

Volver ceros con eliminación hacia atrás

```
# Tercera columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1     5 -8.0  1.0000000 -71.000000
## R2     0  2.8 -8.6000000 105.600000
## R3     0  0.0  0.7857143 -7.857143
```

```
m4 <- - Ab[2,3] / Ab[3,3]
m5 <- - Ab[1,3] / Ab[3,3]
R1 <- m5 * Ab[3,] + Ab[1,]
R2 <- m4 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1     5 -8.0  0.0000000 -61.000000
## R2     0  2.8  0.0000000  19.600000
## R3     0  0.0  0.7857143 -7.857143
```

```
# Segunda columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1     5 -8.0  0.0000000 -61.000000
## R2     0  2.8  0.0000000  19.600000
## R3     0  0.0  0.7857143 -7.857143
```

```
m6 <- - Ab[1,2] / Ab[2,2]
```

```
R1 <- m6 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1      5  0.0 0.0000000 -5.000000
## R2      0  2.8 0.0000000 19.600000
## R3      0  0.0 0.7857143 -7.857143
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]
## R1      5  0.0 0.0000000 -5.000000
## R2      0  2.8 0.0000000 19.600000
## R3      0  0.0 0.7857143 -7.857143
```

```
x1 <- Ab[1,4] / Ab[1,1]
x2 <- Ab[2,4] / Ab[2,2]
x3 <- Ab[3,4] / Ab[3,3]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] -71
```

```
II <- b[2,]; (II)
```

```
## [1] 134
```

```
III <- b[3,]; (III)
```

```
## [1] -58
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: -71"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 134"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: -58"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: -71 técnicamente es: -71"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 134 técnicamente es: 134"
```

```
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)
```

```
## [1] "En la ecuación 3: -58 tecnicamente es: -58"
```

```
cat("Soluciones del sistema:")
```

2.2.2.3.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = -1
```

```
cat("x2 =", x2)
```

```
## x2 = 7
```

```
cat("x3 =", x3)
```

```
## x3 = -10
```

2.2.2.4. Sistema 3

$$0,003000x_1 + 59,14x_2 = 59,17$$

$$5,291x_1 - 6,130x_2 = 46,78$$

Utilice redondeo a 4 cifras significativas

```
A <- matrix(c(0.003000, 59.14,
              5.291, -6.130),
            nrow = 2, byrow = TRUE)
b <- matrix(c(59.17, 46.78), nrow = 2)
Ab <- cbind(A, b); (Ab)
```

2.2.2.4.1. Procedimiento

```
##      [,1] [,2] [,3]
## [1,] 0.003 59.14 59.17
## [2,] 5.291 -6.13 46.78
```

```
# Primer pivoteo
```

```
NuevaA <- Ab[c(2, 1), ]
(NuevaA)
```

2.2.2.4.1.1. Primera eliminación

```
##      [,1] [,2] [,3]
## [1,] 5.291 -6.13 46.78
## [2,] 0.003 59.14 59.17
```

Volver ceros

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
R1 <- Ab[1,]
R2 <- round(m1 * Ab[1,] + Ab[2,], 4)
NuevaA <- rbind(R1, R2)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]
## R1 5.291 -6.1300 46.7800
## R2 0.000 59.1435 59.1435
```

Volver ceros con eliminación hacia atrás

Segunda columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]
## R1 5.291 -6.1300 46.7800
## R2 0.000 59.1435 59.1435
```

```
m2 <- - Ab[1,2] / Ab[2,2]
R1 <- round(m2 * Ab[2,] + Ab[1,], 4)
R3 <- Ab[2,]
```

```
NuevaA <- rbind(R1, R2)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]
## R1 5.291 0.0000 52.9100
## R2 0.000 59.1435 59.1435
```

Soluciones

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]
## R1 5.291 0.0000 52.9100
## R2 0.000 59.1435 59.1435
```

```
x1 <- round(Ab[1,3] / Ab[1,1], 4)
x2 <- round(Ab[2,3] / Ab[2,2], 4)
```

Comprobación

```
I <- b[1,]; (I)
```

```
## [1] 59.17
```

```
II <- b[2,]; (II)
```

```
## [1] 46.78
```

```
c1 <- round((A[1,1] * x1) + (A[1,2] * x2), 4); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 59.17"
c2 <- round((A[2,1] * x1) + (A[2,2] * x2), 4); paste0("El resultado es: ", c2)

## [1] "El resultado es: 46.78"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 59.17 tecnicamente es: 59.17"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: 46.78 tecnicamente es: 46.78"

cat("Soluciones del sistema:")
```

2.2.2.4.2. Resultados

Soluciones del sistema:

```
cat("x1 =", x1)

## x1 = 10
cat("x2 =", x2)

## x2 = 1
```

2.2.2.5. Sistema 4

$$-x_2 + 4x_3 - x_4 = -1$$

$$-x_1 + 4x_2 - x_3 = 2$$

$$-x_1 - x_3 + x_4 = 4$$

$$4x_1 - x_2 = 10$$

```
A <- matrix(c(0, -1, 4, -1,
              -1, 4, -1, 0,
              -1, 0, -1, 4,
              4, -1, 0, 0),
            nrow = 4, byrow = TRUE)
b <- matrix(c(-1, 2, 4, 10), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.2.2.5.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    0   -1    4   -1   -1
## [2,]   -1    4   -1    0    2
## [3,]   -1    0   -1    4    4
## [4,]    4   -1    0    0   10
```


Primer pivoteo

```
NuevaA <- Ab[c(4, 3, 2, 1), ]
(NuevaA)
```

2.2.2.5.1.1. Primera eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    4   -1    0    0   10
## [2,]   -1    0   -1    4    4
## [3,]   -1    4   -1    0    2
## [4,]    0   -1    4   -1   -1
```

Volver ceros

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
R4 <- Ab[4,]
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## R1    4 -1.00    0    0 10.0
## R2    0 -0.25   -1    4  6.5
## R3    0  3.75   -1    0  4.5
## R4    0 -1.00    4   -1 -1.0
```

Segundo pivoteo

```
Ab<- NuevaA[c(1,3,2,4), ]
(Ab)
```

2.2.2.5.1.2. Segunda eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## R1    4 -1.00    0    0 10.0
## R3    0  3.75   -1    0  4.5
## R2    0 -0.25   -1    4  6.5
## R4    0 -1.00    4   -1 -1.0
```

Volver ceros

```
m3 <- - Ab[3,2] / Ab[2,2]
m4 <- - Ab[4,2] / Ab[2,2]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m3 * Ab[2,] + Ab[3,]
R4 <- m4 * Ab[2,] + Ab[4,]
NuevaA <- rbind(R1, R2, R3, R4)
```

```
(NuevaA)
```

```
##      [,1] [,2]      [,3] [,4] [,5]
## R1     4 -1.00  0.000000    0 10.0
## R2     0  3.75 -1.000000    0  4.5
## R3     0  0.00 -1.066667    4  6.8
## R4     0  0.00  3.733333   -1  0.2
```

```
# Segundo pivoteo
```

```
Ab<- NuevaA[c(1, 2, 4, 3), ]
(Ab)
```

2.2.2.5.1.3. Tercera eliminación

```
##      [,1] [,2]      [,3] [,4] [,5]
## R1     4 -1.00  0.000000    0 10.0
## R2     0  3.75 -1.000000    0  4.5
## R4     0  0.00  3.733333   -1  0.2
## R3     0  0.00 -1.066667    4  6.8
```

```
# Volver ceros
```

```
m5 <- - Ab[4,3] / Ab[3,3]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- m5 * Ab[3,] + Ab[4,]
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## R1     4 -1.00  0.000000  0.000000 10.000000
## R2     0  3.75 -1.000000  0.000000  4.500000
## R3     0  0.00  3.733333 -1.000000  0.200000
## R4     0  0.00  0.000000  3.714286  6.857143
```

Volver ceros con eliminación hacia atrás

```
# Cuarta columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## R1     4 -1.00  0.000000  0.000000 10.000000
## R2     0  3.75 -1.000000  0.000000  4.500000
## R3     0  0.00  3.733333 -1.000000  0.200000
## R4     0  0.00  0.000000  3.714286  6.857143
```

```
m6 <- - Ab[3,4] / Ab[4,4]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m6 * Ab[4,] + Ab[3,]
R4 <- Ab[4,]
```

```
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)

##      [,1] [,2]      [,3]      [,4]      [,5]
## R1      4 -1.00  0.000000  0.000000 10.000000
## R2      0  3.75 -1.000000  0.000000  4.500000
## R3      0  0.00  3.733333  0.000000  2.046154
## R4      0  0.00  0.000000  3.714286  6.857143
```

Tercera columna

```
Ab <- NuevaA; (Ab)

##      [,1] [,2]      [,3]      [,4]      [,5]
## R1      4 -1.00  0.000000  0.000000 10.000000
## R2      0  3.75 -1.000000  0.000000  4.500000
## R3      0  0.00  3.733333  0.000000  2.046154
## R4      0  0.00  0.000000  3.714286  6.857143
```

```
m7 <- - Ab[2,3] / Ab[3,3]
```

```
R1 <- Ab[1,]
R2 <- m7 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## R1      4 -1.00  0.000000  0.000000 10.000000
## R2      0  3.75  0.000000  0.000000  5.048077
## R3      0  0.00  3.733333  0.000000  2.046154
## R4      0  0.00  0.000000  3.714286  6.857143
```

Segunda columna

```
Ab <- NuevaA; (Ab)

##      [,1] [,2]      [,3]      [,4]      [,5]
## R1      4 -1.00  0.000000  0.000000 10.000000
## R2      0  3.75  0.000000  0.000000  5.048077
## R3      0  0.00  3.733333  0.000000  2.046154
## R4      0  0.00  0.000000  3.714286  6.857143
```

```
m8 <- - Ab[1,2] / Ab[2,2]
```

```
R1 <- m8 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## R1      4  0.00  0.000000  0.000000 11.346154
```

```
## R2    0 3.75 0.000000 0.000000 5.048077
## R3    0 0.00 3.733333 0.000000 2.046154
## R4    0 0.00 0.000000 3.714286 6.857143
```

```
# Soluciones
```

```
Ab<- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## R1    4 0.00 0.000000 0.000000 11.346154
## R2    0 3.75 0.000000 0.000000 5.048077
## R3    0 0.00 3.733333 0.000000 2.046154
## R4    0 0.00 0.000000 3.714286 6.857143
```

```
x1<- Ab[1,5] / Ab[1,1]
x2<- Ab[2,5] / Ab[2,2]
x3<- Ab[3,5] / Ab[3,3]
x4<- Ab[4,5] / Ab[4,4]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] -1
```

```
II <- b[2,]; (II)
```

```
## [1] 2
```

```
III <- b[3,]; (III)
```

```
## [1] 4
```

```
IV <- b[4,]; (IV)
```

```
## [1] 10
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: -1"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 2"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 4"
```

```
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)
```

```
## [1] "El resultado es: 10"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: -1 técnicamente es: -1"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 2 técnicamente es: 2"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 4 técnicamente es: 4"
```

```
paste0("En la ecuación 3: ", c4, " técnicamente es: ", IV)
```

```
## [1] "En la ecuación 3: 10 técnicamente es: 10"
```

```
cat("Soluciones del sistema:")
```

2.2.2.5.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 2.836538
```

```
cat("x2 =", x2)
```

```
## x2 = 1.346154
```

```
cat("x3 =", x3)
```

```
## x3 = 0.5480769
```

```
cat("x4 =", x4)
```

```
## x4 = 1.846154
```

2.2.2.6. Sistema 5

$$0,00031000x_1 + 1,000000x_2 = 3,000000$$

$$1,00045534x_1 - 1,00034333x_2 = 7,000$$

```
A <- matrix(c(0.00031000, 1.000000,
              1.00045534, 1.00034333),
            nrow = 2, byrow = TRUE)
b <- matrix(c(3.000000, 7.000), nrow = 2)
Ab <- cbind(A, b); (Ab)
```

2.2.2.6.1. Procedimiento

```
##           [,1]      [,2] [,3]
## [1,] 0.000310 1.000000    3
## [2,] 1.000455 1.000343    7
```

```
# Primer pivoteo
```

```
NuevaA <- Ab[c(2, 1), ]
(NuevaA)
```

2.2.2.6.1.1. Primera eliminación

```
##          [,1]      [,2] [,3]
## [1,] 1.000455 1.000343    7
## [2,] 0.000310 1.000000    3
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
NuevaA <- rbind(R1, R2)
(NuevaA)
```

```
##          [,1]      [,2]      [,3]
## R1 1.000455 1.000343 7.000000
## R2 0.000000 0.999690 2.997831
```

Volver ceros con eliminación hacia atrás

```
# Segunda columna
```

```
Ab <- NuevaA; (Ab)
```

```
##          [,1]      [,2]      [,3]
## R1 1.000455 1.000343 7.000000
## R2 0.000000 0.999690 2.997831
```

```
m2 <- - Ab[1,2] / Ab[2,2]
R1 <- m2 * Ab[2,] + Ab[1,]
R3 <- Ab[2,]
```

```
NuevaA <- rbind(R1, R2)
(NuevaA)
```

```
##          [,1]      [,2]      [,3]
## R1 1.000455 0.000000 4.000210
## R2 0.000000 0.99969 2.997831
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##          [,1]      [,2]      [,3]
## R1 1.000455 0.000000 4.000210
## R2 0.000000 0.99969 2.997831
```

```
x1 <- Ab[1,3] / Ab[1,1]
x2 <- Ab[2,3] / Ab[2,2]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 3
```

```
II <- b[2,]; (II)
```

```
## [1] 7
```



```
c1 <- (A[1,1] * x1) + (A[1,2] * x2); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 3"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2); paste0("El resultado es: ", c2)
```

```
## [1] "El resultado es: 7"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 3 técnicamente es: 3"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 7 técnicamente es: 7"
```

```
cat("Soluciones del sistema:")
```

2.2.2.6.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 3.998389
```

```
cat("x2 =", x2)
```

```
## x2 = 2.99876
```

2.2.2.7. Sistema 6

$$A = \begin{bmatrix} 14 & 14 & -9 & 3 & -5 \\ 14 & 52 & -15 & 2 & -32 \\ -9 & -15 & 36 & -5 & 16 \\ 3 & 2 & -15 & 47 & 49 \\ -5 & 32 & 16 & 49 & 79 \end{bmatrix}$$

$$b = \begin{bmatrix} -15 \\ -100 \\ 106 \\ 329 \\ 463 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

Resolver para A

```
A <- matrix(c(14, 14, -9, 3, -5,
              14, 52, -15, 2, -32,
              -9, -15, 36, -5, 16,
```

```

      3, 2, -5, 47, 49,
      -5, 32, 16, 49, 79),
  nrow = 5, byrow = TRUE)
b <- matrix(c(-15, -100, 106, 329, 463), nrow = 5)

Ab <- cbind(A, b); (Ab)

```

2.2.2.7.1. Procedimiento

```

##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]  14   14  -9    3  -5  -15
## [2,]  14   52 -15    2 -32 -100
## [3,]  -9  -15  36   -5  16  106
## [4,]   3    2  -5   47  49  329
## [5,]  -5   32  16   49  79  463

```

2.2.2.7.1.1. Primera eliminación No se necesita realizar pivoteo

Volver ceros

```

m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
m3 <- - Ab[4,1] / Ab[1,1]
m4 <- - Ab[5,1] / Ab[1,1]

R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
R4 <- m3 * Ab[1,] + Ab[4,]
R5 <- m4 * Ab[1,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)

```

```

##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1  14   14 -9.000000  3.000000 -5.00000 -15.00000
## R2   0   38 -6.000000 -1.000000 -27.00000 -85.00000
## R3   0   -6 30.214286 -3.071429 12.78571  96.35714
## R4   0   -1 -3.071429 46.357143 50.07143 332.21429
## R5   0   37 12.785714 50.071429 77.21429 457.64286

```

2.2.2.7.1.2. Segunda eliminación No se necesita realizar pivoteo

Volver ceros

```
Ab <- NuevaA; (Ab)
```

```

##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1  14   14 -9.000000  3.000000 -5.00000 -15.00000
## R2   0   38 -6.000000 -1.000000 -27.00000 -85.00000
## R3   0   -6 30.214286 -3.071429 12.78571  96.35714
## R4   0   -1 -3.071429 46.357143 50.07143 332.21429
## R5   0   37 12.785714 50.071429 77.21429 457.64286

```

```

m5 <- - Ab[3,2] / Ab[2,2]
m6 <- - Ab[4,2] / Ab[2,2]
m7 <- - Ab[5,2] / Ab[2,2]

R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m5 * Ab[2,] + Ab[3,]
R4 <- m6 * Ab[2,] + Ab[4,]
R5 <- m7 * Ab[2,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)

```

```

##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R3     0     0 29.266917 -3.229323  8.522556 82.93609
## R4     0     0 -3.229323 46.330827 49.360902 329.97744
## R5     0     0 18.627820 51.045113 103.503759 540.40602

```

Primer pivoteo

```

Ab <- NuevaA[c(1, 2, 5, 4, 3), ]
(Ab)

```

2.2.2.7.1.3. Tercera eliminación

```

##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R5     0     0 18.627820 51.045113 103.503759 540.40602
## R4     0     0 -3.229323 46.330827 49.360902 329.97744
## R3     0     0 29.266917 -3.229323  8.522556 82.93609

```

Volver ceros

```

Ab <- NuevaA; (Ab)

```

```

##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R3     0     0 29.266917 -3.229323  8.522556 82.93609
## R4     0     0 -3.229323 46.330827 49.360902 329.97744
## R5     0     0 18.627820 51.045113 103.503759 540.40602

```

```

m5 <- - Ab[3,2] / Ab[2,2]
m6 <- - Ab[4,2] / Ab[2,2]
m7 <- - Ab[5,2] / Ab[2,2]

R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m5 * Ab[2,] + Ab[3,]
R4 <- m6 * Ab[2,] + Ab[4,]
R5 <- m7 * Ab[2,] + Ab[5,]

```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R3     0     0 29.266917 -3.229323  8.522556  82.93609
## R4     0     0 -3.229323 46.330827 49.360902 329.97744
## R5     0     0 18.627820 51.045113 103.503759 540.40602
```

2.2.2.7.1.4. Cuarta eliminación No se necesita hacer pivoteo

```
# Volver ceros
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R3     0     0 29.266917 -3.229323  8.522556  82.93609
## R4     0     0 -3.229323 46.330827 49.360902 329.97744
## R5     0     0 18.627820 51.045113 103.503759 540.40602
```

```
m8 <- - Ab[4,3] / Ab[3,3]
```

```
m9 <- - Ab[5,3] / Ab[3,3]
```

```
R1 <- Ab[1,]
```

```
R2 <- Ab[2,]
```

```
R3 <- Ab[3,]
```

```
R4 <- m8 * Ab[3,] + Ab[4,]
```

```
R5 <- m9 * Ab[3,] + Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
```

```
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R3     0     0 29.26692 -3.229323  8.522556  82.93609
## R4     0     0 0.000000 45.974502 50.301285 339.12864
## R5     0     0 0.000000 53.100514 98.079319 487.61882
```

```
# Segundo pivoteo
```

```
Ab <- NuevaA[c(1, 2, 3, 5, 4), ]
```

```
(Ab)
```

2.2.2.7.1.5. Quinta eliminación

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    14 -9.000000  3.000000 -5.000000 -15.00000
## R2     0    38 -6.000000 -1.000000 -27.000000 -85.00000
## R3     0     0 29.26692 -3.229323  8.522556  82.93609
## R5     0     0 0.000000 53.100514 98.079319 487.61882
```

```
## R4    0    0  0.00000 45.974502  50.301285 339.12864
```

```
# Volver ceros
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14   14 -9.00000   3.000000  -5.000000 -15.00000
## R2     0   38 -6.00000  -1.000000 -27.000000 -85.00000
## R3     0    0 29.26692 -3.229323   8.522556  82.93609
## R4     0    0  0.00000 45.974502  50.301285 339.12864
## R5     0    0  0.00000 53.100514  98.079319 487.61882
```

```
m10 <- - Ab[5,4] / Ab[4,4]
```

```
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
R5 <- m10 * Ab[4,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14   14 -9.00000   3.000000  -5.000000 -15.00000
## R2     0   38 -6.00000  -1.000000 -27.000000 -85.00000
## R3     0    0 29.26692 -3.229323   8.522556  82.93609
## R4     0    0  0.00000 45.974502  50.301285 339.12864
## R5     0    0  0.00000  0.000000  39.981375  95.92550
```

Volver ceros con eliminación hacia atrás

```
# Quinta columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14   14 -9.00000   3.000000  -5.000000 -15.00000
## R2     0   38 -6.00000  -1.000000 -27.000000 -85.00000
## R3     0    0 29.26692 -3.229323   8.522556  82.93609
## R4     0    0  0.00000 45.974502  50.301285 339.12864
## R5     0    0  0.00000  0.000000  39.981375  95.92550
```

```
m11 <- - Ab[1,5] / Ab[5,5]
m12 <- - Ab[2,5] / Ab[5,5]
m13 <- - Ab[3,5] / Ab[5,5]
m14 <- - Ab[4,5] / Ab[5,5]
R1 <- m11 * Ab[5,] + Ab[1,]
R2 <- m12 * Ab[5,] + Ab[2,]
R3 <- m13 * Ab[5,] + Ab[3,]
R4 <- m14 * Ab[5,] + Ab[4,]
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1   14   14 -9.00000  3.000000 -8.881784e-16 -3.003727
## R2    0   38 -6.00000 -1.000000  0.000000e+00 -20.220124
## R3    0    0 29.26692 -3.229323  0.000000e+00  62.488307
## R4    0    0  0.00000 45.974502  0.000000e+00 218.443053
## R5    0    0  0.00000  0.000000  3.998138e+01  95.925501
```

```
# Cuarta columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1   14   14 -9.00000  3.000000 -8.881784e-16 -3.003727
## R2    0   38 -6.00000 -1.000000  0.000000e+00 -20.220124
## R3    0    0 29.26692 -3.229323  0.000000e+00  62.488307
## R4    0    0  0.00000 45.974502  0.000000e+00 218.443053
## R5    0    0  0.00000  0.000000  3.998138e+01  95.925501
```

```
m15 <- - Ab[1,4] / Ab[4,4]
m16 <- - Ab[2,4] / Ab[4,4]
m17 <- - Ab[3,4] / Ab[4,4]
R1 <- m15 * Ab[4,] + Ab[1,]
R2 <- m16 * Ab[4,] + Ab[2,]
R3 <- m17 * Ab[4,] + Ab[3,]
R4 <- Ab[4,]
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1   14   14 -9.00000  0.000000e+00 -8.881784e-16 -17.25791
## R2    0   38 -6.00000  0.000000e+00  0.000000e+00 -15.46873
## R3    0    0 29.26692  4.440892e-16  0.000000e+00  77.83210
## R4    0    0  0.00000  4.597450e+01  0.000000e+00 218.44305
## R5    0    0  0.00000  0.000000e+00  3.998138e+01  95.92550
```

```
# Tercera columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1   14   14 -9.00000  0.000000e+00 -8.881784e-16 -17.25791
## R2    0   38 -6.00000  0.000000e+00  0.000000e+00 -15.46873
## R3    0    0 29.26692  4.440892e-16  0.000000e+00  77.83210
## R4    0    0  0.00000  4.597450e+01  0.000000e+00 218.44305
## R5    0    0  0.00000  0.000000e+00  3.998138e+01  95.92550
```

```
m18 <- - Ab[1,3] / Ab[3,3]
m19 <- - Ab[2,3] / Ab[3,3]
R1 <- m18 * Ab[3,] + Ab[1,]
R2 <- m19 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
R5 <- Ab[5,]
```



```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14   14  0.00000 1.365638e-16 -8.881784e-16  6.676582
## R2     0   38  0.00000 9.104257e-17  0.000000e+00  0.487602
## R3     0    0 29.26692 4.440892e-16  0.000000e+00 77.832100
## R4     0    0  0.00000 4.597450e+01  0.000000e+00 218.443053
## R5     0    0  0.00000 0.000000e+00  3.998138e+01 95.925501
```

```
# Segunda columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14   14  0.00000 1.365638e-16 -8.881784e-16  6.676582
## R2     0   38  0.00000 9.104257e-17  0.000000e+00  0.487602
## R3     0    0 29.26692 4.440892e-16  0.000000e+00 77.832100
## R4     0    0  0.00000 4.597450e+01  0.000000e+00 218.443053
## R5     0    0  0.00000 0.000000e+00  3.998138e+01 95.925501
```

```
m20 <- - Ab[1,2] / Ab[2,2]
R1 <- m20 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    0  0.00000 1.030219e-16 -8.881784e-16  6.496939
## R2     0   38  0.00000 9.104257e-17  0.000000e+00  0.487602
## R3     0    0 29.26692 4.440892e-16  0.000000e+00 77.832100
## R4     0    0  0.00000 4.597450e+01  0.000000e+00 218.443053
## R5     0    0  0.00000 0.000000e+00  3.998138e+01 95.925501
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]      [,6]
## R1    14    0  0.00000 1.030219e-16 -8.881784e-16  6.496939
## R2     0   38  0.00000 9.104257e-17  0.000000e+00  0.487602
## R3     0    0 29.26692 4.440892e-16  0.000000e+00 77.832100
## R4     0    0  0.00000 4.597450e+01  0.000000e+00 218.443053
## R5     0    0  0.00000 0.000000e+00  3.998138e+01 95.925501
```

```
X <- matrix(c(1, 2, 3, 4, 5), nrow = 5)
```

```
X[1,] <- Ab[1,6] / Ab[1,1]
X[2,] <- Ab[2,6] / Ab[2,2]
X[3,] <- Ab[3,6] / Ab[3,3]
```

```
X[4,] <- Ab[4,6] / Ab[4,4]
X[5,] <- Ab[5,6] / Ab[5,5]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] -15
```

```
II <- b[2,]; (II)
```

```
## [1] -100
```

```
III <- b[3,]; (III)
```

```
## [1] 106
```

```
IV <- b[4,]; (IV)
```

```
## [1] 329
```

```
V <- b[5,]; (V)
```

```
## [1] 463
```

```
c1 <- (A[1,1] * X[1,]) + (A[1,2] * X[2,]) + (A[1,3] * X[3,]) + (A[1,4] * X[4,]) + (A[1,5] * X[5,]); paste0
```

```
## [1] "El resultado es: -15"
```

```
c2 <- (A[2,1] * X[1,]) + (A[2,2] * X[2,]) + (A[2,3] * X[3,]) + (A[2,4] * X[4,]) + (A[2,5] * X[5,]); paste0
```

```
## [1] "El resultado es: -100"
```

```
c3 <- (A[3,1] * X[1,]) + (A[3,2] * X[2,]) + (A[3,3] * X[3,]) + (A[3,4] * X[4,]) + (A[3,5] * X[5,]); paste0
```

```
## [1] "El resultado es: 106"
```

```
c4 <- (A[4,1] * X[1,]) + (A[4,2] * X[2,]) + (A[4,3] * X[3,]) + (A[4,4] * X[4,]) + (A[4,5] * X[5,]); paste0
```

```
## [1] "El resultado es: 329"
```

```
c5 <- (A[5,1] * X[1,]) + (A[5,2] * X[2,]) + (A[5,3] * X[3,]) + (A[5,4] * X[4,]) + (A[5,5] * X[5,]); paste0
```

```
## [1] "El resultado es: 463"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: -15 técnicamente es: -15"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -100 técnicamente es: -100"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 106 técnicamente es: 106"
```

```
paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)
```

```
## [1] "En la ecuación 4: 329 técnicamente es: 329"
```

```
paste0("En la ecuación 5: ", c5, " tecnicamente es: ", V)
```

```
## [1] "En la ecuación 5: 463 tecnicamente es: 463"
```

```
cat("Soluciones del sistema:")
```

2.2.2.7.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 = ", X[1,])
```

```
## x1 = 0.4640671
```

```
cat("x2 = ", X[2,])
```

```
## x2 = 0.01283163
```

```
cat("x3 = ", X[3,])
```

```
## x3 = 2.659388
```

```
cat("x4 = ", X[4,])
```

```
## x4 = 4.751396
```

```
cat("x5 = ", X[5,])
```

```
## x5 = 2.399255
```

2.2.2.8. Sistema 7

$$A = \begin{bmatrix} \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{2} & 1 & 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 9 \\ 8 \\ 8 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Resolver para A

```
A <- matrix(c(1/4, 1/5, 1/6,
              1/3, 1/4, 1/5,
              1/2, 1, 2),
            nrow = 3, byrow = TRUE)
b <- matrix(c(9, 8, 8), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.2.8.1. Procedimiento

```
##           [,1] [,2]      [,3] [,4]
## [1,] 0.2500000 0.20 0.1666667    9
## [2,] 0.3333333 0.25 0.2000000    8
## [3,] 0.5000000 1.00 2.0000000    8
```

```
# Primer pivoteo
```

```
Ab<- Ab[c(3, 2, 1), ]
(Ab)
```

2.2.2.8.1.1. Primera eliminación

```
##           [,1] [,2]      [,3] [,4]
## [1,] 0.5000000 1.00 2.0000000    8
## [2,] 0.3333333 0.25 0.2000000    8
## [3,] 0.2500000 0.20 0.1666667    9
```

```
# Volver ceros
```

```
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]

R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##           [,1]      [,2]      [,3]      [,4]
## R1  0.5  1.0000000  2.0000000  8.0000000
## R2  0.0 -0.4166667 -1.1333333  2.6666667
## R3  0.0 -0.3000000 -0.8333333  5.0000000
```

2.2.2.8.1.2. Segunda eliminación No se necesita realizar pivoteo

```
# Volver ceros
```

```
Ab <- NuevaA; (Ab)
```

```
##           [,1]      [,2]      [,3]      [,4]
## R1  0.5  1.0000000  2.0000000  8.0000000
## R2  0.0 -0.4166667 -1.1333333  2.6666667
## R3  0.0 -0.3000000 -0.8333333  5.0000000
```

```
m3 <- - Ab[3,2] / Ab[2,2]
```

```
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m3 * Ab[2,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##           [,1]      [,2]      [,3]      [,4]
## R1  0.5  1.0000000  2.0000000  8.0000000
```

```
## R2  0.0 -0.4166667 -1.1333333 2.666667
## R3  0.0  0.0000000 -0.0173333 3.080000
```

Volver ceros con eliminación hacia atrás

Tercera columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1  0.5  1.0000000  2.0000000  8.000000
## R2  0.0 -0.4166667 -1.1333333  2.666667
## R3  0.0  0.0000000 -0.0173333  3.080000
```

```
m4 <- - Ab[1,3] / Ab[3,3]
m5 <- - Ab[2,3] / Ab[3,3]
R1 <- m4 * Ab[3,] + Ab[1,]
R2 <- m5 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1  0.5  1.0000000  0.000000e+00  363.3846
## R2  0.0 -0.4166667 -2.220446e-16 -198.7179
## R3  0.0  0.0000000 -1.733333e-02   3.0800
```

Segunda columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1  0.5  1.0000000  0.000000e+00  363.3846
## R2  0.0 -0.4166667 -2.220446e-16 -198.7179
## R3  0.0  0.0000000 -1.733333e-02   3.0800
```

```
m6 <- - Ab[1,2] / Ab[2,2]
R1 <- m6 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1  0.5  0.0000000 -5.329071e-16 -113.5385
## R2  0.0 -0.4166667 -2.220446e-16 -198.7179
## R3  0.0  0.0000000 -1.733333e-02   3.0800
```

Soluciones

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1  0.5  0.0000000 -5.329071e-16 -113.5385
## R2  0.0 -0.4166667 -2.220446e-16 -198.7179
```

```
## R3 0.0 0.0000000 -1.733333e-02 3.0800
X <- matrix(c(1, 2, 3), nrow = 3)

X[1,] <- Ab[1,4] / Ab[1,1]
X[2,] <- Ab[2,4] / Ab[2,2]
X[3,] <- Ab[3,4] / Ab[3,3]

# Comprobación

I <- b[1,]; (I)

## [1] 9
II <- b[2,]; (II)

## [1] 8
III <- b[3,]; (III)

## [1] 8
c1 <- (A[1,1] * X[1,]) + (A[1,2] * X[2,]) + (A[1,3] * X[3,]); paste0("El resultado es: ", c1)

## [1] "El resultado es: 9.000000000000002"
c2 <- (A[2,1] * X[1,]) + (A[2,2] * X[2,]) + (A[2,3] * X[3,]); paste0("El resultado es: ", c2)

## [1] "El resultado es: 8.000000000000003"
c3 <- (A[3,1] * X[1,]) + (A[3,2] * X[2,]) + (A[3,3] * X[3,]); paste0("El resultado es: ", c3)

## [1] "El resultado es: 8.000000000000006"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 9.000000000000002 tecnicamente es: 9"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: 8.000000000000003 tecnicamente es: 8"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: 8.000000000000006 tecnicamente es: 8"

cat("Soluciones del sistema:")
```

2.2.2.8.2. Resultados

Soluciones del sistema:

```
cat("x1 = ", X[1,])
```

```
## x1 = -227.0769
```

```
cat("x2 = ", X[2,])
```

```
## x2 = 476.9231
```



```
cat("x3 = ", X[3,])
```

```
## x3 = -177.6923
```

2.2.2.9. Sistema 8

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{7} \\ \frac{1}{8} \\ \frac{1}{9} \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Resolver para A

```
A <- matrix(c(1, 1/2, 1/3, 1/4,
              1/2, 1/3, 1/4, 1/5,
              1/3, 1/4, 1/5, 1/6,
              1/4, 1/5, 1/6, 1/7),
            nrow = 4, byrow = TRUE)
b <- matrix(c(1/6, 1/7, 1/8, 1/9), nrow = 4)
Ab <- cbind(A, b); (Ab)
```

2.2.2.9.1. Procedimiento

```
##           [,1]      [,2]      [,3]      [,4]      [,5]
## [1,] 1.0000000 0.5000000 0.3333333 0.2500000 0.1666667
## [2,] 0.5000000 0.3333333 0.2500000 0.2000000 0.1428571
## [3,] 0.3333333 0.2500000 0.2000000 0.1666667 0.1250000
## [4,] 0.2500000 0.2000000 0.1666667 0.1428571 0.1111111
```

2.2.2.9.1.1. Primera eliminación No se necesita pivoteo

Volver ceros

```
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
m3 <- - Ab[4,1] / Ab[1,1]

R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
R4 <- m3 * Ab[1,] + Ab[4,]
```

```
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1 0.50000000 0.33333333 0.25000000 0.16666667
## R2      0 0.08333333 0.08333333 0.07500000 0.05952381
## R3      0 0.08333333 0.08888889 0.08333333 0.06944444
## R4      0 0.07500000 0.08333333 0.08035714 0.06944444
```

2.2.2.9.1.2. Segunda eliminación No se necesita realizar pivoteo

```
# Volver ceros
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1 0.50000000 0.33333333 0.25000000 0.16666667
## R2      0 0.08333333 0.08333333 0.07500000 0.05952381
## R3      0 0.08333333 0.08888889 0.08333333 0.06944444
## R4      0 0.07500000 0.08333333 0.08035714 0.06944444
```

```
m4 <- - Ab[3,2] / Ab[2,2]
```

```
m5 <- - Ab[4,2] / Ab[2,2]
```

```
R1 <- Ab[1,]
```

```
R2 <- Ab[2,]
```

```
R3 <- m4 * Ab[2,] + Ab[3,]
```

```
R4 <- m5 * Ab[2,] + Ab[4,]
```

```
NuevaA <- rbind(R1, R2, R3, R4)
```

```
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1 5.000000e-01 0.33333333 0.25000000 0.16666667
## R2      0 8.333333e-02 0.08333333 0.07500000 0.05952381
## R3      0 -1.387779e-17 0.00555556 0.00833333 0.009920635
## R4      0 0.000000e+00 0.00833333 0.01285714 0.015873016
```

2.2.2.9.1.3. Tercera eliminación No se necesita realizar pivoteo

```
# Volver ceros
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1 5.000000e-01 0.33333333 0.25000000 0.16666667
## R2      0 8.333333e-02 0.08333333 0.07500000 0.05952381
## R3      0 -1.387779e-17 0.00555556 0.00833333 0.009920635
## R4      0 0.000000e+00 0.00833333 0.01285714 0.015873016
```

```
m6 <- - Ab[4,3] / Ab[3,3]
```

```
R1 <- Ab[1,]
```

```
R2 <- Ab[2,]
```

```
R3 <- Ab[3,]
R4 <- m6 * Ab[3,] + Ab[4,]
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1    1  5.000000e-01  0.333333333  0.250000000  0.166666667
## R2    0  8.333333e-02  0.083333333  0.075000000  0.0595238095
## R3    0 -1.387779e-17  0.005555556  0.008333333  0.0099206349
## R4    0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

Volver ceros con eliminación hacia atrás

Cuarta columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1    1  5.000000e-01  0.333333333  0.250000000  0.166666667
## R2    0  8.333333e-02  0.083333333  0.075000000  0.0595238095
## R3    0 -1.387779e-17  0.005555556  0.008333333  0.0099206349
## R4    0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

```
m7 <- - Ab[1,4] / Ab[4,4]
m8 <- - Ab[2,4] / Ab[4,4]
m9 <- - Ab[3,4] / Ab[4,4]
R1 <- m7 * Ab[4,] + Ab[1,]
R2 <- m8 * Ab[4,] + Ab[2,]
R3 <- m9 * Ab[4,] + Ab[3,]
R4 <- Ab[4,]
```

```
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1    1  5.000000e-01  0.333333333  0.000000000 -0.5277777778
## R2    0  8.333333e-02  0.083333333  0.000000000 -0.1488095238
## R3    0 -4.996004e-16  0.005555556  0.000000000 -0.0132275132
## R4    0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

Tercera columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1    1  5.000000e-01  0.333333333  0.000000000 -0.5277777778
## R2    0  8.333333e-02  0.083333333  0.000000000 -0.1488095238
## R3    0 -4.996004e-16  0.005555556  0.000000000 -0.0132275132
## R4    0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

```
m10 <- - Ab[1,3] / Ab[3,3]
m11 <- - Ab[2,3] / Ab[3,3]
R1 <- m10 * Ab[3,] + Ab[1,]
R2 <- m11 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
```

```
R4 <- Ab[4,]

NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)

##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1  5.000000e-01  0.000000000  0.000000000  0.2658730159
## R2      0  8.333333e-02  0.000000000  0.000000000  0.0496031746
## R3      0 -4.996004e-16  0.005555556  0.000000000 -0.0132275132
## R4      0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

Segunda columna

```
Ab <- NuevaA; (Ab)

##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1  5.000000e-01  0.000000000  0.000000000  0.2658730159
## R2      0  8.333333e-02  0.000000000  0.000000000  0.0496031746
## R3      0 -4.996004e-16  0.005555556  0.000000000 -0.0132275132
## R4      0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

```
m12 <- - Ab[1,2] / Ab[2,2]
R1 <- m12 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
```

```
NuevaA <- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1  0.000000e+00  0.000000000  0.000000000 -0.0317460317
## R2      0  8.333333e-02  0.000000000  0.000000000  0.0496031746
## R3      0 -4.996004e-16  0.005555556  0.000000000 -0.0132275132
## R4      0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

Soluciones

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## R1      1  0.000000e+00  0.000000000  0.000000000 -0.0317460317
## R2      0  8.333333e-02  0.000000000  0.000000000  0.0496031746
## R3      0 -4.996004e-16  0.005555556  0.000000000 -0.0132275132
## R4      0  2.081668e-17  0.000000000  0.0003571429  0.0009920635
```

```
X <- matrix(c(1, 2, 3, 4), nrow = 4)
```

```
X[1,] <- Ab[1,5] / Ab[1,1]
X[2,] <- Ab[2,5] / Ab[2,2]
X[3,] <- Ab[3,5] / Ab[3,3]
X[4,] <- Ab[4,5] / Ab[4,4]
```

Comprobación

```

I <- b[1,]; (I)

## [1] 0.1666667
II <- b[2,]; (II)

## [1] 0.1428571
III <- b[3,]; (III)

## [1] 0.125
IV <- b[4,]; (IV)

## [1] 0.1111111
c1 <- (A[1,1] * X[1,]) + (A[1,2] * X[2,]) + (A[1,3] * X[3,]) + (A[1,4] * X[4,]); paste0("El resultado es:
## [1] "El resultado es: 0.16666666666666657"
c2 <- (A[2,1] * X[1,]) + (A[2,2] * X[2,]) + (A[2,3] * X[3,]) + (A[2,4] * X[4,]); paste0("El resultado es:
## [1] "El resultado es: 0.142857142857136"
c3 <- (A[3,1] * X[1,]) + (A[3,2] * X[2,]) + (A[3,3] * X[3,]) + (A[3,4] * X[4,]); paste0("El resultado es:
## [1] "El resultado es: 0.124999999999995"
c4 <- (A[4,1] * X[1,]) + (A[4,2] * X[2,]) + (A[4,3] * X[3,]) + (A[4,4] * X[4,]); paste0("El resultado es:
## [1] "El resultado es: 0.111111111111107"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 0.16666666666666657 técnicamente es: 0.1666666666666667"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 0.142857142857136 técnicamente es: 0.142857142857143"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 0.124999999999995 técnicamente es: 0.125"
paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)

## [1] "En la ecuación 4: 0.111111111111107 técnicamente es: 0.111111111111111"

cat("Soluciones del sistema:")

2.2.2.9.2. Resultados

## Soluciones del sistema:
cat("x1 = ", X[1,])

## x1 = -0.03174603
cat("x2 = ", X[2,])

## x2 = 0.5952381

```

```
cat("x3 = ", X[3,])
```

```
## x3 = -2.380952
```

```
cat("x4 = ", X[4,])
```

```
## x4 = 2.777778
```

2.2.2.10. Sistema 9

$$2x_1 + x_2 - x_3 + x_4 - 3x_5 = 7$$

$$x_1 + 2x_3 - x_4 + x_5 = 2$$

$$-2x_2 - x_3 + x_4 - x_5 = -5$$

$$3x_1 + x_2 - 4x_3 + 5x_5 = 6$$

$$x_1 - x_2 - x_3 - x_4 + x_5 = 3$$

```
A <- matrix(c(2, 1, -1, 1, -3,
              1, 0, 2, -1, 1,
              0, -2, -1, 1, -1,
              3, 1, -4, 0, 5,
              1, -1, -1, -1, 1),
            nrow = 5, byrow = TRUE)
b <- matrix(c(7, 2, -5, 6, 3), nrow = 5)

Ab <- cbind(A, b); (Ab)
```

2.2.2.10.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]    2    1   -1    1   -3    7
## [2,]    1    0    2   -1    1    2
## [3,]    0   -2   -1    1   -1   -5
## [4,]    3    1   -4    0    5    6
## [5,]    1   -1   -1   -1    1    3
```

Primer pivoteo

```
NuevaA <- Ab[c(4,2, 3, 1, 5), ]
(NuevaA)
```

2.2.2.10.1.1. Primera eliminación

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]    3    1   -4    0    5    6
## [2,]    1    0    2   -1    1    2
## [3,]    0   -2   -1    1   -1   -5
## [4,]    2    1   -1    1   -3    7
## [5,]    1   -1   -1   -1    1    3
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
m3 <- - Ab[4,1] / Ab[1,1]
m4 <- - Ab[5,1] / Ab[1,1]
R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
R4 <- m3 * Ab[1,] + Ab[4,]
R5 <- m4 * Ab[1,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2]      [,3] [,4]      [,5] [,6]
## R1    3  1.0000000 -4.0000000  0  5.0000000  6
## R2    0 -0.3333333  3.3333333 -1 -0.6666667  0
## R3    0 -2.0000000 -1.0000000  1 -1.0000000 -5
## R4    0  0.3333333  1.6666667  1 -6.3333333  3
## R5    0 -1.3333333  0.3333333 -1 -0.6666667  1
```

```
# Segundo pivoteo
```

```
Ab <- NuevaA[c(1,3, 2, 4, 5), ]
(Ab)
```

2.2.2.10.1.2. Segunda Eliminación

```
##      [,1]      [,2]      [,3] [,4]      [,5] [,6]
## R1    3  1.0000000 -4.0000000  0  5.0000000  6
## R3    0 -2.0000000 -1.0000000  1 -1.0000000 -5
## R2    0 -0.3333333  3.3333333 -1 -0.6666667  0
## R4    0  0.3333333  1.6666667  1 -6.3333333  3
## R5    0 -1.3333333  0.3333333 -1 -0.6666667  1
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[3,2] / Ab[2,2]
m6 <- - Ab[4,2] / Ab[2,2]
m7 <- - Ab[5,2] / Ab[2,2]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m5 * Ab[2,] + Ab[3,]
R4 <- m6 * Ab[2,] + Ab[4,]
R5 <- m7 * Ab[2,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5] [,6]
## R1    3  1.0000000 -4.000000  0.000000e+00  5.0000000  6
```



```
## R2    0 -0.3333333  3.333333 -1.000000e+00 -0.6666667    0
## R3    0  0.0000000 -21.000000  7.000000e+00  3.0000000   -5
## R4    0  0.0000000  5.000000 -2.220446e-16 -7.0000000    3
## R5    0  0.0000000 -13.000000  3.000000e+00  2.0000000    1
```

2.2.2.10.1.3. Tercera Eliminación No necesita pivoteo

Volver ceros

```
Ab <- NuevaA
m8 <- - Ab[4,3] / Ab[3,3]
m9 <- - Ab[5,3] / Ab[3,3]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- m8 * Ab[3,] + Ab[4,]
R5 <- m9 * Ab[3,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## R1     3  1.0000000 -4.000000  0.000000  5.0000000  6.000000
## R2     0 -0.3333333  3.333333 -1.000000 -0.6666667  0.000000
## R3     0  0.0000000 -21.000000  7.000000  3.0000000 -5.000000
## R4     0  0.0000000  0.000000  1.666667 -6.2857143  1.809524
## R5     0  0.0000000  0.000000 -1.333333  0.1428571  4.095238
```

2.2.2.10.1.4. Cuarta Eliminación No necesita pivoteo

Volver ceros

```
Ab <- NuevaA
m10 <- - Ab[5,4] / Ab[4,4]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
R4 <- Ab[4,]
R5 <- m10 * Ab[4,] + Ab[5,]
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## R1     3  1.0000000 -4.000000  0.000000  5.0000000  6.000000
## R2     0 -0.3333333  3.333333 -1.000000 -0.6666667  0.000000
## R3     0  0.0000000 -21.000000  7.000000  3.0000000 -5.000000
## R4     0  0.0000000  0.000000  1.666667 -6.2857143  1.809524
## R5     0  0.0000000  0.000000  0.000000 -4.8857143  5.542857
```

Volver ceros con eliminación hacia atrás

Quinta columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
```

```
## R1    3  1.0000000 -4.000000  0.000000  5.0000000  6.000000
## R2    0 -0.3333333  3.333333 -1.000000 -0.6666667  0.000000
## R3    0  0.0000000 -21.000000  7.000000  3.0000000 -5.000000
## R4    0  0.0000000  0.000000  1.666667 -6.2857143  1.809524
## R5    0  0.0000000  0.000000  0.000000 -4.8857143  5.542857
```

```
m11 <- - Ab[1,5] / Ab[5,5]
m12 <- - Ab[2,5] / Ab[5,5]
m13 <- - Ab[3,5] / Ab[5,5]
m14 <- - Ab[4,5] / Ab[5,5]
R1 <- m11 * Ab[5,] + Ab[1,]
R2 <- m12 * Ab[5,] + Ab[2,]
R3 <- m13 * Ab[5,] + Ab[3,]
R4 <- m14 * Ab[5,] + Ab[4,]
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## R1     3  1.0000000 -4.000000  0.000000  8.881784e-16 11.6725146
## R2     0 -0.3333333  3.333333 -1.000000  0.000000e+00 -0.7563353
## R3     0  0.0000000 -21.000000  7.000000  0.000000e+00 -1.5964912
## R4     0  0.0000000  0.000000  1.666667  0.000000e+00 -5.3216374
## R5     0  0.0000000  0.000000  0.000000 -4.885714e+00  5.5428571
```

```
# Cuarta columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## R1     3  1.0000000 -4.000000  0.000000  8.881784e-16 11.6725146
## R2     0 -0.3333333  3.333333 -1.000000  0.000000e+00 -0.7563353
## R3     0  0.0000000 -21.000000  7.000000  0.000000e+00 -1.5964912
## R4     0  0.0000000  0.000000  1.666667  0.000000e+00 -5.3216374
## R5     0  0.0000000  0.000000  0.000000 -4.885714e+00  5.5428571
```

```
m15 <- - Ab[1,4] / Ab[4,4]
m16 <- - Ab[2,4] / Ab[4,4]
m17 <- - Ab[3,4] / Ab[4,4]
R1 <- m15 * Ab[4,] + Ab[1,]
R2 <- m16 * Ab[4,] + Ab[2,]
R3 <- m17 * Ab[4,] + Ab[3,]
R4 <- Ab[4,]
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## R1     3  1.0000000 -4.000000  0.000000e+00 8.881784e-16 11.672515
## R2     0 -0.3333333  3.333333 -1.110223e-16 0.000000e+00 -3.949318
## R3     0  0.0000000 -21.000000  0.000000e+00 0.000000e+00 20.754386
## R4     0  0.0000000  0.000000  1.666667e+00 0.000000e+00 -5.321637
```

```
## R5      0  0.0000000  0.000000  0.000000e+00 -4.885714e+00  5.542857
```

```
# Tercera columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## R1      3  1.0000000 -4.000000  0.000000e+00  8.881784e-16  11.672515
## R2      0 -0.3333333  3.333333 -1.110223e-16  0.000000e+00 -3.949318
## R3      0  0.0000000 -21.000000  0.000000e+00  0.000000e+00  20.754386
## R4      0  0.0000000  0.000000  1.666667e+00  0.000000e+00 -5.321637
## R5      0  0.0000000  0.000000  0.000000e+00 -4.885714e+00  5.542857
```

```
m18 <- - Ab[1,3] / Ab[3,3]
```

```
m19 <- - Ab[2,3] / Ab[3,3]
```

```
R1 <- m18 * Ab[3,] + Ab[1,]
```

```
R2 <- m19 * Ab[3,] + Ab[2,]
```

```
R3 <- Ab[3,]
```

```
R4 <- Ab[4,]
```

```
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]      [,6]
## R1      3  1.0000000  0  0.000000e+00  8.881784e-16  7.7192982
## R2      0 -0.3333333  0 -1.110223e-16  0.000000e+00 -0.6549708
## R3      0  0.0000000 -21  0.000000e+00  0.000000e+00  20.7543860
## R4      0  0.0000000  0  1.666667e+00  0.000000e+00 -5.3216374
## R5      0  0.0000000  0  0.000000e+00 -4.885714e+00  5.5428571
```

```
# Segunda columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]      [,6]
## R1      3  1.0000000  0  0.000000e+00  8.881784e-16  7.7192982
## R2      0 -0.3333333  0 -1.110223e-16  0.000000e+00 -0.6549708
## R3      0  0.0000000 -21  0.000000e+00  0.000000e+00  20.7543860
## R4      0  0.0000000  0  1.666667e+00  0.000000e+00 -5.3216374
## R5      0  0.0000000  0  0.000000e+00 -4.885714e+00  5.5428571
```

```
m20 <- - Ab[1,2] / Ab[2,2]
```

```
R1 <- m20 * Ab[2,] + Ab[1,]
```

```
R2 <- Ab[2,]
```

```
R3 <- Ab[3,]
```

```
R4 <- Ab[4,]
```

```
R5 <- Ab[5,]
```

```
NuevaA <- rbind(R1, R2, R3, R4, R5)
(NuevaA)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]      [,6]
## R1      3  0.0000000  0 -3.330669e-16  8.881784e-16  5.7543860
## R2      0 -0.3333333  0 -1.110223e-16  0.000000e+00 -0.6549708
```

```
## R3    0  0.0000000 -21  0.000000e+00  0.000000e+00 20.7543860
## R4    0  0.0000000  0  1.666667e+00  0.000000e+00 -5.3216374
## R5    0  0.0000000  0  0.000000e+00 -4.885714e+00  5.5428571
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]      [,6]
## R1     3  0.0000000  0 -3.330669e-16  8.881784e-16  5.7543860
## R2     0 -0.3333333  0 -1.110223e-16  0.000000e+00 -0.6549708
## R3     0  0.0000000 -21  0.000000e+00  0.000000e+00 20.7543860
## R4     0  0.0000000  0  1.666667e+00  0.000000e+00 -5.3216374
## R5     0  0.0000000  0  0.000000e+00 -4.885714e+00  5.5428571
```

```
X <- matrix(c(1, 2, 3, 4, 5), nrow = 5)
```

```
X[1,] <- Ab[1,6] / Ab[1,1]
X[2,] <- Ab[2,6] / Ab[2,2]
X[3,] <- Ab[3,6] / Ab[3,3]
X[4,] <- Ab[4,6] / Ab[4,4]
X[5,] <- Ab[5,6] / Ab[5,5]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 7
```

```
II <- b[2,]; (II)
```

```
## [1] 2
```

```
III <- b[3,]; (III)
```

```
## [1] -5
```

```
IV <- b[4,]; (IV)
```

```
## [1] 6
```

```
V <- b[5,]; (V)
```

```
## [1] 3
```

```
c1 <- (A[1,1] * X[1,]) + (A[1,2] * X[2,]) + (A[1,3] * X[3,]) + (A[1,4] * X[4,]) + (A[1,5] * X[5,]); paste0
```

```
## [1] "El resultado es: 7"
```

```
c2 <- (A[2,1] * X[1,]) + (A[2,2] * X[2,]) + (A[2,3] * X[3,]) + (A[2,4] * X[4,]) + (A[2,5] * X[5,]); paste0
```

```
## [1] "El resultado es: 2"
```

```
c3 <- - (A[3,1] * X[1,]) + (A[3,2] * X[2,]) + (A[3,3] * X[3,]) + (A[3,4] * X[4,]) + (A[3,5] * X[5,]); paste0
```

```
## [1] "El resultado es: -5"
```

```
c4 <- (A[4,1] * X[1,]) + (A[4,2] * X[2,]) + (A[4,3] * X[3,]) + (A[4,4] * X[4,]) + (A[4,5] * X[5,]); paste0
```

```
## [1] "El resultado es: 6"
```

```

c5 <- (A[5,1] * X[1,]) + (A[5,2] * X[2,]) + (A[5,3] * X[3,]) + (A[5,4] * X[4,]) + (A[5,5] * X[5,]); paste0

## [1] "El resultado es: 3"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 7 tecnicamente es: 7"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: 2 tecnicamente es: 2"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: -5 tecnicamente es: -5"
paste0("En la ecuación 4: ", c4, " tecnicamente es: ", IV)

## [1] "En la ecuación 4: 6 tecnicamente es: 6"
paste0("En la ecuación 5: ", c5, " tecnicamente es: ", V)

## [1] "En la ecuación 5: 3 tecnicamente es: 3"

```

```
cat("Soluciones del sistema:")
```

2.2.2.10.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 = ", X[1,])
```

```
## x1 = 1.918129
```

```
cat("x2 = ", X[2,])
```

```
## x2 = 1.964912
```

```
cat("x3 = ", X[3,])
```

```
## x3 = -0.9883041
```

```
cat("x4 = ", X[4,])
```

```
## x4 = -3.192982
```

```
cat("x5 = ", X[5,])
```

```
## x5 = -1.134503
```

2.2.2.11. Sistema 10

$$3,333x_1 + 15920x_2 - 10,333x_3 = 15913$$

$$2,222x_1 + 16,71x_2 + 9,612x_3 = 28,544$$

$$1,5611x_1 + 5,1791x_2 + 1,6852x_3 = 8,4254$$

```
A <- matrix(c(3.333, 15920, -10.333,
              2.222, 16.71, 9.612,
              1.5611, 5.1791, 1.6852),
            nrow = 3, byrow = TRUE)
b <- matrix(c(15913, 28.544, 8.4254), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.2.11.1. Procedimiento

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 3.3330 15920.0000 -10.3330 15913.0000
## [2,] 2.2220   16.7100   9.6120   28.5440
## [3,] 1.5611    5.1791   1.6852    8.4254
```

2.2.2.11.1.1. Primera eliminación No necesita pivoteo

Volver ceros

```
m1 <- - Ab[2,1] / Ab[1,1]
m2 <- - Ab[3,1] / Ab[1,1]
R1 <- Ab[1,]
R2 <- m1 * Ab[1,] + Ab[2,]
R3 <- m2 * Ab[1,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333   15920.00 -10.333000 15913.000
## R2 0.000 -10596.62  16.500667 -10580.123
## R3 0.000 -7451.38   6.524938 -7444.855
```

2.2.2.11.1.2. Segunda Eliminación No necesita pivoteo

Volver ceros

```
Ab <- NuevaA
m3 <- - Ab[3,2] / Ab[2,2]
R1 <- Ab[1,]
R2 <- Ab[2,]
R3 <- m3 * Ab[2,] + Ab[3,]
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333   15920.00 -10.333000 15913.000000
## R2 0.000 -10596.62  16.500667 -10580.122667
## R3 0.000    0.00  -5.078073   -5.078073
```

Volver ceros con eliminación hacia atrás

Tercera columna

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333 15920.00 -10.333000 15913.000000
## R2 0.000 -10596.62 16.500667 -10580.122667
## R3 0.000      0.00 -5.078073      -5.078073
```

```
m4 <- - Ab[1,3] / Ab[3,3]
m5 <- - Ab[2,3] / Ab[3,3]
R1 <- m4 * Ab[3,] + Ab[1,]
R2 <- m5 * Ab[3,] + Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333 15920.00 0.000000 15923.333000
## R2 0.000 -10596.62 0.000000 -10596.623333
## R3 0.000      0.00 -5.078073      -5.078073
```

```
# Segunda columna
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333 15920.00 0.000000 15923.333000
## R2 0.000 -10596.62 0.000000 -10596.623333
## R3 0.000      0.00 -5.078073      -5.078073
```

```
m6 <- - Ab[1,2] / Ab[2,2]
R1 <- m6 * Ab[2,] + Ab[1,]
R2 <- Ab[2,]
R3 <- Ab[3,]
```

```
NuevaA <- rbind(R1, R2, R3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333      0.00 0.000000      3.333000
## R2 0.000 -10596.62 0.000000 -10596.623333
## R3 0.000      0.00 -5.078073      -5.078073
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1]      [,2]      [,3]      [,4]
## R1 3.333      0.00 0.000000      3.333000
## R2 0.000 -10596.62 0.000000 -10596.623333
## R3 0.000      0.00 -5.078073      -5.078073
```

```
X <- matrix(c(1, 2, 3), nrow = 3)
```

```
X[1,] <- Ab[1,4] / Ab[1,1]
X[2,] <- Ab[2,4] / Ab[2,2]
X[3,] <- Ab[3,4] / Ab[3,3]
```



```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 15913
```

```
II <- b[2,]; (II)
```

```
## [1] 28.544
```

```
III <- b[3,]; (III)
```

```
## [1] 8.4254
```

```
c1 <- (A[1,1] * X[1,]) + (A[1,2] * X[2,]) + (A[1,3] * X[3,]); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 15913"
```

```
c2 <- (A[2,1] * X[1,]) + (A[2,2] * X[2,]) + (A[2,3] * X[3,]); paste0("El resultado es: ", c2)
```

```
## [1] "El resultado es: 28.54400000000006"
```

```
c3 <- (A[3,1] * X[1,]) + (A[3,2] * X[2,]) + (A[3,3] * X[3,]); paste0("El resultado es: ", c3)
```

```
## [1] "El resultado es: 8.425399999999965"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 15913 técnicamente es: 15913"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 28.54400000000006 técnicamente es: 28.544"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 8.425399999999965 técnicamente es: 8.4254"
```

```
cat("Soluciones del sistema:")
```

2.2.2.11.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 = ", X[1,])
```

```
## x1 = 1
```

```
cat("x2 = ", X[2,])
```

```
## x2 = 1
```

```
cat("x3 = ", X[3,])
```

```
## x3 = 1
```

7.5 Gauss-Jordan

$$\begin{aligned} 5,10x_1 + 2x_2 - x_3 &= 27 \\ -3x_1 - 6x_2 + 2x_3 &= -61,5 \\ x_1 + x_2 + 5x_3 &= -21,5 \end{aligned} \Rightarrow \begin{bmatrix} 10 & 2 & -1 & 27 \\ -3 & -6 & 2 & -61,5 \\ 1 & 1 & 5 & -21,5 \end{bmatrix}$$

Normalicamos $\rightarrow$ to R₁

$$\text{to R}_1 = \left[\left(\frac{1}{10}\right)(10), \left(\frac{1}{10}\right)(2), \left(\frac{1}{10}\right)(-1), \left(\frac{1}{10}\right)(27) \right] = \left[1, \frac{1}{5}, -\frac{1}{10}, \frac{27}{10} \right]$$

$$\begin{bmatrix} 1 & \frac{1}{5} & -\frac{1}{10} & \frac{27}{10} \\ -3 & -6 & 2 & -61,5 \\ 1 & 1 & 5 & -21,5 \end{bmatrix}$$

Figura 2.16: Ejercicio 5

2.2.3. 7.5. Eliminación Gauss-Jordan (Método directo)

2.2.3.1. Ejercicios a mano

2.2.3.2. Sistema 1

$$x_1 + 2x_2 + 3x_3 = 1$$

$$-0,4x_1 + 2x_2 - x_3 = 10$$

$$0,5x_1 - 3x_2 + x_3 = 15$$

```
A <- matrix(c(1, 2, 3,
              -0.4, 2, -1,
              0.5, -3, 1),
            nrow = 3, byrow = TRUE)
b <- matrix(c(1, 10, 15), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.3.2.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,] 1.0   2   3   1
## [2,] -0.4  2  -1  10
## [3,] 0.5  -3   1  15
```

```
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

2.2.3.2.1.1. Primera eliminación

```
## [1] 0.4
```

$m_1 = -(-2)/1 = 3$
 $m_2 = -(1)/1 = -1$
 $m_1 R_1 + R_2 = [(3)(1) - 3, (3)(\frac{1}{5}) - 6, (3)(-\frac{1}{10}) + 2, (3)(\frac{23}{10}) - 6.5] = [0, \frac{23}{5}, \frac{17}{10}, \frac{267}{5}]$
 $m_2 R_1 + R_3 = [(-1)(1) + 1, (-1)(\frac{1}{5}) + 1, (-1)(-\frac{1}{10}) + 5, (-1)(\frac{23}{10}) - 21.5] = [0, \frac{4}{5}, \frac{51}{10}, \frac{-121}{5}]$

$$\begin{bmatrix} 1 & \frac{1}{5} & -\frac{1}{10} & \frac{23}{10} \\ 0 & \frac{23}{5} & \frac{17}{10} & \frac{267}{5} \\ 0 & \frac{4}{5} & \frac{51}{10} & \frac{-121}{5} \end{bmatrix}$$
 Normalización
 $(\frac{5}{23}) R_2 = [0, (\frac{5}{23})(\frac{23}{5}), (\frac{5}{23})(\frac{17}{10}), (\frac{5}{23})(\frac{267}{5})]$
 $= [0, 1, \frac{17}{54}, \frac{89}{9}]$

$$\begin{bmatrix} 1 & \frac{1}{5} & -\frac{1}{10} & \frac{23}{10} \\ 0 & 1 & \frac{17}{54} & \frac{89}{9} \\ 0 & \frac{4}{5} & \frac{51}{10} & \frac{-121}{5} \end{bmatrix}$$
 $m_3 = -(\frac{4}{5})/1 = -\frac{4}{5}$
 $m_3 R_2 + R_3 = [0, (\frac{4}{5})(1) + \frac{4}{5}, (\frac{4}{5})(\frac{17}{54}) + \frac{51}{10}, (\frac{4}{5})(\frac{89}{9}) - \frac{121}{5}]$
 $= [0, 0, \frac{289}{54}, \frac{-289}{9}]$

$$\begin{bmatrix} 1 & \frac{1}{5} & -\frac{1}{10} & \frac{23}{10} \\ 0 & 1 & \frac{17}{54} & \frac{89}{9} \\ 0 & 0 & \frac{289}{54} & \frac{-289}{9} \end{bmatrix}$$
 Normalización
 $(\frac{54}{289}) R_3 = [0, 0, (\frac{54}{289})(\frac{289}{54}), (\frac{54}{289})(\frac{-289}{9})]$
 $= [0, 0, 1, -6]$

$$\begin{bmatrix} 1 & \frac{1}{5} & -\frac{1}{10} & \frac{23}{10} \\ 0 & 1 & \frac{17}{54} & \frac{89}{9} \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

Volver ceros
 $m_4 = -(-\frac{1}{10})/1 = 1/10$
 $m_5 = -(\frac{17}{54})/1 = \frac{17}{54}$
 $m_4 R_3 + R_1 = [1, \frac{1}{5}, (\frac{1}{10})(1) - \frac{1}{10}, (\frac{1}{10})(-6) + \frac{23}{10}] = [1, \frac{1}{5}, 0, \frac{21}{10}]$
 $m_5 R_3 + R_2 = [0, 1, (\frac{17}{54})(1) - \frac{17}{54}, (\frac{17}{54})(-6) + \frac{89}{9}] = [0, 1, 0, 8]$

$$\begin{bmatrix} 1 & \frac{1}{5} & 0 & \frac{21}{10} \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$
 $m_6 = -(\frac{1}{5})/1 = -1/5$
 $m_6 R_2 + R_1 = [1, (-\frac{1}{5})(1) + \frac{1}{5}, 0, (-\frac{1}{5})(8) + \frac{21}{10}]$
 $= [1, 0, 0, \frac{1}{2}]$

$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

Figura 2.17: Ejercicio 5-2

Soluciones

$$x_1 = \frac{1}{2} = 0.5$$

$$x_2 = 8$$

$$x_3 = -6$$

Comprobación

$$10(0.5) + 2(8) - 1(-6) = 27 \Rightarrow 5 + 16 + 6 = 27 \Rightarrow 27 = 27$$

$$-3(0.5) - 6(8) + 2(-6) = -61.5 \Rightarrow -1.5 - 48 - 12 = -61.5 \Rightarrow -61.5 = -61.5$$

$$0.5 + 8 + 5(-6) = -21.5 \Rightarrow 0.5 + 8 - 30 = -21.5 \Rightarrow -21.5 = -21.5$$

Soluciones: $x_1 = 0.5, x_2 = 8, x_3 = -6$

8. $x + 2y + 3z + 4w = 1$
 $x - 4y + z + 11w = 2$
 $-x + 8y + 7z + 6w = -2$
 $16x + 8y - 5z + 6w = 11$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 & 1 \\ 1 & -4 & 1 & 11 & 2 \\ -1 & 8 & 7 & 6 & -2 \\ 16 & 8 & -5 & 6 & 11 \end{bmatrix}$$

Pivote $R_4 \leftrightarrow R_1$

Normalizamos $\frac{1}{16}R_1 = [\frac{1}{16}(16), \frac{1}{16}(8), \frac{1}{16}(-5), \frac{1}{16}(6), \frac{1}{16}(11)]$

$$R_1 = [1, \frac{1}{2}, -\frac{5}{16}, \frac{3}{8}, \frac{11}{16}]$$

$$\begin{bmatrix} 1 & \frac{1}{2} & -\frac{5}{16} & \frac{3}{8} & \frac{11}{16} \\ 1 & -4 & 1 & 11 & 2 \\ -1 & 8 & 7 & 6 & -2 \\ 1 & 2 & 3 & 4 & 1 \end{bmatrix}$$

$$-R_1 + R_2 = [-1 + 1, -\frac{1}{2} + 4, \frac{5}{16} + 1, -\frac{3}{8} + 11, -\frac{11}{16} + 2]$$

$$-R_1 + R_3 = [-1 + 1, -\frac{1}{2} + 2, \frac{5}{16} + 3, -\frac{3}{8} + 4, -\frac{11}{16} + 1]$$

$$-R_1 + R_4 = [-1 + 1, -\frac{1}{2} + 2, \frac{5}{16} + 3, -\frac{3}{8} + 4, -\frac{11}{16} + 1]$$

$$R_1 + R_2 = [0, \frac{7}{2}, \frac{19}{16}, \frac{35}{8}, \frac{21}{16}]$$

$$R_1 + R_3 = [0, \frac{7}{2}, \frac{19}{16}, \frac{35}{8}, \frac{21}{16}]$$

$$R_1 + R_4 = [0, \frac{7}{2}, \frac{19}{16}, \frac{35}{8}, \frac{21}{16}]$$

Pivote $R_3 \leftrightarrow R_2$

Normalizamos $-\frac{2}{7}R_2$

Figura 2.18: Ejercicio 8

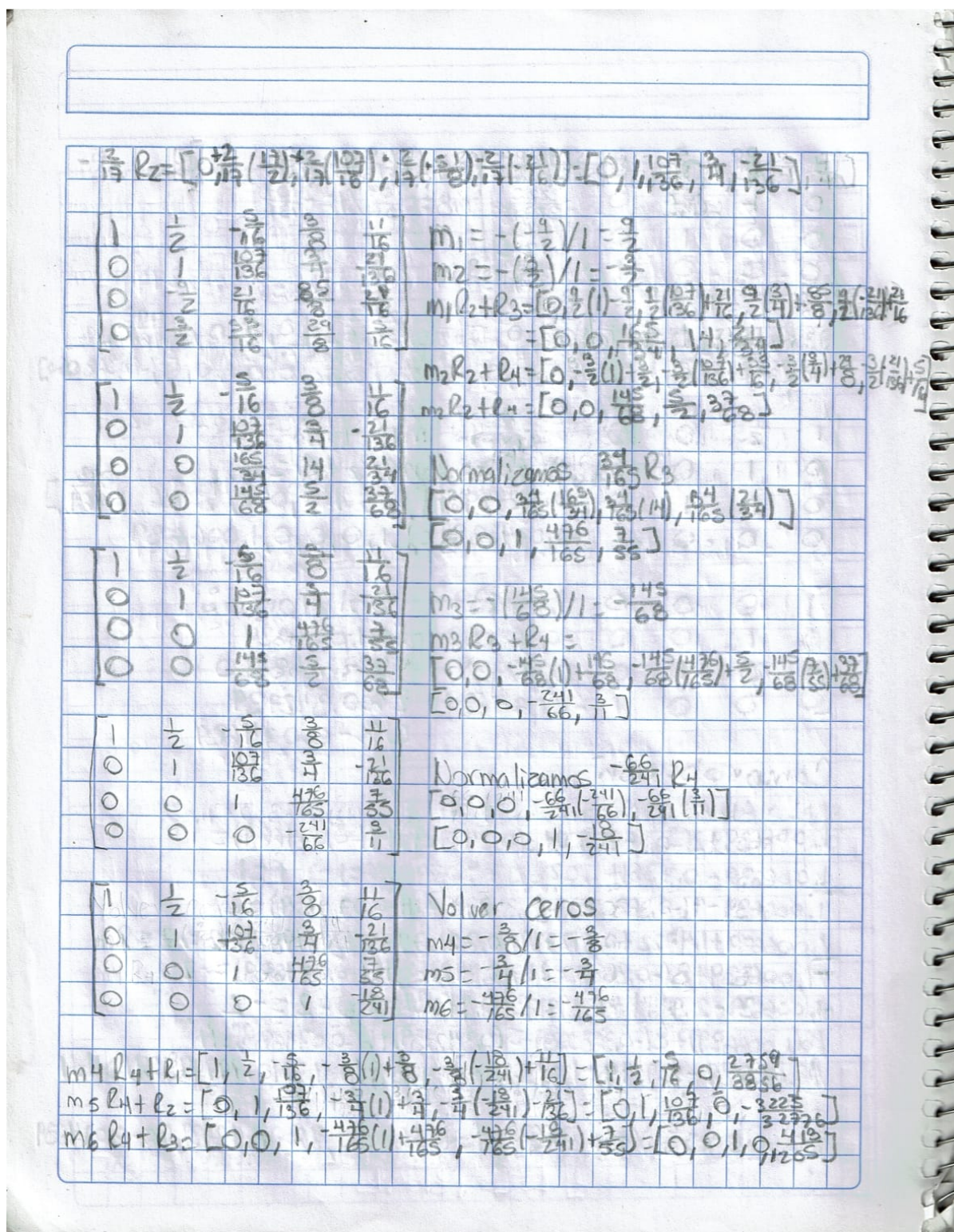


Figura 2.19: Ejercicio 8-2

$$\begin{bmatrix} 1 & \frac{1}{2} & -\frac{5}{16} & 0 & \frac{2789}{5856} \\ 0 & 1 & \frac{107}{136} & 0 & \frac{3223}{32736} \\ 0 & 0 & 1 & 0 & \frac{483}{1205} \\ 0 & 0 & 0 & 1 & -\frac{18}{241} \end{bmatrix} \quad \begin{aligned} m_7 &= -(-\frac{5}{16})/1 = \frac{5}{16} \\ m_8 &= -\frac{107}{136}/1 = -\frac{107}{136} \end{aligned}$$

$$\begin{aligned} m_7 R_3 + R_1 &= [0, \frac{1}{2}, \frac{5}{16}(1) - \frac{5}{16}, 0, \frac{2789}{5856} + \frac{483}{1205}] = [0, \frac{1}{2}, 0, 0, \frac{793}{964}] \\ m_8 R_3 + R_2 &= [0, 1, -\frac{107}{136}(1) + \frac{107}{136}, 0, \frac{3223}{32736} - \frac{483}{1205}] = [0, 1, 0, 0, -\frac{368050}{964}]
 \end{aligned}$$

$$\begin{bmatrix} 1 & \frac{1}{2} & 0 & 0 & \frac{793}{964} \\ 0 & 1 & 0 & 0 & -\frac{368050}{964} \\ 0 & 0 & 1 & 0 & \frac{0.342739}{1} \\ 0 & 0 & 0 & 1 & -\frac{0.074689}{1} \end{bmatrix} \quad \begin{aligned} m_4 &= -(\frac{1}{2})/1 = -\frac{1}{2} \\ m_9 R_2 + R_1 &= [1, -\frac{1}{2}(1) + \frac{1}{2}, 0, 0, \frac{793}{964} - \frac{368050}{964}] \\ &= [1, 0, 0, 0, 1.006639] \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1.006639 \\ 0 & 1 & 0 & 0 & -0.368050 \\ 0 & 0 & 1 & 0 & 0.342739 \\ 0 & 0 & 0 & 1 & -0.074689 \end{bmatrix} \quad \begin{aligned} &\text{Soluciones:} \\ X &\approx 1.006639 \\ Y &\approx -0.368050 \\ Z &\approx 0.342739 \\ W &\approx -0.074689 \end{aligned}$$

Comprobación

$$\begin{aligned} 1.006639 + 2(-0.368050) + 3(0.342739) + 4(-0.074689) &= 1 \\ 1.006639 - 0.7361 + 1.028217 - 0.298756 &= 1 \Rightarrow 1 = 1 \\ 1.006639 - 4(-0.368050) + 0.342739 + 11(-0.074689) &= 2 \\ 1.006639 + 1.4722 + 0.342739 - 0.821579 &= 2 \Rightarrow 1.999999 \approx 2 \Rightarrow 2 = 2 \\ -1.006639 + 8(-0.368050) + 7(0.342739) + 6(-0.074689) &= -2 \\ -1.006639 - 2.9444 + 2.399173 - 0.448134 &= -2 \Rightarrow -2 = -2 \\ 16(1.006639) + 8(-0.368050) - 5(0.342739) + 6(-0.074689) &= 11 \\ 16.106224 - 2.9444 - 1.713695 - 0.448134 &= 11 \Rightarrow 10.999995 \approx 11 \Rightarrow 11 = 11 \end{aligned}$$

Soluciones: $X \approx 1.006639, Y \approx -0.368050, Z \approx 0.342739, W \approx -0.074689$

Figura 2.20: Ejercicio 8-3

```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## [1] -0.5
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 1 2 3 1
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.0 2.8 0.2 10.4
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.0 -4.0 -0.5 14.5
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
```

```
## A1      1  2.0  3.0  1.0
```

```
## A2      0  2.8  0.2 10.4
```

```
## A3      0 -4.0 -0.5 14.5
```

```
# Primer pivote
```

```
Ab <- NuevaA[c(1, 3, 2), ]
(Ab)
```

```
##      [,1] [,2] [,3] [,4]
```

```
## A1      1  2.0  3.0  1.0
```

```
## A3      0 -4.0 -0.5 14.5
```

```
## A2      0  2.8  0.2 10.4
```

```
# Primera normalización
```

```
n1 <- 1 / Ab[2,2]
```

```
A1 <- Ab[1,]
```

```
A2 <- n1 * Ab[2,]
```

```
(A2)
```

```
## [1] 0.000 1.000 0.125 -3.625
```

```
A3 <- Ab[3,]
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
```

```
## A1      1  2.0 3.000 1.000
```

```
## A2      0  1.0 0.125 -3.625
```

```
## A3      0  2.8 0.200 10.400
```



```
Ab <- NuevaA
m3 <- - Ab[1,2] / Ab[2,2]
(m3)
```

2.2.3.2.1.2. Segunda eliminación

```
## A1
## -2
```

```
m4 <- - Ab[3,2] / Ab[2,2]
(m4)
```

```
## A3
## -2.8
```

```
A1 <- m3 * Ab[2,] + Ab[1,]
A2 <- Ab[2,]
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1] 0.00 0.00 -0.15 20.55
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1   0 2.750 8.250
## A2      0   1 0.125 -3.625
## A3      0   0 -0.150 20.550
```

Segunda normalización

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

```
## [1] 0 0 1 -137
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1   0 2.750 8.250
## A2      0   1 0.125 -3.625
## A3      0   0 1.000 -137.000
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

2.2.3.2.1.3. Tercera eliminación

```
##      A1
## -2.75
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
##      A2
## -0.125
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
(A3)
```

```
## [1]      0      0      1 -137
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1      0      0 385.0
## A2      0      1      0 13.5
## A3      0      0      1 -137.0
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1      0      0 385.0
## A2      0      1      0 13.5
## A3      0      0      1 -137.0
```

```
x1 <- Ab[1,4] / Ab[1,1]
x2 <- Ab[2,4] / Ab[2,2]
x3 <- Ab[3,4] / Ab[3,3]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 1
```

```
II <- b[2,]; (II)
```

```
## [1] 10
```

```
III <- b[3,]; (III)
```

```
## [1] 15
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 1"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 10"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 15"
```

```
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 1 tecnicamente es: 1"
```

```
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 10 tecnicamente es: 10"
```

```
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 15 tecnicamente es: 15"
```

```
cat("Soluciones del sistema:")
```

2.2.3.2.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 385
```

```
cat("x2 =", x2)
```

```
## x2 = 13.5
```

```
cat("x3 =", x3)
```

```
## x3 = -137
```

2.2.3.3. Sistema 2

$$2x_1 - 0,9x_2 + 3x_3 = -3,61$$

$$-0,5x_1 + 0,1x_2 - x_3 = 2,035$$

$$x_1 - 6,35x_2 - 0,45x_3 = 15,401$$

```
A <- matrix(c(2, -0.9, 3,
              -0.5, 0.1, -1,
              1, -6.35, -0.45),
            nrow = 3, byrow = TRUE)
b <- matrix(c(-3.61, 2.035, 15.401), nrow = 3)
```

```
Ab <- cbind(A, b); (Ab)
```

2.2.3.3.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  2.0 -0.90  3.00 -3.610
## [2,] -0.5  0.10 -1.00  2.035
## [3,]  1.0 -6.35 -0.45 15.401
```

```
# Primera normalización
```

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

2.2.3.3.1.1. Primera eliminación

```
## [1] 1.000 -0.450 1.500 -1.805
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1  1.0 -0.45 1.50 -1.805
## A2 -0.5 0.10 -1.00 2.035
## A3 1.0 -6.35 -0.45 15.401
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## 0.5
```

```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## -1
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 1.000 -0.450 1.500 -1.805
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.0000 -0.1250 -0.2500 1.1325
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.000 -5.900 -1.950 17.206
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1  1 -0.450 1.50 -1.8050
## A2  0 -0.125 -0.25 1.1325
## A3  0 -5.900 -1.95 17.2060
```

```
# Primer pivote
```

```
Ab <- NuevaA[c(1, 3, 2), ]
(Ab)
```

2.2.3.3.1.2. Segunda eliminación

```
##      [,1] [,2] [,3] [,4]
## A1      1 -0.450  1.50 -1.8050
## A3      0 -5.900 -1.95 17.2060
## A2      0 -0.125 -0.25  1.1325
```

```
# Segunda normalización
```

```
n1 <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n1 * Ab[2,]
(A2)
```

```
## [1]  0.0000000  1.0000000  0.3305085 -2.9162712
```

```
A3 <- Ab[3,]
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1 -0.450  1.5000000 -1.805000
## A2      0  1.000  0.3305085 -2.916271
## A3      0 -0.125 -0.2500000  1.132500
```

```
# Volver ceros
```

```
Ab <- NuevaA
m3 <- - Ab[1,2] / Ab[2,2]
(m3)
```

```
##      A1
## 0.45
```

```
m4 <- - Ab[3,2] / Ab[2,2]
(m4)
```

```
##      A3
## 0.125
```

```
A1 <- m3 * Ab[2,] + Ab[1,]
A2 <- Ab[2,]
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1]  0.0000000  0.0000000 -0.2086864  0.7679661
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
```

```
## A1    1    0  1.6487288 -3.1173220
## A2    0    1  0.3305085 -2.9162712
## A3    0    0 -0.2086864  0.7679661
```

```
# Tercera normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.3.1.3. Tercera eliminación

```
## [1] 0.00 0.00 1.00 -3.68
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## A1     1    0 1.6487288 -3.117322
## A2     0    1 0.3305085 -2.916271
## A3     0    0 1.0000000 -3.680000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
##      A1
## -1.648729
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
##      A2
## -0.3305085
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
(A3)
```

```
## [1] 0.00 0.00 1.00 -3.68
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1     1    0    0  2.95
## A2     0    1    0 -1.70
## A3     0    0    1 -3.68
```


Soluciones

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1   0   0  2.95
## A2      0   1   0 -1.70
## A3      0   0   1 -3.68
```

```
x1 <- Ab[1,4] / Ab[1,1]
x2 <- Ab[2,4] / Ab[2,2]
x3 <- Ab[3,4] / Ab[3,3]
```

Comprobación

```
I <- b[1,]; (I)
```

```
## [1] -3.61
```

```
II <- b[2,]; (II)
```

```
## [1] 2.035
```

```
III <- b[3,]; (III)
```

```
## [1] 15.401
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: -3.61"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 2.035"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 15.401"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: -3.61 técnicamente es: -3.61"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 2.035 técnicamente es: 2.035"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 15.401 técnicamente es: 15.401"
```

```
cat("Soluciones del sistema:")
```

2.2.3.3.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 2.95
cat("x2 =", x2)
```

```
## x2 = -1.7
cat("x3 =", x3)
```

```
## x3 = -3.68
```

2.2.3.4. Sistema 3

$$0,7x_1 + 2,7x_2 - 6x_3 + 0,7x_4 = 1,6487$$

$$2x_1 - 0,8x_2 + 3x_3 - x_4 = -2,342$$

$$-x_1 - 1,5x_2 + 1,4x_3 + 3x_4 = -4,189$$

$$7x_2 - 1,56x_3 + x_4 = 15,792$$

```
A <- matrix(c(0.7, 2.7, -6, 0.7,
              2, -0.8, 3, -1,
              -1, -1.5, 1.4, 3,
              0, 7, -1.56, 1),
            nrow = 4, byrow = TRUE)
b <- matrix(c(1.6487, -2.342, -4.189, 15.792), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.2.3.4.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]  0.7  2.7 -6.00  0.7  1.6487
## [2,]  2.0 -0.8  3.00 -1.0 -2.3420
## [3,] -1.0 -1.5  1.40  3.0 -4.1890
## [4,]  0.0  7.0 -1.56  1.0 15.7920
```

Primera normalización

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

2.2.3.4.1.1. Primera eliminación

```
## [1]  1.000000  3.857143 -8.571429  1.000000  2.355286
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1  3.857143 -8.571429  1  2.355286
## A2      2 -0.800000  3.000000 -1 -2.342000
```

```
## A3   -1 -1.500000  1.400000    3 -4.189000
## A4    0  7.000000 -1.560000    1 15.792000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## -2
```

```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## 1
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1]  1.000000  3.857143 -8.571429  1.000000  2.355286
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1]  0.000000 -8.514286 20.142857 -3.000000 -7.052571
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1]  0.000000  2.357143 -7.171429  4.000000 -1.833714
```

```
A4 <- Ab[4,]
(A4)
```

```
## [1]  0.000  7.000 -1.560  1.000 15.792
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3] [,4]      [,5]
## A1     1  3.857143 -8.571429    1  2.355286
## A2     0 -8.514286 20.142857   -3 -7.052571
## A3     0  2.357143 -7.171429    4 -1.833714
## A4     0  7.000000 -1.560000    1 15.792000
```

```
# Segunda normalización
```

```
Ab <- NuevaA
n <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n * Ab[2,]
(A2)
```

2.2.3.4.1.2. Segunda eliminación

```
## [1]  0.0000000  1.0000000 -2.3657718  0.3523490  0.8283221
```

```

A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

##      [,1]      [,2]      [,3]      [,4]      [,5]
## A1      1 3.857143 -8.571429 1.000000 2.3552857
## A2      0 1.000000 -2.365772 0.352349 0.8283221
## A3      0 2.357143 -7.171429 4.000000 -1.8337143
## A4      0 7.000000 -1.560000 1.000000 15.7920000

# Volver ceros

Ab <- NuevaA
m3 <- - Ab[1,2] / Ab[2,2]
(m3)

##      A1
## -3.857143

m4 <- - Ab[3,2] / Ab[2,2]
(m4)

##      A3
## -2.357143

m5 <- - Ab[4,2] / Ab[2,2]
(m5)

## A4
## -7

A1 <- m3 * Ab[2,] + Ab[1,]
A2 <- Ab[2,]
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)

## [1] 0.000000 0.000000 -1.594966 3.169463 -3.786188

A4 <- m5 * Ab[2,] + Ab[4,]
(A4)

## [1] 0.000000 0.000000 15.000403 -1.466443 9.993745

NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1  0 0.5536913 -0.3590604 -0.8396711
## A2      0  1 -2.3657718 0.3523490 0.8283221
## A3      0  0 -1.5949664 3.1694631 -3.7861879
## A4      0  0 15.0004027 -1.4664430 9.9937450

# Primer pivote

```

```
Ab <- NuevaA[c(1, 2, 4, 3), ]
(Ab)
```

2.2.3.4.1.3. Tercera eliminación

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1      0  0.5536913 -0.3590604 -0.8396711
## A2      0      1 -2.3657718  0.3523490  0.8283221
## A4      0      0 15.0004027 -1.4664430  9.9937450
## A3      0      0 -1.5949664  3.1694631 -3.7861879
```

Tercera normalización

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

```
## [1] 0.000000 0.000000 1.000000 -1.987166 2.373835
```

```
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1      0  0.5536913 -0.3590604 -0.8396711
## A2      0      1 -2.3657718  0.3523490  0.8283221
## A3      0      0  1.0000000 -1.9871660  2.3738355
## A4      0      0 15.0004027 -1.4664430  9.9937450
```

Volver ceros

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
##      A1
## -0.5536913
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
##      A2
## 2.365772
```

```
m7 <- - Ab[4,3] / Ab[3,3]
(m7)
```

```
##      A4
## -15.0004
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
A4 <- m7 * Ab[3,] + Ab[4,]
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      0 0.7412161 -2.154043
## A2      0      1      0 -4.3488323  6.444275
## A3      0      0      1 -1.9871660  2.373835
## A4      0      0      0 28.3418473 -25.614743
```

```
# Cuarta normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[4,4]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- n2 * Ab[4,]
(A4)
```

2.2.3.4.1.4. Cuarta eliminación

```
## [1] 0.0000000 0.0000000 0.0000000 1.0000000 -0.9037782
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      0 0.7412161 -2.1540431
## A2      0      1      0 -4.3488323  6.4442752
## A3      0      0      1 -1.9871660  2.3738355
## A4      0      0      0 1.0000000 -0.9037782
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,4] / Ab[4,4]
(m5)
```

```
##      A1
## -0.7412161
```

```
m6 <- - Ab[2,4] / Ab[4,4]
(m6)
```

```
##      A2
## 4.348832
```

```
m7 <- - Ab[3,4] / Ab[4,4]
(m7)
```

```
##      A3
## 1.987166
```

```
A1 <- m5 * Ab[4,] + Ab[1,]
A2 <- m6 * Ab[4,] + Ab[2,]
```



```
A3 <- m7 * Ab[4,] + Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      1      0      0      0 -1.4841482
## A2      0      1      0      0  2.5138955
## A3      0      0      1      0  0.5778782
## A4      0      0      0      1 -0.9037782
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      1      0      0      0 -1.4841482
## A2      0      1      0      0  2.5138955
## A3      0      0      1      0  0.5778782
## A4      0      0      0      1 -0.9037782
```

```
x1 <- Ab[1,5] / Ab[1,1]
x2 <- Ab[2,5] / Ab[2,2]
x3 <- Ab[3,5] / Ab[3,3]
x4 <- Ab[4,5] / Ab[4,4]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 1.6487
```

```
II <- b[2,]; (II)
```

```
## [1] -2.342
```

```
III <- b[3,]; (III)
```

```
## [1] -4.189
```

```
IV <- b[4,]; (IV)
```

```
## [1] 15.792
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 1.6487"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -2.342"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: -4.189"
```

```
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)
```

```
## [1] "El resultado es: 15.792"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 1.6487 técnicamente es: 1.6487"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: -2.342 técnicamente es: -2.342"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: -4.189 técnicamente es: -4.189"
paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)

## [1] "En la ecuación 4: 15.792 técnicamente es: 15.792"
```

```
cat("Soluciones del sistema:")
```

2.2.3.4.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = -1.484148
```

```
cat("x2 =", x2)
```

```
## x2 = 2.513895
```

```
cat("x3 =", x3)
```

```
## x3 = 0.5778782
```

```
cat("x4 =", x4)
```

```
## x4 = -0.9037782
```

2.2.3.5. Sistema 4

$$3x_1 - 0,1x_2 - 0,2x_3 = 7,85$$

$$0,1x_1 + 7x_2 - 0,3x_3 = -19,3$$

$$0,3x_1 - 0,2x_2 + 10x_3 = 71,4$$

```
A <- matrix(c(3, - 0.1, -0.2,
              0.1, 7, -0.3,
              0.3, -0.2, 10),
            nrow = 3, byrow = TRUE)
b <- matrix(c(7.85, -19.3, 71.4), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.3.5.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  3.0 -0.1 -0.2  7.85
```

```
## [2,] 0.1 7.0 -0.3 -19.30
## [3,] 0.3 -0.2 10.0 71.40
```

```
# Primera normalización
```

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

2.2.3.5.1.1. Primera eliminación

```
## [1] 1.00000000 -0.03333333 -0.06666667 2.61666667
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## A1  1.0 -0.03333333 -0.06666667 2.616667
## A2  0.1 7.00000000 -0.30000000 -19.300000
## A3  0.3 -0.20000000 10.00000000 71.400000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## -0.1
```

```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## -0.3
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 1.00000000 -0.03333333 -0.06666667 2.61666667
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.00000000 7.00333333 -0.29333333 -19.5616667
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.000 -0.190 10.020 70.615
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## A1  1 -0.03333333 -0.06666667 2.616667
```

```
## A2    0  7.00333333 -0.29333333 -19.561667
## A3    0 -0.19000000 10.02000000  70.615000
```

```
# Segunda normalización
```

```
Ab <- NuevaA
n1 <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n1 * Ab[2,]
(A2)
```

2.2.3.5.1.2. Segunda eliminación

```
## [1]  0.00000000  1.00000000 -0.04188482 -2.79319372
```

```
A3 <- Ab[3,]
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]
## A1    1 -0.03333333 -0.06666667  2.616667
## A2    0  1.00000000 -0.04188482 -2.793194
## A3    0 -0.19000000 10.02000000 70.615000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m3 <- - Ab[1,2] / Ab[2,2]
(m3)
```

```
##      A1
## 0.03333333
m4 <- - Ab[3,2] / Ab[2,2]
(m4)
```

```
##      A3
## 0.19
A1 <- m3 * Ab[2,] + Ab[1,]
A2 <- Ab[2,]
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1]  0.00000  0.00000 10.01204 70.08429
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## A1    1    0 -0.06806283  2.523560
## A2    0    1 -0.04188482 -2.793194
## A3    0    0 10.01204188 70.084293
```

```
# Tercera normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.5.1.3. Tercera eliminación

```
## [1] 0 0 1 7
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## A1      1      0 -0.06806283  2.523560
## A2      0      1 -0.04188482 -2.793194
## A3      0      0  1.00000000  7.000000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
##          A1
## 0.06806283
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
##          A2
## 0.04188482
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
(A3)
```

```
## [1] 0 0 1 7
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1      0      0  3.0
## A2      0      1      0 -2.5
## A3      0      0      1  7.0
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1      0      0  3.0
```

```
## A2    0    1    0 -2.5
## A3    0    0    1  7.0
```

```
x1 <- Ab[1,4] / Ab[1,1]
x2 <- Ab[2,4] / Ab[2,2]
x3 <- Ab[3,4] / Ab[3,3]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 7.85
```

```
II <- b[2,]; (II)
```

```
## [1] -19.3
```

```
III <- b[3,]; (III)
```

```
## [1] 71.4
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 7.85"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -19.3"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 71.4"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 7.85 técnicamente es: 7.85"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -19.3 técnicamente es: -19.3"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 71.4 técnicamente es: 71.4"
```

```
cat("Soluciones del sistema:")
```

2.2.3.5.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 3
```

```
cat("x2 =", x2)
```

```
## x2 = -2.5
```

```
cat("x3 =", x3)
```

```
## x3 = 7
```

2.2.3.6. Sistema 5

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61,5$$

$$x_1 + x_2 + 5x_3 = -21,5$$

```
A <- matrix(c(10, 2, -1,
              -3, -6, 2,
              1, 1, 5),
            nrow = 3, byrow = TRUE)
b <- matrix(c(27, -61.5, -21.5), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.2.3.6.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  10   2  -1  27.0
## [2,]  -3  -6   2 -61.5
## [3,]   1   1   5 -21.5
```

Primera normalización

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

2.2.3.6.1.1. Primera eliminación

```
## [1]  1.0  0.2 -0.1  2.7
A2 <- Ab[2,]
A3 <- Ab[3,]
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1  0.2 -0.1  2.7
## A2     -3 -6.0  2.0 -61.5
## A3      1  1.0  5.0 -21.5
```

Voular ceros

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## 3
```



```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## -1
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 1.0 0.2 -0.1 2.7
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.0 -5.4 1.7 -53.4
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.0 0.8 5.1 -24.2
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1 0.2 -0.1 2.7
## A2      0 -5.4 1.7 -53.4
## A3      0 0.8 5.1 -24.2
```

```
# Segunda normalización
```

```
Ab <- NuevaA
n1 <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n1 * Ab[2,]
(A2)
```

2.2.3.6.1.2. Segunda eliminación

```
## [1] 0.0000000 1.0000000 -0.3148148 9.8888889
```

```
A3 <- Ab[3,]
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]
## A1      1 0.2 -0.1000000 2.700000
## A2      0 1.0 -0.3148148 9.888889
## A3      0 0.8 5.1000000 -24.200000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m3 <- - Ab[1,2] / Ab[2,2]
(m3)
```

```
## A1
```

```
## -0.2
m4 <- - Ab[3,2] / Ab[2,2]
(m4)

## A3
## -0.8
A1 <- m3 * Ab[2,] + Ab[1,]
A2 <- Ab[2,]
A3 <- m4 * Ab[2,] + Ab[3,]
(A3)

## [1] 0.000000 0.000000 5.351852 -32.111111
NuevaA <- rbind(A1, A2, A3)
(NuevaA)

## [,1] [,2] [,3] [,4]
## A1 1 0 -0.03703704 0.7222222
## A2 0 1 -0.31481481 9.8888889
## A3 0 0 5.35185185 -32.1111111
```

Tercera normalización

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.6.1.3. Tercera eliminación

```
## [1] 0 0 1 -6
NuevaA <- rbind(A1, A2, A3)
(NuevaA)

## [,1] [,2] [,3] [,4]
## A1 1 0 -0.03703704 0.7222222
## A2 0 1 -0.31481481 9.8888889
## A3 0 0 1.00000000 -6.0000000
```

Volver ceros

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
## A1
## 0.03703704
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
## A2
```

```
## 0.3148148
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
(A3)
```

```
## [1] 0 0 1 -6
```

```
NuevaA <- rbind(A1, A2, A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1    0    0 0.5
## A2      0    1    0 8.0
## A3      0    0    1 -6.0
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4]
## A1      1    0    0 0.5
## A2      0    1    0 8.0
## A3      0    0    1 -6.0
```

```
x1 <- Ab[1,4] / Ab[1,1]
x2 <- Ab[2,4] / Ab[2,2]
x3 <- Ab[3,4] / Ab[3,3]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 27
```

```
II <- b[2,]; (II)
```

```
## [1] -61.5
```

```
III <- b[3,]; (III)
```

```
## [1] -21.5
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 27"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -61.5"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: -21.5"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 27 técnicamente es: 27"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -61.5 técnicamente es: -61.5"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: -21.5 técnicamente es: -21.5"
```

```
cat("Soluciones del sistema:")
```

2.2.3.6.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 0.5
```

```
cat("x2 =", x2)
```

```
## x2 = 8
```

```
cat("x3 =", x3)
```

```
## x3 = -6
```

2.2.3.7. Sistema 6

$$A = \begin{bmatrix} 1 & 3 & -2 & 1 \\ 1 & 3 & -1 & 2 \\ 0 & 1 & -1 & 4 \\ 2 & 6 & 1 & 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 4 \\ 1 \\ 5 \\ 2 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Resolver para A

```
A <- matrix(c(1, 3, -2, 1,
              1, 3, -1, 2,
              0, 1, -1, 4,
              2, 6, 1, 2),
            nrow = 4, byrow = TRUE)
b <- matrix(c(4, 1, 5, 2), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.2.3.7.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    3   -2    1    4
## [2,]    1    3   -1    2    1
## [3,]    0    1   -1    4    5
## [4,]    2    6    1    2    2
```

```
# Primer pivote
```

```
Ab <- Ab[c(4, 2, 3, 1), ]
(Ab)
```

2.2.3.7.1.1. Primera eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2    6    1    2    2
## [2,]    1    3   -1    2    1
## [3,]    0    1   -1    4    5
## [4,]    1    3   -2    1    4
```

```
# Primera normalización
```

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

```
## [1] 1.0 3.0 0.5 1.0 1.0
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1     1    3  0.5    1    1
## A2     1    3 -1.0    2    1
## A3     0    1 -1.0    4    5
## A4     1    3 -2.0    1    4
```

```
# Vouler ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## -1
```

```
m2 <- - Ab[4,1] / Ab[1,1]
(m2)
```

```
## A4
## -1
```

```
A1 <- Ab[1,]
(A1)

## [1] 1.0 3.0 0.5 1.0 1.0
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)

## [1] 0.0 0.0 -1.5 1.0 0.0
A3 <- Ab[3,]
(A3)
```

```
## [1] 0 1 -1 4 5
A4 <- m2 * Ab[1,] + Ab[4,]
(A4)
```

```
## [1] 0.0 0.0 -2.5 0.0 3.0
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1    3  0.5    1    1
## A2      0    0 -1.5    1    0
## A3      0    1 -1.0    4    5
## A4      0    0 -2.5    0    3
```

```
# Segundo pivote
```

```
Ab <- NuevaA[c(1, 3, 2, 4), ]
(Ab)
```

2.2.3.7.1.2. Segunda eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1    3  0.5    1    1
## A3      0    1 -1.0    4    5
## A2      0    0 -1.5    1    0
## A4      0    0 -2.5    0    3
```

```
# Volver ceros
```

```
m1 <- - Ab[1,2] / Ab[2,2]
(m1)
```

```
## A1
## -3
A1 <- m1 * Ab[2,] + Ab[1,]
(A1)
```

```
## [1] 1.0 0.0 3.5 -11.0 -14.0
A2 <- Ab[2,]
(A2)
```

```
## [1] 0 1 -1 4 5
```

```
A3 <- Ab[3,]
(A3)
```

```
## [1] 0.0 0.0 -1.5 1.0 0.0
```

```
A4 <- Ab[4,]
(A4)
```

```
## [1] 0.0 0.0 -2.5 0.0 3.0
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1    0  3.5 -11 -14
## A2      0    1 -1.0  4    5
## A3      0    0 -1.5  1    0
## A4      0    0 -2.5  0    3
```

```
# Tercer pivote
```

```
Ab <- NuevaA[c(1, 2, 4, 3), ]
(Ab)
```

2.2.3.7.1.3. Tercera eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1    0  3.5 -11 -14
## A2      0    1 -1.0  4    5
## A4      0    0 -2.5  0    3
## A3      0    0 -1.5  1    0
```

```
# Tercera normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

```
## [1] 0.0000000 0.0000000 1.0000000 -0.6666667 0.0000000
```

```
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4] [,5]
## A1      1    0  3.5 -11.0000000 -14
## A2      0    1 -1.0  4.0000000    5
## A3      0    0  1.0 -0.6666667    0
## A4      0    0 -2.5  0.0000000    3
```



```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
## A1
## -3.5
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
## A2
## 1
```

```
m7 <- - Ab[4,3] / Ab[3,3]
(m7)
```

```
## A4
## 2.5
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
A4 <- m7 * Ab[3,] + Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4] [,5]
## A1      1    0    0 -8.6666667 -14
## A2      0    1    0  3.3333333  5
## A3      0    0    1 -0.6666667  0
## A4      0    0    0 -1.6666667  3
```

```
# Cuarta normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[4,4]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- n2 * Ab[4,]
(A4)
```

2.2.3.7.1.4. Cuarta eliminación

```
## [1] 0.0 0.0 0.0 1.0 -1.8
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4] [,5]
## A1      1    0    0 -8.6666667 -14.0
## A2      0    1    0  3.3333333  5.0
## A3      0    0    1 -0.6666667  0.0
```

```
## A4      0      0      0  1.0000000 -1.8
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,4] / Ab[4,4]
(m5)
```

```
##          A1
## 8.666667
m6 <- - Ab[2,4] / Ab[4,4]
(m6)
```

```
##          A2
## -3.333333
m7 <- - Ab[3,4] / Ab[4,4]
(m7)
```

```
##          A3
## 0.666667
A1 <- m5 * Ab[4,] + Ab[1,]
A2 <- m6 * Ab[4,] + Ab[2,]
A3 <- m7 * Ab[4,] + Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1      0      0      0 -29.6
## A2      0      1      0      0  11.0
## A3      0      0      1      0 -1.2
## A4      0      0      0      1 -1.8
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1      0      0      0 -29.6
## A2      0      1      0      0  11.0
## A3      0      0      1      0 -1.2
## A4      0      0      0      1 -1.8
```

```
x1 <- Ab[1,5] / Ab[1,1]
x2 <- Ab[2,5] / Ab[2,2]
x3 <- Ab[3,5] / Ab[3,3]
x4 <- Ab[4,5] / Ab[4,4]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 4
```

```

II <- b[2,]; (II)

## [1] 1
III <- b[3,]; (III)

## [1] 5
IV <- b[4,]; (IV)

## [1] 2
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)

## [1] "El resultado es: 4"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)

## [1] "El resultado es: 0.999999999999995"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)

## [1] "El resultado es: 5"
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)

## [1] "El resultado es: 1.999999999999999"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 4 técnicamente es: 4"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 0.999999999999995 técnicamente es: 1"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 5 técnicamente es: 5"
paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)

## [1] "En la ecuación 4: 1.999999999999999 técnicamente es: 2"

```

```
cat("Soluciones del sistema:")
```

2.2.3.7.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = -29.6
```

```
cat("x2 =", x2)
```

```
## x2 = 11
```

```
cat("x3 =", x3)
```

```
## x3 = -1.2
```

```
cat("x4 =", x4)
```

```
## x4 = -1.8
```

2.2.3.8. Sistema 7

$$6x_1 - x_2 - x_3 + 4x_4 = 17$$

$$x_1 - 10x_2 + 2x_3 - x_4 = -17$$

$$3x_1 - 2x_2 + 8x_3 - x_4 = 19$$

$$x_1 + x_2 + x_3 - 5x_4 = -14$$

```
A <- matrix(c(6, -1, -1, 4,
              1, -10, 2, -1,
              3, -2, 8, -1,
              1, 1, 1, -5),
            nrow = 4, byrow = TRUE)
b <- matrix(c(17, -17, 19, -14), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.2.3.8.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    6   -1   -1    4   17
## [2,]    1  -10    2   -1  -17
## [3,]    3   -2    8   -1   19
## [4,]    1    1    1   -5  -14
```

Primera normalización

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

2.2.3.8.1.1. Primera eliminación

```
## [1] 1.0000000 -0.1666667 -0.1666667 0.6666667 2.8333333
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## A1    1 -0.1666667 -0.1666667 0.6666667 2.8333333
## A2    1 -10.0000000  2.0000000 -1.0000000 -17.0000000
## A3    3  -2.0000000  8.0000000 -1.0000000 19.0000000
## A4    1  1.0000000  1.0000000 -5.0000000 -14.0000000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## -1
```

```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## -3
```

```
m3 <- - Ab[4,1] / Ab[1,1]
(m3)
```

```
## A4
## -1
```

```
A1 <- Ab[1,]
(A1)
```

```
## [1] 1.0000000 -0.1666667 -0.1666667 0.6666667 2.8333333
```

```
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1] 0.0000000 -9.8333333 2.1666667 -1.6666667 -19.8333333
```

```
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1] 0.0 -1.5 8.5 -3.0 10.5
```

```
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)
```

```
## [1] 0.0000000 1.1666667 1.1666667 -5.6666667 -16.8333333
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## A1      1 -0.1666667 -0.1666667 0.6666667 2.8333333
## A2      0 -9.8333333 2.1666667 -1.6666667 -19.8333333
## A3      0 -1.5000000 8.5000000 -3.0000000 10.5000000
## A4      0 1.1666667 1.1666667 -5.6666667 -16.8333333
```

```
# Segunda normalización
```

```
Ab <- NuevaA
n1 <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n1 * Ab[2,]
```

```
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

2.2.3.8.1.2. Segunda eliminación

```
##      [,1]      [,2]      [,3]      [,4]      [,5]
## A1      1 -0.1666667 -0.1666667  0.6666667  2.833333
## A2      0  1.0000000 -0.2203390  0.1694915  2.016949
## A3      0 -1.5000000  8.5000000 -3.0000000 10.500000
## A4      0  1.1666667  1.1666667 -5.6666667 -16.833333
```

Volver ceros

```
Ab <- NuevaA
m1 <- - Ab[1,2] / Ab[2,2]
(m1)
```

```
##      A1
## 0.1666667
```

```
m2 <- - Ab[3,2] / Ab[2,2]
(m2)
```

```
##      A3
## 1.5
```

```
m3 <- - Ab[4,2] / Ab[2,2]
(m3)
```

```
##      A4
## -1.1666667
```

```
A1 <- m1 * Ab[2,] + Ab[1,]
(A1)
```

```
## [1]  1.0000000  0.0000000 -0.2033898  0.6949153  3.1694915
```

```
A2 <- Ab[2,]
(A2)
```

```
## [1]  0.0000000  1.0000000 -0.2203390  0.1694915  2.0169492
```

```
A3 <- m2 * Ab[2,] + Ab[3,]
(A3)
```

```
## [1]  0.0000000  0.0000000  8.169492 -2.745763 13.525424
```

```
A4 <- m3 * Ab[2,] + Ab[4,]
(A4)
```

```
## [1]  0.0000000  0.0000000  1.423729 -5.864407 -19.186441
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1   0 -0.2033898  0.6949153  3.169492
```

```
## A2    0    1 -0.2203390  0.1694915  2.016949
## A3    0    0  8.1694915 -2.7457627 13.525424
## A4    0    0  1.4237288 -5.8644068 -19.186441
```

```
# Tercera normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.8.1.3. Tercera eliminación

```
## [1] 0.0000000 0.0000000 1.0000000 -0.3360996 1.6556017
```

```
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1    0 -0.2033898  0.6949153  3.169492
## A2      0    1 -0.2203390  0.1694915  2.016949
## A3      0    0  1.0000000 -0.3360996  1.655602
## A4      0    0  1.4237288 -5.8644068 -19.186441
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
##      A1
## 0.2033898
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
##      A2
## 0.220339
```

```
m7 <- - Ab[4,3] / Ab[3,3]
(m7)
```

```
##      A4
## -1.423729
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
A4 <- m7 * Ab[3,] + Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
```



```
## A1      1      0      0  0.62655602  3.506224
## A2      0      1      0  0.09543568  2.381743
## A3      0      0      1 -0.33609959  1.655602
## A4      0      0      0 -5.38589212 -21.543568
```

```
# Cuarta normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[4,4]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- n2 * Ab[4,]
(A4)
```

2.2.3.8.1.4. Cuarta eliminación

```
## [1] 0 0 0 1 4
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      0  0.62655602  3.506224
## A2      0      1      0  0.09543568  2.381743
## A3      0      0      1 -0.33609959  1.655602
## A4      0      0      0  1.00000000  4.000000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,4] / Ab[4,4]
(m5)
```

```
##      A1
## -0.626556
```

```
m6 <- - Ab[2,4] / Ab[4,4]
(m6)
```

```
##      A2
## -0.09543568
```

```
m7 <- - Ab[3,4] / Ab[4,4]
(m7)
```

```
##      A3
## 0.3360996
```

```
A1 <- m5 * Ab[4,] + Ab[1,]
A2 <- m6 * Ab[4,] + Ab[2,]
A3 <- m7 * Ab[4,] + Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1      0      0      0      1
## A2      0      1      0      0      2
## A3      0      0      1      0      3
## A4      0      0      0      1      4
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1      1      0      0      0      1
## A2      0      1      0      0      2
## A3      0      0      1      0      3
## A4      0      0      0      1      4
```

```
x1 <- Ab[1,5] / Ab[1,1]
x2 <- Ab[2,5] / Ab[2,2]
x3 <- Ab[3,5] / Ab[3,3]
x4 <- Ab[4,5] / Ab[4,4]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 17
```

```
II <- b[2,]; (II)
```

```
## [1] -17
```

```
III <- b[3,]; (III)
```

```
## [1] 19
```

```
IV <- b[4,]; (IV)
```

```
## [1] -14
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 17"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -17"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 19"
```

```
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)
```

```
## [1] "El resultado es: -14"
```

```
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 17 tecnicamente es: 17"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -17 técnicamente es: -17"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 19 técnicamente es: 19"
```

```
paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)
```

```
## [1] "En la ecuación 4: -14 técnicamente es: -14"
```

```
cat("Soluciones del sistema:")
```

2.2.3.8.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 1
```

```
cat("x2 =", x2)
```

```
## x2 = 2
```

```
cat("x3 =", x3)
```

```
## x3 = 3
```

```
cat("x4 =", x4)
```

```
## x4 = 4
```

2.2.3.9. Sistema 8

$$x + 2y + 3z + 4w = 1$$

$$x - 4y + z + 11w = 2$$

$$-x + 8y + 7z + 6w = -2$$

$$16x + 8y - 5z + 6w = 11$$

```
A <- matrix(c(1, 2, 3, 4,
              1, -4, 1, 11,
              -1, 8, 7, 6,
              16, 8, -5, 6),
            nrow = 4, byrow = TRUE)
b <- matrix(c(1, 2, -2, 11), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.2.3.9.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    2    3    4    1
```

```
## [2,]    1   -4    1   11    2
## [3,]   -1    8    7    6   -2
## [4,]   16    8   -5    6   11
```

```
# Primer pivote
```

```
Ab <- Ab[c(4, 2, 3, 1), ]
(Ab)
```

2.2.3.9.1.1. Primera eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]   16    8   -5    6   11
## [2,]    1   -4    1   11    2
## [3,]   -1    8    7    6   -2
## [4,]    1    2    3    4    1
```

```
# Primera normalización
```

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

```
## [1]  1.0000  0.5000 -0.3125  0.3750  0.6875
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## A1     1  0.5 -0.3125 0.375 0.6875
## A2     1 -4.0  1.0000 11.000 2.0000
## A3    -1  8.0  7.0000  6.000 -2.0000
## A4     1  2.0  3.0000  4.000  1.0000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## -1
```

```
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## 1
```

```
m3 <- - Ab[4,1] / Ab[1,1]
(m3)
```

```
## A4
```

```
## -1
A1 <- Ab[1,]
(A1)

## [1] 1.0000 0.5000 -0.3125 0.3750 0.6875
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)

## [1] 0.0000 -4.5000 1.3125 10.6250 1.3125
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)

## [1] 0.0000 8.5000 6.6875 6.3750 -1.3125
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)

## [1] 0.0000 1.5000 3.3125 3.6250 0.3125
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1 0.5 -0.3125 0.375 0.6875
## A2      0 -4.5 1.3125 10.625 1.3125
## A3      0 8.5 6.6875 6.375 -1.3125
## A4      0 1.5 3.3125 3.625 0.3125
```

Segundo pivote

```
Ab <- NuevaA[c(1, 3, 2, 4), ]
(Ab)
```

2.2.3.9.1.2. Segunda eliminación

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1 0.5 -0.3125 0.375 0.6875
## A3      0 8.5 6.6875 6.375 -1.3125
## A2      0 -4.5 1.3125 10.625 1.3125
## A4      0 1.5 3.3125 3.625 0.3125
```

Segunda normalización

```
n1 <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n1 * Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1 0.5 -0.3125000 0.375 0.6875000
## A2      0 1.0 0.7867647 0.750 -0.1544118
```

```

## A3      0 -4.5  1.3125000 10.625  1.3125000
## A4      0  1.5  3.3125000  3.625  0.3125000

# Volver ceros

Ab <- NuevaA
m1 <- - Ab[1,2] / Ab[2,2]
(m1)

##      A1
## -0.5

m2 <- - Ab[3,2] / Ab[2,2]
(m2)

##      A3
##  4.5

m3 <- - Ab[4,2] / Ab[2,2]
(m3)

##      A4
## -1.5

A1 <- m1 * Ab[2,] + Ab[1,]
(A1)

## [1]  1.0000000  0.0000000 -0.7058824  0.0000000  0.7647059

A2 <- Ab[2,]
(A2)

## [1]  0.0000000  1.0000000  0.7867647  0.7500000 -0.1544118

A3 <- m2 * Ab[2,] + Ab[3,]
(A3)

## [1]  0.0000000  0.0000000  4.8529412 14.0000000  0.6176471

A4 <- m3 * Ab[2,] + Ab[4,]
(A4)

## [1]  0.0000000  0.0000000  2.1323529 2.5000000  0.5441176

NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

##      [,1] [,2]      [,3] [,4]      [,5]
## A1      1   0 -0.7058824  0.00  0.7647059
## A2      0   1  0.7867647  0.75 -0.1544118
## A3      0   0  4.8529412 14.00  0.6176471
## A4      0   0  2.1323529  2.50  0.5441176

# Tercera normalización

Ab <- NuevaA
n2 <- 1 / Ab[3,3]

```

```
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.9.1.3. Tercera eliminación

```
## [1] 0.0000000 0.0000000 1.0000000 2.8848485 0.1272727
```

```
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2]      [,3]      [,4]      [,5]
## A1      1    0 -0.7058824 0.0000000 0.7647059
## A2      0    1  0.7867647 0.7500000 -0.1544118
## A3      0    0  1.0000000 2.8848484 0.1272727
## A4      0    0  2.1323529 2.5000000 0.5441176
```

Volver ceros

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
##      A1
## 0.7058824
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
##      A2
## -0.7867647
```

```
m7 <- - Ab[4,3] / Ab[3,3]
(m7)
```

```
##      A4
## -2.132353
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
A4 <- m7 * Ab[3,] + Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1    0    0  2.036364  0.8545455
## A2      0    1    0 -1.519697 -0.2545455
## A3      0    0    1  2.884848  0.1272727
## A4      0    0    0 -3.651515  0.2727273
```

Cuarta normalización


```
Ab <- NuevaA
n2 <- 1 / Ab[4,4]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- n2 * Ab[4,]
(A4)
```

2.2.3.9.1.4. Cuarta eliminación

```
## [1] 0.0000000 0.0000000 0.0000000 1.0000000 -0.0746888
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      0 2.036364 0.8545455
## A2      0      1      0 -1.519697 -0.2545455
## A3      0      0      1 2.884848 0.1272727
## A4      0      0      0 1.000000 -0.0746888
```

Volver ceros

```
Ab <- NuevaA
m5 <- - Ab[1,4] / Ab[4,4]
(m5)
```

```
##      A1
## -2.036364
```

```
m6 <- - Ab[2,4] / Ab[4,4]
(m6)
```

```
##      A2
## 1.519697
```

```
m7 <- - Ab[3,4] / Ab[4,4]
(m7)
```

```
##      A3
## -2.884848
```

```
A1 <- m5 * Ab[4,] + Ab[1,]
A2 <- m6 * Ab[4,] + Ab[2,]
A3 <- m7 * Ab[4,] + Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      1      0      0      0 1.0066390
## A2      0      1      0      0 -0.3680498
## A3      0      0      1      0 0.3427386
## A4      0      0      0      1 -0.0746888
```

Soluciones

```

Ab <- NuevaA; (Ab)

##      [,1] [,2] [,3] [,4]      [,5]
## A1      1      0      0      0 1.0066390
## A2      0      1      0      0 -0.3680498
## A3      0      0      1      0 0.3427386
## A4      0      0      0      1 -0.0746888

x <- Ab[1,5] / Ab[1,1]
y <- Ab[2,5] / Ab[2,2]
z <- Ab[3,5] / Ab[3,3]
w <- Ab[4,5] / Ab[4,4]

# Comprobación

I <- b[1,]; (I)

## [1] 1
II <- b[2,]; (II)

## [1] 2
III <- b[3,]; (III)

## [1] -2
IV <- b[4,]; (IV)

## [1] 11
c1 <- (A[1,1] * x) + (A[1,2] * y) + (A[1,3] * z) + (A[1,4] * w); paste0("El resultado es: ",c1)

## [1] "El resultado es: 1"
c2 <- (A[2,1] * x) + (A[2,2] * y) + (A[2,3] * z) + (A[2,4] * w); paste0("El resultado es: ",c2)

## [1] "El resultado es: 2"
c3 <- (A[3,1] * x) + (A[3,2] * y) + (A[3,3] * z) + (A[3,4] * w); paste0("El resultado es: ",c3)

## [1] "El resultado es: -2"
c4 <- (A[4,1] * x) + (A[4,2] * y) + (A[4,3] * z) + (A[4,4] * w); paste0("El resultado es: ",c4)

## [1] "El resultado es: 11"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 1 técnicamente es: 1"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 2 técnicamente es: 2"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: -2 técnicamente es: -2"
paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)

```

```
## [1] "En la ecuación 4: 11 tecnicamente es: 11"
```

```
cat("Soluciones del sistema:")
```

2.2.3.9.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x =", x)
```

```
## x = 1.006639
```

```
cat("y =", y)
```

```
## y = -0.3680498
```

```
cat("z =", z)
```

```
## z = 0.3427386
```

```
cat("w =", w)
```

```
## w = -0.0746888
```

2.2.3.10. Sistema 9

$$x_1 + x_2 = 3$$

$$x_1 + 2x_2 + x_3 = -1$$

$$x_2 + 3x_3 + x_4 = 2$$

$$x_3 + 4x_4 + x_5 = 1$$

$$x_4 + 5x_5 = 3$$

```
A <- matrix(c(1, 1, 0, 0, 0,
              1, 2, 1, 0, 0,
              0, 1, 3, 1, 0,
              0, 0, 1, 4, 1,
              0, 0, 0, 1, 5),
            nrow = 5, byrow = TRUE)
b <- matrix(c(3, -1, 2, 1, 3), nrow = 5)

Ab <- cbind(A, b); (Ab)
```

2.2.3.10.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]    1    1    0    0    0    3
## [2,]    1    2    1    0    0   -1
## [3,]    0    1    3    1    0    2
## [4,]    0    0    1    4    1    1
## [5,]    0    0    0    1    5    3
```

```
#Volver ceros
```

```
A1 <- Ab[1,]  
(A1)
```

2.2.3.10.1.1. Primera eliminación

```
## [1] 1 1 0 0 0 3
```

```
A2 <- Ab[1,] - Ab[2,]  
(A2)
```

```
## [1] 0 -1 -1 0 0 4
```

```
A3 <- Ab[3,]  
(A3)
```

```
## [1] 0 1 3 1 0 2
```

```
A4 <- Ab[4,]  
(A4)
```

```
## [1] 0 0 1 4 1 1
```

```
A5 <- Ab[5,]  
(A5)
```

```
## [1] 0 0 0 1 5 3
```

```
NuevaA <- rbind(A1, A2, A3, A4, A5)  
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]  
## A1      1      1      0      0      0      3  
## A2      0     -1     -1      0      0      4  
## A3      0      1      3      1      0      2  
## A4      0      0      1      4      1      1  
## A5      0      0      0      1      5      3
```

```
# Primera normalización
```

```
Ab <- NuevaA  
n1 <- -1  
A1 <- Ab[1,]  
A2 <- n1 * Ab[2,]  
A3 <- Ab[3,]  
A4 <- Ab[4,]  
A5 <- Ab[5,]  
NuevaA <- rbind(A1, A2, A3, A4, A5)  
(NuevaA)
```

2.2.3.10.1.2. Segunda eliminación

```
##      [,1] [,2] [,3] [,4] [,5] [,6]  
## A1      1      1      0      0      0      3
```

```
## A2    0    1    1    0    0   -4
## A3    0    1    3    1    0    2
## A4    0    0    1    4    1    1
## A5    0    0    0    1    5    3
```

```
# Volver ceros
```

```
Ab <- NuevaA
(Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## A1     1    1    0    0    0    3
## A2     0    1    1    0    0   -4
## A3     0    1    3    1    0    2
## A4     0    0    1    4    1    1
## A5     0    0    0    1    5    3
```

```
A1 <- Ab[1,] - Ab[2,]
(A1)
```

```
## [1]  1  0 -1  0  0  7
```

```
A2 <- Ab[2,]
(A2)
```

```
## [1]  0  1  1  0  0 -4
```

```
A3 <- Ab[2,] - Ab[3,]
(A3)
```

```
## [1]  0  0 -2 -1  0 -6
```

```
A4 <- Ab[4,]
(A4)
```

```
## [1]  0  0  1  4  1  1
```

```
A5 <- Ab[5,]
(A5)
```

```
## [1]  0  0  0  1  5  3
```

```
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## A1     1    0   -1    0    0    7
## A2     0    1    1    0    0   -4
## A3     0    0   -2   -1    0   -6
## A4     0    0    1    4    1    1
## A5     0    0    0    1    5    3
```

```
# Tercera normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
```

```
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.10.1.3. Tercera eliminación

```
## [1] 0.0 0.0 1.0 0.5 0.0 3.0
```

```
A4 <- Ab[4,]
A5 <- Ab[5,]
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## A1      1    0  -1  0.0    0    7
## A2      0    1    1  0.0    0   -4
## A3      0    0    1  0.5    0    3
## A4      0    0    1  4.0    1    1
## A5      0    0    0  1.0    5    3
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
## A1
## 1
```

```
m6 <- - Ab[2,3] / Ab[3,3]
(m6)
```

```
## A2
## -1
```

```
m7 <- - Ab[4,3] / Ab[3,3]
(m7)
```

```
## A4
## -1
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- m6 * Ab[3,] + Ab[2,]
A3 <- Ab[3,]
A4 <- m7 * Ab[3,] + Ab[4,]
A5 <- Ab[5,]
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6]
## A1      1    0    0  0.5    0   10
## A2      0    1    0 -0.5    0   -7
## A3      0    0    1  0.5    0    3
## A4      0    0    0  3.5    1   -2
## A5      0    0    0  1.0    5    3
```

```
# Cuarta normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[4,4]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- n2 * Ab[4,]
(A4)
```

2.2.3.10.1.4. Cuarta eliminación

```
## [1] 0.0000000 0.0000000 0.0000000 1.0000000 0.2857143 -0.5714286
```

```
A5 <- Ab[5,]
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]
## A1      1    0    0  0.5 0.0000000 10.0000000
## A2      0    1    0 -0.5 0.0000000 -7.0000000
## A3      0    0    1  0.5 0.0000000  3.0000000
## A4      0    0    0  1.0 0.2857143 -0.5714286
## A5      0    0    0  1.0 5.0000000  3.0000000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,4] / Ab[4,4]
(m5)
```

```
##      A1
## -0.5
```

```
m6 <- - Ab[2,4] / Ab[4,4]
(m6)
```

```
##      A2
## 0.5
```

```
m7 <- - Ab[3,4] / Ab[4,4]
(m7)
```

```
##      A3
## -0.5
```

```
A1 <- m5 * Ab[4,] + Ab[1,]
A2 <- m6 * Ab[4,] + Ab[2,]
A3 <- m7 * Ab[4,] + Ab[3,]
A4 <- Ab[4,]
A5 <- Ab[5,] - Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]
```

```
## A1      1      0      0      0 -0.1428571 10.2857143
## A2      0      1      0      0  0.1428571 -7.2857143
## A3      0      0      1      0 -0.1428571  3.2857143
## A4      0      0      0      1  0.2857143 -0.5714286
## A5      0      0      0      0  4.7142857  3.5714286
```

Quinta normalización

```
Ab <- NuevaA
n2 <- 1 / Ab[5,5]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
(A4)
```

2.2.3.10.1.5. Quinta eliminación

```
## [1] 0.0000000 0.0000000 0.0000000 1.0000000 0.2857143 -0.5714286
```

```
A5 <- n2 * Ab[5,]
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]
## A1      1      0      0      0 -0.1428571 10.2857143
## A2      0      1      0      0  0.1428571 -7.2857143
## A3      0      0      1      0 -0.1428571  3.2857143
## A4      0      0      0      1  0.2857143 -0.5714286
## A5      0      0      0      0  1.0000000  0.7575758
```

Volver ceros

```
Ab <- NuevaA
m5 <- - Ab[1,5] / Ab[5,5]
(m5)
```

```
##      A1
## 0.1428571
```

```
m6 <- - Ab[2,5] / Ab[5,5]
(m6)
```

```
##      A2
## -0.1428571
```

```
m7 <- - Ab[3,5] / Ab[5,5]
(m7)
```

```
##      A3
## 0.1428571
```

```
m8 <- - Ab[4,5] / Ab[5,5]
(m8)
```

```
##      A4
```



```
## -0.2857143
```

```
A1 <- m5 * Ab[5,] + Ab[1,]
A2 <- m6 * Ab[5,] + Ab[2,]
A3 <- m7 * Ab[5,] + Ab[3,]
A4 <- m8 * Ab[5,] + Ab[4,]
A5 <- Ab[5,]
NuevaA <- rbind(A1, A2, A3, A4, A5)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]      [,6]
## A1      1      0      0      0      0 10.3939394
## A2      0      1      0      0      0 -7.3939394
## A3      0      0      1      0      0  3.3939394
## A4      0      0      0      1      0 -0.7878788
## A5      0      0      0      0      1  0.7575758
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]      [,6]
## A1      1      0      0      0      0 10.3939394
## A2      0      1      0      0      0 -7.3939394
## A3      0      0      1      0      0  3.3939394
## A4      0      0      0      1      0 -0.7878788
## A5      0      0      0      0      1  0.7575758
```

```
x1 <- Ab[1,6] / Ab[1,1]
x2 <- Ab[2,6] / Ab[2,2]
x3 <- Ab[3,6] / Ab[3,3]
x4 <- Ab[4,6] / Ab[4,4]
x5 <- Ab[5,6] / Ab[5,5]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 3
```

```
II <- b[2,]; (II)
```

```
## [1] -1
```

```
III <- b[3,]; (III)
```

```
## [1] 2
```

```
IV <- b[4,]; (IV)
```

```
## [1] 1
```

```
V <- b[5,]; (V)
```

```
## [1] 3
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4) + (A[1,5] * x5); paste0("El resultado es: ")
```

```
## [1] "El resultado es: 3"
```

```

c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4) + (A[2,5] * x5); paste0("El resultado es: ", c2)

## [1] "El resultado es: -0.999999999999999"

c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4) + (A[3,5] * x5); paste0("El resultado es: ", c3)

## [1] "El resultado es: 2"

c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4) + (A[4,5] * x5); paste0("El resultado es: ", c4)

## [1] "El resultado es: 1"

c5 <- (A[5,1] * x1) + (A[5,2] * x2) + (A[5,3] * x3) + (A[5,4] * x4) + (A[5,5] * x5); paste0("El resultado es: ", c5)

## [1] "El resultado es: 3"

paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 3 técnicamente es: 3"

paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: -0.999999999999999 técnicamente es: -1"

paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 2 técnicamente es: 2"

paste0("En la ecuación 4: ", c4, " técnicamente es: ", IV)

## [1] "En la ecuación 4: 1 técnicamente es: 1"

paste0("En la ecuación 5: ", c5, " técnicamente es: ", V)

## [1] "En la ecuación 5: 3 técnicamente es: 3"

```

```
cat("Soluciones del sistema:")
```

2.2.3.10.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 = ", x1)
```

```
## x1 = 10.39394
```

```
cat("x2 = ", x2)
```

```
## x2 = -7.393939
```

```
cat("x3 = ", x3)
```

```
## x3 = 3.393939
```

```
cat("x4 = ", x4)
```

```
## x4 = -0.7878788
```

```
cat("x5 = ", x5)
```

```
## x5 = 0.7575758
```

2.2.3.11. Sistema 10

$$15x_1 - 18x_2 + 15x_3 - 3x_4 = 11$$

$$-18x_1 + 24x_2 - 18x_3 + 4x_4 = 10$$

$$15x_1 - 18x_2 + 18x_3 - 3x_4 = 11$$

$$-3x_1 + 4x_2 - 3x_3 + x_4 = 13$$

```
A <- matrix(c(15, -18, 15, -3,
              -18, 24, -18, 4,
              15, -18, 18, -3,
              -3, 4, -3, 1),
            nrow = 4, byrow = TRUE)
b <- matrix(c(11, 10, 11, 13), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.2.3.11.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]  15 -18  15  -3  11
## [2,] -18  24 -18   4  10
## [3,]  15 -18  18  -3  11
## [4,]  -3   4  -3   1  13
```

Primer pivote

```
Ab <- Ab[c(2, 1, 3, 4), ]
(Ab)
```

2.2.3.11.1.1. Primera eliminación

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,] -18  24 -18   4  10
## [2,]  15 -18  15  -3  11
## [3,]  15 -18  18  -3  11
## [4,]  -3   4  -3   1  13
```

Primera normalización

```
n1 <- 1 / Ab[1,1]
A1 <- n1 * Ab[1,]
(A1)
```

```
## [1]  1.0000000 -1.3333333  1.0000000 -0.2222222 -0.5555556
```

```
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]
```

```
## A1    1 -1.333333    1 -0.2222222 -0.5555556
## A2   15 -18.000000   15 -3.0000000 11.0000000
## A3   15 -18.000000   18 -3.0000000 11.0000000
## A4    -3  4.000000   -3  1.0000000 13.0000000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m1 <- - Ab[2,1] / Ab[1,1]
(m1)
```

```
## A2
## -15
m2 <- - Ab[3,1] / Ab[1,1]
(m2)
```

```
## A3
## -15
m3 <- - Ab[4,1] / Ab[1,1]
(m3)
```

```
## A4
## 3
A1 <- Ab[1,]
(A1)
```

```
## [1]  1.0000000 -1.3333333  1.0000000 -0.2222222 -0.5555556
A2 <- m1 * Ab[1,] + Ab[2,]
(A2)
```

```
## [1]  0.0000000  2.0000000  0.0000000  0.3333333 19.3333333
A3 <- m2 * Ab[1,] + Ab[3,]
(A3)
```

```
## [1]  0.0000000  2.0000000  3.0000000  0.3333333 19.3333333
A4 <- m3 * Ab[1,] + Ab[4,]
(A4)
```

```
## [1]  0.0000000  0.0000000  0.0000000  0.3333333 11.3333333
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1]      [,2] [,3]      [,4]      [,5]
## A1     1 -1.333333    1 -0.2222222 -0.5555556
## A2     0  2.000000    0  0.3333333 19.3333333
## A3     0  2.000000    3  0.3333333 19.3333333
## A4     0  0.000000    0  0.3333333 11.3333333
```

```
# Segunda normalización
```

```

Ab <- NuevaA
n1 <- 1 / Ab[2,2]
A1 <- Ab[1,]
A2 <- n1 * Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)

```

2.2.3.11.1.2. Segunda eliminación

```

##      [,1]      [,2] [,3]      [,4]      [,5]
## A1      1 -1.333333      1 -0.2222222 -0.5555556
## A2      0  1.000000      0  0.1666667  9.6666667
## A3      0  2.000000      3  0.3333333 19.3333333
## A4      0  0.000000      0  0.3333333 11.3333333

```

Volver ceros

```

Ab <- NuevaA
m1 <- - Ab[1,2] / Ab[2,2]
(m1)

```

```

##      A1
## 1.333333

```

```

m2 <- - Ab[3,2] / Ab[2,2]
(m2)

```

```

## A3
## -2

```

```

m3 <- - Ab[4,2] / Ab[2,2]
(m3)

```

```

## A4
## 0

```

```

A1 <- m1 * Ab[2,] + Ab[1,]
(A1)

```

```

## [1] 1.000000e+00 0.000000e+00 1.000000e+00 -1.942890e-16 1.233333e+01

```

```

A2 <- Ab[2,]
(A2)

```

```

## [1] 0.0000000 1.0000000 0.0000000 0.1666667 9.6666667

```

```

A3 <- m2 * Ab[2,] + Ab[3,]
(A3)

```

```

## [1] 0 0 3 0 0

```

```

A4 <- Ab[4,]
(A4)

```

```

## [1] 0.0000000 0.0000000 0.0000000 0.3333333 11.3333333

```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      1 -1.942890e-16 12.333333
## A2      0      1      0  1.666667e-01  9.666667
## A3      0      0      3  0.000000e+00  0.000000
## A4      0      0      0  3.333333e-01 11.333333
```

```
# Tercera normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[3,3]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- n2 * Ab[3,]
(A3)
```

2.2.3.11.1.3. Tercera eliminación

```
## [1] 0 0 1 0 0
```

```
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      1 -1.942890e-16 12.333333
## A2      0      1      0  1.666667e-01  9.666667
## A3      0      0      1  0.000000e+00  0.000000
## A4      0      0      0  3.333333e-01 11.333333
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,3] / Ab[3,3]
(m5)
```

```
## A1
## -1
```

```
A1 <- m5 * Ab[3,] + Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      0 -1.942890e-16 12.333333
## A2      0      1      0  1.666667e-01  9.666667
## A3      0      0      1  0.000000e+00  0.000000
## A4      0      0      0  3.333333e-01 11.333333
```

```
# Cuarta normalización
```

```
Ab <- NuevaA
n2 <- 1 / Ab[4,4]
A1 <- Ab[1,]
A2 <- Ab[2,]
A3 <- Ab[3,]
A4 <- n2 * Ab[4,]
(A4)
```

2.2.3.11.1.4. Cuarta eliminación

```
## [1] 0 0 0 1 34
```

```
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3]      [,4]      [,5]
## A1      1      0      0 -1.942890e-16 12.333333
## A2      0      1      0  1.666667e-01  9.666667
## A3      0      0      1  0.000000e+00  0.000000
## A4      0      0      0  1.000000e+00 34.000000
```

```
# Volver ceros
```

```
Ab <- NuevaA
m5 <- - Ab[1,4] / Ab[4,4]
(m5)
```

```
##          A1
## 1.94289e-16
```

```
m6 <- - Ab[2,4] / Ab[4,4]
(m6)
```

```
##          A2
## -0.1666667
```

```
m7 <- - Ab[3,4] / Ab[4,4]
(m7)
```

```
## A3
## 0
```

```
A1 <- m5 * Ab[4,] + Ab[1,]
A2 <- m6 * Ab[4,] + Ab[2,]
A3 <- m7 * Ab[4,] + Ab[3,]
A4 <- Ab[4,]
NuevaA <- rbind(A1, A2, A3, A4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      1      0      0      0 12.33333
## A2      0      1      0      0  4.00000
## A3      0      0      1      0  0.00000
```

```
## A4      0      0      0      1 34.00000
```

```
# Soluciones
```

```
Ab <- NuevaA; (Ab)
```

```
##      [,1] [,2] [,3] [,4]      [,5]
## A1      1      0      0      0 12.33333
## A2      0      1      0      0  4.00000
## A3      0      0      1      0  0.00000
## A4      0      0      0      1 34.00000
```

```
x1 <- Ab[1,5] / Ab[1,1]
x2 <- Ab[2,5] / Ab[2,2]
x3 <- Ab[3,5] / Ab[3,3]
x4 <- Ab[4,5] / Ab[4,4]
```

```
# Comprobación
```

```
I <- b[1,]; (I)
```

```
## [1] 11
```

```
II <- b[2,]; (II)
```

```
## [1] 10
```

```
III <- b[3,]; (III)
```

```
## [1] 11
```

```
IV <- b[4,]; (IV)
```

```
## [1] 13
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 11"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 10"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 11"
```

```
c4 <- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)
```

```
## [1] "El resultado es: 13"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 11 técnicamente es: 11"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 10 técnicamente es: 10"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```



```
## [1] "En la ecuación 3: 11 tecnicamente es: 11"
paste0("En la ecuación 4: ", c4, " tecnicamente es: ", IV)
```

```
## [1] "En la ecuación 4: 13 tecnicamente es: 13"
```

```
cat("Soluciones del sistema:")
```

2.2.3.11.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 12.33333
```

```
cat("x2 =", x2)
```

```
## x2 = 4
```

```
cat("x3 =", x3)
```

```
## x3 = 0
```

```
cat("x4 =", x4)
```

```
## x4 = 34
```

2.3. Solucion de Sistemas de Ecuaciones Lineales por métodos iterativos

2.3.1. 7.6. Eliminación Gauss-Seidel (Método iterativo)

2.3.1.1. Ejercicios a mano

2.3.1.2. Sistema 1

$$3x_1 - 0,2x_2 - 0,5x_3 = 8$$

$$0,1x_1 + 7x_2 + 0,4x_3 = -19,5$$

$$0,4x_1 - 0,1x_2 + 10x_3 = 72,4$$

```
A <- matrix(c(3, -0.2, -0.5,
              0.1, 7, 0.4,
              0.4, -0.1, 10),
            nrow = 3, byrow = TRUE)
b <- c(8, -19.5, 72.4)

Ab <- cbind(A, b); (Ab)
```

2.3.1.2.1. Procedimiento

```
##               b
## [1,] 3.0 -0.2 -0.5  8.0
## [2,] 0.1  7.0  0.4 -19.5
```

7.6 Gauss-Seidel

$$\begin{cases} 3x_1 - 0.2x_2 - 0.5x_3 = 8 \\ 0.1x_1 + 7x_2 + 0.4x_3 = -19.5 \\ 0.4x_1 - 0.1x_2 + 10x_3 = 72.4 \end{cases} \Rightarrow \begin{bmatrix} 3 & -0.2 & -0.5 & 8 \\ 0.1 & 7 & 0.4 & -19.5 \\ 0.4 & -0.1 & 10 & 72.4 \end{bmatrix}$$

Verificamos si la matriz tiene una diagonal dominante

$$\begin{cases} |3| > |-0.2| + |-0.5| \Rightarrow |3| > |0.7| \\ |7| > |0.1| + |0.4| \Rightarrow |7| > |0.5| \\ |10| > |0.4| + |-0.1| \Rightarrow |10| > |0.5| \end{cases} \left. \vphantom{\begin{cases} |3| > |-0.2| + |-0.5| \\ |7| > |0.1| + |0.4| \\ |10| > |0.4| + |-0.1| \end{cases}} \right\} \text{Si es dominante}$$

$$x_1 = \frac{8 + 0.2x_2 + 0.5x_3}{3} \quad x_2 = \frac{-19.5 - 0.1x_1 - 0.4x_3}{7}$$

$$x_3 = \frac{72.4 - 0.4x_1 + 0.1x_2}{10}$$

$$x^{(0)} = (0, 0, 0)$$

$$x_1^{(1)} = \frac{8 + 0.2(0) + 0.5(0)}{3} = \frac{8}{3} \quad x_2^{(1)} = \frac{-19.5 - 0.1(\frac{8}{3}) - 0.4(0)}{7} = \frac{-593}{210}$$

$$x_3^{(1)} = \frac{72.4 + 0.4(\frac{8}{3}) + 0.1(-\frac{593}{210})}{10} \approx 7.10509524$$

$$x_1^{(2)} = \frac{8 + 0.2(-\frac{593}{210}) + 0.5(7.10509524)}{3} \approx 3.66259524$$

$$x_2^{(2)} = \frac{-19.5 - 0.1(3.66259524) - 0.4(7.10509524)}{7} \approx -3.24404252$$

$$x_3^{(2)} = \frac{72.4 - 0.4(3.66259524) + 0.1(-3.24404252)}{10} \approx 7.0605577$$

Figura 2.21: Ejercicio 1

Se comprueba el error en X_1

$$E = [3.66259524 - (-\frac{593}{210})] = 6.48640476$$

$$X_1^{(3)} = \frac{8 + 0.2(-3.2404252) + 0.5(7.06105577)}{3} \approx 3.62723979$$

$$X_2^{(3)} = \frac{-19.5 - 0.1(3.62723979) - 0.4(7.06105577)}{7} \approx -3.24102090$$

$$X_3^{(3)} = \frac{72.4 - 0.4(3.62723979) + 0.1(-3.24102090)}{10} \approx 7.06250020$$

$$X_1^{(4)} = \frac{8 + 0.2(-3.24102090) + 0.5(7.06250020)}{3} \approx 3.62809864$$

$$X_2^{(4)} = \frac{-19.5 - 0.1(3.62809864) - 0.4(7.06250020)}{7} \approx -3.24111571$$

$$X_3^{(4)} = \frac{72.4 - 0.4(3.62809864) + 0.1(-3.24111571)}{10} \approx 7.06246490$$

$$E = [3.62809864 - 3.62723979] = 0.00085885$$

$$X_1^{(5)} = \frac{8 + 0.2(-3.24111571) + 0.5(7.06250020)}{3} \approx 3.627676$$

$$X_2^{(5)} = \frac{-19.5 - 0.1(3.627676) - 0.4(7.06246490)}{7} \approx -3.241108$$

$$X_3^{(5)} = \frac{72.4 - 0.4(3.627676) + 0.1(-3.241108)}{10} \approx 7.062482$$

$$X_1^{(6)} = \frac{8 + 0.2(-3.241108) + 0.5(7.062482)}{3} \approx 3.627673$$

$$X_2^{(6)} = \frac{-19.5 - 0.1(3.627673) - 0.4(7.062482)}{7} \approx -3.241109$$

$$X_3^{(6)} = \frac{72.4 - 0.4(3.627673) + 0.1(-3.241109)}{10} \approx 7.062482$$

$$E = [3.627673 - 3.627676] = -0.000003$$

Figura 2.22: Ejercicio 1-2

$$\begin{aligned}
 x_1^{(7)} &= \frac{8 + 0.2(-3.241109) + 0.5(7.062482)}{3} = 3.627673 \\
 x_2^{(7)} &= \frac{-19.5 - 0.1(3.627673) - 0.4(7.062482)}{7} = -3.241109 \\
 x_3^{(7)} &= \frac{72.4 - 0.4(3.627673) + 0.1(-3.241109)}{10} = 7.062482 \\
 x_1^{(8)} &= \frac{8 + 0.2(-3.241109) + 0.5(7.062482)}{3} = 3.627673 \\
 x_2^{(8)} &= \frac{-19.5 - 0.1(3.627673) - 0.4(7.062482)}{7} = -3.241109 \\
 x_3^{(8)} &= \frac{72.4 - 0.4(3.627673) + 0.1(-3.241109)}{10} = 7.062482 \\
 E &= |3.627673 - 3.627673| = 0 \\
 &= |-3.241109 - (-3.241109)| = 0 \\
 &= |7.062482 - 7.062482| = 0
 \end{aligned}$$

Comprobación

$$\begin{aligned}
 3(3.627673) - 0.2(-3.241109) - 0.5(7.062482) &= 8 \\
 10.883019 + 0.648222 - 3.531241 &\Rightarrow 8 = 8 \\
 0.1(3.627673) + 7(-3.241109) + 0.4(7.062482) &= -19.5 \\
 0.362767 - 22.687763 + 2.824993 &= -19.5 \Rightarrow -19.500003 \approx -19.5 \\
 0.4(3.627673) - 0.1(-3.241109) + 10(7.062482) &= 72.4 \\
 1.451069 + 0.324111 + 70.62482 &= 72.4 \Rightarrow 72.4 = 72.4
 \end{aligned}$$

Soluciones: $x_1 \approx 3.627673$
 $x_2 \approx -3.241109$
 $x_3 \approx 7.062482$

Figura 2.23: Ejercicio 1-3

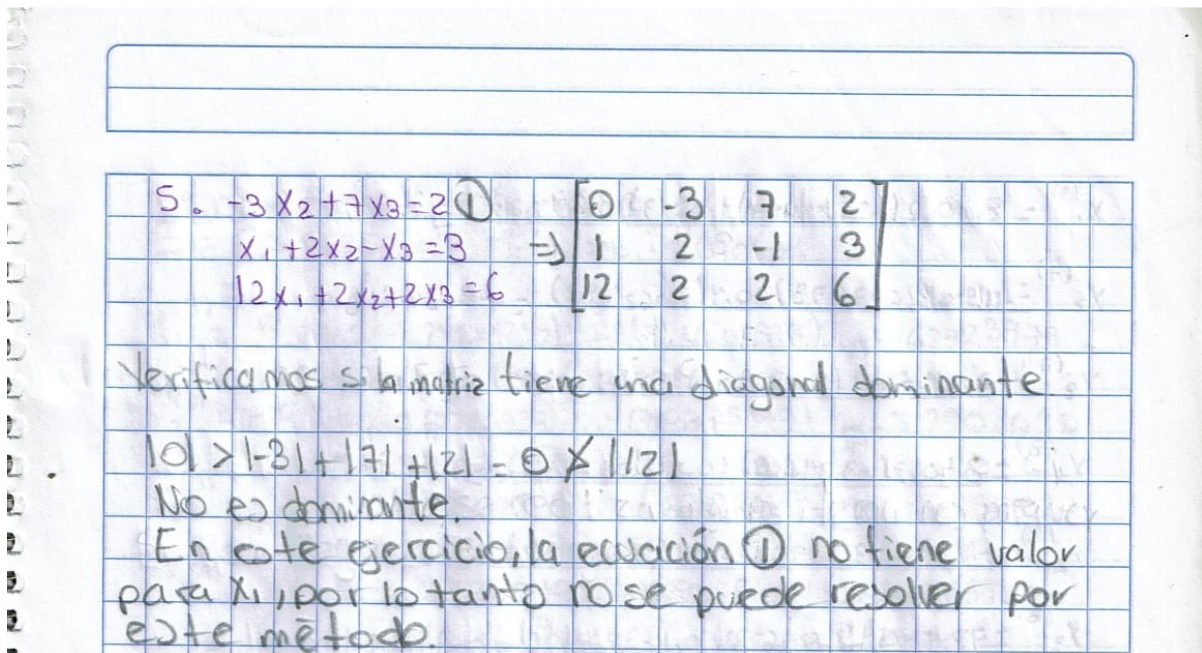


Figura 2.24: Ejercicio 5

```
## [3,] 0.4 -0.1 10.0 72.4
```

```
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.1. Iteración 1

```
## x1^(1) = (8.0 - (-0.2)*0.00000000 - (-0.5)*0.00000000)/3.0 = 2.66666667
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(1) = (-19.5 - (0.1)*2.66666667 - (0.4)*0.00000000)/7.0 = -2.82380952
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(1) = (72.4 - (0.4)*2.66666667 - (-0.1)*-2.82380952)/10.0 = 7.10509524
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(1) = 2.66666667, -2.82380952, 7.10509524
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 7.10509524e+00 | ||r||_inf = 2.98778571e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.2.1.2. Iteración 2

```
## x1^(2) = (8.0 - (-0.2)*-2.82380952 - (-0.5)*7.10509524)/3.0 = 3.66259524
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(2) = (-19.5 - (0.1)*3.66259524 - (0.4)*7.10509524)/7.0 = -3.24404252
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(2) = (72.4 - (0.4)*3.66259524 - (-0.1)*-3.24404252)/10.0 = 7.06105577
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(2) = 3.66259524, -3.24404252, 7.06105577
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 9.95928571e-01 | ||r||_inf = 1.06066335e-01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```



```
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.3. Iteración 3

```
## x1^(3) = (8.0 - (-0.2)*-3.24404252 - (-0.5)*7.06105577)/3.0 = 3.62723979
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(3) = (-19.5 - (0.1)*3.62723979 - (0.4)*7.06105577)/7.0 = -3.24102090
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(3) = (72.4 - (0.4)*3.62723979 - (-0.1)*-3.24102090)/10.0 = 7.06250020
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(3) = 3.62723979, -3.24102090, 7.06250020
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.53554450e-02 | ||r||_inf = 1.32654081e-03
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.4. Iteración 4

```
## x1^(4) = (8.0 - (-0.2)*-3.24102090 - (-0.5)*7.06250020)/3.0 = 3.62768197
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(4) = (-19.5 - (0.1)*3.62768197 - (0.4)*7.06250020)/7.0 = -3.24110975
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(4) = (72.4 - (0.4)*3.62768197 - (-0.1)*-3.24110975)/10.0 = 7.06248162
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(4) = 3.62768197, -3.24110975, 7.06248162
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 4.42180271e-04 | ||r||_inf = 2.70590744e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.5. Iteración 5

```
## x1^(5) = (8.0 - (-0.2)*-3.24110975 - (-0.5)*7.06248162)/3.0 = 3.62767295
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(5) = (-19.5 - (0.1)*3.62767295 - (0.4)*7.06248162)/7.0 = -3.24110856
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(5) = (72.4 - (0.4)*3.62767295 - (-0.1)*-3.24110856)/10.0 = 7.06248200
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(5) = 3.62767295, -3.24110856, 7.06248200
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 9.01969147e-06 | ||r||_inf = 4.24410521e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.6. Iteración 6

```
## x1^(6) = (8.0 - (-0.2)*-3.24110856 - (-0.5)*7.06248200)/3.0 = 3.62767310
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(6) = (-19.5 - (0.1)*3.62767310 - (0.4)*7.06248200)/7.0 = -3.24110859
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(6) = (72.4 - (0.4)*3.62767310 - (-0.1)*-3.24110859)/10.0 = 7.06248199
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(6) = 3.62767310, -3.24110859, 7.06248199
```



```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.41470173e-07 | ||r||_inf = 7.60951657e-09

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.7. Iteración 7

```
## x1^(7) = (8.0 - (-0.2)*-3.24110859 - (-0.5)*7.06248199)/3.0 = 3.62767309
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(7) = (-19.5 - (0.1)*3.62767309 - (0.4)*7.06248199)/7.0 = -3.24110859
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(7) = (72.4 - (0.4)*3.62767309 - (-0.1)*-3.24110859)/10.0 = 7.06248199
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(7) = 3.62767309, -3.24110859, 7.06248199
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.53650523e-09 | ||r||_inf = 1.27178268e-10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.8. Iteración 8

```
## x1^(8) = (8.0 - (-0.2)*-3.24110859 - (-0.5)*7.06248199)/3.0 = 3.62767309
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(8) = (-19.5 - (0.1)*3.62767309 - (0.4)*7.06248199)/7.0 = -3.24110859
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(8) = (72.4 - (0.4)*3.62767309 - (-0.1)*-3.24110859)/10.0 = 7.06248199
#Error o tolerancia

r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = 3.62767309, -3.24110859, 7.06248199

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.23927560e-11 | ||r||_inf = 2.20445884e-12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.9. Iteración 9

```
## x1^(9) = (8.0 - (-0.2)*-3.24110859 - (-0.5)*7.06248199)/3.0 = 3.62767309
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]))

## x2^(9) = (-19.5 - (0.1)*3.62767309 - (0.4)*7.06248199)/7.0 = -3.24110859
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(9) = (72.4 - (0.4)*3.62767309 - (-0.1)*-3.24110859)/10.0 = 7.06248199
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = 3.62767309, -3.24110859, 7.06248199

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 7.34967642e-13 | ||r||_inf = 3.64153152e-14
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.2.1.10. Iteración 10

```
## x1^(10) = (8.0 - (-0.2)*-3.24110859 - (-0.5)*7.06248199)/3.0 = 3.62767309
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]))

## x2^(10) = (-19.5 - (0.1)*3.62767309 - (0.4)*7.06248199)/7.0 = -3.24110859
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(10) = (72.4 - (0.4)*3.62767309 - (-0.1)*-3.24110859)/10.0 = 7.06248199
```

```

#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(10) = 3.62767309, -3.24110859, 7.06248199
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.24344979e-14 | ||r||_inf = 0.00000000e+00

# Soluciones
x_old <- x
x1 <- x_old[1]
x2 <- x_old[2]
x3 <- x_old[3]

# Comprobación

I <- b[1]; (I)

## [1] 8
II <- b[2]; (II)

## [1] -19.5
III <- b[3]; (III)

## [1] 72.4
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: 8"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: -19.5"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 72.4"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 8 técnicamente es: 8"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: -19.5 técnicamente es: -19.5"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 72.4 técnicamente es: 72.4"

cat("Soluciones del sistema:")

```

2.3.1.2.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 3.627673
```

```
cat("x2 =", x2)
```

```
## x2 = -3.241109
```

```
cat("x3 =", x3)
```

```
## x3 = 7.062482
```

2.3.1.3. Sistema 2

$$-5x_1 + 1,4x_2 - 2,7x_3 = 94,2$$

$$0,7x_1 - 2,5x_2 + 15x_3 = -6$$

$$3,3x_1 - 11x_2 + 4,4x_3 = -27,5$$

```
A <- matrix(c(-5, 1.4, -2.7,
              0.7, -2.5, 15,
              3.3, -11, 4.4),
            nrow = 3, byrow = TRUE)
b <- c(94.2, -6, -27.5)

Ab <- cbind(A, b); (Ab)
```

2.3.1.3.1. Procedimiento

```
##              b
## [1,] -5.0    1.4 -2.7  94.2
## [2,]  0.7   -2.5 15.0  -6.0
## [3,]  3.3 -11.0  4.4 -27.5

# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## No es dominante
```

Aunque no es una matriz con diagonal dominante, se intenta resolverlo con 10 iteraciones para ver si puede o no llegar a converger.

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=",paste(sprintf("%.8f",x), collapse=" "), "\n")
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.1. Iteración 1

```
## x1^(1) = (94.2 - (1.4)*0.00000000 - (-2.7)*0.00000000)/-5.0 = -18.84000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(1) = (-6.0 - (0.7)*-18.84000000 - (15.0)*0.00000000)/-2.5 = -2.87520000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(1) = (-27.5 - (3.3)*-18.84000000 - (-11.0)*-2.87520000)/4.4 = 0.69200000
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(1) = -18.84000000, -2.87520000, 0.69200000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.88400000e+01 | ||r||_inf = 1.03800000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.2. Iteración 2

```
## x1^(2) = (94.2 - (1.4)*-2.87520000 - (-2.7)*0.69200000)/-5.0 = -20.01873600
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(2) = (-6.0 - (0.7)*-20.01873600 - (15.0)*0.69200000)/-2.5 = 0.94675392
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(2) = (-27.5 - (3.3)*-20.01873600 - (-11.0)*0.94675392)/4.4 = 11.13093680
```

#Error o tolerancia

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(2) = -20.01873600, 0.94675392, 11.13093680

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.04389368e+01 | ||r||_inf = 1.56584052e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.3. Iteración 3

```
## x1^(3) = (94.2 - (1.4)*0.94675392 - (-2.7)*11.13093680)/-5.0 = -24.58561477
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], A[2,2]))

## x2^(3) = (-6.0 - (0.7)*-24.58561477 - (15.0)*11.13093680)/-2.5 = 62.30164866
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], A[3,3]))

## x3^(3) = (-27.5 - (3.3)*-24.58561477 - (-11.0)*62.30164866)/4.4 = 167.94333274
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(3) = -24.58561477, 62.30164866, 167.94333274

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.56812396e+02 | ||r||_inf = 2.35218594e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.4. Iteración 4

```
## x1^(4) = (94.2 - (1.4)*62.30164866 - (-2.7)*167.94333274)/-5.0 = -92.08493805
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], A[2,2]))

## x2^(4) = (-6.0 - (0.7)*-92.08493805 - (15.0)*167.94333274)/-2.5 = 984.27621378
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], A[3,3]))

## x3^(4) = (-27.5 - (3.3)*-92.08493805 - (-11.0)*984.27621378)/4.4 = 2523.50423798
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(4) = -92.08493805, 984.27621378, 2523.50423798
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.35556091e+03 | ||r||_inf = 3.53334136e+04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.5. Iteración 5

```
## x1^(5) = (94.2 - (1.4)*984.27621378 - (-2.7)*2523.50423798)/-5.0 = -1105.93494865
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(5) = (-6.0 - (0.7)*-1105.93494865 - (15.0)*2523.50423798)/-2.5 = 14833.76364228
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(5) = (-27.5 - (3.3)*-1105.93494865 - (-11.0)*14833.76364228)/4.4 = 37907.61031718
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(5) = -1105.93494865, 14833.76364228, 37907.61031718
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.53841061e+04 | ||r||_inf = 5.30761591e+05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.6. Iteración 6

```
## x1^(6) = (94.2 - (1.4)*14833.76364228 - (-2.7)*37907.61031718)/-5.0 = -16335.49575144
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(6) = (-6.0 - (0.7)*-16335.49575144 - (15.0)*37907.61031718)/-2.5 = 222874.12309270
```



```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(6) = (-27.5 - (3.3)*-16335.49575144 - (-11.0)*222874.12309270)/4.4 = 569430.67954534
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(6) = -16335.49575144, 222874.12309270, 569430.67954534
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 5.31523069e+05 | ||r||_inf = 7.97284604e+06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.7. Iteración 7

```
## x1^(7) = (94.2 - (1.4)*222874.12309270 - (-2.7)*569430.67954534)/-5.0 = -245106.65248853
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(7) = (-6.0 - (0.7)*-245106.65248853 - (15.0)*569430.67954534)/-2.5 = 3347956.61457526
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(7) = (-27.5 - (3.3)*-245106.65248853 - (-11.0)*3347956.61457526)/4.4 = 8553715.27580454
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(7) = -245106.65248853, 3347956.61457526, 8553715.27580454
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 7.98428460e+06 | ||r||_inf = 1.19764269e+08
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.3.1.8. Iteración 8

```
## x1^(8) = (94.2 - (1.4)*3347956.61457526 - (-2.7)*8553715.27580454)/-5.0 = -3681597.23685338
```

```

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(8) = (-6.0 - (0.7)*-3681597.23685338 - (15.0)*8553715.27580454)/-2.5 = 50291446.82850830
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(8) = (-27.5 - (3.3)*-3681597.23685338 - (-11.0)*50291446.82850830)/4.4 = 128489808.74891074
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = -3681597.23685338, 50291446.82850830, 128489808.74891074
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.19936093e+08 | ||r||_inf = 1.79904140e+09

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))

```

2.3.1.3.1.9. Iteración 9

```

## x1^(9) = (94.2 - (1.4)*50291446.82850830 - (-2.7)*128489808.74891074)/-5.0 = -55302910.45242949
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(9) = (-6.0 - (0.7)*-55302910.45242949 - (15.0)*128489808.74891074)/-2.5 = 755454039.96678424
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(9) = (-27.5 - (3.3)*-55302910.45242949 - (-11.0)*755454039.96678424)/4.4 = 1930112276.50628257
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = -55302910.45242949, 755454039.96678424, 1930112276.50628257
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.80162247e+09 | ||r||_inf = 2.70243370e+10

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))

```

2.3.1.3.1.10. Iteración 10

```
## x1^(10) = (94.2 - (1.4)*755454039.96678424 - (-2.7)*1930112276.50628257)/-5.0 = -830733516.96269298
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o

## x2^(10) = (-6.0 - (0.7)*-830733516.96269298 - (15.0)*1930112276.50628257)/-2.5 = 11348068276.68814087
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(10) = (-27.5 - (3.3)*-830733516.96269298 - (-11.0)*11348068276.68814087)/4.4 = 28993220823.19236755
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(10) = -830733516.96269298, 11348068276.68814087, 28993220823.19236755
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.70631085e+10 | ||r||_inf = 4.05946628e+11
```

2.3.1.3.2. Resultados Después de 10 iteraciones, se observa que el sistema no converge, al contrario, hay divergencia.

2.3.1.4. Sistema 3

$$3x_1 - 0,5x_2 + 0,6x_3 = 5,24$$

$$0,3x_1 - 4x_2 - x_3 = -0,387$$

$$-0,7x_1 + 2x_2 + 7x_3 = 14,803$$

```
A <- matrix(c(3, -0.5, 0.6,
              0.3, -4, -1,
              -0.7, 2, 7),
            nrow = 3, byrow = TRUE)
b <- c(5.24, -0.387, 14.803)

Ab <- cbind(A, b); (Ab)
```

2.3.1.4.1. Procedimiento

```
##                b
## [1,]  3.0 -0.5  0.6  5.240
## [2,]  0.3 -4.0 -1.0 -0.387
## [3,] -0.7  2.0  7.0 14.803
# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) #  $x^0 = (0, 0, 0)$ 
cat("x0=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x0 = 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.1. Iteración 1

```
## x1(1) = (5.2 - (-0.5)*0.00000000 - (0.6)*0.00000000)/3.0 = 1.74666667
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2(1) = (-0.4 - (0.3)*1.74666667 - (-1.0)*0.00000000)/-4.0 = 0.22775000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3(1) = (14.8 - (-0.7)*1.74666667 - (2.0)*0.22775000)/7.0 = 2.22430952
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x1 = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x1 = 1.74666667, 0.22775000, 2.22430952
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.22430952e+00 | ||r||_inf = 2.22430952e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.2. Iteración 2

```
## x1(2) = (5.2 - (-0.5)*0.22775000 - (0.6)*2.22430952)/3.0 = 1.33976310
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
##  $x_2^{(2)} = (-0.4 - (0.3)*1.33976310 - (-1.0)*2.22430952)/-4.0 = -0.35884515$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
##  $x_3^{(2)} = (14.8 - (-0.7)*1.33976310 - (2.0)*-0.35884515)/7.0 = 2.35121778$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
##  $x^{(2)} = 1.33976310, -0.35884515, 2.35121778$ 
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 5.86595149e-01 | ||r||_inf = 3.69442528e-01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.3. Iteración 3

```
##  $x_1^{(3)} = (5.2 - (-0.5)*-0.35884515 - (0.6)*2.35121778)/3.0 = 1.21661559$ 
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old
```

```
##  $x_2^{(3)} = (-0.4 - (0.3)*1.21661559 - (-1.0)*2.35121778)/-4.0 = -0.39980828$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
##  $x_3^{(3)} = (14.8 - (-0.7)*1.21661559 - (2.0)*-0.39980828)/7.0 = 2.35060678$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
##  $x^{(3)} = 1.21661559, -0.39980828, 2.35060678$ 
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.23147509e-01 | ||r||_inf = 2.01149636e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.4. Iteración 4

```
##  $x_1^{(4)} = (5.2 - (-0.5)*-0.39980828 - (0.6)*2.35060678)/3.0 = 1.20991060$ 
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(4) = (-0.4 - (0.3)*1.20991060 - (-1.0)*2.35060678)/-4.0 = -0.40015840

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(4) = (14.8 - (-0.7)*1.20991060 - (2.0)*-0.40015840)/7.0 = 2.35003632

#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = 1.20991060, -0.40015840, 2.35003632

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 6.70498785e-03 | ||r||_inf = 5.70463350e-04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.4.1.5. Iteración 5

```
## x1^(5) = (5.2 - (-0.5)*-0.40015840 - (0.6)*2.35003632)/3.0 = 1.20996634

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(5) = (-0.4 - (0.3)*1.20996634 - (-1.0)*2.35003632)/-4.0 = -0.40001160

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(5) = (14.8 - (-0.7)*1.20996634 - (2.0)*-0.40001160)/7.0 = 2.34999995

#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = 1.20996634, -0.40001160, 2.34999995

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.46796237e-04 | ||r||_inf = 9.52188680e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.4.1.6. Iteración 6

```
## x1^(6) = (5.2 - (-0.5)*-0.40001160 - (0.6)*2.34999995)/3.0 = 1.20999808
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(6) = (-0.4 - (0.3)*1.20999808 - (-1.0)*2.34999995)/-4.0 = -0.40000013
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(6) = (14.8 - (-0.7)*1.20999808 - (2.0)*-0.40000013)/7.0 = 2.34999985
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(6) = 1.20999808, -0.40000013, 2.34999985
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.17396227e-05 | ||r||_inf = 5.79855374e-06
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.7. Iteración 7

```
## x1^(7) = (5.2 - (-0.5)*-0.40000013 - (0.6)*2.34999985)/3.0 = 1.21000001
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(7) = (-0.4 - (0.3)*1.21000001 - (-1.0)*2.34999985)/-4.0 = -0.39999996
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(7) = (14.8 - (-0.7)*1.21000001 - (2.0)*-0.39999996)/7.0 = 2.34999999
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(7) = 1.21000001, -0.39999996, 2.34999999
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.93285125e-06 | ||r||_inf = 1.44446830e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.8. Iteración 8

```
## x1^(8) = (5.2 - (-0.5)*-0.39999996 - (0.6)*2.34999999)/3.0 = 1.21000001
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(8) = (-0.4 - (0.3)*1.21000001 - (-1.0)*2.34999999)/-4.0 = -0.40000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(8) = (14.8 - (-0.7)*1.21000001 - (2.0)*-0.40000000)/7.0 = 2.35000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(8) = 1.21000001, -0.40000000, 2.35000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.61417344e-08 | ||r||_inf = 2.42425715e-08
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.9. Iteración 9

```
## x1^(9) = (5.2 - (-0.5)*-0.40000000 - (0.6)*2.35000000)/3.0 = 1.21000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(9) = (-0.4 - (0.3)*1.21000000 - (-1.0)*2.35000000)/-4.0 = -0.40000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(9) = (14.8 - (-0.7)*1.21000000 - (2.0)*-0.40000000)/7.0 = 2.35000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = 1.21000000, -0.40000000, 2.35000000
```



```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 8.08085709e-09 | ||r||_inf = 1.64868563e-09
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.4.1.10. Iteración 10

```
## x1^(10) = (5.2 - (-0.5)*-0.40000000 - (0.6)*2.35000000)/3.0 = 1.21000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(10) = (-0.4 - (0.3)*1.21000000 - (-1.0)*2.35000000)/-4.0 = -0.40000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(10) = (14.8 - (-0.7)*1.21000000 - (2.0)*-0.40000000)/7.0 = 2.35000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 1.21000000, -0.40000000, 2.35000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 5.49561952e-10 | ||r||_inf = 3.60516061e-11
```

```
# Soluciones
```

```
x_old <- x
x1 <- x_old[1]
x2 <- x_old[2]
x3 <- x_old[3]
```

```
# Comprobación
```

```
I <- b[1]; (I)
```

```
## [1] 5.24
```

```
II <- b[2]; (II)
```

```
## [1] -0.387
```

```
III <- b[3]; (III)
```

```
## [1] 14.803
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 5.24000000001145"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: -0.386999999963948"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 14.803"
```

```
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 5.24000000001145 tecnicamente es: 5.24"
```

```
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)
```

```
## [1] "En la ecuación 2: -0.386999999963948 tecnicamente es: -0.387"
```

```
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 14.803 tecnicamente es: 14.803"
```

```
cat("Soluciones del sistema:")
```

2.3.1.4.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 1.21
```

```
cat("x2 =", x2)
```

```
## x2 = -0.4
```

```
cat("x3 =", x3)
```

```
## x3 = 2.35
```

2.3.1.5. Sistema 4

$$5x_1 - 0,2x_2 + x_3 = 1,5$$

$$0,1x_1 + 3x_2 - 0,5x_3 = -2,7$$

$$-0,3x_1 + x_2 - 7x_3 = 9,5$$

```
A <- matrix(c(5, -0.2, 1,
              0.1, 3, -0.5,
              -0.3, 1, -7),
            nrow = 3, byrow = TRUE)
b <- c(1.5, -2.7, 9.5)
```

```
Ab <- cbind(A, b); (Ab)
```

2.3.1.5.1. Procedimiento

```
##                                     b
## [1,]  5.0 -0.2  1.0  1.5
```

```
## [2,] 0.1 3.0 -0.5 -2.7
## [3,] -0.3 1.0 -7.0 9.5
```

```
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) #  $x^{(0)} = (0, 0, 0)$ 
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.1. Iteración 1

```
##  $x_1^{(1)} = (1.5 - (-0.2)*0.00000000 - (1.0)*0.00000000)/5.0 = 0.30000000$ 
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
##  $x_2^{(1)} = (-2.7 - (0.1)*0.30000000 - (-0.5)*0.00000000)/3.0 = -0.91000000$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
##  $x_3^{(1)} = (9.5 - (-0.3)*0.30000000 - (1.0)*-0.91000000)/-7.0 = -1.50000000$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(1) = 0.30000000, -0.91000000, -1.50000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.50000000e+00 | ||r||_inf = 1.31800000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.2. Iteración 2

```
## x1^(2) = (1.5 - (-0.2)*-0.91000000 - (1.0)*-1.50000000)/5.0 = 0.56360000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(2) = (-2.7 - (0.1)*0.56360000 - (-0.5)*-1.50000000)/3.0 = -1.16878667
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(2) = (9.5 - (-0.3)*0.56360000 - (1.0)*-1.16878667)/-7.0 = -1.54826667
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(2) = 0.56360000, -1.16878667, -1.54826667
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.63600000e-01 | ||r||_inf = 2.41333333e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.3. Iteración 3

```
## x1^(3) = (1.5 - (-0.2)*-1.16878667 - (1.0)*-1.54826667)/5.0 = 0.56290187
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(3) = (-2.7 - (0.1)*0.56290187 - (-0.5)*-1.54826667)/3.0 = -1.17680784
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(3) = (9.5 - (-0.3)*0.56290187 - (1.0)*-1.17680784)/-7.0 = -1.54938263
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(3) = 0.56290187, -1.17680784, -1.54938263
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 8.02117333e-03 | ||r||_inf = 5.57980952e-04

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.4. Iteración 4

```
## x1^(4) = (1.5 - (-0.2)*-1.17680784 - (1.0)*-1.54938263)/5.0 = 0.56280421
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(4) = (-2.7 - (0.1)*0.56280421 - (-0.5)*-1.54938263)/3.0 = -1.17699058
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(4) = (9.5 - (-0.3)*0.56280421 - (1.0)*-1.17699058)/-7.0 = -1.54940455
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = 0.56280421, -1.17699058, -1.54940455
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.82738499e-04 | ||r||_inf = 1.46273950e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.5. Iteración 5

```
## x1^(5) = (1.5 - (-0.2)*-1.17699058 - (1.0)*-1.54940455)/5.0 = 0.56280129
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(5) = (-2.7 - (0.1)*0.56280129 - (-0.5)*-1.54940455)/3.0 = -1.17699413
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(5) = (9.5 - (-0.3)*0.56280129 - (1.0)*-1.17699413)/-7.0 = -1.54940493
#Error o tolerancia

r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(5) = 0.56280129, -1.17699413, -1.54940493

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.55586816e-06 | ||r||_inf = 3.28570138e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.6. Iteración 6

```
## x1^(6) = (1.5 - (-0.2)*-1.17699413 - (1.0)*-1.54940493)/5.0 = 0.56280122
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]))

## x2^(6) = (-2.7 - (0.1)*0.56280122 - (-0.5)*-1.54940493)/3.0 = -1.17699420
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(6) = (9.5 - (-0.3)*0.56280122 - (1.0)*-1.17699420)/-7.0 = -1.54940494
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(6) = 0.56280122, -1.17699420, -1.54940494

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 6.57140277e-08 | ||r||_inf = 6.33498898e-09
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.7. Iteración 7

```
## x1^(7) = (1.5 - (-0.2)*-1.17699420 - (1.0)*-1.54940494)/5.0 = 0.56280122
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]))

## x2^(7) = (-2.7 - (0.1)*0.56280122 - (-0.5)*-1.54940494)/3.0 = -1.17699420
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(7) = (9.5 - (-0.3)*0.56280122 - (1.0)*-1.17699420)/-7.0 = -1.54940494
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(7) = 0.56280122, -1.17699420, -1.54940494
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.26699762e-09 | ||r||_inf = 1.08842713e-10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.8. Iteración 8

```
## x1^(8) = (1.5 - (-0.2)*-1.17699420 - (1.0)*-1.54940494)/5.0 = 0.56280122
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(8) = (-2.7 - (0.1)*0.56280122 - (-0.5)*-1.54940494)/3.0 = -1.17699420
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(8) = (9.5 - (-0.3)*0.56280122 - (1.0)*-1.17699420)/-7.0 = -1.54940494
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(8) = 0.56280122, -1.17699420, -1.54940494
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.17685869e-11 | ||r||_inf = 1.67288405e-12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.5.1.9. Iteración 9

```
## x1^(9) = (1.5 - (-0.2)*-1.17699420 - (1.0)*-1.54940494)/5.0 = 0.56280122
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(9) = (-2.7 - (0.1)*0.56280122 - (-0.5)*-1.54940494)/3.0 = -1.17699420
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
x[3]))

## x3^(9) = (9.5 - (-0.3)*0.56280122 - (1.0)*-1.17699420)/-7.0 = -1.54940494
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(9) = 0.56280122, -1.17699420, -1.54940494
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.34621220e-13 | ||r||_inf = 2.19824159e-14
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
x[1]))
```

2.3.1.5.1.10. Iteración 10

```
## x1^(10) = (1.5 - (-0.2)*-1.17699420 - (1.0)*-1.54940494)/5.0 = 0.56280122
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
x[2]))

## x2^(10) = (-2.7 - (0.1)*0.56280122 - (-0.5)*-1.54940494)/3.0 = -1.17699420
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
x[3]))

## x3^(10) = (9.5 - (-0.3)*0.56280122 - (1.0)*-1.17699420)/-7.0 = -1.54940494
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(10) = 0.56280122, -1.17699420, -1.54940494
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.44089210e-15 | ||r||_inf = 4.44089210e-16
```

```
# Soluciones
x_old <- x
x1 <- x_old[1]
x2 <- x_old[2]
x3 <- x_old[3]
```

```
# Comprobación
```

```
I <- b[1]; (I)
```



```
## [1] 1.5
II <- b[2]; (II)

## [1] -2.7
III <- b[3]; (III)

## [1] 9.5
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: 1.5"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: -2.7"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 9.5"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 1.5 tecnicamente es: 1.5"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: -2.7 tecnicamente es: -2.7"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: 9.5 tecnicamente es: 9.5"

cat("Soluciones del sistema:")
```

2.3.1.5.2. Resultados

Soluciones del sistema:

```
cat("x1 =", x1)
```

x1 = 0.5628012

```
cat("x2 =", x2)
```

x2 = -1.176994

```
cat("x3 =", x3)
```

x3 = -1.549405

2.3.1.6. Sistema 5

$$-3x_2 + 7x_3 = 2$$

$$x_1 + 2x_2 - x_3 = 3$$

$$12x_1 + 2x_2 + 2x_3 = 6$$

```
A <- matrix(c(0, -3, 7,
              1, 2, -1,
              12, 2, 2),
            nrow = 3, byrow = TRUE)
b <- c(2, 3, 6)

Ab <- cbind(A, b); (Ab)
```

2.3.1.6.1. Procedimiento

```
##                b
## [1,]  0 -3  7  2
## [2,]  1  2 -1  3
## [3,] 12  2  2  6

# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

No es dominante

Aunque no es una matriz con diagonal dominante, se intenta resolverlo con 10 iteraciones para ver si puede o no llegar a converger.

Asignar valores iniciales

```
x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

x^(0)= 0.00000000, 0.00000000, 0.00000000

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.6.1.1. Iteración 1

x1^(1) = (2.0 - (-3.0)*0.00000000 - (7.0)*0.00000000)/0.0 = Inf

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

x2^(1) = (3.0 - (1.0)*Inf - (-1.0)*0.00000000)/2.0 = -Inf

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(1) = (6.0 - (12.0)*Inf - (2.0)*-Inf)/2.0 = NaN
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(1) = Inf, -Inf, NaN
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = NaN | ||r||_inf = NaN
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.6.1.2. Iteración 2

```
## x1^(2) = (2.0 - (-3.0)*-Inf - (7.0)*NaN)/0.0 = NaN
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(2) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(2) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(2) = NaN, NaN, NaN
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = NaN | ||r||_inf = NaN
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.6.1.3. Iteración 3

```
## x1^(3) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(3) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
```

```

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(3) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(3) = NaN, NaN, NaN
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.1.6.1.4. Iteración 4

```

## x1^(4) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]

## x2^(4) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(4) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = NaN, NaN, NaN
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.1.6.1.5. Iteración 5

```

## x1^(5) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN

```

```

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(5) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(5) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = NaN, NaN, NaN
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))

```

2.3.1.6.1.6. Iteración 6

```

## x1^(6) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(6) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(6) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(6) = NaN, NaN, NaN
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))

```

2.3.1.6.1.7. Iteración 7

```
## x1^(7) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(7) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))

## x3^(7) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(7) = NaN, NaN, NaN
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.6.1.8. Iteración 8

```
## x1^(8) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(8) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))

## x3^(8) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = NaN, NaN, NaN
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.6.1.9. Iteración 9

```
## x1^(9) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(9) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(9) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = NaN, NaN, NaN
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = NaN | ||r||_inf = NaN
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.6.1.10. Iteración 10

```
## x1^(10) = (2.0 - (-3.0)*NaN - (7.0)*NaN)/0.0 = NaN
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(10) = (3.0 - (1.0)*NaN - (-1.0)*NaN)/2.0 = NaN
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(10) = (6.0 - (12.0)*NaN - (2.0)*NaN)/2.0 = NaN
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(10) = NaN, NaN, NaN
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = NaN | ||r||_inf = NaN
```

2.3.1.6.2. Resultados Después de 10 iteraciones, se observa que el sistema no converge, y se puede notar desde el principio porque en la ecuación 1 se indefinir al dividirse entre 0.

2.3.1.7. Sistema 6

$$0,15x_1 + 2,11x_2 + 30,75x_3 = -26,38$$

$$0,64x_1 + 1,21x_2 + 2,05x_3 = 1,01$$

$$3,21x_1 + 1,53x_2 + 1,04x_3 = 5,23$$

```
A <- matrix(c(0.15, 2.11, 30.75,
              0.64, 1.21, 2.05,
              3.21, 1.53, 1.04),
            nrow = 3, byrow = TRUE)
b <- c(-26.38, 1.01, 5.23)

Ab <- cbind(A, b); (Ab)
```

2.3.1.7.1. Procedimiento

```
##                               b
## [1,] 0.15 2.11 30.75 -26.38
## [2,] 0.64 1.21 2.05 1.01
## [3,] 3.21 1.53 1.04 5.23

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## No es dominante
```

Aunque no es una matriz con diagonal dominante, se intenta resolverlo con 10 iteraciones para ver si puede o no llegar a converger.

```
# Asignar valores iniciales

x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```



```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.1. Iteración 1

```
## x1^(1) = (-26.4 - (2.1)*0.00000000 - (30.8)*0.00000000)/0.1 = -175.86666667
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old
```

```
## x2^(1) = (1.0 - (0.6)*-175.86666667 - (2.0)*0.00000000)/1.2 = 93.85509642
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(1) = (5.2 - (3.2)*-175.86666667 - (1.5)*93.85509642)/1.0 = 409.77279085
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(1) = -175.86666667, 93.85509642, 409.77279085
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 4.09772791e+02 | ||r||_inf = 1.27985476e+04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.2. Iteración 2

```
## x1^(2) = (-26.4 - (2.1)*93.85509642 - (30.8)*409.77279085)/0.1 = -85499.51714629
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old
```

```
## x2^(2) = (1.0 - (0.6)*-85499.51714629 - (2.0)*409.77279085)/1.2 = 44529.47665487
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(2) = (5.2 - (3.2)*-85499.51714629 - (1.5)*44529.47665487)/1.0 = 198392.86611311
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(2) = -85499.51714629, 44529.47665487, 198392.86611311
```

```

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.97983093e+05 | ||r||_inf = 6.18173928e+06

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
x_old[3], x[1]))

## x1^(3) = (-26.4 - (2.1)*44529.47665487 - (30.8)*198392.86611311)/0.1 = -41297094.72479962
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
x_old[2], x[2]))

## x2^(3) = (1.0 - (0.6)*-41297094.72479962 - (2.0)*198392.86611311)/1.2 = 21506972.11433048
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
x[3]))

## x3^(3) = (5.2 - (3.2)*-41297094.72479962 - (1.5)*21506972.11433048)/1.0 = 95825011.50161648
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(3) = -41297094.72479962, 21506972.11433048, 95825011.50161648
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 9.56266186e+07 | ||r||_inf = 2.98580428e+09

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
x_old[3], x[1]))

## x1^(4) = (-26.4 - (2.1)*21506972.11433048 - (30.8)*95825011.50161648)/0.1 = -19946658941.43962479
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
x_old[2], x[2]))

## x2^(4) = (1.0 - (0.6)*-19946658941.43962479 - (2.0)*95825011.50161648)/1.2 = 10387950785.08516312
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
x[3]))

## x3^(4) = (5.2 - (3.2)*-19946658941.43962479 - (1.5)*10387950785.08516312)/1.0 = 46283856255.83740234
#Error o tolerancia

r <- b - A %*% x

```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(4) = -19946658941.43962479, 10387950785.08516312, 46283856255.83740234
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.61880312e+10 | ||r||_inf = 1.44215516e+12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.5. Iteración 5

```
## x1^(5) = (-26.4 - (2.1)*10387950785.08516312 - (30.8)*46283856255.83740234)/0.1 = -9634314373666.064453
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]))

## x2^(5) = (1.0 - (0.6)*-9634314373666.06445312 - (2.0)*46283856255.83740234)/1.2 = 5017420903985.80566406
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(5) = (5.2 - (3.2)*-9634314373666.06445312 - (1.5)*5017420903985.80566406)/1.0 = 22355283804206.74609375
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(5) = -9634314373666.06445312, 5017420903985.80566406, 22355283804206.74609375
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.23089999e+13 | ||r||_inf = 6.96566588e+14
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.6. Iteración 6

```
## x1^(6) = (-26.4 - (2.1)*5017420903985.80566406 - (30.8)*22355283804206.74609375)/0.1 = -4653411567245293
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3]))

## x2^(6) = (1.0 - (0.6)*-4653411567245293.00000000 - (2.0)*22355283804206.74609375)/1.2 = 2423433943172202
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(6) = (5.2 - (3.2)*-4653411567245293.00000000 - (1.5)*2423433943172202.00000000)/1.0 = 1079768961327
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(6) = -4653411567245293.00000000, 2423433943172202.00000000, 10797689613273008.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.07753343e+16 | ||r||_inf = 3.36444389e+17
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.7. Iteración 7

```
## x1^(7) = (-26.4 - (2.1)*2423433943172202.00000000 - (30.8)*10797689613273008.00000000)/0.1 = -2247616008188256000.00000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(7) = (1.0 - (0.6)*-2247616008188256000.00000000 - (2.0)*10797689613273008.00000000)/1.2 = 1170528083911796736.00000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(7) = (5.2 - (3.2)*-2247616008188256000.00000000 - (1.5)*1170528083911796736.00000000)/1.0 = 5215326363364665344.00000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = -2247616008188256000.00000000, 1170528083911796736.00000000, 5215326363364665344.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 5.20452867e+18 | ||r||_inf = 1.62503958e+20
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.8. Iteración 8

```
## x1^(8) = (-26.4 - (2.1)*1170528083911796736.00000000 - (30.8)*5215326363364665344.00000000)/0.1 = -1085607332870115688448.00000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(8) = (1.0 - (0.6)*-1085607332870115688448.00000000 - (2.0)*5215326363364665344.00000000)/1.2 = 565326363364665344.00000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
## x3^(8) = (5.2 - (3.2)*-1085607332870115688448.00000000 - (1.5)*565369647927253270528.00000000)/1.0 = 25
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = -1085607332870115688448.00000000, 565369647927253270528.00000000, 2519023054984974958592.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.51380773e+21 | ||r||_inf = 7.84900478e+22
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.9. Iteración 9

```
## x1^(9) = (-26.4 - (2.1)*565369647927253270528.00000000 - (30.8)*2519023054984974958592.00000000)/0.1 =
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
## x2^(9) = (1.0 - (0.6)*-524352592652763289092096.00000000 - (2.0)*2519023054984974958592.00000000)/1.2 =
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
## x3^(9) = (5.2 - (3.2)*-524352592652763289092096.00000000 - (1.5)*273075753747974648233984.00000000)/1.0
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = -524352592652763289092096.00000000, 273075753747974648233984.00000000, 1216697999212469987311616.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.21417898e+24 | ||r||_inf = 3.79110004e+25
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.7.1.10. Iteración 10

```
## x1^(10) = (-26.4 - (2.1)*273075753747974648233984.00000000 - (30.8)*1216697999212469987311616.00000000)/0.1 =
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o

## x2^(10) = (1.0 - (0.6)*-253264355441277898684628992.00000000 - (2.0)*1216697999212469987311616.00000000

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(10) = (5.2 - (3.2)*-253264355441277898684628992.00000000 - (1.5)*131896658333910991953199104.00000000

#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ", ")))

## x^(10) = -253264355441277898684628992.00000000, 131896658333910991953199104.00000000, 58766989780347902

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 5.86453200e+26 | ||r||_inf = 1.83111617e+28
```

2.3.1.7.2. Resultados Después de 10 iteraciones, se observa que el sistema no converge, al contrario, diverge.

2.3.1.8. Sistema 7

$$x_1 + x_2 - x_3 = -3$$

$$6x_1 + 2x_2 + 2x_3 = 2$$

$$-3x_1 + 4x_2 + x_3 = 1$$

```
A <- matrix(c(1, 1, -1,
              6, 2, 2,
              -3, 4, 1),
            nrow = 3, byrow = TRUE)
b <- c(-3, 2, 1)

Ab <- cbind(A, b); (Ab)
```

2.3.1.8.1. Procedimiento

```
##                b
## [1,]  1 1 -1 -3
## [2,]  6 2  2  2
## [3,] -3 4  1  1

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
}
```

```

} else {
  cat("No es dominante")
}

```

No es dominante

La diagonal de la matriz no es dominante, de igual forma se realizan 10 iteraciones para ver si puede llegar a converger.

Asignar valores iniciales

```

x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")

```

x^(0)= 0.00000000, 0.00000000, 0.00000000

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.1.8.1.1. Iteración 1

$x_1^{(1)} = (-3.0 - (1.0)*0.00000000 - (-1.0)*0.00000000)/1.0 = -3.00000000$

```

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],

```

$x_2^{(1)} = (2.0 - (6.0)*-3.00000000 - (2.0)*0.00000000)/2.0 = 10.00000000$

```

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],

```

$x_3^{(1)} = (1.0 - (-3.0)*-3.00000000 - (4.0)*10.00000000)/1.0 = -48.00000000$

#Error o tolerancia

```

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

```

x^(1) = -3.00000000, 10.00000000, -48.00000000

```

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

```

||dx||_inf = 4.80000000e+01 | ||r||_inf = 9.60000000e+01

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.1.8.1.2. Iteración 2

$x_1^{(2)} = (-3.0 - (1.0)*10.00000000 - (-1.0)*-48.00000000)/1.0 = -61.00000000$

```

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],

```

```
##  $x_2^{(2)} = (2.0 - (6.0)*-61.00000000 - (2.0)*-48.00000000)/2.0 = 232.00000000$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
##  $x_3^{(2)} = (1.0 - (-3.0)*-61.00000000 - (4.0)*232.00000000)/1.0 = -1110.00000000$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
##  $x^{(2)} = -61.00000000, 232.00000000, -1110.00000000$ 
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.06200000e+03 | ||r||_inf = 2.12400000e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.8.1.3. Iteración 3

```
##  $x_1^{(3)} = (-3.0 - (1.0)*232.00000000 - (-1.0)*-1110.00000000)/1.0 = -1345.00000000$ 
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old
```

```
##  $x_2^{(3)} = (2.0 - (6.0)*-1345.00000000 - (2.0)*-1110.00000000)/2.0 = 5146.00000000$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
##  $x_3^{(3)} = (1.0 - (-3.0)*-1345.00000000 - (4.0)*5146.00000000)/1.0 = -24618.00000000$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
##  $x^{(3)} = -1345.00000000, 5146.00000000, -24618.00000000$ 
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.35080000e+04 | ||r||_inf = 4.70160000e+04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.8.1.4. Iteración 4

```
##  $x_1^{(4)} = (-3.0 - (1.0)*5146.00000000 - (-1.0)*-24618.00000000)/1.0 = -29767.00000000$ 
```



```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(4) = (2.0 - (6.0)*-29767.00000000 - (2.0)*-24618.00000000)/2.0 = 113920.00000000

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(4) = (1.0 - (-3.0)*-29767.00000000 - (4.0)*113920.00000000)/1.0 = -544980.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(4) = -29767.00000000, 113920.00000000, -544980.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 5.20362000e+05 | ||r||_inf = 1.04072400e+06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.8.1.5. Iteración 5

```
## x1^(5) = (-3.0 - (1.0)*113920.00000000 - (-1.0)*-544980.00000000)/1.0 = -658903.00000000

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(5) = (2.0 - (6.0)*-658903.00000000 - (2.0)*-544980.00000000)/2.0 = 2521690.00000000

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(5) = (1.0 - (-3.0)*-658903.00000000 - (4.0)*2521690.00000000)/1.0 = -12063468.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(5) = -658903.00000000, 2521690.00000000, -12063468.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.15184880e+07 | ||r||_inf = 2.30369760e+07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.8.1.6. Iteración 6

```
## x1^(6) = (-3.0 - (1.0)*2521690.00000000 - (-1.0)*-12063468.00000000)/1.0 = -14585161.00000000
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(6) = (2.0 - (6.0)*-14585161.00000000 - (2.0)*-12063468.00000000)/2.0 = 55818952.00000000
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(6) = (1.0 - (-3.0)*-14585161.00000000 - (4.0)*55818952.00000000)/1.0 = -267031290.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(6) = -14585161.00000000, 55818952.00000000, -267031290.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.54967822e+08 | ||r||_inf = 5.09935644e+08
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.8.1.7. Iteración 7

```
## x1^(7) = (-3.0 - (1.0)*55818952.00000000 - (-1.0)*-267031290.00000000)/1.0 = -322850245.00000000
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(7) = (2.0 - (6.0)*-322850245.00000000 - (2.0)*-267031290.00000000)/2.0 = 1235582026.00000000
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(7) = (1.0 - (-3.0)*-322850245.00000000 - (4.0)*1235582026.00000000)/1.0 = -5910878838.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(7) = -322850245.00000000, 1235582026.00000000, -5910878838.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 5.64384755e+09 | ||r||_inf = 1.12876951e+10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.8.1.8. Iteración 8

```
## x1^(8) = (-3.0 - (1.0)*1235582026.00000000 - (-1.0)*-5910878838.00000000)/1.0 = -7146460867.00000000
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(8) = (2.0 - (6.0)*-7146460867.00000000 - (2.0)*-5910878838.00000000)/2.0 = 27350261440.00000000
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(8) = (1.0 - (-3.0)*-7146460867.00000000 - (4.0)*27350261440.00000000)/1.0 = -130840428360.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = -7146460867.00000000, 27350261440.00000000, -130840428360.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.24929550e+11 | ||r||_inf = 2.49859099e+11
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.8.1.9. Iteración 9

```
## x1^(9) = (-3.0 - (1.0)*27350261440.00000000 - (-1.0)*-130840428360.00000000)/1.0 = -158190689803.00000000
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(9) = (2.0 - (6.0)*-158190689803.00000000 - (2.0)*-130840428360.00000000)/2.0 = 605412497770.00000000
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(9) = (1.0 - (-3.0)*-158190689803.00000000 - (4.0)*605412497770.00000000)/1.0 = -2896222060488.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = -158190689803.00000000, 605412497770.00000000, -2896222060488.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.76538163e+12 | ||r||_inf = 5.53076326e+12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.8.1.10. Iteración 10

```
## x1^(10) = (-3.0 - (1.0)*605412497770.00000000 - (-1.0)*-2896222060488.00000000)/1.0 = -3501634558261.00000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(10) = (2.0 - (6.0)*-3501634558261.00000000 - (2.0)*-2896222060488.00000000)/2.0 = 13401125735272.00000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(10) = (1.0 - (-3.0)*-3501634558261.00000000 - (4.0)*13401125735272.00000000)/1.0 = -64109406615870.00000000
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = -3501634558261.00000000, 13401125735272.00000000, -64109406615870.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 6.12131846e+13 | ||r||_inf = 1.22426369e+14
```

2.3.1.8.2. Resultados Después de 10 iteraciones, se observa que el sistema no converge, al contrario, diverge.

2.3.1.9. Sistema 8

$$2x_1 + x_2 - x_3 = 1$$

$$5x_1 + 2x_2 + 2x_3 = -4$$

$$3x_1 + x_2 + x_3 = 5$$

```
A <- matrix(c(2, 1, -1,
              5, 2, 2,
              3, 1, 1),
            nrow = 3, byrow = TRUE)
b <- c(1, -4, 5)

Ab <- cbind(A, b); (Ab)
```

2.3.1.9.1. Procedimiento

```
##          b
## [1,] 2 1 -1 1
## [2,] 5 2 2 -4
## [3,] 3 1 1 5
```

```
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## No es dominante
```

La matriz no tiene una diagonal dominante, aún así se intentan 10 iteraciones para ver si puede llegar a converger.

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) #  $x^0 = (0, 0, 0)$ 
cat("x^0=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^0= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.1. Iteración 1

```
## x1^1) = (1.0 - (1.0)*0.00000000 - (-1.0)*0.00000000)/2.0 = 0.50000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^1) = (-4.0 - (5.0)*0.50000000 - (2.0)*0.00000000)/2.0 = -3.25000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^1) = (5.0 - (3.0)*0.50000000 - (1.0)*-3.25000000)/1.0 = 6.75000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^1) = 0.50000000, -3.25000000, 6.75000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 6.75000000e+00 | ||r||_inf = 1.35000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.2. Iteración 2

```
## x1^(2) = (1.0 - (1.0)*-3.25000000 - (-1.0)*6.75000000)/2.0 = 5.50000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(2) = (-4.0 - (5.0)*5.50000000 - (2.0)*6.75000000)/2.0 = -22.50000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(2) = (5.0 - (3.0)*5.50000000 - (1.0)*-22.50000000)/1.0 = 11.00000000
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(2) = 5.50000000, -22.50000000, 11.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.92500000e+01 | ||r||_inf = 2.35000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.3. Iteración 3

```
## x1^(3) = (1.0 - (1.0)*-22.50000000 - (-1.0)*11.00000000)/2.0 = 17.25000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(3) = (-4.0 - (5.0)*17.25000000 - (2.0)*11.00000000)/2.0 = -56.12500000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(3) = (5.0 - (3.0)*17.25000000 - (1.0)*-56.12500000)/1.0 = 9.37500000
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(3) = 17.25000000, -56.12500000, 9.37500000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.36250000e+01 | ||r||_inf = 3.20000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.4. Iteración 4

```
## x1^(4) = (1.0 - (1.0)*-56.12500000 - (-1.0)*9.37500000)/2.0 = 33.25000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(4) = (-4.0 - (5.0)*33.25000000 - (2.0)*9.37500000)/2.0 = -94.50000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(4) = (5.0 - (3.0)*33.25000000 - (1.0)*-94.50000000)/1.0 = -0.25000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(4) = 33.25000000, -94.50000000, -0.25000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.83750000e+01 | ||r||_inf = 2.87500000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.5. Iteración 5

```
## x1^(5) = (1.0 - (1.0)*-94.50000000 - (-1.0)*-0.25000000)/2.0 = 47.62500000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(5) = (-4.0 - (5.0)*47.62500000 - (2.0)*-0.25000000)/2.0 = -120.81250000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(5) = (5.0 - (3.0)*47.62500000 - (1.0)*-120.81250000)/1.0 = -17.06250000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(5) = 47.62500000, -120.81250000, -17.06250000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.63125000e+01 | ||r||_inf = 3.36250000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.6. Iteración 6

```
## x1^(6) = (1.0 - (1.0)*-120.81250000 - (-1.0)*-17.06250000)/2.0 = 52.37500000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(6) = (-4.0 - (5.0)*52.37500000 - (2.0)*-17.06250000)/2.0 = -115.87500000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(6) = (5.0 - (3.0)*52.37500000 - (1.0)*-115.87500000)/1.0 = -36.25000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(6) = 52.37500000, -115.87500000, -36.25000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.91875000e+01 | ||r||_inf = 3.83750000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.7. Iteración 7

```
## x1^(7) = (1.0 - (1.0)*-115.87500000 - (-1.0)*-36.25000000)/2.0 = 40.31250000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(7) = (-4.0 - (5.0)*40.31250000 - (2.0)*-36.25000000)/2.0 = -66.53125000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(7) = (5.0 - (3.0)*40.31250000 - (1.0)*-66.53125000)/1.0 = -49.40625000
```



```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(7) = 40.31250000, -66.53125000, -49.40625000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 4.93437500e+01 | ||r||_inf = 6.25000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.8. Iteración 8

```
## x1^(8) = (1.0 - (1.0)*-66.53125000 - (-1.0)*-49.40625000)/2.0 = 9.06250000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(8) = (-4.0 - (5.0)*9.06250000 - (2.0)*-49.40625000)/2.0 = 24.75000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(8) = (5.0 - (3.0)*9.06250000 - (1.0)*24.75000000)/1.0 = -46.93750000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(8) = 9.06250000, 24.75000000, -46.93750000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 9.12812500e+01 | ||r||_inf = 8.88125000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.9.1.9. Iteración 9

```
## x1^(9) = (1.0 - (1.0)*24.75000000 - (-1.0)*-46.93750000)/2.0 = -35.34375000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(9) = (-4.0 - (5.0)*-35.34375000 - (2.0)*-46.93750000)/2.0 = 133.29687500
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
x[3]))

## x3^(9) = (5.0 - (3.0)*-35.34375000 - (1.0)*133.29687500)/1.0 = -22.26562500
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = -35.34375000, 133.29687500, -22.26562500
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.08546875e+02 | ||r||_inf = 8.38750000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
x[1]))
```

2.3.1.9.1.10. Iteración 10

```
## x1^(10) = (1.0 - (1.0)*133.29687500 - (-1.0)*-22.26562500)/2.0 = -77.28125000
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
x[2]))

## x2^(10) = (-4.0 - (5.0)*-77.28125000 - (2.0)*-22.26562500)/2.0 = 213.46875000
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
x[3]))

## x3^(10) = (5.0 - (3.0)*-77.28125000 - (1.0)*213.46875000)/1.0 = 23.37500000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(10) = -77.28125000, 213.46875000, 23.37500000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 8.01718750e+01 | ||r||_inf = 9.12812500e+01
```

```
# Soluciones
x_old <- x
x1 <- x_old[1]
x2 <- x_old[2]
x3 <- x_old[3]
```

```
# Comprobación
```

```
I <- b[1]; (I)
```

```
## [1] 1
II <- b[2]; (II)

## [1] -4
III <- b[3]; (III)

## [1] 5
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: 35.53125"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: 87.28125"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 5"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 35.53125 técnicamente es: 1"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 87.28125 técnicamente es: -4"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 5 técnicamente es: 5"
```

2.3.1.9.2. Resultados Después de 10 iteraciones se aprecia que la matriz en vez de converger, diverge.

```
cat("Soluciones del sistema:")
```

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = -77.28125
```

```
cat("x2 =", x2)
```

```
## x2 = 213.4688
```

```
cat("x3 =", x3)
```

```
## x3 = 23.375
```

2.3.1.10. Sistema 9

$$3x - 0,1y - 0,2z = 7,85$$

$$0,1x + 7y - 0,3z = -19,3$$

$$0,3x - 0,2y + 10z = 71,4$$

```
A <- matrix(c(3, -0.1, -0.2,
              0.1, 7, -0.3,
              0.3, -0.2, 10),
            nrow = 3, byrow = TRUE)
b <- c(7.85, -19.3, 71.4)

Ab <- cbind(A, b); (Ab)
```

2.3.1.10.1. Procedimiento

```
##                               b
## [1,] 3.0 -0.1 -0.2    7.85
## [2,] 0.1  7.0 -0.3 -19.30
## [3,] 0.3 -0.2 10.0   71.40
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) #  $x^0 = (0, 0, 0)$ 
cat("x^0=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^0= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.1. Iteración 1

```
## x1^(1) = (7.8 - (-0.1)*0.00000000 - (-0.2)*0.00000000)/3.0 = 2.61666667
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(1) = (-19.3 - (0.1)*2.61666667 - (-0.3)*0.00000000)/7.0 = -2.79452381
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(1) = (71.4 - (0.3)*2.61666667 - (-0.2)*-2.79452381)/10.0 = 7.00560952
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(1) = 2.61666667, -2.79452381, 7.00560952
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 7.00560952e+00 | ||r||_inf = 2.10168286e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.2. Iteración 2

```
## x1^(2) = (7.8 - (-0.1)*-2.79452381 - (-0.2)*7.00560952)/3.0 = 2.99055651
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(2) = (-19.3 - (0.1)*2.99055651 - (-0.3)*7.00560952)/7.0 = -2.49962468
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(2) = (71.4 - (0.3)*2.99055651 - (-0.2)*-2.49962468)/10.0 = 7.00029081
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(2) = 2.99055651, -2.49962468, 7.00029081
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 3.73889841e-01 | ||r||_inf = 2.84261699e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.3. Iteración 3

```
## x1^(3) = (7.8 - (-0.1)*-2.49962468 - (-0.2)*7.00029081)/3.0 = 3.00003190
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(3) = (-19.3 - (0.1)*3.00003190 - (-0.3)*7.00029081)/7.0 = -2.49998799
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(3) = (71.4 - (0.3)*3.00003190 - (-0.2)*-2.49998799)/10.0 = 6.99999928
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(3) = 3.00003190, -2.49998799, 6.99999928
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 9.47538997e-03 | ||r||_inf = 9.46363246e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.4. Iteración 4

```
## x1^(4) = (7.8 - (-0.1)*-2.49998799 - (-0.2)*6.99999928)/3.0 = 3.00000035
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3])

## x2^(4) = (-19.3 - (0.1)*3.00000035 - (-0.3)*6.99999928)/7.0 = -2.50000004
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2])

## x3^(4) = (71.4 - (0.3)*3.00000035 - (-0.2)*-2.50000004)/10.0 = 6.99999999
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = 3.00000035, -2.50000004, 6.99999999
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.15454415e-05 | ||r||_inf = 1.06324111e-06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.5. Iteración 5

```
## x1^(5) = (7.8 - (-0.1)*-2.50000004 - (-0.2)*6.99999999)/3.0 = 3.00000000
```

```

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(5) = (-19.3 - (0.1)*3.00000000 - (-0.3)*6.99999999)/7.0 = -2.50000000

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(5) = (71.4 - (0.3)*3.00000000 - (-0.2)*-2.50000000)/10.0 = 7.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = 3.00000000, -2.50000000, 7.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.54413705e-07 | ||r||_inf = 5.79753312e-09

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))

```

2.3.1.10.1.6. Iteración 6

```

## x1^(6) = (7.8 - (-0.1)*-2.50000000 - (-0.2)*7.00000000)/3.0 = 3.00000000

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(6) = (-19.3 - (0.1)*3.00000000 - (-0.3)*7.00000000)/7.0 = -2.50000000

x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]))

## x3^(6) = (71.4 - (0.3)*3.00000000 - (-0.2)*-2.50000000)/10.0 = 7.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(6) = 3.00000000, -2.50000000, 7.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.93251104e-09 | ||r||_inf = 3.60689256e-11

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))

```

2.3.1.10.1.7. Iteración 7

```
##  $x_1^{(7)} = (7.8 - (-0.1)*-2.50000000 - (-0.2)*7.00000000)/3.0 = 3.00000000$ 
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))
```

```
##  $x_2^{(7)} = (-19.3 - (0.1)*3.00000000 - (-0.3)*7.00000000)/7.0 = -2.50000000$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))
```

```
##  $x_3^{(7)} = (71.4 - (0.3)*3.00000000 - (-0.2)*-2.50000000)/10.0 = 7.00000000$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
##  $x^{(7)} = 3.00000000, -2.50000000, 7.00000000$ 
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.20237154e-11 | ||r||_inf = 3.08197912e-13
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.10.1.8. Iteración 8

```
##  $x_1^{(8)} = (7.8 - (-0.1)*-2.50000000 - (-0.2)*7.00000000)/3.0 = 3.00000000$ 
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))
```

```
##  $x_2^{(8)} = (-19.3 - (0.1)*3.00000000 - (-0.3)*7.00000000)/7.0 = -2.50000000$ 
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))
```

```
##  $x_3^{(8)} = (71.4 - (0.3)*3.00000000 - (-0.2)*-2.50000000)/10.0 = 7.00000000$ 
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
##  $x^{(8)} = 3.00000000, -2.50000000, 7.00000000$ 
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.03028697e-13 | ||r||_inf = 3.55271368e-15
```



```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.9. Iteración 9

```
## x1^(9) = (7.8 - (-0.1)*-2.50000000 - (-0.2)*7.00000000)/3.0 = 3.00000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(9) = (-19.3 - (0.1)*3.00000000 - (-0.3)*7.00000000)/7.0 = -2.50000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(9) = (71.4 - (0.3)*3.00000000 - (-0.2)*-2.50000000)/10.0 = 7.00000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = 3.00000000, -2.50000000, 7.00000000
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 4.44089210e-16 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.10.1.10. Iteración 10

```
## x1^(10) = (7.8 - (-0.1)*-2.50000000 - (-0.2)*7.00000000)/3.0 = 3.00000000
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(10) = (-19.3 - (0.1)*3.00000000 - (-0.3)*7.00000000)/7.0 = -2.50000000
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(10) = (71.4 - (0.3)*3.00000000 - (-0.2)*-2.50000000)/10.0 = 7.00000000
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 3.00000000, -2.50000000, 7.00000000
```

```

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
# Soluciones
x_old <- x
x1 <- x_old[1]
x2 <- x_old[2]
x3 <- x_old[3]

# Comprobación
I <- b[1]; (I)

## [1] 7.85
II <- b[2]; (II)

## [1] -19.3
III <- b[3]; (III)

## [1] 71.4
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: 7.85"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: -19.3"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 71.4"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 7.85 tecnicamente es: 7.85"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: -19.3 tecnicamente es: -19.3"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: 71.4 tecnicamente es: 71.4"

```

```
cat("Soluciones del sistema:")
```

2.3.1.10.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x =", x1)
```

```
## x = 3
```

```
cat("y =", x2)
```

```
## y = -2.5
cat("z =", x3)
```

```
## z = 7
```

2.3.1.11. Sistema 10

$$17x_1 - 2x_2 - 3x_3 = 500$$

$$-5x_1 + 21x_2 - 2x_3 = 200$$

$$-5x_1 - 5x_2 + 22x_3 = 30$$

```
A <- matrix(c(17, -2, -3,
              -5, 21, -2,
              -5, -5, 22),
            nrow = 3, byrow = TRUE)
b <- c(500, 200, 30)

Ab <- cbind(A, b); (Ab)
```

2.3.1.11.1. Procedimiento

```
##          b
## [1,] 17 -2 -3 500
## [2,] -5 21 -2 200
## [3,] -5 -5 22 30

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}

## Sí es dominante

# Asignar valores iniciales

x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))

## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f ) / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.1. Iteración 1

```
## x1^(1) = (500.0 - (-2.0)*0.00000000 - (-3.0)*0.00000000)/17.0 = 29.41176471
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(1) = (200.0 - (-5.0)*29.41176471 - (-2.0)*0.00000000)/21.0 = 16.52661064
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))

## x3^(1) = (30.0 - (-5.0)*29.41176471 - (-5.0)*16.52661064)/22.0 = 11.80417622
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(1) = 29.41176471, 16.52661064, 11.80417622
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.94117647e+01 | ||r||_inf = 6.84657499e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.11.1.2. Iteración 2

```
## x1^(2) = (500.0 - (-2.0)*16.52661064 - (-3.0)*11.80417622)/17.0 = 33.43916176
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(2) = (200.0 - (-5.0)*33.43916176 - (-2.0)*11.80417622)/21.0 = 18.60972196
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))

## x3^(2) = (30.0 - (-5.0)*33.43916176 - (-5.0)*18.60972196)/22.0 = 13.19292812
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(2) = 33.43916176, 18.60972196, 13.19292812
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.02739706e+00 | ||r||_inf = 8.33247835e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.3. Iteración 3

```
## x1^(3) = (500.0 - (-2.0)*18.60972196 - (-3.0)*13.19292812)/17.0 = 33.92930755
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(3) = (200.0 - (-5.0)*33.92930755 - (-2.0)*13.19292812)/21.0 = 18.85868543
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(3) = (30.0 - (-5.0)*33.92930755 - (-5.0)*18.85868543)/22.0 = 13.36090749
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(3) = 33.92930755, 18.85868543, 13.36090749
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 4.90145785e-01 | ||r||_inf = 1.00186505e+00
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.4. Iteración 4

```
## x1^(4) = (500.0 - (-2.0)*18.85868543 - (-3.0)*13.36090749)/17.0 = 33.98824078
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(4) = (200.0 - (-5.0)*33.98824078 - (-2.0)*13.36090749)/21.0 = 18.88871519
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(4) = (30.0 - (-5.0)*33.98824078 - (-5.0)*18.88871519)/22.0 = 13.38112636
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(4) = 33.98824078, 18.88871519, 13.38112636
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 5.89332383e-02 | ||r||_inf = 1.20716107e-01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.5. Iteración 5

```
## x1^(5) = (500.0 - (-2.0)*18.88871519 - (-3.0)*13.38112636)/17.0 = 33.99534173
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(5) = (200.0 - (-5.0)*33.99534173 - (-2.0)*13.38112636)/21.0 = 18.89233149
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(5) = (30.0 - (-5.0)*33.99534173 - (-5.0)*18.89233149)/22.0 = 13.38356210
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) =%s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(5) = 33.99534173, 18.89233149, 13.38356210
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 7.10094749e-03 | ||r||_inf = 1.45398351e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.6. Iteración 6

```
## x1^(6) = (500.0 - (-2.0)*18.89233149 - (-3.0)*13.38356210)/17.0 = 33.99619702
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(6) = (200.0 - (-5.0)*33.99619702 - (-2.0)*13.38356210)/21.0 = 18.89276711
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(6) = (30.0 - (-5.0)*33.99619702 - (-5.0)*18.89276711)/22.0 = 13.38385548
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) =%s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(6) = 33.99619702, 18.89276711, 13.38385548
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 8.55284416e-04 | ||r||_inf = 1.75138700e-03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.7. Iteración 7

```
## x1^(7) = (500.0 - (-2.0)*18.89276711 - (-3.0)*13.38385548)/17.0 = 33.99630004
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(7) = (200.0 - (-5.0)*33.99630004 - (-2.0)*13.38385548)/21.0 = 18.89281958
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(7) = (30.0 - (-5.0)*33.99630004 - (-5.0)*18.89281958)/22.0 = 13.38389082
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = 33.99630004, 18.89281958, 13.38389082
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.03022765e-04 | ||r||_inf = 2.10959837e-04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.8. Iteración 8

```
## x1^(8) = (500.0 - (-2.0)*18.89281958 - (-3.0)*13.38389082)/17.0 = 33.99631245
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_ol
```

```
## x2^(8) = (200.0 - (-5.0)*33.99631245 - (-2.0)*13.38389082)/21.0 = 18.89282590
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(8) = (30.0 - (-5.0)*33.99631245 - (-5.0)*18.89282590)/22.0 = 13.38389508
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(8) = 33.99631245, 18.89282590, 13.38389508
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.24094022e-05 | ||r||_inf = 2.54107987e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.9. Iteración 9

```
## x1^(9) = (500.0 - (-2.0)*18.89282590 - (-3.0)*13.38389508)/17.0 = 33.99631394
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(9) = (200.0 - (-5.0)*33.99631394 - (-2.0)*13.38389508)/21.0 = 18.89282666
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(9) = (30.0 - (-5.0)*33.99631394 - (-5.0)*18.89282666)/22.0 = 13.38389559
```

#Error o tolerancia

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = 33.99631394, 18.89282666, 13.38389559
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.49475287e-06 | ||r||_inf = 3.06081233e-06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.10. Iteración 10

```
## x1^(10) = (500.0 - (-2.0)*18.89282666 - (-3.0)*13.38389559)/17.0 = 33.99631412
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3],
```

```
## x2^(10) = (200.0 - (-5.0)*33.99631412 - (-2.0)*13.38389559)/21.0 = 18.89282675
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2],
```

```
## x3^(10) = (30.0 - (-5.0)*33.99631412 - (-5.0)*18.89282675)/22.0 = 13.38389565
```



```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 33.99631412, 18.89282675, 13.38389565
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 1.80047785e-07 | ||r||_inf = 3.68684653e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.11. Iteración 11

```
## x1^(11) = (500.0 - (-2.0)*18.89282675 - (-3.0)*13.38389565)/17.0 = 33.99631415
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(11) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389565)/21.0 = 18.89282676
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(11) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(11) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(11) = 33.99631415, 18.89282676, 13.38389566
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 2.16873346e-08 | ||r||_inf = 4.44092052e-08
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.12. Iteración 12

```
## x1^(12) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(12) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))

## x3^(12) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(12) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(12) = 33.99631415, 18.89282676, 13.38389566
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.61231037e-09 | ||r||_inf = 5.34924993e-09
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.11.1.13. Iteración 13

```
## x1^(13) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_old[3], x[2]))

## x2^(13) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2], x[3]))

## x3^(3) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(13) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(13) = 33.99631415, 18.89282676, 13.38389566
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.14663851e-10 | ||r||_inf = 6.44263309e-10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], x[1]))
```

2.3.1.11.1.14. Iteración 14

```
## x1^(14) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
```

```

x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o

## x2^(14) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(14) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(14) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(14) = 33.99631415, 18.89282676, 13.38389566
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.79003495e-11 | ||r||_inf = 7.75344233e-11

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.1.11.1.15. Iteración 15

```

## x1^(15) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o

## x2^(15) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(15) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(15) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(15) = 33.99631415, 18.89282676, 13.38389566
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.56168436e-12 | ||r||_inf = 9.37916411e-12

```

```

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.1.11.1.16. Iteración 16

```
## x1^(16) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(16) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(16) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(1) = 33.99631415, 18.89282676, 13.38389566
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 5.47117907e-13 | ||r||_inf = 1.19371180e-12
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.17. Iteración 17

```
## x1^(17) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(17) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
```

```
cat(sprintf("x3^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(17) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(17) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(17) = 33.99631415, 18.89282676, 13.38389566
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 6.39488462e-14 | ||r||_inf = 2.27373675e-13
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.18. Iteración 18

```
## x1^(18) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o

## x2^(18) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(18) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(18) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(18) = 33.99631415, 18.89282676, 13.38389566
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.42108547e-14 | ||r||_inf = 5.68434189e-14
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.19. Iteración 19

```
## x1^(19) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o

## x2^(19) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]

## x3^(19) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(19) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(19) = 33.99631415, 18.89282676, 13.38389566
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.68434189e-14
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.1.11.1.20. Iteración 20

```
## x1^(20) = (500.0 - (-2.0)*18.89282676 - (-3.0)*13.38389566)/17.0 = 33.99631415
```

```
x[2] <- (b[2] - A[2,1]*x[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x[1], A[2,3], x_o
```

```
## x2^(20) = (200.0 - (-5.0)*33.99631415 - (-2.0)*13.38389566)/21.0 = 18.89282676
```

```
x[3] <- (b[3] - A[3,1]*x[1] - A[3,2]*x[2]) / A[3,3]
cat(sprintf("x3^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x[1], A[3,2], x[2]
```

```
## x3^(20) = (30.0 - (-5.0)*33.99631415 - (-5.0)*18.89282676)/22.0 = 13.38389566
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(20) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(20) = 33.99631415, 18.89282676, 13.38389566
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.68434189e-14
```

```
# Soluciones
```

```
x_old <- x
x1 <- x_old[1]
x2 <- x_old[2]
x3 <- x_old[3]
```

```
# Comprobación
```

```
I <- b[1]; (I)
```

```
## [1] 500
```

```
II <- b[2]; (II)
```

```
## [1] 200
```

```
III <- b[3]; (III)
```

```
## [1] 30
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ", c1)
```

```
## [1] "El resultado es: 500"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 200"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 30"
```

```
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 500 tecnicamente es: 500"
```

```
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 200 tecnicamente es: 200"
```

```
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 30 tecnicamente es: 30"
```

```
cat("Soluciones del sistema:")
```

2.3.1.11.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 33.99631
```

```
cat("x2 =", x2)
```

```
## x2 = 18.89283
```

```
cat("x3 =", x3)
```

```
## x3 = 13.3839
```

2.3.2. 7.7. Eliminación por Jacobi (Método iterativo)

2.3.2.1. Ejercicios a mano

2.3.2.2. Sistema 1

$$A = \left[\begin{array}{cccc|c} 10 & 2 & -1 & 0 & 26 \\ 1 & 20 & -2 & 3 & -15 \\ -2 & 1 & 30 & 0 & 53 \\ 1 & 2 & 3 & 20 & 47 \end{array} \right]$$

```
A <- matrix(c(10, 2, -1, 0,
              1, 20, -2, 3,
              -2, 1, 30, 0,
              1, 2, 3, 20),
            nrow = 4, byrow = TRUE)
b <- matrix(c(26, -15, 53, 47), nrow = 4)
Ab <- cbind(A, b); (Ab)
```


7.7 JACOBI

7. $x_1 + 2x_2 - 2x_3 = 7$
 $x_1 + x_2 + x_3 = 2 \Rightarrow$
 $2x_1 + 2x_2 + x_3 = 5$

$$\begin{bmatrix} 1 & 2 & -2 & 7 \\ 1 & 1 & 1 & 2 \\ 2 & 2 & 1 & 5 \end{bmatrix}$$

Verificamos si la matriz tiene una diagonal dominante

$|1| > |2| + |-2| + |7| \Rightarrow |1| \nless |11|$
 No es dominante.

Realizamos algunas iteraciones para ver si puede llegar a converger.

$x_1 = 7 - 2x_2 + 2x_3 \quad | \quad x_2 = 2 - x_1 - x_3 \quad | \quad x_3 = 5 - 2x_1 - 2x_2$

$x^{(0)} = (0, 0, 0)$

$x_1^{(1)} = 7 - 2(0) + 2(0) = 7 \quad | \quad x_1^{(2)} = 7 - 2(2) + 2(5) = 7 - 4 + 10 = 13$
 $x_2^{(1)} = 2 - 0 - 0 = 2 \quad | \quad x_2^{(2)} = 2 - 7 - 5 = -10$
 $x_3^{(1)} = 5 - 2(0) - 2(0) = 5 \quad | \quad x_3^{(2)} = 5 - 2(7) - 2(2) = 5 - 14 - 4 = -13$
 $E = |13 - 7| = 6$

Figura 2.25: Ejercicio 7

$$\begin{aligned}
 x_1^{(3)} &= 7 - 2(-10) + 2(-13) = 7 + 20 - 26 = 1 \\
 x_2^{(3)} &= 2 - 13 - (-13) = 2 - 13 + 13 = 2 \\
 x_3^{(3)} &= 5 - 2(13) - 2(-10) = 5 - 26 + 20 = -1 \\
 \\
 x_1^{(4)} &= 7 - 2(2) + 2(-1) = 7 - 4 - 2 = 1 \\
 x_2^{(4)} &= 2 - 1 - (-1) = 2 - 1 + 1 = 2 \\
 x_3^{(4)} &= 5 - 2(1) - 2(2) = 5 - 2 - 4 = -1 \\
 \\
 E = \begin{cases} [1-1] = 0 \\ [2-2] = 0 \\ [-1-(-1)] = 0 \end{cases} \\
 \\
 \text{Comprobación} \\
 (1) + 2(2) - 2(-1) = 7 \Rightarrow 1 + 4 + 2 = 7 \Rightarrow 7 = 7 \\
 (1) + (2) + (-1) = 2 \Rightarrow 1 + 2 - 1 = 2 \Rightarrow 2 = 2 \\
 2(1) + 2(2) + (-1) = 5 \Rightarrow 2 + 4 - 1 = 5 \Rightarrow 5 = 5 \\
 \\
 \text{Soluciones: } x_1 = 1, x_2 = 2, x_3 = -1 \\
 \\
 9. \begin{cases} x + y + 2z = 1 \\ x + 2y + z = 1 \\ 2x + y + z = 1 \end{cases} \Rightarrow \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 1 \end{bmatrix} \\
 \\
 \text{Verificamos si tiene matriz con diagonal dominante} \\
 |1| > |1| + |2| \Rightarrow |1| > |3| \\
 \\
 \text{No es dominante.} \\
 \text{Realizamos algunas iteraciones para ver si puede llegar a converger.}
 \end{aligned}$$

Figura 2.26: Ejercicio 9

$$\begin{aligned}
 &X = 1 - Y - 2Z \quad | \quad Y = 1 - X - Z \quad | \quad Z = 1 - 2X - Y \\
 &X^{(0)} = (0, 0, 0) \\
 &X^{(1)} = 1 - 0 - 2(0) = 1 \quad | \quad Y = \frac{1 - 0 - 0}{2} = 0.5 \quad | \quad Z = 1 - 2(0) - 0 = 1 \\
 &X^{(2)} = 1 - 0.5 - 2(1) = -1.5 \quad | \quad Y = \frac{1 - 1 - 1}{2} = -0.5 \quad | \quad Z = 1 - 2(1) - 0.5 = -1.5 \\
 &E = [1 - 1 - 1.5] = 2.5 \\
 &X^{(3)} = 1 - (-0.5) - 2(-1.5) = 4.5 \quad | \quad Y = \frac{1 - (-1.5) - (-1.5)}{2} = 1.5 \quad | \quad Z = 1 - 2(-1.5) - (-0.5) = 4.5 \\
 &X^{(4)} = 1 - 2 - 2(4.5) = -10 \quad | \quad Y = \frac{1 - 4.5 - 4.5}{2} = -4 \quad | \quad Z = 1 - 2(4.5) - 2 = -10 \\
 &E = [-10 - 4.5] = 14.5 \\
 &X^{(5)} = 1 - (-4) - 2(-10) = 25 \quad | \quad Y = \frac{1 - (-10) - (-10)}{2} = 10.5 \quad | \quad Z = 1 - 2(-10) - (-4) = 25 \\
 &X^{(6)} = 1 - 10.5 - 2(25) = -59.5 \quad | \quad Y = \frac{1 - 25 - 25}{2} = -24.5 \quad | \quad Z = 1 - 2(25) - 10.5 = -59.5 \\
 &E = [-59.5 - 25] = 84.5
 \end{aligned}$$

Se puede apreciar que cada vez el error o la tolerancia crece más por lo que este sistema en vez de converger, va a diverger.

Figura 2.27: Ejercicio 9-2

2.3.2.2.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]  10    2  -1    0   26
## [2,]   1   20  -2    3  -15
## [3,]  -2    1   30    0   53
## [4,]   1    2    3   20   47
```

```
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]) + abs(A[1,4]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]) + abs(A[2,4]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]) + abs(A[3,4]))
F4 <- abs(A[4,4]) > (abs(A[4,1]) + abs(A[4,2]) + abs(A[4,3]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 4) #  $x^0 = (0, 0, 0)$ 
cat("x0=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x0 = 0.00000000, 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x11 = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.1. Iteración 1

```
## x11 = (26.0 - (2.0)*0.00000000 - (-1.0)*0.00000000 - (0.0)*0.00000000)/10.0 = 2.60000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x21 = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x21 = (-15.0 - (1.0)*0.00000000 - (-2.0)*0.00000000 - (3.0)*0.00000000)/20.0 = -0.75000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x31 = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x31 = (53.0 - (-2.0)*0.00000000 - (1.0)*0.00000000 - (0.0)*0.00000000)/30.0 = 1.76666667
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x41 = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x41 = (47.0 - (1.0)*0.00000000 - (2.0)*0.00000000 - (3.0)*0.00000000)/20.0 = 2.35000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 2.6
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -0.75
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 1.766667
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2.35
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(1) = 2.60000000, -0.75000000, 1.76666667, 2.35000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.40000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.2. Iteración 2

```
## x1^(2) = (26.0 - (2.0)*-0.75000000 - (-1.0)*1.76666667 - (0.0)*2.35000000)/10.0 = 2.92666667
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(2) = (-15.0 - (1.0)*2.60000000 - (-2.0)*1.76666667 - (3.0)*2.35000000)/20.0 = -1.05583333
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(2) = (53.0 - (-2.0)*2.60000000 - (1.0)*-0.75000000 - (0.0)*2.35000000)/30.0 = 1.96500000
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(2) = (47.0 - (1.0)*2.60000000 - (2.0)*-0.75000000 - (3.0)*2.35000000)/20.0 = 2.03000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 2.926667
```

```

x2 <- x_old[2]; (x2)

## [1] -1.055833
x3 <- x_old[3]; (x3)

## [1] 1.965
x4 <- x_old[4]; (x4)

## [1] 2.03
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(2) = 2.92666667, -1.05583333, 1.96500000, 2.03000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.03000000e+00

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.2.1.3. Iteración 3

```

## x1^(3) = (26.0 - (2.0)*-1.05583333 - (-1.0)*1.96500000 - (0.0)*2.03000000)/10.0 = 3.00766667
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(3) = (-15.0 - (1.0)*2.92666667 - (-2.0)*1.96500000 - (3.0)*2.03000000)/20.0 = -1.00433333
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(3) = (53.0 - (-2.0)*2.92666667 - (1.0)*-1.05583333 - (0.0)*2.03000000)/30.0 = 1.99697222
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(3) = (47.0 - (1.0)*2.92666667 - (2.0)*-1.05583333 - (3.0)*2.03000000)/20.0 = 2.01450000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```

## [1] 3.007667
x2 <- x_old[2]; (x2)

## [1] -1.004333
x3 <- x_old[3]; (x3)

```

```
## [1] 1.996972
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2.0145
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(3) = 3.00766667, -1.00433333, 1.99697222, 2.01450000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.79916667e-01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.4. Iteración 4

```
## x1^(4) = (26.0 - (2.0)*-1.00433333 - (-1.0)*1.99697222 - (0.0)*2.01450000)/10.0 = 3.00056389
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(4) = (-15.0 - (1.0)*3.00766667 - (-2.0)*1.99697222 - (3.0)*2.01450000)/20.0 = -1.00286111
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(4) = (53.0 - (-2.0)*3.00766667 - (1.0)*-1.00433333 - (0.0)*2.01450000)/30.0 = 2.00065556
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(4) = (47.0 - (1.0)*3.00766667 - (2.0)*-1.00433333 - (3.0)*2.01450000)/20.0 = 2.00050417
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.000564
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1.002861
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2.000656
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2.000504
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(4) = 3.00056389, -1.00286111, 2.00065556, 2.00050417
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.64569444e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.5. Iteración 5

```
## x1^(5) = (26.0 - (2.0)*-1.00286111 - (-1.0)*2.00065556 - (0.0)*2.00050417)/10.0 = 3.00063778
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(5) = (-15.0 - (1.0)*3.00056389 - (-2.0)*2.00065556 - (3.0)*2.00050417)/20.0 = -1.00003826
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(5) = (53.0 - (-2.0)*3.00056389 - (1.0)*-1.00286111 - (0.0)*2.00050417)/30.0 = 2.00013296
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(5) = (47.0 - (1.0)*3.00056389 - (2.0)*-1.00286111 - (3.0)*2.00050417)/20.0 = 2.00015958
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.000638
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1.000038
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2.000133
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2.00016
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```



```
## x^(5) = 3.00063778, -1.00003826, 2.00013296, 2.00015958
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.16828704e-03

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

2.3.2.2.1.6. Iteración 6

## x1^(6) = (26.0 - (2.0)*-1.00003826 - (-1.0)*2.00013296 - (0.0)*2.00015958)/10.0 = 3.00002095
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(6) = (-15.0 - (1.0)*3.00063778 - (-2.0)*2.00013296 - (3.0)*2.00015958)/20.0 = -1.00004253
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(6) = (53.0 - (-2.0)*3.00063778 - (1.0)*-1.00003826 - (0.0)*2.00015958)/30.0 = 2.00004379
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(6) = (47.0 - (1.0)*3.00063778 - (2.0)*-1.00003826 - (3.0)*2.00015958)/20.0 = 1.99995199
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3.000021
x2 <- x_old[2]; (x2)

## [1] -1.000043
x3 <- x_old[3]; (x3)

## [1] 2.000044
x4 <- x_old[4]; (x4)

## [1] 1.999952
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(6) = 3.00002095, -1.00004253, 2.00004379, 1.99995199
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.22939120e-03
```



```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.7. Iteración 7

```
## x1^(7) = (26.0 - (2.0)*-1.00004253 - (-1.0)*2.00004379 - (0.0)*1.99995199)/10.0 = 3.00001289
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(7) = (-15.0 - (1.0)*3.00002095 - (-2.0)*2.00004379 - (3.0)*1.99995199)/20.0 = -0.99998947
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(7) = (53.0 - (-2.0)*3.00002095 - (1.0)*-1.00004253 - (0.0)*1.99995199)/30.0 = 2.00000281
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(7) = (47.0 - (1.0)*3.00002095 - (2.0)*-1.00004253 - (3.0)*1.99995199)/20.0 = 1.99999664
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.000013
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -0.9999895
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2.000003
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.999997
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = 3.00001289, -0.99998947, 2.00000281, 1.99999664
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.07825965e-04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.8. Iteración 8

```
## x1^(8) = (26.0 - (2.0)*-0.99998947 - (-1.0)*2.00000281 - (0.0)*1.99999664)/10.0 = 2.99999817
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_o

## x2^(8) = (-15.0 - (1.0)*3.00001289 - (-2.0)*2.00000281 - (3.0)*1.99999664)/20.0 = -0.99999986
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_o

## x3^(8) = (53.0 - (-2.0)*3.00001289 - (1.0)*-0.99998947 - (0.0)*1.99999664)/30.0 = 2.00000051
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_o

## x4^(8) = (47.0 - (1.0)*3.00001289 - (2.0)*-0.99998947 - (3.0)*1.99999664)/20.0 = 1.99999788
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 2.999998
x2 <- x_old[2]; (x2)

## [1] -0.9999999
x3 <- x_old[3]; (x3)

## [1] 2.000001
x4 <- x_old[4]; (x4)

## [1] 1.999998
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = 2.99999817, -0.99999986, 2.00000051, 1.99999788
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.24122222e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_o
```

2.3.2.2.1.9. Iteración 9

```
## x1^(9) = (26.0 - (2.0)*-0.99999986 - (-1.0)*2.00000051 - (0.0)*1.99999788)/10.0 = 3.00000002
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_o
```

```

##  $x_2^{(9)} = (-15.0 - (1.0)*2.99999817 - (-2.0)*2.00000051 - (3.0)*1.99999788)/20.0 = -0.99999954$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

##  $x_3^{(9)} = (53.0 - (-2.0)*2.99999817 - (1.0)*-0.99999986 - (0.0)*1.99999788)/30.0 = 1.99999987$ 
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

##  $x_4^{(9)} = (47.0 - (1.0)*2.99999817 - (2.0)*-0.99999986 - (3.0)*1.99999788)/20.0 = 2.00000000$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3
x2 <- x_old[2]; (x2)

## [1] -0.9999995
x3 <- x_old[3]; (x3)

## [1] 2
x4 <- x_old[4]; (x4)

## [1] 2
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(9)} = 3.00000002, -0.99999954, 1.99999987, 2.00000000$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 9.47811664e-06

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.2.1.10. Iteración 10

```

##  $x_1^{(10)} = (26.0 - (2.0)*-0.99999954 - (-1.0)*1.99999987 - (0.0)*2.00000000)/10.0 = 2.99999990$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

##  $x_2^{(10)} = (-15.0 - (1.0)*3.00000002 - (-2.0)*1.99999987 - (3.0)*2.00000000)/20.0 = -1.00000001$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

##  $x_3^{(10)} = (53.0 - (-2.0)*3.00000002 - (1.0)*-0.99999954 - (0.0)*2.00000000)/30.0 = 1.99999999$ 

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(10) = (47.0 - (1.0)*3.00000002 - (2.0)*-0.99999954 - (3.0)*2.00000000)/20.0 = 1.99999997
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3
x2 <- x_old[2]; (x2)

## [1] -1
x3 <- x_old[3]; (x3)

## [1] 2
x4 <- x_old[4]; (x4)

## [1] 2
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(10) = 2.99999990, -1.00000001, 1.99999999, 1.99999997
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.06037600e-06

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.2.1.11. Iteración 11

```

## x1^(11) = (26.0 - (2.0)*-1.00000001 - (-1.0)*1.99999999 - (0.0)*1.99999997)/10.0 = 3.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(11) = (-15.0 - (1.0)*2.99999990 - (-2.0)*1.99999999 - (3.0)*1.99999997)/20.0 = -0.99999999
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(11) = (53.0 - (-2.0)*2.99999990 - (1.0)*-1.00000001 - (0.0)*1.99999997)/30.0 = 1.99999999
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(11) = (47.0 - (1.0)*2.99999990 - (2.0)*-1.00000001 - (3.0)*1.99999997)/20.0 = 2.00000001

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(11) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(11) = 3.00000000, -0.99999999, 1.99999999, 2.00000001
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.01972467e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(12) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.2.1.12. Iteración 12

```
## x1^(12) = (26.0 - (2.0)*-0.99999999 - (-1.0)*1.99999999 - (0.0)*2.00000001)/10.0 = 3.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(12) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(12) = (-15.0 - (1.0)*3.00000000 - (-2.0)*1.99999999 - (3.0)*2.00000001)/20.0 = -1.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(12) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(12) = (53.0 - (-2.0)*3.00000000 - (1.0)*-0.99999999 - (0.0)*2.00000001)/30.0 = 2.00000000
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(12) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(12) = (47.0 - (1.0)*3.00000000 - (2.0)*-0.99999999 - (3.0)*2.00000001)/20.0 = 2.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3
```

```

x2 <- x_old[2]; (x2)

## [1] -1
x3 <- x_old[3]; (x3)

## [1] 2
x4 <- x_old[4]; (x4)

## [1] 2
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(12) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(12) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.21280006e-08

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_

```

2.3.2.2.1.13. Iteración 13

```

## x1^(13) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_

## x2^(13) = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_

## x3^(13) = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_

## x4^(13) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3
x2 <- x_old[2]; (x2)

## [1] -1
x3 <- x_old[3]; (x3)

```

```
## [1] 2
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(13) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(13) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.14371140e-09
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
```

```
cat(sprintf("x1^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3], x_old[3], A[1,4], x_old[4], A[1,1], x[1]))
```

2.3.2.2.1.14. Iteración 14

```
## x1^(14) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
```

```
cat(sprintf("x2^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3], x_old[3], A[2,4], x_old[4], A[2,2], x[2]))
```

```
## x2^(14) = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
```

```
cat(sprintf("x3^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2], x_old[2], A[3,4], x_old[4], A[3,3], x[3]))
```

```
## x3^(14) = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
```

```
cat(sprintf("x4^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_old[1], A[4,2], x_old[2], A[4,3], x_old[3], A[4,4], x[4]))
```

```
## x4^(14) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 3
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(14) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(14) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.69694125e-09
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.2.1.15. Iteración 15

```
## x1^(15) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(15) = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(15) = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(15) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(15) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```



```
## x^(15) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.91839433e-10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.2.1.16. Iteración 16

```
## x1^(16) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(16) = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(16) = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(16) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(16) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(16) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.10906864e-11
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.2.1.17. Iteración 17

```
## x1^(17) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(17) = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(17) = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(17) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(17) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(17) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.64552874e-12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.2.1.18. Iteración 18

```
## x1^(18) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(18) = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(18) = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(18) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3
x2 <- x_old[2]; (x2)

## [1] -1
x3 <- x_old[3]; (x3)

## [1] 2
x4 <- x_old[4]; (x4)

## [1] 2
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(18) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(18) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.28608235e-12

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_


```

2.3.2.2.1.19. Iteración 19

```
## x1^(19) = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_


```

```

##  $x_2^{(19)} = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
##  $x_3^{(19)} = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000$ 
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
##  $x_4^{(19)} = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3
x2 <- x_old[2]; (x2)

## [1] -1
x3 <- x_old[3]; (x3)

## [1] 2
x4 <- x_old[4]; (x4)

## [1] 2
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(19) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(19)} = 3.00000000, -1.00000000, 2.00000000, 2.00000000$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.19744231e-13

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_

```

2.3.2.2.1.20. Iteración 20

```

##  $x_1^{(20)} = (26.0 - (2.0)*-1.00000000 - (-1.0)*2.00000000 - (0.0)*2.00000000)/10.0 = 3.00000000$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
##  $x_2^{(20)} = (-15.0 - (1.0)*3.00000000 - (-2.0)*2.00000000 - (3.0)*2.00000000)/20.0 = -1.00000000$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
##  $x_3^{(20)} = (53.0 - (-2.0)*3.00000000 - (1.0)*-1.00000000 - (0.0)*2.00000000)/30.0 = 2.00000000$ 

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(20) = (47.0 - (1.0)*3.00000000 - (2.0)*-1.00000000 - (3.0)*2.00000000)/20.0 = 2.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3
x2 <- x_old[2]; (x2)

## [1] -1
x3 <- x_old[3]; (x3)

## [1] 2
x4 <- x_old[4]; (x4)

## [1] 2
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(20) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(20) = 3.00000000, -1.00000000, 2.00000000, 2.00000000
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.55271368e-14
# Comprobación
I <- b[1]; (I)

## [1] 26
II <- b[2]; (II)

## [1] -15
III <- b[3]; (III)

## [1] 53
IV <- b[4]; (IV)

## [1] 47
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)

## [1] "El resultado es: 26"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)

## [1] "El resultado es: -15"

```

```

c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)

## [1] "El resultado es: 53"
c4<- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)

## [1] "El resultado es: 47"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 26 tecnicamente es: 26"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: -15 tecnicamente es: -15"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: 53 tecnicamente es: 53"
paste0("En la ecuación 3: ", c4, " tecnicamente es: ", IV)

## [1] "En la ecuación 3: 47 tecnicamente es: 47"

```

```
cat("Soluciones del sistema:")
```

2.3.2.2.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 3
```

```
cat("x2 =", x2)
```

```
## x2 = -1
```

```
cat("x3 =", x3)
```

```
## x3 = 2
```

```
cat("x4 =", x4)
```

```
## x4 = 2
```

2.3.2.3. Sistema 2

$$A = \left[\begin{array}{ccc|c} -1 & 2 & 10 & 26 \\ 11 & -1 & 2 & 12 \\ 1 & 5 & 2 & 8 \end{array} \right]$$

```

A <- matrix(c(-1, 2, 10,
              11, -1, 2,
              1, 5, 2),
            nrow = 3, byrow = TRUE)
b <- matrix(c(11, 2, 8), nrow = 3)

```

```
Ab <- cbind(A, b); (Ab)
```

2.3.2.3.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  -1   2  10  11
## [2,]  11  -1   2   2
## [3,]   1   5   2   8
```

```
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## No es dominante
```

La matriz no tiene diagonal dominante pero se hacen 10 iteraciones para ver si puede llegar a converger.

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) #  $x^0 = (0, 0, 0)$ 
cat("x^0=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^0= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.3.1.1. Iteración 1

```
## x1^1) = (11.0 - (2.0)*0.00000000 - (10.0)*0.00000000 / -1.0 = -11.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^1) = (2.0 - (11.0)*0.00000000 - (2.0)*0.00000000 / -1.0 = -2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^1) = (8.0 - (1.0)*0.00000000 - (5.0)*0.00000000 / 2.0 = 4.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -11
```

```

x2 <- x_old[2]; (x2)

## [1] -2
x3 <- x_old[3]; (x3)

## [1] 4
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(1) = -11.00000000, -2.00000000, 4.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.13000000e+02

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
2.3.2.3.1.2. Iteración 2

## x1^(2) = (11.0 - (2.0)*-2.00000000 - (10.0)*4.00000000 / -1.0 = 25.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(2) = (2.0 - (11.0)*-11.00000000 - (2.0)*4.00000000 / -1.0 = -115.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(2) = (8.0 - (1.0)*-11.00000000 - (5.0)*-2.00000000 / 2.0 = 14.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 25
x2 <- x_old[2]; (x2)

## [1] -115
x3 <- x_old[3]; (x3)

## [1] 14.5
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

```



```
## x^(2) = 25.00000000, -115.00000000, 14.50000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.29000000e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.3.1.3. Iteración 3

```
## x1^(3) = (11.0 - (2.0)*-115.00000000 - (10.0)*14.50000000 / -1.0 = -96.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(3) = (2.0 - (11.0)*25.00000000 - (2.0)*14.50000000 / -1.0 = 302.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(3) = (8.0 - (1.0)*25.00000000 - (5.0)*-115.00000000 / 2.0 = 279.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -96
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 302
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 279
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(3) = -96.00000000, 302.00000000, 279.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.47900000e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.3.1.4. Iteración 4

```
## x1^(4) = (11.0 - (2.0)*302.00000000 - (10.0)*279.00000000 / -1.0 = 3383.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(4) = (2.0 - (11.0)*-96.00000000 - (2.0)*279.00000000 / -1.0 = -500.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(4) = (8.0 - (1.0)*-96.00000000 - (5.0)*302.00000000 / 2.0 = -703.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3383
x2 <- x_old[2]; (x2)

## [1] -500
x3 <- x_old[3]; (x3)

## [1] -703
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = 3383.00000000, -500.00000000, -703.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.63050000e+04

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.3.1.5. Iteración 5

```

## x1^(5) = (11.0 - (2.0)*-500.00000000 - (10.0)*-703.00000000 / -1.0 = -8041.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(5) = (2.0 - (11.0)*3383.00000000 - (2.0)*-703.00000000 / -1.0 = 35805.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(5) = (8.0 - (1.0)*3383.00000000 - (5.0)*-500.00000000 / 2.0 = -437.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] -8041
x2 <- x_old[2]; (x2)

## [1] 35805
x3 <- x_old[3]; (x3)

## [1] -437.5
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(5) = -8041.00000000, 35805.00000000, -437.50000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.70101000e+05

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

2.3.2.3.1.6. Iteración 6

## x1^(6) = (11.0 - (2.0)*35805.00000000 - (10.0)*-437.50000000 / -1.0 = 67224.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(6) = (2.0 - (11.0)*-8041.00000000 - (2.0)*-437.50000000 / -1.0 = -89328.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(6) = (8.0 - (1.0)*-8041.00000000 - (5.0)*35805.00000000 / 2.0 = -85488.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 67224
x2 <- x_old[2]; (x2)

## [1] -89328
x3 <- x_old[3]; (x3)

## [1] -85488
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(6) = 67224.00000000, -89328.00000000, -85488.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.10077100e+06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.3.1.7. Iteración 7

```
## x1^(7) = (11.0 - (2.0)*-89328.00000000 - (10.0)*-85488.00000000 / -1.0 = -1033547.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(7) = (2.0 - (11.0)*67224.00000000 - (2.0)*-85488.00000000 / -1.0 = 568486.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(7) = (8.0 - (1.0)*67224.00000000 - (5.0)*-89328.00000000 / 2.0 = 189712.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -1033547
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 568486
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 189712
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = -1033547.00000000, 568486.00000000, 189712.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.15580810e+07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.3.1.8. Iteración 8

```
## x1^(8) = (11.0 - (2.0)*568486.00000000 - (10.0)*189712.00000000 / -1.0 = 3034081.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(8) = (2.0 - (11.0)*-1033547.00000000 - (2.0)*189712.00000000 / -1.0 = -10989595.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(8) = (8.0 - (1.0)*-1033547.00000000 - (5.0)*568486.00000000 / 2.0 = -904437.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3034081
x2 <- x_old[2]; (x2)

## [1] -10989595
x3 <- x_old[3]; (x3)

## [1] -904437.5
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = 3034081.00000000, -10989595.00000000, -904437.50000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.37227770e+07

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.3.1.9. Iteración 9

```

## x1^(9) = (11.0 - (2.0)*-10989595.00000000 - (10.0)*-904437.50000000 / -1.0 = -31023576.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(9) = (2.0 - (11.0)*3034081.00000000 - (2.0)*-904437.50000000 / -1.0 = 31566014.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(9) = (8.0 - (1.0)*3034081.00000000 - (5.0)*-10989595.00000000 / 2.0 = 25956951.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] -31023576
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 31566014
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 25956951
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = -31023576.00000000, 31566014.00000000, 25956951.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.53725103e+08
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.3.1.10. Iteración 10

```
## x1^(10) = (11.0 - (2.0)*31566014.00000000 - (10.0)*25956951.00000000 / -1.0 = 322701527.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(10) = (2.0 - (11.0)*-31023576.00000000 - (2.0)*25956951.00000000 / -1.0 = -289345436.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(10) = (8.0 - (1.0)*-31023576.00000000 - (5.0)*31566014.00000000 / 2.0 = -63403243.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 322701527
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -289345436
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -63403243
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 322701527.00000000, -289345436.00000000, -63403243.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.71225574e+09
```

2.3.2.3.2. Resultados La matriz no converge, de hecho, diverge de forma rápida.

2.3.2.4. Sistema 3

$$A = \left[\begin{array}{ccc|c} 8 & 2 & 3 & 51 \\ 2 & 5 & 1 & 23 \\ -3 & 1 & 6 & 20 \end{array} \right]$$

```
A <- matrix(c(8, 2, 3,
              2, 5, 1,
              -3, 1, 6),
            nrow = 3, byrow = TRUE)
b <- matrix(c(51, 23, 20), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.3.2.4.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]    8    2    3    51
## [2,]    2    5    1    23
## [3,]   -3    1    6    20

# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.1. Iteración 1

```
## x1^(1) = (51.0 - (2.0)*0.00000000 - (3.0)*0.00000000 / 8.0 = 6.37500000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(1) = (23.0 - (2.0)*0.00000000 - (1.0)*0.00000000 / 5.0 = 4.60000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(1) = (20.0 - (-3.0)*0.00000000 - (1.0)*0.00000000 / 6.0 = 3.33333333
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 6.375
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 4.6
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 3.333333
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(1) = 6.37500000, 4.60000000, 3.33333333
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.92000000e+01
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.2. Iteración 2

```
## x1^(2) = (51.0 - (2.0)*4.60000000 - (3.0)*3.33333333 / 8.0 = 3.97500000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(2) = (23.0 - (2.0)*6.37500000 - (1.0)*3.33333333 / 5.0 = 1.38333333
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(2) = (20.0 - (-3.0)*6.37500000 - (1.0)*4.60000000 / 6.0 = 5.75416667
```



```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.975
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.383333
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5.754167
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(2) = 3.97500000, 1.38333333, 5.75416667
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.98333333e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.3. Iteración 3

```
## x1^(3) = (51.0 - (2.0)*1.38333333 - (3.0)*5.75416667) / 8.0 = 3.87135417
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(3) = (23.0 - (2.0)*3.97500000 - (1.0)*5.75416667) / 5.0 = 1.85916667
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(3) = (20.0 - (-3.0)*3.97500000 - (1.0)*1.38333333) / 6.0 = 5.09027778
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.871354
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.859167
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5.090278
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(3) = 3.87135417, 1.85916667, 5.09027778
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.04000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.4. Iteración 4

```
## x1^(4) = (51.0 - (2.0)*1.85916667 - (3.0)*5.09027778 / 8.0 = 4.00135417
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(4) = (23.0 - (2.0)*3.87135417 - (1.0)*5.09027778 / 5.0 = 2.03340278
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(4) = (20.0 - (-3.0)*3.87135417 - (1.0)*1.85916667 / 6.0 = 4.95914931
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.001354
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2.033403
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 4.959149
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(4) = 4.00135417, 2.03340278, 4.95914931
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.15763889e-01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.5. Iteración 5

```
## x1^(5) = (51.0 - (2.0)*2.03340278 - (3.0)*4.95914931 / 8.0 = 4.00696832
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(5) = (23.0 - (2.0)*4.00135417 - (1.0)*4.95914931 / 5.0 = 2.00762847
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(5) = (20.0 - (-3.0)*4.00135417 - (1.0)*2.03340278 / 6.0 = 4.99510995
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.006968
x2 <- x_old[2]; (x2)
```

```
## [1] 2.007628
x3 <- x_old[3]; (x3)
```

```
## [1] 4.99511
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(5) = 4.00696832, 2.00762847, 4.99510995
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.63333333e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.6. Iteración 6

```
## x1^(6) = (51.0 - (2.0)*2.00762847 - (3.0)*4.99510995 / 8.0 = 3.99992665
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(6) = (23.0 - (2.0)*4.00696832 - (1.0)*4.99510995 / 5.0 = 1.99819068
```

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(6) = (20.0 - (-3.0)*4.00696832 - (1.0)*2.00762847 / 6.0 = 5.00221275
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3.999927
x2 <- x_old[2]; (x2)

## [1] 1.998191
x3 <- x_old[3]; (x3)

## [1] 5.002213
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(6) = 3.99992665, 1.99819068, 5.00221275
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.16872106e-02

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.4.1.7. Iteración 7

```

## x1^(7) = (51.0 - (2.0)*1.99819068 - (3.0)*5.00221275 / 8.0 = 3.99962255
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(7) = (23.0 - (2.0)*3.99992665 - (1.0)*5.00221275 / 5.0 = 1.99958679
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(7) = (20.0 - (-3.0)*3.99992665 - (1.0)*1.99819068 / 6.0 = 5.00026488
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3.999623
x2 <- x_old[2]; (x2)

## [1] 1.999587

```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5.000265
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = 3.99962255, 1.99958679, 5.00026488
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.05138889e-03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.8. Iteración 8

```
## x1^(8) = (51.0 - (2.0)*1.99958679 - (3.0)*5.00026488 / 8.0 = 4.00000397
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(8) = (23.0 - (2.0)*3.99962255 - (1.0)*5.00026488 / 5.0 = 2.00009800
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(8) = (20.0 - (-3.0)*3.99962255 - (1.0)*1.99958679 / 6.0 = 4.99988014
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.000004
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2.000098
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 4.99988
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(8) = 4.00000397, 2.00009800, 4.99988014
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.33057243e-04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.9. Iteración 9

```
## x1^(9) = (51.0 - (2.0)*2.00009800 - (3.0)*4.99988014 / 8.0 = 4.00002045
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(9) = (23.0 - (2.0)*4.00000397 - (1.0)*4.99988014 / 5.0 = 2.00002238
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(9) = (20.0 - (-3.0)*4.00000397 - (1.0)*2.00009800 / 6.0 = 4.99998565
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.00002
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2.000022
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 4.999986
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(9) = 4.00002045, 2.00002238, 4.99998565
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.65283565e-04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.10. Iteración 10

```
## x1^(10) = (51.0 - (2.0)*2.00002238 - (3.0)*4.99998565 / 8.0 = 3.99999978
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```

##  $x_2^{(10)} = (23.0 - (2.0)*4.00002045 - (1.0)*4.99998565) / 5.0 = 1.99999469$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
x_old[3], x[3]))

##  $x_3^{(10)} = (20.0 - (-3.0)*4.00002045 - (1.0)*2.00002238) / 6.0 = 5.00000649$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 1.999995
x3 <- x_old[3]; (x3)

## [1] 5.000006
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(10)} = 3.99999978, 1.99999469, 5.00000649$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.42906007e-05

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
x_old[3], x[1]))

2.3.2.4.1.11. Iteración 11

##  $x_1^{(11)} = (51.0 - (2.0)*1.99999469 - (3.0)*5.00000649) / 8.0 = 3.99999889$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
x_old[3], x[2]))

##  $x_2^{(11)} = (23.0 - (2.0)*3.99999978 - (1.0)*5.00000649) / 5.0 = 1.99999879$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
x_old[3], x[3]))

##  $x_3^{(11)} = (20.0 - (-3.0)*3.99999978 - (1.0)*1.99999469) / 6.0 = 5.00000078$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3.999999
x2 <- x_old[2]; (x2)

```

```
## [1] 1.999999
x3 <- x_old[3]; (x3)
```

```
## [1] 5.000001
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(11) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(11) = 3.99999889, 1.99999879, 5.00000078
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.95285976e-06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.12. Iteración 12

```
## x1^(12) = (51.0 - (2.0)*1.99999879 - (3.0)*5.00000078 / 8.0 = 4.00000001
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(12) = (23.0 - (2.0)*3.99999889 - (1.0)*5.00000078 / 5.0 = 2.00000029
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(12) = (20.0 - (-3.0)*3.99999889 - (1.0)*1.99999879 / 6.0 = 4.99999965
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(12) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(12) = 4.00000001, 2.00000029, 4.99999965
```



```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.85740754e-06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.13. Iteración 13

```
## x1^(13) = (51.0 - (2.0)*2.000000029 - (3.0)*4.999999965 / 8.0 = 4.000000006
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(13) = (23.0 - (2.0)*4.000000001 - (1.0)*4.999999965 / 5.0 = 2.000000007
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(13) = (20.0 - (-3.0)*4.000000001 - (1.0)*2.000000029 / 6.0 = 4.999999996
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(13) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(13) = 4.000000006, 2.000000007, 4.999999996
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.84946582e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.14. Iteración 14

```
## x1^(14) = (51.0 - (2.0)*2.000000007 - (3.0)*4.999999996 / 8.0 = 4.000000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(14) = (23.0 - (2.0)*4.00000006 - (1.0)*4.99999996 / 5.0 = 1.99999998
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(14) = (20.0 - (-3.0)*4.00000006 - (1.0)*2.00000007 / 6.0 = 5.00000002
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(14) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(14) = 4.00000000, 1.99999998, 5.00000002
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.00609576e-07

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.4.1.15. Iteración 15

```

## x1^(15) = (51.0 - (2.0)*1.99999998 - (3.0)*5.00000002 / 8.0 = 4.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(15) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000002 / 5.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(15) = (20.0 - (-3.0)*4.00000000 - (1.0)*1.99999998 / 6.0 = 5.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(15) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(15) = 4.00000000, 2.00000000, 5.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.62679407e-08
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.16. Iteración 16

```
## x1^(16) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 / 8.0 = 4.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(16) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 / 5.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(16) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 / 6.0 = 5.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(16) = %s\n", paste(sprintf("%.8f", x), collapse=",")))
```

```
## x^(16) = 4.00000000, 2.00000000, 5.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.44968515e-09
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.17. Iteración 17

```
## x1^(17) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 / 8.0 = 4.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(17) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 / 5.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(17) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 / 6.0 = 5.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(17) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(17) = 4.00000000, 2.00000000, 5.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.42284762e-09
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.18. Iteración 18

```
## x1^(18) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 / 8.0 = 4.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f =%.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(18) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 /5.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f =%.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(18) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 /6.0 = 5.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(18) =%s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(18) = 4.00000000, 2.00000000, 5.00000000
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.95191427e-10

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f =%.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.4.1.19. Iteración 19

```

## x1^(19) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 /8.0 = 4.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f =%.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(19) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 /5.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(19) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f =%.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(19) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 /6.0 = 5.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(19) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(19) = 4.00000000, 2.00000000, 5.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.70725705e-11
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.20. Iteración 20

```
## x1^(20) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 / 8.0 = 4.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(20) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 / 5.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(20) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 / 6.0 = 5.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(20) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(20) = 4.00000000, 2.00000000, 5.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.59872116e-11
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.21. Iteración 21

```
## x1^(21) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 / 8.0 = 4.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(21) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 / 5.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(21) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 / 6.0 = 5.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(21) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(21) = 4.00000000, 2.00000000, 5.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.17088586e-12
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.22. Iteración 22

```
## x1^(22) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 / 8.0 = 4.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(22) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 /5.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(22) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 /6.0 = 5.00000000
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(22) =%s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(22) = 4.00000000, 2.00000000, 5.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 6.98996416e-13 | ||r||_inf = 8.66862138e-13

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.4.1.23. Iteración 23

```

## x1^(23) = (51.0 - (2.0)*2.00000000 - (3.0)*5.00000000 /8.0 = 4.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(23) = (23.0 - (2.0)*4.00000000 - (1.0)*5.00000000 /5.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(23) = (20.0 - (-3.0)*4.00000000 - (1.0)*2.00000000 /6.0 = 5.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5
#Error o tolerancia

r <- b - A %*% x

```



```

dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(23) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(23) = 4.000000000, 2.000000000, 5.000000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.20268248e-13
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
2.3.2.4.1.24. Iteración 24

## x1^(24) = (51.0 - (2.0)*2.000000000 - (3.0)*5.000000000 / 8.0 = 4.000000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(24) = (23.0 - (2.0)*4.000000000 - (1.0)*5.000000000 / 5.0 = 2.000000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(24) = (20.0 - (-3.0)*4.000000000 - (1.0)*2.000000000 / 6.0 = 5.000000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5
#Error o tolerancia
r <- b - A %*% x

```

```
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(24) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(24) = 4.000000000, 2.000000000, 5.000000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.61852778e-14
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.4.1.25. Iteración 25

```
## x1^(25) = (51.0 - (2.0)*2.000000000 - (3.0)*5.000000000 / 8.0 = 4.000000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(25) = (23.0 - (2.0)*4.000000000 - (1.0)*5.000000000 / 5.0 = 2.000000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(25) = (20.0 - (-3.0)*4.000000000 - (1.0)*2.000000000 / 6.0 = 5.000000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 4
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```

dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(25) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(25) = 4.00000000, 2.00000000, 5.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.42108547e-14
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5
# Comprobación
I <- b[1]; (I)

## [1] 51
II <- b[2]; (II)

## [1] 23
III <- b[3]; (III)

## [1] 20
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: 51"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: 23"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 20"
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 51 técnicamente es: 51"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 23 técnicamente es: 23"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 20 técnicamente es: 20"

```

```
# Comprobación
```

```
I <- b[1]; (I)
```

```
## [1] 51
```

```
II <- b[2]; (II)
```

```
## [1] 23
```

```
III <- b[3]; (III)
```

```
## [1] 20
```

```
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)
```

```
## [1] "El resultado es: 51"
```

```
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)
```

```
## [1] "El resultado es: 23"
```

```
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)
```

```
## [1] "El resultado es: 20"
```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)
```

```
## [1] "En la ecuación 1: 51 técnicamente es: 51"
```

```
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)
```

```
## [1] "En la ecuación 2: 23 técnicamente es: 23"
```

```
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)
```

```
## [1] "En la ecuación 3: 20 técnicamente es: 20"
```

```
cat("Soluciones del sistema:")
```

2.3.2.4.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 4
```

```
cat("x2 =", x2)
```

```
## x2 = 2
```

```
cat("x3 =", x3)
```

```
## x3 = 5
```

2.3.2.5. Sistema 4

$$A = \left[\begin{array}{cccc|c} 2 & -1 & 1 & 3 & 10 \\ 2 & 2 & 2 & 2 & 1 \\ -1 & -1 & 2 & 2 & -5 \\ 3 & 1 & -1 & 4 & 6 \end{array} \right]$$

```
A <- matrix(c(2, -1, 1, 3,
              2, 2, 2, 2,
              -1, -1, 2, 2,
              3, 1, -1, 4),
            nrow = 4, byrow = TRUE)
b <- matrix(c(10, 1, -5, 6), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.3.2.5.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2   -1    1    3   10
## [2,]    2    2    2    2    1
## [3,]   -1   -1    2    2   -5
## [4,]    3    1   -1    4    6

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]) + abs(A[1,4]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]) + abs(A[2,4]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]) + abs(A[3,4]))
F4 <- abs(A[4,4]) > (abs(A[4,1]) + abs(A[4,2]) + abs(A[4,3]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

No es dominante

La matriz no tiene diagonal dominante pero se hacen 20 iteraciones para ver si es que puede llegar a converger.

Asignar valores iniciales

```
x <- rep(0, 4) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))
```

x^(0)= 0.00000000, 0.00000000, 0.00000000, 0.00000000

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.5.1.1. Iteración 1

```

## x1^(1) = (10.0 - (-1.0)*0.00000000 - (1.0)*0.00000000 - (3.0)*0.00000000)/2.0 = 5.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(1) = (1.0 - (2.0)*0.00000000 - (2.0)*0.00000000 - (2.0)*0.00000000)/2.0 = 0.50000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(1) = (-5.0 - (-1.0)*0.00000000 - (-1.0)*0.00000000 - (2.0)*0.00000000)/2.0 = -2.50000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(1) = (6.0 - (3.0)*0.00000000 - (1.0)*0.00000000 - (-1.0)*0.00000000)/4.0 = 1.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 5
x2 <- x_old[2]; (x2)

## [1] 0.5
x3 <- x_old[3]; (x3)

## [1] -2.5
x4 <- x_old[4]; (x4)

## [1] 1.5
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(1) = 5.00000000, 0.50000000, -2.50000000, 1.50000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.80000000e+01

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.5.1.2. Iteración 2

```

## x1^(2) = (10.0 - (-1.0)*0.50000000 - (1.0)*-2.50000000 - (3.0)*1.50000000)/2.0 = 4.25000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(2) = (1.0 - (2.0)*5.00000000 - (2.0)*-2.50000000 - (2.0)*1.50000000)/2.0 = -3.50000000

```

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(2) = (-5.0 - (-1.0)*5.00000000 - (-1.0)*0.50000000 - (2.0)*1.50000000)/2.0 = -1.25000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(2) = (6.0 - (3.0)*5.00000000 - (1.0)*0.50000000 - (-1.0)*1.50000000)/4.0 = -3.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4.25
x2 <- x_old[2]; (x2)

## [1] -3.5
x3 <- x_old[3]; (x3)

## [1] -1.25
x4 <- x_old[4]; (x4)

## [1] -3
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(2) = 4.25000000, -3.50000000, -1.25000000, -3.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.25000000e+00

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.5.1.3. Iteración 3

```

## x1^(3) = (10.0 - (-1.0)*-3.50000000 - (1.0)*-1.25000000 - (3.0)*-3.00000000)/2.0 = 8.37500000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(3) = (1.0 - (2.0)*4.25000000 - (2.0)*-1.25000000 - (2.0)*-3.00000000)/2.0 = 0.50000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(3) = (-5.0 - (-1.0)*4.25000000 - (-1.0)*-3.50000000 - (2.0)*-3.00000000)/2.0 = 0.87500000

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(3) = (6.0 - (3.0)*4.25000000 - (1.0)*-3.50000000 - (-1.0)*-3.00000000)/4.0 = -1.12500000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 8.375
x2 <- x_old[2]; (x2)

## [1] 0.5
x3 <- x_old[3]; (x3)

## [1] 0.875
x4 <- x_old[4]; (x4)

## [1] -1.125
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(3) = 8.37500000, 0.50000000, 0.87500000, -1.12500000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.62500000e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.5.1.4. Iteración 4

```

## x1^(4) = (10.0 - (-1.0)*0.50000000 - (1.0)*0.87500000 - (3.0)*-1.12500000)/2.0 = 6.50000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(4) = (1.0 - (2.0)*8.37500000 - (2.0)*0.87500000 - (2.0)*-1.12500000)/2.0 = -7.62500000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(4) = (-5.0 - (-1.0)*8.37500000 - (-1.0)*0.50000000 - (2.0)*-1.12500000)/2.0 = 3.06250000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(4) = (6.0 - (3.0)*8.37500000 - (1.0)*0.50000000 - (-1.0)*-1.12500000)/4.0 = -4.68750000

```



```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 6.5
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -7.625
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 3.0625
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -4.6875
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(4) = 6.50000000, -7.62500000, 3.06250000, -4.68750000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.59375000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.5.1.5. Iteración 5

```
## x1^(5) = (10.0 - (-1.0)*-7.62500000 - (1.0)*3.06250000 - (3.0)*-4.68750000)/2.0 = 6.68750000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(5) = (1.0 - (2.0)*6.50000000 - (2.0)*3.06250000 - (2.0)*-4.68750000)/2.0 = -4.37500000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(5) = (-5.0 - (-1.0)*6.50000000 - (-1.0)*-7.62500000 - (2.0)*-4.68750000)/2.0 = 1.62500000
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(5) = (6.0 - (3.0)*6.50000000 - (1.0)*-7.62500000 - (-1.0)*-4.68750000)/4.0 = -0.70312500
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 6.6875
```

```

x2 <- x_old[2]; (x2)

## [1] -4.375

x3 <- x_old[3]; (x3)

## [1] 1.625

x4 <- x_old[4]; (x4)

## [1] -0.703125

#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = 6.68750000, -4.37500000, 1.62500000, -0.70312500
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.26562500e+00

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.5.1.6. Iteración 6

```

## x1^(6) = (10.0 - (-1.0)*-4.37500000 - (1.0)*1.62500000 - (3.0)*-0.70312500)/2.0 = 3.05468750
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(6) = (1.0 - (2.0)*6.68750000 - (2.0)*1.62500000 - (2.0)*-0.70312500)/2.0 = -7.10937500
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(6) = (-5.0 - (-1.0)*6.68750000 - (-1.0)*-4.37500000 - (2.0)*-0.70312500)/2.0 = -0.64062500
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(6) = (6.0 - (3.0)*6.68750000 - (1.0)*-4.37500000 - (-1.0)*-0.70312500)/4.0 = -2.01562500
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3.054688
x2 <- x_old[2]; (x2)

## [1] -7.109375
x3 <- x_old[3]; (x3)

```

```
## [1] -0.640625
x4 <- x_old[4]; (x4)
```

```
## [1] -2.015625
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(6) = 3.05468750, -7.10937500, -0.64062500, -2.01562500
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.44218750e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.5.1.7. Iteración 7

```
## x1^(7) = (10.0 - (-1.0)*-7.10937500 - (1.0)*-0.64062500 - (3.0)*-2.01562500)/2.0 = 4.78906250
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(7) = (1.0 - (2.0)*3.05468750 - (2.0)*-0.64062500 - (2.0)*-2.01562500)/2.0 = 0.10156250
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(7) = (-5.0 - (-1.0)*3.05468750 - (-1.0)*-7.10937500 - (2.0)*-2.01562500)/2.0 = -2.51171875
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(7) = (6.0 - (3.0)*3.05468750 - (1.0)*-7.10937500 - (-1.0)*-2.01562500)/4.0 = 0.82617188
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.789062
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 0.1015625
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -2.511719
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 0.8261719
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = 4.78906250, 0.10156250, -2.51171875, 0.82617188
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.42851562e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.5.1.8. Iteración 8

```
## x1^(8) = (10.0 - (-1.0)*0.10156250 - (1.0)*-2.51171875 - (3.0)*0.82617188)/2.0 = 5.06738281
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(8) = (1.0 - (2.0)*4.78906250 - (2.0)*-2.51171875 - (2.0)*0.82617188)/2.0 = -2.60351562
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(8) = (-5.0 - (-1.0)*4.78906250 - (-1.0)*0.10156250 - (2.0)*0.82617188)/2.0 = -0.88085938
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(8) = (6.0 - (3.0)*4.78906250 - (1.0)*0.10156250 - (-1.0)*0.82617188)/4.0 = -2.74511719
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 5.067383
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -2.603516
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.8808594
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -2.745117
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(8) = 5.06738281, -2.60351562, -0.88085938, -2.74511719
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.37792969e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.5.1.9. Iteración 9

```
## x1^(9) = (10.0 - (-1.0)*-2.60351562 - (1.0)*-0.88085938 - (3.0)*-2.74511719)/2.0 = 8.25634766
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(9) = (1.0 - (2.0)*5.06738281 - (2.0)*-0.88085938 - (2.0)*-2.74511719)/2.0 = -0.94140625
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(9) = (-5.0 - (-1.0)*5.06738281 - (-1.0)*-2.60351562 - (2.0)*-2.74511719)/2.0 = 1.47705078
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(9) = (6.0 - (3.0)*5.06738281 - (1.0)*-2.60351562 - (-1.0)*-2.74511719)/4.0 = -1.86987305
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 8.256348
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -0.9414062
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 1.477051
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -1.869873
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = 8.25634766, -0.94140625, 1.47705078, -1.86987305
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.28442383e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.10. Iteración 10

```
## x1^(10) = (10.0 - (-1.0)*-0.94140625 - (1.0)*1.47705078 - (3.0)*-1.86987305)/2.0 = 6.59558105
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(10) = (1.0 - (2.0)*8.25634766 - (2.0)*1.47705078 - (2.0)*-1.86987305)/2.0 = -7.36352539
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(10) = (-5.0 - (-1.0)*8.25634766 - (-1.0)*-0.94140625 - (2.0)*-1.86987305)/2.0 = 3.02734375
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(10) = (6.0 - (3.0)*8.25634766 - (1.0)*-0.94140625 - (-1.0)*-1.86987305)/4.0 = -4.08764648
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 6.595581
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -7.363525
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 3.027344
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -4.087646
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 6.59558105, -7.36352539, 3.02734375, -4.08764648
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.29547119e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.11. Iteración 11

```
## x1^(11) = (10.0 - (-1.0)*-7.36352539 - (1.0)*3.02734375 - (3.0)*-4.08764648)/2.0 = 5.93603516
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(11) = (1.0 - (2.0)*6.59558105 - (2.0)*3.02734375 - (2.0)*-4.08764648)/2.0 = -5.03527832
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(11) = (-5.0 - (-1.0)*6.59558105 - (-1.0)*-7.36352539 - (2.0)*-4.08764648)/2.0 = 1.20367432
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(11) = (6.0 - (3.0)*6.59558105 - (1.0)*-7.36352539 - (-1.0)*-4.08764648)/4.0 = -0.84896851
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 5.936035
x2 <- x_old[2]; (x2)

## [1] -5.035278
x3 <- x_old[3]; (x3)

## [1] 1.203674
x4 <- x_old[4]; (x4)

## [1] -0.8489685
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(11) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(11) = 5.93603516, -5.03527832, 1.20367432, -0.84896851
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.56411743e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.12. Iteración 12

```
## x1^(12) = (10.0 - (-1.0)*-5.03527832 - (1.0)*1.20367432 - (3.0)*-0.84896851)/2.0 = 3.15397644
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```

##  $x_2^{(12)} = (1.0 - (2.0)*5.93603516 - (2.0)*1.20367432 - (2.0)*-0.84896851)/2.0 = -5.79074097$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
##  $x_3^{(12)} = (-5.0 - (-1.0)*5.93603516 - (-1.0)*-5.03527832 - (2.0)*-0.84896851)/2.0 = -1.20065308$ 
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
##  $x_4^{(12)} = (6.0 - (3.0)*5.93603516 - (1.0)*-5.03527832 - (-1.0)*-0.84896851)/4.0 = -1.39228821$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 3.153976
x2 <- x_old[2]; (x2)

## [1] -5.790741
x3 <- x_old[3]; (x3)

## [1] -1.200653
x4 <- x_old[4]; (x4)

## [1] -1.392288
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(12) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(12)} = 3.15397644, -5.79074097, -1.20065308, -1.39228821$ 
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.14594116e+01

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.5.1.13. Iteración 13

```

##  $x_1^{(13)} = (10.0 - (-1.0)*-5.79074097 - (1.0)*-1.20065308 - (3.0)*-1.39228821)/2.0 = 4.79338837$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
##  $x_2^{(13)} = (1.0 - (2.0)*3.15397644 - (2.0)*-1.20065308 - (2.0)*-1.39228821)/2.0 = -0.06103516$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
##  $x_3^{(13)} = (-5.0 - (-1.0)*3.15397644 - (-1.0)*-5.79074097 - (2.0)*-1.39228821)/2.0 = -2.42609406$ 

```



```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(13) = (6.0 - (3.0)*3.15397644 - (1.0)*-5.79074097 - (-1.0)*-1.39228821)/4.0 = 0.28203964
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 4.793388
x2 <- x_old[2]; (x2)

## [1] -0.06103516
x3 <- x_old[3]; (x3)

## [1] -2.426094
x4 <- x_old[4]; (x4)

## [1] 0.2820396
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(13) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(13) = 4.79338837, -0.06103516, -2.42609406, 0.28203964
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.18733826e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.5.1.14. Iteración 14

```

## x1^(14) = (10.0 - (-1.0)*-0.06103516 - (1.0)*-2.42609406 - (3.0)*0.28203964)/2.0 = 5.75946999
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(14) = (1.0 - (2.0)*4.79338837 - (2.0)*-2.42609406 - (2.0)*0.28203964)/2.0 = -2.14933395
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(14) = (-5.0 - (-1.0)*4.79338837 - (-1.0)*-0.06103516 - (2.0)*0.28203964)/2.0 = -0.41586304
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(14) = (6.0 - (3.0)*4.79338837 - (1.0)*-0.06103516 - (-1.0)*0.28203964)/4.0 = -2.68630600

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 5.75947
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -2.149334
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.415863
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -2.686306
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(14) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(14) = 5.75946999, -2.14933395, -0.41586304, -2.68630600
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.81447411e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.15. Iteración 15

```
## x1^(15) = (10.0 - (-1.0)*-2.14933395 - (1.0)*-0.41586304 - (3.0)*-2.68630600)/2.0 = 8.16272354
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(15) = (1.0 - (2.0)*5.75946999 - (2.0)*-0.41586304 - (2.0)*-2.68630600)/2.0 = -2.15730095
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(15) = (-5.0 - (-1.0)*5.75946999 - (-1.0)*-2.14933395 - (2.0)*-2.68630600)/2.0 = 1.99137402
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(15) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(15) = (6.0 - (3.0)*5.75946999 - (1.0)*-2.14933395 - (-1.0)*-2.68630600)/4.0 = -2.38623476
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 8.162724
```

```

x2 <- x_old[2]; (x2)

## [1] -2.157301
x3 <- x_old[3]; (x3)

## [1] 1.991374
x4 <- x_old[4]; (x4)

## [1] -2.386235
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(15) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(15) = 8.16272354, -2.15730095, 1.99137402, -2.38623476
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.02211237e+01

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
## x1^(16) = (10.0 - (-1.0)*-2.15730095 - (1.0)*1.99137402 - (3.0)*-2.38623476)/2.0 = 6.50501466
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
## x2^(16) = (1.0 - (2.0)*8.16272354 - (2.0)*1.99137402 - (2.0)*-2.38623476)/2.0 = -7.26786280
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
## x3^(16) = (-5.0 - (-1.0)*8.16272354 - (-1.0)*-2.15730095 - (2.0)*-2.38623476)/2.0 = 2.88894606
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
## x4^(16) = (6.0 - (3.0)*8.16272354 - (1.0)*-2.15730095 - (-1.0)*-2.38623476)/4.0 = -3.58487391
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 6.505015
x2 <- x_old[2]; (x2)

## [1] -7.267863
x3 <- x_old[3]; (x3)

```

```
## [1] 2.888946
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -3.584874
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(16) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(16) = 6.50501466, -7.26786280, 2.88894606, -3.58487391
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.09812605e+01
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
```

```
cat(sprintf("x1^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.17. Iteración 17

```
## x1^(17) = (10.0 - (-1.0)*-7.26786280 - (1.0)*2.88894606 - (3.0)*-3.58487391)/2.0 = 5.29890645
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
```

```
cat(sprintf("x2^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(17) = (1.0 - (2.0)*6.50501466 - (2.0)*2.88894606 - (2.0)*-3.58487391)/2.0 = -5.30908680
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
```

```
cat(sprintf("x3^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(17) = (-5.0 - (-1.0)*6.50501466 - (-1.0)*-7.26786280 - (2.0)*-3.58487391)/2.0 = 0.70344985
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
```

```
cat(sprintf("x4^(17) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(17) = (6.0 - (3.0)*6.50501466 - (1.0)*-7.26786280 - (-1.0)*-3.58487391)/4.0 = -0.83955878
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 5.298906
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -5.309087
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 0.7034498
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -0.8395588
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(17) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(17) = 5.29890645, -5.30908680, 0.70344985, -0.83955878
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.73796248e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(18) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.18. Iteración 18

```
## x1^(18) = (10.0 - (-1.0)*-5.30908680 - (1.0)*0.70344985 - (3.0)*-0.83955878)/2.0 = 3.25306985
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(18) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(18) = (1.0 - (2.0)*5.29890645 - (2.0)*0.70344985 - (2.0)*-0.83955878)/2.0 = -4.66279751
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(18) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(18) = (-5.0 - (-1.0)*5.29890645 - (-1.0)*-5.30908680 - (2.0)*-0.83955878)/2.0 = -1.66553140
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(18) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(18) = (6.0 - (3.0)*5.29890645 - (1.0)*-5.30908680 - (-1.0)*-0.83955878)/4.0 = -0.97104567
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.25307
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -4.662798
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -1.665531
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -0.9710457
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(18) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(18) = 3.25306985, -4.66279751, -1.66553140, -0.97104567
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 9.09260947e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.19. Iteración 19

```
## x1^(19) = (10.0 - (-1.0)*-4.66279751 - (1.0)*-1.66553140 - (3.0)*-0.97104567)/2.0 = 4.95793545
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(19) = (1.0 - (2.0)*3.25306985 - (2.0)*-1.66553140 - (2.0)*-0.97104567)/2.0 = -0.11649278
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(19) = (-5.0 - (-1.0)*3.25306985 - (-1.0)*-4.66279751 - (2.0)*-0.97104567)/2.0 = -2.23381816
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(19) = (6.0 - (3.0)*3.25306985 - (1.0)*-4.66279751 - (-1.0)*-0.97104567)/4.0 = -0.19048586
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.957935
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -0.1164928
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -2.233818
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -0.1904859
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(19) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(19) = 4.95793545, -0.11649278, -2.23381816, -0.19048586
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.02291883e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.5.1.20. Iteración 20

```
## x1^(20) = (10.0 - (-1.0)*-0.11649278 - (1.0)*-2.23381816 - (3.0)*-0.19048586)/2.0 = 6.34439148
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(20) = (1.0 - (2.0)*4.95793545 - (2.0)*-2.23381816 - (2.0)*-0.19048586)/2.0 = -2.03363144
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(20) = (-5.0 - (-1.0)*4.95793545 - (-1.0)*-0.11649278 - (2.0)*-0.19048586)/2.0 = 0.11120719
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(20) = (6.0 - (3.0)*4.95793545 - (1.0)*-0.11649278 - (-1.0)*-0.19048586)/4.0 = -2.74778293
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 6.344391
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -2.033631
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 0.1112072
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -2.747783
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(20) =%s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(20) = 6.34439148, -2.03363144, 0.11120719, -2.74778293
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.58391152e+00
```

2.3.2.5.2. Resultados A pesar de que la matriz no es dominante, esta si convergió, sin embargo, no lo hizo en el lugar correct. Por lo tanto, por medio de Jacobi, no se puede resolver este sistema.

2.3.2.6. Sistema 5

$$A = \left[\begin{array}{cccc|c} 3 & 1 & 1 & -1 & 5 \\ 0 & 2 & 1 & 4 & 0 \\ 1 & 1 & -1 & 9 & 1 \\ 2 & 4 & 6 & 3 & 0 \end{array} \right]$$

```
A <- matrix(c(3, 1, 1, -1,
              0, 2, 1, 4,
              1, 1, -1, 9,
              2, 4, 6, 3),
            nrow = 4, byrow = TRUE)
b <- matrix(c(5, 0, 1, 0), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.3.2.6.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    3    1    1   -1    5
## [2,]    0    2    1    4    0
## [3,]    1    1   -1    9    1
## [4,]    2    4    6    3    0

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]) + abs(A[1,4]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]) + abs(A[2,4]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]) + abs(A[3,4]))
F4 <- abs(A[4,4]) > (abs(A[4,1]) + abs(A[4,2]) + abs(A[4,3]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

No es dominante

La matriz no tiene diagonal dominante pero se hacen 10 iteraciones para ver si es que puede llegar a converger.

Asignar valores iniciales

```
x <- rep(0, 4) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=", "), "\n")
```

x^(0)= 0.00000000, 0.00000000, 0.00000000, 0.00000000

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.6.1.1. Iteración 1


```

## x1^(1) = (5.0 - (1.0)*0.00000000 - (1.0)*0.00000000 - (-1.0)*0.00000000)/3.0 = 1.66666667
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(1) = (0.0 - (0.0)*0.00000000 - (1.0)*0.00000000 - (4.0)*0.00000000)/2.0 = 0.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(1) = (1.0 - (1.0)*0.00000000 - (1.0)*0.00000000 - (9.0)*0.00000000)/-1.0 = -1.00000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(1) = (0.0 - (2.0)*0.00000000 - (4.0)*0.00000000 - (6.0)*0.00000000)/3.0 = 0.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1.666667
x2 <- x_old[2]; (x2)

## [1] 0
x3 <- x_old[3]; (x3)

## [1] -1
x4 <- x_old[4]; (x4)

## [1] 0
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(1) = 1.66666667, 0.00000000, -1.00000000, 0.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.66666667e+00

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.6.1.2. Iteración 2

```

## x1^(2) = (5.0 - (1.0)*0.00000000 - (1.0)*-1.00000000 - (-1.0)*0.00000000)/3.0 = 2.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(2) = (0.0 - (0.0)*1.66666667 - (1.0)*-1.00000000 - (4.0)*0.00000000)/2.0 = 0.50000000

```

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(2) = (1.0 - (1.0)*1.66666667 - (1.0)*0.00000000 - (9.0)*0.00000000)/-1.0 = 0.66666667
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(2) = (0.0 - (2.0)*1.66666667 - (4.0)*0.00000000 - (6.0)*0.00000000)/3.0 = 0.88888889
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 2
x2 <- x_old[2]; (x2)

## [1] 0.5
x3 <- x_old[3]; (x3)

## [1] 0.6666667
x4 <- x_old[4]; (x4)

## [1] 0.8888889
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(2) = 2.00000000, 0.50000000, 0.66666667, 0.88888889
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.26666667e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.6.1.3. Iteración 3

```

## x1^(3) = (5.0 - (1.0)*0.50000000 - (1.0)*0.66666667 - (-1.0)*0.88888889)/3.0 = 1.57407407
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(3) = (0.0 - (0.0)*2.00000000 - (1.0)*0.66666667 - (4.0)*0.88888889)/2.0 = -2.11111111
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(3) = (1.0 - (1.0)*2.00000000 - (1.0)*0.50000000 - (9.0)*0.88888889)/-1.0 = 9.50000000

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(3) = (0.0 - (2.0)*2.00000000 - (4.0)*0.50000000 - (6.0)*0.88888889)/3.0 = -3.33333333
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1.574074
x2 <- x_old[2]; (x2)

## [1] -2.111111
x3 <- x_old[3]; (x3)

## [1] 9.5
x4 <- x_old[4]; (x4)

## [1] -3.333333
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(3) = 1.57407407, -2.11111111, 9.50000000, -3.33333333
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.17037037e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.6.1.4. Iteración 4

```

## x1^(4) = (5.0 - (1.0)*-2.11111111 - (1.0)*9.50000000 - (-1.0)*-3.33333333)/3.0 = -1.90740741
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(4) = (0.0 - (0.0)*1.57407407 - (1.0)*9.50000000 - (4.0)*-3.33333333)/2.0 = 1.91666667
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(4) = (1.0 - (1.0)*1.57407407 - (1.0)*-2.11111111 - (9.0)*-3.33333333)/-1.0 = -31.53703704
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(4) = (0.0 - (2.0)*1.57407407 - (4.0)*-2.11111111 - (6.0)*-3.33333333)/3.0 = -17.23456790

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -1.907407
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.916667
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -31.53704
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -17.23457
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(4) = -1.90740741, 1.91666667, -31.53703704, -17.23456790
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.37074074e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.6.1.5. Iteración 5

```
## x1^(5) = (5.0 - (1.0)*1.91666667 - (1.0)*-31.53703704 - (-1.0)*-17.23456790)/3.0 = 5.79526749
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(5) = (0.0 - (0.0)*-1.90740741 - (1.0)*-31.53703704 - (4.0)*-17.23456790)/2.0 = 50.23765432
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(5) = (1.0 - (1.0)*-1.90740741 - (1.0)*1.91666667 - (9.0)*-17.23456790)/-1.0 = -156.10185185
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(5) = (0.0 - (2.0)*-1.90740741 - (4.0)*1.91666667 - (6.0)*-17.23456790)/3.0 = 61.79012346
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 5.795267
```

```

x2 <- x_old[2]; (x2)

## [1] 50.23765
x3 <- x_old[3]; (x3)

## [1] -156.1019
x4 <- x_old[4]; (x4)

## [1] 61.79012
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = 5.79526749, 50.23765432, -156.10185185, 61.79012346
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.67245885e+02

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.6.1.6. Iteración 6

```

## x1^(6) = (5.0 - (1.0)*50.23765432 - (1.0)*-156.10185185 - (-1.0)*61.79012346)/3.0 = 57.55144033
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(6) = (0.0 - (0.0)*5.79526749 - (1.0)*-156.10185185 - (4.0)*61.79012346)/2.0 = -45.52932099
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(6) = (1.0 - (1.0)*5.79526749 - (1.0)*50.23765432 - (9.0)*61.79012346)/-1.0 = 611.14403292
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(6) = (0.0 - (2.0)*5.79526749 - (4.0)*50.23765432 - (6.0)*61.79012346)/3.0 = 241.35665295
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 57.55144
x2 <- x_old[2]; (x2)

## [1] -45.52932
x3 <- x_old[3]; (x3)

```

```
## [1] 611.144
x4 <- x_old[4]; (x4)
```

```
## [1] 241.3567
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(6) = 57.55144033, -45.52932099, 611.14403292, 241.35665295
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.32391975e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.6.1.7. Iteración 7

```
## x1^(7) = (5.0 - (1.0)*-45.52932099 - (1.0)*611.14403292 - (-1.0)*241.35665295)/3.0 = -106.41935299
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(7) = (0.0 - (0.0)*57.55144033 - (1.0)*611.14403292 - (4.0)*241.35665295)/2.0 = -788.28532236
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(7) = (1.0 - (1.0)*57.55144033 - (1.0)*-45.52932099 - (9.0)*241.35665295)/-1.0 = 2183.23199588
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(7) = (0.0 - (2.0)*57.55144033 - (4.0)*-45.52932099 - (6.0)*241.35665295)/3.0 = -1199.94993141
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -106.4194
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -788.2853
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2183.232
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -1199.95
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = -106.41935299, -788.28532236, 2183.23199588, -1199.94993141
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.38784861e+04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.6.1.8. Iteración 8

```
## x1^(8) = (5.0 - (1.0)*-788.28532236 - (1.0)*2183.23199588 - (-1.0)*-1199.94993141)/3.0 = -863.29886831
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(8) = (0.0 - (0.0)*-106.41935299 - (1.0)*2183.23199588 - (4.0)*-1199.94993141)/2.0 = 1308.28386488
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(8) = (1.0 - (1.0)*-106.41935299 - (1.0)*-788.28532236 - (9.0)*-1199.94993141)/-1.0 = -11695.2540580
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(8) = (0.0 - (2.0)*-106.41935299 - (4.0)*-788.28532236 - (6.0)*-1199.94993141)/3.0 = -3244.47065996
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -863.2989
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1308.284
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -11695.25
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -3244.471
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(8) = -863.29886831, 1308.28386488, -11695.25405807, -3244.47065996
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.63983986e+04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.6.1.9. Iteración 9

```
## x1^(9) = (5.0 - (1.0)*1308.28386488 - (1.0)*-11695.25405807 - (-1.0)*-3244.47065996)/3.0 = 2382.4998444
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(9) = (0.0 - (0.0)*-863.29886831 - (1.0)*-11695.25405807 - (4.0)*-3244.47065996)/2.0 = 12336.5683489
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(9) = (1.0 - (1.0)*-863.29886831 - (1.0)*1308.28386488 - (9.0)*-3244.47065996)/-1.0 = -28756.2509430
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(9) = (0.0 - (2.0)*-863.29886831 - (4.0)*1308.28386488 - (6.0)*-3244.47065996)/3.0 = 22221.66220850
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 2382.5
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 12336.57
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -28756.25
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 22221.66
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = 2382.49984441, 12336.56834896, -28756.25094307, 22221.66220850
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.43469279e+05
```



```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.6.1.10. Iteración 10

```
## x1^(10) = (5.0 - (1.0)*12336.56834896 - (1.0)*-28756.25094307 - (-1.0)*22221.66220850)/3.0 = 12882.1149
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(10) = (0.0 - (0.0)*2382.49984441 - (1.0)*-28756.25094307 - (4.0)*22221.66220850)/2.0 = -30065.19894
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(10) = (1.0 - (1.0)*2382.49984441 - (1.0)*12336.56834896 - (9.0)*22221.66220850)/-1.0 = 214713.02806
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(10) = (0.0 - (2.0)*2382.49984441 - (4.0)*12336.56834896 - (6.0)*22221.66220850)/3.0 = 39475.4108579
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 12882.11
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -30065.2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 214713
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 39475.41
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 12882.11493421, -30065.19894547, 214713.02806991, 39475.41085793
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.31220784e+06
```

2.3.2.6.2. Resultados La matriz no se puede resolver por este método, ya que diverge.

2.3.2.7. Sistema 6

$$A = \left[\begin{array}{cccc|c} 10 & -1 & 2 & 0 & 6 \\ -1 & 11 & -1 & 3 & 25 \\ 2 & -1 & 10 & -1 & -11 \\ 0 & 2 & -1 & 8 & 15 \end{array} \right]$$

```
A <- matrix(c(10, -1, 2, 0,
              -1, 11, -1, 3,
              2, -1, 10, -1,
              0, 2, -1, 8),
            nrow = 4, byrow = TRUE)
b <- matrix(c(6, 25, -11, 15), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.3.2.7.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]  10   -1   2   0   6
## [2,]  -1  11  -1   3  25
## [3,]   2  -1  10  -1 -11
## [4,]   0   2  -1   8  15

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]) + abs(A[1,4]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]) + abs(A[2,4]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]) + abs(A[3,4]))
F4 <- abs(A[4,4]) > (abs(A[4,1]) + abs(A[4,2]) + abs(A[4,3]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## Sí es dominante
```

```
# Asignar valores iniciales
```

```
x <- rep(0, 4) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.7.1.1. Iteración 1

```
## x1^(1) = (6.0 - (-1.0)*0.00000000 - (2.0)*0.00000000 - (0.0)*0.00000000)/10.0 = 0.60000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(1) = (25.0 - (-1.0)*0.00000000 - (-1.0)*0.00000000 - (3.0)*0.00000000)/11.0 = 2.27272727
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(1) = (-11.0 - (2.0)*0.00000000 - (-1.0)*0.00000000 - (-1.0)*0.00000000)/10.0 = -1.10000000
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(1) = (15.0 - (0.0)*0.00000000 - (2.0)*0.00000000 - (-1.0)*0.00000000)/8.0 = 1.87500000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.6
x2 <- x_old[2]; (x2)

## [1] 2.272727
x3 <- x_old[3]; (x3)

## [1] -1.1
x4 <- x_old[4]; (x4)

## [1] 1.875
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(1) = 0.60000000, 2.27272727, -1.10000000, 1.87500000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.12500000e+00

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.7.1.2. Iteración 2

```

## x1^(2) = (6.0 - (-1.0)*2.27272727 - (2.0)*-1.10000000 - (0.0)*1.87500000)/10.0 = 1.04727273
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(2) = (25.0 - (-1.0)*0.60000000 - (-1.0)*-1.10000000 - (3.0)*1.87500000)/11.0 = 1.71590909

```

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(2) = (-11.0 - (2.0)*0.60000000 - (-1.0)*2.27272727 - (-1.0)*1.87500000)/10.0 = -0.80522727
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(2) = (15.0 - (0.0)*0.60000000 - (2.0)*2.27272727 - (-1.0)*1.87500000)/8.0 = 1.16931818
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1.047273
x2 <- x_old[2]; (x2)

## [1] 1.715909
x3 <- x_old[3]; (x3)

## [1] -0.8052273
x4 <- x_old[4]; (x4)

## [1] 1.169318
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(2) = 1.04727273, 1.71590909, -0.80522727, 1.16931818
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.85909091e+00

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.7.1.3. Iteración 3

```

## x1^(3) = (6.0 - (-1.0)*1.71590909 - (2.0)*-0.80522727 - (0.0)*1.16931818)/10.0 = 0.93263636
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(3) = (25.0 - (-1.0)*1.04727273 - (-1.0)*-0.80522727 - (3.0)*1.16931818)/11.0 = 1.97582645
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(3) = (-11.0 - (2.0)*1.04727273 - (-1.0)*1.71590909 - (-1.0)*1.16931818)/10.0 = -1.02093182

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(3) = (15.0 - (0.0)*1.04727273 - (2.0)*1.71590909 - (-1.0)*1.16931818)/8.0 = 1.34536932
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9326364
x2 <- x_old[2]; (x2)

## [1] 1.975826
x3 <- x_old[3]; (x3)

## [1] -1.020932
x4 <- x_old[4]; (x4)

## [1] 1.345369
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=",")))

## x^(3) = 0.93263636, 1.97582645, -1.02093182, 1.34536932
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.58494318e-01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.7.1.4. Iteración 4

```

## x1^(4) = (6.0 - (-1.0)*1.97582645 - (2.0)*-1.02093182 - (0.0)*1.34536932)/10.0 = 1.00176901
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(4) = (25.0 - (-1.0)*0.93263636 - (-1.0)*-1.02093182 - (3.0)*1.34536932)/11.0 = 1.89778151
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(4) = (-11.0 - (2.0)*0.93263636 - (-1.0)*1.97582645 - (-1.0)*1.34536932)/10.0 = -0.95440770
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(4) = (15.0 - (0.0)*0.93263636 - (2.0)*1.97582645 - (-1.0)*1.34536932)/8.0 = 1.25342691

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 1.001769
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.897782
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9544077
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.253427
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(4) = 1.00176901, 1.89778151, -0.95440770, 1.25342691
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.11483988e-01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.7.1.5. Iteración 5

```
## x1^(5) = (6.0 - (-1.0)*1.89778151 - (2.0)*-0.95440770 - (0.0)*1.25342691)/10.0 = 0.98065969
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(5) = (25.0 - (-1.0)*1.00176901 - (-1.0)*-0.95440770 - (3.0)*1.25342691)/11.0 = 1.93518914
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(5) = (-11.0 - (2.0)*1.00176901 - (-1.0)*1.89778151 - (-1.0)*1.25342691)/10.0 = -0.98523296
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(5) = (15.0 - (0.0)*1.00176901 - (2.0)*1.89778151 - (-1.0)*1.25342691)/8.0 = 1.28125366
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9806597
```

```

x2 <- x_old[2]; (x2)

## [1] 1.935189
x3 <- x_old[3]; (x3)

## [1] -0.985233
x4 <- x_old[4]; (x4)

## [1] 1.281254
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = 0.98065969, 1.93518914, -0.98523296, 1.28125366
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.35414831e-01

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.7.1.6. Iteración 6

```

## x1^(6) = (6.0 - (-1.0)*1.93518914 - (2.0)*-0.98523296 - (0.0)*1.28125366)/10.0 = 0.99056551
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(6) = (25.0 - (-1.0)*0.98065969 - (-1.0)*-0.98523296 - (3.0)*1.28125366)/11.0 = 1.92287870
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(6) = (-11.0 - (2.0)*0.98065969 - (-1.0)*1.93518914 - (-1.0)*1.28125366)/10.0 = -0.97448766
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(6) = (15.0 - (0.0)*0.98065969 - (2.0)*1.93518914 - (-1.0)*1.28125366)/8.0 = 1.26804859
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```

## [1] 0.9905655
x2 <- x_old[2]; (x2)

## [1] 1.922879
x3 <- x_old[3]; (x3)

```

```
## [1] -0.9744877
x4 <- x_old[4]; (x4)
```

```
## [1] 1.268049
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(6) = 0.99056551, 1.92287870, -0.97448766, 1.26804859
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.02663186e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.7.1.7. Iteración 7

```
## x1^(7) = (6.0 - (-1.0)*1.92287870 - (2.0)*-0.97448766 - (0.0)*1.26804859)/10.0 = 0.98718540
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(7) = (25.0 - (-1.0)*0.99056551 - (-1.0)*-0.97448766 - (3.0)*1.26804859)/11.0 = 1.92835746
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(7) = (-11.0 - (2.0)*0.99056551 - (-1.0)*1.92287870 - (-1.0)*1.26804859)/10.0 = -0.97902037
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(7) = (15.0 - (0.0)*0.99056551 - (2.0)*1.92287870 - (-1.0)*1.26804859)/8.0 = 1.27246937
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9871854
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.928357
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9790204
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.272469
```



```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = 0.98718540, 1.92835746, -0.97902037, 1.27246937
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.11751359e-02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.7.1.8. Iteración 8

```
## x1^(8) = (6.0 - (-1.0)*1.92835746 - (2.0)*-0.97902037 - (0.0)*1.27246937)/10.0 = 0.98863982
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(8) = (25.0 - (-1.0)*0.98718540 - (-1.0)*-0.97902037 - (3.0)*1.27246937)/11.0 = 1.92643245
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(8) = (-11.0 - (2.0)*0.98718540 - (-1.0)*1.92835746 - (-1.0)*1.27246937)/10.0 = -0.97735440
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(8) = (15.0 - (0.0)*0.98718540 - (2.0)*1.92835746 - (-1.0)*1.27246937)/8.0 = 1.27053309
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9886398
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.926432
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9773544
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.270533
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(8) = 0.98863982, 1.92643245, -0.97735440, 1.27053309
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.92922700e-03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.7.1.9. Iteración 9

```
## x1^(9) = (6.0 - (-1.0)*1.92643245 - (2.0)*-0.97735440 - (0.0)*1.27053309)/10.0 = 0.98811412
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(9) = (25.0 - (-1.0)*0.98863982 - (-1.0)*-0.97735440 - (3.0)*1.27053309)/11.0 = 1.92724420
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(9) = (-11.0 - (2.0)*0.98863982 - (-1.0)*1.92643245 - (-1.0)*1.27053309)/10.0 = -0.97803141
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(9) = (15.0 - (0.0)*0.98863982 - (2.0)*1.92643245 - (-1.0)*1.27053309)/8.0 = 1.27122259
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9881141
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927244
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9780314
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.271223
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = 0.98811412, 1.92724420, -0.97803141, 1.27122259
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.27120814e-03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.10. Iteración 10

```
## x1^(10) = (6.0 - (-1.0)*1.92724420 - (2.0)*-0.97803141 - (0.0)*1.27122259)/10.0 = 0.98833070
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(10) = (25.0 - (-1.0)*0.98811412 - (-1.0)*-0.97803141 - (3.0)*1.27122259)/11.0 = 1.92694681
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(10) = (-11.0 - (2.0)*0.98811412 - (-1.0)*1.92724420 - (-1.0)*1.27122259)/10.0 = -0.97777615
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(10) = (15.0 - (0.0)*0.98811412 - (2.0)*1.92724420 - (-1.0)*1.27122259)/8.0 = 1.27093502
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9883307
x2 <- x_old[2]; (x2)
```

```
## [1] 1.926947
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9777761
x4 <- x_old[4]; (x4)
```

```
## [1] 1.270935
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = 0.98833070, 1.92694681, -0.97777615, 1.27093502
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.33453202e-03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.11. Iteración 11

```
## x1^(11) = (6.0 - (-1.0)*1.92694681 - (2.0)*-0.97777615 - (0.0)*1.27093502)/10.0 = 0.98824991
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(11) = (25.0 - (-1.0)*0.98833070 - (-1.0)*-0.97777615 - (3.0)*1.27093502)/11.0 = 1.92706813
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(11) = (-11.0 - (2.0)*0.98833070 - (-1.0)*1.92694681 - (-1.0)*1.27093502)/10.0 = -0.97787796
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(11) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(11) = (15.0 - (0.0)*0.98833070 - (2.0)*1.92694681 - (-1.0)*1.27093502)/8.0 = 1.27104128
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882499
x2 <- x_old[2]; (x2)

## [1] 1.927068
x3 <- x_old[3]; (x3)

## [1] -0.977878
x4 <- x_old[4]; (x4)

## [1] 1.271041
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(11) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(11) = 0.98824991, 1.92706813, -0.97787796, 1.27104128
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.01362037e-04
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.12. Iteración 12

```
## x1^(12) = (6.0 - (-1.0)*1.92706813 - (2.0)*-0.97787796 - (0.0)*1.27104128)/10.0 = 0.98828240
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```

##  $x_2^{(12)} = (25.0 - (-1.0)*0.98824991 - (-1.0)*-0.97787796 - (3.0)*1.27104128)/11.0 = 1.92702256$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
##  $x_3^{(12)} = (-11.0 - (2.0)*0.98824991 - (-1.0)*1.92706813 - (-1.0)*1.27104128)/10.0 = -0.97783904$ 
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(12) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
##  $x_4^{(12)} = (15.0 - (0.0)*0.98824991 - (2.0)*1.92706813 - (-1.0)*1.27104128)/8.0 = 1.27099822$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882824
x2 <- x_old[2]; (x2)

## [1] 1.927023
x3 <- x_old[3]; (x3)

## [1] -0.977839
x4 <- x_old[4]; (x4)

## [1] 1.270998
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(12) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(12)} = 0.98828240, 1.92702256, -0.97783904, 1.27099822$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.00579408e-04

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.13. Iteración 13

```

##  $x_1^{(13)} = (6.0 - (-1.0)*1.92702256 - (2.0)*-0.97783904 - (0.0)*1.27099822)/10.0 = 0.98827006$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
##  $x_2^{(13)} = (25.0 - (-1.0)*0.98828240 - (-1.0)*-0.97783904 - (3.0)*1.27099822)/11.0 = 1.92704079$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
##  $x_3^{(13)} = (-11.0 - (2.0)*0.98828240 - (-1.0)*1.92702256 - (-1.0)*1.27099822)/10.0 = -0.97785440$ 

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(13) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(13) = (15.0 - (0.0)*0.98828240 - (2.0)*1.92702256 - (-1.0)*1.27099822)/8.0 = 1.27101448
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882701
x2 <- x_old[2]; (x2)

## [1] 1.927041
x3 <- x_old[3]; (x3)

## [1] -0.9778544
x4 <- x_old[4]; (x4)

## [1] 1.271014
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(13) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(13) = 0.98827006, 1.92704079, -0.97785440, 1.27101448
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.64804446e-05

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.14. Iteración 14

```

## x1^(14) = (6.0 - (-1.0)*1.92704079 - (2.0)*-0.97785440 - (0.0)*1.27101448)/10.0 = 0.98827496
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(14) = (25.0 - (-1.0)*0.98827006 - (-1.0)*-0.97785440 - (3.0)*1.27101448)/11.0 = 1.92703384
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(14) = (-11.0 - (2.0)*0.98827006 - (-1.0)*1.92704079 - (-1.0)*1.27101448)/10.0 = -0.97784849
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(14) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(14) = (15.0 - (0.0)*0.98827006 - (2.0)*1.92704079 - (-1.0)*1.27101448)/8.0 = 1.27100800

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.988275
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927034
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778485
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.271008
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(14) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(14) = 0.98827496, 1.92703384, -0.97784849, 1.27100800
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.02501981e-05
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(15) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.15. Iteración 15

```
## x1^(15) = (6.0 - (-1.0)*1.92703384 - (2.0)*-0.97784849 - (0.0)*1.27100800)/10.0 = 0.98827308
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(15) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(15) = (25.0 - (-1.0)*0.98827496 - (-1.0)*-0.97784849 - (3.0)*1.27100800)/11.0 = 1.92703659
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(15) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(15) = (-11.0 - (2.0)*0.98827496 - (-1.0)*1.92703384 - (-1.0)*1.27100800)/10.0 = -0.97785081
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(15) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(15) = (15.0 - (0.0)*0.98827496 - (2.0)*1.92703384 - (-1.0)*1.27100800)/8.0 = 1.27101048
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882731
```

```

x2 <- x_old[2]; (x2)

## [1] 1.927037
x3 <- x_old[3]; (x3)

## [1] -0.9778508
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(15) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(15) = 0.98827308, 1.92703659, -0.97785081, 1.27101048
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.16347937e-05

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.16. Iteración 16

```

## x1^(16) = (6.0 - (-1.0)*1.92703659 - (2.0)*-0.97785081 - (0.0)*1.27101048)/10.0 = 0.98827382
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_

```

x2^(16) = (25.0 - (-1.0)*0.98827308 - (-1.0)*-0.97785081 - (3.0)*1.27101048)/11.0 = 1.92703553

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_

```

x3^(16) = (-11.0 - (2.0)*0.98827308 - (-1.0)*1.92703659 - (-1.0)*1.27101048)/10.0 = -0.97784991

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(16) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_

```

x4^(16) = (15.0 - (0.0)*0.98827308 - (2.0)*1.92703659 - (-1.0)*1.27101048)/8.0 = 1.27100950

```

# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882738
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

```



```
## [1] -0.9778499
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(16) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(16) = 0.98827382, 1.92703553, -0.97784991, 1.27100950
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.57141448e-06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(17) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.17. Iteración 17

```
## x1^(17) = (6.0 - (-1.0)*1.92703553 - (2.0)*-0.97784991 - (0.0)*1.27100950)/10.0 = 0.98827353
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(17) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(17) = (25.0 - (-1.0)*0.98827382 - (-1.0)*-0.97784991 - (3.0)*1.27100950)/11.0 = 1.92703595
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(17) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(17) = (-11.0 - (2.0)*0.98827382 - (-1.0)*1.92703553 - (-1.0)*1.27100950)/10.0 = -0.97785026
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(17) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(17) = (15.0 - (0.0)*0.98827382 - (2.0)*1.92703553 - (-1.0)*1.27100950)/8.0 = 1.27100988
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882735
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778503
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(17) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(17) = 0.98827353, 1.92703595, -0.97785026, 1.27100988
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.76716221e-06
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.18. Iteración 18

```
## x1^(18) = (6.0 - (-1.0)*1.92703595 - (2.0)*-0.97785026 - (0.0)*1.27100988)/10.0 = 0.98827365
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(18) = (25.0 - (-1.0)*0.98827353 - (-1.0)*-0.97785026 - (3.0)*1.27100988)/11.0 = 1.92703579
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(18) = (-11.0 - (2.0)*0.98827353 - (-1.0)*1.92703595 - (-1.0)*1.27100988)/10.0 = -0.97785012
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(18) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(18) = (15.0 - (0.0)*0.98827353 - (2.0)*1.92703595 - (-1.0)*1.27100988)/8.0 = 1.27100973
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778501
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(18) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(18) = 0.98827365, 1.92703579, -0.97785012, 1.27100973
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.91655519e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.19. Iteración 19

```
## x1^(19) = (6.0 - (-1.0)*1.92703579 - (2.0)*-0.97785012 - (0.0)*1.27100973)/10.0 = 0.98827360
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(19) = (25.0 - (-1.0)*0.98827365 - (-1.0)*-0.97785012 - (3.0)*1.27100973)/11.0 = 1.92703585
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(19) = (-11.0 - (2.0)*0.98827365 - (-1.0)*1.92703579 - (-1.0)*1.27100973)/10.0 = -0.97785018
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(19) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(19) = (15.0 - (0.0)*0.98827365 - (2.0)*1.92703579 - (-1.0)*1.27100973)/8.0 = 1.27100979
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(19) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(19) = 0.98827360, 1.92703585, -0.97785018, 1.27100979
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.68159805e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.20. Iteración 20

```
## x1^(20) = (6.0 - (-1.0)*1.92703585 - (2.0)*-0.97785018 - (0.0)*1.27100979)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(20) = (25.0 - (-1.0)*0.98827360 - (-1.0)*-0.97785018 - (3.0)*1.27100979)/11.0 = 1.92703582
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(20) = (-11.0 - (2.0)*0.98827360 - (-1.0)*1.92703585 - (-1.0)*1.27100979)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(20) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(20) = (15.0 - (0.0)*0.98827360 - (2.0)*1.92703585 - (-1.0)*1.27100979)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(20) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(20) = 0.98827362, 1.92703582, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.04720357e-07
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.21. Iteración 21

```
## x1^(21) = (6.0 - (-1.0)*1.92703582 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827361
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(21) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(21) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703582 - (-1.0)*1.27100977)/10.0 = -0.97785017
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(21) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(21) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703582 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(21) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(21) = 0.98827361, 1.92703583, -0.97785017, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.06704110e-08
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.22. Iteración 22

```
## x1^(22) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785017 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```

## x2^(22) = (25.0 - (-1.0)*0.98827361 - (-1.0)*-0.97785017 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(22) = (-11.0 - (2.0)*0.98827361 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(22) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(22) = (15.0 - (0.0)*0.98827361 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(22) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(22) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.58616587e-08

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.23. Iteración 23

```

## x1^(23) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(23) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(23) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(23) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(23) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(23) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(23) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.16635631e-09

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.24. Iteración 24

```

## x1^(24) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(24) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(24) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(24) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(24) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(24) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(24) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.40308040e-09
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.25. Iteración 25

```
## x1^(25) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(25) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(25) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(25) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(25) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```



```

x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(25) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(25) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 9.34761601e-10

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(26) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.26. Iteración 26

```

## x1^(26) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(26) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_

## x2^(26) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(26) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_

## x3^(26) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(26) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_

## x4^(26) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

```

```
## [1] -0.9778502
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(26) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(26) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.64124730e-10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(27) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.27. Iteración 27

```
## x1^(27) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(27) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(27) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(27) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(27) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(27) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(27) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(27) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(27) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.41689327e-10
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(28) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.28. Iteración 28

```
## x1^(28) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(28) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(28) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(28) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(28) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(28) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(28) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(28) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(28) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.51771961e-11
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(29) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.29. Iteración 29

```
## x1^(29) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(29) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(29) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(29) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(29) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(29) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(29) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(29) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(29) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.14726015e-11
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(30) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.30. Iteración 30

```
## x1^(30) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(30) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(30) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(30) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(30) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(30) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(30) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(30) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(30) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.35953529e-12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(31) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.31. Iteración 31

```
## x1^(31) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(31) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(31) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(31) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(31) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(31) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(31) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(31) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(31) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.25073302e-12
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(32) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.32. Iteración 32

```
## x1^(32) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(32) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```

##  $x_2^{(32)} = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(32) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
##  $x_3^{(32)} = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016$ 
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(32) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
##  $x_4^{(32)} = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(32) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(32)} = 0.98827362, 1.92703583, -0.97785016, 1.27100977$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.27187150e-12

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3] - A[1,4]*x_old[4]) / A[1,1]
cat(sprintf("x1^(33) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.33. Iteración 33

```

##  $x_1^{(33)} = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(33) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
##  $x_2^{(33)} = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(33) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
##  $x_3^{(33)} = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016$ 

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(33) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(33) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(33) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(33) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.97379915e-13

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(34) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.34. Iteración 34

```

## x1^(34) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(34) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(34) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(34) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(34) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(34) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(34) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977

```



```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(34) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(34) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.91846539e-13
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(35) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.35. Iteración 35

```
## x1^(35) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(35) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(35) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(35) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(35) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(35) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(35) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```

x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(35) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(35) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.46069873e-14

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(36) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
2.3.2.7.1.36. Iteración 36

## x1^(36) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(36) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_

## x2^(36) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(36) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_

## x3^(36) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(36) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_

## x4^(36) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

```

```
## [1] -0.9778502
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(36) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(36) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.48689958e-14
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(37) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.37. Iteración 37

```
## x1^(37) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(37) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(37) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(37) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(37) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(37) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(37) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(37) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(37) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 7.10542736e-15
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(38) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.38. Iteración 38

```
## x1^(38) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(38) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(38) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(38) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(38) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(38) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(38) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(38) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(38) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.55271368e-15
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(39) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.39. Iteración 39

```
## x1^(39) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(39) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(39) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(39) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(39) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(39) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(39) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(39) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(39) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.88178420e-16
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(40) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.40. Iteración 40

```
## x1^(40) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(40) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(40) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(40) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
```

```
## x3^(40) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(40) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
```

```
## x4^(40) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(40) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(40) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(41) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.41. Iteración 41

```
## x1^(41) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(41) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(41) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(41) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(41) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(41) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(41) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(41) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(41) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(42) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.42. Iteración 42

```
## x1^(42) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(42) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
```

```
## x2^(42) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(42) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(42) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(42) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(42) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(42) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(42) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(43) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.43. Iteración 43

```
## x1^(43) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(43) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(43) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(43) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(43) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```



```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(43) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(43) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(43) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(43) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(44) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

```

2.3.2.7.1.44. Iteración 44

```

## x1^(44) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(44) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(44) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(44) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(44) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(44) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(44) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(44) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(44) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(45) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.45. Iteración 45

```
## x1^(45) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(45) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(45) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(45) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(45) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(45) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(45) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```

x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(45) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(45) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(46) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
2.3.2.7.1.46. Iteración 46

## x1^(46) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(46) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_

## x2^(46) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(46) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_

## x3^(46) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(46) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_

## x4^(46) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

```

```
## [1] -0.9778502
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(46) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(46) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(47) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.47. Iteración 47

```
## x1^(47) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(47) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(47) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(47) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(47) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(47) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(47) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(47) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(47) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(48) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.48. Iteración 48

```
## x1^(48) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(48) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(48) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(48) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(48) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(48) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(48) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(48) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(48) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(49) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_
```

2.3.2.7.1.49. Iteración 49

```
## x1^(49) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(49) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_
```

```
## x2^(49) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(49) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_
```

```
## x3^(49) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(49) = ( %.1f - ( %.1f)*%.8f - ( %.1f)*%.8f - ( %.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_
```

```
## x4^(49) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 0.9882736
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 1.927036
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.9778502
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.27101
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(49) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(49) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(50) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[1], A[1,2], x_

2.3.2.7.1.50. Iteración 50

## x1^(50) = (6.0 - (-1.0)*1.92703583 - (2.0)*-0.97785016 - (0.0)*1.27100977)/10.0 = 0.98827362
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(50) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_

## x2^(50) = (25.0 - (-1.0)*0.98827362 - (-1.0)*-0.97785016 - (3.0)*1.27100977)/11.0 = 1.92703583
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(50) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_

## x3^(50) = (-11.0 - (2.0)*0.98827362 - (-1.0)*1.92703583 - (-1.0)*1.27100977)/10.0 = -0.97785016
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(50) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_

## x4^(50) = (15.0 - (0.0)*0.98827362 - (2.0)*1.92703583 - (-1.0)*1.27100977)/8.0 = 1.27100977
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.9882736
x2 <- x_old[2]; (x2)

## [1] 1.927036
x3 <- x_old[3]; (x3)

## [1] -0.9778502
x4 <- x_old[4]; (x4)

## [1] 1.27101
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(50) =%s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(50) = 0.98827362, 1.92703583, -0.97785016, 1.27100977
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
# Comprobación

I <- b[1]; (I)

## [1] 6

```

```

II <- b[2]; (II)

## [1] 25
III <- b[3]; (III)

## [1] -11
IV <- b[4]; (IV)

## [1] 15
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3) + (A[1,4] * x4); paste0("El resultado es: ",c1)

## [1] "El resultado es: 6"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3) + (A[2,4] * x4); paste0("El resultado es: ",c2)

## [1] "El resultado es: 25"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3) + (A[3,4] * x4); paste0("El resultado es: ",c3)

## [1] "El resultado es: -11"
c4<- (A[4,1] * x1) + (A[4,2] * x2) + (A[4,3] * x3) + (A[4,4] * x4); paste0("El resultado es: ",c4)

## [1] "El resultado es: 15"
paste0("En la ecuación 1: ", c1, " tecnicamente es: ", I)

## [1] "En la ecuación 1: 6 tecnicamente es: 6"
paste0("En la ecuación 2: ", c2, " tecnicamente es: ", II)

## [1] "En la ecuación 2: 25 tecnicamente es: 25"
paste0("En la ecuación 3: ", c3, " tecnicamente es: ", III)

## [1] "En la ecuación 3: -11 tecnicamente es: -11"
paste0("En la ecuación 3: ", c4, " tecnicamente es: ", IV)

## [1] "En la ecuación 3: 15 tecnicamente es: 15"

cat("Soluciones del sistema:")

```

2.3.2.7.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 0.9882736
```

```
cat("x2 =", x2)
```

```
## x2 = 1.927036
```

```
cat("x3 =", x3)
```

```
## x3 = -0.9778502
```



```
cat("x4 =", x4)
```

```
## x4 = 1.27101
```

2.3.2.8. Sistema 7

$$x_1 + 2x_2 - 2x_3 = 7 \quad x_1 + x_2 + x_3 = 2 \quad 2x_1 + 2x_2 + x_3 = 5$$

```
A <- matrix(c(1, 2, -2,
              1, 1, 1,
              2, 2, 1),
            nrow = 3, byrow = TRUE)
b <- matrix(c(7, 2, 5), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.3.2.8.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    2   -2    7
## [2,]    1    1    1    2
## [3,]    2    2    1    5
```

```
# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## No es dominante
```

La matriz no tiene diagonal dominante pero se hacen 10 iteraciones para ver si puede llegar a converger.

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.8.1.1. Iteración 1

```
## x1^(1) = (7.0 - (2.0)*0.00000000 - (-2.0)*0.00000000 / 1.0 = 7.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(1) = (2.0 - (1.0)*0.00000000 - (1.0)*0.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(1) = (5.0 - (2.0)*0.00000000 - (2.0)*0.00000000 / 1.0 = 5.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 7
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] 5
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(1) = 7.00000000, 2.00000000, 5.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.80000000e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.8.1.2. Iteración 2

```

## x1^(2) = (7.0 - (2.0)*2.00000000 - (-2.0)*5.00000000 / 1.0 = 13.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(2) = (2.0 - (1.0)*7.00000000 - (1.0)*5.00000000 / 1.0 = -10.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(2) = (5.0 - (2.0)*7.00000000 - (2.0)*2.00000000 / 1.0 = -13.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] 13
x2 <- x_old[2]; (x2)

## [1] -10
x3 <- x_old[3]; (x3)

## [1] -13
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(2) = 13.00000000, -10.00000000, -13.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.20000000e+01

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

2.3.2.8.1.3. Iteración 3

## x1^(3) = (7.0 - (2.0)*-10.00000000 - (-2.0)*-13.00000000 / 1.0 = 1.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(3) = (2.0 - (1.0)*13.00000000 - (1.0)*-13.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(3) = (5.0 - (2.0)*13.00000000 - (2.0)*-10.00000000 / 1.0 = -1.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(3) = 1.00000000, 2.00000000, -1.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.8.1.4. Iteración 4

```
## x1^(4) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(4) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(4) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 1
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -1
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(4) = 1.00000000, 2.00000000, -1.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.8.1.5. Iteración 5

```
## x1^(5) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(5) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(5) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(5) = 1.00000000, 2.00000000, -1.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.8.1.6. Iteración 6

```

## x1^(6) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(6) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(6) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(6) = 1.00000000, 2.00000000, -1.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

2.3.2.8.1.7. Iteración 7

## x1^(7) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(7) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(7) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(7) = 1.00000000, 2.00000000, -1.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.8.1.8. Iteración 8

```
## x1^(8) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(8) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(8) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] 1
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -1
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(8) = 1.00000000, 2.00000000, -1.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.8.1.9. Iteración 9

```
## x1^(9) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(9) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(9) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(9) = 1.00000000, 2.00000000, -1.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.8.1.10. Iteración 10

```

## x1^(10) = (7.0 - (2.0)*2.00000000 - (-2.0)*-1.00000000 / 1.0 = 1.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(10) = (2.0 - (1.0)*1.00000000 - (1.0)*-1.00000000 / 1.0 = 2.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(10) = (5.0 - (2.0)*1.00000000 - (2.0)*2.00000000 / 1.0 = -1.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```



```

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(10) = 1.00000000, 2.00000000, -1.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 0.00000000e+00
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 2
x3 <- x_old[3]; (x3)

## [1] -1
# Comprobación

I <- b[1]; (I)

## [1] 7
II <- b[2]; (II)

## [1] 2
III <- b[3]; (III)

## [1] 5
c1 <- (A[1,1] * x1) + (A[1,2] * x2) + (A[1,3] * x3); paste0("El resultado es: ",c1)

## [1] "El resultado es: 7"
c2 <- (A[2,1] * x1) + (A[2,2] * x2) + (A[2,3] * x3); paste0("El resultado es: ",c2)

## [1] "El resultado es: 2"
c3 <- (A[3,1] * x1) + (A[3,2] * x2) + (A[3,3] * x3); paste0("El resultado es: ",c3)

## [1] "El resultado es: 5"

```

```
paste0("En la ecuación 1: ", c1, " técnicamente es: ", I)

## [1] "En la ecuación 1: 7 técnicamente es: 7"
paste0("En la ecuación 2: ", c2, " técnicamente es: ", II)

## [1] "En la ecuación 2: 2 técnicamente es: 2"
paste0("En la ecuación 3: ", c3, " técnicamente es: ", III)

## [1] "En la ecuación 3: 5 técnicamente es: 5"
```

```
cat("Soluciones del sistema:")
```

2.3.2.8.2. Resultados

```
## Soluciones del sistema:
```

```
cat("x1 =", x1)
```

```
## x1 = 1
```

```
cat("x2 =", x2)
```

```
## x2 = 2
```

```
cat("x3 =", x3)
```

```
## x3 = -1
```

2.3.2.9. Sistema 8

$$-4x_1 + 14x_2 = 10 \quad -5x_1 + 13x_2 = 8 \quad -x_1 + 2x_3 = 1$$

```
A <- matrix(c(-4, 14, 0,
              -5, 13, 0,
              -1, 0, 2),
            nrow = 3, byrow = TRUE)
b <- matrix(c(10, 8, 1), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.3.2.9.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]  -4   14   0   10
## [2,]  -5   13   0    8
## [3,]  -1    0   2    1

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}

```

No es dominante

La matriz no tiene diagonal dominante pero se hacen 10 iteraciones para ver si puede llegar a converger.

Asignar valores iniciales

```

x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))

```

x^(0)= 0.00000000, 0.00000000, 0.00000000

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.9.1.1. Iteración 1

x1^(1) = (10.0 - (14.0)*0.00000000 - (0.0)*0.00000000 / -4.0 = -2.50000000

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

```

x2^(1) = (8.0 - (-5.0)*0.00000000 - (0.0)*0.00000000 / 13.0 = 0.61538462

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

```

x3^(1) = (1.0 - (-1.0)*0.00000000 - (0.0)*0.00000000 / 2.0 = 0.50000000

Imprime los valores de la iteración

```

x_old <- x
x1 <- x_old[1]; (x1)

```

[1] -2.5

```

x2 <- x_old[2]; (x2)

```

[1] 0.6153846

```

x3 <- x_old[3]; (x3)

```

[1] 0.5

#Error o tolerancia

```

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

```

x^(1) = -2.50000000, 0.61538462, 0.50000000

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.25000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.9.1.2. Iteración 2

```
## x1^(2) = (10.0 - (14.0)*0.61538462 - (0.0)*0.50000000 / -4.0 = -0.34615385
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(2) = (8.0 - (-5.0)*-2.50000000 - (0.0)*0.50000000 / 13.0 = -0.34615385
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(2) = (1.0 - (-1.0)*-2.50000000 - (0.0)*0.61538462 / 2.0 = -0.75000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -0.3461538
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -0.3461538
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -0.75
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(2) = -0.34615385, -0.34615385, -0.75000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.34615385e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.9.1.3. Iteración 3

```
## x1^(3) = (10.0 - (14.0)*-0.34615385 - (0.0)*-0.75000000 / -4.0 = -3.71153846
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(3) = (8.0 - (-5.0)*-0.34615385 - (0.0)*-0.75000000 / 13.0 = 0.48224852
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(3) = (1.0 - (-1.0)*-0.34615385 - (0.0)*-0.34615385 / 2.0 = 0.32692308
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -3.711538
x2 <- x_old[2]; (x2)

## [1] 0.4822485
x3 <- x_old[3]; (x3)

## [1] 0.3269231
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(3) = -3.71153846, 0.48224852, 0.32692308
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.68269231e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.9.1.4. Iteración 4

```

## x1^(4) = (10.0 - (14.0)*0.48224852 - (0.0)*0.32692308 / -4.0 = -0.81213018
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(4) = (8.0 - (-5.0)*-3.71153846 - (0.0)*0.32692308 / 13.0 = -0.81213018
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(4) = (1.0 - (-1.0)*-3.71153846 - (0.0)*0.48224852 / 2.0 = -1.35576923
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```

```
## [1] -0.8121302
x2 <- x_old[2]; (x2)

## [1] -0.8121302
x3 <- x_old[3]; (x3)

## [1] -1.355769
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(4) = -0.81213018, -0.81213018, -1.35576923
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.81213018e+01

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.9.1.5. Iteración 5

```
## x1^(5) = (10.0 - (14.0)*-0.81213018 - (0.0)*-1.35576923 / -4.0 = -5.34245562
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(5) = (8.0 - (-5.0)*-0.81213018 - (0.0)*-1.35576923 / 13.0 = 0.30302685
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(5) = (1.0 - (-1.0)*-0.81213018 - (0.0)*-0.81213018 / 2.0 = 0.09393491
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -5.342456
x2 <- x_old[2]; (x2)

## [1] 0.3030269
x3 <- x_old[3]; (x3)

## [1] 0.09393491
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(5) = -5.34245562, 0.30302685, 0.09393491
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.26516272e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.9.1.6. Iteración 6

```
## x1^(6) = (10.0 - (14.0)*0.30302685 - (0.0)*0.09393491 / -4.0 = -1.43940601
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(6) = (8.0 - (-5.0)*-5.34245562 - (0.0)*0.09393491 / 13.0 = -1.43940601
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(6) = (1.0 - (-1.0)*-5.34245562 - (0.0)*0.30302685 / 2.0 = -2.17122781
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -1.439406
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -1.439406
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -2.171228
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(6) = -1.43940601, -1.43940601, -2.17122781
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.43940601e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.9.1.7. Iteración 7

```
## x1^(7) = (10.0 - (14.0)*-1.43940601 - (0.0)*-2.17122781 / -4.0 = -7.53792103
```

```

x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(7) = (8.0 - (-5.0)*-1.43940601 - (0.0)*-2.17122781 / 13.0 = 0.06176692
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(7) = (1.0 - (-1.0)*-1.43940601 - (0.0)*-1.43940601 / 2.0 = -0.21970300
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -7.537921
x2 <- x_old[2]; (x2)

## [1] 0.06176692
x3 <- x_old[3]; (x3)

## [1] -0.219703
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(7) = -7.53792103, 0.06176692, -0.21970300
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.04925751e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.9.1.8. Iteración 8

```

## x1^(8) = (10.0 - (14.0)*0.06176692 - (0.0)*-0.21970300 / -4.0 = -2.28381578
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
## x2^(8) = (8.0 - (-5.0)*-7.53792103 - (0.0)*-0.21970300 / 13.0 = -2.28381578
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(8) = (1.0 - (-1.0)*-7.53792103 - (0.0)*0.06176692 / 2.0 = -3.26896051
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

```



```
## [1] -2.283816
x2 <- x_old[2]; (x2)

## [1] -2.283816
x3 <- x_old[3]; (x3)

## [1] -3.268961
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = -2.28381578, -2.28381578, -3.26896051
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.28381578e+01

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

2.3.2.9.1.9. Iteración 9

## x1^(9) = (10.0 - (14.0)*-2.28381578 - (0.0)*-3.26896051 / -4.0 = -10.49335523
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(9) = (8.0 - (-5.0)*-2.28381578 - (0.0)*-3.26896051 / 13.0 = -0.26300607
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(9) = (1.0 - (-1.0)*-2.28381578 - (0.0)*-2.28381578 / 2.0 = -0.64190789
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -10.49336
x2 <- x_old[2]; (x2)

## [1] -0.2630061
x3 <- x_old[3]; (x3)

## [1] -0.6419079
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(9) = -10.49335523, -0.26300607, -0.64190789
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.10476973e+01
```

```
x_old <- x
```

```
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
```

```
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.9.1.10. Iteración 10

```
## x1^(10) = (10.0 - (14.0)*-0.26300607 - (0.0)*-0.64190789 / -4.0 = -3.42052124
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
```

```
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(10) = (8.0 - (-5.0)*-10.49335523 - (0.0)*-0.64190789 / 13.0 = -3.42052124
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
```

```
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(10) = (1.0 - (-1.0)*-10.49335523 - (0.0)*-0.26300607 / 2.0 = -4.74667762
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] -3.420521
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -3.420521
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -4.746678
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
```

```
dx_inf <- max(abs(x - x_old))
```

```
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(10) = -3.42052124, -3.42052124, -4.74667762
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.42052124e+01
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
```

```
x1 <- x_old[1]; (x1)
```

```
## [1] -3.420521
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -3.420521
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -4.746678
```

2.3.2.9.2. Resultados El sistema no llega a converger, diverge.

2.3.2.10. Sistema 9

$$x + y + 2z = 1 \quad x + 2y + z = 1 \quad 2x + y + z = 1$$

```
A <- matrix(c(1, 1, 2,
              1, 2, 1,
              2, 1, 1),
            nrow = 3, byrow = TRUE)
b <- matrix(c(1, 1, 1), nrow = 3)

Ab <- cbind(A, b); (Ab)
```

2.3.2.10.1. Procedimiento

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    2    1
## [2,]    1    2    1    1
## [3,]    2    1    1    1

# Checar si converge o no con la diagonal dominante
```

```
F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]))
```

```
if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

```
## No es dominante
```

La matriz no tiene diagonal dominante pero se hacen 10 iteraciones para ver si puede llegar a converger.

```
# Asignar valores iniciales
```

```
x <- rep(0, 3) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.1. Iteración 1

```
##  $x_1^{(1)} = (1.0 - (1.0)*0.00000000 - (2.0)*0.00000000) / 1.0 = 1.00000000$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
x[2]), "\n")

##  $x_2^{(1)} = (1.0 - (1.0)*0.00000000 - (1.0)*0.00000000) / 2.0 = 0.50000000$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
x[3]), "\n")

##  $x_3^{(1)} = (1.0 - (2.0)*0.00000000 - (1.0)*0.00000000) / 1.0 = 1.00000000$ 
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1
x2 <- x_old[2]; (x2)

## [1] 0.5
x3 <- x_old[3]; (x3)

## [1] 1
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(1)} = 1.00000000, 0.50000000, 1.00000000$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.50000000e+00

x_old <- x
x[1] <- (b[1] - A[1,2]*x_old[2] - A[1,3]*x_old[3]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
x[1]), "\n")
```

2.3.2.10.1.2. Iteración 2

```
##  $x_1^{(2)} = (1.0 - (1.0)*0.50000000 - (2.0)*1.00000000) / 1.0 = -1.50000000$ 
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
x[2]), "\n")

##  $x_2^{(2)} = (1.0 - (1.0)*1.00000000 - (1.0)*1.00000000) / 2.0 = -0.50000000$ 
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
x[3]), "\n")

##  $x_3^{(2)} = (1.0 - (2.0)*1.00000000 - (1.0)*0.50000000) / 1.0 = -1.50000000$ 
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -1.5
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -0.5
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -1.5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(2) = -1.50000000, -0.50000000, -1.50000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.00000000e+00
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.3. Iteración 3

```
## x1^(3) = (1.0 - (1.0)*-0.50000000 - (2.0)*-1.50000000) / 1.0 = 4.50000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(3) = (1.0 - (1.0)*-1.50000000 - (1.0)*-1.50000000) / 2.0 = 2.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f) / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(3) = (1.0 - (2.0)*-1.50000000 - (1.0)*-0.50000000) / 1.0 = 4.50000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 4.5
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 2
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 4.5
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(3) = 4.50000000, 2.00000000, 4.50000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.45000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.4. Iteración 4

```
## x1^(4) = (1.0 - (1.0)*2.00000000 - (2.0)*4.50000000 / 1.0 = -10.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(4) = (1.0 - (1.0)*4.50000000 - (1.0)*4.50000000 / 2.0 = -4.00000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(4) = (1.0 - (2.0)*4.50000000 - (1.0)*2.00000000 / 1.0 = -10.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -10
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -4
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -10
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(4) = -10.00000000, -4.00000000, -10.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.50000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.5. Iteración 5

```
## x1^(5) = (1.0 - (1.0)*-4.00000000 - (2.0)*-10.00000000 / 1.0 = 25.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(5) = (1.0 - (1.0)*-10.00000000 - (1.0)*-10.00000000 / 2.0 = 10.50000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(5) = (1.0 - (2.0)*-10.00000000 - (1.0)*-4.00000000 / 1.0 = 25.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 25
x2 <- x_old[2]; (x2)
```

```
## [1] 10.5
x3 <- x_old[3]; (x3)
```

```
## [1] 25
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(5) = 25.00000000, 10.50000000, 25.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.45000000e+01
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.6. Iteración 6

```
## x1^(6) = (1.0 - (1.0)*10.50000000 - (2.0)*25.00000000 / 1.0 = -59.50000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(6) = (1.0 - (1.0)*25.00000000 - (1.0)*25.00000000 / 2.0 = -24.50000000
```

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(6) = (1.0 - (2.0)*25.00000000 - (1.0)*10.50000000 / 1.0 = -59.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -59.5
x2 <- x_old[2]; (x2)

## [1] -24.5
x3 <- x_old[3]; (x3)

## [1] -59.5
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(6) = -59.50000000, -24.50000000, -59.50000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.04000000e+02

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

```

2.3.2.10.1.7. Iteración 7

```

## x1^(7) = (1.0 - (1.0)*-24.50000000 - (2.0)*-59.50000000 / 1.0 = 144.50000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(7) = (1.0 - (1.0)*-59.50000000 - (1.0)*-59.50000000 / 2.0 = 60.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(7) = (1.0 - (2.0)*-59.50000000 - (1.0)*-24.50000000 / 1.0 = 144.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 144.5
x2 <- x_old[2]; (x2)

## [1] 60

```



```

x3 <- x_old[3]; (x3)

## [1] 144.5
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(7) = 144.50000000, 60.00000000, 144.50000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.92500000e+02

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],

2.3.2.10.1.8. Iteración 8

## x1^(8) = (1.0 - (1.0)*60.00000000 - (2.0)*144.50000000 / 1.0 = -348.00000000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],

## x2^(8) = (1.0 - (1.0)*144.50000000 - (1.0)*144.50000000 / 2.0 = -144.00000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],

## x3^(8) = (1.0 - (2.0)*144.50000000 - (1.0)*60.00000000 / 1.0 = -348.00000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -348
x2 <- x_old[2]; (x2)

## [1] -144
x3 <- x_old[3]; (x3)

## [1] -348
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(8) = -348.00000000, -144.00000000, -348.00000000
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.18900000e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.9. Iteración 9

```
## x1^(9) = (1.0 - (1.0)*-144.00000000 - (2.0)*-348.00000000 / 1.0 = 841.00000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(9) = (1.0 - (1.0)*-348.00000000 - (1.0)*-348.00000000 / 2.0 = 348.50000000
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[3], A[3,1], x_old[1], A[3,2],
```

```
## x3^(9) = (1.0 - (2.0)*-348.00000000 - (1.0)*-144.00000000 / 1.0 = 841.00000000
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 841
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 348.5
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 841
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(9) = 841.00000000, 348.50000000, 841.00000000
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.87050000e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[1], A[1,2], x_old[2], A[1,3],
```

2.3.2.10.1.10. Iteración 10

```
## x1^(10) = (1.0 - (1.0)*348.50000000 - (2.0)*841.00000000 / 1.0 = -2029.50000000
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f / %.1f = %.8f\n", b[2], A[2,1], x_old[1], A[2,3],
```

```
## x2^(10) = (1.0 - (1.0)*841.00000000 - (1.0)*841.00000000 /2.0 = -840.50000000
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f /%.1f =%.8f\n", b[3], A[3,1], x_old[1], A[3,2],
## x3^(10) = (1.0 - (2.0)*841.00000000 - (1.0)*348.50000000 /1.0 = -2029.50000000
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -2029.5
x2 <- x_old[2]; (x2)

## [1] -840.5
x3 <- x_old[3]; (x3)

## [1] -2029.5
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) =%s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(10) = -2029.50000000, -840.50000000, -2029.50000000
cat(sprintf("||dx||_inf =%.8e | ||r||_inf =%.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 6.93000000e+03
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -2029.5
x2 <- x_old[2]; (x2)

## [1] -840.5
x3 <- x_old[3]; (x3)

## [1] -2029.5
```

2.3.2.10.2. Resultados El sistema no llega a converger, diverge.

2.3.2.11. Sistema 10

$$6x_1 - 2x_2 + 2x_3 + 4x_4 = 10$$

$$12x_1 - 8x_2 + 6x_3 + 10x_4 = 20$$

$$3x_1 - 13x_2 + 9x_3 + 3x_4 = 2$$

$$-6x_1 + 4x_2 + x_3 - 18x_4 = -19$$

```
A <- matrix(c(6, -2, 2, 4,
              12, -8, 6, 10,
              3, -13, 9, 3,
              -6, 4, 1, -18),
            nrow = 4, byrow = TRUE)
b <- matrix(c(10, 20, 2, -19), nrow = 4)

Ab <- cbind(A, b); (Ab)
```

2.3.2.11.1. Procedimiento

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    6   -2    2    4    10
## [2,]   12   -8    6   10   20
## [3,]    3  -13    9    3     2
## [4,]   -6    4    1  -18  -19

# Checar si converge o no con la diagonal dominante

F1 <- abs(A[1,1]) > (abs(A[1,2]) + abs(A[1,3]) + abs(A[1,4]))
F2 <- abs(A[2,2]) > (abs(A[2,1]) + abs(A[2,3]) + abs(A[2,4]))
F3 <- abs(A[3,3]) > (abs(A[3,1]) + abs(A[3,2]) + abs(A[3,4]))
F4 <- abs(A[4,4]) > (abs(A[4,1]) + abs(A[4,2]) + abs(A[4,3]))

if(F1 == TRUE & F2 == TRUE & F3 == TRUE) {
  cat("Sí es dominante")
} else {
  cat("No es dominante")
}
```

No es dominante

La matriz no tiene diagonal dominante pero se hacen 10 iteraciones para ver si puede llegar a converger.

Asignar valores iniciales

```
x <- rep(0, 4) # x^(0)=(0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" ", "\n"))
```

x^(0)= 0.00000000, 0.00000000, 0.00000000, 0.00000000

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.1. Iteración 1

x1^(1) = (10.0 - (-2.0)*0.00000000 - (2.0)*0.00000000 - (4.0)*0.00000000)/6.0 = 1.66666667

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

x2^(1) = (20.0 - (12.0)*0.00000000 - (6.0)*0.00000000 - (10.0)*0.00000000)/-8.0 = -2.50000000

```

x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(1) = (2.0 - (3.0)*0.00000000 - (-13.0)*0.00000000 - (3.0)*0.00000000)/9.0 = 0.22222222
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(1) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(1) = (-19.0 - (-6.0)*0.00000000 - (4.0)*0.00000000 - (1.0)*0.00000000)/-18.0 = 1.05555556
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 1.666667
x2 <- x_old[2]; (x2)

## [1] -2.5
x3 <- x_old[3]; (x3)

## [1] 0.2222222
x4 <- x_old[4]; (x4)

## [1] 1.055556
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(1) = 1.66666667, -2.50000000, 0.22222222, 1.05555556
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 4.06666667e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.11.1.2. Iteración 2

```

## x1^(2) = (10.0 - (-2.0)*-2.50000000 - (2.0)*0.22222222 - (4.0)*1.05555556)/6.0 = 0.05555556
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(2) = (20.0 - (12.0)*1.66666667 - (6.0)*0.22222222 - (10.0)*1.05555556)/-8.0 = 1.48611111
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(2) = (2.0 - (3.0)*1.66666667 - (-13.0)*-2.50000000 - (3.0)*1.05555556)/9.0 = -4.29629630

```

```

x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(2) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(2) = (-19.0 - (-6.0)*1.66666667 - (4.0)*-2.50000000 - (1.0)*1.05555556)/-18.0 = -0.04320988
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 0.05555556
x2 <- x_old[2]; (x2)

## [1] 1.486111
x3 <- x_old[3]; (x3)

## [1] -4.296296
x4 <- x_old[4]; (x4)

## [1] -0.04320988
#Error o tolerancia
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(2) = 0.05555556, 1.48611111, -4.29629630, -0.04320988
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.99490741e+01

```

```

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.11.1.3. Iteración 3

```

## x1^(3) = (10.0 - (-2.0)*1.48611111 - (2.0)*-4.29629630 - (4.0)*-0.04320988)/6.0 = 3.62294239
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(3) = (20.0 - (12.0)*0.05555556 - (6.0)*-4.29629630 - (10.0)*-0.04320988)/-8.0 = -5.69290123
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(3) = (2.0 - (3.0)*0.05555556 - (-13.0)*1.48611111 - (3.0)*-0.04320988)/9.0 = 2.36471193
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(3) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(3) = (-19.0 - (-6.0)*0.05555556 - (4.0)*1.48611111 - (1.0)*-0.04320988)/-18.0 = 1.12860082

```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 3.622942
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -5.692901
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 2.364712
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 1.128601
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(3) = 3.62294239, -5.69290123, 2.36471193, 1.12860082
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.07544753e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.4. Iteración 4

```
## x1^(4) = (10.0 - (-2.0)*-5.69290123 - (2.0)*2.36471193 - (4.0)*1.12860082)/6.0 = -1.77160494
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(4) = (20.0 - (12.0)*3.62294239 - (6.0)*2.36471193 - (10.0)*1.12860082)/-8.0 = 6.11869856
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(4) = (2.0 - (3.0)*3.62294239 - (-13.0)*-5.69290123 - (3.0)*1.12860082)/9.0 = -9.58470508
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(4) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(4) = (-19.0 - (-6.0)*3.62294239 - (4.0)*-5.69290123 - (1.0)*1.12860082)/-18.0 = -1.28580818
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -1.771605
```

```

x2 <- x_old[2]; (x2)

## [1] 6.118699
x3 <- x_old[3]; (x3)

## [1] -9.584705
x4 <- x_old[4]; (x4)

## [1] -1.285808
#Error o tolerancia

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = -1.77160494, 6.11869856, -9.58470508, -1.28580818
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.76977666e+02

x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o

```

2.3.2.11.1.5. Iteración 5

```

## x1^(5) = (10.0 - (-2.0)*6.11869856 - (2.0)*-9.58470508 - (4.0)*-1.28580818)/6.0 = 7.75834000
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o

## x2^(5) = (20.0 - (12.0)*-1.77160494 - (6.0)*-9.58470508 - (10.0)*-1.28580818)/-8.0 = -13.95319644
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o

## x3^(5) = (2.0 - (3.0)*-1.77160494 - (-13.0)*6.11869856 - (3.0)*-1.28580818)/9.0 = 10.07948007
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(5) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o

## x4^(5) = (-19.0 - (-6.0)*-1.77160494 - (4.0)*6.11869856 - (1.0)*-1.28580818)/-18.0 = 2.47331771
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] 7.75834
x2 <- x_old[2]; (x2)

## [1] -13.9532
x3 <- x_old[3]; (x3)

```



```
## [1] 10.07948
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 2.473318
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(5) = 7.75834000, -13.95319644, 10.07948007, 2.47331771
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 3.00801848e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.6. Iteración 6

```
## x1^(6) = (10.0 - (-2.0)*-13.95319644 - (2.0)*10.07948007 - (4.0)*2.47331771)/6.0 = -7.99310398
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(6) = (20.0 - (12.0)*7.75834000 - (6.0)*10.07948007 - (10.0)*2.47331771)/-8.0 = 19.78876719
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(6) = (2.0 - (3.0)*7.75834000 - (-13.0)*-13.95319644 - (3.0)*2.47331771)/9.0 = -23.34294744
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(6) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(6) = (-19.0 - (-6.0)*7.75834000 - (4.0)*-13.95319644 - (1.0)*2.47331771)/-18.0 = -4.07129698
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -7.993104
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 19.78877
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -23.34295
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -4.071297
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(6) = -7.99310398, 19.78876719, -23.34294744, -4.07129698
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 5.05533703e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.7. Iteración 7

```
## x1^(7) = (10.0 - (-2.0)*19.78876719 - (2.0)*-23.34294744 - (4.0)*-4.07129698)/6.0 = 18.75810287
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(7) = (20.0 - (12.0)*-7.99310398 - (6.0)*-23.34294744 - (10.0)*-4.07129698)/-8.0 = -37.08598778
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(7) = (2.0 - (3.0)*-7.99310398 - (-13.0)*19.78876719 - (3.0)*-4.07129698)/9.0 = 32.82746405
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(7) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(7) = (-19.0 - (-6.0)*-7.99310398 - (4.0)*19.78876719 - (1.0)*-4.07129698)/-18.0 = 6.82059696
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 18.7581
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -37.08599
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 32.82746
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 6.820597
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(7) = 18.75810287, -37.08598778, 32.82746405, 6.82059696
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 8.52301117e+02
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.8. Iteración 8

```
## x1^(8) = (10.0 - (-2.0)*-37.08598778 - (2.0)*32.82746405 - (4.0)*6.82059696)/6.0 = -26.18488191
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(8) = (20.0 - (12.0)*18.75810287 - (6.0)*32.82746405 - (10.0)*6.82059696)/-8.0 = 58.78349853
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(8) = (2.0 - (3.0)*18.75810287 - (-13.0)*-37.08598778 - (3.0)*6.82059696)/9.0 = -61.87266006
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(8) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(8) = (-19.0 - (-6.0)*18.75810287 - (4.0)*-37.08598778 - (1.0)*6.82059696)/-18.0 = -11.61472801
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] -26.18488
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] 58.7835
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] -61.87266
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] -11.61473
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))
```

```
## x^(8) = -26.18488191, 58.78349853, -61.87266006, -11.61472801
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 1.43643825e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.9. Iteración 9

```
## x1^(9) = (10.0 - (-2.0)*58.78349853 - (2.0)*-61.87266006 - (4.0)*-11.61472801)/6.0 = 49.62853821
```

```
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[2], A[2,1], x_o
```

```
## x2^(9) = (20.0 - (12.0)*-26.18488191 - (6.0)*-61.87266006 - (10.0)*-11.61472801)/-8.0 = -102.70022793
```

```
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[3], A[3,1], x_o
```

```
## x3^(9) = (2.0 - (3.0)*-26.18488191 - (-13.0)*58.78349853 - (3.0)*-11.61472801)/9.0 = 97.73159007
```

```
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(9) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[4], A[4,1], x_o
```

```
## x4^(9) = (-19.0 - (-6.0)*-26.18488191 - (4.0)*58.78349853 - (1.0)*-11.61472801)/-18.0 = 19.40947920
```

```
# Imprime los valores de la iteración
```

```
x_old <- x
x1 <- x_old[1]; (x1)
```

```
## [1] 49.62854
```

```
x2 <- x_old[2]; (x2)
```

```
## [1] -102.7002
```

```
x3 <- x_old[3]; (x3)
```

```
## [1] 97.73159
```

```
x4 <- x_old[4]; (x4)
```

```
## [1] 19.40948
```

```
#Error o tolerancia
```

```
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))
```

```
## x^(9) = 49.62853821, -102.70022793, 97.73159007, 19.40947920
```

```
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))
```

```
## ||dx||_inf = 0.00000000e+00 | ||r||_inf = 2.41980133e+03
```

```
x_old <- x
x[1] <- (b[1] - A[1,2] * x_old[2] - A[1,3] * x_old[3] - A[1,4] * x_old[4]) / A[1,1]
cat(sprintf("x1^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f = %.8f\n", b[1], A[1,2], x_o
```

2.3.2.11.1.10. Iteración 10

```
## x1^(10) = (10.0 - (-2.0)*-102.70022793 - (2.0)*97.73159007 - (4.0)*19.40947920)/6.0 = -78.08359213
x[2] <- (b[2] - A[2,1]*x_old[1] - A[2,3]*x_old[3] - A[2,4]*x_old[4]) / A[2,2]
cat(sprintf("x2^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[2], A[2,1], x_
## x2^(10) = (20.0 - (12.0)*49.62853821 - (6.0)*97.73159007 - (10.0)*19.40947920)/-8.0 = 169.50334886
x[3] <- (b[3] - A[3,1]*x_old[1] - A[3,2]*x_old[2] - A[3,4]*x_old[4]) / A[3,3]
cat(sprintf("x3^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[3], A[3,1], x_
## x3^(10) = (2.0 - (3.0)*49.62853821 - (-13.0)*-102.70022793 - (3.0)*19.40947920)/9.0 = -171.13522392
x[4] <- (b[4] - A[4,1]*x_old[1] - A[4,2]*x_old[2] - A[4,3]*x_old[3]) / A[4,4]
cat(sprintf("x4^(10) = (%.1f - (%.1f)*%.8f - (%.1f)*%.8f - (%.1f)*%.8f)/%.1f =%.8f\n", b[4], A[4,1], x_
## x4^(10) = (-19.0 - (-6.0)*49.62853821 - (4.0)*-102.70022793 - (1.0)*19.40947920)/-18.0 = -32.88003061
# Imprime los valores de la iteración
x_old <- x
x1 <- x_old[1]; (x1)

## [1] -78.08359
x2 <- x_old[2]; (x2)

## [1] 169.5033
x3 <- x_old[3]; (x3)

## [1] -171.1352
x4 <- x_old[4]; (x4)

## [1] -32.88003
```

2.3.2.11.2. Resultados La matriz no converge. Comienza a diverger.

2.4. Raíces de funciones

2.4.1. Método de Bisección

2.4.1.1. Ejercicios a mano

2.4.1.2. Código para implementación

```
# Condición para los intervalos

condicion <- function(a, b, f, inicio = 1, fin = 10, paso = 1) {
  fa <- f(a)
  fb <- f(b)
  producto <- fa * fb

  # Resultados
  cat("f(a) = f(", a, ") = ", fa, "\n")
  cat("f(b) = f(", b, ") = ", fb, "\n")
```

Redondeamos a 6 cifras

11. $f(x) = -2x^4 + 4x^3 + 6x^2 - 3x + 2$ en $[-2, -1]$

Verificamos la condición:

$f(a) \cdot f(b) < 0$

$f(-2) = -32$ $f(-2)f(-1) < 0$ $-160 < 0$

$f(-1) = 5$ $(-32)(5) < 0$

$c_1 = \frac{-2 + (-1)}{2} = -1.5$

$f(a) = -$ $f(b) = +$ $f(c) = -$

$[-1.5, -1]$

$c_2 = \frac{-1.5 + (-1)}{2} = -1.25$

$f(a) = -$ $f(b) = +$ $f(c) = + \rightarrow b$

$[-1.5, -1.25]$

$c_3 = \frac{-1.5 + (-1.25)}{2} = -1.375$

$f(a) = -$ $f(b) = +$ $f(c) = - \rightarrow a$

$[-1.375, -1.25]$

$c_4 = \frac{-1.375 - 1.25}{2} = -1.3125$

$f(a) = -$ $f(b) = +$ $f(c) = + \rightarrow b$

$[-1.375, -1.3125]$

$c_5 = \frac{-1.375 - 1.3125}{2} = -1.34375$

$f(a) = -$ $f(b) = +$ $f(c) = + \rightarrow b$

$[-1.375, -1.34375]$

$c_6 = \frac{-1.375 - 1.34375}{2} = -1.359375$

$f(a) = -$ $f(b) = +$ $f(c) = + \rightarrow b$

$[-1.375, -1.359375]$

Figura 2.28: Ejercicio 11-1

$$c_7 = \frac{-1.375 - 1.359375}{2} = -1.367188$$

$$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$$

$$[-1.375, -1.367188]$$

$$c_8 = \frac{-1.375 - 1.367188}{2} = -1.371094$$

$$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$$

$$[-1.375, -1.371094]$$

$$c_9 = \frac{-1.375 - 1.371094}{2} = -1.373047$$

$$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$$

$$[-1.373047, -1.371094]$$

$$c_{10} = \frac{-1.373047 - 1.371094}{2} = -1.372071$$

$$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$$

$$[-1.372071, -1.371094]$$

$$c_{11} = \frac{-1.372071 - 1.371094}{2} = -1.371582$$

$$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$$

$$[-1.372071, -1.371582]$$

$$c_{12} = \frac{-1.372071 - 1.371582}{2} = -1.371827$$

$$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$$

$$[-1.371827, -1.371582]$$

$$c_{13} = \frac{-1.371827 - 1.371582}{2} = -1.371704$$

$$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$$

$$[-1.371827, -1.371704]$$

$$c_{14} = \frac{-1.371827 - 1.371704}{2} = -1.371766$$

$$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$$

Figura 2.29: Ejercicio 11-2

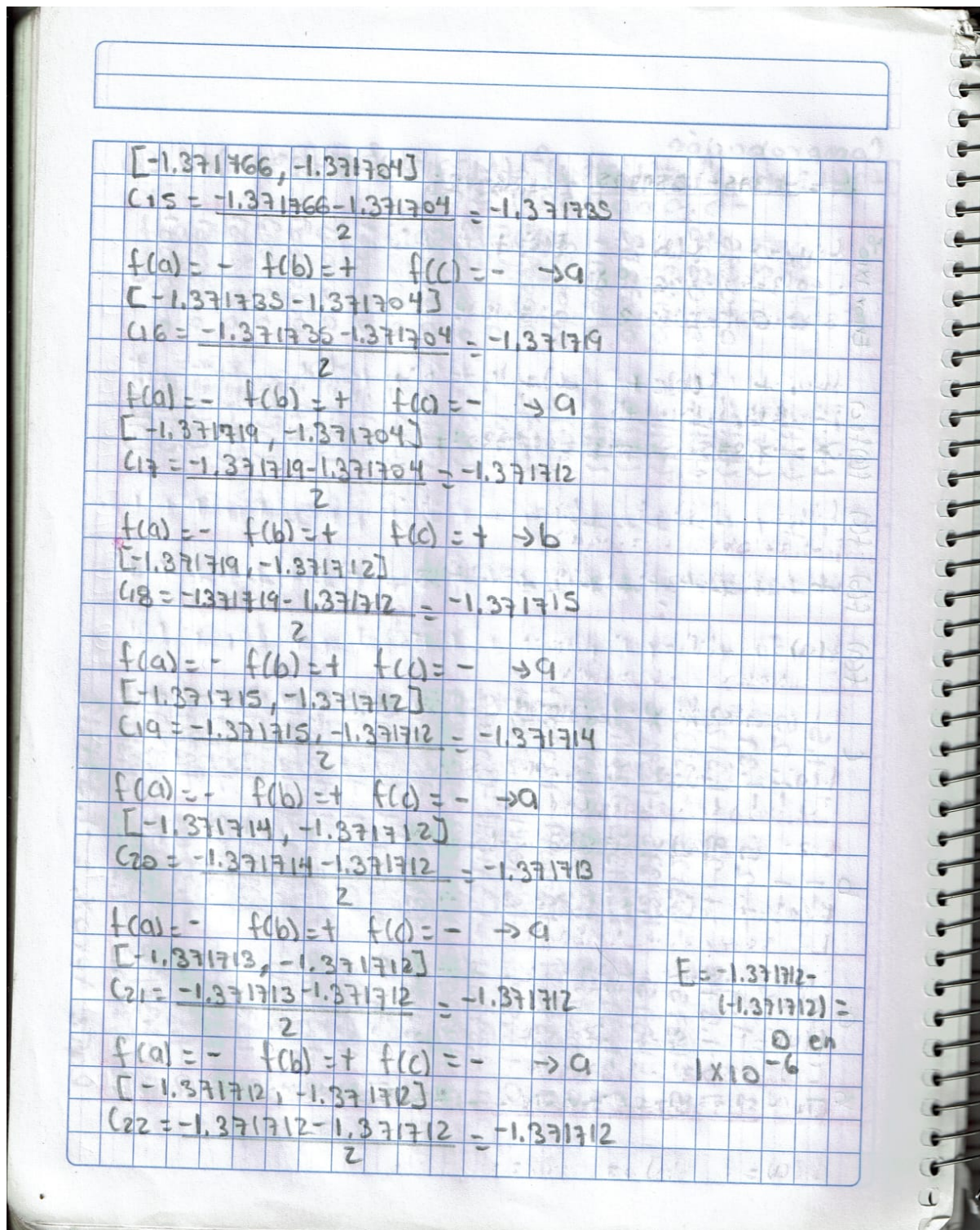


Figura 2.30: Ejercicio 11-3

Comprobación

$$-2(-1.371712)^4 + 4(-1.371712)^3 + 6(-1.371712)^2 - 3(-1.371712) + 2 = 0$$

n	a	b	c	f(a)	f(b)	f(c)	f(a)f(c)	Error $\times 10^{-6}$
1	-2	-1	-1.5	-	+	-	$(-)(-)=+$	
2	-1.5	-1	-1.25	-	+	+	$(-)(+)=$	0.25
3	-1.5	-1.25	-1.375	-	+	-	$(-)(-)=+$	0.125
4	-1.375	-1.25	-1.3125	-	+	+	$(-)(+)=$	0.0625
5	-1.375	-1.3125	-1.34375	-	+	+	$(-)(+)=$	0.03125
6	-1.375	-1.34375	-1.359375	-	+	+	$(-)(+)=$	0.015625
7	-1.375	-1.359375	-1.3671875	-	+	+	$(-)(+)=$	0.0078125
8	-1.375	-1.3671875	-1.37094	-	+	+	$(-)(+)=$	0.00391
9	-1.375	-1.37094	-1.373047	-	+	-	$(-)(-)=+$	0.001953
10	-1.372047	-1.371044	-1.372071	-	+	-	$(-)(-)=+$	0.000976
11	-1.372071	-1.371094	-1.371582	-	+	+	$(-)(+)=$	0.000489
12	-1.372071	-1.371582	-1.371827	-	+	-	$(-)(-)=+$	0.000245
13	-1.371827	-1.37152	-1.371704	-	+	+	$(-)(+)=$	0.000123
14	-1.371827	-1.371704	-1.371766	-	+	-	$(-)(-)=+$	0.000062
15	-1.371716	-1.371704	-1.371735	-	+	-	$(-)(-)=+$	0.000031
16	-1.371735	-1.371704	-1.371719	-	+	+	$(-)(+)=+$	0.000016
17	-1.371719	-1.371704	-1.371712	-	+	+	$(-)(+)=$	0.000007
18	-1.371719	-1.371712	-1.371715	-	+	-	$(-)(-)=+$	0.000003
19	-1.371715	-1.371712	-1.371714	-	+	-	$(-)(-)=+$	0.000001
20	-1.371715	-1.371712	-1.371713	-	+	-	$(-)(-)=+$	0.000001
21	-1.371713	-1.371712	-1.371712	-	+	-	$(-)(-)=+$	0.000001
22	-1.371712	-1.371712	-1.371712	-	+	-	$(-)(-)=+$	0
Dato Aprox. = -1.371712								

Figura 2.31: Ejercicio 11 TABLA

13. $f(x) = x^4 + 2x^3 - 10x^2 - x + 5$ en $[0, 1]$
 Verificamos la condición:
 $f(a) \cdot f(b) < 0$ $f(0)f(1) < 0$
 $f(0) = 5$ $(5)(-3) < 0$
 $f(1) = -3$ $-15 < 0$

$c_1 = \frac{0+1}{2} = 0.5$
 $f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.5, 1]$
 $c_2 = \frac{0.5+1}{2} = 0.75$
 $f(a) = +$ $f(b) = -$ $f(c) = - \rightarrow b$
 $[0.5, 0.75]$
 $c_3 = \frac{0.5+0.75}{2} = 0.625$
 $f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.625, 0.75]$
 $c_4 = \frac{0.625+0.75}{2} = 0.6875$
 $f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.6875, 0.75]$
 $c_5 = \frac{0.6875+0.75}{2} = 0.71875$
 $f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.71875, 0.75]$
 $c_6 = \frac{0.71875+0.75}{2} = 0.734375$
 $f(a) = +$ $f(b) = -$ $f(c) = - \rightarrow b$
 $[0.71875, 0.734375]$

Figura 2.32: Ejercicio 13-1

$$C_7 = \frac{0.71875 + 0.734375}{2} = 0.726563$$

$f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.726563, 0.734375]$

$$C_8 = \frac{0.726563 + 0.734375}{2} = 0.730469$$

$f(a) = +$ $f(b) = -$ $f(c) = - \rightarrow b$
 $[0.726563, 0.730469]$

$$C_9 = \frac{0.726563 + 0.730469}{2} = 0.728516$$

$f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.728516, 0.730469]$

$$C_{10} = \frac{0.728516 + 0.730469}{2} = 0.729492$$

$f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.729492, 0.730469]$

$$C_{11} = \frac{0.729492 + 0.730469}{2} = 0.729981$$

$f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.729981, 0.730469]$

$$C_{12} = \frac{0.729981 + 0.730469}{2} = 0.730225$$

$f(a) = +$ $f(b) = -$ $f(c) = + \rightarrow a$
 $[0.730225, 0.730469]$

$$C_{13} = \frac{0.730225 + 0.730469}{2} = 0.730347$$

$f(a) = +$ $f(b) = -$ $f(c) = - \rightarrow b$
 $[0.730225, 0.730347]$

$$C_{14} = \frac{0.730225 + 0.730347}{2} = 0.730286$$

$f(a) = +$ $f(b) = -$ $f(c) = - \rightarrow b$

Figura 2.33: Ejercicio 13-2

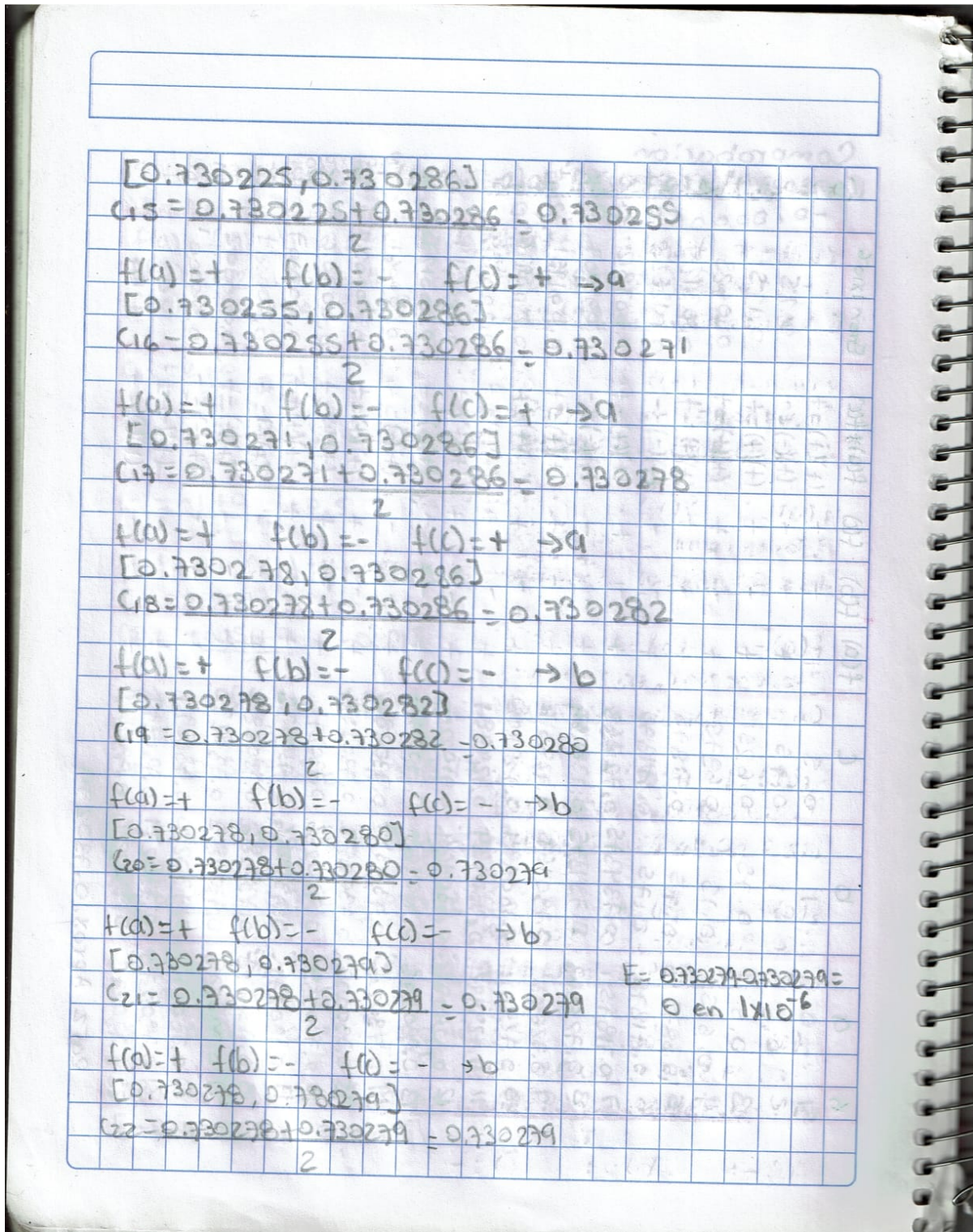


Figura 2.34: Ejercicio 13-3

Comprobación

$$(0.730279)^4 + 2(0.730279)^3 + 10(0.730279)^2 - (0.730279) + 5 = -0.000009460420$$

k	a	b	c	f(a)	f(b)	f(a)*f(b)	Error $\times 10^6$
1	0	1	0.5	+	-	+	
2	0.5	1	0.75	+	-	-	0.15
3	0.5	0.75	0.625	+	-	+	0.125
4	0.625	0.75	0.6875	+	-	+	0.0625
5	0.6875	0.75	0.71875	+	-	+	0.03125
6	0.71875	0.75	0.734375	+	-	-	0.015625
7	0.71875	0.734375	0.726563	+	-	+	0.007812
8	0.726563	0.734375	0.730469	+	-	-	0.003904
9	0.726563	0.730469	0.728516	+	-	+	0.001951
10	0.728516	0.730469	0.729492	+	-	+	0.000976
11	0.729492	0.730469	0.729981	+	-	+	0.000489
12	0.729981	0.730469	0.730225	+	-	+	0.000244
13	0.730225	0.730469	0.730347	+	-	-	0.000122
14	0.730225	0.730347	0.730286	+	-	-	0.000061
15	0.730225	0.730286	0.730255	+	-	+	0.000031
16	0.730255	0.730286	0.730271	+	-	+	0.000016
17	0.730271	0.730286	0.730278	+	-	+	0.000007
18	0.730278	0.730286	0.730282	+	-	-	0.000004
19	0.730278	0.730282	0.730280	+	-	-	0.000002
20	0.730278	0.730280	0.730279	+	-	-	0.000001
21	0.730278	0.730279	0.730279	+	-	-	0
22	0.730278	0.730279	0.730279	+	-	-	0
Rat2 Aprox. 0.730279							

Figura 2.35: Ejercicio 13 TABLA

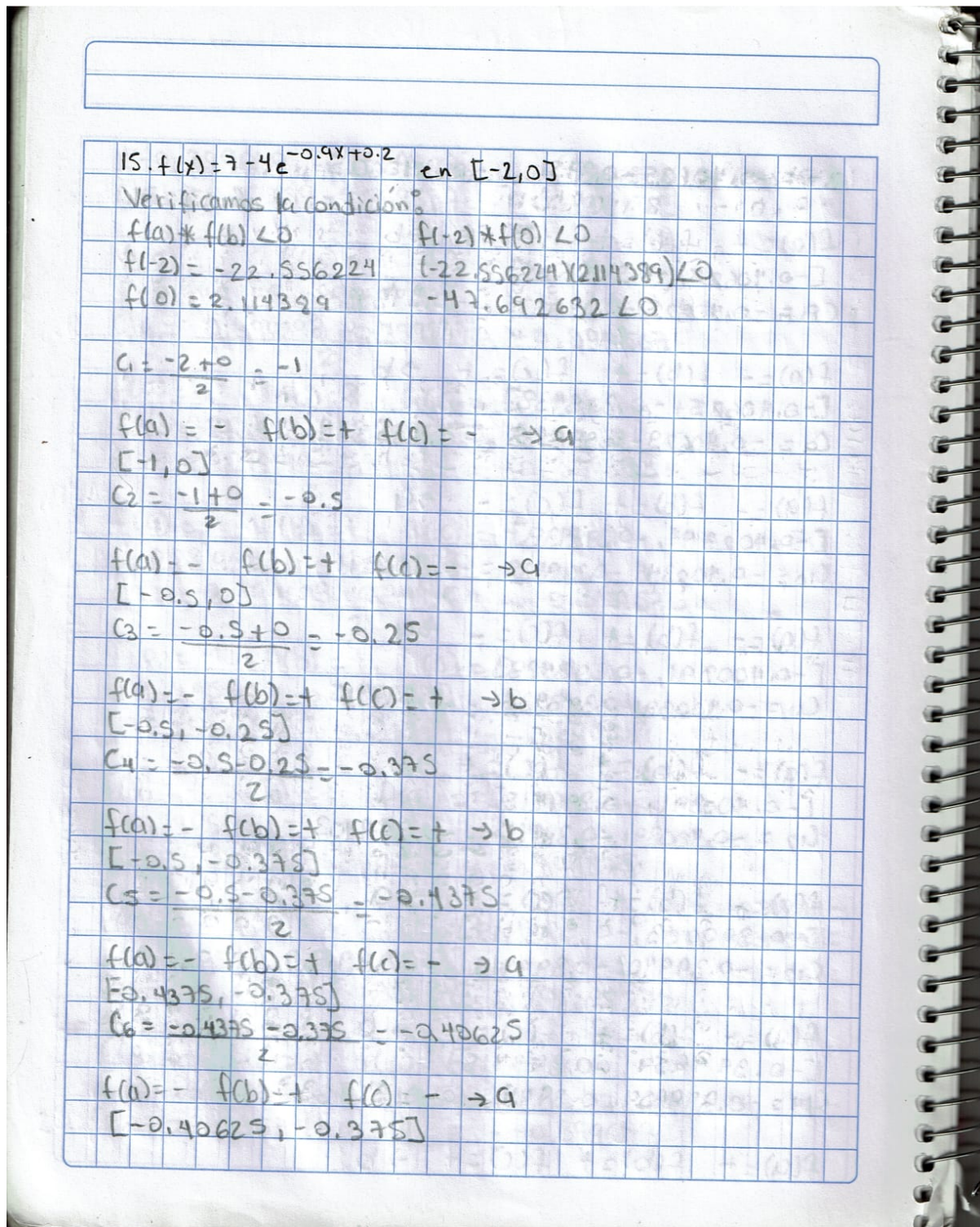


Figura 2.36: Ejercicio 15-1

$c_7 = \frac{-0.40625 - 0.395}{2} = -0.390625$	$f(a) = -$	$f(b) = +$	$f(c) = + \rightarrow b$
	$[-0.40625, -0.390625]$		
$c_8 = \frac{-0.40625 - 0.390625}{2} = -0.398438$	$f(a) = -$	$f(b) = +$	$f(c) = + \rightarrow b$
	$[-0.40625, -0.398438]$		
$c_9 = \frac{-0.40625 - 0.398438}{2} = -0.402344$	$f(a) = -$	$f(b) = +$	$f(c) = - \rightarrow a$
	$[-0.402344, -0.398438]$		
$c_{10} = \frac{-0.402344 - 0.398438}{2} = -0.400391$	$f(a) = -$	$f(b) = +$	$f(c) = - \rightarrow a$
	$[-0.400391, -0.398438]$		
$c_{11} = \frac{-0.400391 - 0.398438}{2} = -0.399415$	$f(a) = -$	$f(b) = +$	$f(c) = + \rightarrow b$
	$[-0.400391, -0.399415]$		
$c_{12} = \frac{-0.400391 - 0.399415}{2} = -0.399903$	$f(a) = -$	$f(b) = +$	$f(c) = - \rightarrow a$
	$[-0.399903, -0.399415]$		
$c_{13} = \frac{-0.399903 - 0.399415}{2} = -0.399659$	$f(a) = -$	$f(b) = +$	$f(c) = - \rightarrow a$
	$[-0.399659, -0.399415]$		
$c_{14} = \frac{-0.399659 - 0.399415}{2} = -0.399537$	$f(a) = -$	$f(b) = +$	$f(c) = + \rightarrow b$

Figura 2.37: Ejercicio 15-2

$[-0.399659, -0.399537]$	
$c15 = \frac{-0.399659 - 0.399537}{2} = -0.399598$	
$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$	
$[-0.399598, -0.399537]$	
$c16 = \frac{-0.399598 - 0.399537}{2} = -0.399567$	
$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$	
$[-0.399598, -0.399567]$	
$c17 = \frac{-0.399598 - 0.399567}{2} = -0.399583$	
$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$	
$[-0.399583, -0.399567]$	
$c18 = \frac{-0.399583 - 0.399567}{2} = -0.399575$	
$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$	
$[-0.399575, -0.399567]$	
$c19 = \frac{-0.399575 - 0.399567}{2} = -0.399571$	
$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$	
$[-0.399575, -0.399571]$	
$c20 = \frac{-0.399575 - 0.399571}{2} = -0.399573$	
$f(a) = - \quad f(b) = + \quad f(c) = + \rightarrow b$	$E = -0.399574 - 0.399573 =$
$[-0.399575, -0.399573]$	
$c21 = \frac{-0.399575 - 0.399573}{2} = -0.399574$	
$f(a) = - \quad f(b) = + \quad f(c) = - \rightarrow a$	
$[-0.399574, -0.399573]$	
$c22 = \frac{-0.399574 - 0.399573}{2} = -0.399573$	

Figura 2.38: Ejercicio 15-3

k	a _i	b	C	f(a)	f(b)	f(c)	f(a)*f(c)	Error x δ _c
1	-2	0	-1	-	+	-	(-)(-) = +	
2	-1	0	-0.5	-	+	-	(-)(-) = +	0.5
3	-0.5	0	-0.25	-	+	+	(-)(+) = -	0.25
4	-0.5	-0.25	-0.375	-	+	+	(-)(+) = -	0.125
5	-0.5	-0.375	-0.4375	-	+	-	(-)(-) = +	0.0625
6	-0.4375	-0.375	-0.40625	-	+	-	(-)(-) = +	0.03125
7	-0.40625	-0.375	-0.31625	-	+	+	(-)(+) = -	0.015625
8	-0.40625	-0.31625	-0.348438	-	+	+	(-)(+) = -	0.037813
9	-0.40625	-0.348438	-0.402344	-	+	-	(-)(-) = +	0.07964
10	-0.402344	-0.348438	-0.400391	-	+	-	(-)(-) = +	0.001953
11	-0.400391	-0.348438	-0.344415	-	+	+	(-)(+) = -	0.006976
12	-0.400391	-0.344415	-0.319903	-	+	-	(-)(-) = +	0.004458
13	-0.349903	-0.344415	-0.34659	-	+	-	(-)(-) = +	0.000244
14	-0.349903	-0.344415	-0.399537	-	+	+	(-)(+) = -	0.000122
15	-0.349903	-0.399537	-0.34898	-	+	-	(-)(-) = +	0.000061
16	-0.349903	-0.399537	-0.399567	-	+	+	(-)(+) = -	0.000031
17	-0.349903	-0.399567	-0.349583	-	+	-	(-)(-) = +	0.000016
18	-0.349903	-0.349583	-0.399575	-	+	-	(-)(-) = +	0.000008
19	-0.349575	-0.349583	-0.399571	-	+	+	(-)(+) = -	0.000004
20	-0.349575	-0.349571	-0.399573	-	+	+	(-)(+) = -	0.000002
21	-0.349575	-0.349573	-0.399574	-	+	-	(-)(-) = +	0.000001
22	-0.349574	-0.399573	-0.399573	-	+	+	(-)(+) = -	0.000000
Rat2	Approx. 2	-0.399574	-0.399573	-	+	+	(-)(+) = -	0.000000

(See hand in book entire bs dos.)

Figura 2.39: Ejercicio 15 TABLA

```

cat("f(a) · f(b) = ", producto, "\n")
cat(producto, "< 0", "\n\n")

if(producto < 0) {
  cat("Sí cumple la condición.\n")
  # Asignar globalmente
  a <- a
  b <- b
  return(TRUE)
} else {
  cat("NO cumple la condición.\n")
  cat("Se buscará un nuevo intervalo.\n\n")

  # Buscar automáticamente un intervalo que sí funcione
  x <- seq(inicio, fin, paso)
  for (i in 1:(length(x)-1)) {
    a_nuevo <- x[i]
    b_nuevo <- x[i+1]
    if (f(a_nuevo) * f(b_nuevo) < 0) {
      cat("Nuevo intervalo encontrado: [", a_nuevo, ", ", b_nuevo, "]\n")
      fa <- f(a_nuevo)
      fb <- f(b_nuevo)
      producto <- fa * fb
      cat("f(a) = f(", a_nuevo, ") = ", fa, "\n")
      cat("f(b) = f(", b_nuevo, ") = ", fb, "\n")
      cat("f(a) · f(b) = ", producto, "\n")
      cat(producto, "< 0", "\n\n")
      cat("Sí cumple la condición.\n")
      # Asignar globalmente
      a <- a_nuevo
      b <- b_nuevo
      return(TRUE)
    }
  }
  cat("No se encontró ningún intervalo con cambio de signo\n")
  return(FALSE)
}
}

# Método de Bisección que guarda los valores para la tabla

biseccion <- function(f, a, b, tol = 1e-5, iter_max = 100){
  if(f(a)*f(b) > 0){
    cat("Error: no hay cambio de signo en el intervalo [a,b].\n")
    return(NA)
  }

  ## Vectores para almacenar los datos de las iteraciones

  iter_vec <- numeric(0)
  a_vec <- numeric(0)
  b_vec <- numeric(0)

```

```

c_vec <- numeric(0)
fa_vec <- numeric(0)
fb_vec <- numeric(0)
fc_vec <- numeric(0)
signFAFC_vec <- numeric(0)
signFBFC_vec <- numeric(0)
error_vec <- numeric(0)

iter <- 0
error <- abs(b - a)
c <- a

while(error > tol && iter < iter_max){
  iter <- iter + 1
  c <- (a + b)/2
  fc <- f(c)

  if(fc == 0){
    iter_vec <- c(iter_vec, iter)
    a_vec <- c(a_vec, a)
    b_vec <- c(b_vec, b)
    c_vec <- c(c_vec, c)
    fa_vec <- c(fa_vec, f(a))
    fb_vec <- c(fb_vec, f(b))
    fc_vec <- c(fc_vec, fc)
    signFAFC_vec <- c(signFAFC_vec, sign(f(a) * fc))
    signFBFC_vec <- c(signFBFC_vec, sign(f(b) * fc))
    error_vec <- c(error_vec, error)
    break
  }

  fa <- f(a)
  fb <- f(b)

# Almacena los datos de cada iteración
  iter_vec <- c(iter_vec, iter)
  a_vec <- c(a_vec, a)
  b_vec <- c(b_vec, b)
  c_vec <- c(c_vec, c)
  fa_vec <- c(fa_vec, fa)
  fb_vec <- c(fb_vec, fb)
  fc_vec <- c(fc_vec, fc)
  signFAFC_vec <- c(signFAFC_vec, sign(fa * fc))
  signFBFC_vec <- c(signFBFC_vec, sign(fb * fc))
  error_vec <- c(error_vec, error)

  if(fa * fc < 0){
    b <- c
  } else {
    a <- c
  }
}

```

```

    error <- abs(b - a)
  }

# Crea la tabla de las iteraciones
TAB <- data.frame(
  k = iter_vec,
  a = a_vec,
  b = b_vec,
  c = c_vec,
  "f(a)" = fa_vec,
  "f(b)" = fb_vec,
  "f(c)" = fc_vec,
  "Signo(f(a)*f(c))" = signFAFC_vec,
  "Signo(f(b)*f(c))" = signFBFC_vec,
  Error = error_vec,
  check.names = FALSE
)

cat("Raíz aproximada:", c, "\n")
cat("Iteraciones:", iter, "\n\n")

return(list(raiz = c, iteraciones = iter, tabla = TAB))
}

# Tabla "bonita"

tabla <- function(resultado) {
  library(DT)

  datatable(
    resultado$tabla,
    options = list(
      scrollX = TRUE,
      scrollY = "400px",
      paging = FALSE,
      dom = 't',
      autoWidth = FALSE,
      columnDefs = list(
        list(className = 'dt-center', targets = 0:5)
      ),
      rownames = FALSE,
      class = 'cell-border stripe',
      width = "100%",
      caption = "Tabla de iteraciones en el método de la secante",
      escape = FALSE
    ) %>%
    formatRound(columns = c(2, 3, 4, 5, 6, 7, 10), digits = 6)
}

# Gráficas

grafica <- function(a, b, f, resultado) {

```

```

tabla_iteraciones <- resultado$tabla
total_iteraciones <- nrow(tabla_iteraciones)
filas <- ceiling(total_iteraciones / 4)

par(mfrow = c(filas, 4), mar = c(2, 2, 1.5, 1))

x_vals <- seq(a - 0.5, b + 0.5, length.out = 1000)
y_vals <- f(x_vals)

for (i in 1:total_iteraciones) {
  plot(x_vals, y_vals, type = "l", main = paste("Iteración", i),
       xlab = "x", ylab = "f(x)", col = "blue", lwd = 1.2)
  abline(h = 0, col = "gray", lty = 2)

  a_actual <- tabla_iteraciones$a[i]
  b_actual <- tabla_iteraciones$b[i]
  c_actual <- tabla_iteraciones$c[i]

  points(c(a_actual, b_actual), c(f(a_actual), f(b_actual)),
        col = c("red", "green"), pch = 19, cex = 0.8)
  points(c(c_actual, f(c_actual)), col = "orange", pch = 19, cex = 0.6)

  abline(v = a_actual, col = "red", lty = 2)
  abline(v = b_actual, col = "green", lty = 2)
  abline(v = c_actual, col = "orange", lty = 3)

  legend("topright", legend = paste("Iter", i), bty = "n")
}

par(mfrow = c(1, 1))
}

```

2.4.1.3. Sistema 1

$f(x) = x^2 - 5$ en $[2, 3]$

Datos

```

a <- 2
b <- 3
f <- function(x) x^2 - 5

```

2.4.1.3.1. Intervalo Para cada función $f(x)$, encuentre un intervalo $[a, b]$ tal que $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```

## f(a) = f( 2 ) = -1
## f(b) = f( 3 ) = 4
## f(a) · f(b) = -4
## -4 < 0
##

```

```
## Sí cumple la condición.
```

```
## [1] TRUE
```

2.4.1.3.2. Iteraciones Realice al menos 8 iteraciones del método de bisección.

```
resultado <- biseccion(f, a, b)
```

```
## Raíz aproximada: 2.236061
```

```
## Iteraciones: 17
```

2.4.1.3.3. Tabla Registre los valores ak , bk , ck , $f(ak)$, $f(bk)$, $f(ck)$, $\text{signo}(f(ak)f(ck))$ y el error $|bk - ak|$.

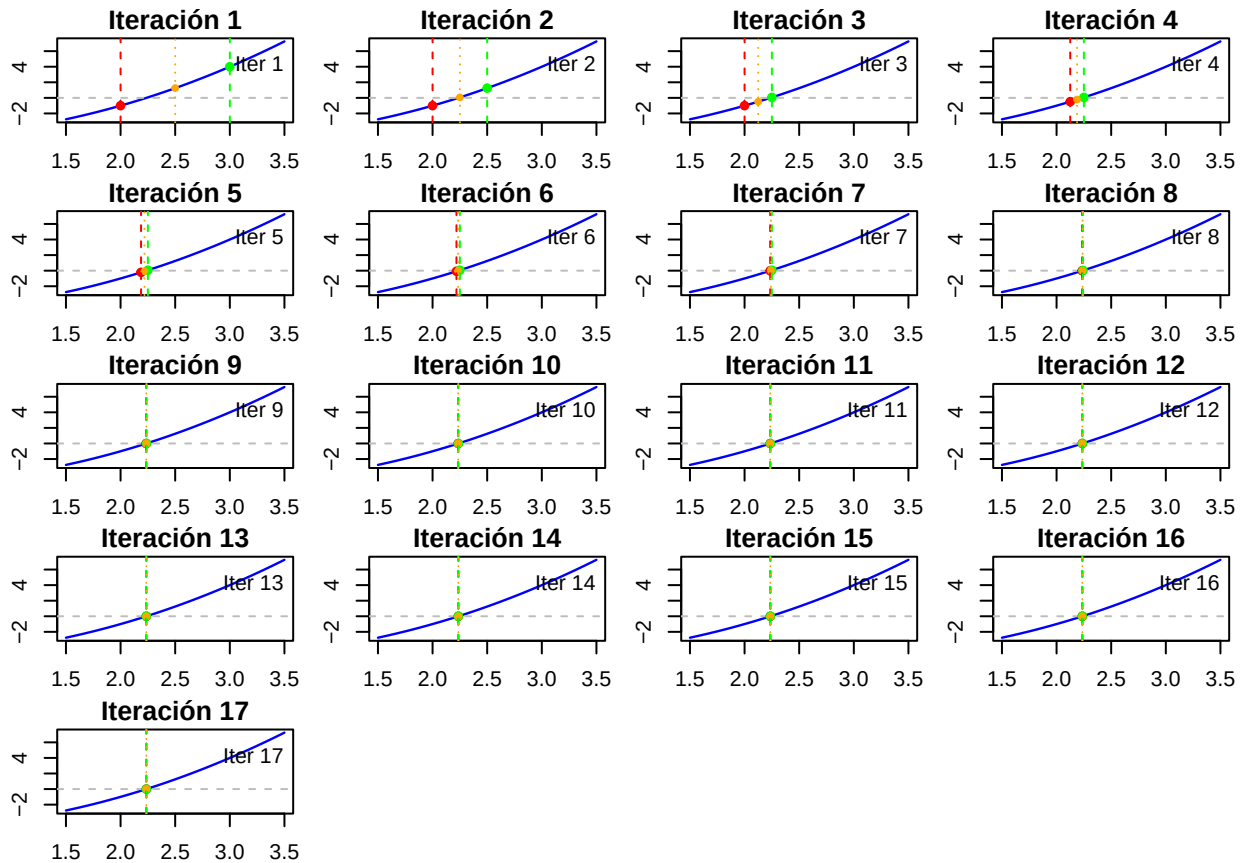
```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	a	b	c	f(a)	f(b)	f(c)	Signo(f(a)*f(c))
1	2.000000	3.000000	2.500000	-1.000000	4.000000	1.250000	
2	2.000000	2.500000	2.250000	-1.000000	1.250000	0.062500	
3	2.000000	2.250000	2.125000	-1.000000	0.062500	-0.484375	
4	2.125000	2.250000	2.187500	-0.484375	0.062500	-0.214844	
5	2.187500	2.250000	2.218750	-0.214844	0.062500	-0.077148	
6	2.218750	2.250000	2.234375	-0.077148	0.062500	-0.007568	
7	2.234375	2.250000	2.242188	-0.007568	0.062500	0.027405	
8	2.234375	2.242188	2.238281	-0.007568	0.027405	0.009903	
9	2.234375	2.238281	2.236328	-0.007568	0.009903	0.001163	
10	2.234375	2.236328	2.235352	-0.007568	0.001163	-0.003203	
11	2.235352	2.236328	2.235840	-0.003203	0.001163	-0.001020	
12	2.235840	2.236328	2.236084	-0.001020	0.001163	-0.000072	

2.4.1.3.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.


```
grafica(a, b, f, resultado)
```



2.4.1.4. Sistema 3

$$f(x) = x^4 - 8x + 2 \text{ en } [0, 3]$$

```
# Datos
```

```
a <- 0
b <- 3
f <- function(x) x^4 - 8*x + 2
```

2.4.1.4.1. Intervalo Para cada función $f(x)$, encuentre un intervalo $[a, b]$ tal que $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```
## f(a) = f( 0 ) = 2
## f(b) = f( 3 ) = 59
## f(a) · f(b) = 118
## 118 < 0
##
## NO cumple la condición.
## Se buscará un nuevo intervalo.
##
```

```
## Nuevo intervalo encontrado: [ 1 , 2 ]
## f(a) = f( 1 ) = -5
## f(b) = f( 2 ) = 2
## f(a) · f(b) = -10
## -10 < 0
##
## Sí cumple la condición.
## [1] TRUE
```

2.4.1.4.2. Iteraciones Realice al menos 8 iteraciones del método de bisección.

```
resultado <- biseccion(f, a, b)
```

```
## Raíz aproximada: 1.908562
## Iteraciones: 17
```

2.4.1.4.3. Tabla Registre los valores ak , bk , ck , $f(ak)$, $f(bk)$, $f(ck)$, $\text{signo}(f(ak)f(ck))$ y el error $|bk - ak|$.

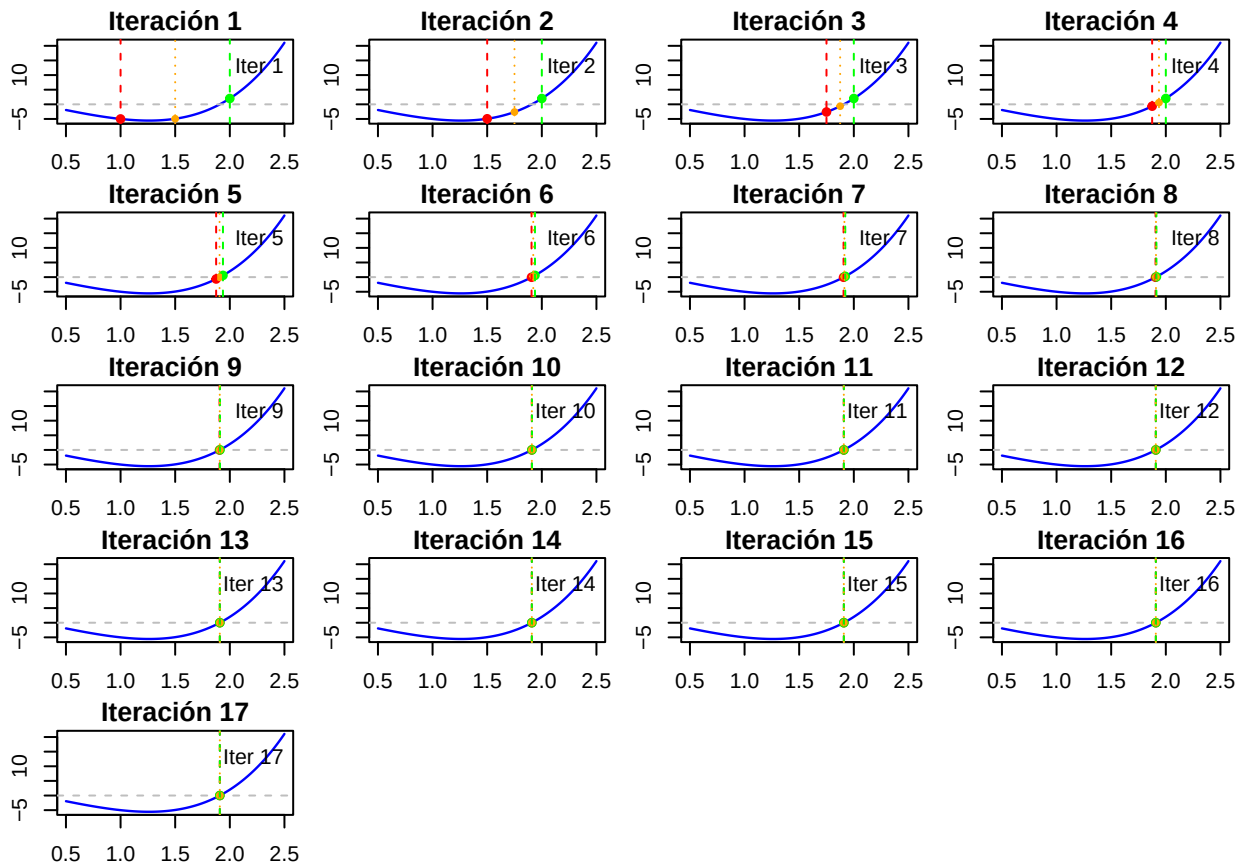
```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	a	b	c	f(a)	f(b)	f(c)	Signo(f(a)*f(c))
1	1.000000	2.000000	1.500000	-5.000000	2.000000	-4.937500	
2	1.500000	2.000000	1.750000	-4.937500	2.000000	-2.621094	
3	1.750000	2.000000	1.875000	-2.621094	2.000000	-0.640381	
4	1.875000	2.000000	1.937500	-0.640381	2.000000	0.591812	
5	1.875000	1.937500	1.906250	-0.640381	0.591812	-0.045577	
6	1.906250	1.937500	1.921875	-0.045577	0.591812	0.267707	
7	1.906250	1.921875	1.914063	-0.045577	0.267707	0.109723	
8	1.906250	1.914063	1.910156	-0.045577	0.109723	0.031739	
9	1.906250	1.910156	1.908203	-0.045577	0.031739	-0.007002	
10	1.908203	1.910156	1.909180	-0.007002	0.031739	0.012348	
11	1.908203	1.909180	1.908691	-0.007002	0.012348	0.002667	
12	1.908203	1.908691	1.908447	-0.007002	0.002667	0.002160	

2.4.1.4.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.


```
grafica(a, b, f, resultado)
```



2.4.1.5. Sistema 5

$f(x) = \cos(x) - x^2$ en $[0, 1]$

```
# Datos
```

```
a <- 0
b <- 1
f <- function(x) cos(x) - x^2
```

2.4.1.5.1. Intervalo Para cada función $f(x)$, encuentre un intervalo $[a, b]$ tal que $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```
## f(a) = f( 0 ) = 1
## f(b) = f( 1 ) = -0.4596977
## f(a) · f(b) = -0.4596977
## -0.4596977 < 0
##
## SÍ cumple la condición.
## [1] TRUE
```

2.4.1.5.2. Iteraciones Realice al menos 8 iteraciones del método de bisección.

```
resultado <- biseccion(f, a, b)
```

```
## Raíz aproximada: 0.8241348
```

```
## Iteraciones: 17
```

2.4.1.5.3. Tabla Registre los valores ak , bk , ck , $f(ak)$, $f(bk)$, $f(ck)$, $\text{signo}(f(ak)f(ck))$ y el error $|bk - ak|$.

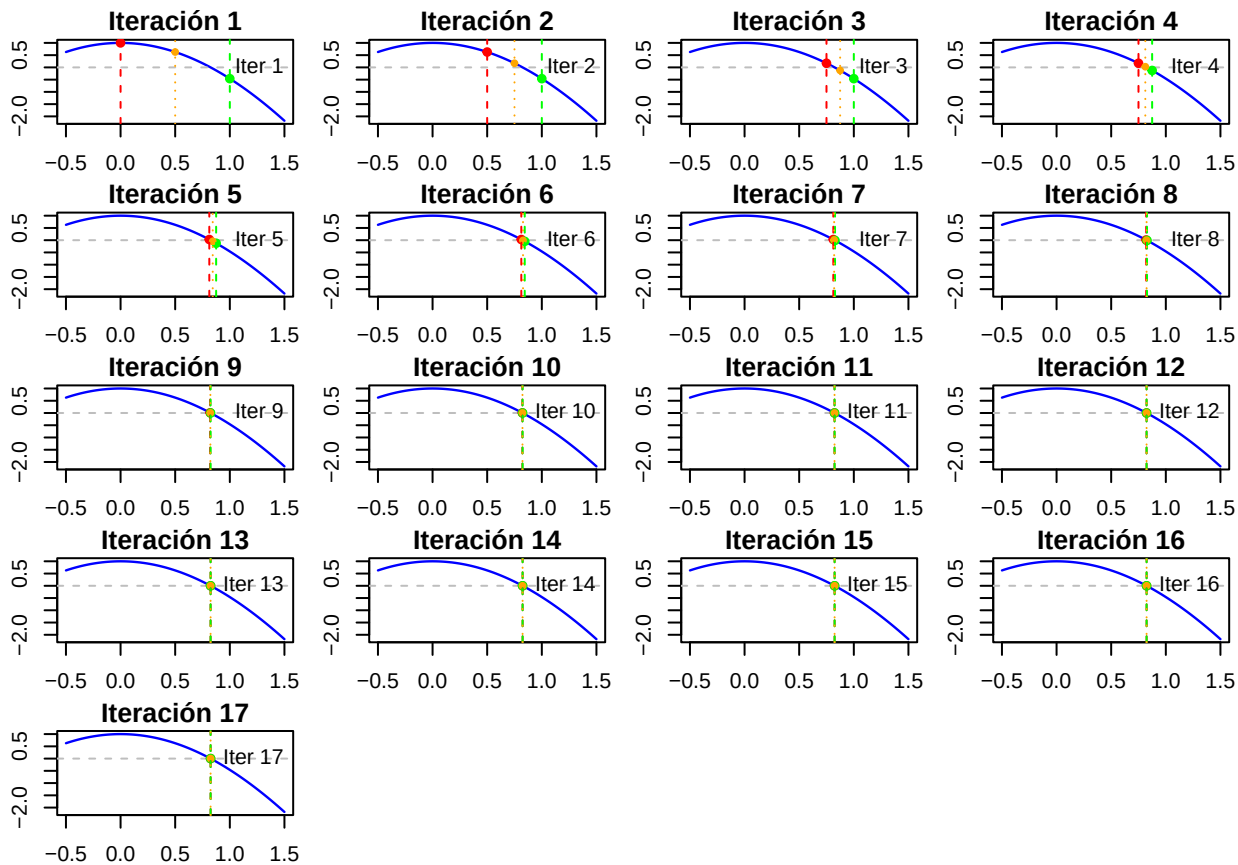
```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	a	b	c	f(a)	f(b)	f(c)	Signo(f(a)*f(c))
1	0.000000	1.000000	0.500000	1.000000	-0.459698	0.627583	
2	0.500000	1.000000	0.750000	0.627583	-0.459698	0.169189	
3	0.750000	1.000000	0.875000	0.169189	-0.459698	-0.124628	
4	0.750000	0.875000	0.812500	0.169189	-0.124628	0.027529	
5	0.812500	0.875000	0.843750	0.027529	-0.124628	-0.047248	
6	0.812500	0.843750	0.828125	0.027529	-0.047248	-0.009533	
7	0.812500	0.828125	0.820313	0.027529	-0.009533	0.009080	
8	0.820313	0.828125	0.824219	0.009080	-0.009533	-0.000206	
9	0.820313	0.824219	0.822266	0.009080	-0.000206	0.004442	
10	0.822266	0.824219	0.823242	0.004442	-0.000206	0.002119	
11	0.823242	0.824219	0.823730	0.002119	-0.000206	0.000957	
12	0.823730	0.824219	0.823975	0.000957	-0.000206	0.000276	

2.4.1.5.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(a, b, f, resultado)
```



2.4.1.6. Sistema 7

$f(x) = -4x^2 + 5x + 6$ en $[-1, 0]$.

Datos

```
a <- -1
b <- 0
f <- function(x) -4*x^2 + 5*x + 6
```

2.4.1.6.1. Intervalo Para cada función $f(x)$, encuentre un intervalo $[a, b]$ tal que $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```
## f(a) = f( -1 ) = -3
## f(b) = f( 0 ) = 6
## f(a) · f(b) = -18
## -18 < 0
##
## Sí cumple la condición.
## [1] TRUE
```

2.4.1.6.2. Iteraciones Realice al menos 8 iteraciones del método de bisección.

```
resultado <- biseccion(f, a, b)
```

```
## Raíz aproximada: -0.75
## Iteraciones: 2
```

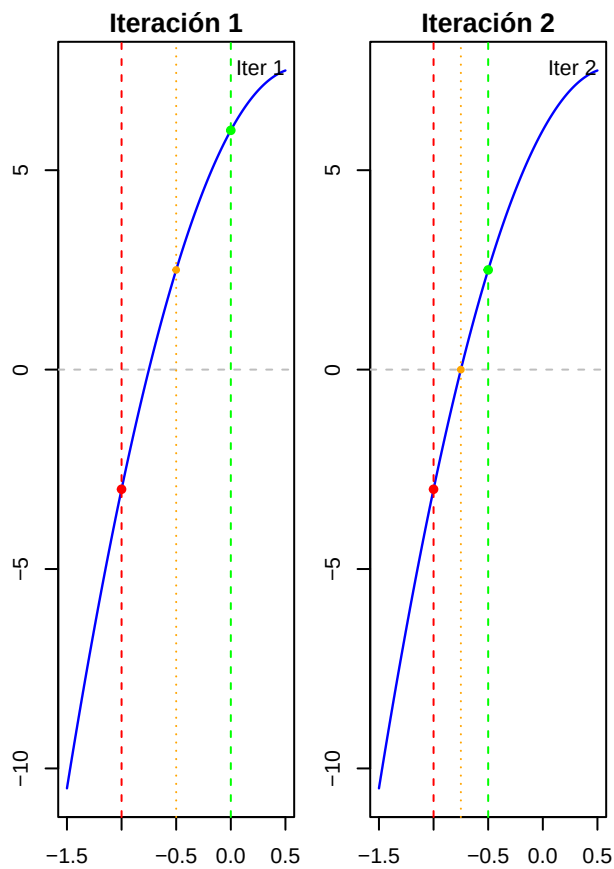
2.4.1.6.3. **Tabla** Registre los valores ak , bk , ck , $f(ak)$, $f(bk)$, $f(ck)$, $signo(f(ak)f(ck))$ y el error $|bk - ak|$.

```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante							
k	a	b	c	f(a)	f(b)	f(c)	Signo(f(a)*f(b))
1	-1.000000	0.000000	-0.500000	-3.000000	6.000000	2.500000	
2	-1.000000	-0.500000	-0.750000	-3.000000	2.500000	0.000000	

2.4.1.6.4. **Gráficas** Grafique la función y los intervalos de reducción en cada paso.

```
grafica(a, b, f, resultado)
```



En este caso, se encontró la raíz exacta en la iteración 2 por lo que graficamente no se puede visualizar pues como tal, no hubo una aproximación.

2.4.1.7. Sistema 9

$f(x) = x^3 - 5x + 1$ en $[0, 1]$.

Datos

```
a <- 0
b <- 1
f <- function(x) x^3 - 5*x + 1
```

2.4.1.7.1. Intervalo Para cada función $f(x)$, encuentre un intervalo $[a, b]$ tal que $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```
## f(a) = f( 0 ) = 1
## f(b) = f( 1 ) = -3
## f(a) · f(b) = -3
## -3 < 0
##
## Sí cumple la condición.
## [1] TRUE
```

2.4.1.7.2. Iteraciones Realice al menos 8 iteraciones del método de bisección.

```
resultado <- biseccion(f, a, b)
```

```
## Raíz aproximada: 0.2016373
```

```
## Iteraciones: 17
```

2.4.1.7.3. Tabla Registre los valores ak , bk , ck , $f(ak)$, $f(bk)$, $f(ck)$, $\text{signo}(f(ak)f(ck))$ y el error $|bk - ak|$.

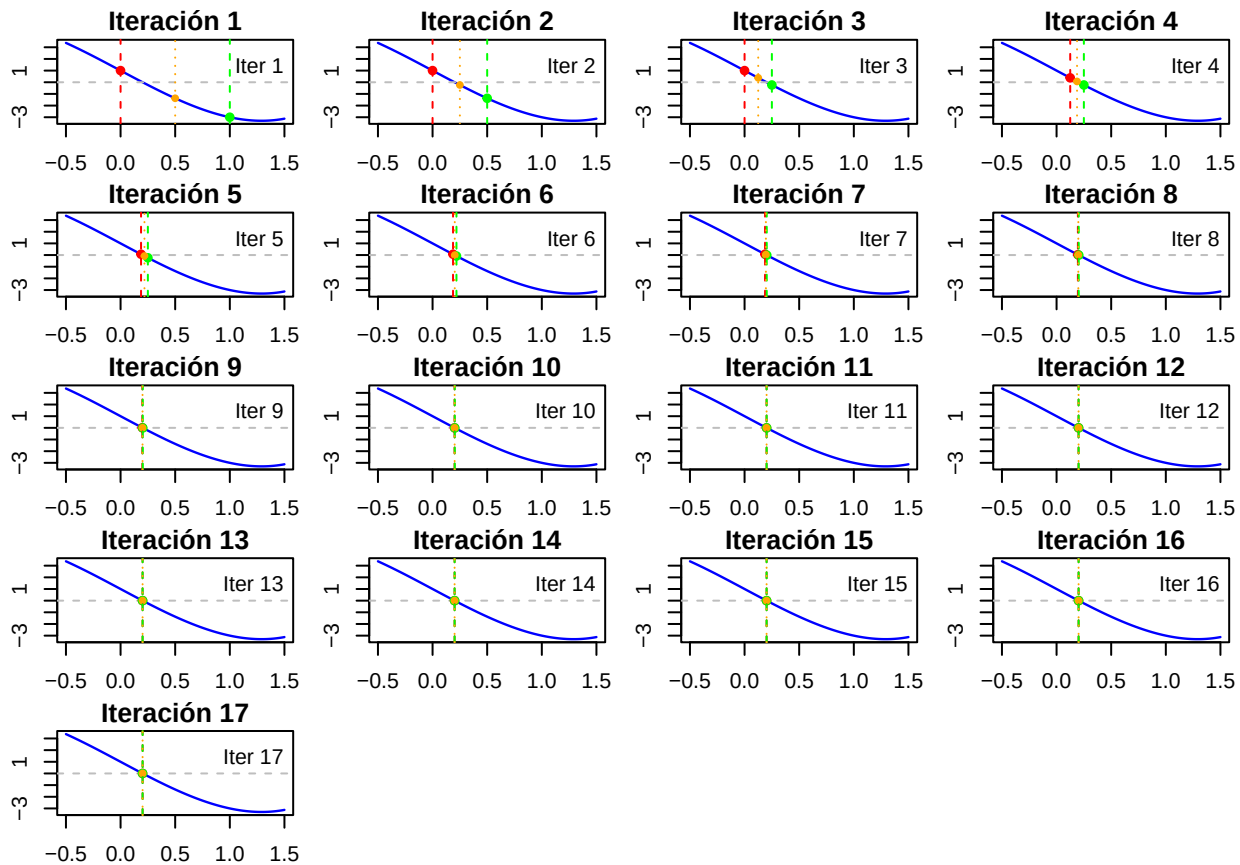
```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	a	b	c	f(a)	f(b)	f(c)	Signo(f(a)*f(c))
1	0.000000	1.000000	0.500000	1.000000	-3.000000	-1.375000	
2	0.000000	0.500000	0.250000	1.000000	-1.375000	-0.234375	
3	0.000000	0.250000	0.125000	1.000000	-0.234375	0.376953	
4	0.125000	0.250000	0.187500	0.376953	-0.234375	0.069092	
5	0.187500	0.250000	0.218750	0.069092	-0.234375	-0.083282	
6	0.187500	0.218750	0.203125	0.069092	-0.083282	-0.007244	
7	0.187500	0.203125	0.195313	0.069092	-0.007244	0.030888	
8	0.195313	0.203125	0.199219	0.030888	-0.007244	0.011813	
9	0.199219	0.203125	0.201172	0.011813	-0.007244	0.002282	
10	0.201172	0.203125	0.202148	0.002282	-0.007244	-0.002482	
11	0.201172	0.202148	0.201660	0.002282	-0.002482	-0.000100	
12	0.201172	0.201660	0.201416	0.002282	-0.000100	-0.001001	

2.4.1.7.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(a, b, f, resultado)
```



2.4.2. Método de la Secante

2.4.2.1. Ejercicios a mano

2.4.2.2. Código para implementación

Método de la secante que guarda los valores para la tabla

```
secante <- function(f, x0, x1, tol = 1e-5, iter_max = 100){
```

Vectores para almacenar los datos de las iteraciones

```
  iter_vec <- numeric(0)
  x0_vec <- numeric(0)
  x1_vec <- numeric(0)
  x2_vec <- numeric(0)
  fx0_vec <- numeric(0)
  fx1_vec <- numeric(0)
  fx2_vec <- numeric(0)
  error_vec <- numeric(0)
```

```
  iter <- 0
  error <- 1
```

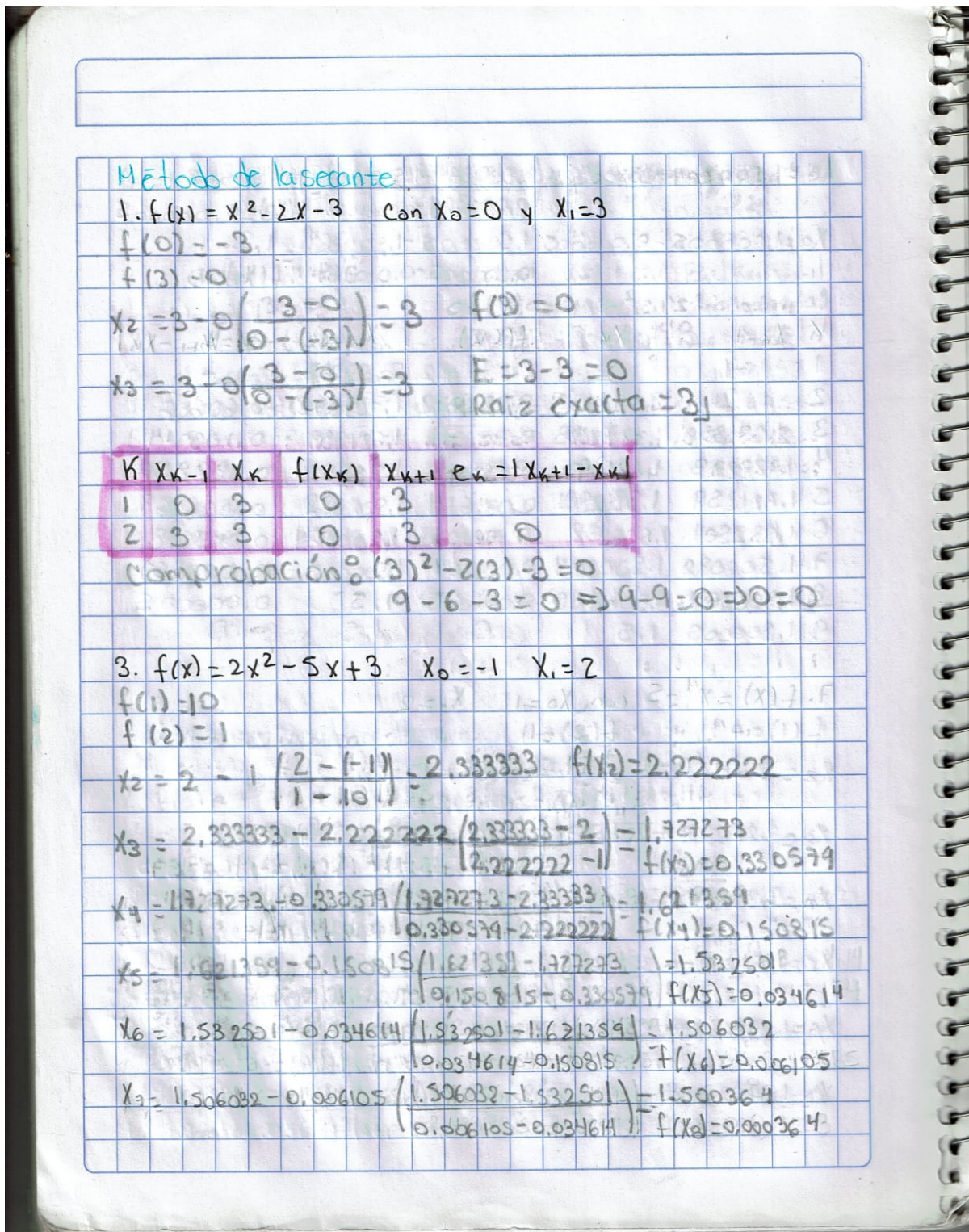


Figura 2.40: Ejercicio 1 y 3

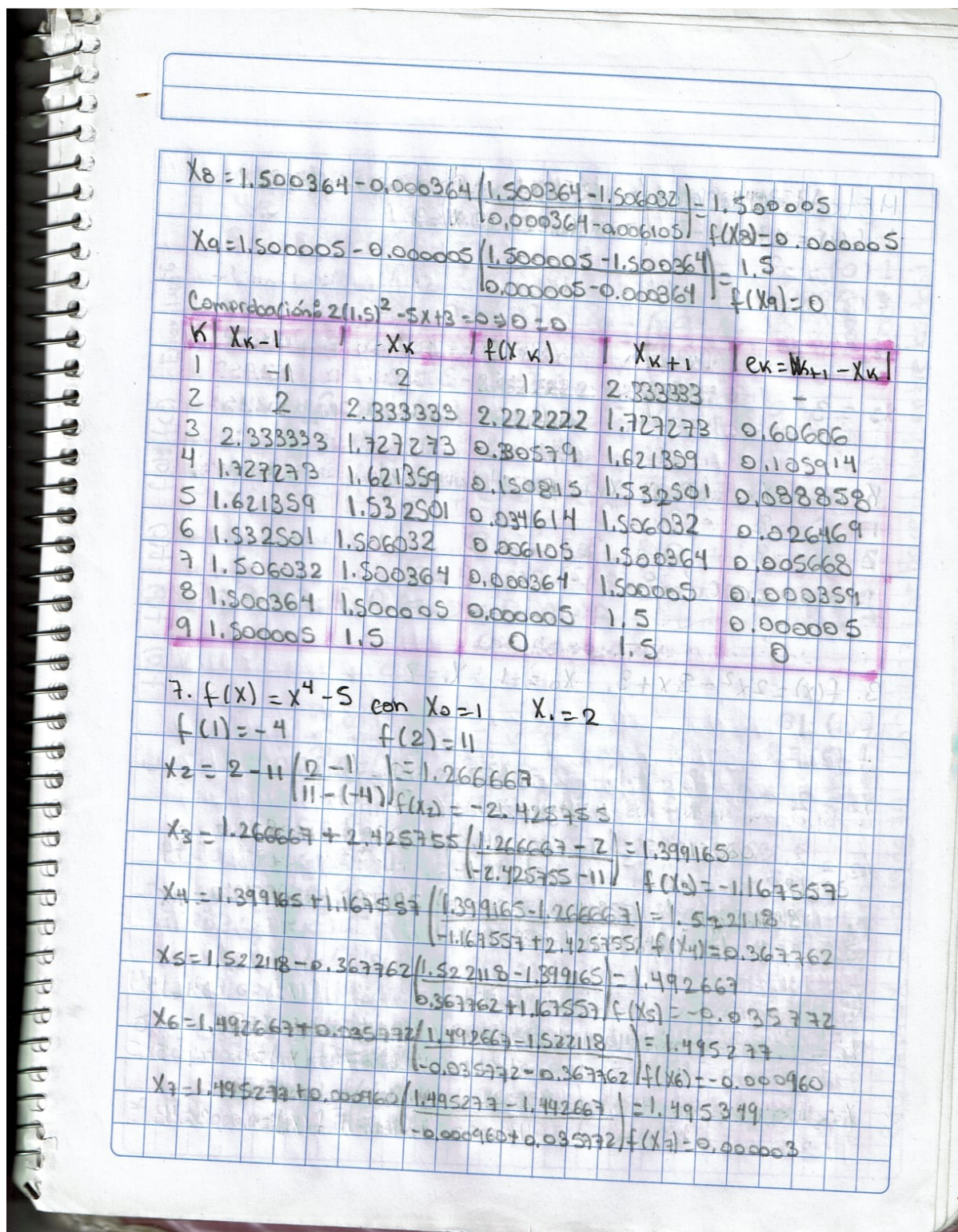


Figura 2.41: Ejercicio 3-2 y 7

$$X_8 = 1.495349 - 0.000003 \frac{1.495349 - 1.495277}{0.000003 + 0.000960} = 1.495349$$

$$0.000003 + 0.000960 \quad f(X_8) = 0.000003$$

k	X_{k-1}	X_k	$f(X_k)$	X_{k+1}	$E_k = X_{k+1} - X_k $
1	1	2	11	1.266667	-
2	2	1.266667	-2.425755	1.399165	0.132498
3	1.266667	1.399165	-1.167557	1.522118	0.122953
4	1.399165	1.522118	0.367762	1.492667	0.029451
5	1.522118	1.492667	-0.035772	1.495277	0.002618
6	1.492667	1.495277	-0.000960	1.495349	0.000072
7	1.495277	1.495349	0.000003	1.495349	0
8	1.495349	1.495349	0.000003	-	-

Comprobación:

$$(1.495349)^4 - 5 = 0 \Rightarrow 0.000003 \approx 0$$

Figura 2.42: Ejercicio 7-2


```

while(error > tol && iter < iter_max){
  fx0 <- f(x0)
  fx1 <- f(x1)

  # Evitar división por cero

  if(abs(fx1 - fx0) < 1e-12){
    cat("Error: División por cero (f(x1) - f(x0) approx 0)\n")
    return(NA)
  }

  x2 <- x1 - fx1 * (x1 - x0) / (fx1 - fx0)
  error <- abs(x2 - x1)

  # Almacena los datos de cada iteración

  iter <- iter + 1
  iter_vec <- c(iter_vec, iter)
  x0_vec <- c(x0_vec, x0)
  x1_vec <- c(x1_vec, x1)
  x2_vec <- c(x2_vec, x2)
  fx0_vec <- c(fx0_vec, fx0)
  fx1_vec <- c(fx1_vec, fx1)
  fx2_vec <- c(fx2_vec, f(x2))
  error_vec <- c(error_vec, error)

  # Actualizar valores para siguiente iteración

  x0 <- x1
  x1 <- x2
}

# Crea la tabla de las iteraciones

TAB <- data.frame(
  "k" = iter_vec,
  "x " = x0_vec,
  "x " = x1_vec,
  "f(x)" = fx1_vec,
  "x " = x2_vec,
  "e = |x - x|" = error_vec,
  check.names = FALSE
)

cat("Raíz aproximada:", x2, "\n")
cat("Iteraciones:", iter, "\n\n")

return(list(raiz = x2, iteraciones = iter, tabla = TAB))
}

# Tabla "bonita"

```

```

tabla <- function(resultado) {
  library(DT)

  datatable(
    resultado$tabla,
    options = list(
      scrollX = TRUE,
      scrollY = "400px",
      paging = FALSE,
      dom = 't',
      autoWidth = FALSE,
      columnDefs = list(
        list(className = 'dt-center', targets = 0:5))
    ),
    rownames = FALSE,
    class = 'cell-border stripe',
    width = "100%",
    caption = "Tabla de iteraciones en el método de la secante",
    escape = FALSE
  ) %>%
  formatRound(columns = c(2, 3, 4, 5, 6), digits = 6)
}

# Iteraciones en el método de bisección

n_biseccion <- function(x0, x1) {
  n <- ceiling(log(abs(x0 - x1) / 1e-5) / log(2))
  return(n)
}

# Gráficas

grafica <- function(x0, x1, f, resultado) {
  tabla_iteraciones <- resultado$tabla
  total_iteraciones <- nrow(tabla_iteraciones)
  filas <- ceiling(total_iteraciones / 4)

  par(mfrow = c(filas, 4), mar = c(2, 2, 1.5, 1))

  # Calcular rango dinámico

  x_min <- min(tabla_iteraciones[,2]) - 0.5
  x_max <- max(tabla_iteraciones[,3]) + 0.5
  x_vals <- seq(x_min, x_max, length.out = 1000)
  y_vals <- f(x_vals)

  for (i in 1:total_iteraciones) {
    plot(x_vals, y_vals, type = "l", main = paste("Iteración", i),
      xlab = "x", ylab = "f(x)", col = "blue", lwd = 1.2)
    abline(h = 0, col = "gray", lty = 2)

    x_anterior <- tabla_iteraciones[i, 2] #  $x_{k-1}$ 
  }
}

```

```

x_actual <- tabla_iteraciones[i, 3]      #  $x_k$ 
x_siguiente <- tabla_iteraciones[i, 4]   #  $x_{k+1}$ 

# Calcula la pendiente para la gráfica

f_anterior <- f(x_anterior)
f_actual <- f(x_actual)
pendiente <- (f_actual - f_anterior) / (x_actual - x_anterior)

# Grafica la línea secante

x_secante <- seq(x_min, x_max, length.out = 100)
y_secante <- f_anterior + pendiente * (x_secante - x_anterior)
lines(x_secante, y_secante, col = "purple", lty = 2, lwd = 1.5)

# Puntos de la iteración

points(c(x_anterior, x_actual), c(f_anterior, f_actual),
       col = c("red", "green"), pch = 19, cex = 0.8)

# Punto de intersección con eje x ( $x_{k+1}$ )

points(x_siguiente, 0, col = "orange", pch = 17, cex = 1.2)

# Líneas verticales para los puntos importantes

abline(v = x_anterior, col = "red", lty = 3, lwd = 0.8)
abline(v = x_actual, col = "green", lty = 3, lwd = 0.8)
abline(v = x_siguiente, col = "orange", lty = 3, lwd = 1.2)

# Leyenda explicativa

legend("topright",
      legend = c("f(x)", "Secante", "x_{k-1}", "x_k", "x_{k+1}"),
      col = c("blue", "purple", "red", "green", "orange"),
      lty = c(1, 2, NA, NA, NA), pch = c(NA, NA, 19, 19, 17),
      cex = 0.7, bty = "n")
}

par(mfrow = c(1, 1))
}

```

2.4.2.3. Sistema 2

$$f(x) = x^4 - 10x^2 + 9$$

$$x_0 = 0$$

$$x_1 = 2$$

Datos

$x_0 <- 0$

```
x1 <- 2
f <- function(x) x^4 - 10*x^2 + 9
```

2.4.2.3.1. Puntos iniciales Para cada función $f(x)$, elija dos puntos iniciales x_0, x_1 cercanos a la raíz.

Nota: En la información dada ya nos da los puntos iniciales:

$$x_0 = 0$$

$$x_1 = 2$$

2.4.2.3.2. Iteraciones Aplique iterativamente: $x_{k+1} = x_k - f(x_k) \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$

```
resultado <- secante(f, x0, x1)
```

```
## Raíz aproximada: 1
```

```
## Iteraciones: 5
```

2.4.2.3.3. Comparación Compare el número de iteraciones con el método de bisección.

```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	x_{k-1}	x_k	$f(x_k)$	x_{k+1}	$e_k =$ $ x_{k+1} - x_k $
1	0.000000	2.000000	-15.000000	0.750000	1.250000
2	2.000000	0.750000	3.691406	0.996865	0.246865
3	0.750000	0.996865	0.050117	1.000263	0.003398
4	0.996865	1.000263	-0.004208	1.000000	0.000263
5	1.000263	1.000000	0.000003	1.000000	0.000000

```
t_biseccion <- n_biseccion(x0, x1)
cat("En el método de bisección:", t_biseccion, "iteraciones\n")

## En el método de bisección: 18 iteraciones

cat("En el método de la secante:", resultado$iteraciones, "iteraciones\n")

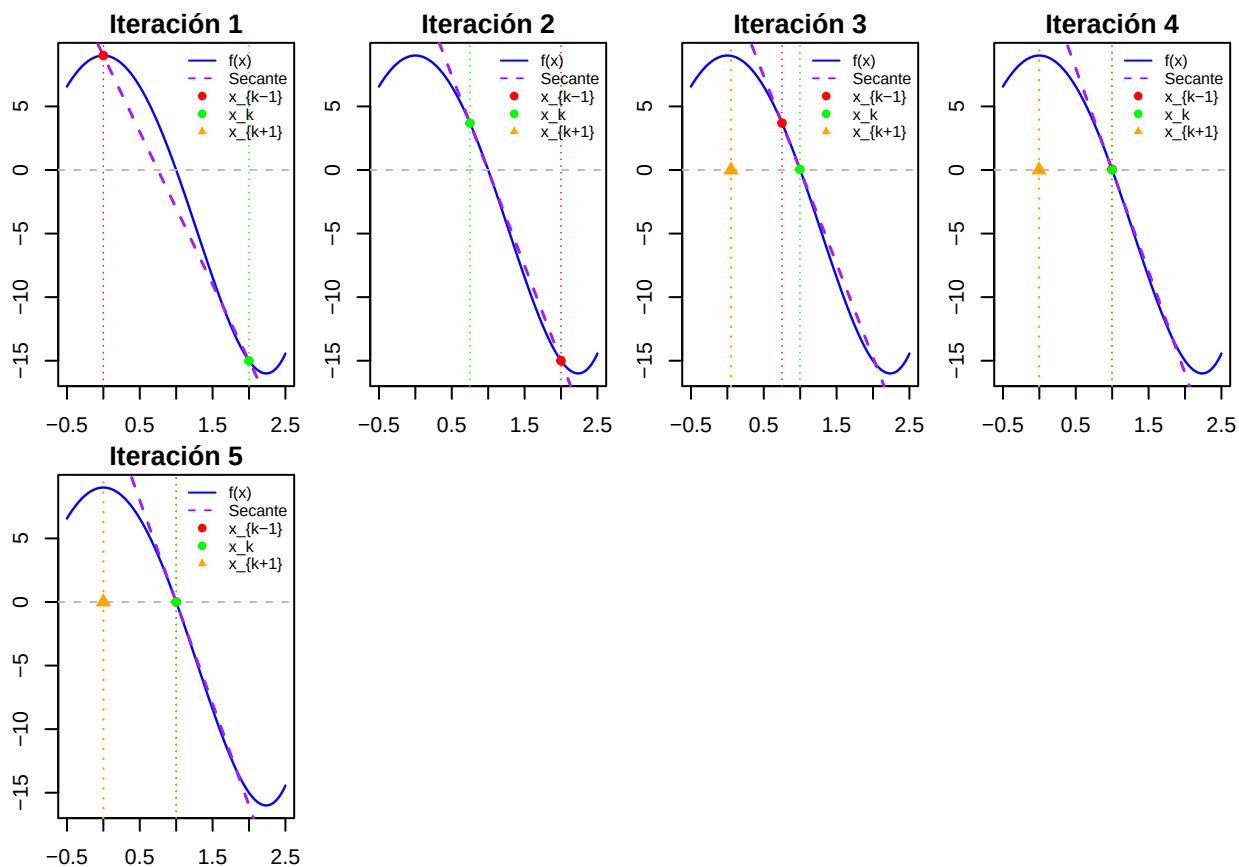
## En el método de la secante: 5 iteraciones

cat("Ventaja:", t_biseccion - resultado$iteraciones, "iteraciones menos\n")

## Ventaja: 13 iteraciones menos
```

2.4.2.3.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(x0, x1, f, resultado)
```



2.4.2.4. Sistema 4

$$f(x) = -3x^2 + 4x + 2$$

$$x_0 = -1$$

$$x_1 = 3$$

Datos

```
x0 <- -1
```

```
x1 <- 3
f <- function(x) -3*x^2 + 4*x + 2
```

2.4.2.4.1. Puntos iniciales Para cada función $f(x)$, elija dos puntos iniciales x_0, x_1 cercanos a la raíz.

Nota: En la información dada ya nos da los puntos iniciales:

$$x_0 = -1$$

$$x_1 = 3$$

2.4.2.4.2. Iteraciones Aplique iterativamente: $x_{k+1} = x_k - f(x_k) \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$

```
resultado <- secante(f, x0, x1)
```

```
## Raíz aproximada: 1.720759
```

```
## Iteraciones: 14
```

2.4.2.4.3. Comparación Compare el número de iteraciones con el método de bisección.

```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	x_{k-1}	x_k	$f(x_k)$	x_{k+1}	$e_k = x_{k+1} - x_k $
1	-1.000000	3.000000	-13.000000	-3.500000	6.500000
2	3.000000	-3.500000	-48.750000	5.363636	8.863636
3	-3.500000	5.363636	-62.851240	-34.142857	39.506494
4	5.363636	-34.142857	-3,631.775510	6.059373	40.202230
5	-34.142857	6.059373	-83.910518	7.010196	0.950823
6	6.059373	7.010196	-117.387750	3.676141	3.334055
7	7.010196	3.676141	-23.837477	2.826593	0.849548
8	3.676141	2.826593	-10.662514	2.139053	0.687540
9	2.826593	2.139053	-3.170430	1.848106	0.290947
10	2.139053	1.848106	-0.854063	1.740832	0.107274
11	1.848106	1.740832	-0.128157	1.721892	0.018939
12	1.740832	1.721892	-0.007171	1.720759	0.001133


```

t_biseccion <- n_biseccion(x0, x1)
cat("En el método de bisección:", t_biseccion, "iteraciones\n")

## En el método de bisección: 19 iteraciones

cat("En el método de la secante:", resultado$iteraciones, "iteraciones\n")

## En el método de la secante: 14 iteraciones

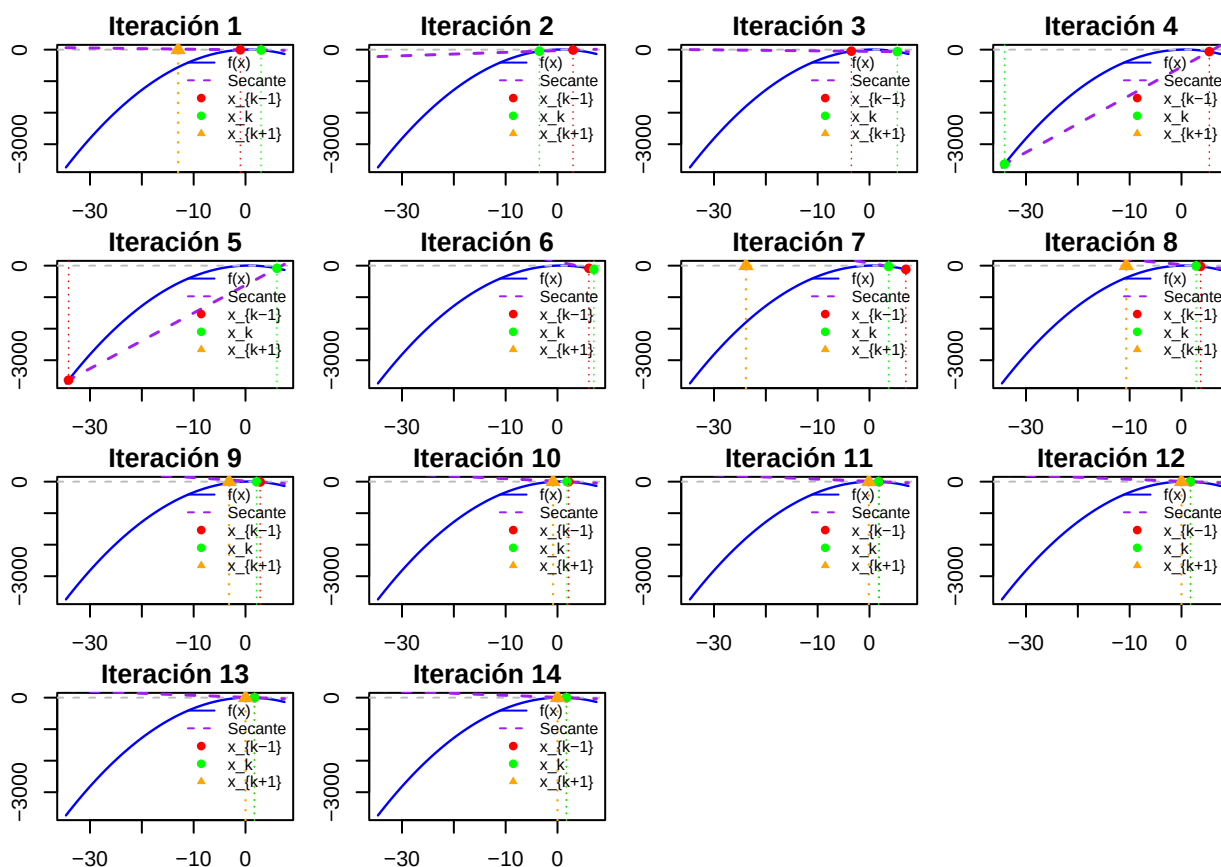
cat("Ventaja:", t_biseccion - resultado$iteraciones, "iteraciones menos\n")

## Ventaja: 5 iteraciones menos

```

2.4.2.4.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(x0, x1, f, resultado)
```



2.4.2.5. Sistema 6

$$f(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$$

$$x_0 = 0$$

$$x_1 = 3$$

Datos

```
x0 <- 0
```

```
x1 <- 3
f <- function(x) x^4 - 5*x^3 + 6*x^2 + 4*x - 8
```

2.4.2.5.1. Puntos iniciales Para cada función $f(x)$, elija dos puntos iniciales x_0, x_1 cercanos a la raíz.

Nota: En la información dada ya nos da los puntos iniciales:

$$x_0 = 0$$

$$x_1 = 3$$

2.4.2.5.2. Iteraciones Aplique iterativamente: $x_{k+1} = x_k - f(x_k) \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$

```
resultado <- secante(f, x0, x1)
```

```
## Raíz aproximada: 2
```

```
## Iteraciones: 2
```

2.4.2.5.3. Comparación Compare el número de iteraciones con el método de bisección.

```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k ♦	x_{k-1} ♦	x_k ♦	$f(x_k)$ ♦	x_{k+1} ♦	$e_k =$ $ x_{k+1} - x_k $ ♦
1	0.000000	3.000000	3.000000	4.000000	2.000000
2	3.000000	2.000000	2.000000	0.000000	2.000000

```
t_biseccion <- n_biseccion(x0, x1)
cat("En el método de bisección:", t_biseccion, "iteraciones\n")

## En el método de bisección: 19 iteraciones

cat("En el método de la secante:", resultado$iteraciones, "iteraciones\n")

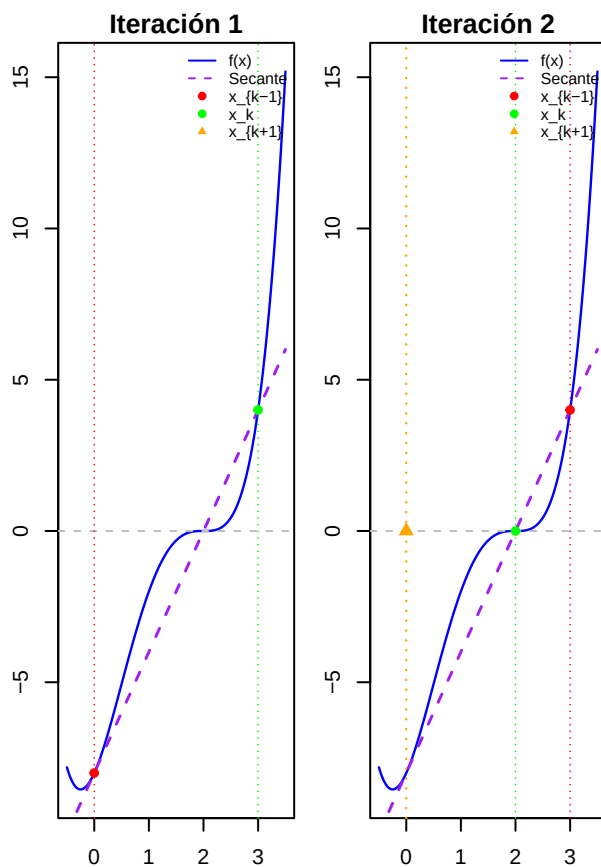
## En el método de la secante: 2 iteraciones

cat("Ventaja:", t_biseccion - resultado$iteraciones, "iteraciones menos\n")

## Ventaja: 17 iteraciones menos
```

2.4.2.5.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(x0, x1, f, resultado)
```



2.4.2.6. Sistema 8

$$f(x) = -2x^4 + 3x^3 + 7x^2 - 5x + 1$$

$$x_0 = -1$$

$$x_1 = -2$$

Datos

```
x0 <- -1
```

```
x1 <- -2
f <- function(x) -2*x^4 + 3*x^3 + 7*x^2 - 5*x + 1
```

2.4.2.6.1. Puntos iniciales Para cada función $f(x)$, elija dos puntos iniciales x_0, x_1 cercanos a la raíz.

Nota: En la información dada ya nos da los puntos iniciales:

$$x_0 = -1$$

$$x_1 = -2$$

2.4.2.6.2. Iteraciones Aplique iterativamente: $x_{k+1} = x_k - f(x_k) \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$

```
resultado <- secante(f, x0, x1)
```

```
## Raíz aproximada: -1.65022
```

```
## Iteraciones: 8
```

2.4.2.6.3. Comparación Compare el número de iteraciones con el método de bisección.

```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	x_{k-1}	x_k	$f(x_k)$	x_{k+1}	$e_k = x_{k+1} - x_k $
1	-1.000000	-2.000000	-17.000000	-1.320000	0.680000
2	-2.000000	-1.320000	6.824980	-1.514795	0.194795
3	-1.320000	-1.514795	3.678225	-1.742490	0.227695
4	-1.514795	-1.742490	-3.343546	-1.634069	0.108421
5	-1.742490	-1.634069	0.512104	-1.648469	0.014400
6	-1.634069	-1.648469	0.056527	-1.650256	0.001787
7	-1.648469	-1.650256	-0.001164	-1.650220	0.000036
8	-1.650256	-1.650220	0.000003	-1.650220	0.000000

```
t_biseccion <- n_biseccion(x0, x1)
cat("En el método de bisección:", t_biseccion, "iteraciones\n")

## En el método de bisección: 17 iteraciones

cat("En el método de la secante:", resultado$iteraciones, "iteraciones\n")

## En el método de la secante: 8 iteraciones

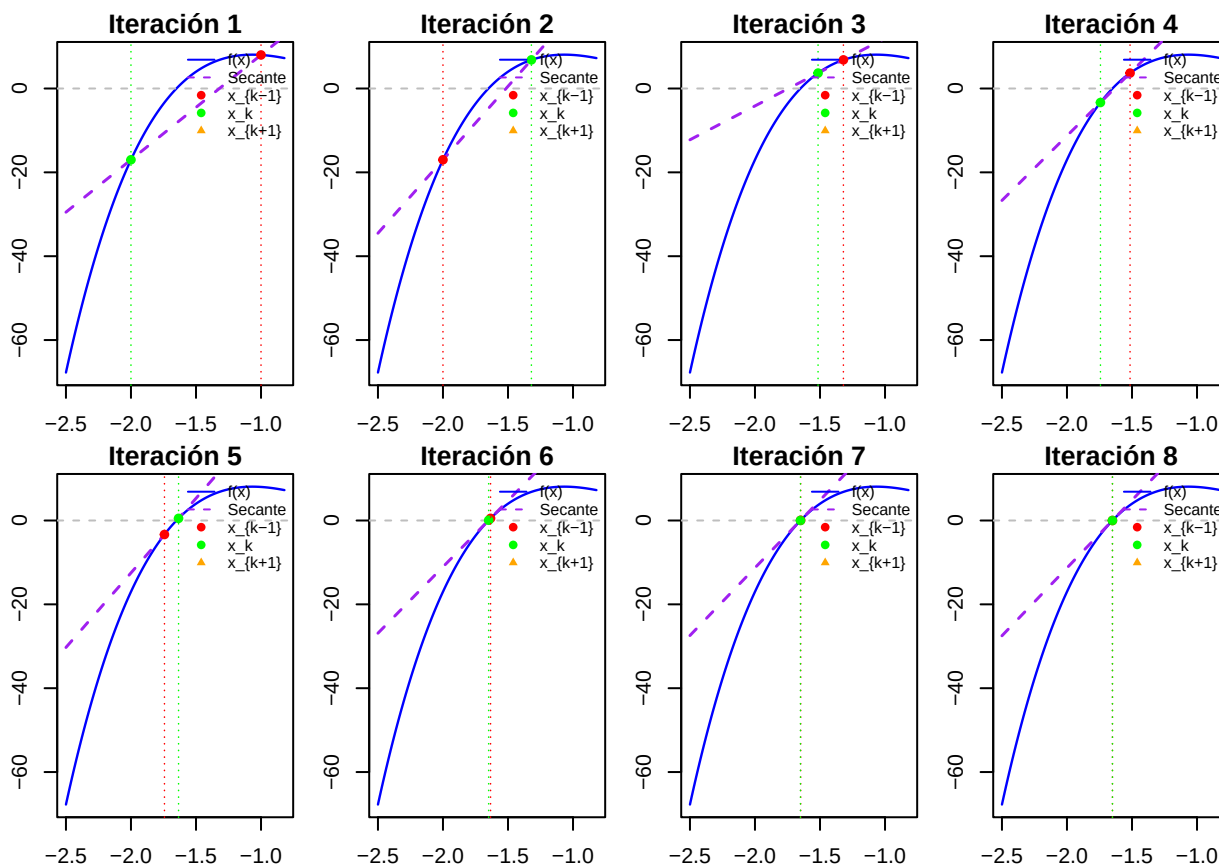
cat("Ventaja:", t_biseccion - resultado$iteraciones, "iteraciones menos\n")

## Ventaja: 9 iteraciones menos
```

En este caso, este método no es muy conveniente.

2.4.2.6.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(x0, x1, f, resultado)
```



2.4.2.7. Sistema 12

$$f(x) = (3x + 3,1)\cos(2x + \frac{\pi}{6})$$

$$x_0 = 0,3$$

$$x_1 = 0,8$$

```
# Datos
x0 <- 0.3
x1 <- 0.8
f <- function(x) (3*x + 3.1) * cos(2*x + (pi/6))
```

2.4.2.7.1. Puntos iniciales Para cada función $f(x)$, elija dos puntos iniciales x_0, x_1 cercanos a la raíz.

Nota: En la información dada ya nos da los puntos iniciales:

$$x_0 = 0,3$$

$$x_1 = 0,8$$

2.4.2.7.2. Iteraciones Aplique iterativamente: $x_{k+1} = x_k - f(x_k) \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$

```
resultado <- secante(f, x0, x1)
```

```
## Raíz aproximada: 0.5235988
## Iteraciones: 5
```

2.4.2.7.3. Comparación Compare el número de iteraciones con el método de bisección.

```
tabla(resultado)
```

Tabla de iteraciones en el método de la secante

k	x_{k-1}	x_k	$f(x_k)$	x_{k+1}	$e_k = x_{k+1} - x_k $
1	0.300000	0.800000	-2.887909	0.487298	0.312702
2	0.800000	0.487298	0.330909	0.519445	0.032147
3	0.487298	0.519445	0.038697	0.523702	0.004257
4	0.519445	0.523702	-0.000968	0.523598	0.000104
5	0.523702	0.523598	0.000003	0.523599	0.000000

```

t_biseccion <- n_biseccion(x0, x1)
cat("En el método de bisección:", t_biseccion, "iteraciones\n")

## En el método de bisección: 16 iteraciones
cat("En el método de la secante:", resultado$iteraciones, "iteraciones\n")

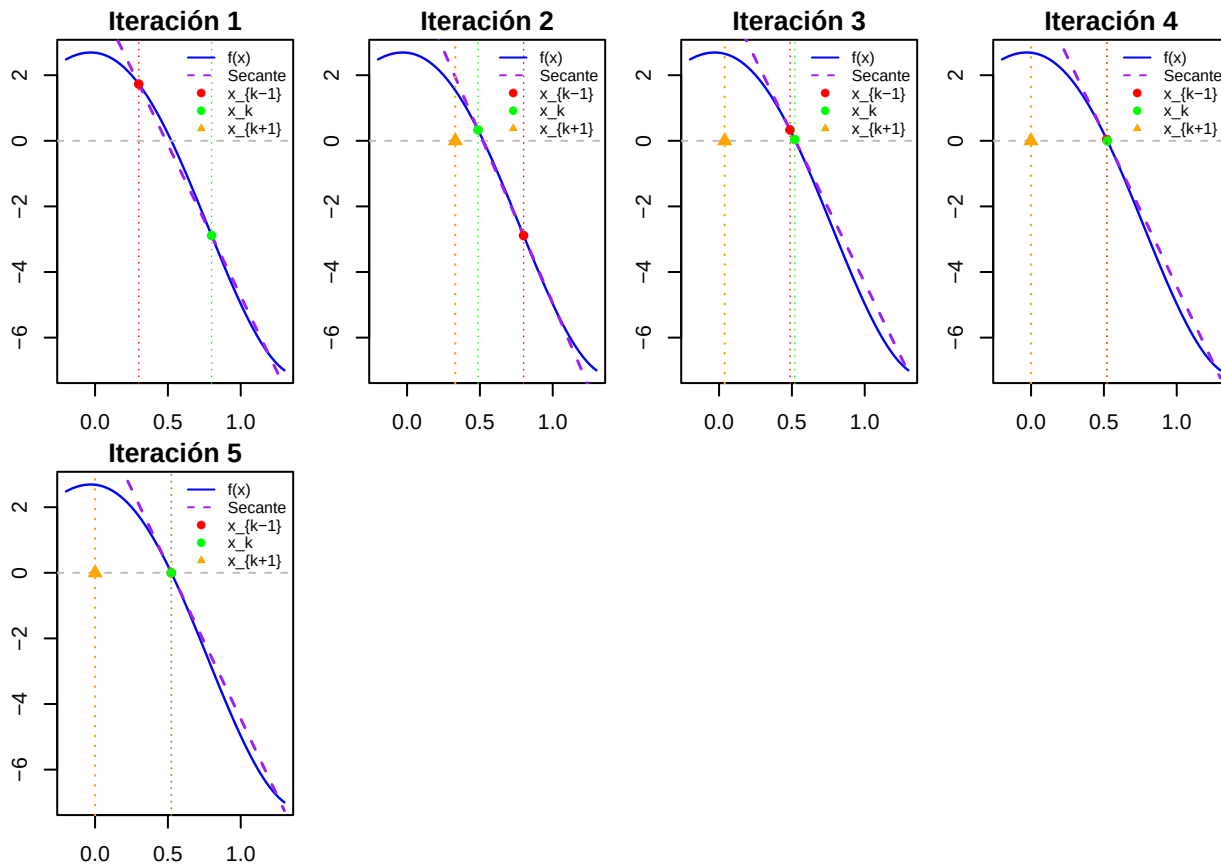
## En el método de la secante: 5 iteraciones
cat("Ventaja:", t_biseccion - resultado$iteraciones, "iteraciones menos\n")

## Ventaja: 11 iteraciones menos

```

2.4.2.7.4. Gráficas Grafique la función y los intervalos de reducción en cada paso.

```
grafica(x0, x1, f, resultado)
```



2.4.3. Método de Newton-Raphson

2.4.3.1. Ejercicios a mano

2.4.3.2. Código para implementación

Condición para los intervalos

```
condicion <- function(a, b, f, inicio = 0, fin = 10, paso = 1) {
  fa <- f(a)
  fb <- f(b)
  producto <- fa * fb
```

Resultados

```
cat("f(a) = f(", a, ") = ", fa, "\n")
cat("f(b) = f(", b, ") = ", fb, "\n")
cat("f(a) · f(b) = ", producto, "\n")
cat(producto, "< 0", "\n\n")
```

```
if(producto < 0) {
  cat("Sí cumple la condición.\n")
  # Asignar globalmente
  a <- a
  b <- b
```


Método de Newton-Raphson

2. $f(x) = x^3 + 2x^2 - 5$ $[a, b] = [1, 2]$

Verificamos condición:

$f(a) \cdot f(b) < 0$	$f(1) \cdot f(2) < 0$	$f'(x) = 3x^2 + 4x$
$f(1) = -2$	$(-2)(11) < 0$	$X_0 = \frac{1+2}{2} = 1.5$
$f(2) = 11$	$-22 < 0$	

$f(1.5) = 2.875$	$f'(1.5) = 12.75$	
$X_1 = 1.5 - \frac{2.875}{12.75} = 1.274510$	$f(X_1) = 0.319035$	
	$f'(X_1) = 9.971167$	
$X_2 = 1.274510 - \frac{0.319035}{9.971167} = 1.272514$	$f(X_2) = 0.005926$	
	$f'(X_2) = 9.601579$	

Figura 2.43: Ejercicio 2-1

```

return(TRUE)
} else {
  cat("NO cumple la condición.\n")
  cat("Se buscará un nuevo intervalo.\n\n")

  # Buscar automáticamente un intervalo que sí funcione
  x <- seq(inicio, fin, paso)
  for (i in 1:(length(x)-1)) {
    a_nuevo <- x[i]
    b_nuevo <- x[i+1]
    if (f(a_nuevo) * f(b_nuevo) < 0) {
      cat("Nuevo intervalo encontrado: [", a_nuevo, ", ", b_nuevo, "]\n")
      fa <- f(a_nuevo)
      fb <- f(b_nuevo)
      producto <- fa * fb
      cat("f(a) = f(", a_nuevo, ") = ", fa, "\n")
      cat("f(b) = f(", b_nuevo, ") = ", fb, "\n")
      cat("f(a) · f(b) = ", producto, "\n")
      cat(producto, "< 0", "\n\n")
      cat("Sí cumple la condición.\n")
      # Asignar globalmente
      a <- a_nuevo
      b <- b_nuevo
      return(TRUE)
    }
  }
}

```

$x_3 = 1.242514 - \frac{0.000004}{9.601579} = 1.241897$	$f(x_3) = 0.000004$ $f'(x_3) = 9.594512$
$x_4 = 1.241897 - \frac{0.000004}{9.594512} = 1.241897$	$f(x_4) = 0.000004$ $f'(x_4) = 9.594512$

k	x_k	$f(x_k)$	$f'(x_k)$	$ x_{k+1} - x_k $
1	1.5	2.875	12.75	1.274510
2	1.274510	0.319035	9.971167	0.031996
3	1.242514	0.005926	9.601579	0.000617
4	1.241897	0.00004	9.594512	0

4. $f(x) = \sin(x) - \frac{x}{3}$, $[a, b] = [0, 2]$
 Verificamos la condición:
 $f(a) \cdot f(b) < 0$ $f(0) \cdot f(2) < 0$
 $f(0) = 0$ $(0)(0.242631) < 0$
 $f(2) = 0.242631$ $0 < 0$

No se cumple la condición, además en $f(0)$ el resultado ya es 0 por lo que es una raíz, además en el intervalo dado, es única raíz, así que aunque se quisiera buscar otra, en este intervalo, no la hay.

8. $f(x) = \cos(x) - 2x$, $[a, b] = [0, 1]$
 Verificamos condición:
 $f(a) \cdot f(b) < 0$ $f(0) \cdot f(1) < 0$
 $f(0) = 1$ $(1)(-1.459698) < 0$
 $f(1) = -1.459698$ $-1.459698 < 0$

$x_1 = 0.5 - \frac{-0.122417}{-2.479426} = 0.450627$
 $x_2 = 0.450627 - \frac{-0.001080}{-2.435530} = 0.450184$

$f'(x) = -\sin(x) - 2$
 $x_0 = \frac{0+1}{2} = \frac{1}{2} = 0.5$
 $f(0.5) = -0.122417$
 $f'(0.5) = -2.479426$
 $f(x_1) = -0.001080$
 $f'(x_1) = -2.435530$
 $f(x_2) = -0.000001$
 $f'(x_2) = -2.435301$

k	x_k	$f(x_k)$	$f'(x_k)$	$ x_{k+1} - x_k $
1	0.5	-0.122417	-2.435131	0.049373
2	0.450629	-0.001080	-2.435131	0.000443
3	0.450184	-0.00001	-2.435131	0

Figura 2.45: Ejercicio 8-2

```

cat("No se encontró ningún intervalo con cambio de signo\n")
return(FALSE)
}
}

```

Método de Newton que guarda los valores para la tabla

```
newton <- function(f, df, x0, tol = 1e-5, iter_max = 100){
```

Vectores para almacenar los datos de las iteraciones

```

iter_vec <- numeric(0)
x0_vec <- numeric(0)
x1_vec <- numeric(0)
fx0_vec <- numeric(0)
dfx0_vec <- numeric(0)
error_vec <- numeric(0)

```

```

iter <- 0
error <- 1

```

```
while(error > tol && iter < iter_max){
```

```

  fx0 <- f(x0)
  dfx0 <- df(x0)

```

Evitar división por cero

```

  if(abs(dfx0) < 1e-12){
    cat("Error: Derivada cercana a cero\n")
    return(NA)
  }

```

```

    }

# Aplicar fórmula de Newton

    x1 <- x0 - fx0 / dfx0
    error <- abs(x1 - x0)

# Almacena los datos de cada iteración

    iter <- iter + 1
    iter_vec <- c(iter_vec, iter)
    x0_vec <- c(x0_vec, x0)
    fx0_vec <- c(fx0_vec, fx0)
    dfx0_vec <- c(dfx0_vec, dfx0)
    error_vec <- c(error_vec, error)

# Actualizar para siguiente iteración

    x0 <- x1
  }

# Crea la tabla de las iteraciones

  TAB <- data.frame(
    "k" = iter_vec,
    "x" = x0_vec,
    "f(x)" = fx0_vec,
    "f'(x)" = dfx0_vec,
    "e = |x - x|" = error_vec,
    check.names = FALSE
  )

  cat("Raíz aproximada:", x1, "\n")
  cat("Iteraciones:", iter, "\n\n")

  return(list(raiz = x1, iteraciones = iter, tabla = TAB))
}

```

```

# Tabla "bonita"

```

```

tabla <- function(resultado) {
  library(DT)

  datatable(
    resultado$tabla,
    options = list(
      scrollX = TRUE,
      scrollY = "400px",
      paging = FALSE,
      dom = 't',
      autoWidth = FALSE,
      columnDefs = list(

```

```

    list(className = 'dt-center', targets = 0:4))
  ),
  rownames = FALSE,
  class = 'cell-border stripe',
  width = "100%",
  caption = "Tabla de iteraciones en el método de la Newton-Raphson",
  escape = FALSE
) %>%
  formatRound(columns = c(2:5), digits = 9)
}

```

Gráficas

```

grafica <- function(x0, f, df, resultado) {
  tabla_iteraciones <- resultado$tabla
  total_iteraciones <- nrow(tabla_iteraciones)
  filas <- ceiling(total_iteraciones / 4)

  par(mfrow = c(filas, 4), mar = c(2, 2, 1.5, 1))

  # Calcular rango dinámico

  x_min <- min(tabla_iteraciones[,2]) - 0.5
  x_max <- max(tabla_iteraciones[,2]) + 0.5
  x_vals <- seq(x_min, x_max, length.out = 1000)
  y_vals <- f(x_vals)

  for (i in 1:total_iteraciones) {
    plot(x_vals, y_vals, type = "l", main = paste("Iteración", i),
         xlab = "x", ylab = "f(x)", col = "blue", lwd = 1.2)
    abline(h = 0, col = "gray", lty = 2)

    x_actual <- tabla_iteraciones[i, 2] #  $x_k$ 
    f_actual <- tabla_iteraciones[i, 3] #  $f(x_k)$ 
    df_actual <- tabla_iteraciones[i, 4] #  $f'(x_k)$ 

    # Calcular recta tangente

    pendiente <- df_actual
    intercepto <- f_actual - pendiente * x_actual
    y_tangente <- pendiente * x_vals + intercepto

    # Graficar recta tangente

    lines(x_vals, y_tangente, col = "purple", lty = 2, lwd = 1.5)

    # Puntos de la iteración

    points(x_actual, f_actual, col = "red", pch = 19, cex = 0.8)

    # Línea vertical

```

```

abline(v = x_actual, col = "red", lty = 3, lwd = 0.8)

# Leyenda

legend("topright",
      legend = c("f(x)", "Tangente", "x_k"),
      col = c("blue", "purple", "red"),
      lty = c(1, 2, NA), pch = c(NA, NA, 19),
      cex = 0.7, bty = "n")
}

par(mfrow = c(1, 1))
}

```

2.4.3.3. Sistema 1

$f(x) = x^3 - 7x + 6$ en $[a, b] = [0, 1]$

Datos

```

a <- 0
b <- 1
x0 <- (a + b) / 2
f <- function(x) x^3 - 7*x + 6
df <- function(x) 3*x^2 - 7

```

2.4.3.3.1. Intervalo Verifica que exista un intervalo $[a, b]$ donde $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```

## f(a) = f( 0 ) = 6
## f(b) = f( 1 ) = 0
## f(a) · f(b) = 0
## 0 < 0
##
## NO cumple la condición.
## Se buscará un nuevo intervalo.
##
## No se encontró ningún intervalo con cambio de signo
## [1] FALSE

```

En este caso la raíz está justo en el extremo del intervalo en $x = 1$ por lo cual, la condición da exactamente 0.

2.4.3.3.2. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- newton(f, df, x0)
```

```

## Raíz aproximada: 1
## Iteraciones: 5

```

2.4.3.3.3. Tabla Registre los valores x_k , $f(x_k)$, $f'(x_k)$, y el error $|b_k - a_k|$.

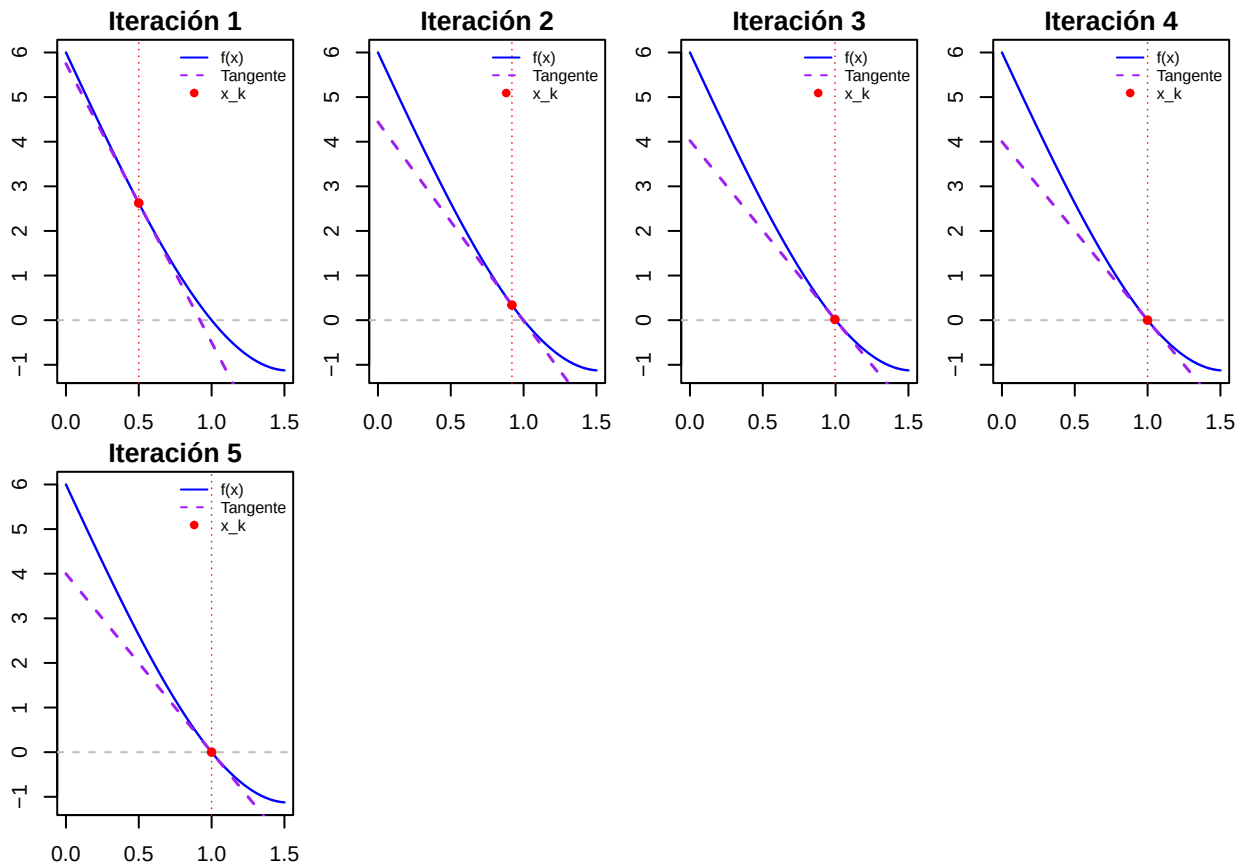
```
tabla(resultado)
```

Tabla de iteraciones en el método de la Newton-Raphson

k	x_k	$f(x_k)$	$f'(x_k)$	$e_k = x_{k+1} - x_k $
1	0.500000000	2.625000000	-6.250000000	0.420000000
2	0.920000000	0.338688000	-4.460800000	0.075925395
3	0.995925395	0.016348161	-4.024397825	0.004062263
4	0.999987657	0.000049371	-4.000074056	0.000012343
5	1.000000000	0.000000000	-4.000000001	0.000000000

2.4.3.3.4. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(x0, f, df, resultado)
```



2.4.3.4. Sistema 3

$$f(x) = x^4 - 3x^3 + 2 \text{ en } [a, b] = [0, 1]$$

Datos

```
a <- 0
b <- 1
x0 <- (a + b) / 2
f <- function(x) x^4 - 3*x^3 + 2
df <- function(x) 4*x^3 - 9*x^2
```

2.4.3.4.1. Intervalo Verifica que exista un intervalo $[a, b]$ donde $f(a) \cdot f(b) < 0$.

Nota: En la información dada ya nos da un intervalo, por lo que solo se revisa si cumple con la condición:

```
condicion(a, b, f)
```

```
## f(a) = f( 0 ) = 2
## f(b) = f( 1 ) = 0
## f(a) · f(b) = 0
## 0 < 0
##
## NO cumple la condición.
## Se buscará un nuevo intervalo.
##
```



```
## Nuevo intervalo encontrado: [ 2 , 3 ]
## f(a) = f( 2 ) = -6
## f(b) = f( 3 ) = 2
## f(a) · f(b) = -12
## -12 < 0
##
## Sí cumple la condición.
## [1] TRUE
```

En este caso la raíz está justo en el extremo del intervalo en $x = 1$ por lo cual, la condición da exactamente 0. Y al buscar el nuevo intervalo, nos da aquel en donde se encuentra otra de las raíces.

2.4.3.4.2. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- newton(f, df, x0)
```

```
## Raíz aproximada: 1
## Iteraciones: 5
```

2.4.3.4.3. Tabla Registre los valores x_k , $f(x_k)$, $f'(x_k)$, y el error $|b_k - a_k|$.

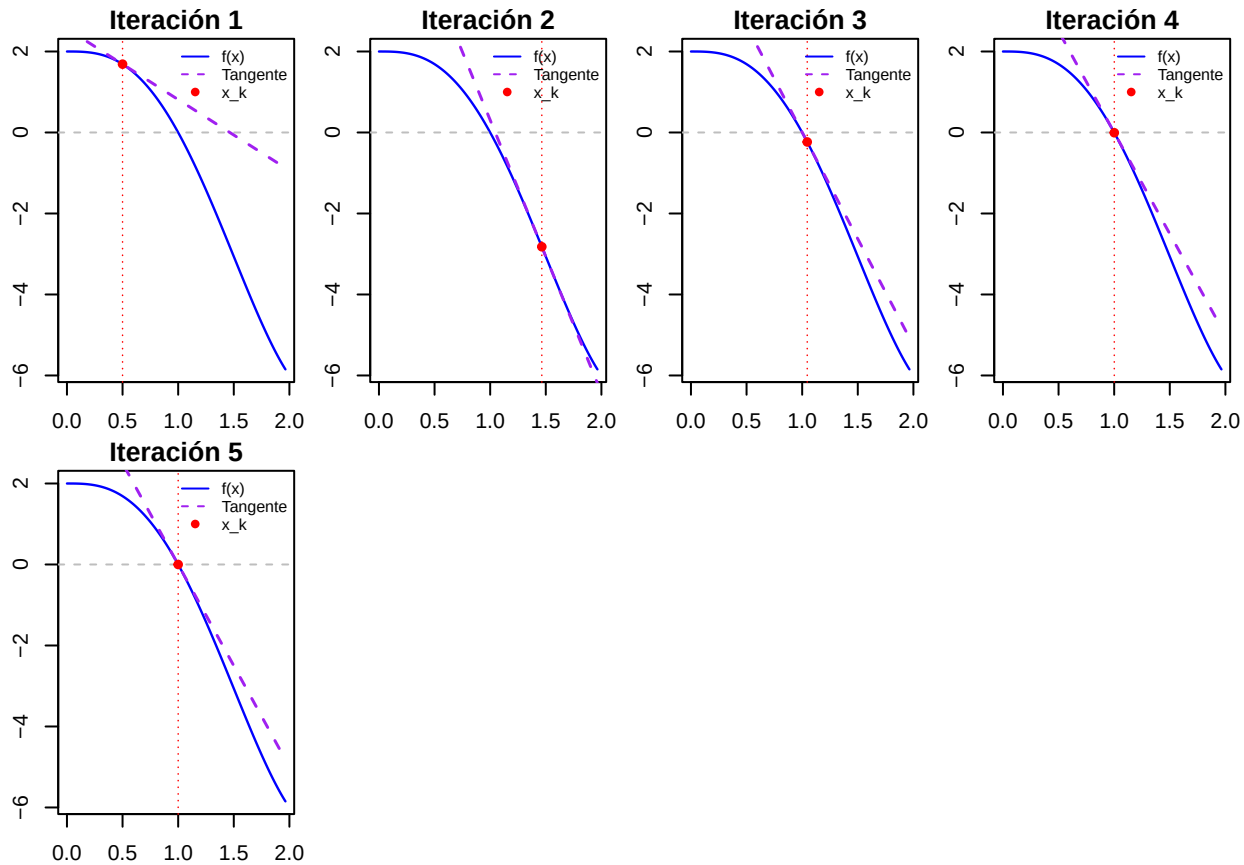
```
tabla(resultado)
```

Tabla de iteraciones en el método de la Newton-Raphson

k ♦	x_k ♦	$f(x_k)$ ♦	$f'(x_k)$ ♦	$e_k = x_{k+1} - x_k $ ♦
1	0.500000000	1.687500000	-1.750000000	0.964285714
2	1.464285714	-2.821563606	-6.738702624	0.418710212
3	1.045575502	-0.234009909	-5.266842968	0.044430774
4	1.001144728	-0.005727571	-5.006864432	0.001143944
5	1.000000785	-0.000003923	-5.000004707	0.000000785

2.4.3.4.4. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(x0, f, df, resultado)
```



2.4.3.5. Sistema 5

$f(x) = e^{-x} - x$ en $[a, b] = [0, 1]$

```
# Datos
```

```
a <- 0
b <- 1
x0 <- (a + b) / 2
f <- function(x) exp(-x) - x
df <- function(x) -exp(-x) - 1
```

2.4.3.5.1. Intervalo Verifica que exista un intervalo $[a, b]$ donde $f(a) \cdot f(b) < 0$.

```
condicion(a, b, f)
```

```
## f(a) = f( 0 ) = 1
## f(b) = f( 1 ) = -0.6321206
## f(a) · f(b) = -0.6321206
## -0.6321206 < 0
##
## Sí cumple la condición.
```

```
## [1] TRUE
```

2.4.3.5.2. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- newton(f, df, x0)
```

```
## Raíz aproximada: 0.5671433
```

```
## Iteraciones: 3
```

2.4.3.5.3. Tabla Registre los valores x_k , $f(x_k)$, $f'(x_k)$, y el error $|b_k - a_k|$.

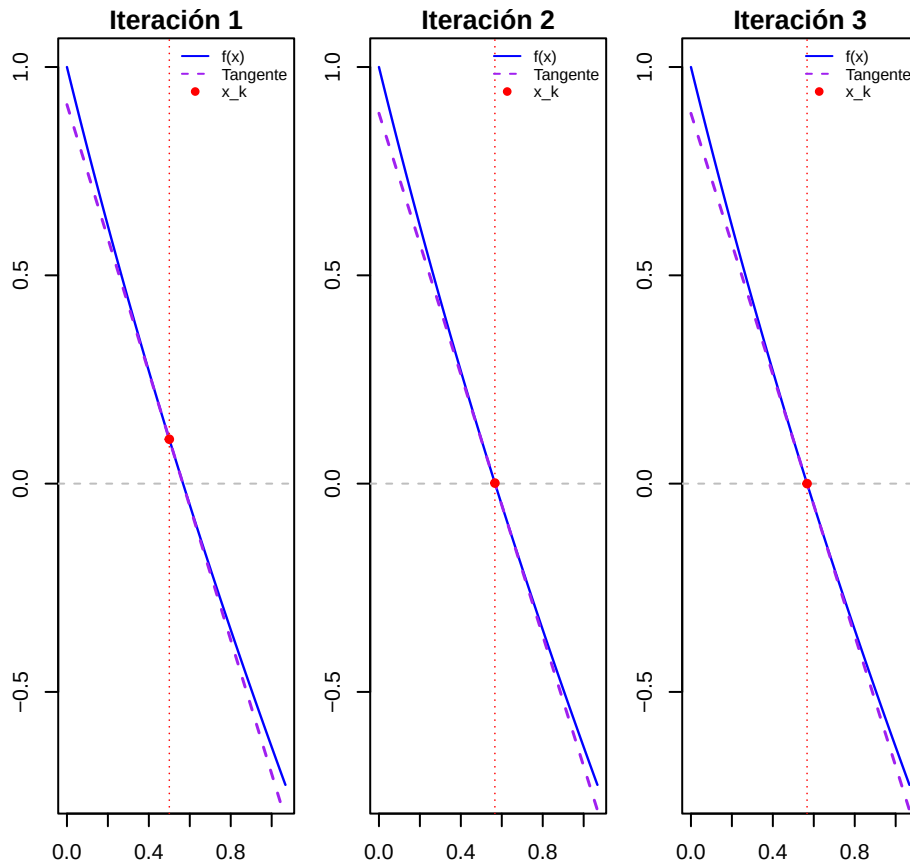
```
tabla(resultado)
```

Tabla de iteraciones en el método de la Newton-Raphson

k	x_k	$f(x_k)$	$f'(x_k)$	$e_k = x_{k+1} - x_k $
1	0.500000000	0.106530660	-1.606530660	0.066311003
2	0.566311003	0.001304510	-1.567615513	0.000832162
3	0.567143165	0.000000196	-1.567143362	0.000000125

2.4.3.5.4. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(x0, f, df, resultado)
```



2.4.3.6. Sistema 7

$f(x) = x^5 - 3x + 1$ en $[a, b] = [0, 1]$

```
# Datos
```

```
a <- 0
b <- 1
x0 <- (a + b) / 2
f <- function(x) x^5 - 3*x + 1
df <- function(x) 5*x^4 - 3
```

2.4.3.6.1. Intervalo

Verifica que exista un intervalo $[a, b]$ donde $f(a) \cdot f(b) < 0$.

```
condicion(a, b, f)
```

```
## f(a) = f( 0 ) = 1
## f(b) = f( 1 ) = -1
## f(a) · f(b) = -1
## -1 < 0
##
## Sí cumple la condición.
## [1] TRUE
```

2.4.3.6.2. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- newton(f, df, x0)
```

```
## Raíz aproximada: 0.3347341
```

```
## Iteraciones: 4
```

2.4.3.6.3. Tabla Registre los valores x_k , $f(x_k)$, $f'(x_k)$, y el error $|b_k - a_k|$.

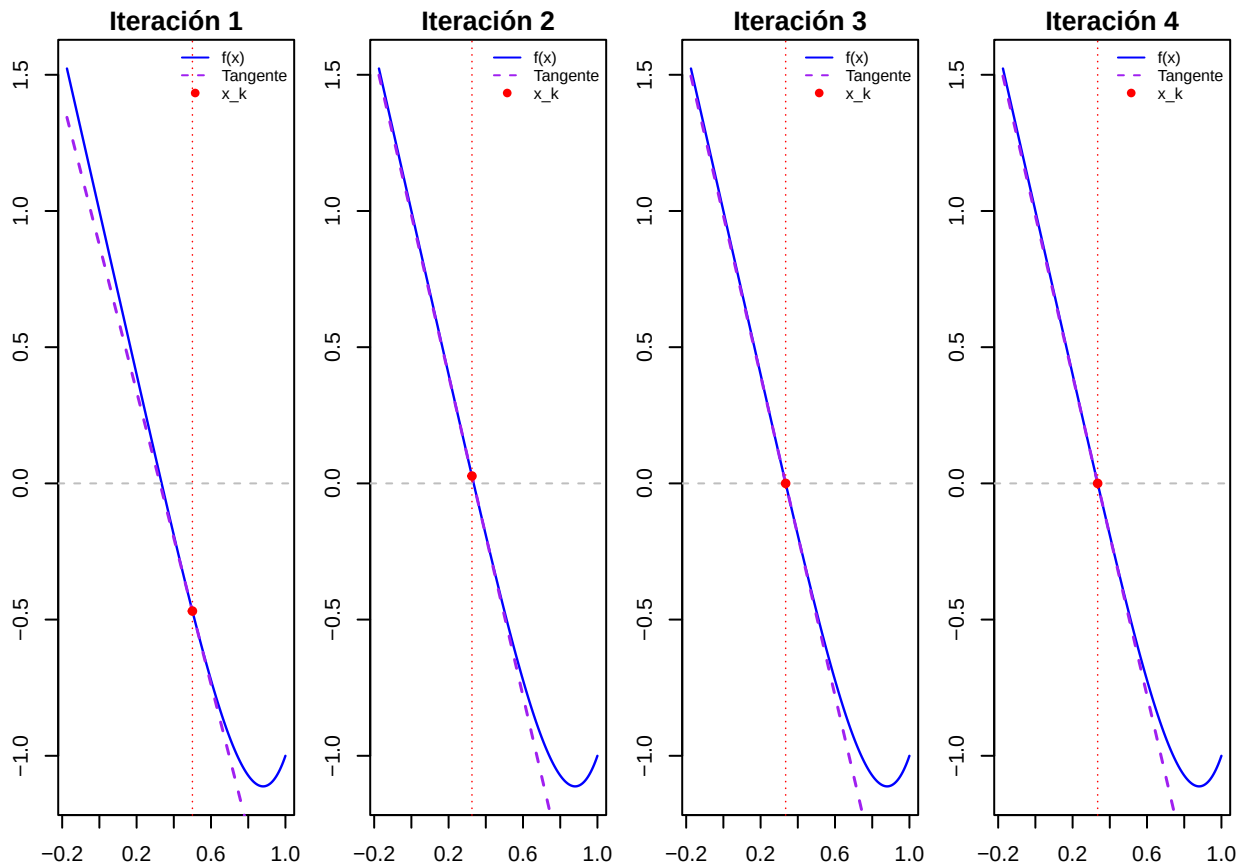
```
tabla(resultado)
```

Tabla de iteraciones en el método de la Newton-Raphson

k	x_k	$f(x_k)$	$f'(x_k)$	$e_k = x_{k+1} - x_k $
1	0.500000000	-0.468750000	-2.687500000	0.174418605
2	0.325581395	0.026914270	-2.943816560	0.009142645
3	0.334724041	0.000029670	-2.937234987	0.000010101
4	0.334734142	0.000000000	-2.937227410	0.000000000

2.4.3.6.4. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(x0, f, df, resultado)
```



2.4.3.7. Sistema 9

$f(x) = \ln(x + 2) + x - 1$ en $[a, b] = [-1, 0]$

Datos

```
a <- -1
b <- 0
x0 <- (a + b) / 2
f <- function(x) log(x + 2) + x - 1
df <- function(x) 1/(x + 2) + 1
```

2.4.3.7.1. Intervalo Verifica que exista un intervalo $[a, b]$ donde $f(a) \cdot f(b) < 0$.

condicion(a, b, f)

```
## f(a) = f( -1 ) = -2
## f(b) = f( 0 ) = -0.3068528
## f(a) · f(b) = 0.6137056
## 0.6137056 < 0
##
## NO cumple la condición.
## Se buscará un nuevo intervalo.
##
## Nuevo intervalo encontrado: [ 0 , 1 ]
```

```
## f(a) = f( 0 ) = -0.3068528
## f(b) = f( 1 ) = 1.098612
## f(a) · f(b) = -0.3371123
## -0.3371123 < 0
##
## Sí cumple la condición.
## [1] TRUE
```

2.4.3.7.2. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- newton(f, df, x0)
```

```
## Raíz aproximada: 0.20794
## Iteraciones: 4
```

2.4.3.7.3. Tabla Registre los valores x_k , $f(x_k)$, $f'(x_k)$, y el error $|b_k - a_k|$.

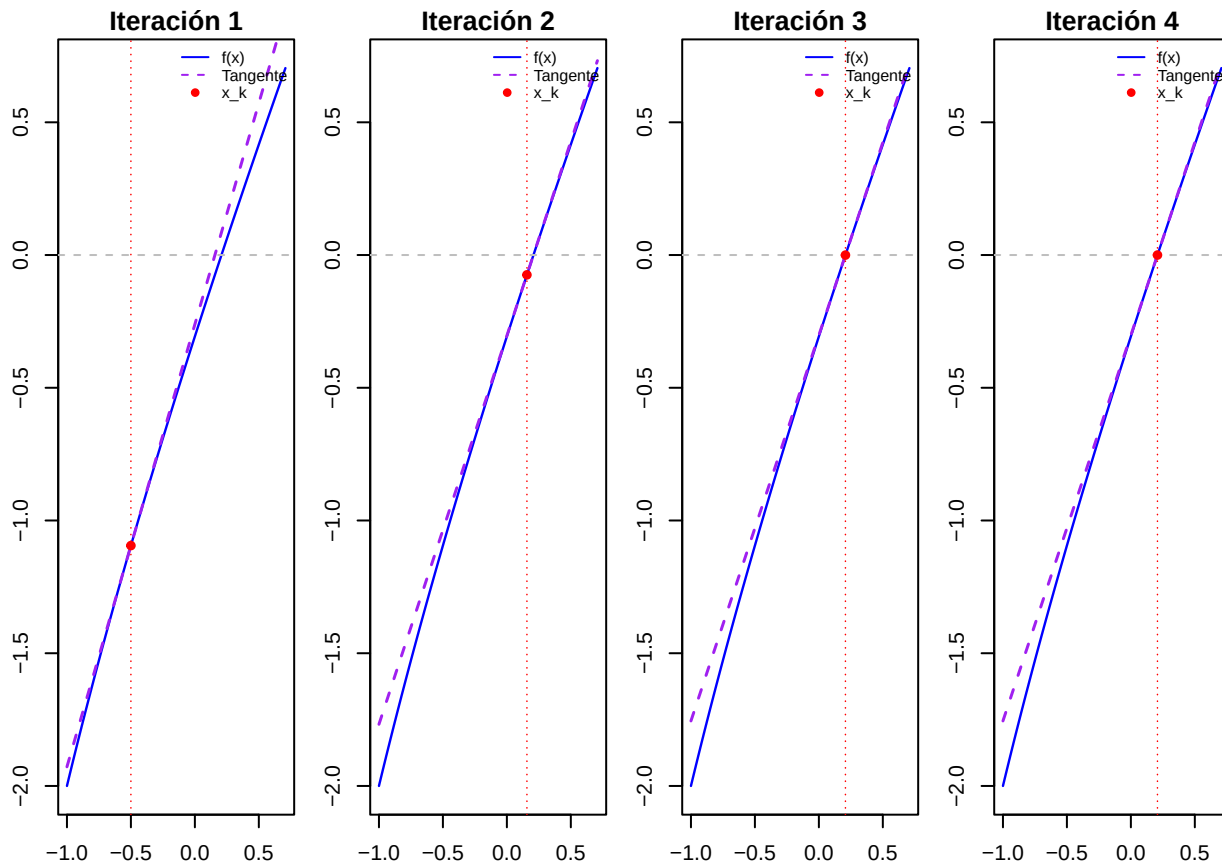
```
tabla(resultado)
```

Tabla de iteraciones en el método de la Newton-Raphson

k	x_k	$f(x_k)$	$f'(x_k)$	$e_k = x_{k+1} - x_k $
1	-0.500000000	-1.094534892	1.666666667	0.656720935
2	0.156720935	-0.074690082	1.463666849	0.051029428
3	0.207750363	-0.000275575	1.452949761	0.000189666
4	0.207940029	-0.000000004	1.452910852	0.000000003

2.4.3.7.4. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(x0, f, df, resultado)
```



2.4.4. Método del Punto Fijo

2.4.4.1. Ejercicios a mano

2.4.4.2. Código para implementación

```
# Método del Punto Fijo que guarda los valores para la tabla
```

```
fijo <- function(g, x0, tol = 1e-5, iter_max = 100){
```

```
# Vectores para almacenar los datos de las iteraciones
```

```
  iter_vec <- numeric(0)
  x0_vec <- numeric(0)
  x1_vec <- numeric(0)
  gx0_vec <- numeric(0)
  error_vec <- numeric(0)
```

```
  iter <- 0
  error <- 1
```

```
  while(error > tol && iter < iter_max){
```


Método del punto fijo

9. $f(x) = x - \sin(x) - 1$ $g(x) = \sin(x) + 1$ $x_0 = 1$

$g(1) = 1.841471$ $g(1.138324) = 1.933219$ $g(1.934624) = 1.934542$
 $g(1.841471) = 1.963591$ $g(1.933219) = 1.935041$ $g(1.934542) = 1.934571$
 $g(1.963591) = 1.923843$ $g(1.935041) = 1.934393$ $g(1.934571) = 1.934560$
 $g(1.923843) = 1.938324$ $g(1.934393) = 1.934624$ $g(1.934560) = 1.934564$
 $g(1.938324) = 1.934563$ $g(1.934624) = 1.934563$ $g(1.934564) = 1.934563$

k	x_k	$g(x_k)$	$ x_{k+1} - x_k $
1	1	1.841471	
2	1.841471	1.963591	0.841471
3	1.963591	1.923843	0.12212
4	1.923843	1.938324	0.034348
5	1.938324	1.933219	0.014481
6	1.933219	1.935041	0.005122
7	1.935041	1.934393	0.001822
8	1.934393	1.934624	0.000648
9	1.934624	1.934542	0.000231
10	1.934542	1.934571	0.000082
11	1.934571	1.934560	0.000029
12	1.934560	1.934564	0.000011
13	1.934564	1.934563	0.000004
14	1.934563	1.934561	0.000001
15	1.934561	1.934563	0.000000

Comprobación $(1.934563) - \sin(1.934563) - 1 = -0.000000735$

$f(x) = x - \sin(x) - 1 = 0$ 20

Figura 2.46: Ejercicio 9

11. $f(x) = 2x - e^{-x}$	$g(x) = \frac{e^{-x}}{2}$	$X_0 = 0$		
$g(0) = 0.5$	Comprobación: $2(0.351734) - e^{-0.351734} = 0.0000007 \approx 0$			
$g(0.5) = 0.303265$	k	X_k	$g(X_k)$	$ X_{k+1} - X_k $
$g(0.303265) = 0.369202$	1	0	0.5	0.196735
$g(0.369202) = 0.345643$	2	0.5	0.303265	0.196736
$g(0.345643) = 0.353883$	3	0.303265	0.369202	0.065937
$g(0.353883) = 0.350979$	4	0.369202	0.345643	0.023559
$g(0.350979) = 0.351993$	5	0.345643	0.353883	0.00824
$g(0.351993) = 0.351643$	6	0.353883	0.350979	0.002904
$g(0.351643) = 0.351766$	7	0.350979	0.351993	0.001014
$g(0.351766) = 0.351722$	8	0.351993	0.351643	0.00035
$g(0.351722) = 0.351732$	9	0.351643	0.351766	0.000123
$g(0.351732) = 0.351732$	10	0.351766	0.351722	0.000044
$g(0.351732) = 0.351734$	11	0.351722	0.351738	0.000016
$g(0.351734) = 0.351734$	12	0.351738	0.351732	0.000006
$g(0.351734) = 0.351734$	13	0.351732	0.351734	0.000002
	14	0.351734	0.351734	0

4. $f(x) = x^2 - 4x + 1$	$g(x) = \frac{x^2 + 1}{4}$	$X_0 = 1$		
$g(1) = 0.5$	Comprobación: $(0.267949)^2 - 4(0.267949) + 1 = 0.0000006 \approx 0$			
$g(0.5) = 0.3125$	k	X_k	$g(X_k)$	$ X_{k+1} - X_k $
$g(0.3125) = 0.274414$	1	1	0.5	0.1875
$g(0.274414) = 0.268826$	2	0.5	0.3125	0.038086
$g(0.268826) = 0.268067$	3	0.3125	0.274414	0.005588
$g(0.268067) = 0.267965$	4	0.274414	0.268826	0.000759
$g(0.267965) = 0.267951$	5	0.268826	0.268067	0.000102
$g(0.267951) = 0.267949$	6	0.268067	0.267965	0.000014
$g(0.267949) = 0.267949$	7	0.267965	0.267951	0.000002
$g(0.267949) = 0.267949$	8	0.267951	0.267949	0
	9	0.267949	0.267949	0

Figura 2.47: Ejercicio 11 y 4

```
# Aplicar fórmula de punto fijo

x1 <- g(x0)
error <- abs(x1 - x0)

# Almacena los datos de cada iteración

iter <- iter + 1
iter_vec <- c(iter_vec, iter)
x0_vec <- c(x0_vec, x0)
x1_vec <- c(x1_vec, x1)
gx0_vec <- c(gx0_vec, g(x0))
error_vec <- c(error_vec, error)

# Actualizar para siguiente iteración

x0 <- x1
}

# Crea la tabla de las iteraciones

TAB <- data.frame(
  "k" = iter_vec,
  "x" = x0_vec,
  "g(x)" = gx0_vec,
  "x" = x1_vec,
  "e = |x - x|" = error_vec,
  check.names = FALSE
)

cat("Raíz aproximada:", x1, "\n")
cat("Iteraciones:", iter, "\n\n")

return(list(raiz = x1, iteraciones = iter, tabla = TAB))
}
```

```
# Tabla "bonita" para Punto Fijo

tabla <- function(resultado) {
  library(DT)

  datatable(
    resultado$tabla[, -4],
    options = list(
      scrollX = TRUE,
      scrollY = "400px",
      paging = FALSE,
      dom = 't',
      autoWidth = FALSE,
      columnDefs = list(
        list(className = 'dt-center', targets = 0:3))
    )
  )
}
```

```

),
rownames = FALSE,
class = 'cell-border stripe',
width = "100%",
caption = "Tabla de iteraciones en el método del Punto Fijo",
escape = FALSE
) %>%
  formatRound(columns = c(2:4), digits = 6)
}

# Gráfica para Punto Fijo

grafica <- function(g, resultado) {

  # Obtener la raíz aproximada

  raiz <- resultado$raiz

  # Calcular rango alrededor de la raíz

  x_min <- raiz - 1
  x_max <- raiz + 1
  x_vals <- seq(x_min, x_max, length.out = 1000)
  y_g <- g(x_vals)
  y_x <- x_vals # y = x

  # Crear la gráfica

  plot(x_vals, y_g, type = "l", main = "Punto de Intersección: y = g(x) y y = x",
       xlab = "x", ylab = "y", col = "blue", lwd = 2, ylim = range(c(y_g, y_x)))
  lines(x_vals, y_x, col = "red", lwd = 2, lty = 2)

  # Marcar el punto de intersección (raíz)

  points(raiz, raiz, col = "green", pch = 19, cex = 1.5)

  # Líneas para mostrar que es la intersección

  abline(v = raiz, col = "green", lty = 3, lwd = 1)
  abline(h = raiz, col = "green", lty = 3, lwd = 1)

  # Leyenda

  legend("topright",
        legend = c("y = g(x)", "y = x", "Punto fijo"),
        col = c("blue", "red", "green"),
        lty = c(1, 2, NA), pch = c(NA, NA, 19),
        lwd = c(2, 2, NA), cex = 0.8, bty = "n")
}

```

2.4.4.3. Sistema 2

$$f(x) = x^3 + x - 1$$

$$g(x) = 1 - x^3$$

$$x_0 = 0,6$$

Datos

```
x0 <- 0.6
f <- function(x) x^3 + x - 1
g <- function(x) 1 - x^3
dg <- function(x) -3*x^2
```

2.4.4.3.1. Forma $x = g(x)$ Reescriba la ecuación $f(x) = 0$ en forma $x = g(x)$.

Ya se nos da la sugerencia:

$$g(x) = 1 - x^3$$

2.4.4.3.2. Condición Verifique que la función $g(x)$ cumpla la condición de convergencia local $|g'(x)| < 1$ en el intervalo considerado.

```
if(abs(dg(x0)) < 1) {
  cat("Condición cumplida\n")
} else {
  cat("Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.\n")
}
```

Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.

2.4.4.3.3. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- fijo(g, x0)
```

Raíz aproximada: 0

Iteraciones: 100

2.4.4.3.4. Tabla Registre los valores x_k , $g(x_k)$, y el error $|x_{k+1} - x_k|$.

```
tabla(resultado)
```

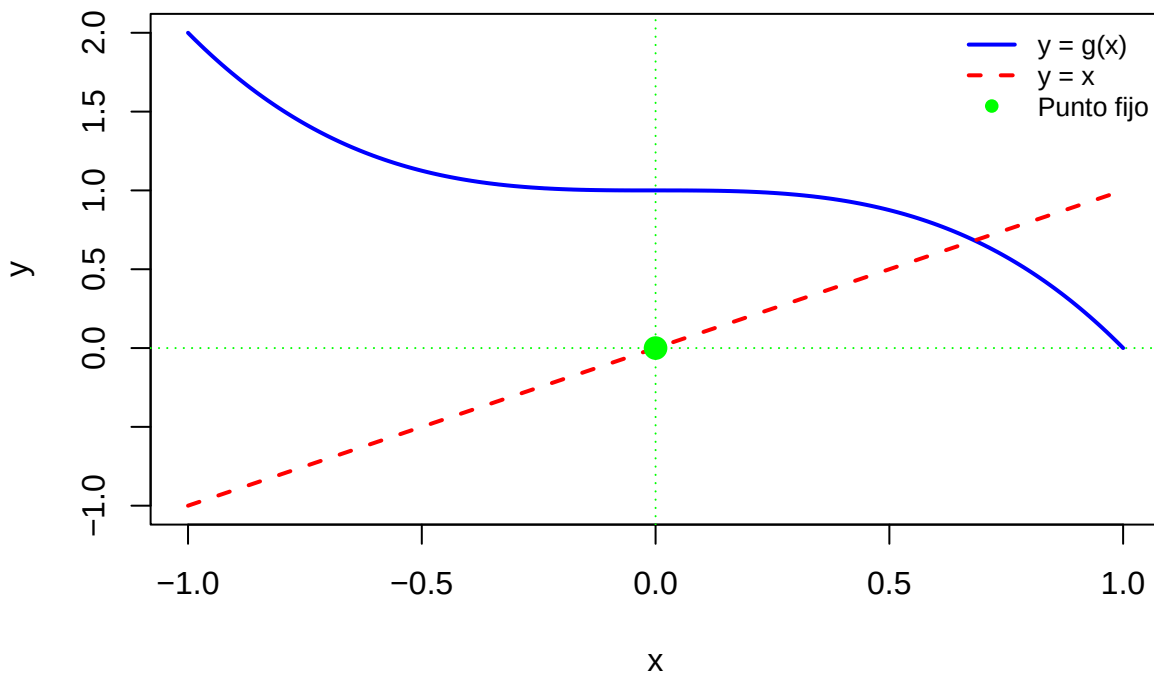
Tabla de iteraciones en el método del Punto Fijo

k	x_k	$g(x_k)$	$e_k = x_{k+1} - x_k $
1	0.600000	0.784000	0.184000
2	0.784000	0.518110	0.265890
3	0.518110	0.860920	0.342810
4	0.860920	0.361901	0.499019
5	0.361901	0.952601	0.590700
6	0.952601	0.135563	0.817038
7	0.135563	0.997509	0.861945
8	0.997509	0.007455	0.990053
9	0.007455	1.000000	0.992544
10	1.000000	0.000001	0.999998
11	0.000001	1.000000	0.999999
12	1.000000	0.000000	1.000000

2.4.4.3.5. Gráficas Grafica la función y marca el punto donde converge la raíz.

`grafica(g, resultado)`

Punto de Intersección: $y = g(x)$ y $y = x$



Para este sistema, el método no es el adecuado ya que en las 100 iteraciones que se hicieron, la función no convergió.

2.4.4.4. Sistema 4

$$f(x) = x^2 - 4^x + 1$$

$$g(x) = \frac{x^2 + 1}{4}$$

$$x_0 = 1$$

Datos

```
x0 <- 1
f <- function(x) x^2 - 4^x + 1
g <- function(x) (x^2 + 1) / 4
dg <- function(x) (1 / 2) * x
```

2.4.4.4.1. Forma $x = g(x)$ Reescriba la ecuación $f(x) = 0$ en forma $x = g(x)$.

Ya se nos da la sugerencia:

$$g(x) = \frac{x^2 + 1}{4}$$

2.4.4.4.2. Condición Verifique que la función $g(x)$ cumpla la condición de convergencia local $|g'(x)| < 1$ en el intervalo considerado.

```
if(abs(dg(x0)) < 1) {
  cat("Condición cumplida\n")
} else {
```

```
cat("Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.\n")
}
```

```
## Condición cumplida
```

2.4.4.4.3. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- fijo(g, x0)
```

```
## Raíz aproximada: 0.2679495
```

```
## Iteraciones: 8
```

2.4.4.4.4. Tabla Registre los valores x_k , $g(x_k)$, y el error $|x_{k+1} - x_k|$.

```
tabla(resultado)
```

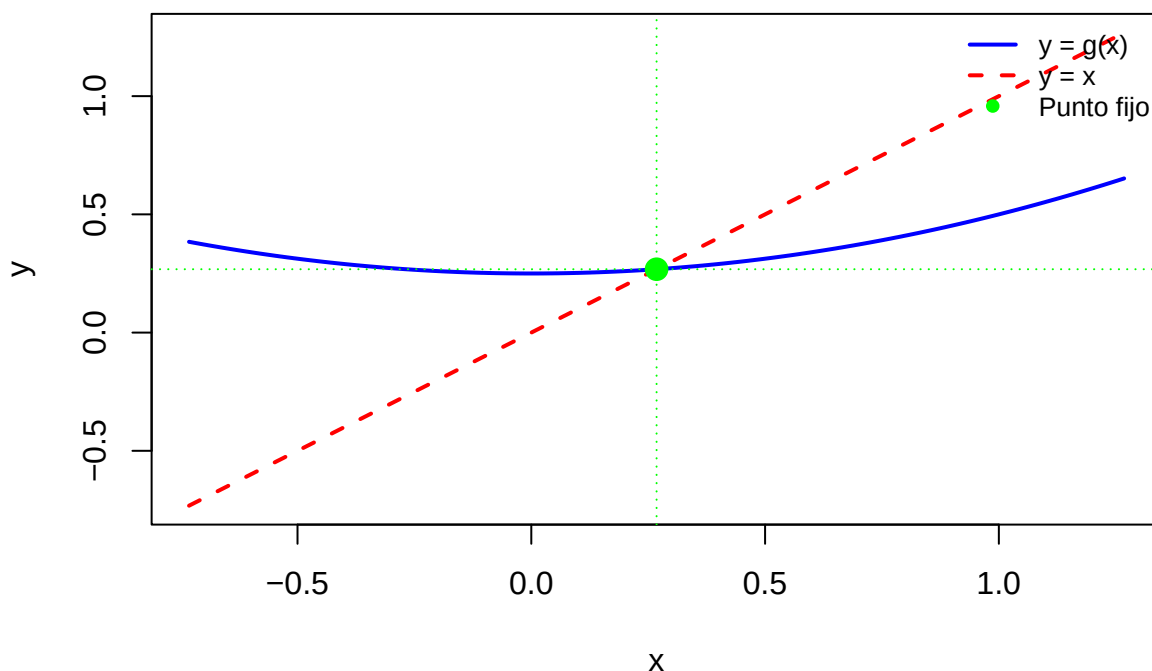
Tabla de iteraciones en el método del Punto Fijo

k	x_k	$g(x_k)$	$e_k = x_{k+1} - x_k $
1	1.000000	0.500000	0.500000
2	0.500000	0.312500	0.187500
3	0.312500	0.274414	0.038086
4	0.274414	0.268826	0.005588
5	0.268826	0.268067	0.000759
6	0.268067	0.267965	0.000102
7	0.267965	0.267951	0.000014
8	0.267951	0.267949	0.000002

2.4.4.4.5. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(g, resultado)
```


Punto de Intersección: $y = g(x)$ y $y = x$



2.4.4.5. Sistema 8

$$f(x) = x^3 - 3x + 1$$

$$g(x) = \sqrt[3]{3x - 1}$$

$$x_0 = 0,8$$

Datos

```
x0 <- 0.8
f <- function(x) x^3 - 3*x + 1
g <- function(x) (3*x - 1)^(1/3)
dg <- function(x) (3*x - 1)^(-2/3)
```

2.4.4.5.1. Forma $x = g(x)$ Reescriba la ecuación $f(x) = 0$ en forma $x = g(x)$.

Ya se nos da la sugerencia:

$$g(x) = \sqrt[3]{3x - 1}$$

2.4.4.5.2. Condición Verifique que la función $g(x)$ cumpla la condición de convergencia local $|g'(x)| < 1$ en el intervalo considerado.

```
if(abs(dg(x0)) < 1) {
  cat("Condición cumplida\n")
} else {
  cat("Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.\n")
}
```

```
## Condición cumplida
```

2.4.4.5.3. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- fijo(g, x0)
```

```
## Raíz aproximada: 1.532085
```

```
## Iteraciones: 15
```

2.4.4.5.4. Tabla Registre los valores x_k , $g(x_k)$, y el error $|x_{k+1} - x_k|$.

```
tabla(resultado)
```

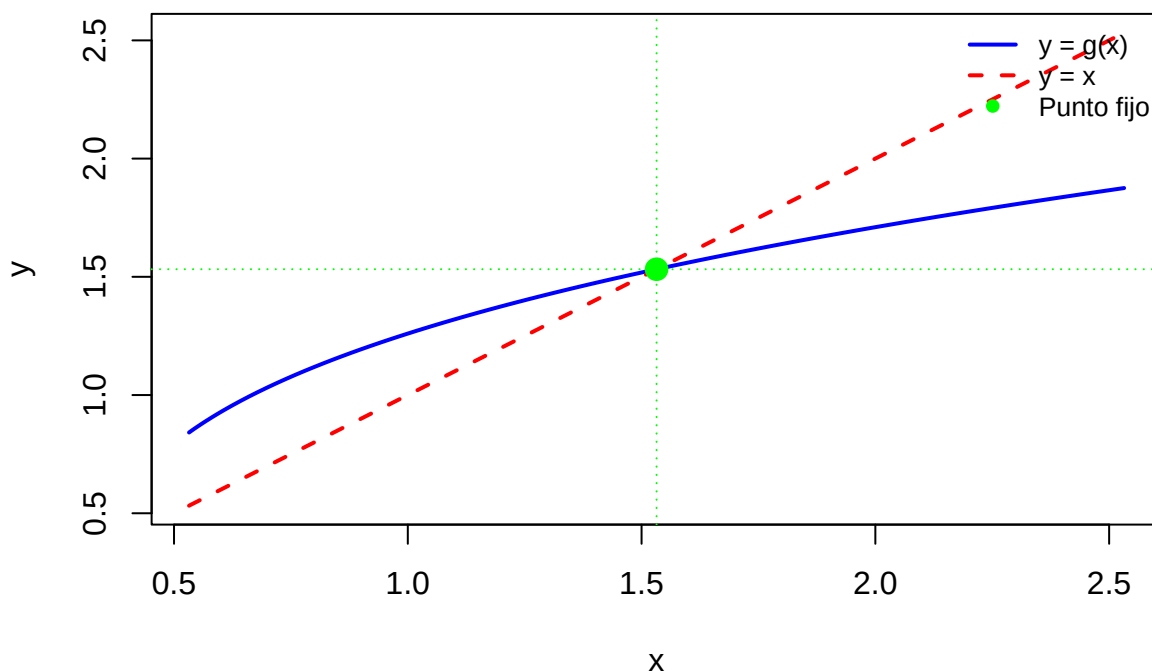
Tabla de iteraciones en el método del Punto Fijo

k	x_k	$g(x_k)$	$e_k = x_{k+1} - x_k $
1	0.800000	1.118689	0.318689
2	1.118689	1.330646	0.211957
3	1.330646	1.440956	0.110310
4	1.440956	1.492237	0.051280
5	1.492237	1.514919	0.022682
6	1.514919	1.524739	0.009820
7	1.524739	1.528951	0.004212
8	1.528951	1.530751	0.001800
9	1.530751	1.531519	0.000768
10	1.531519	1.531846	0.000327
11	1.531846	1.531985	0.000139
12	1.531985	1.532045	0.000059

2.4.4.5.5. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(g, resultado)
```

Punto de Intersección: $y = g(x)$ y $y = x$



2.4.4.6. Sistema 10

$$f(x) = x^2 - 3$$

$$g(x) = \frac{3}{x}$$

$$x_0 = 1,5$$

Datos

```
x0 <- 1.5
f <- function(x) x^2 - 3
g <- function(x) 3 / x
dg <- function(x) - 3 / x^2
```

2.4.4.6.1. Forma $x = g(x)$ Reescriba la ecuación $f(x) = 0$ en forma $x = g(x)$.

Ya se nos da la sugerencia:

$$g(x) = \frac{3}{x}$$

2.4.4.6.2. Condición Verifique que la función $g(x)$ cumpla la condición de convergencia local $|g'(x)| < 1$ en el intervalo considerado.

```
if(abs(dg(x0)) < 1) {
  cat("Condición cumplida\n")
} else {
  cat("Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.\n")
}
```

Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.

2.4.4.6.3. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- fijo(g, x0)
```

Raíz aproximada: 1.5

Iteraciones: 100

2.4.4.6.4. Tabla Registre los valores x_k , $g(x_k)$, y el error $|x_{k+1} - x_k|$.

```
tabla(resultado)
```

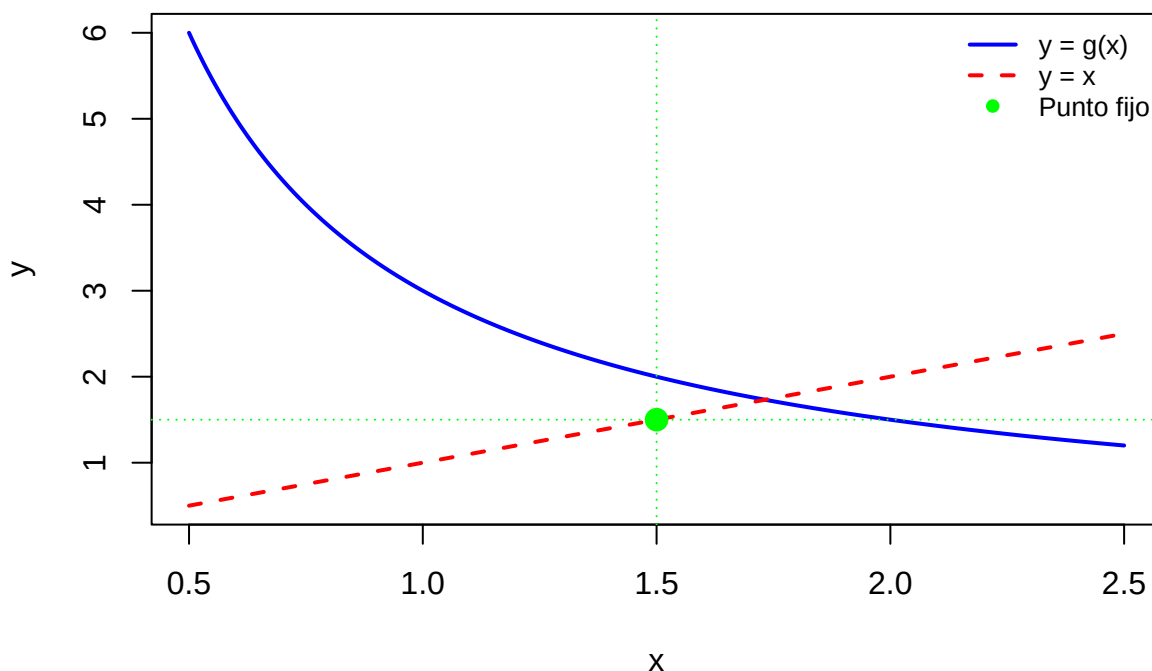
Tabla de iteraciones en el método del Punto Fijo

k	x_k	$g(x_k)$	$e_k = x_{k+1} - x_k $
1	1.500000	2.000000	0.500000
2	2.000000	1.500000	0.500000
3	1.500000	2.000000	0.500000
4	2.000000	1.500000	0.500000
5	1.500000	2.000000	0.500000
6	2.000000	1.500000	0.500000
7	1.500000	2.000000	0.500000
8	2.000000	1.500000	0.500000
9	1.500000	2.000000	0.500000
10	2.000000	1.500000	0.500000
11	1.500000	2.000000	0.500000
12	2.000000	1.500000	0.500000

2.4.4.6.5. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(g, resultado)
```

Punto de Intersección: $y = g(x)$ y $y = x$



Para este sistema, el método no es el adecuado ya que en las 100 iteraciones que se hicieron, la función no convergió.

2.4.4.7. Sistema 14

$$f(x) = x - e^{-x^2}$$

$$g(x) = e^{-x^2}$$

$$x_0 = 0,5$$

Datos

```
x0 <- 0.5
f <- function(x) x - exp(-x^2)
g <- function(x) exp(-x^2)
dg <- function(x) - 2 * exp(-x^2)
```

2.4.4.7.1. Forma $x = g(x)$ Reescriba la ecuación $f(x) = 0$ en forma $x = g(x)$.

Ya se nos da la sugerencia:

$$g(x) = e^{-x^2}$$

2.4.4.7.2. Condición Verifique que la función $g(x)$ cumpla la condición de convergencia local $|g'(x)| < 1$ en el intervalo considerado.

```
if(abs(dg(x0)) < 1) {
  cat("Condición cumplida\n")
} else {
```

```
cat("Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.\n")
}
```

Condición no cumplida, es probable que en las primeras 100 iteraciones no converga.

2.4.4.7.3. Iteraciones Realice al menos 8 iteraciones del método.

```
resultado <- fijo(g, x0)
```

Raíz aproximada: 0.6529231

Iteraciones: 65

2.4.4.7.4. Tabla Registre los valores x_k , $g(x_k)$, y el error $|x_{k+1} - x_k|$.

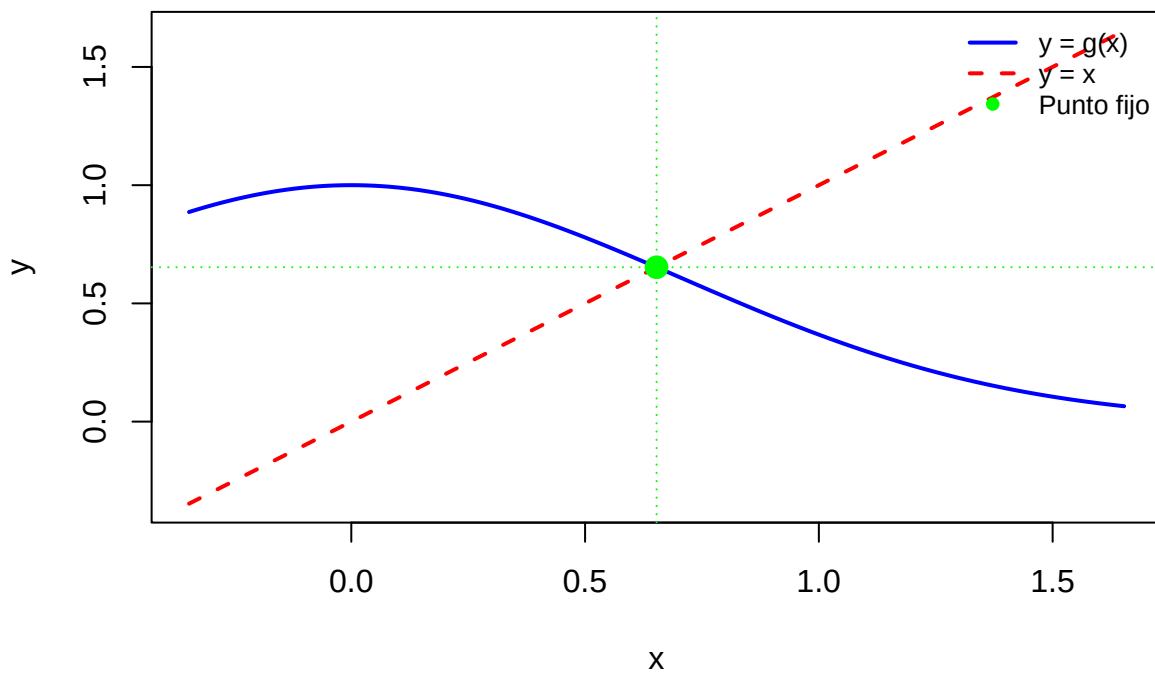
```
tabla(resultado)
```

Tabla de iteraciones en el método del Punto Fijo

k	x_k	$g(x_k)$	$e_k = x_{k+1} - x_k $
1	0.500000	0.778801	0.278801
2	0.778801	0.545239	0.233562
3	0.545239	0.742832	0.197592
4	0.742832	0.575913	0.166919
5	0.575913	0.717720	0.141807
6	0.717720	0.597428	0.120292
7	0.597428	0.699829	0.102401
8	0.699829	0.612773	0.087055
9	0.612773	0.686952	0.074178
10	0.686952	0.623814	0.063138
11	0.623814	0.677637	0.053822
12	0.677637	0.621794	0.045842

2.4.4.7.5. Gráficas Grafica la función y marca el punto donde converge la raíz.

```
grafica(g, resultado)
```

Punto de Intersección: $y = g(x)$ y $y = x$ 

Para este sistema, el método sí convergió aunque la condición no se haya cumplido, sin embargo, para ello tomo 80 iteraciones.

2.5. Interpolación y Aproximación**2.5.1. Ejercicio 1****2.5.1.1. 1,1****2.5.1.1.1. a) Polinomio interpolante****2.5.1.1.2. b) Tabla****2.5.1.1.3. c) Comparación gráfica**

1. Dado el conjunto de puntos $(0, 1), (1, 3), (2, 11)$

a) Polinomio interpolante

$$P_n(x) = \sum_{i=0}^n f_i L_i(x) \quad L_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j}$$

$$L_0 = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2} \quad L_1 = \frac{(x-0)(x-2)}{(1-0)(1-2)} = \frac{(x-0)(x-2)}{-1} = -x(2-x)$$

$$L_2 = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{(x-0)(x-1)}{2}$$

$$P_2(x) = 1 \left(\frac{(x-1)(x-2)}{2} \right) + 3 \left(\frac{(x-0)(x-2)}{-1} \right) + 11 \left(\frac{(x-0)(x-1)}{2} \right) =$$

$$\frac{(x-1)(x-2)}{2} - 3x(2-x) + \frac{11x(x-1)}{2} = \frac{(x-1)(x-2) + 11x(x-1) - 6x(2-x)}{2}$$

$$P_2(x) = \frac{(x-1)(2x-2) + 11x(2-x) - 6x(2-x)}{2} = \frac{(x-1)(2x-2) + 5x(2-x)}{2}$$

$$P_2(x) = \frac{2x^2 - 2x - 2x + 2 + 10x - 5x^2}{2} = \frac{-3x^2 + 6x + 2}{2} = -\frac{3}{2}x^2 + 3x + 1$$

$$P_2(x) = 3x^2 - X + 1$$

Figura 2.48: Ejercicio 1.1 a

b) Tabla de diferencias divididas y método de Newton

i	x_i	$f(x_i)$	$f(x_i, x_{i+1})$	$f(x_i, x_{i+1}, x_{i+2})$
0	0	1		
1	1	3	2	
2	2	11	8	3

$\frac{3-1}{1-0} = \frac{2}{1} = 2$
 $\frac{11-3}{2-1} = \frac{8}{1} = 8$
 $\frac{8-2}{2-0} = \frac{6}{2} = 3$

Polinomio de Newton:

$$P_2(x) = 1 + 2(x-0) + 3(x-0)(x-1) = 1 + (x-0)(2+3x-3) = 1 + x(3x-1)$$

$$P_2(x) = 1 + 3x^2 - x = 3x^2 - X + 1$$

Figura 2.49: Ejercicio 1.1 b

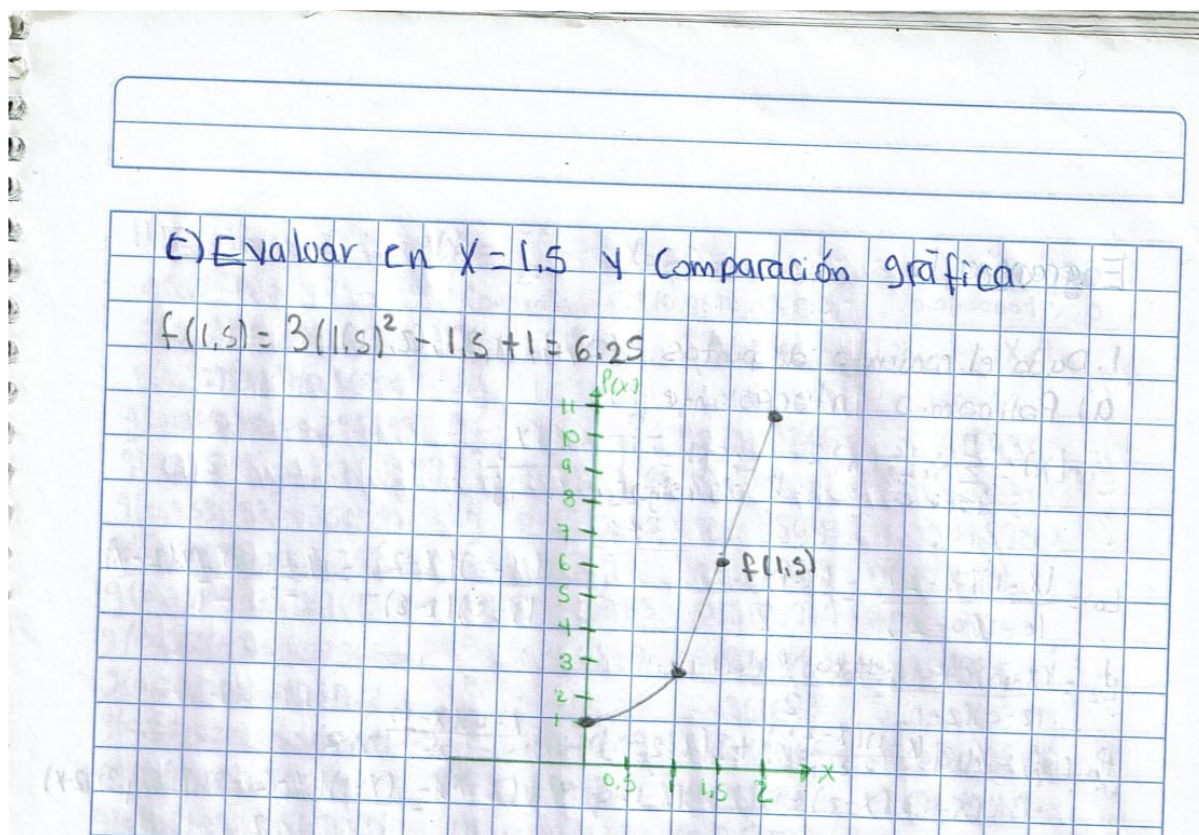


Figura 2.50: Ejercicio 1.1 c

2.5.1.2. 1,2

2. Con los puntos (1,2), (2,5), (3,10), (4,17)
a) Polinomio interpolante grado 3 de Newton

2.5.1.2.1. a) Polinomio interpolante grado 3

$$\begin{aligned} P_3(x) &= 2 + 3(x-1) + 1(x-1)(x-2) + 0(x-1)(x-2)(x-3) = 2 + 3(x-1) + (x-1)(x-2) \\ &= 2 + (x-1)(3+x-2) = 2 + (x-1)(x+1) = 2 + x^2 + x - x - 1 = x^2 + 1 \\ P_3(x) &= x^2 + 1 \end{aligned}$$

i	x_i	$f(x_i)$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$	b)
0	1	2	3			
1	2	5	5	1		
2	3	10	7	1	0	
3	4	17				

$$\frac{5-2}{2-1} = 3 \quad \frac{5-3}{3-1} = 1 \quad \frac{1-1}{4-1} = 0$$

$$\frac{10-5}{3-2} = 5 \quad \frac{7-5}{4-2} = 1$$

$$\frac{17-10}{4-3} = 7$$

Figura 2.51: Ejercicio 1.2 b

2.5.1.2.2. b) Tabla

c) Evaluar en $x = 2,5$

$$f(2,5) = (2,5)^2 + 1 = 7,25$$

Figura 2.52: Ejercicio 1.2 c

2.5.1.2.3. c) Evaluar en $x = 2,5$

2.5.1.2.4. d) Verificar y comparar

d) Verificar y comparar

$$f(1) = (1)^2 + 1 = 2$$

$$f(2) = (2)^2 + 1 = 5$$

$$f(3) = (3)^2 + 1 = 10$$

$$f(4) = (4)^2 + 1 = 17$$

SI coincide

Figura 2.53: Ejercicio 1.2 d

2.5.1.3. 1,3

3. Dado los puntos $(-1, 4), (0, 1), (2, 3)$

a) Polinomio de Lagrange

$$L_0 = \frac{(x-0)(x-2)}{(-1-0)(-1-2)} = \frac{(x-0)(x-2)}{3}$$

$$L_1 = \frac{(x+1)(x-2)}{(0+1)(0-2)} = \frac{(x+1)(x-2)}{-2}$$

$$L_2 = \frac{(x+1)(x-0)}{(2+1)(2-0)} = \frac{(x+1)(x-0)}{6}$$

$$P_2(x) = 4\left(\frac{x(x-2)}{3}\right) + 1\left(\frac{(x+1)(x-2)}{-2}\right) + 3\left(\frac{(x+1)x}{6}\right) = (x-2)\left(\frac{4}{3}x - \frac{x+1}{2}\right) + \frac{3}{2}(x+1)x$$

$$= (x-2)\left(\frac{8x-3x-3}{6}\right) + \frac{3x(x+1)}{2} = (x-2)\left(\frac{5x-3}{6}\right) + \frac{3x(x+1)}{2}$$

Figura 2.54: Ejercicio 1.3 a

2.5.1.3.1. a) Polinomio de Lagrange

2.5.1.3.2. b) Expanda y simplifique

2.5.1.3.3. c) Verifique los valores

2.5.2. Ejercicio 2

2.5.2.1. 2,1

2.5.2.1.1. a) Tabla

2.5.2.1.2. b) Polinomio

2.5.2.1.3. c) Calcule $f(2,5)$ y comparación

2.5.2.2. 2,2

2.5.2.2.1. a) Tabla

2.5.2.2.2. b) Newton hacia atrás

$$\begin{aligned}
 &= \frac{(x-2)(8x-3x-3)}{6} + \frac{x(x+1)}{2} = \frac{(x-2)(5x-3)}{6} + \frac{x(x+1)}{2} \\
 &= \frac{5x^2-3x-10x+6}{6} + \frac{x^2+x}{2} = \frac{5x^2-13x+6}{6} + \frac{x^2+x}{2} \\
 &= \frac{10x^2-26x+12+6x^2+6x}{12} = \frac{16x^2-20x+12}{12} = \frac{4x^2-5x+3}{3} \\
 P_2(x) &= \frac{4x^2-5x+3}{3} \\
 \text{b) Expandir y Simplificar} \\
 P_2(x) &= \frac{4}{3}x^2 - \frac{5}{3}x + 1
 \end{aligned}$$

Figura 2.55: Ejercicio 1.3 b

$$\begin{aligned}
 \text{c) Verifique los valores} \\
 P_2(-1) &= \frac{4}{3}(-1)^2 - \frac{5}{3}(-1) + 1 = 4 \\
 P_2(0) &= \frac{4}{3}(0)^2 - \frac{5}{3}(0) + 1 = 1 \\
 P_2(2) &= \frac{4}{3}(2)^2 - \frac{5}{3}(2) + 1 = 3
 \end{aligned}$$

Figura 2.56: Ejercicio 1.3 c

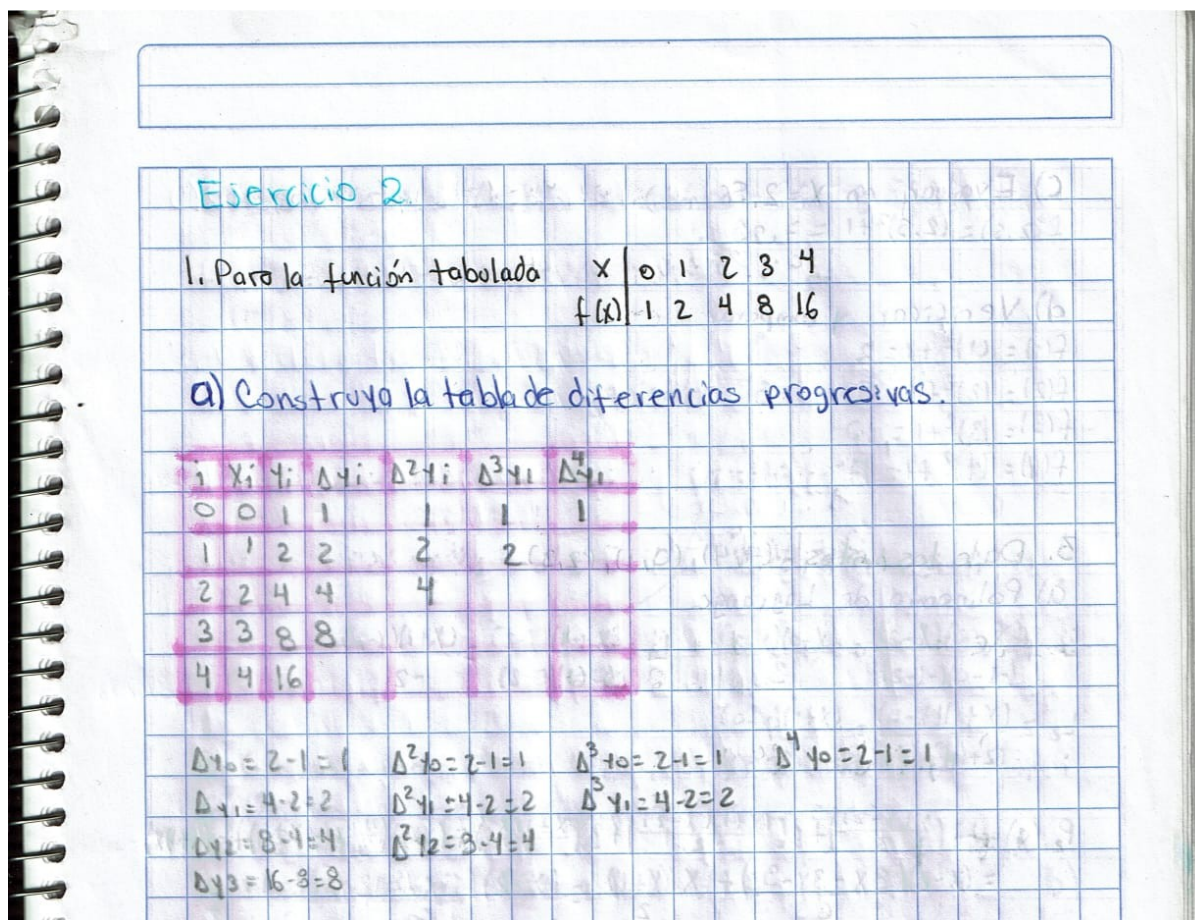


Figura 2.57: Ejercicio 2.1 a

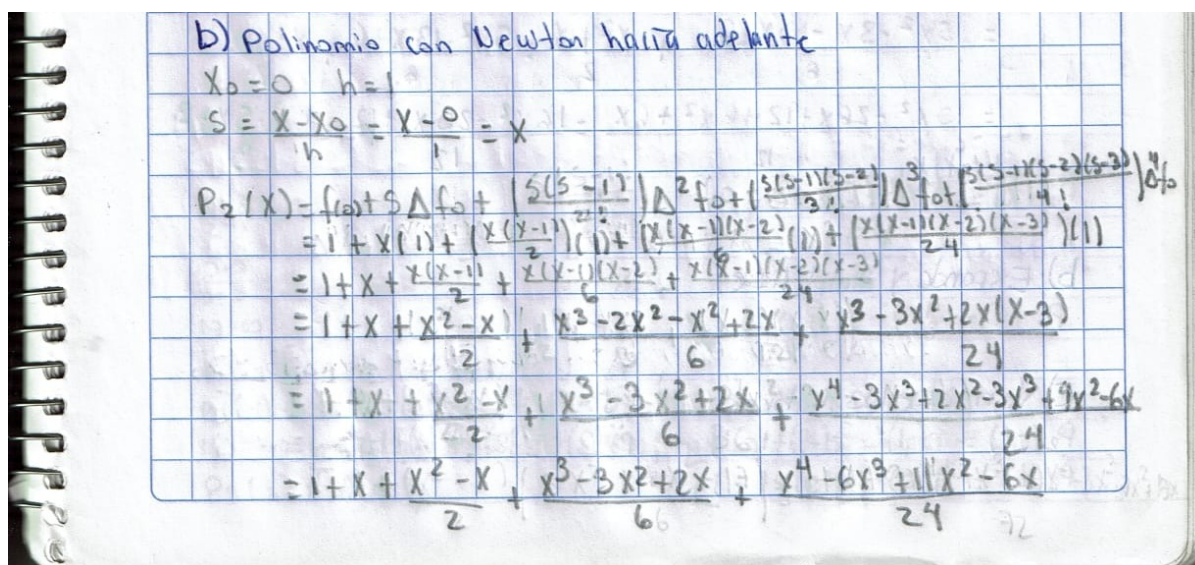


Figura 2.58: Ejercicio 2.1 b1

$$\begin{aligned}
 &= \frac{24}{24} + \frac{24x}{24} + \frac{12x^2 - 12x}{24} + \frac{4x^3 - 12x^2 + 8x}{24} + \frac{x^4 - 6x^3 + 11x^2 - 6x}{24} \\
 &= \frac{x^4 - 2x^3 + 11x^2 + 14x + 24}{24} = \frac{1}{24}x^4 - \frac{1}{12}x^3 + \frac{11}{24}x^2 + \frac{7}{12}x + 1
 \end{aligned}$$

Figura 2.59: Ejercicio 2.1 b2

c) Calcule $f(2.5)$ y compare con $2^{2.5}$

$$\begin{aligned}
 f(2.5) &= \frac{(2.5)^4}{24} - \frac{(2.5)^3}{12} + \frac{11(2.5)^2}{24} + \frac{7(2.5)}{12} + 1 = 5.648438 \\
 2^{2.5} &= 5.656854 \\
 5.656854 - 5.648438 &= 0.008416 \rightarrow \text{Diferencia o error}
 \end{aligned}$$

Figura 2.60: Ejercicio 2.1 c

2. Para los valores

x	1	2	3	4
f(x)	1	8	27	64

a) Tabla de diferencias regresivas ∇y_i :

i	y_i	∇y_i	$\nabla^2 y_i$	$\nabla^3 y_i$
0	1			
1	8	7		
2	27	19	12	
3	64	37	18	6

$$\begin{aligned}
 \nabla y_1 &= 8 - 1 = 7 & \nabla^2 y_2 &= 19 - 7 = 12 & \nabla^3 y_3 &= 18 - 6 = 12 \\
 \nabla y_2 &= 27 - 8 = 19 & \nabla^2 y_3 &= 37 - 19 = 18 \\
 \nabla y_3 &= 64 - 27 = 37
 \end{aligned}$$

Figura 2.61: Ejercicio 2.2 a

b) Newton hacia atrás para $f(3.4)$
 $x_n = 4$ $h = 1$ $q = 3.4 - 4 = -0.6$

$$P(3.4) = y_3 + q \nabla y_3 + \frac{q(q+1)}{2!} \nabla^2 y_3 = 64 - 0.6(37) + \frac{-0.6(-0.6+1)}{2!}(18)$$

$$+ \frac{-0.6(-0.6+1)(-0.6+2)}{3!}(6) = 41.8 - 2.16 + 0.336 = 39.304$$

Figura 2.62: Ejercicio 2.2 b

c) Compare con $f(x) = x^3$

$$f(3.4) = (3.4)^3 = 39.304 \quad \text{Coinciden igual}$$

Figura 2.63: Ejercicio 2.2 c

2.5.2.2.3. c) Comparación con $f(x) = x^3$

2.5.2.3. 2,3

3. Para los datos experimentales

X	10	20	30	40	50
Y	20,5	24,0	32,0	42,5	51,0

a) Nodos equiespaciados

$x_k = x_0 + kh$ $h=10$ $40 = 10 + 3(10) = 40$
 $10 = 10 + 0(10) = 10$ $50 = 10 + 4(10) = 50$
 $20 = 10 + 1(10) = 20$
 $30 = 10 + 2(10) = 30$

Figura 2.64: Ejercicio 2.3 a

2.5.2.3.1. a) Nodos equiespaciados

b) Calcule las diferencias progresivas hasta el tercer orden.

i	X _i	Y _i	ΔY_i	$\Delta^2 Y_i$	$\Delta^3 Y_i$
0	10	20,5	3,5	4,5	-2
1	20	24,0	8	2,5	-4,5
2	30	32,0	10,5	-2	
3	40	42,5	8,5		
4	50	51,0			

$\Delta Y_0 = 24 - 20,5 = 3,5$	$\Delta^2 Y_0 = 8 - 3,5 = 4,5$	$\Delta^3 Y_0 = 2,5 - 4,5 = -2$
$\Delta Y_1 = 32 - 24 = 8$	$\Delta^2 Y_1 = 10,5 - 8 = 2,5$	$\Delta^3 Y_1 = -2 - 2,5 = -4,5$
$\Delta Y_2 = 42,5 - 32 = 10,5$	$\Delta^2 Y_2 = 8,5 - 10,5 = -2$	
$\Delta Y_3 = 51 - 42,5 = 8,5$		

Figura 2.65: Ejercicio 2.3 b

2.5.2.3.2. b) Tabla

2.5.2.3.3. c) Estimación $y(25)$

2.5.2.3.4. d) Mejora o no

c) Estime $y(2.5)$ con Newton hacia adelante.

$$h = \frac{x - x_0}{n} = \frac{2.5 - 1.0}{10} = \frac{1.5}{10} = 1.5$$

Figura 2.66: Ejercicio 2.3 c1

$$\begin{aligned}
 P_3(2.5) &= y_0 + S \Delta y_0 + \frac{S(S-1)}{2!} \Delta^2 y_0 + \frac{S(S-1)(S-2)}{3!} \Delta^3 y_0 \\
 &= 20.5 + 1.5(3.5) + \frac{1.5(1.5-1)}{2} (4.5) + \frac{1.5(1.5-1)(1.5-2)}{6} (-2) \\
 &= 20.5 + 5.25 + \frac{1.5(0.5)}{2} (4.5) + \frac{1.5(0.5)(-0.5)}{6} (-2) \\
 &= 25.75 + 1.6875 + 0.125 = 27.5625
 \end{aligned}$$

Figura 2.67: Ejercicio 2.3 c2

d) Comente si el uso de un polinomio de orden superior puede mejorar o empeorar la estabilidad numérica.

Los polinomios de orden más alto, mejoran o aumentan la exactitud de donde pasan los puntos pero la oscilación entre ellos es más grande.

Figura 2.68: Ejercicio 2.3 d

Ejercicio 3. Polinomio general interpolante
Puntos $(0, 1), (1, 3), (2, 11)$

$$P_2(x) = a_0 + a_1x + a_2x^2$$

$$\begin{cases} a_0 = 1 \\ a_0 + a_1 + a_2 = 3 \rightarrow 1 + a_1 + a_2 = 3 \Rightarrow a_1 + a_2 = 2 \\ a_0 + 2a_1 + 4a_2 = 11 \rightarrow 1 + 2a_1 + 4a_2 = 11 \Rightarrow 2a_1 + 4a_2 = 10 \end{cases}$$

$$\begin{cases} a_1 + a_2 = 2 \\ 2a_1 + 4a_2 = 10 \end{cases}$$

$$P_2(x) = a_0 + a_1x + a_2x^2 = 1 + (-1)x + 3x^2$$

$$P_2(x) = 3x^2 - x + 1$$

$$\begin{array}{c|c} \begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 4 & 10 \end{array} & \begin{array}{c} \frac{1}{2} R_2 \\ \hline \begin{array}{cc|c} 1 & 1 & 2 \\ 1 & 2 & 5 \end{array} \end{array} \\ \hline \begin{array}{c} R_2 - R_1 \\ \hline \begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 3 \end{array} \end{array} \\ \hline \begin{array}{c} R_1 - R_2 \\ \hline \begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 3 \end{array} \end{array} \end{array}$$

$$\begin{aligned} a_1 &= -1 \\ a_2 &= 3 \end{aligned}$$

Figura 2.69: Ejercicio 3.1

2.5.3. Ejercicio 3

2.5.3.1. Puntos $(0, 1), (1, 3), (2, 11)$ 2.5.3.2. Puntos $(1, 2), (2, 5), (3, 10), (4, 17)$

Puntos $(1, 2), (2, 5), (3, 10), (4, 17)$

$$P_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

$$\begin{aligned} a_0 + a_1 + a_2 + a_3 &= 2 \\ a_0 + 2a_1 + 4a_2 + 8a_3 &= 5 \\ a_0 + 3a_1 + 9a_2 + 27a_3 &= 10 \\ a_0 + 4a_1 + 16a_2 + 64a_3 &= 17 \end{aligned}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 4 & 8 & 5 \\ 1 & 3 & 9 & 27 & 10 \\ 1 & 4 & 16 & 64 & 17 \end{array} \right) \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 7 & 3 \\ 0 & 2 & 8 & 26 & 8 \\ 0 & 3 & 15 & 63 & 15 \end{array} \right) \begin{array}{l} \\ \frac{1}{3}R_4 - R_2 \\ \frac{1}{3}R_4 - R_2 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 7 & 3 \\ 0 & 0 & 1 & 6 & 1 \\ 0 & 0 & 2 & 14 & 2 \end{array} \right) \begin{array}{l} \\ \\ \frac{1}{2}R_4 - R_3 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 3 & 7 & 3 \\ 0 & 0 & 1 & 6 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right) \begin{array}{l} \\ \\ \\ \end{array}$$

$a_3 = 0$

$a_2 = 1 - 7(0) = 1$

$a_1 = 3 - 7(0) - 3(1) = 0$

$a_0 = 2 - 0 - 1 - 0 = 1$

$$P_3(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

$$P_3(x) = 1 + 0x + 1(x^2) + 0x^3$$

$$P_3(x) = x^2 + 1$$

Figura 2.70: Ejercicio 3.2

2.5.3.3. Puntos $(-1, 4), (0, 1), (2, 3)$ 2.5.3.4. Puntos $(0, 2, 0), (1, 2, 7), (2, 4, 8), (3, 8, 9), (4, 16, 0)$

2.5.4. Ejercicio 4

2.5.4.0.1. a) Datos experimentales equiespaciados

Puntos $(-1, 4), (0, 1), (2, 3)$

$$P_2(x) = a_0 + a_1x + a_2x^2$$

$$\begin{cases} a_0 - a_1 + a_2 = 4 \\ a_0 = 1 \\ a_0 + 2a_1 + 4a_2 = 3 \end{cases} \Rightarrow \begin{cases} -a_1 + a_2 = 3 \\ 2a_1 + 4a_2 = 2 \end{cases}$$

$$\begin{pmatrix} -1 & 1 & 3 \\ 2 & 4 & 2 \end{pmatrix} \xrightarrow{R_2 \sim R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 4 \end{pmatrix}$$

$$a_2 = \frac{4}{3}$$

$$a_1 = \frac{4}{3} - 3 = -\frac{5}{3}$$

$$a_0 = 1$$

$$P_2(x) = 1 - \frac{5}{3}x + \frac{4}{3}x^2$$

$$P_2(x) = \frac{4}{3}x^2 - \frac{5}{3}x + 1$$

Figura 2.71: Ejercicio 3.3

2.6. Prácticas de laboratorio

2.6.1. Sesión 1

```
#---- SESION 1 ----
```

```
##---- OPERADORES LOGICOS ----
```

```
17<5
```

```
## [1] FALSE
```

```
17>5
```

```
## [1] TRUE
```

```
17<=5
```

```
## [1] FALSE
```

```
17>=5
```

```
## [1] TRUE
```

```
17!=5
```

```
## [1] TRUE
```

```
17==5
```

```
## [1] FALSE
```

```
##---- OPERADORES ARITMETICOS ----
```

```
# SUMA, RESTA, MULTIPLICACION, DIVISION, POTENCIA, MODULO, DIVISION ENTERA
```

```
17+5
```


$x = \{0, 1, 2, 3, 4\}$ $y = \{2.0, 2.7, 4.8, 8.9, 16.0\}$

$P_4(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$

$a_0 = 2$

$a_1 + a_2 + a_3 + a_4 = 2.7 \Rightarrow a_1 + a_2 + a_3 + a_4 = 0.7$

$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = 4.8 \Rightarrow 2a_1 + 4a_2 + 8a_3 + 16a_4 = 2.8$

$a_0 + 3a_1 + 9a_2 + 27a_3 + 81a_4 = 8.9 \Rightarrow 3a_1 + 9a_2 + 27a_3 + 81a_4 = 6.9$

$a_0 + 4a_1 + 16a_2 + 64a_3 + 256a_4 = 16 \Rightarrow 4a_1 + 16a_2 + 64a_3 + 256a_4 = 14$

1	1	1	1	0.7	$\frac{1}{2}R_2 + R_1$	1	1	1	0.7	$\frac{1}{3}R_3 - R_2$	1	1	1	0.7		
2	4	8	16	2.8	$\frac{1}{3}R_3 - R_1$	0	1	3	7	0.7	$\frac{1}{3}R_4 - R_2$	0	1	3	7	0.7
3	9	27	81	6.9	$\frac{1}{4}R_4 - R_1$	0	2	8	26	1.6	0	0	1	6	0.1	
4	16	64	256	14		0	3	15	63	2.8	0	0	2	14	$\frac{7}{30}$	

$a_4 = \frac{1}{60} \approx 0.016667$

$a_3 = 0.1 - 6\left(\frac{1}{60}\right) = 0$

$a_2 = 0.7 - 7\left(\frac{1}{60}\right) - 3(0) = \frac{7}{12} \approx 0.583333$

$a_1 = 0.7 - \frac{1}{60} - 0 - \frac{7}{12} = 0.1$

$P_4(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$

$P_4(x) = 2 + 0.1x + 0.583333x^2 + 0x^3 + 0.016667x^4$

$P_4(x) = 0.016667x^4 + 0.583333x^2 + 0.1x + 2$

Figura 2.72: Ejercicio 3.4

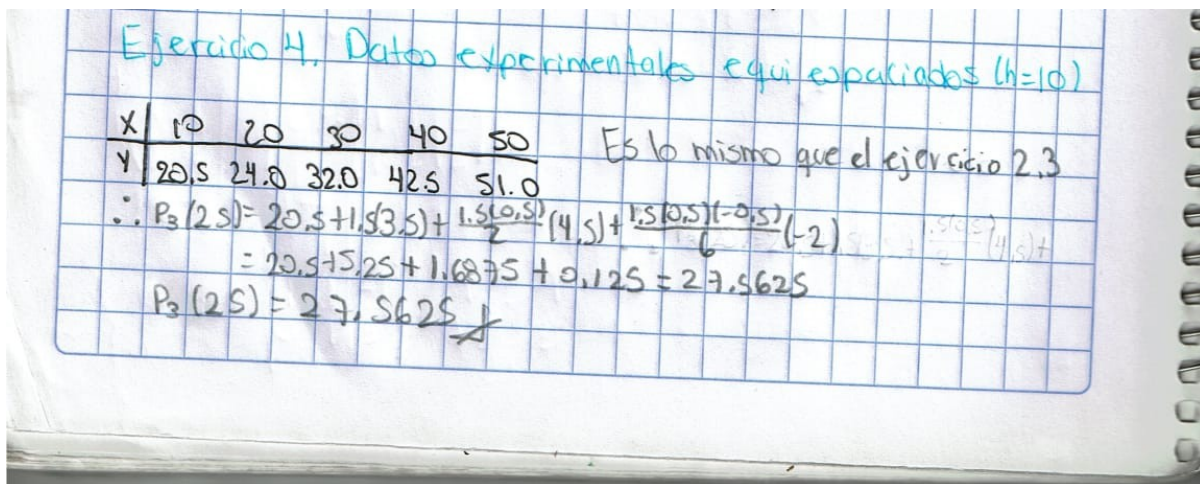


Figura 2.73: Ejercicio 4 a

[1] 22

17*5

[1] 85

17*5

[1] 85

17^5

[1] 1419857

17%/%5

[1] 3

17%%5

[1] 2

##---- LOGARITMOS Y EXPONENCIALES ----

log(1)

[1] 0

log(12)

[1] 2.484907

log(12,2)

[1] 3.584963

exp(12)

[1] 162754.8

exp(1)

```
## [1] 2.718282
##---- FUNCIONES TRIGONOMETRICAS ----
sin(45)

## [1] 0.8509035
cos(45)

## [1] 0.525322
tan(45)

## [1] 1.619775
asin(0.96)

## [1] 1.287002
acos(0.97)

## [1] 0.2455655
atan(0.45)

## [1] 0.4228539
##---- FUNCIONES VARIAS ----
abs(-34)

## [1] 34
sqrt(8)

## [1] 2.828427
floor(1.56)

## [1] 1
ceiling(1.56)

## [1] 2
max(4,7,2,12)

## [1] 12
min(4,7,2,12)

## [1] 2
sign(-45)

## [1] -1
##---- EJERCICIOS DE PRACTICA ----
# calcular la expresion
#cos(pi/6+pi/2)+e^2
# calcular la expresion
#cos(pi/6+pi/2)+e^2*log(5)+a(1/raiz(2))
# introducir las siguientes expresiones:
1/7
```

```
## [1] 0.1428571
```

```
options(digits=3); 1/7
```

```
## [1] 0.143
```

```
# c) options(digits=6); 1/7
```

```
# d) round(67.45)
```

```
# e) round(75.324568,2)
```

```
# f) options(digits=7);
```

```
# g) signif(56.345458234234,2)
```

```
# h) signif(56.345458234234)
```

```
# i) exp(-30)
```

```
# j) options(scipen= 999)
```

```
# k) exp(-30)
```

```
# l) options(scipen=0)
```

```
####---- MIS EJERCICIOS DE PRACTICA SESION 1 ----
```

```
##---- OPERADORES LOGICOS ----
```

```
20<10
```

```
## [1] FALSE
```

```
30>5
```

```
## [1] TRUE
```

```
46<=7
```

```
## [1] FALSE
```

```
36>=6
```

```
## [1] TRUE
```

```
26!=5
```

```
## [1] TRUE
```

```
26==5
```

```
## [1] FALSE
```

```
##---- OPERADORES ARITMETICOS ----
```

```
# SUMA, RESTA, MULTIPLICACION, DIVISION, POTENCIA, MODULO, DIVISION ENTERA
```

```
15+1
```

```
## [1] 16
```

```
16*2
```

```
## [1] 32
```

```
17*3
```

```
## [1] 51
```

```
18^4
```

```
## [1] 104976
```

```
19%/%5
```

```
## [1] 3
```



```
20%%6
```

```
## [1] 2
```

```
##----- LOGARITMOS Y EXPONENCIALES -----
```

```
log(2)
```

```
## [1] 0.693
```

```
log(6)
```

```
## [1] 1.79
```

```
log(7,3)
```

```
## [1] 1.77
```

```
exp(2)
```

```
## [1] 7.39
```

```
exp(6)
```

```
## [1] 403
```

```
##----- FUNCIONES TRIGONOMETRICAS -----
```

```
sin(90)
```

```
## [1] 0.894
```

```
cos(180)
```

```
## [1] -0.598
```

```
tan(90)
```

```
## [1] -2
```

```
asin(0.74)
```

```
## [1] 0.833
```

```
acos(0.79)
```

```
## [1] 0.66
```

```
atan(0.54)
```

```
## [1] 0.495
```

```
##----- FUNCIONES VARIAS -----
```

```
abs(-15)
```

```
## [1] 15
```

```
sqrt(9)
```

```
## [1] 3
```

```
floor(1.76)
```

```
## [1] 1
```

```
ceiling(1.76)

## [1] 2
max(15,215,3,12)

## [1] 215
min(1,9,4,2)

## [1] 1
sign(-69)

## [1] -1
##---- EJERCICIOS DE PRACTICA ----
# calcular la expresion
cos(pi/6+pi/2)+exp(1)^2

## [1] 6.89
# calcular la expresion
cos(pi/6+pi/2)+exp(1)^2*log(5)+acos(1/sqrt(2))

## [1] 12.2
# introducir las siguientes expresiones:
7/1

## [1] 7
options(digits=4); 2/8

## [1] 0.25
options(digits=6); 53/3

## [1] 17.6667
round(99.99)

## [1] 100
round(99.98765,3)

## [1] 99.988
options(digits=2);
signif(77.77777,4)

## [1] 78
signif(pi)

## [1] 3.1
exp(63)

## [1] 2.3e+27
```

```
options(scipen= 999)
exp(63)
```

```
## [1] 2293783159469609954760982528
```

```
options(scipen=0)
exp(63)
```

```
## [1] 2.3e+27
```

2.6.2. Sesión 2

```
##### SESION 2 #####
##### DEFINICION DE CONSTANTES #####
e = exp(1);
x = 0.0034
e <- exp(1)
x <- 0.034;
x0 = e^(2*x)
##### CONCATENAR Y PEGAR EXPRESIONES #####
txt = "El valor de x0 es _"
cat(txt, x0)
```

```
## El valor de x0 es _ 1.1
```

```
paste(txt,x0)
```

```
## [1] "El valor de x0 es _ 1.07036530847877"
```

```
paste0(txt,x0)
```

```
## [1] "El valor de x0 es _1.07036530847877"
```

```
##### ASIGNACION E IMPRESION #####
x0 <- 1
x1 <- x0 - pi*x0 + 1
(x1 <- x0 - pi*x0 + 1 )
```

```
## [1] -1.1
```

```
print(x1)
```

```
## [1] -1.1
```

```
##### LISTADO DE OBJETOS DEFINIDOS #####
ls()
```

```
## [1] "a"          "A"          "A1"         "A2"         "A3"
## [6] "A4"         "A5"         "Ab"         "b"          "base"
## [11] "binario"    "biseccion"  "c1"         "c2"         "c3"
## [16] "c4"         "c5"         "condicion"  "decimal"    "df"
## [21] "dg"         "dx_inf"     "e"          "f"          "F1"
## [26] "F2"         "F3"         "F4"         "fijo"       "g"
## [31] "grafica"    "I"          "II"         "III"        "IV"
## [36] "m1"         "m10"        "m11"        "m12"        "m13"
## [41] "m14"        "m15"        "m16"        "m17"        "m18"
```

```
## [46] "m19"      "m2"      "m20"      "m3"      "m4"
## [51] "m5"      "m6"      "m7"      "m8"      "m9"
## [56] "n"       "n_biseccion" "n1"      "n2"      "newton"
## [61] "NuevaA"  "numero"   "r"       "R1"      "R2"
## [66] "R3"      "R4"      "R5"      "resultado" "resultado1"
## [71] "resultado2" "secante" "t_biseccion" "tabla"   "txt"
## [76] "v"       "w"       "x"       "X"       "x_old"
## [81] "x0"      "x1"      "x2"      "x3"      "x4"
## [86] "x5"      "y"       "z"
```

```
# Eliminar todos los objetos
```

```
rm(list= ls())
ls()
```

```
## character(0)
```

```
##---- IMPRIMIR PEGAR AVANZADO ----
```

```
x0 <- 1
x1 <- x0 - pi*x0 + 1
cat("x0 =", x0, "\n", "x1 =", x1)
```

```
## x0 = 1
## x1 = -1.1
```

```
##---- EJERCICIOS DE PRACTICA ----
```

```
##---- TRABAJO EN CLASE ----
```

```
a <- 5
b <- 6
c <- 3
raiz <- ((b^2)-(4*a*c))
disc <- abs(raiz)
abs_disc <- abs(disc)
texto <- "El valor del discriminante es:"
cat(texto,disc)
```

```
## El valor del discriminante es: 24
```

```
###---- DISTINTAS MANERAS DE ESCRIBIR----
```

```
paste (texto,disc)
```

```
## [1] "El valor del discriminante es: 24"
```

```
paste0(texto,disc)
```

```
## [1] "El valor del discriminante es:24"
```

```
x0 <- 1
x1 <- x0 - pi*x0 + 1
texto <- "El valor de x0 es:"
paste (texto,x0)
```

```
## [1] "El valor de x0 es: 1"
```

```
texto <- "El valor de x1 es:"
cat(texto,x1)
```

```
## El valor de x1 es: -1.1
```

```
x1_2 <- x1^2
x1_3 <- x1^3
x1_4 <- x1^4
texto1 <- "El valor de"
texto2 <- ",el cuadrado es:"
paste(texto1,x1,texto2,x1_2)
```

```
## [1] "El valor de -1.14159265358979 ,el cuadrado es: 1.30323378673019"
```

```
ls()
```

```
## [1] "a"          "abs_disc" "b"          "c"          "disc"       "raiz"
## [7] "texto"      "texto1"   "texto2"     "x0"         "x1"         "x1_2"
## [13] "x1_3"      "x1_4"
```

```
rm(list=ls())
x0 <- 1
x1 <- x0 - pi*x0 + 1
cat("x0 es",x0,"\nx1 es",x1)
```

```
## x0 es 1
```

```
## x1 es -1.1
```

```
cat("x0 es",x0,"\n", "x1 es",x1)
```

```
## x0 es 1
```

```
## x1 es -1.1
```

```
###---- FUNCIONES TRIGONOMETRICAS----
```

```
a <- 1
b <- 2
c <- 3
alfa <- 35
beta <- 55
gama <- 90
seno_alfa <- a/c
coseno_alfa <- b/c
tangente_alfa <- a/b
texto <- "El valor del seno de alfa es:"
paste(texto,seno_alfa)
```

```
## [1] "El valor del seno de alfa es: 0.333333333333333"
```

```
texto <- "El valor del coseno de alfa es:"
paste(texto,coseno_alfa)
```

```
## [1] "El valor del coseno de alfa es: 0.666666666666667"
```

```
texto <- "El valor de la tangente de alfa es:"
paste(texto,tangente_alfa)
```

```
## [1] "El valor de la tangente de alfa es: 0.5"
```

2.6.3. Sesión 3

```
x1 <- c(1.2,1-5,1-9,1-10)
(x1)
```

```
## [1] 1.2 -4.0 -8.0 -9.0
```

```
x2 <- 1:20
(x2)
```

```
## [1] 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
```

```
x3 <- seq(1,20,by=0.1)
(x3)
```

```
## [1] 1.0 1.1 1.2 1.3 1.4 1.5 1.6 1.7 1.8 1.9 2.0 2.1 2.2 2.3 2.4
## [16] 2.5 2.6 2.7 2.8 2.9 3.0 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9
## [31] 4.0 4.1 4.2 4.3 4.4 4.5 4.6 4.7 4.8 4.9 5.0 5.1 5.2 5.3 5.4
## [46] 5.5 5.6 5.7 5.8 5.9 6.0 6.1 6.2 6.3 6.4 6.5 6.6 6.7 6.8 6.9
## [61] 7.0 7.1 7.2 7.3 7.4 7.5 7.6 7.7 7.8 7.9 8.0 8.1 8.2 8.3 8.4
## [76] 8.5 8.6 8.7 8.8 8.9 9.0 9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9
## [91] 10.0 10.1 10.2 10.3 10.4 10.5 10.6 10.7 10.8 10.9 11.0 11.1 11.2 11.3 11.4
## [106] 11.5 11.6 11.7 11.8 11.9 12.0 12.1 12.2 12.3 12.4 12.5 12.6 12.7 12.8 12.9
## [121] 13.0 13.1 13.2 13.3 13.4 13.5 13.6 13.7 13.8 13.9 14.0 14.1 14.2 14.3 14.4
## [136] 14.5 14.6 14.7 14.8 14.9 15.0 15.1 15.2 15.3 15.4 15.5 15.6 15.7 15.8 15.9
## [151] 16.0 16.1 16.2 16.3 16.4 16.5 16.6 16.7 16.8 16.9 17.0 17.1 17.2 17.3 17.4
## [166] 17.5 17.6 17.7 17.8 17.9 18.0 18.1 18.2 18.3 18.4 18.5 18.6 18.7 18.8 18.9
## [181] 19.0 19.1 19.2 19.3 19.4 19.5 19.6 19.7 19.8 19.9 20.0
```

```
x4 <- seq(20,1,by=-0.1)
(x4)
```

```
## [1] 20.0 19.9 19.8 19.7 19.6 19.5 19.4 19.3 19.2 19.1 19.0 18.9 18.8 18.7 18.6
## [16] 18.5 18.4 18.3 18.2 18.1 18.0 17.9 17.8 17.7 17.6 17.5 17.4 17.3 17.2 17.1
## [31] 17.0 16.9 16.8 16.7 16.6 16.5 16.4 16.3 16.2 16.1 16.0 15.9 15.8 15.7 15.6
## [46] 15.5 15.4 15.3 15.2 15.1 15.0 14.9 14.8 14.7 14.6 14.5 14.4 14.3 14.2 14.1
## [61] 14.0 13.9 13.8 13.7 13.6 13.5 13.4 13.3 13.2 13.1 13.0 12.9 12.8 12.7 12.6
## [76] 12.5 12.4 12.3 12.2 12.1 12.0 11.9 11.8 11.7 11.6 11.5 11.4 11.3 11.2 11.1
## [91] 11.0 10.9 10.8 10.7 10.6 10.5 10.4 10.3 10.2 10.1 10.0 9.9 9.8 9.7 9.6
## [106] 9.5 9.4 9.3 9.2 9.1 9.0 8.9 8.8 8.7 8.6 8.5 8.4 8.3 8.2 8.1
## [121] 8.0 7.9 7.8 7.7 7.6 7.5 7.4 7.3 7.2 7.1 7.0 6.9 6.8 6.7 6.6
## [136] 6.5 6.4 6.3 6.2 6.1 6.0 5.9 5.8 5.7 5.6 5.5 5.4 5.3 5.2 5.1
## [151] 5.0 4.9 4.8 4.7 4.6 4.5 4.4 4.3 4.2 4.1 4.0 3.9 3.8 3.7 3.6
## [166] 3.5 3.4 3.3 3.2 3.1 3.0 2.9 2.8 2.7 2.6 2.5 2.4 2.3 2.2 2.1
## [181] 2.0 1.9 1.8 1.7 1.6 1.5 1.4 1.3 1.2 1.1 1.0
```

```
x5 <- rep(1,times=10)
(x5)
```

```
## [1] 1 1 1 1 1 1 1 1 1 1
```

```
n1 <- length(x1); (n1)
```

```
## [1] 4
```

```
n2 <- length(x1); (n2)
```

```
## [1] 4
```

```
n3 <- length(x1); (n3)
```

```
## [1] 4
```

```
n4 <- length(x1); (n4)
```

```
## [1] 4
```

```
n5 <- length(x1); (n5)
```

```
## [1] 4
```

```
x <- runif(100,-3,3)
(x)
```

```
## [1] 2.693 -1.436 0.888 0.927 1.106 0.163 -1.994 1.374 1.164 -1.901
## [11] 0.867 -0.021 -1.608 -0.519 -1.273 2.949 1.811 -2.218 1.766 -0.660
## [21] 1.301 2.605 -2.908 -2.256 0.100 1.416 -1.660 -0.320 1.033 1.767
## [31] -0.911 -0.324 -2.882 -1.659 -2.391 1.425 -0.706 0.355 -2.433 -1.699
## [41] -2.137 -2.812 -0.435 -1.941 0.375 -1.551 0.566 1.296 -0.298 2.571
## [51] 2.134 0.168 1.995 0.976 1.188 1.388 1.257 2.866 -1.157 -0.295
## [61] -2.487 1.440 2.783 -1.444 -0.868 1.967 -1.874 -1.292 -1.562 1.296
## [71] -2.001 0.666 -0.672 0.623 -2.814 -1.610 -1.958 2.619 0.539 2.729
## [81] -0.606 1.074 -1.245 1.056 1.226 2.567 0.197 -1.697 2.579 2.599
## [91] 2.626 0.963 -0.650 -0.540 1.805 -2.393 0.332 1.035 -0.160 -0.603
```

Ejemplo de redondeo y truncamiento

```
x <- runif(20,-1,1);(x)
```

```
## [1] 0.021 -0.253 -0.806 0.291 0.813 0.699 -0.979 -0.917 0.513 -0.199
## [11] 0.400 0.547 0.226 0.526 -0.593 0.401 0.584 0.999 -0.191 -0.038
```

```
xx <- 100*x; (xx)
```

```
## [1] 2.1 -25.3 -80.6 29.1 81.3 69.9 -97.9 -91.7 51.3 -19.9 40.0 54.7
## [13] 22.6 52.6 -59.3 40.1 58.4 99.9 -19.1 -3.8
```

Fijamos la semilla para generar números aleatorios

```
set.seed(123) #Sirve para que cuando comparta mi archivo con alguien, vea exactamente los mismos números q
x <- runif(20,-1,1)
(x)
```

```
## [1] -0.425 0.577 -0.182 0.766 0.881 -0.909 0.056 0.785 0.103 -0.087
## [11] 0.914 -0.093 0.355 0.145 -0.794 0.800 -0.508 -0.916 -0.344 0.909
```

```
xx <- 100*x; (xx)
```

```
## [1] -42.5 57.7 -18.2 76.6 88.1 -90.9 5.6 78.5 10.3 -8.7 91.4 -9.3
## [13] 35.5 14.5 -79.4 80.0 -50.8 -91.6 -34.4 90.9
```

Redondeamos a tres cifras significativas:

```
x_red2 <- round(xx,2); (x_red2)
```

```
## [1] -42.5 57.7 -18.2 76.6 88.1 -90.9 5.6 78.5 10.3 -8.7 91.4 -9.3
## [13] 35.5 14.5 -79.4 80.0 -50.8 -91.6 -34.4 90.9
```



```
x_red3 <- round(xx,3); (x_red3)
```

```
## [1] -42.5  57.7 -18.2  76.6  88.1 -90.9   5.6  78.5  10.3 -8.7  91.4 -9.3
## [13]  35.5  14.5 -79.4  80.0 -50.8 -91.6 -34.4  90.9
```

Generamos una nueva variable con 5000 datos, con una distribución uniforme entre -10 y 10, con 3 cifras significativas, posteriormente obtendremos una muestra de tamaño 500 con el comando `sample(datos_origen,tamaño_de_la_muestra)` y se almacenaran en la variable `muestra_1`:

```
set.seed(123)
```

```
x5000 <- runif(5000,-10,10); (x5000) #sirve para crear una lista de números aleatorios con distribución un
```

```
## [1] -4.2484  5.7661 -1.8205  7.6603  8.8093 -9.0889  0.5621  7.8484
## [9]  1.0287 -0.8677  9.1367 -0.9333  3.5514  1.4527 -7.9415  7.9965
## [17] -5.0782 -9.1588 -3.4416  9.0901  7.7908  3.8561  2.8101  9.8854
## [25]  3.1141  4.1706  0.8813  1.8828 -4.2168 -7.0577  9.2605  8.0460
## [33]  3.8141  5.9093 -9.5077 -0.4441  5.1692 -5.6718 -3.6364 -5.3675
## [41] -7.1440 -1.7091 -1.7255 -2.6231 -6.9511 -7.2239 -5.3393 -0.6808
## [49] -4.6805  7.1566 -9.0834 -1.1560  5.9785 -7.5620  1.2190 -5.8694
## [57] -7.4494  5.0662  7.9009 -2.5107  3.3023 -8.1032 -2.3206 -4.5123
## [65]  6.2928 -1.0297  6.2013  6.2478  5.8868 -1.2034  5.0895  2.5844
## [73]  4.2036 -9.9875 -0.4937 -5.5976 -2.4037  2.2554 -2.9640 -7.7773
## [81] -5.1276  3.3611 -1.6471  5.7639 -7.9427 -1.3021  9.6991  7.8610
## [89]  7.7294 -6.4989 -7.3861  3.0620 -3.1297  3.1352 -3.5925 -6.2462
## [97]  5.6459 -8.1281 -0.6644  0.2301  1.9998 -3.3435 -0.2277  9.0895
## [105] -0.3420  7.8070  8.2888  2.1747 -1.7862 -7.0581  8.7060 -3.9754
## [113] -8.7856  8.9545  4.4119 -7.1541  0.9857  9.0818  1.7097 -1.9098
## [121]  2.9579 -3.6036 -3.8456 -5.6046 -2.6102  9.6844 -6.9160 -8.1791
## [129] -7.1619  3.8001  2.3851  7.8279  3.4600  4.7416  0.4227  3.1968
## [137]  6.4361  5.7256  9.5964 -1.2114 -3.7660 -1.8105 -9.7907 -6.3230
## [145]  6.8546 -5.3768 -5.2180 -8.4662 -5.0855  4.6427  6.9491 -0.0495
## [153] -2.2418 -5.0710 -7.7781 -2.2001  1.4387 -5.6621 -1.1046 -5.6402
## [161]  0.0460 -2.9219  2.9997 -2.5057 -2.8911  0.6738  4.8067 -5.5779
## [169] -1.7451 -4.6863  2.5995 -6.3234  7.2729  4.9314  3.3657  2.3604
## [177] -2.5552  0.5967  7.4936  1.6350  6.7954 -3.7510  4.1658 -4.6996
## [185]  1.8869 -0.3742 -4.6993  1.2918  8.2638  8.0375 -4.5167 -3.5703
## [193]  9.7128  2.3999  8.7463 -0.6693 -1.8633  3.1846 -6.9531  1.4573
## [201] -5.2255  9.2472  2.0273  0.3006 -1.9485  7.6049 -2.7182 -4.2352
## [209] -6.5871 -6.5566 -0.3591 -4.9407 -5.6749  3.4875 -9.0467  4.0171
## [217] -2.9622 -1.8211  6.4190  8.3771 -4.3494  9.2221  4.5679  3.7275
## [225] -8.9431 -2.0956 -0.4431  1.2051  3.9652  8.3137  2.3670 -1.4316
## [233]  0.8416 -8.8304 -4.7829 -2.0570 -6.0451  6.6386 -6.9423  6.0684
## [241]  0.9365  3.2464 -6.5660  2.6611 -3.7626  4.4911 -2.0212  9.3871
## [249]  9.3480  4.5341 -4.8557 -5.5642  1.8609 -4.6496  0.6214  5.7058
## [257] -6.6388 -1.9120 -0.5685  7.3621  8.5142  7.6396  3.4837  9.0033
## [265]  0.3289  1.5304 -3.2734 -3.0535 -9.5995  0.0563  7.4209 -9.8740
## [273] -8.5589 -6.7158  5.4067  4.7037  9.4375 -0.6706 -8.5123  2.9764
## [281]  5.1719 -7.2579 -2.0683 -5.5003 -8.8408 -2.0821 -8.7014 -5.4823
## [289] -8.9074  3.4056 -4.0452 -7.9856 -8.5619  7.6088  5.0849  6.3321
## [297]  9.6428 -7.9280 -8.0192  5.9766  5.6915 -9.8114  5.5813  4.5878
## [305]  2.6026 -0.3818 -6.8673 -9.8357 -0.9508 -0.1541 -2.2083 -0.7067
## [313]  4.2656 -8.8940 -2.9043  6.0562  6.7142 -5.2450 -2.9203  7.1377
## [321]  7.0753 -4.0821 -7.0590  4.0798 -7.9239 -9.3254  9.9881 -9.3025
```

##	[329]	-3.2322	8.3013	2.3447	-4.2743	4.7559	6.6811	-3.7146	-0.1487
##	[337]	3.9475	2.8292	2.8785	9.5571	-1.7053	-7.6119	0.5206	-5.4985
##	[345]	-0.2718	-2.5957	9.6670	-2.2336	-5.4151	2.4660	-7.2692	9.3494
##	[353]	0.3014	-6.7386	2.4380	9.7191	3.3754	-1.6217	-3.5331	6.7051
##	[361]	-7.1237	-6.1437	7.9348	-3.8376	-2.7340	5.6789	-6.1324	-9.6447
##	[369]	-1.8678	-0.3366	-1.5631	-3.1438	7.3297	-0.8978	0.6753	9.2769
##	[377]	5.4918	-5.8225	-3.8243	9.4268	1.6980	5.2165	-2.5458	5.3839
##	[385]	0.7535	8.2799	-6.2941	-4.3556	-8.1008	-5.7903	9.5420	-4.0740
##	[393]	4.5197	5.7138	-7.8916	-5.2081	-4.5891	-7.9788	-7.6417	9.8247
##	[401]	9.7211	-7.2587	8.1062	1.5260	-2.0910	-1.0040	4.1300	-8.3499
##	[409]	-3.2137	3.6158	-3.6610	6.6314	-5.6966	-0.0410	-4.4790	-6.1595
##	[417]	9.0124	-3.5655	-0.4309	-9.4401	0.9492	2.8848	1.9253	-3.5613
##	[425]	7.8223	2.5251	-3.9419	-2.2359	-6.7905	7.2510	9.0620	1.2729
##	[433]	-3.4091	9.9323	-5.3036	2.2534	-7.8364	-0.2593	-8.0111	-6.7767
##	[441]	-4.3401	1.6774	4.6342	-6.6896	7.3294	4.1715	5.2080	-7.0583
##	[449]	-2.8389	3.4666	0.4765	-3.0040	-5.1894	-8.8362	-5.2676	7.8016
##	[457]	6.2365	4.9503	-6.9018	-7.5052	9.4945	-1.2774	-0.7197	-6.6940
##	[465]	1.6987	-4.5844	-5.3981	3.8242	-4.3430	6.2080	-8.1217	6.4406
##	[473]	-1.4514	5.1177	3.2477	-1.1095	2.5429	-9.9907	-5.6551	4.0974
##	[481]	-5.6967	6.2787	-3.8447	3.7549	8.6536	-7.6844	-7.4459	3.5645
##	[489]	-1.4210	6.6880	9.4288	-8.5902	-0.8043	4.0317	-8.2612	9.8589
##	[497]	-4.9380	-9.0092	3.7265	5.7385	-2.9279	-2.6712	-4.2580	-8.4005
##	[505]	-2.6909	-6.4397	0.7211	0.0790	8.9007	-3.1736	-0.7057	-8.3494
##	[513]	7.2021	-2.0868	4.7180	-6.5651	-0.9048	5.4041	-8.7470	6.3016
##	[521]	-3.9771	-2.7066	-3.7577	-9.2526	0.3761	3.5803	8.0647	-9.4895
##	[529]	9.7816	-3.9422	8.7828	3.7522	-1.0593	6.3296	-9.2100	4.7806
##	[537]	-3.0256	6.5850	0.7108	-4.5091	6.0190	-8.1671	6.6422	-4.4629
##	[545]	5.0622	9.2831	-8.3707	7.0873	6.0448	-2.2965	-3.4481	-5.9012
##	[553]	1.3877	7.7611	0.5943	1.7392	3.3147	0.5978	0.1969	-9.6768
##	[561]	-9.0458	8.5870	5.3857	-5.9784	3.0052	3.0753	-2.0949	6.2461
##	[569]	0.9405	7.7027	1.0663	8.1210	1.7492	-1.5307	8.9917	4.1808
##	[577]	-1.7339	-9.6327	1.3347	-0.1987	7.5735	6.2571	7.0820	-2.6421
##	[585]	7.4790	-6.9732	-4.3638	3.3341	9.5477	1.6548	0.5318	-8.7844
##	[593]	9.3808	-7.5953	-8.2327	7.6153	0.1674	-3.2501	7.8867	-9.3606
##	[601]	-5.2554	3.7298	-5.4836	-3.6301	-6.5203	6.0286	-7.0744	6.4543
##	[609]	-3.3800	-2.5166	2.5949	-8.0673	-9.5601	9.8609	1.6788	5.6365
##	[617]	7.8379	5.0973	9.5841	-9.1171	8.0680	7.3100	5.5082	-2.4637
##	[625]	-9.1578	-2.7118	-4.5250	7.0093	-2.7520	-3.9104	5.1827	6.8967
##	[633]	-0.8416	4.5926	-7.9184	-5.6003	9.0790	5.0408	6.3791	-1.6445
##	[641]	1.8768	5.9765	7.7682	-2.2962	-8.1936	2.5149	4.9127	-8.2924
##	[649]	-3.9896	2.2912	5.0586	8.3411	-0.4850	1.3424	4.7324	7.1489
##	[657]	8.1829	-8.8724	0.0582	-2.9891	6.9111	6.1287	-7.6534	4.2537
##	[665]	-5.2946	-8.5009	8.7129	-6.8566	2.9411	-6.5296	-9.5985	0.4263
##	[673]	-8.2744	-4.3400	-1.5913	1.7456	6.1339	-5.9601	-0.8115	-1.0371
##	[681]	4.6750	4.2951	6.6244	7.7313	9.0529	1.0123	0.4467	-6.5860
##	[689]	-0.4146	-4.9236	-9.2044	2.6960	0.7912	-7.1979	-4.3258	1.6606
##	[697]	-6.6995	-8.0740	-1.4277	-2.8848	6.8987	-4.7974	-9.5371	7.2480
##	[705]	-3.3082	2.6358	0.9285	-2.4711	-6.2825	-1.4212	2.6155	0.4168
##	[713]	3.1924	4.5872	-0.2635	-2.3109	-9.8633	-9.9263	9.8987	-7.8423
##	[721]	-1.6266	4.3577	4.8508	7.4400	2.1574	5.1241	6.9448	2.2556
##	[729]	5.8645	-9.5429	-1.6617	7.4976	2.9469	8.4896	-6.5555	-3.6837
##	[737]	6.0851	9.7793	-3.5523	-9.9762	9.8434	-7.0330	-9.0127	1.9471

##	[745]	-3.5734	0.5586	5.9084	-8.6213	3.9004	8.9726	-4.4446	-5.0119
##	[753]	-6.8113	-8.5674	1.1488	-0.2033	-0.1142	7.7674	-9.2659	-5.9251
##	[761]	0.2756	-5.2732	1.5087	-0.3548	1.3871	-7.1170	-7.0858	-1.7543
##	[769]	3.6602	3.0686	8.3197	6.2623	3.5178	6.1500	-7.4387	-4.9844
##	[777]	-3.3691	-1.8546	2.7074	6.1722	-4.8240	6.3893	-9.6904	3.0880
##	[785]	6.2344	-0.7726	-5.9317	-9.6246	-4.3373	8.3055	8.6739	0.8506
##	[793]	-5.6115	-0.2114	6.0230	-1.8474	-7.9258	-4.3887	-2.7707	-4.8151
##	[801]	-0.5864	-2.6831	-7.5746	-9.0601	-4.7441	9.3728	-0.2301	-0.4436
##	[809]	4.9759	3.3528	-9.0117	3.9021	-2.7348	7.6827	5.5059	-7.2159
##	[817]	-4.0998	-7.4783	1.7980	1.2335	3.7744	-3.7746	2.1117	9.8207
##	[825]	4.8641	-8.4829	-0.9766	-8.9293	-3.2089	4.6790	-9.9179	5.4382
##	[833]	-0.7405	4.4168	3.3301	1.4415	4.0763	3.1444	-4.2130	-8.0552
##	[841]	9.2484	4.7267	2.2545	-7.6014	1.0052	-4.7449	7.9672	-9.8164
##	[849]	-5.2753	-7.3991	-3.4753	4.5280	9.8349	4.3027	0.0888	-1.2790
##	[857]	8.9765	-7.5964	-8.4971	7.7804	-1.5110	-9.1528	2.9488	-0.6276
##	[865]	2.3585	-4.5837	-6.8541	-7.7149	0.1537	0.9606	-7.1871	-6.6046
##	[873]	5.2397	0.5479	7.2198	3.4711	-9.7392	3.8640	7.8343	2.6370
##	[881]	-7.8541	8.4213	3.5072	-7.0281	4.9068	8.8520	-1.5843	-4.0455
##	[889]	-4.8115	-5.5424	1.3131	5.1330	3.3921	0.9305	6.2293	5.1833
##	[897]	-9.5986	-2.3819	-8.9824	5.9581	8.4740	0.8520	7.0473	1.6713
##	[905]	3.3665	0.2263	5.2550	8.0672	6.4095	-8.5714	-9.9221	-8.9560
##	[913]	7.3312	1.5249	-3.7231	9.1893	1.8239	0.6282	-2.3213	-3.6089
##	[921]	6.1677	-9.1616	-2.7251	7.1319	3.9589	3.6897	-3.0397	1.0936
##	[929]	-7.2551	5.6986	7.7373	-5.9181	5.4125	1.9273	9.1534	-6.8262
##	[937]	0.5195	7.4630	7.3941	-9.5262	9.5178	-0.1955	-2.2166	-1.6489
##	[945]	-8.1415	-6.7638	-1.8917	-3.1637	-1.6949	-3.9190	1.2056	-6.8825
##	[953]	9.1316	-9.1207	-2.5568	9.2523	2.9086	-8.7747	-1.8011	-1.4819
##	[961]	0.1632	-1.0080	2.4652	-7.2004	8.1589	1.3889	0.9656	-7.6634
##	[969]	5.2406	-0.4326	5.6394	-9.0795	6.3969	-4.6118	-4.3430	2.8644
##	[977]	8.9624	-9.8600	-2.9676	-1.6191	-0.8425	4.2339	8.3970	2.5422
##	[985]	8.0436	5.1466	-7.2428	-6.9349	-6.1738	-1.3363	-8.2556	-5.5244
##	[993]	1.4430	-1.9966	1.3093	6.5925	2.8423	-2.1700	4.1916	-7.8235
##	[1001]	-4.5275	1.8773	-6.7963	7.0686	6.9548	-0.4423	5.4738	-4.0920
##	[1009]	-8.6874	-1.1893	-0.7580	-3.1813	-6.2972	0.1400	-9.6162	5.4737
##	[1017]	1.1933	2.8293	-0.6314	-4.4155	-0.7757	5.8548	1.7113	-2.1048
##	[1025]	6.2247	-3.6287	1.6289	-0.8902	-4.6239	1.1035	-5.3754	0.8647
##	[1033]	-1.0486	-7.3942	2.4345	0.9599	-7.0151	1.1909	-1.4623	-1.0454
##	[1041]	6.6659	4.3917	-0.8623	0.4253	-5.1509	-8.4828	-2.1741	-9.7378
##	[1049]	7.3500	9.8523	4.1771	6.5343	7.8379	0.7882	1.1063	-4.4712
##	[1057]	3.1840	-5.1313	-4.2347	4.3156	4.5507	-6.5391	-3.8738	-2.6019
##	[1065]	-8.4456	4.2630	1.0185	-7.2827	9.4833	1.2393	5.7682	6.9783
##	[1073]	7.5608	6.0095	2.8030	2.5786	-4.5718	-9.4012	-1.6027	-2.8913
##	[1081]	-2.1534	-6.5613	-3.6072	-1.7289	5.1867	-4.5622	-7.6884	-9.4133
##	[1089]	6.1371	-0.3975	6.1250	-8.4199	-7.6933	-7.4117	4.7750	5.1420
##	[1097]	9.8491	2.4911	-4.2262	-6.0018	6.0184	4.5553	-5.6590	4.8848
##	[1105]	7.3331	0.5881	1.9728	8.7526	9.0145	3.2106	-8.5543	6.5266
##	[1113]	-0.4091	-2.1754	-4.0175	-5.1138	-1.5638	8.7554	-4.7103	1.3714
##	[1121]	-5.9563	-6.1439	4.3722	-3.7296	-7.2322	-8.8394	8.6220	1.3668
##	[1129]	-7.6446	1.2959	0.8051	6.3937	4.0596	1.3550	4.4262	5.0489
##	[1137]	-2.3715	-4.2398	0.6337	-4.8071	6.6358	-1.8927	-8.5474	9.0752
##	[1145]	-5.6563	-1.8875	2.5134	2.5681	-3.4352	-7.2218	8.2931	1.9823
##	[1153]	4.9921	8.3791	0.5753	-2.5604	-8.6838	7.0874	0.2082	-5.8000

## [1161]	-2.4831	-5.8527	-0.8152	-5.1700	-7.6262	-8.5506	3.8198	-4.2608
## [1169]	-7.0118	9.4486	-1.7897	-1.9637	2.9711	8.3261	-5.6668	0.9600
## [1177]	4.4010	-2.0551	-8.1200	2.6979	-9.9504	-5.7895	3.5805	7.3238
## [1185]	5.9939	-7.3001	-2.2500	5.7694	3.8583	1.8855	-7.5228	1.1016
## [1193]	-1.0118	5.3233	-9.4780	-6.0966	7.6247	-7.8040	9.3710	-2.2960
## [1201]	7.1718	7.7477	-0.2182	4.3618	-0.2659	9.7742	-8.7050	-6.8467
## [1209]	5.7070	0.8438	-1.6691	9.9777	-4.8865	0.1575	-8.4206	6.2734
## [1217]	-2.3564	6.0437	-6.0415	8.9280	-3.0866	0.4410	-7.7662	7.7195
## [1225]	9.0851	-9.1924	-0.1277	-5.4787	7.1760	0.6187	-9.9072	-4.4488
## [1233]	-3.4959	1.7741	-5.0063	-9.1377	-7.7864	4.0751	8.7804	-3.7766
## [1241]	-8.4301	-3.5651	2.4981	-1.1952	6.0269	-4.4143	1.4143	-9.1574
## [1249]	-6.1857	4.5417	6.5338	0.2144	1.3545	-9.9769	-7.1244	7.3193
## [1257]	-8.3488	-5.1086	9.6309	1.5516	-5.0255	2.2991	-9.3649	-7.0715
## [1265]	4.0615	-8.6888	2.4361	8.7465	-8.2421	9.4561	-9.4798	4.5046
## [1273]	-0.9259	1.3236	7.4546	1.0913	4.7530	2.4626	-3.7794	-2.1561
## [1281]	-5.9583	7.0556	2.1366	5.1220	1.2503	-4.4820	4.7112	0.9856
## [1289]	-3.0775	0.2993	6.3154	0.5255	-5.9392	6.9622	-2.5901	-3.9335
## [1297]	5.4119	4.6690	6.7781	1.3810	-9.4744	-0.2775	0.8536	6.7332
## [1305]	4.5733	1.1023	-8.5317	-7.2700	3.6922	-8.0076	-5.9243	-2.6804
## [1313]	6.9239	-5.2625	4.0976	-7.8478	5.5814	-4.8979	0.9609	-9.3220
## [1321]	6.1222	-3.2449	8.3257	-2.8579	9.5073	-2.3124	-0.2265	-0.1845
## [1329]	-9.7549	2.8295	0.2515	-3.5930	1.6287	2.3469	-1.2345	-6.5926
## [1337]	4.3016	4.9557	6.8783	0.6952	-3.9553	-0.8696	-2.3131	-5.0191
## [1345]	3.0925	9.1065	-4.1783	3.0038	0.7272	1.9490	-9.5020	-9.9032
## [1353]	-7.3724	-9.5037	-8.1574	-0.4888	-6.0669	6.2935	-5.1188	-0.2238
## [1361]	2.9776	-6.8257	6.7396	-1.2681	-5.3301	-4.0229	-6.8116	1.7113
## [1369]	-7.0244	-6.4188	-3.2156	-6.3220	-1.8892	2.2712	3.7134	6.4705
## [1377]	-2.5204	9.0568	-9.6234	2.2475	-0.7285	-2.9054	7.6485	-8.4839
## [1385]	7.6658	-6.2356	-5.6987	7.3906	-8.7832	-5.3020	9.8603	3.5640
## [1393]	-1.2903	-2.4538	-0.7763	6.0370	3.2593	-0.6799	-8.9348	-5.2560
## [1401]	-5.3352	-5.3869	-8.7655	-0.0576	-5.1175	5.1637	0.9461	8.9613
## [1409]	-8.2766	8.9577	4.4480	8.4401	2.2768	-6.7807	-6.3412	4.6037
## [1417]	1.7905	-3.7504	5.4559	-3.2740	7.1135	1.6106	-1.6855	8.6515
## [1425]	-0.7460	-8.0131	-0.6910	9.6817	8.5055	-1.2920	7.3944	-3.9929
## [1433]	5.9595	6.0102	-2.2615	-2.3682	2.9103	7.2333	3.1327	0.3265
## [1441]	-7.0244	2.2553	-9.1607	-0.0328	4.7902	5.0780	-8.7381	-1.1504
## [1449]	0.0134	-4.5951	1.9760	4.7168	4.2716	-7.6701	1.5024	5.9626
## [1457]	-5.3635	4.4971	6.7605	-7.6232	9.1790	4.9553	6.2178	-6.2016
## [1465]	-9.4806	7.2756	8.3836	-7.2474	-0.4470	-0.7408	4.0036	-8.6885
## [1473]	4.6608	-2.0814	-0.7702	4.5060	-0.5999	9.9905	6.9187	6.5641
## [1481]	5.2329	-7.7579	6.7795	6.2770	9.8282	-9.0337	4.9358	-1.6137
## [1489]	1.6429	6.6147	-9.7294	-9.7696	9.9289	5.3138	-3.7067	6.2402
## [1497]	9.8244	-5.9848	2.9207	-1.2490	8.7606	9.7601	-0.8736	-5.3877
## [1505]	3.9098	1.1326	1.6942	-1.3274	-1.4765	1.9371	-0.9583	9.1322
## [1513]	6.8886	-5.5446	-1.5966	-1.6003	-9.5842	-2.3910	-1.5530	-6.1715
## [1521]	4.1066	-9.7296	4.6241	-1.3455	4.6248	-9.7212	-9.0937	-1.6317
## [1529]	2.8768	-5.6041	6.6689	3.8953	7.9825	-1.7152	-1.7799	-4.9997
## [1537]	-2.5247	-5.3621	8.6321	-0.4731	-9.0467	5.4706	-3.3777	1.0575
## [1545]	3.5266	-4.4533	-8.9429	-1.4848	2.2024	9.5209	9.3960	-8.2303
## [1553]	-0.0324	-2.0862	-2.1934	-3.6300	3.6515	-4.3077	-2.1981	9.8034
## [1561]	5.8401	-3.7519	6.3424	-5.5818	0.0215	-6.6606	-7.6068	7.9522
## [1569]	-8.1257	7.3012	-4.4682	-9.4543	5.7477	-2.8268	-9.4979	4.9219

## [1577]	-9.4075	-1.3227	-4.8675	-0.0397	3.0671	-8.3269	-6.3906	-7.9107
## [1585]	6.2479	2.5500	2.6114	-2.1822	-1.3549	-3.0273	5.8712	7.7387
## [1593]	3.0225	1.8573	-3.4436	-8.4862	1.8277	-6.4333	4.8270	-4.1225
## [1601]	2.7837	-7.5036	-4.8947	6.4115	6.0756	-9.0833	7.5106	-7.4398
## [1609]	2.1766	-0.6087	1.1458	4.6079	-0.6028	9.7722	9.6934	-2.1343
## [1617]	2.1769	-3.0903	-1.2577	-5.4174	-9.8786	-9.5225	-8.8248	5.2915
## [1625]	-0.6191	7.0166	1.6344	-6.2622	2.1812	7.8668	5.8864	0.0819
## [1633]	-1.5394	3.0876	7.7548	7.8403	-6.4322	-4.8500	-7.7312	5.9432
## [1641]	-7.8092	6.5814	5.4209	8.8738	-0.6608	8.7971	5.7037	7.4922
## [1649]	-2.1107	-8.0060	-4.4608	4.4444	-2.8739	-2.3458	-9.3606	0.4289
## [1657]	-7.5784	7.7037	-8.5071	5.6846	7.0826	-9.2564	-4.4971	2.8414
## [1665]	5.7016	-1.9987	8.7060	-2.1897	-1.5208	-4.2716	2.2350	1.1763
## [1673]	-9.2007	4.9056	-5.8703	3.1299	0.4483	2.6687	2.3513	-4.3021
## [1681]	-5.5239	0.7804	9.9274	-3.4217	-3.5333	-9.2935	0.5122	1.9361
## [1689]	4.8417	9.0567	-0.2076	-8.3343	-4.8014	7.4969	7.0412	6.8581
## [1697]	8.9381	3.0536	-0.2369	-6.2449	4.2607	-8.2986	-0.7760	8.7181
## [1705]	6.9059	4.3237	-7.5239	1.8965	9.7971	2.3082	-4.5384	-8.7259
## [1713]	-8.5539	7.4702	-2.7435	8.9124	1.1664	0.2532	8.9554	-7.8040
## [1721]	6.3782	-2.4532	1.1328	-3.7226	-8.3531	5.5988	-6.1353	3.1117
## [1729]	-1.2971	-6.0570	9.8932	-1.0651	-9.3715	8.7403	7.4197	2.0249
## [1737]	-4.0197	6.1869	9.0428	7.0411	-7.4538	9.3776	1.1420	-9.8927
## [1745]	-7.2847	-9.8831	6.3375	5.6433	8.6213	-2.6262	9.4891	6.4749
## [1753]	5.7490	2.4565	9.3471	-0.0016	7.8739	-1.9791	-1.0492	8.7156
## [1761]	3.6666	-9.6367	-6.6895	-0.0833	-1.4693	9.8411	7.7790	-3.0849
## [1769]	4.1169	8.1745	3.5259	2.7820	-7.0079	2.0225	-4.5450	-0.8429
## [1777]	-5.5549	-4.9016	3.0738	-0.5601	-6.7807	-4.5584	4.2591	2.2671
## [1785]	-7.3576	-8.5999	9.3253	-2.6346	3.5491	6.9190	-5.1670	-9.2970
## [1793]	1.9048	6.7238	-2.7550	-3.3562	2.8947	9.1723	1.9241	0.2056
## [1801]	-6.8947	6.9170	-5.7124	3.3975	2.3551	-9.0000	8.9880	1.6510
## [1809]	7.2191	0.3693	-4.6775	1.9476	5.9115	3.7857	-0.3874	-0.6814
## [1817]	2.3682	-1.0146	2.0542	-9.9078	9.8999	-1.8708	-7.6410	7.3908
## [1825]	0.8039	7.6320	-9.2492	7.6287	4.4470	5.3962	7.2719	0.0148
## [1833]	8.5173	-5.8385	-9.4579	-3.6841	3.2022	-3.7062	8.8892	5.5794
## [1841]	-3.2114	-0.3671	-1.6801	5.5267	-0.2915	-9.6940	2.8490	-5.3034
## [1849]	4.9405	-4.0879	8.1244	7.0665	-0.7607	-8.5730	1.5561	1.5916
## [1857]	9.8715	-3.2945	3.3360	8.7711	1.4971	3.8342	-8.2048	7.3292
## [1865]	0.0228	-5.5349	-1.7516	7.7261	-0.0881	3.4551	-4.3499	6.3412
## [1873]	-5.0780	-2.6071	-5.2895	8.7151	-1.6985	-7.2510	8.2873	-3.0472
## [1881]	7.0586	-7.1473	-2.8113	-5.0899	-9.0819	5.6910	-6.0166	-2.1630
## [1889]	-3.5288	2.0421	0.8254	9.8165	-4.9242	4.7054	9.5522	1.1609
## [1897]	-2.1458	6.4938	-7.7523	0.7156	-1.1234	-4.5209	-6.8528	7.4368
## [1905]	1.2409	1.0138	1.8473	4.5563	2.7785	2.2890	-8.9517	1.3032
## [1913]	1.7468	2.5713	2.4378	-8.3546	-8.4471	3.7419	6.6061	-7.0818
## [1921]	5.6706	-0.2710	9.7853	-0.0179	1.2452	-2.4349	0.3727	-8.5143
## [1929]	0.7694	9.2212	0.5562	-1.2568	-9.3547	-9.2453	-9.0530	-6.7808
## [1937]	-0.6181	6.1185	-4.3880	5.0957	0.4365	-4.2211	-9.6525	3.3670
## [1945]	-8.6605	6.7277	-7.2931	3.1176	6.7590	-4.8470	-7.2801	5.2698
## [1953]	-5.7601	-4.6406	0.6366	-0.5177	-2.5311	5.5070	1.1641	-6.8953
## [1961]	9.7884	-5.1111	-7.3924	-5.7464	-2.3999	-2.8076	-3.9468	-9.1120
## [1969]	8.6957	1.3743	-5.5839	7.5592	-0.6542	0.8470	5.6880	2.9574
## [1977]	-7.1021	-3.5673	9.0211	-6.2360	5.0081	-0.8168	-7.1727	-5.8961
## [1985]	3.5059	-7.2394	-1.6891	-0.9751	2.4589	-4.9765	-0.7169	-6.6898

## [1993]	7.1561	-6.8690	-8.2334	7.8684	-3.9876	5.0417	-1.9679	-1.0288
## [2001]	-6.8065	-7.1097	-7.0164	0.2887	-0.1435	2.3269	-1.0515	-8.8865
## [2009]	-9.8921	-5.5633	7.0193	-4.6531	1.9720	2.1720	9.8432	-6.1762
## [2017]	5.0678	-5.1523	-3.4510	-9.9293	9.9485	-1.2388	9.9538	-6.0656
## [2025]	-7.7705	-4.9853	3.6095	1.5451	-1.6731	4.5836	1.4838	1.5527
## [2033]	1.8000	-5.5414	-7.9302	4.7301	2.2481	9.0242	9.1975	4.8967
## [2041]	-1.3034	-6.5045	-1.2942	5.2492	8.3806	8.1340	6.3103	-6.0049
## [2049]	-9.0075	-0.4210	1.8079	-8.0676	9.0178	-6.7451	8.4301	-7.4445
## [2057]	3.2548	5.6808	5.2923	3.4514	-7.6867	7.0260	2.3592	6.7282
## [2065]	-6.6014	-9.7944	-3.5306	-4.6063	6.5052	8.6732	-7.4442	8.2668
## [2073]	2.1047	-1.9943	3.3159	5.3520	0.4379	-2.1098	9.3157	-9.0203
## [2081]	-6.9338	3.9368	4.5557	-7.8324	-0.7086	2.8425	-2.0578	9.8086
## [2089]	3.5742	-3.6936	-6.9246	-3.9750	-8.1097	-9.4625	-3.7903	4.8488
## [2097]	9.2018	-1.6839	0.4447	-8.6496	2.5968	-2.9317	-1.5057	9.2754
## [2105]	3.6200	4.3693	-6.5874	-5.9278	7.5310	6.7505	4.4125	-4.3787
## [2113]	-5.8002	-0.1170	0.4351	-5.6034	5.2621	-9.5096	4.2297	6.8178
## [2121]	8.6164	3.2418	-4.6012	-5.6353	7.3564	-9.5086	7.0005	0.4787
## [2129]	-1.2907	8.6259	-6.7090	-2.4242	-7.2391	-7.0056	3.5289	1.2423
## [2137]	-0.5670	6.2220	9.2478	0.3082	4.0745	-6.3824	-2.9007	-9.0019
## [2145]	-6.9489	-8.0146	-4.3970	-2.2510	2.6832	4.1413	-3.4796	9.3037
## [2153]	3.9291	-6.7784	-2.6188	-3.9556	-0.8407	6.4657	-5.3492	-1.9536
## [2161]	9.4323	-7.5116	2.6058	4.0773	1.8240	9.9303	-9.0933	9.8015
## [2169]	4.9061	-1.2650	-6.9399	-1.0227	-6.2735	-1.4690	6.4156	-8.9924
## [2177]	-3.4924	6.8482	-9.1960	5.8375	9.2326	3.2184	-9.8257	-0.6842
## [2185]	4.3323	-9.4364	6.9097	4.9283	-4.7194	4.9183	3.4316	-6.5589
## [2193]	3.3943	8.6071	7.0195	5.6215	3.7168	1.8274	-1.9345	-7.4330
## [2201]	6.4033	4.3344	5.7665	-9.6337	-6.5102	1.9730	-8.3872	4.1463
## [2209]	1.2734	-4.3438	-2.1589	3.4546	-6.7564	-7.8300	0.6689	-7.9360
## [2217]	-8.1316	2.7985	1.2808	-3.8893	-4.6798	2.6500	6.6152	-8.9729
## [2225]	9.8467	-9.5583	-6.4032	-3.4262	7.0979	-6.3128	5.9071	7.1770
## [2233]	-0.5596	2.9641	-3.4992	-0.6773	8.8481	-5.0907	-9.5509	9.3414
## [2241]	-4.7990	3.0821	-8.4924	9.3849	8.3708	7.3631	-1.5124	5.6206
## [2249]	-4.0019	-4.2109	9.9855	9.7407	-0.3991	5.2455	-3.4278	-5.2752
## [2257]	2.3572	-3.1910	-8.8322	9.6389	3.7608	7.2436	-0.7661	-7.4795
## [2265]	3.6059	-9.6010	-6.7408	7.9163	-6.9368	-4.8272	-5.1195	-3.7627
## [2273]	-5.5751	-9.3891	8.0607	9.1879	8.8570	-0.9340	-1.2467	-8.9967
## [2281]	-2.8036	7.9329	-2.5787	0.0657	0.5520	-6.5946	0.7720	-7.7901
## [2289]	2.2821	0.8995	-5.4474	8.0302	-6.0295	-1.5638	7.6872	-5.2360
## [2297]	-4.1674	7.5279	2.3814	7.1460	-0.4543	7.8846	-6.6186	2.4214
## [2305]	4.4552	-6.1957	1.3743	3.4306	8.3842	-3.4888	0.9350	-9.9002
## [2313]	-2.5977	2.7524	2.1921	6.3658	-7.6422	3.2672	-5.9679	1.7372
## [2321]	3.9014	0.5658	-2.6096	3.9694	-0.5259	-5.3838	-0.9173	-8.4162
## [2329]	-4.8464	1.0816	-9.5648	-5.7780	2.7279	4.3395	5.5372	-8.5358
## [2337]	-8.0250	8.9279	8.6177	8.4290	2.9988	-6.8730	8.2029	8.4412
## [2345]	-6.6152	-8.5327	1.3246	-2.6919	-0.7967	-0.7053	5.5781	6.4733
## [2353]	-4.3529	4.5996	-0.0806	-8.6746	-9.2470	-5.2483	-5.6321	-1.5523
## [2361]	0.9950	4.2748	-5.2001	7.9136	-0.3475	-4.7431	-3.6016	3.2860
## [2369]	4.5610	1.4287	7.5739	-1.8159	-5.8921	-0.5836	-2.4118	-5.0487
## [2377]	8.5007	-0.6765	-9.7212	8.8645	-2.5091	-9.8392	9.6105	-2.4492
## [2385]	9.7174	-7.2077	1.2444	5.8872	-6.1220	4.2140	1.3142	-9.7977
## [2393]	-4.8610	-0.5804	8.5373	9.9058	-5.8004	0.9037	2.9079	8.0222
## [2401]	4.6464	2.1942	-5.5126	8.3234	6.0552	-3.7518	-5.4550	3.1519

## [2409]	4.7146	-0.6981	7.2721	3.5000	-6.7706	-9.7582	7.3214	-7.9857
## [2417]	-3.7554	-1.3101	2.3151	-3.2155	1.5996	6.5935	-3.3076	-2.3596
## [2425]	2.1138	-0.0853	-7.8146	6.9186	-1.0829	5.0075	5.9108	-2.1860
## [2433]	-9.7030	4.8683	-8.6323	0.6404	-7.5474	3.5154	-8.8796	5.9602
## [2441]	3.2521	-5.1152	-6.7926	-8.8400	-9.6958	6.6671	-4.7625	-8.8917
## [2449]	-3.0393	1.6044	6.0844	-3.1947	-7.3030	5.4723	7.5448	9.7594
## [2457]	3.0964	-4.9195	2.8291	-0.9878	-6.0608	2.1237	9.4926	-6.0165
## [2465]	-1.3048	-3.2092	-9.3198	-8.0160	-1.6153	-9.0532	-9.4694	-6.5298
## [2473]	-2.4383	1.7665	-3.2714	-0.1090	5.0465	-6.6188	6.8726	-3.5939
## [2481]	-5.3223	-3.8001	5.7982	-4.9098	8.4565	-7.0329	-5.6704	2.5218
## [2489]	-4.8586	4.1435	4.8481	-4.7529	1.4524	8.3130	4.1684	6.8871
## [2497]	8.3988	-4.1073	3.0131	-7.1647	7.0706	3.3201	4.6665	-3.7052
## [2505]	3.3544	-0.7223	3.0057	5.7155	8.0341	0.5482	-6.8378	-4.6076
## [2513]	-7.3092	-5.2488	4.4611	2.5425	-0.9527	2.6861	0.5899	-3.3979
## [2521]	5.4147	2.6998	5.4873	-6.4230	-2.0720	9.2543	-2.4537	-8.4489
## [2529]	0.5867	8.9945	7.1223	0.9887	3.2991	0.0090	8.4814	-8.1614
## [2537]	-0.0609	5.6445	7.3969	8.0310	6.2276	-6.9254	4.5952	5.1901
## [2545]	3.2164	8.1607	-2.2043	9.1396	-0.8683	-6.0998	1.8144	-6.8920
## [2553]	0.3899	-1.7432	6.5443	1.2619	-0.8705	-6.1530	-0.5611	-5.3500
## [2561]	-3.7184	7.3370	-1.1003	-1.3214	-0.5482	6.6680	-9.7932	-3.8174
## [2569]	-8.2769	6.4601	-0.5779	0.6467	-2.0921	-0.9113	-7.7020	-9.3782
## [2577]	-9.5358	-8.4496	-2.7694	5.2282	4.8656	-8.0462	9.2381	-7.1359
## [2585]	-0.3068	1.2982	8.6461	6.4005	0.6617	-2.3106	0.9198	5.9346
## [2593]	2.5469	-7.9209	-6.1601	6.8422	-0.4125	5.5452	6.3636	-4.0613
## [2601]	-5.4658	2.4072	-2.5225	7.7194	-7.4894	-6.8401	2.7693	-6.2774
## [2609]	3.2902	0.2405	4.8324	-3.2697	-7.7746	-1.9542	0.8540	4.8052
## [2617]	-6.5486	1.7297	-0.0031	8.7272	8.2061	-6.9419	-3.8540	3.9659
## [2625]	5.2643	0.1974	-5.4655	-6.6828	-3.6773	-9.1012	3.3914	4.9255
## [2633]	-5.0293	-0.6609	-9.1143	-8.0347	7.9419	2.3030	2.8142	-0.7800
## [2641]	-4.4054	8.8990	-9.5386	-6.4537	9.4286	5.9948	9.9930	-4.7851
## [2649]	1.6437	-8.5541	6.0457	7.6113	7.7926	-9.7735	-5.1720	2.1454
## [2657]	-2.7456	4.1842	-9.1585	3.1434	-5.3862	-4.5405	4.6633	-6.4349
## [2665]	-2.2873	3.6871	-1.6962	0.8432	-0.8962	-7.4587	-9.3763	-6.6869
## [2673]	5.9776	-7.1522	-8.5112	-9.4285	-1.6522	7.0806	-3.3902	-1.1113
## [2681]	-1.7303	-2.9650	8.5570	8.6011	-4.3947	4.5862	-5.6643	8.8050
## [2689]	-8.7140	9.7734	-5.7958	-3.6294	-7.5646	1.6843	3.1669	9.9233
## [2697]	-6.1186	6.3299	2.3890	9.7121	-1.1642	4.4837	-8.4830	2.6069
## [2705]	-5.7083	8.3826	6.2390	7.1223	6.3351	-1.6580	9.5510	-0.0127
## [2713]	-0.0286	-3.1306	-8.6530	8.1473	-5.5776	-4.0467	3.1708	1.3324
## [2721]	9.4258	-7.9014	0.8760	8.3074	7.4587	8.0908	5.5756	-6.6566
## [2729]	-7.5723	-7.6341	-1.3586	5.5477	8.0820	-1.4707	6.1902	-0.6325
## [2737]	3.5721	5.1745	3.6672	6.9516	-3.7686	-6.3447	-8.1296	-9.7041
## [2745]	8.0483	5.9211	-8.4446	-7.9651	-6.5222	0.5736	4.7063	-7.6356
## [2753]	-7.9312	1.0453	-4.1896	-1.1912	-7.6311	8.4324	4.6518	-0.7694
## [2761]	3.4463	-4.2301	3.2689	-2.4986	3.4185	-1.7455	4.2264	-4.3785
## [2769]	4.7741	1.4984	-9.5101	-2.7461	-1.1830	-1.9320	0.8954	7.2639
## [2777]	5.3135	-8.6916	-7.6496	-3.8042	-3.8264	4.6467	-7.1702	9.5515
## [2785]	7.0960	1.6785	7.9913	-9.8131	5.6911	8.1560	-7.7844	-0.6836
## [2793]	3.4870	-0.6910	7.5004	-1.9783	1.3337	3.0741	-4.2868	7.8389
## [2801]	-7.2257	1.1439	-4.9414	-8.5279	5.2530	2.1274	-3.3401	8.1933
## [2809]	1.8005	8.9285	8.0466	-2.2606	4.3696	6.5469	8.2764	-7.3876
## [2817]	-9.1117	9.7647	-2.2104	4.2211	0.5188	-2.2620	4.3713	7.6036

## [2825]	-7.5768	-2.6590	5.7986	-5.6880	2.4145	-2.4456	2.0807	1.2904
## [2833]	3.8889	6.2829	-0.9263	-2.6007	6.4532	0.2525	4.8316	6.9868
## [2841]	-1.1940	-1.8818	7.0184	4.5405	2.3014	4.0000	4.9635	-4.7630
## [2849]	-4.4761	-6.0107	8.8599	-4.5771	-0.0318	-9.2662	6.0347	-1.8955
## [2857]	-0.7975	8.6230	-2.2024	4.9163	5.6722	-5.8509	-8.8695	-3.3697
## [2865]	3.6761	5.3555	3.0610	8.6761	-9.9297	-2.4322	2.8741	-8.0185
## [2873]	-9.6987	-7.0287	4.9062	7.0549	-3.5011	-4.6666	-5.1291	9.1250
## [2881]	-0.0546	3.8890	8.3026	2.1784	-4.7478	8.3786	4.2323	-9.1234
## [2889]	2.6702	7.0546	-7.6181	9.0408	-9.1832	-6.5880	-9.5369	-2.2018
## [2897]	4.1855	3.1151	-0.2771	-8.3855	9.3577	1.2055	4.3381	-4.3315
## [2905]	6.0451	2.5844	8.1792	-2.0775	-3.8343	4.6339	3.9002	-1.2940
## [2913]	6.1504	-3.8877	8.2911	2.5966	5.5433	4.3429	3.2635	4.0618
## [2921]	-6.1473	1.9616	5.3434	-5.2725	3.8311	-8.1552	4.7408	8.8936
## [2929]	3.2818	3.3483	-1.5996	-7.2407	-0.6127	-8.1249	5.0815	-2.6734
## [2937]	1.3627	-2.3396	-5.9352	0.7053	-2.2831	7.2457	-8.1287	-8.7310
## [2945]	-6.6647	8.7131	-1.1498	3.4879	-9.2784	-0.6316	-9.0869	2.4007
## [2953]	7.2978	-6.0826	-4.3580	7.3037	-9.3593	5.6471	-0.8986	-8.3976
## [2961]	8.1352	-6.2199	4.9219	-1.3687	3.4106	-6.2510	7.6349	-9.0227
## [2969]	-5.5959	-4.7640	5.3442	-2.2807	-4.4285	-8.4477	0.0061	-0.9710
## [2977]	9.7320	-7.1573	6.6767	-9.4094	5.5752	-5.4265	-7.3228	-6.1878
## [2985]	-5.6827	8.6834	9.7764	-3.7995	-7.2574	3.2197	1.6983	-3.1187
## [2993]	6.2791	-4.3363	6.9168	-7.2596	7.2390	-8.6576	-1.2958	3.6027
## [3001]	-5.8835	8.8508	-2.4135	2.5248	-6.3300	3.1842	4.6939	-0.0864
## [3009]	7.3744	6.6368	9.6660	3.3370	2.8572	-9.0971	-6.1831	-9.3483
## [3017]	6.9439	-1.5109	6.3441	-6.9668	-1.8831	4.4502	-8.8073	-3.3930
## [3025]	5.5353	6.8092	7.4116	9.0608	-2.3209	9.0126	4.0821	-1.5604
## [3033]	-2.1737	-5.7887	5.0496	-1.0984	-0.9337	-8.3350	-2.6966	7.8991
## [3041]	0.8888	-9.3835	-2.2313	-7.7321	-4.4549	4.3904	-2.4795	2.6892
## [3049]	-0.0650	6.1497	1.6438	-6.5597	8.7458	-8.9238	-8.2554	6.8312
## [3057]	-1.5822	4.3497	4.7230	9.7494	9.2193	-7.1641	3.2990	-9.3369
## [3065]	-0.1097	5.8270	-2.4158	8.9660	3.2135	-1.3447	6.7737	6.0940
## [3073]	-1.4588	-3.4122	1.3008	7.2246	1.7174	2.9024	5.3417	3.8571
## [3081]	7.3360	-8.1060	2.1988	-5.1825	-0.6072	-4.3216	8.3688	-2.0051
## [3089]	1.3069	-8.3825	8.8540	4.7156	-0.4938	-6.3631	4.6073	2.2925
## [3097]	-8.7792	-8.9341	-0.8949	-6.3808	-0.1739	9.6374	-5.5176	4.2506
## [3105]	-6.9975	0.5145	6.5695	-7.4628	6.3937	-5.0799	-8.0596	-6.7260
## [3113]	3.2871	3.5486	-7.3410	9.5905	-7.0678	8.8363	4.0070	1.3593
## [3121]	-5.0730	-3.1826	6.7952	8.4304	0.3073	7.7622	4.0814	-1.5247
## [3129]	-3.9946	9.2677	-7.5036	3.6068	6.3968	-5.6745	1.8817	-6.7661
## [3137]	-1.8895	-5.2223	0.5520	3.4191	8.1513	-3.3441	-5.1493	-3.8002
## [3145]	-5.4639	-7.5399	-4.6449	8.4731	-8.8703	0.0291	5.8757	6.7585
## [3153]	1.5255	-5.1630	1.6412	-7.2497	-0.3458	-8.1263	-3.9006	3.5341
## [3161]	-5.8973	-2.1016	6.0571	-5.4336	-8.4618	5.5645	3.4028	0.2222
## [3169]	-5.7200	-6.2461	2.2229	-9.2252	-5.4112	6.4788	-3.9926	-9.6782
## [3177]	9.2317	-8.2451	4.9605	6.7202	-9.6804	7.0572	1.0059	-3.2063
## [3185]	-3.4860	2.7844	-7.4431	-3.7776	1.6617	5.1839	1.0341	3.7557
## [3193]	-2.5747	8.1349	9.5147	5.3219	-9.7547	0.4521	5.9809	-5.8509
## [3201]	-2.2744	0.8477	4.8851	-3.3334	-3.5017	-1.5251	-4.4740	2.5218
## [3209]	-9.1280	2.1307	-1.6590	-9.1276	9.8682	2.5312	6.7768	-2.0583
## [3217]	9.0618	-2.4320	6.3979	-9.8555	7.4749	-8.5211	6.3341	1.7209
## [3225]	3.6549	4.2832	-4.3979	3.4703	-4.0592	1.4209	-8.9059	-3.0478
## [3233]	-0.0464	0.6338	6.3154	0.2065	6.9789	1.8839	0.7579	2.5408

## [3241]	-4.1605	1.3448	9.9900	8.4091	5.3869	6.8747	8.4456	2.0158
## [3249]	-2.6440	4.9401	-0.3028	-9.9038	3.4628	-1.9270	9.0636	-7.3126
## [3257]	-2.4456	7.9971	6.0668	0.1285	-9.6381	-4.9945	-3.8802	3.2345
## [3265]	-8.0365	9.3058	7.3445	-4.3027	-1.3063	-3.1810	-3.0374	-3.9165
## [3273]	8.2903	-4.3284	7.3569	6.3338	-6.3083	-5.4555	3.3125	-0.6384
## [3281]	-7.8072	-7.0870	-3.6565	2.1441	8.9403	-1.5483	8.5298	-8.0072
## [3289]	7.3258	-4.7902	-4.3194	-6.6828	-9.4009	-8.0376	-7.6028	-6.1170
## [3297]	6.5919	-9.6937	-2.2862	8.2871	-9.6943	-3.8070	-1.4299	1.2426
## [3305]	8.4153	0.3384	4.8008	2.3569	-5.8850	0.1738	-8.7788	5.4900
## [3313]	-5.9151	7.1415	0.2754	-4.9676	6.2550	-8.6599	-9.4761	1.9251
## [3321]	6.9379	-0.9520	0.0435	-5.8068	4.3117	0.7276	-9.0172	6.4543
## [3329]	-4.9501	0.4290	-3.4602	-7.7926	7.0346	2.4928	6.9613	-7.4798
## [3337]	3.3650	2.1587	8.9133	1.6125	-3.9373	8.6956	6.8872	-1.5085
## [3345]	-0.2867	5.3704	-4.9545	6.2408	6.4419	-1.6533	8.3275	-8.5825
## [3353]	6.2019	-0.2819	-3.8953	2.3575	-3.6149	3.0467	8.3170	1.3816
## [3361]	2.7578	-1.6786	5.9322	7.2455	-2.6517	-0.2412	7.9270	-0.7235
## [3369]	-5.8134	3.5251	4.6826	3.0539	-5.8617	9.2154	-9.8621	8.0433
## [3377]	-8.2071	6.7419	1.0841	2.0683	6.2272	-4.7405	-9.2011	-7.2343
## [3385]	-7.8922	-6.4919	9.2236	-6.1194	-3.3564	3.8460	3.8693	-4.2779
## [3393]	-4.0176	-1.4009	-2.3474	8.1363	0.6952	2.1481	0.0875	0.0882
## [3401]	-1.5270	-5.6493	-3.6150	-5.1676	-9.9770	-5.5237	9.3833	-1.3519
## [3409]	9.2661	-4.8836	-7.2914	1.2518	-1.3350	7.2579	8.3095	2.3662
## [3417]	6.8186	-4.2349	7.9826	-7.1779	-7.1617	1.5914	9.9004	0.6696
## [3425]	-7.4298	-6.4576	5.4910	3.0102	1.1235	4.9101	-3.1357	-2.1752
## [3433]	-2.9562	2.4558	-6.7851	-5.0713	8.9534	9.7668	0.7133	-9.0938
## [3441]	2.0453	4.4336	1.6605	0.5295	-2.9247	-8.3988	-7.4321	-2.6825
## [3449]	2.0029	-5.2431	-9.6371	-3.8025	-8.4397	-4.9818	-7.1825	-2.5977
## [3457]	-7.1927	1.3599	0.8536	-6.7261	8.9007	2.5179	-8.7031	-6.3816
## [3465]	3.3548	8.1345	6.3164	-1.1505	-6.7760	-6.6845	-0.4211	-2.0827
## [3473]	5.8984	5.3967	-2.0167	2.6039	-1.7466	-2.6922	2.4171	9.1192
## [3481]	-0.4785	2.4093	-4.2155	-1.8486	-1.1025	-4.3293	9.4028	7.6520
## [3489]	6.1770	4.1645	-6.3960	-8.7205	-7.8684	6.9637	-2.8114	5.7000
## [3497]	8.1608	2.6685	2.3023	-8.6492	-5.1701	-1.7844	6.2235	-2.4172
## [3505]	-1.0610	1.4102	-7.3758	-9.7396	3.5534	6.7031	8.7253	-8.8310
## [3513]	3.3605	-2.2829	1.5226	-7.2237	-4.8833	-3.7388	4.3024	-6.2264
## [3521]	-7.1564	1.4822	-9.0170	-1.2998	-0.3458	-0.7037	-0.2760	4.7517
## [3529]	9.8197	6.8082	-7.7615	7.2284	1.3570	3.6656	5.7921	-1.9495
## [3537]	7.0655	-6.2079	-0.0856	-7.5991	-5.4267	8.1100	-0.5245	0.8095
## [3545]	9.4759	9.4486	3.7072	5.5992	-9.5909	4.5693	8.4540	-1.1908
## [3553]	4.1115	6.3345	-0.2678	6.8600	-0.1425	-9.6466	-1.1907	-0.4415
## [3561]	2.0248	-1.1351	-3.8565	3.4598	2.1823	9.4736	-6.4833	9.4645
## [3569]	1.5863	4.7625	7.3164	2.9497	3.8939	9.9181	6.9875	5.3164
## [3577]	-7.2467	-8.0962	0.4370	-6.1508	4.6293	-0.1083	-7.1448	6.8243
## [3585]	8.8261	-4.4407	-6.9924	4.1335	1.3068	-8.1900	1.2068	2.3052
## [3593]	0.9684	-2.1302	-1.6666	8.0982	3.4724	9.0958	-6.9754	1.1948
## [3601]	-8.0273	2.4690	-4.8725	-0.4147	-0.4571	6.6098	7.9116	1.5375
## [3609]	8.8759	-4.1633	2.5064	2.4037	2.9724	-5.3018	-1.9249	-6.5810
## [3617]	0.3833	1.6872	-8.3984	4.5781	2.2813	3.4179	-4.0773	-2.7886
## [3625]	2.6213	9.2021	-2.7064	-6.0402	-4.9178	-0.2939	-1.7344	-3.4351
## [3633]	-2.8631	3.6772	7.2630	-0.6206	1.6571	-0.0288	3.3462	3.7575
## [3641]	-7.8534	-7.7558	8.6556	-1.5625	1.2662	9.1248	-6.0816	-5.5251
## [3649]	2.4271	6.6913	6.1594	-6.9785	-8.3354	5.5786	9.0899	2.3680

## [3657]	-1.2557	3.2884	7.3778	0.0524	0.5761	5.4149	-2.6043	8.8101
## [3665]	-9.3348	-3.2345	-7.2874	-3.2575	-0.3298	5.1431	5.9225	-1.6708
## [3673]	-9.4012	8.7670	8.3101	2.2423	8.3818	2.5252	-7.4709	-7.1472
## [3681]	3.6329	8.4605	-6.9870	7.6969	-0.9963	-9.0666	6.4694	9.3257
## [3689]	6.1479	-4.0749	-6.0422	-5.9665	-3.0097	-5.5183	-6.7102	-8.1454
## [3697]	2.6580	5.6958	-8.8102	-8.3140	-9.9144	-7.1556	8.5839	-6.7623
## [3705]	9.7700	4.0845	-3.5091	-3.4383	5.5340	-0.0279	-0.0599	-4.6725
## [3713]	8.5960	3.5509	4.2052	-4.3641	1.6138	9.0851	-9.2800	0.2047
## [3721]	7.2117	-4.4502	-2.7430	8.3304	-8.3953	0.2545	-1.5992	-6.8390
## [3729]	-1.0088	-6.6076	7.1049	3.7627	-7.5695	-7.6177	-4.2291	8.8047
## [3737]	8.6328	-6.3414	8.2575	-8.6708	7.5526	-8.8238	9.0339	-6.7986
## [3745]	1.6167	4.2283	-4.3935	-9.4674	4.2124	-3.7451	7.1027	-0.5636
## [3753]	9.8421	-8.4427	-9.5058	1.5034	2.1533	-9.9127	4.8740	-9.0645
## [3761]	-0.4354	8.7711	-8.8072	9.0728	5.4161	7.0310	-5.6407	9.8895
## [3769]	3.8705	-4.2573	6.3588	9.1885	8.0084	-7.4632	-8.8073	6.3683
## [3777]	7.2063	-9.0960	-5.5094	4.0964	-8.0261	-9.6822	8.1126	-9.0193
## [3785]	7.4831	-7.5886	-1.9167	-5.4015	5.5247	1.5265	-6.1039	-1.5049
## [3793]	-1.2005	8.1769	-0.7467	8.6320	-2.2109	6.1053	5.3711	-3.1819
## [3801]	8.7579	0.0766	-3.5145	-1.1471	-0.2605	6.8321	8.9832	-0.5926
## [3809]	-2.5788	-5.4818	-9.9079	4.1887	3.9274	-9.7061	-6.2463	0.7371
## [3817]	-6.8198	-5.0798	8.4463	-2.7379	9.8563	4.5145	-7.0156	1.8380
## [3825]	6.9747	4.6799	-0.7501	0.2217	1.1277	-5.9841	7.7838	-6.9123
## [3833]	1.6920	-0.3813	-6.0543	9.5685	-3.6194	-9.1042	5.0737	1.3372
## [3841]	-9.8051	-3.9849	7.2659	-5.3570	-8.8292	4.2210	0.1750	-1.9011
## [3849]	-1.5097	8.2793	8.0401	-5.6387	1.4979	-9.1869	0.8120	-7.7434
## [3857]	5.1491	-7.7771	-5.9177	5.3541	-1.5531	0.9391	7.6104	4.5526
## [3865]	5.0569	2.0148	-7.9787	6.5245	4.7650	7.6595	-8.3167	6.5895
## [3873]	-4.5037	-3.3157	7.7407	-6.1735	-9.7562	4.1069	-9.2476	6.4505
## [3881]	8.0687	-6.6708	-5.8481	5.9176	7.8639	-5.1263	-1.3361	-2.0369
## [3889]	3.3119	-3.8961	5.8394	-3.1755	-4.8509	-9.9208	-2.3932	1.9885
## [3897]	-3.2193	-6.4843	9.9514	-4.6920	1.0038	-7.2951	3.6364	-0.9930
## [3905]	-2.6671	-2.2774	-1.9106	6.5513	8.3203	5.2241	-2.6530	3.8278
## [3913]	0.1607	-9.6115	2.9380	2.4602	-3.3406	-9.4743	9.4989	9.8255
## [3921]	-8.4684	0.7589	9.5571	-7.8065	-2.7486	-0.0036	8.2526	1.4412
## [3929]	-9.5441	7.1046	6.6097	7.5630	6.1920	-9.9987	-7.9635	6.6965
## [3937]	-5.5771	-9.3917	1.5345	5.1376	7.4731	-8.7927	-5.5632	-1.7195
## [3945]	-1.7734	1.5878	8.8390	9.7520	-7.4040	-4.5025	6.5677	9.8252
## [3953]	3.4640	-8.4522	-7.3119	-3.2381	-4.9578	-4.9835	5.6654	-6.5243
## [3961]	1.9661	3.8317	8.4587	3.7995	-4.5268	-9.7811	-9.1356	-6.6420
## [3969]	7.0528	-9.0817	-4.5722	7.7789	0.9579	6.0964	9.1439	3.7949
## [3977]	-1.9773	9.4208	8.7875	0.3656	-7.2742	2.5347	6.4530	-7.2747
## [3985]	1.9494	9.6593	-5.3874	-1.6526	-7.9946	-4.4298	0.6109	-3.7176
## [3993]	2.0140	9.7420	2.1856	-7.0853	4.0863	3.5836	-3.5483	-1.5337
## [4001]	-3.9107	6.6564	1.8730	6.1439	-4.1190	-7.1783	7.7724	-9.8343
## [4009]	1.3824	9.3510	-4.6162	9.6792	-9.2922	-4.8405	-4.8022	8.9638
## [4017]	9.5925	4.6877	-4.2504	-1.8208	-5.9684	5.3317	4.8621	-0.0308
## [4025]	3.4100	5.0493	5.4773	2.1034	-2.1897	3.4212	7.3842	-1.2719
## [4033]	-7.5905	2.3791	-0.3898	8.4779	-5.1865	5.5296	5.0398	3.5296
## [4041]	1.0344	1.9572	7.3288	-0.0932	9.6046	-8.6125	1.1186	-6.2125
## [4049]	-4.0709	6.8201	-5.8979	1.6544	-2.0748	-3.8886	-0.5549	4.8650
## [4057]	9.5361	-6.8747	-7.4115	1.6112	-7.2570	-5.9601	-3.1670	-4.6820
## [4065]	4.4317	3.9098	5.8842	-4.5106	-6.3609	-0.7479	1.0109	2.4544

## [4073]	-0.3421	-7.4034	7.6799	9.8264	7.2393	7.0450	-2.4517	2.3415
## [4081]	0.3668	-9.2614	4.8606	6.8486	-9.0177	7.8658	-2.4365	3.9113
## [4089]	4.3462	3.6505	-3.9919	-3.9475	-0.4778	-9.0304	-0.1676	1.7725
## [4097]	-5.3072	9.2928	-6.8001	1.5048	-1.5842	-7.5514	-2.7143	9.9745
## [4105]	5.9577	-3.0030	8.7395	4.2639	-0.0951	6.2449	3.0987	-2.7851
## [4113]	-0.5614	1.9672	4.5240	-2.8800	1.7296	0.3321	1.8703	7.7654
## [4121]	7.3426	6.6240	-6.7656	-3.3720	8.5234	8.5008	2.7300	5.8582
## [4129]	1.4349	-9.2125	-4.2469	-0.3832	-1.3017	8.6494	2.9136	-9.9834
## [4137]	-9.6096	-4.0605	-4.5251	6.5894	4.4212	3.6111	-2.3161	-6.2945
## [4145]	-5.7659	4.0738	-4.7502	-5.4730	1.8637	-7.0526	-8.9792	-9.7223
## [4153]	6.1637	6.0513	-9.6991	1.9059	-3.8527	2.1222	-5.6882	8.9574
## [4161]	-7.8819	-7.6102	-7.1892	-4.1429	1.9747	-5.8193	-0.9534	-0.9434
## [4169]	-2.3647	9.7515	-9.7970	7.3097	-8.1460	-6.3582	-1.0447	4.5818
## [4177]	-6.2008	-7.2138	-5.7441	-6.6254	7.0607	4.8003	1.3938	7.1067
## [4185]	-7.0344	-2.4509	-3.6071	-9.4014	-2.2428	4.1888	-5.0368	2.9580
## [4193]	-6.6498	5.4479	-0.4132	-8.7926	5.4142	7.5402	1.7946	2.1954
## [4201]	-1.5863	-9.0297	3.0185	0.0380	5.7276	0.9405	-1.1175	6.7559
## [4209]	3.5147	-8.9668	-7.4803	6.5274	-1.9714	0.1016	-3.2573	2.4768
## [4217]	1.5513	-5.5550	2.7915	-2.8651	-0.9838	-1.9862	-5.5645	-5.0425
## [4225]	-6.4748	-1.3539	2.1939	-1.8189	8.2517	-4.3790	-5.6880	1.0484
## [4233]	-2.9961	8.7609	2.5911	-1.2604	-4.2085	-9.2667	0.9676	-3.5925
## [4241]	-0.3796	6.1861	-5.6240	-8.2125	5.9428	-6.5119	6.0262	-9.2612
## [4249]	6.9423	6.4328	7.9520	-2.5141	-3.8740	8.3793	-3.5770	8.5063
## [4257]	2.0608	1.6233	4.7136	-2.0064	-2.6586	-1.3758	-3.2796	3.9617
## [4265]	-4.6360	7.0698	8.6171	-0.7062	-9.8786	8.8162	-1.3308	6.6640
## [4273]	0.3048	-0.3992	-7.1068	-8.8411	3.0008	-2.5289	-9.5083	-8.5033
## [4281]	6.6041	4.6859	-9.0376	4.5015	-9.7239	2.2191	-5.5487	2.4098
## [4289]	1.2876	0.1494	-4.8532	-2.8218	-4.2403	-6.1997	2.6102	-9.7846
## [4297]	-7.1036	0.1874	-0.6841	0.4209	3.8147	9.9988	8.9774	9.0490
## [4305]	3.6838	4.7229	9.1361	-5.8548	3.4957	-8.0464	-0.0259	5.1610
## [4313]	-9.7596	-6.8025	-7.7930	6.0598	-2.5025	2.6128	5.2370	-5.0276
## [4321]	-2.5294	-3.2105	4.1332	1.4333	-7.1259	-4.0669	-2.1602	0.4441
## [4329]	1.7180	-3.8185	-2.8069	8.5862	0.8984	1.7805	-4.9762	-2.2247
## [4337]	2.9185	-2.6947	-0.6460	-6.4596	-0.3753	8.6316	-9.4868	-2.3363
## [4345]	-4.9948	1.5529	-8.6368	3.0073	-8.9166	-4.1787	-3.7861	-1.1661
## [4353]	-1.2775	9.8436	2.2328	-3.7000	7.2484	-3.2461	3.4325	7.9903
## [4361]	6.8850	-8.2080	-4.0133	0.6174	-2.4180	6.9050	7.2215	3.6268
## [4369]	0.0096	6.0190	-9.0709	-4.7644	-6.0276	6.3230	6.8635	6.6561
## [4377]	-4.5863	-9.6101	2.6120	4.9612	0.1135	-2.7074	-3.8036	-2.8079
## [4385]	-2.7425	0.7730	3.0474	-1.9318	1.6600	3.9155	7.8305	8.9592
## [4393]	-1.5838	2.7321	-6.4410	-8.3937	1.3177	6.3398	5.1940	1.4365
## [4401]	9.7751	9.9034	-3.5713	-6.5596	-2.5584	0.7790	6.2249	3.9737
## [4409]	7.9991	4.4652	-4.8990	-3.8266	4.9367	-5.8951	-4.2803	7.9774
## [4417]	-6.6404	-7.1281	4.6695	0.8567	0.8481	4.0630	3.0297	5.2241
## [4425]	-4.3842	9.6818	9.2734	-9.4382	4.9413	-2.6645	-2.9190	-9.6121
## [4433]	5.1779	-6.8476	-7.7545	4.2763	-1.0956	-9.4772	8.3949	5.6986
## [4441]	-1.4978	-7.4172	6.3744	4.3361	-1.7563	-6.5018	-9.7844	-3.1819
## [4449]	-6.2126	1.2469	1.7626	-4.3649	8.5198	4.0804	-4.4709	4.3036
## [4457]	2.1744	-1.8357	-6.6536	-9.5107	-6.9059	1.7316	-5.6303	-9.1122
## [4465]	8.2902	7.4029	6.5294	2.1934	4.4726	5.9377	7.3535	4.0068
## [4473]	-3.2874	-0.0472	5.5249	-1.6168	1.6604	-3.9889	-7.0587	-4.4889
## [4481]	-5.9406	-5.4961	0.3992	8.4125	1.6292	-4.7700	-2.5206	1.2708

## [4489]	-6.8099	3.4160	-9.6346	-9.7522	3.2933	-5.8887	-2.3221	-6.8099
## [4497]	4.1710	2.2950	-8.7787	-1.8749	-7.8992	-1.9499	-0.8865	2.4929
## [4505]	-8.4229	-3.1603	-9.5263	-3.1697	5.6672	4.8242	6.3234	-8.8887
## [4513]	-6.9697	3.0621	-2.1442	-6.3245	-7.4319	-7.4221	2.4832	0.0401
## [4521]	-0.2368	7.4563	-6.1540	5.1941	-5.6120	-6.8670	-7.1180	7.7271
## [4529]	3.3743	-0.8640	4.3201	4.1431	2.9344	8.3195	-6.0071	-1.3001
## [4537]	-9.2593	-6.8517	4.0450	4.8016	-1.4054	-2.9813	-0.3142	-9.5278
## [4545]	6.9799	3.0529	-6.2842	0.9475	7.4738	-5.4099	-3.2053	5.5236
## [4553]	-5.2427	-8.4717	-1.6301	-6.0111	3.0348	2.4340	-1.6416	-7.0263
## [4561]	-6.3215	-1.4928	-2.2177	-8.8549	-9.8898	7.6600	-3.8891	9.2495
## [4569]	3.5159	7.0860	-1.3441	8.5119	5.0875	-4.4591	3.7302	8.4269
## [4577]	4.8149	3.4501	4.8820	-7.7329	-9.1789	-2.5376	0.0332	9.0077
## [4585]	8.9691	2.8924	-3.6912	0.6473	3.4931	7.1439	1.1402	-8.1121
## [4593]	4.2248	6.7044	-2.1440	-6.1055	-4.8321	-9.2571	0.5737	1.8152
## [4601]	1.6463	7.9409	-1.4703	-5.2484	0.2508	-8.6698	3.1919	4.6924
## [4609]	-5.6236	-6.2037	7.4558	-4.1627	9.7225	4.5829	8.6038	5.0352
## [4617]	-3.5047	-2.4919	-9.3168	7.8965	0.4322	-3.1039	6.2129	5.8365
## [4625]	5.6349	0.6408	-7.8121	7.7668	-7.3436	6.5955	1.0640	1.1454
## [4633]	4.6874	-2.2309	6.1730	7.0421	5.8572	-1.6603	-6.6694	4.4968
## [4641]	-8.1068	-8.8625	-9.5583	3.0458	3.8639	-6.2980	5.9239	5.8160
## [4649]	2.6302	9.1067	-7.1046	-8.7789	8.8305	6.9348	-2.8873	-1.9744
## [4657]	9.2477	-8.0523	4.1864	1.0747	7.6608	4.6596	2.5906	-6.8900
## [4665]	-0.5155	-1.2963	-6.8872	6.6627	-4.2351	2.7177	-0.3757	1.3214
## [4673]	2.2211	-7.8790	-0.2650	-6.7528	-1.4126	3.4095	1.4558	-4.6625
## [4681]	-3.9998	-9.7861	-4.0800	4.0761	-8.5453	9.4929	6.0260	7.8201
## [4689]	0.3743	-3.1319	-0.6664	-7.8272	-5.4126	-2.5025	-1.9569	3.3188
## [4697]	-7.2287	-0.7121	-7.0596	1.0566	-9.4501	6.0996	6.7375	-5.9715
## [4705]	1.0024	2.2590	-7.3591	3.9460	7.5549	9.0079	4.6776	-9.9657
## [4713]	5.4214	-3.1160	-1.7862	-9.9373	-3.2761	-3.5123	-0.7811	6.5065
## [4721]	3.1430	-2.6676	5.7624	-2.3672	8.8805	5.4363	-3.8930	-5.1808
## [4729]	-6.9016	7.7630	0.8034	8.1296	9.9749	1.1041	-3.3180	3.3739
## [4737]	7.4807	9.5587	-1.9207	4.6875	-3.6774	-0.8673	-5.2255	3.8469
## [4745]	-1.6730	3.7960	8.9937	-8.9811	2.3723	4.9330	3.7397	7.4895
## [4753]	5.9838	7.6511	9.6171	0.9718	5.6203	1.5897	-8.4645	-0.6241
## [4761]	-6.9864	1.1255	-8.8804	4.7308	-9.9561	-8.2537	8.0320	9.4832
## [4769]	-3.7850	2.8399	3.0714	2.2456	7.6264	5.6299	-3.9123	-8.9197
## [4777]	-1.0736	-1.4796	2.7385	4.8940	1.7870	-0.7429	-5.6126	-0.9304
## [4785]	5.3511	-4.8352	4.1468	5.5745	-2.2700	-3.8288	-7.7844	-0.4022
## [4793]	4.7348	0.3544	8.4657	-9.7108	8.3257	1.5922	6.1714	-1.3288
## [4801]	-3.7100	-1.8235	-4.0482	-7.5971	-4.4326	-4.4888	-3.1944	-9.6332
## [4809]	5.2179	-1.0567	2.0236	-7.9021	-9.3660	-6.3075	-9.3720	8.7764
## [4817]	-0.1822	6.0497	1.1919	-7.8028	-9.7897	-1.3538	-5.8576	6.1258
## [4825]	0.2199	0.7737	8.5620	-8.4766	-9.5587	1.1565	-7.9068	-4.7382
## [4833]	-7.1999	-6.9403	1.3923	-8.8290	2.5923	4.7200	-2.5155	4.1045
## [4841]	-8.9311	1.6752	-4.7167	-6.3995	-4.2485	2.0768	-1.6054	2.7218
## [4849]	8.9539	0.5167	-5.0145	4.8609	0.6079	-7.2733	-5.1944	-7.6347
## [4857]	-7.8511	3.0156	4.7519	-1.0560	-7.0657	6.8105	9.9377	-9.6109
## [4865]	0.7375	5.9277	0.4805	-5.0680	-0.5306	-4.1565	9.3477	3.9819
## [4873]	4.9326	-9.7274	-1.9818	-0.9227	-7.5646	-4.8560	-7.0107	-1.6604
## [4881]	-5.6754	0.4538	7.7561	0.3327	-1.4989	-4.0952	-5.5107	-8.9364
## [4889]	5.7167	9.1631	8.2143	-4.1618	-5.1265	-3.8447	-3.4363	2.2923
## [4897]	2.7012	-7.8385	-0.0387	-9.0486	3.1553	9.0919	9.1351	8.2073

```
## [4905] -0.4815 -2.6155 -2.2106  8.8156  3.7262 -1.6607 -0.3936  4.9417
## [4913]  4.6984 -0.5764 -1.7722 -1.9584  3.4261  3.8149 -8.8212 -0.7960
## [4921]  0.1109 -6.8566 -3.9430 -5.9957 -7.6617  7.0263 -3.2044 -8.7644
## [4929]  2.9010  5.2738  0.7357  0.0489  5.1199  7.7204  6.5324  0.0177
## [4937]  5.9259 -6.1425 -2.5443 -1.8560 -8.1587 -9.9598 -2.2078 -6.8290
## [4945]  7.5786  1.8096 -0.9669  0.6541 -8.9843  5.0682  3.7684 -7.4413
## [4953]  2.2192  2.9938 -3.1129  3.7781  1.3763 -4.4659  8.2912 -9.0792
## [4961]  8.0612  6.3497 -3.5182 -1.6486  0.0851  1.0418 -8.5408 -3.6154
## [4969] -5.3324 -8.0480  9.5538 -3.1549  8.6587  6.2022 -1.8133  5.9508
## [4977] -5.1135  5.5682 -8.2826 -5.7605  9.6650 -9.0969  0.9126  3.4755
## [4985] -4.4745 -0.1409  7.1900 -0.3042 -7.2849 -3.6567  5.3251 -9.0965
## [4993]  8.5038  3.6584 -1.6647 -7.2933  8.5330 -5.3679  4.7884 -9.8379
```

```
x_redondeo <- round(x5000,3); (x_redondeo) #round sirve para redondear, en este caso a 3 cifras significat
```

```
## [1] -4.248  5.766 -1.820  7.660  8.809 -9.089  0.562  7.848  1.029
## [10] -0.868  9.137 -0.933  3.551  1.453 -7.942  7.996 -5.078 -9.159
## [19] -3.442  9.090  7.791  3.856  2.810  9.885  3.114  4.171  0.881
## [28]  1.883 -4.217 -7.058  9.260  8.046  3.814  5.909 -9.508 -0.444
## [37]  5.169 -5.672 -3.636 -5.367 -7.144 -1.709 -1.726 -2.623 -6.951
## [46] -7.224 -5.339 -0.681 -4.681  7.157 -9.083 -1.156  5.978 -7.562
## [55]  1.219 -5.869 -7.449  5.066  7.901 -2.511  3.302 -8.103 -2.321
## [64] -4.512  6.293 -1.030  6.201  6.248  5.887 -1.203  5.090  2.584
## [73]  4.204 -9.988 -0.494 -5.598 -2.404  2.255 -2.964 -7.777 -5.128
## [82]  3.361 -1.647  5.764 -7.943 -1.302  9.699  7.861  7.729 -6.499
## [91] -7.386  3.062 -3.130  3.135 -3.593 -6.246  5.646 -8.128 -0.664
## [100]  0.230  2.000 -3.344 -0.228  9.089 -0.342  7.807  8.289  2.175
## [109] -1.786 -7.058  8.706 -3.975 -8.786  8.955  4.412 -7.154  0.986
## [118]  9.082  1.710 -1.910  2.958 -3.604 -3.846 -5.605 -2.610  9.684
## [127] -6.916 -8.179 -7.162  3.800  2.385  7.828  3.460  4.742  0.423
## [136]  3.197  6.436  5.726  9.596 -1.211 -3.766 -1.811 -9.791 -6.323
## [145]  6.855 -5.377 -5.218 -8.466 -5.086  4.643  6.949 -0.049 -2.242
## [154] -5.071 -7.778 -2.200  1.439 -5.662 -1.105 -5.640  0.046 -2.922
## [163]  3.000 -2.506 -2.891  0.674  4.807 -5.578 -1.745 -4.686  2.599
## [172] -6.323  7.273  4.931  3.366  2.360 -2.555  0.597  7.494  1.635
## [181]  6.795 -3.751  4.166 -4.700  1.887 -0.374 -4.699  1.292  8.264
## [190]  8.037 -4.517 -3.570  9.713  2.400  8.746 -0.669 -1.863  3.185
## [199] -6.953  1.457 -5.225  9.247  2.027  0.301 -1.949  7.605 -2.718
## [208] -4.235 -6.587 -6.557 -0.359 -4.941 -5.675  3.488 -9.047  4.017
## [217] -2.962 -1.821  6.419  8.377 -4.349  9.222  4.568  3.728 -8.943
## [226] -2.096 -0.443  1.205  3.965  8.314  2.367 -1.432  0.842 -8.830
## [235] -4.783 -2.057 -6.045  6.639 -6.942  6.068  0.937  3.246 -6.566
## [244]  2.661 -3.763  4.491 -2.021  9.387  9.348  4.534 -4.856 -5.564
## [253]  1.861 -4.650  0.621  5.706 -6.639 -1.912 -0.568  7.362  8.514
## [262]  7.640  3.484  9.003  0.329  1.530 -3.273 -3.054 -9.600  0.056
## [271]  7.421 -9.874 -8.559 -6.716  5.407  4.704  9.438 -0.671 -8.512
## [280]  2.976  5.172 -7.258 -2.068 -5.500 -8.841 -2.082 -8.701 -5.482
## [289] -8.907  3.406 -4.045 -7.986 -8.562  7.609  5.085  6.332  9.643
## [298] -7.928 -8.019  5.977  5.692 -9.811  5.581  4.588  2.603 -0.382
## [307] -6.867 -9.836 -0.951 -0.154 -2.208 -0.707  4.266 -8.894 -2.904
## [316]  6.056  6.714 -5.245 -2.920  7.138  7.075 -4.082 -7.059  4.080
## [325] -7.924 -9.325  9.988 -9.303 -3.232  8.301  2.345 -4.274  4.756
```

##	[334]	6.681	-3.715	-0.149	3.947	2.829	2.878	9.557	-1.705	-7.612
##	[343]	0.521	-5.499	-0.272	-2.596	9.667	-2.234	-5.415	2.466	-7.269
##	[352]	9.349	0.301	-6.739	2.438	9.719	3.375	-1.622	-3.533	6.705
##	[361]	-7.124	-6.144	7.935	-3.838	-2.734	5.679	-6.132	-9.645	-1.868
##	[370]	-0.337	-1.563	-3.144	7.330	-0.898	0.675	9.277	5.492	-5.822
##	[379]	-3.824	9.427	1.698	5.216	-2.546	5.384	0.754	8.280	-6.294
##	[388]	-4.356	-8.101	-5.790	9.542	-4.074	4.520	5.714	-7.892	-5.208
##	[397]	-4.589	-7.979	-7.642	9.825	9.721	-7.259	8.106	1.526	-2.091
##	[406]	-1.004	4.130	-8.350	-3.214	3.616	-3.661	6.631	-5.697	-0.041
##	[415]	-4.479	-6.160	9.012	-3.565	-0.431	-9.440	0.949	2.885	1.925
##	[424]	-3.561	7.822	2.525	-3.942	-2.236	-6.790	7.251	9.062	1.273
##	[433]	-3.409	9.932	-5.304	2.253	-7.836	-0.259	-8.011	-6.777	-4.340
##	[442]	1.677	4.634	-6.690	7.329	4.171	5.208	-7.058	-2.839	3.467
##	[451]	0.476	-3.004	-5.189	-8.836	-5.268	7.802	6.237	4.950	-6.902
##	[460]	-7.505	9.495	-1.277	-0.720	-6.694	1.699	-4.584	-5.398	3.824
##	[469]	-4.343	6.208	-8.122	6.441	-1.451	5.118	3.248	-1.109	2.543
##	[478]	-9.991	-5.655	4.097	-5.697	6.279	-3.845	3.755	8.654	-7.684
##	[487]	-7.446	3.564	-1.421	6.688	9.429	-8.590	-0.804	4.032	-8.261
##	[496]	9.859	-4.938	-9.009	3.726	5.739	-2.928	-2.671	-4.258	-8.401
##	[505]	-2.691	-6.440	0.721	0.079	8.901	-3.174	-0.706	-8.349	7.202
##	[514]	-2.087	4.718	-6.565	-0.905	5.404	-8.747	6.302	-3.977	-2.707
##	[523]	-3.758	-9.253	0.376	3.580	8.065	-9.489	9.782	-3.942	8.783
##	[532]	3.752	-1.059	6.330	-9.210	4.781	-3.026	6.585	0.711	-4.509
##	[541]	6.019	-8.167	6.642	-4.463	5.062	9.283	-8.371	7.087	6.045
##	[550]	-2.297	-3.448	-5.901	1.388	7.761	0.594	1.739	3.315	0.598
##	[559]	0.197	-9.677	-9.046	8.587	5.386	-5.978	3.005	3.075	-2.095
##	[568]	6.246	0.940	7.703	1.066	8.121	1.749	-1.531	8.992	4.181
##	[577]	-1.734	-9.633	1.335	-0.199	7.573	6.257	7.082	-2.642	7.479
##	[586]	-6.973	-4.364	3.334	9.548	1.655	0.532	-8.784	9.381	-7.595
##	[595]	-8.233	7.615	0.167	-3.250	7.887	-9.361	-5.255	3.730	-5.484
##	[604]	-3.630	-6.520	6.029	-7.074	6.454	-3.380	-2.517	2.595	-8.067
##	[613]	-9.560	9.861	1.679	5.636	7.838	5.097	9.584	-9.117	8.068
##	[622]	7.310	5.508	-2.464	-9.158	-2.712	-4.525	7.009	-2.752	-3.910
##	[631]	5.183	6.897	-0.842	4.593	-7.918	-5.600	9.079	5.041	6.379
##	[640]	-1.644	1.877	5.976	7.768	-2.296	-8.194	2.515	4.913	-8.292
##	[649]	-3.990	2.291	5.059	8.341	-0.485	1.342	4.732	7.149	8.183
##	[658]	-8.872	0.058	-2.989	6.911	6.129	-7.653	4.254	-5.295	-8.501
##	[667]	8.713	-6.857	2.941	-6.530	-9.599	0.426	-8.274	-4.340	-1.591
##	[676]	1.746	6.134	-5.960	-0.812	-1.037	4.675	4.295	6.624	7.731
##	[685]	9.053	1.012	0.447	-6.586	-0.415	-4.924	-9.204	2.696	0.791
##	[694]	-7.198	-4.326	1.661	-6.699	-8.074	-1.428	-2.885	6.899	-4.797
##	[703]	-9.537	7.248	-3.308	2.636	0.929	-2.471	-6.283	-1.421	2.615
##	[712]	0.417	3.192	4.587	-0.264	-2.311	-9.863	-9.926	9.899	-7.842
##	[721]	-1.627	4.358	4.851	7.440	2.157	5.124	6.945	2.256	5.864
##	[730]	-9.543	-1.662	7.498	2.947	8.490	-6.555	-3.684	6.085	9.779
##	[739]	-3.552	-9.976	9.843	-7.033	-9.013	1.947	-3.573	0.559	5.908
##	[748]	-8.621	3.900	8.973	-4.445	-5.012	-6.811	-8.567	1.149	-0.203
##	[757]	-0.114	7.767	-9.266	-5.925	0.276	-5.273	1.509	-0.355	1.387
##	[766]	-7.117	-7.086	-1.754	3.660	3.069	8.320	6.262	3.518	6.150
##	[775]	-7.439	-4.984	-3.369	-1.855	2.707	6.172	-4.824	6.389	-9.690
##	[784]	3.088	6.234	-0.773	-5.932	-9.625	-4.337	8.305	8.674	0.851
##	[793]	-5.612	-0.211	6.023	-1.847	-7.926	-4.389	-2.771	-4.815	-0.586

##	[802]	-2.683	-7.575	-9.060	-4.744	9.373	-0.230	-0.444	4.976	3.353
##	[811]	-9.012	3.902	-2.735	7.683	5.506	-7.216	-4.100	-7.478	1.798
##	[820]	1.234	3.774	-3.775	2.112	9.821	4.864	-8.483	-0.977	-8.929
##	[829]	-3.209	4.679	-9.918	5.438	-0.740	4.417	3.330	1.441	4.076
##	[838]	3.144	-4.213	-8.055	9.248	4.727	2.254	-7.601	1.005	-4.745
##	[847]	7.967	-9.816	-5.275	-7.399	-3.475	4.528	9.835	4.303	0.089
##	[856]	-1.279	8.977	-7.596	-8.497	7.780	-1.511	-9.153	2.949	-0.628
##	[865]	2.359	-4.584	-6.854	-7.715	0.154	0.961	-7.187	-6.605	5.240
##	[874]	0.548	7.220	3.471	-9.739	3.864	7.834	2.637	-7.854	8.421
##	[883]	3.507	-7.028	4.907	8.852	-1.584	-4.046	-4.811	-5.542	1.313
##	[892]	5.133	3.392	0.930	6.229	5.183	-9.599	-2.382	-8.982	5.958
##	[901]	8.474	0.852	7.047	1.671	3.366	0.226	5.255	8.067	6.409
##	[910]	-8.571	-9.922	-8.956	7.331	1.525	-3.723	9.189	1.824	0.628
##	[919]	-2.321	-3.609	6.168	-9.162	-2.725	7.132	3.959	3.690	-3.040
##	[928]	1.094	-7.255	5.699	7.737	-5.918	5.412	1.927	9.153	-6.826
##	[937]	0.519	7.463	7.394	-9.526	9.518	-0.196	-2.217	-1.649	-8.141
##	[946]	-6.764	-1.892	-3.164	-1.695	-3.919	1.206	-6.883	9.132	-9.121
##	[955]	-2.557	9.252	2.909	-8.775	-1.801	-1.482	0.163	-1.008	2.465
##	[964]	-7.200	8.159	1.389	0.966	-7.663	5.241	-0.433	5.639	-9.079
##	[973]	6.397	-4.612	-4.343	2.864	8.962	-9.860	-2.968	-1.619	-0.842
##	[982]	4.234	8.397	2.542	8.044	5.147	-7.243	-6.935	-6.174	-1.336
##	[991]	-8.256	-5.524	1.443	-1.997	1.309	6.592	2.842	-2.170	4.192
##	[1000]	-7.824	-4.528	1.877	-6.796	7.069	6.955	-0.442	5.474	-4.092
##	[1009]	-8.687	-1.189	-0.758	-3.181	-6.297	0.140	-9.616	5.474	1.193
##	[1018]	2.829	-0.631	-4.416	-0.776	5.855	1.711	-2.105	6.225	-3.629
##	[1027]	1.629	-0.890	-4.624	1.104	-5.375	0.865	-1.049	-7.394	2.435
##	[1036]	0.960	-7.015	1.191	-1.462	-1.045	6.666	4.392	-0.862	0.425
##	[1045]	-5.151	-8.483	-2.174	-9.738	7.350	9.852	4.177	6.534	7.838
##	[1054]	0.788	1.106	-4.471	3.184	-5.131	-4.235	4.316	4.551	-6.539
##	[1063]	-3.874	-2.602	-8.446	4.263	1.018	-7.283	9.483	1.239	5.768
##	[1072]	6.978	7.561	6.010	2.803	2.579	-4.572	-9.401	-1.603	-2.891
##	[1081]	-2.153	-6.561	-3.607	-1.729	5.187	-4.562	-7.688	-9.413	6.137
##	[1090]	-0.397	6.125	-8.420	-7.693	-7.412	4.775	5.142	9.849	2.491
##	[1099]	-4.226	-6.002	6.018	4.555	-5.659	4.885	7.333	0.588	1.973
##	[1108]	8.753	9.014	3.211	-8.554	6.527	-0.409	-2.175	-4.018	-5.114
##	[1117]	-1.564	8.755	-4.710	1.371	-5.956	-6.144	4.372	-3.730	-7.232
##	[1126]	-8.839	8.622	1.367	-7.645	1.296	0.805	6.394	4.060	1.355
##	[1135]	4.426	5.049	-2.372	-4.240	0.634	-4.807	6.636	-1.893	-8.547
##	[1144]	9.075	-5.656	-1.887	2.513	2.568	-3.435	-7.222	8.293	1.982
##	[1153]	4.992	8.379	0.575	-2.560	-8.684	7.087	0.208	-5.800	-2.483
##	[1162]	-5.853	-0.815	-5.170	-7.626	-8.551	3.820	-4.261	-7.012	9.449
##	[1171]	-1.790	-1.964	2.971	8.326	-5.667	0.960	4.401	-2.055	-8.120
##	[1180]	2.698	-9.950	-5.790	3.580	7.324	5.994	-7.300	-2.250	5.769
##	[1189]	3.858	1.885	-7.523	1.102	-1.012	5.323	-9.478	-6.097	7.625
##	[1198]	-7.804	9.371	-2.296	7.172	7.748	-0.218	4.362	-0.266	9.774
##	[1207]	-8.705	-6.847	5.707	0.844	-1.669	9.978	-4.887	0.158	-8.421
##	[1216]	6.273	-2.356	6.044	-6.042	8.928	-3.087	0.441	-7.766	7.719
##	[1225]	9.085	-9.192	-0.128	-5.479	7.176	0.619	-9.907	-4.449	-3.496
##	[1234]	1.774	-5.006	-9.138	-7.786	4.075	8.780	-3.777	-8.430	-3.565
##	[1243]	2.498	-1.195	6.027	-4.414	1.414	-9.157	-6.186	4.542	6.534
##	[1252]	0.214	1.355	-9.977	-7.124	7.319	-8.349	-5.109	9.631	1.552
##	[1261]	-5.025	2.299	-9.365	-7.072	4.061	-8.689	2.436	8.747	-8.242

## [1270]	9.456	-9.480	4.505	-0.926	1.324	7.455	1.091	4.753	2.463
## [1279]	-3.779	-2.156	-5.958	7.056	2.137	5.122	1.250	-4.482	4.711
## [1288]	0.986	-3.077	0.299	6.315	0.525	-5.939	6.962	-2.590	-3.933
## [1297]	5.412	4.669	6.778	1.381	-9.474	-0.277	0.854	6.733	4.573
## [1306]	1.102	-8.532	-7.270	3.692	-8.008	-5.924	-2.680	6.924	-5.263
## [1315]	4.098	-7.848	5.581	-4.898	0.961	-9.322	6.122	-3.245	8.326
## [1324]	-2.858	9.507	-2.312	-0.227	-0.184	-9.755	2.829	0.251	-3.593
## [1333]	1.629	2.347	-1.235	-6.593	4.302	4.956	6.878	0.695	-3.955
## [1342]	-0.870	-2.313	-5.019	3.093	9.106	-4.178	3.004	0.727	1.949
## [1351]	-9.502	-9.903	-7.372	-9.504	-8.157	-0.489	-6.067	6.293	-5.119
## [1360]	-0.224	2.978	-6.826	6.740	-1.268	-5.330	-4.023	-6.812	1.711
## [1369]	-7.024	-6.419	-3.216	-6.322	-1.889	2.271	3.713	6.471	-2.520
## [1378]	9.057	-9.623	2.247	-0.729	-2.905	7.648	-8.484	7.666	-6.236
## [1387]	-5.699	7.391	-8.783	-5.302	9.860	3.564	-1.290	-2.454	-0.776
## [1396]	6.037	3.259	-0.680	-8.935	-5.256	-5.335	-5.387	-8.765	-0.058
## [1405]	-5.117	5.164	0.946	8.961	-8.277	8.958	4.448	8.440	2.277
## [1414]	-6.781	-6.341	4.604	1.791	-3.750	5.456	-3.274	7.114	1.611
## [1423]	-1.686	8.651	-0.746	-8.013	-0.691	9.682	8.505	-1.292	7.394
## [1432]	-3.993	5.959	6.010	-2.261	-2.368	2.910	7.233	3.133	0.327
## [1441]	-7.024	2.255	-9.161	-0.033	4.790	5.078	-8.738	-1.150	0.013
## [1450]	-4.595	1.976	4.717	4.272	-7.670	1.502	5.963	-5.364	4.497
## [1459]	6.760	-7.623	9.179	4.955	6.218	-6.202	-9.481	7.276	8.384
## [1468]	-7.247	-0.447	-0.741	4.004	-8.689	4.661	-2.081	-0.770	4.506
## [1477]	-0.600	9.990	6.919	6.564	5.233	-7.758	6.780	6.277	9.828
## [1486]	-9.034	4.936	-1.614	1.643	6.615	-9.729	-9.770	9.929	5.314
## [1495]	-3.707	6.240	9.824	-5.985	2.921	-1.249	8.761	9.760	-0.874
## [1504]	-5.388	3.910	1.133	1.694	-1.327	-1.476	1.937	-0.958	9.132
## [1513]	6.889	-5.545	-1.597	-1.600	-9.584	-2.391	-1.553	-6.171	4.107
## [1522]	-9.730	4.624	-1.346	4.625	-9.721	-9.094	-1.632	2.877	-5.604
## [1531]	6.669	3.895	7.982	-1.715	-1.780	-5.000	-2.525	-5.362	8.632
## [1540]	-0.473	-9.047	5.471	-3.378	1.058	3.527	-4.453	-8.943	-1.485
## [1549]	2.202	9.521	9.396	-8.230	-0.032	-2.086	-2.193	-3.630	3.652
## [1558]	-4.308	-2.198	9.803	5.840	-3.752	6.342	-5.582	0.021	-6.661
## [1567]	-7.607	7.952	-8.126	7.301	-4.468	-9.454	5.748	-2.827	-9.498
## [1576]	4.922	-9.407	-1.323	-4.867	-0.040	3.067	-8.327	-6.391	-7.911
## [1585]	6.248	2.550	2.611	-2.182	-1.355	-3.027	5.871	7.739	3.023
## [1594]	1.857	-3.444	-8.486	1.828	-6.433	4.827	-4.122	2.784	-7.504
## [1603]	-4.895	6.411	6.076	-9.083	7.511	-7.440	2.177	-0.609	1.146
## [1612]	4.608	-0.603	9.772	9.693	-2.134	2.177	-3.090	-1.258	-5.417
## [1621]	-9.879	-9.522	-8.825	5.292	-0.619	7.017	1.634	-6.262	2.181
## [1630]	7.867	5.886	0.082	-1.539	3.088	7.755	7.840	-6.432	-4.850
## [1639]	-7.731	5.943	-7.809	6.581	5.421	8.874	-0.661	8.797	5.704
## [1648]	7.492	-2.111	-8.006	-4.461	4.444	-2.874	-2.346	-9.361	0.429
## [1657]	-7.578	7.704	-8.507	5.685	7.083	-9.256	-4.497	2.841	5.702
## [1666]	-1.999	8.706	-2.190	-1.521	-4.272	2.235	1.176	-9.201	4.906
## [1675]	-5.870	3.130	0.448	2.669	2.351	-4.302	-5.524	0.780	9.927
## [1684]	-3.422	-3.533	-9.294	0.512	1.936	4.842	9.057	-0.208	-8.334
## [1693]	-4.801	7.497	7.041	6.858	8.938	3.054	-0.237	-6.245	4.261
## [1702]	-8.299	-0.776	8.718	6.906	4.324	-7.524	1.897	9.797	2.308
## [1711]	-4.538	-8.726	-8.554	7.470	-2.744	8.912	1.166	0.253	8.955
## [1720]	-7.804	6.378	-2.453	1.133	-3.723	-8.353	5.599	-6.135	3.112
## [1729]	-1.297	-6.057	9.893	-1.065	-9.371	8.740	7.420	2.025	-4.020

## [1738]	6.187	9.043	7.041	-7.454	9.378	1.142	-9.893	-7.285	-9.883
## [1747]	6.337	5.643	8.621	-2.626	9.489	6.475	5.749	2.457	9.347
## [1756]	-0.002	7.874	-1.979	-1.049	8.716	3.667	-9.637	-6.689	-0.083
## [1765]	-1.469	9.841	7.779	-3.085	4.117	8.174	3.526	2.782	-7.008
## [1774]	2.023	-4.545	-0.843	-5.555	-4.902	3.074	-0.560	-6.781	-4.558
## [1783]	4.259	2.267	-7.358	-8.600	9.325	-2.635	3.549	6.919	-5.167
## [1792]	-9.297	1.905	6.724	-2.755	-3.356	2.895	9.172	1.924	0.206
## [1801]	-6.895	6.917	-5.712	3.397	2.355	-9.000	8.988	1.651	7.219
## [1810]	0.369	-4.677	1.948	5.912	3.786	-0.387	-0.681	2.368	-1.015
## [1819]	2.054	-9.908	9.900	-1.871	-7.641	7.391	0.804	7.632	-9.249
## [1828]	7.629	4.447	5.396	7.272	0.015	8.517	-5.839	-9.458	-3.684
## [1837]	3.202	-3.706	8.889	5.579	-3.211	-0.367	-1.680	5.527	-0.291
## [1846]	-9.694	2.849	-5.303	4.941	-4.088	8.124	7.066	-0.761	-8.573
## [1855]	1.556	1.592	9.872	-3.295	3.336	8.771	1.497	3.834	-8.205
## [1864]	7.329	0.023	-5.535	-1.752	7.726	-0.088	3.455	-4.350	6.341
## [1873]	-5.078	-2.607	-5.290	8.715	-1.698	-7.251	8.287	-3.047	7.059
## [1882]	-7.147	-2.811	-5.090	-9.082	5.691	-6.017	-2.163	-3.529	2.042
## [1891]	0.825	9.816	-4.924	4.705	9.552	1.161	-2.146	6.494	-7.752
## [1900]	0.716	-1.123	-4.521	-6.853	7.437	1.241	1.014	1.847	4.556
## [1909]	2.779	2.289	-8.952	1.303	1.747	2.571	2.438	-8.355	-8.447
## [1918]	3.742	6.606	-7.082	5.671	-0.271	9.785	-0.018	1.245	-2.435
## [1927]	0.373	-8.514	0.769	9.221	0.556	-1.257	-9.355	-9.245	-9.053
## [1936]	-6.781	-0.618	6.118	-4.388	5.096	0.437	-4.221	-9.652	3.367
## [1945]	-8.660	6.728	-7.293	3.118	6.759	-4.847	-7.280	5.270	-5.760
## [1954]	-4.641	0.637	-0.518	-2.531	5.507	1.164	-6.895	9.788	-5.111
## [1963]	-7.392	-5.746	-2.400	-2.808	-3.947	-9.112	8.696	1.374	-5.584
## [1972]	7.559	-0.654	0.847	5.688	2.957	-7.102	-3.567	9.021	-6.236
## [1981]	5.008	-0.817	-7.173	-5.896	3.506	-7.239	-1.689	-0.975	2.459
## [1990]	-4.977	-0.717	-6.690	7.156	-6.869	-8.233	7.868	-3.988	5.042
## [1999]	-1.968	-1.029	-6.807	-7.110	-7.016	0.289	-0.143	2.327	-1.052
## [2008]	-8.886	-9.892	-5.563	7.019	-4.653	1.972	2.172	9.843	-6.176
## [2017]	5.068	-5.152	-3.451	-9.929	9.948	-1.239	9.954	-6.066	-7.770
## [2026]	-4.985	3.610	1.545	-1.673	4.584	1.484	1.553	1.800	-5.541
## [2035]	-7.930	4.730	2.248	9.024	9.198	4.897	-1.303	-6.504	-1.294
## [2044]	5.249	8.381	8.134	6.310	-6.005	-9.007	-0.421	1.808	-8.068
## [2053]	9.018	-6.745	8.430	-7.444	3.255	5.681	5.292	3.451	-7.687
## [2062]	7.026	2.359	6.728	-6.601	-9.794	-3.531	-4.606	6.505	8.673
## [2071]	-7.444	8.267	2.105	-1.994	3.316	5.352	0.438	-2.110	9.316
## [2080]	-9.020	-6.934	3.937	4.556	-7.832	-0.709	2.843	-2.058	9.809
## [2089]	3.574	-3.694	-6.925	-3.975	-8.110	-9.462	-3.790	4.849	9.202
## [2098]	-1.684	0.445	-8.650	2.597	-2.932	-1.506	9.275	3.620	4.369
## [2107]	-6.587	-5.928	7.531	6.750	4.413	-4.379	-5.800	-0.117	0.435
## [2116]	-5.603	5.262	-9.510	4.230	6.818	8.616	3.242	-4.601	-5.635
## [2125]	7.356	-9.509	7.001	0.479	-1.291	8.626	-6.709	-2.424	-7.239
## [2134]	-7.006	3.529	1.242	-0.567	6.222	9.248	0.308	4.074	-6.382
## [2143]	-2.901	-9.002	-6.949	-8.015	-4.397	-2.251	2.683	4.141	-3.480
## [2152]	9.304	3.929	-6.778	-2.619	-3.956	-0.841	6.466	-5.349	-1.954
## [2161]	9.432	-7.512	2.606	4.077	1.824	9.930	-9.093	9.801	4.906
## [2170]	-1.265	-6.940	-1.023	-6.274	-1.469	6.416	-8.992	-3.492	6.848
## [2179]	-9.196	5.837	9.233	3.218	-9.826	-0.684	4.332	-9.436	6.910
## [2188]	4.928	-4.719	4.918	3.432	-6.559	3.394	8.607	7.019	5.621
## [2197]	3.717	1.827	-1.935	-7.433	6.403	4.334	5.766	-9.634	-6.510

## [2206]	1.973	-8.387	4.146	1.273	-4.344	-2.159	3.455	-6.756	-7.830
## [2215]	0.669	-7.936	-8.132	2.799	1.281	-3.889	-4.680	2.650	6.615
## [2224]	-8.973	9.847	-9.558	-6.403	-3.426	7.098	-6.313	5.907	7.177
## [2233]	-0.560	2.964	-3.499	-0.677	8.848	-5.091	-9.551	9.341	-4.799
## [2242]	3.082	-8.492	9.385	8.371	7.363	-1.512	5.621	-4.002	-4.211
## [2251]	9.985	9.741	-0.399	5.246	-3.428	-5.275	2.357	-3.191	-8.832
## [2260]	9.639	3.761	7.244	-0.766	-7.479	3.606	-9.601	-6.741	7.916
## [2269]	-6.937	-4.827	-5.120	-3.763	-5.575	-9.389	8.061	9.188	8.857
## [2278]	-0.934	-1.247	-8.997	-2.804	7.933	-2.579	0.066	0.552	-6.595
## [2287]	0.772	-7.790	2.282	0.900	-5.447	8.030	-6.029	-1.564	7.687
## [2296]	-5.236	-4.167	7.528	2.381	7.146	-0.454	7.885	-6.619	2.421
## [2305]	4.455	-6.196	1.374	3.431	8.384	-3.489	0.935	-9.900	-2.598
## [2314]	2.752	2.192	6.366	-7.642	3.267	-5.968	1.737	3.901	0.566
## [2323]	-2.610	3.969	-0.526	-5.384	-0.917	-8.416	-4.846	1.082	-9.565
## [2332]	-5.778	2.728	4.339	5.537	-8.536	-8.025	8.928	8.618	8.429
## [2341]	2.999	-6.873	8.203	8.441	-6.615	-8.533	1.325	-2.692	-0.797
## [2350]	-0.705	5.578	6.473	-4.353	4.600	-0.081	-8.675	-9.247	-5.248
## [2359]	-5.632	-1.552	0.995	4.275	-5.200	7.914	-0.348	-4.743	-3.602
## [2368]	3.286	4.561	1.429	7.574	-1.816	-5.892	-0.584	-2.412	-5.049
## [2377]	8.501	-0.677	-9.721	8.864	-2.509	-9.839	9.610	-2.449	9.717
## [2386]	-7.208	1.244	5.887	-6.122	4.214	1.314	-9.798	-4.861	-0.580
## [2395]	8.537	9.906	-5.800	0.904	2.908	8.022	4.646	2.194	-5.513
## [2404]	8.323	6.055	-3.752	-5.455	3.152	4.715	-0.698	7.272	3.500
## [2413]	-6.771	-9.758	7.321	-7.986	-3.755	-1.310	2.315	-3.216	1.600
## [2422]	6.593	-3.308	-2.360	2.114	-0.085	-7.815	6.919	-1.083	5.008
## [2431]	5.911	-2.186	-9.703	4.868	-8.632	0.640	-7.547	3.515	-8.880
## [2440]	5.960	3.252	-5.115	-6.793	-8.840	-9.696	6.667	-4.762	-8.892
## [2449]	-3.039	1.604	6.084	-3.195	-7.303	5.472	7.545	9.759	3.096
## [2458]	-4.919	2.829	-0.988	-6.061	2.124	9.493	-6.017	-1.305	-3.209
## [2467]	-9.320	-8.016	-1.615	-9.053	-9.469	-6.530	-2.438	1.766	-3.271
## [2476]	-0.109	5.047	-6.619	6.873	-3.594	-5.322	-3.800	5.798	-4.910
## [2485]	8.456	-7.033	-5.670	2.522	-4.859	4.143	4.848	-4.753	1.452
## [2494]	8.313	4.168	6.887	8.399	-4.107	3.013	-7.165	7.071	3.320
## [2503]	4.667	-3.705	3.354	-0.722	3.006	5.716	8.034	0.548	-6.838
## [2512]	-4.608	-7.309	-5.249	4.461	2.543	-0.953	2.686	0.590	-3.398
## [2521]	5.415	2.700	5.487	-6.423	-2.072	9.254	-2.454	-8.449	0.587
## [2530]	8.994	7.122	0.989	3.299	0.009	8.481	-8.161	-0.061	5.644
## [2539]	7.397	8.031	6.228	-6.925	4.595	5.190	3.216	8.161	-2.204
## [2548]	9.140	-0.868	-6.100	1.814	-6.892	0.390	-1.743	6.544	1.262
## [2557]	-0.870	-6.153	-0.561	-5.350	-3.718	7.337	-1.100	-1.321	-0.548
## [2566]	6.668	-9.793	-3.817	-8.277	6.460	-0.578	0.647	-2.092	-0.911
## [2575]	-7.702	-9.378	-9.536	-8.450	-2.769	5.228	4.866	-8.046	9.238
## [2584]	-7.136	-0.307	1.298	8.646	6.400	0.662	-2.311	0.920	5.935
## [2593]	2.547	-7.921	-6.160	6.842	-0.412	5.545	6.364	-4.061	-5.466
## [2602]	2.407	-2.522	7.719	-7.489	-6.840	2.769	-6.277	3.290	0.241
## [2611]	4.832	-3.270	-7.775	-1.954	0.854	4.805	-6.549	1.730	-0.003
## [2620]	8.727	8.206	-6.942	-3.854	3.966	5.264	0.197	-5.465	-6.683
## [2629]	-3.677	-9.101	3.391	4.926	-5.029	-0.661	-9.114	-8.035	7.942
## [2638]	2.303	2.814	-0.780	-4.405	8.899	-9.539	-6.454	9.429	5.995
## [2647]	9.993	-4.785	1.644	-8.554	6.046	7.611	7.793	-9.773	-5.172
## [2656]	2.145	-2.746	4.184	-9.159	3.143	-5.386	-4.540	4.663	-6.435
## [2665]	-2.287	3.687	-1.696	0.843	-0.896	-7.459	-9.376	-6.687	5.978

## [2674]	-7.152	-8.511	-9.429	-1.652	7.081	-3.390	-1.111	-1.730	-2.965
## [2683]	8.557	8.601	-4.395	4.586	-5.664	8.805	-8.714	9.773	-5.796
## [2692]	-3.629	-7.565	1.684	3.167	9.923	-6.119	6.330	2.389	9.712
## [2701]	-1.164	4.484	-8.483	2.607	-5.708	8.383	6.239	7.122	6.335
## [2710]	-1.658	9.551	-0.013	-0.029	-3.131	-8.653	8.147	-5.578	-4.047
## [2719]	3.171	1.332	9.426	-7.901	0.876	8.307	7.459	8.091	5.576
## [2728]	-6.657	-7.572	-7.634	-1.359	5.548	8.082	-1.471	6.190	-0.632
## [2737]	3.572	5.175	3.667	6.952	-3.769	-6.345	-8.130	-9.704	8.048
## [2746]	5.921	-8.445	-7.965	-6.522	0.574	4.706	-7.636	-7.931	1.045
## [2755]	-4.190	-1.191	-7.631	8.432	4.652	-0.769	3.446	-4.230	3.269
## [2764]	-2.499	3.418	-1.746	4.226	-4.378	4.774	1.498	-9.510	-2.746
## [2773]	-1.183	-1.932	0.895	7.264	5.313	-8.692	-7.650	-3.804	-3.826
## [2782]	4.647	-7.170	9.551	7.096	1.678	7.991	-9.813	5.691	8.156
## [2791]	-7.784	-0.684	3.487	-0.691	7.500	-1.978	1.334	3.074	-4.287
## [2800]	7.839	-7.226	1.144	-4.941	-8.528	5.253	2.127	-3.340	8.193
## [2809]	1.801	8.929	8.047	-2.261	4.370	6.547	8.276	-7.388	-9.112
## [2818]	9.765	-2.210	4.221	0.519	-2.262	4.371	7.604	-7.577	-2.659
## [2827]	5.799	-5.688	2.415	-2.446	2.081	1.290	3.889	6.283	-0.926
## [2836]	-2.601	6.453	0.252	4.832	6.987	-1.194	-1.882	7.018	4.540
## [2845]	2.301	4.000	4.963	-4.763	-4.476	-6.011	8.860	-4.577	-0.032
## [2854]	-9.266	6.035	-1.896	-0.797	8.623	-2.202	4.916	5.672	-5.851
## [2863]	-8.869	-3.370	3.676	5.355	3.061	8.676	-9.930	-2.432	2.874
## [2872]	-8.019	-9.699	-7.029	4.906	7.055	-3.501	-4.667	-5.129	9.125
## [2881]	-0.055	3.889	8.303	2.178	-4.748	8.379	4.232	-9.123	2.670
## [2890]	7.055	-7.618	9.041	-9.183	-6.588	-9.537	-2.202	4.186	3.115
## [2899]	-0.277	-8.386	9.358	1.205	4.338	-4.332	6.045	2.584	8.179
## [2908]	-2.077	-3.834	4.634	3.900	-1.294	6.150	-3.888	8.291	2.597
## [2917]	5.543	4.343	3.264	4.062	-6.147	1.962	5.343	-5.272	3.831
## [2926]	-8.155	4.741	8.894	3.282	3.348	-1.600	-7.241	-0.613	-8.125
## [2935]	5.081	-2.673	1.363	-2.340	-5.935	0.705	-2.283	7.246	-8.129
## [2944]	-8.731	-6.665	8.713	-1.150	3.488	-9.278	-0.632	-9.087	2.401
## [2953]	7.298	-6.083	-4.358	7.304	-9.359	5.647	-0.899	-8.398	8.135
## [2962]	-6.220	4.922	-1.369	3.411	-6.251	7.635	-9.023	-5.596	-4.764
## [2971]	5.344	-2.281	-4.429	-8.448	0.006	-0.971	9.732	-7.157	6.677
## [2980]	-9.409	5.575	-5.427	-7.323	-6.188	-5.683	8.683	9.776	-3.800
## [2989]	-7.257	3.220	1.698	-3.119	6.279	-4.336	6.917	-7.260	7.239
## [2998]	-8.658	-1.296	3.603	-5.883	8.851	-2.414	2.525	-6.330	3.184
## [3007]	4.694	-0.086	7.374	6.637	9.666	3.337	2.857	-9.097	-6.183
## [3016]	-9.348	6.944	-1.511	6.344	-6.967	-1.883	4.450	-8.807	-3.393
## [3025]	5.535	6.809	7.412	9.061	-2.321	9.013	4.082	-1.560	-2.174
## [3034]	-5.789	5.050	-1.098	-0.934	-8.335	-2.697	7.899	0.889	-9.383
## [3043]	-2.231	-7.732	-4.455	4.390	-2.479	2.689	-0.065	6.150	1.644
## [3052]	-6.560	8.746	-8.924	-8.255	6.831	-1.582	4.350	4.723	9.749
## [3061]	9.219	-7.164	3.299	-9.337	-0.110	5.827	-2.416	8.966	3.214
## [3070]	-1.345	6.774	6.094	-1.459	-3.412	1.301	7.225	1.717	2.902
## [3079]	5.342	3.857	7.336	-8.106	2.199	-5.183	-0.607	-4.322	8.369
## [3088]	-2.005	1.307	-8.382	8.854	4.716	-0.494	-6.363	4.607	2.293
## [3097]	-8.779	-8.934	-0.895	-6.381	-0.174	9.637	-5.518	4.251	-6.997
## [3106]	0.515	6.570	-7.463	6.394	-5.080	-8.060	-6.726	3.287	3.549
## [3115]	-7.341	9.590	-7.068	8.836	4.007	1.359	-5.073	-3.183	6.795
## [3124]	8.430	0.307	7.762	4.081	-1.525	-3.995	9.268	-7.504	3.607
## [3133]	6.397	-5.675	1.882	-6.766	-1.889	-5.222	0.552	3.419	8.151

## [3142]	-3.344	-5.149	-3.800	-5.464	-7.540	-4.645	8.473	-8.870	0.029
## [3151]	5.876	6.759	1.525	-5.163	1.641	-7.250	-0.346	-8.126	-3.901
## [3160]	3.534	-5.897	-2.102	6.057	-5.434	-8.462	5.564	3.403	0.222
## [3169]	-5.720	-6.246	2.223	-9.225	-5.411	6.479	-3.993	-9.678	9.232
## [3178]	-8.245	4.960	6.720	-9.680	7.057	1.006	-3.206	-3.486	2.784
## [3187]	-7.443	-3.778	1.662	5.184	1.034	3.756	-2.575	8.135	9.515
## [3196]	5.322	-9.755	0.452	5.981	-5.851	-2.274	0.848	4.885	-3.333
## [3205]	-3.502	-1.525	-4.474	2.522	-9.128	2.131	-1.659	-9.128	9.868
## [3214]	2.531	6.777	-2.058	9.062	-2.432	6.398	-9.855	7.475	-8.521
## [3223]	6.334	1.721	3.655	4.283	-4.398	3.470	-4.059	1.421	-8.906
## [3232]	-3.048	-0.046	0.634	6.315	0.207	6.979	1.884	0.758	2.541
## [3241]	-4.161	1.345	9.990	8.409	5.387	6.875	8.446	2.016	-2.644
## [3250]	4.940	-0.303	-9.904	3.463	-1.927	9.064	-7.313	-2.446	7.997
## [3259]	6.067	0.129	-9.638	-4.995	-3.880	3.234	-8.036	9.306	7.344
## [3268]	-4.303	-1.306	-3.181	-3.037	-3.917	8.290	-4.328	7.357	6.334
## [3277]	-6.308	-5.455	3.313	-0.638	-7.807	-7.087	-3.657	2.144	8.940
## [3286]	-1.548	8.530	-8.007	7.326	-4.790	-4.319	-6.683	-9.401	-8.038
## [3295]	-7.603	-6.117	6.592	-9.694	-2.286	8.287	-9.694	-3.807	-1.430
## [3304]	1.243	8.415	0.338	4.801	2.357	-5.885	0.174	-8.779	5.490
## [3313]	-5.915	7.141	0.275	-4.968	6.255	-8.660	-9.476	1.925	6.938
## [3322]	-0.952	0.044	-5.807	4.312	0.728	-9.017	6.454	-4.950	0.429
## [3331]	-3.460	-7.793	7.035	2.493	6.961	-7.480	3.365	2.159	8.913
## [3340]	1.612	-3.937	8.696	6.887	-1.509	-0.287	5.370	-4.954	6.241
## [3349]	6.442	-1.653	8.328	-8.582	6.202	-0.282	-3.895	2.357	-3.615
## [3358]	3.047	8.317	1.382	2.758	-1.679	5.932	7.246	-2.652	-0.241
## [3367]	7.927	-0.724	-5.813	3.525	4.683	3.054	-5.862	9.215	-9.862
## [3376]	8.043	-8.207	6.742	1.084	2.068	6.227	-4.741	-9.201	-7.234
## [3385]	-7.892	-6.492	9.224	-6.119	-3.356	3.846	3.869	-4.278	-4.018
## [3394]	-1.401	-2.347	8.136	0.695	2.148	0.087	0.088	-1.527	-5.649
## [3403]	-3.615	-5.168	-9.977	-5.524	9.383	-1.352	9.266	-4.884	-7.291
## [3412]	1.252	-1.335	7.258	8.310	2.366	6.819	-4.235	7.983	-7.178
## [3421]	-7.162	1.591	9.900	0.670	-7.430	-6.458	5.491	3.010	1.123
## [3430]	4.910	-3.136	-2.175	-2.956	2.456	-6.785	-5.071	8.953	9.767
## [3439]	0.713	-9.094	2.045	4.434	1.660	0.529	-2.925	-8.399	-7.432
## [3448]	-2.682	2.003	-5.243	-9.637	-3.803	-8.440	-4.982	-7.183	-2.598
## [3457]	-7.193	1.360	0.854	-6.726	8.901	2.518	-8.703	-6.382	3.355
## [3466]	8.135	6.316	-1.151	-6.776	-6.685	-0.421	-2.083	5.898	5.397
## [3475]	-2.017	2.604	-1.747	-2.692	2.417	9.119	-0.479	2.409	-4.216
## [3484]	-1.849	-1.102	-4.329	9.403	7.652	6.177	4.165	-6.396	-8.720
## [3493]	-7.868	6.964	-2.811	5.700	8.161	2.668	2.302	-8.649	-5.170
## [3502]	-1.784	6.224	-2.417	-1.061	1.410	-7.376	-9.740	3.553	6.703
## [3511]	8.725	-8.831	3.361	-2.283	1.523	-7.224	-4.883	-3.739	4.302
## [3520]	-6.226	-7.156	1.482	-9.017	-1.300	-0.346	-0.704	-0.276	4.752
## [3529]	9.820	6.808	-7.761	7.228	1.357	3.666	5.792	-1.949	7.066
## [3538]	-6.208	-0.086	-7.599	-5.427	8.110	-0.525	0.810	9.476	9.449
## [3547]	3.707	5.599	-9.591	4.569	8.454	-1.191	4.111	6.334	-0.268
## [3556]	6.860	-0.143	-9.647	-1.191	-0.442	2.025	-1.135	-3.856	3.460
## [3565]	2.182	9.474	-6.483	9.465	1.586	4.763	7.316	2.950	3.894
## [3574]	9.918	6.987	5.316	-7.247	-8.096	0.437	-6.151	4.629	-0.108
## [3583]	-7.145	6.824	8.826	-4.441	-6.992	4.133	1.307	-8.190	1.207
## [3592]	2.305	0.968	-2.130	-1.667	8.098	3.472	9.096	-6.975	1.195
## [3601]	-8.027	2.469	-4.873	-0.415	-0.457	6.610	7.912	1.538	8.876

## [3610]	-4.163	2.506	2.404	2.972	-5.302	-1.925	-6.581	0.383	1.687
## [3619]	-8.398	4.578	2.281	3.418	-4.077	-2.789	2.621	9.202	-2.706
## [3628]	-6.040	-4.918	-0.294	-1.734	-3.435	-2.863	3.677	7.263	-0.621
## [3637]	1.657	-0.029	3.346	3.757	-7.853	-7.756	8.656	-1.562	1.266
## [3646]	9.125	-6.082	-5.525	2.427	6.691	6.159	-6.979	-8.335	5.579
## [3655]	9.090	2.368	-1.256	3.288	7.378	0.052	0.576	5.415	-2.604
## [3664]	8.810	-9.335	-3.234	-7.287	-3.258	-0.330	5.143	5.922	-1.671
## [3673]	-9.401	8.767	8.310	2.242	8.382	2.525	-7.471	-7.147	3.633
## [3682]	8.460	-6.987	7.697	-0.996	-9.067	6.469	9.326	6.148	-4.075
## [3691]	-6.042	-5.967	-3.010	-5.518	-6.710	-8.145	2.658	5.696	-8.810
## [3700]	-8.314	-9.914	-7.156	8.584	-6.762	9.770	4.084	-3.509	-3.438
## [3709]	5.534	-0.028	-0.060	-4.673	8.596	3.551	4.205	-4.364	1.614
## [3718]	9.085	-9.280	0.205	7.212	-4.450	-2.743	8.330	-8.395	0.254
## [3727]	-1.599	-6.839	-1.009	-6.608	7.105	3.763	-7.569	-7.618	-4.229
## [3736]	8.805	8.633	-6.341	8.257	-8.671	7.553	-8.824	9.034	-6.799
## [3745]	1.617	4.228	-4.394	-9.467	4.212	-3.745	7.103	-0.564	9.842
## [3754]	-8.443	-9.506	1.503	2.153	-9.913	4.874	-9.064	-0.435	8.771
## [3763]	-8.807	9.073	5.416	7.031	-5.641	9.889	3.870	-4.257	6.359
## [3772]	9.188	8.008	-7.463	-8.807	6.368	7.206	-9.096	-5.509	4.096
## [3781]	-8.026	-9.682	8.113	-9.019	7.483	-7.589	-1.917	-5.402	5.525
## [3790]	1.527	-6.104	-1.505	-1.200	8.177	-0.747	8.632	-2.211	6.105
## [3799]	5.371	-3.182	8.758	0.077	-3.514	-1.147	-0.261	6.832	8.983
## [3808]	-0.593	-2.579	-5.482	-9.908	4.189	3.927	-9.706	-6.246	0.737
## [3817]	-6.820	-5.080	8.446	-2.738	9.856	4.514	-7.016	1.838	6.975
## [3826]	4.680	-0.750	0.222	1.128	-5.984	7.784	-6.912	1.692	-0.381
## [3835]	-6.054	9.568	-3.619	-9.104	5.074	1.337	-9.805	-3.985	7.266
## [3844]	-5.357	-8.829	4.221	0.175	-1.901	-1.510	8.279	8.040	-5.639
## [3853]	1.498	-9.187	0.812	-7.743	5.149	-7.777	-5.918	5.354	-1.553
## [3862]	0.939	7.610	4.553	5.057	2.015	-7.979	6.524	4.765	7.660
## [3871]	-8.317	6.589	-4.504	-3.316	7.741	-6.173	-9.756	4.107	-9.248
## [3880]	6.451	8.069	-6.671	-5.848	5.918	7.864	-5.126	-1.336	-2.037
## [3889]	3.312	-3.896	5.839	-3.175	-4.851	-9.921	-2.393	1.989	-3.219
## [3898]	-6.484	9.951	-4.692	1.004	-7.295	3.636	-0.993	-2.667	-2.277
## [3907]	-1.911	6.551	8.320	5.224	-2.653	3.828	0.161	-9.611	2.938
## [3916]	2.460	-3.341	-9.474	9.499	9.825	-8.468	0.759	9.557	-7.807
## [3925]	-2.749	-0.004	8.253	1.441	-9.544	7.105	6.610	7.563	6.192
## [3934]	-9.999	-7.963	6.697	-5.577	-9.392	1.535	5.138	7.473	-8.793
## [3943]	-5.563	-1.719	-1.773	1.588	8.839	9.752	-7.404	-4.503	6.568
## [3952]	9.825	3.464	-8.452	-7.312	-3.238	-4.958	-4.984	5.665	-6.524
## [3961]	1.966	3.832	8.459	3.799	-4.527	-9.781	-9.136	-6.642	7.053
## [3970]	-9.082	-4.572	7.779	0.958	6.096	9.144	3.795	-1.977	9.421
## [3979]	8.788	0.366	-7.274	2.535	6.453	-7.275	1.949	9.659	-5.387
## [3988]	-1.653	-7.995	-4.430	0.611	-3.718	2.014	9.742	2.186	-7.085
## [3997]	4.086	3.584	-3.548	-1.534	-3.911	6.656	1.873	6.144	-4.119
## [4006]	-7.178	7.772	-9.834	1.382	9.351	-4.616	9.679	-9.292	-4.840
## [4015]	-4.802	8.964	9.592	4.688	-4.250	-1.821	-5.968	5.332	4.862
## [4024]	-0.031	3.410	5.049	5.477	2.103	-2.190	3.421	7.384	-1.272
## [4033]	-7.590	2.379	-0.390	8.478	-5.187	5.530	5.040	3.530	1.034
## [4042]	1.957	7.329	-0.093	9.605	-8.613	1.119	-6.213	-4.071	6.820
## [4051]	-5.898	1.654	-2.075	-3.889	-0.555	4.865	9.536	-6.875	-7.411
## [4060]	1.611	-7.257	-5.960	-3.167	-4.682	4.432	3.910	5.884	-4.511
## [4069]	-6.361	-0.748	1.011	2.454	-0.342	-7.403	7.680	9.826	7.239

## [4078]	7.045	-2.452	2.342	0.367	-9.261	4.861	6.849	-9.018	7.866
## [4087]	-2.436	3.911	4.346	3.650	-3.992	-3.948	-0.478	-9.030	-0.168
## [4096]	1.772	-5.307	9.293	-6.800	1.505	-1.584	-7.551	-2.714	9.975
## [4105]	5.958	-3.003	8.740	4.264	-0.095	6.245	3.099	-2.785	-0.561
## [4114]	1.967	4.524	-2.880	1.730	0.332	1.870	7.765	7.343	6.624
## [4123]	-6.766	-3.372	8.523	8.501	2.730	5.858	1.435	-9.212	-4.247
## [4132]	-0.383	-1.302	8.649	2.914	-9.983	-9.610	-4.061	-4.525	6.589
## [4141]	4.421	3.611	-2.316	-6.295	-5.766	4.074	-4.750	-5.473	1.864
## [4150]	-7.053	-8.979	-9.722	6.164	6.051	-9.699	1.906	-3.853	2.122
## [4159]	-5.688	8.957	-7.882	-7.610	-7.189	-4.143	1.975	-5.819	-0.953
## [4168]	-0.943	-2.365	9.751	-9.797	7.310	-8.146	-6.358	-1.045	4.582
## [4177]	-6.201	-7.214	-5.744	-6.625	7.061	4.800	1.394	7.107	-7.034
## [4186]	-2.451	-3.607	-9.401	-2.243	4.189	-5.037	2.958	-6.650	5.448
## [4195]	-0.413	-8.793	5.414	7.540	1.795	2.195	-1.586	-9.030	3.019
## [4204]	0.038	5.728	0.941	-1.117	6.756	3.515	-8.967	-7.480	6.527
## [4213]	-1.971	0.102	-3.257	2.477	1.551	-5.555	2.791	-2.865	-0.984
## [4222]	-1.986	-5.565	-5.043	-6.475	-1.354	2.194	-1.819	8.252	-4.379
## [4231]	-5.688	1.048	-2.996	8.761	2.591	-1.260	-4.208	-9.267	0.968
## [4240]	-3.593	-0.380	6.186	-5.624	-8.212	5.943	-6.512	6.026	-9.261
## [4249]	6.942	6.433	7.952	-2.514	-3.874	8.379	-3.577	8.506	2.061
## [4258]	1.623	4.714	-2.006	-2.659	-1.376	-3.280	3.962	-4.636	7.070
## [4267]	8.617	-0.706	-9.879	8.816	-1.331	6.664	0.305	-0.399	-7.107
## [4276]	-8.841	3.001	-2.529	-9.508	-8.503	6.604	4.686	-9.038	4.502
## [4285]	-9.724	2.219	-5.549	2.410	1.288	0.149	-4.853	-2.822	-4.240
## [4294]	-6.200	2.610	-9.785	-7.104	0.187	-0.684	0.421	3.815	9.999
## [4303]	8.977	9.049	3.684	4.723	9.136	-5.855	3.496	-8.046	-0.026
## [4312]	5.161	-9.760	-6.802	-7.793	6.060	-2.502	2.613	5.237	-5.028
## [4321]	-2.529	-3.210	4.133	1.433	-7.126	-4.067	-2.160	0.444	1.718
## [4330]	-3.818	-2.807	8.586	0.898	1.781	-4.976	-2.225	2.919	-2.695
## [4339]	-0.646	-6.460	-0.375	8.632	-9.487	-2.336	-4.995	1.553	-8.637
## [4348]	3.007	-8.917	-4.179	-3.786	-1.166	-1.277	9.844	2.233	-3.700
## [4357]	7.248	-3.246	3.432	7.990	6.885	-8.208	-4.013	0.617	-2.418
## [4366]	6.905	7.222	3.627	0.010	6.019	-9.071	-4.764	-6.028	6.323
## [4375]	6.863	6.656	-4.586	-9.610	2.612	4.961	0.113	-2.707	-3.804
## [4384]	-2.808	-2.743	0.773	3.047	-1.932	1.660	3.915	7.830	8.959
## [4393]	-1.584	2.732	-6.441	-8.394	1.318	6.340	5.194	1.436	9.775
## [4402]	9.903	-3.571	-6.560	-2.558	0.779	6.225	3.974	7.999	4.465
## [4411]	-4.899	-3.827	4.937	-5.895	-4.280	7.977	-6.640	-7.128	4.670
## [4420]	0.857	0.848	4.063	3.030	5.224	-4.384	9.682	9.273	-9.438
## [4429]	4.941	-2.664	-2.919	-9.612	5.178	-6.848	-7.754	4.276	-1.096
## [4438]	-9.477	8.395	5.699	-1.498	-7.417	6.374	4.336	-1.756	-6.502
## [4447]	-9.784	-3.182	-6.213	1.247	1.763	-4.365	8.520	4.080	-4.471
## [4456]	4.304	2.174	-1.836	-6.654	-9.511	-6.906	1.732	-5.630	-9.112
## [4465]	8.290	7.403	6.529	2.193	4.473	5.938	7.353	4.007	-3.287
## [4474]	-0.047	5.525	-1.617	1.660	-3.989	-7.059	-4.489	-5.941	-5.496
## [4483]	0.399	8.412	1.629	-4.770	-2.521	1.271	-6.810	3.416	-9.635
## [4492]	-9.752	3.293	-5.889	-2.322	-6.810	4.171	2.295	-8.779	-1.875
## [4501]	-7.899	-1.950	-0.886	2.493	-8.423	-3.160	-9.526	-3.170	5.667
## [4510]	4.824	6.323	-8.889	-6.970	3.062	-2.144	-6.325	-7.432	-7.422
## [4519]	2.483	0.040	-0.237	7.456	-6.154	5.194	-5.612	-6.867	-7.118
## [4528]	7.727	3.374	-0.864	4.320	4.143	2.934	8.319	-6.007	-1.300
## [4537]	-9.259	-6.852	4.045	4.802	-1.405	-2.981	-0.314	-9.528	6.980

## [4546]	3.053	-6.284	0.948	7.474	-5.410	-3.205	5.524	-5.243	-8.472
## [4555]	-1.630	-6.011	3.035	2.434	-1.642	-7.026	-6.321	-1.493	-2.218
## [4564]	-8.855	-9.890	7.660	-3.889	9.250	3.516	7.086	-1.344	8.512
## [4573]	5.088	-4.459	3.730	8.427	4.815	3.450	4.882	-7.733	-9.179
## [4582]	-2.538	0.033	9.008	8.969	2.892	-3.691	0.647	3.493	7.144
## [4591]	1.140	-8.112	4.225	6.704	-2.144	-6.105	-4.832	-9.257	0.574
## [4600]	1.815	1.646	7.941	-1.470	-5.248	0.251	-8.670	3.192	4.692
## [4609]	-5.624	-6.204	7.456	-4.163	9.722	4.583	8.604	5.035	-3.505
## [4618]	-2.492	-9.317	7.897	0.432	-3.104	6.213	5.836	5.635	0.641
## [4627]	-7.812	7.767	-7.344	6.595	1.064	1.145	4.687	-2.231	6.173
## [4636]	7.042	5.857	-1.660	-6.669	4.497	-8.107	-8.863	-9.558	3.046
## [4645]	3.864	-6.298	5.924	5.816	2.630	9.107	-7.105	-8.779	8.831
## [4654]	6.935	-2.887	-1.974	9.248	-8.052	4.186	1.075	7.661	4.660
## [4663]	2.591	-6.890	-0.515	-1.296	-6.887	6.663	-4.235	2.718	-0.376
## [4672]	1.321	2.221	-7.879	-0.265	-6.753	-1.413	3.410	1.456	-4.662
## [4681]	-4.000	-9.786	-4.080	4.076	-8.545	9.493	6.026	7.820	0.374
## [4690]	-3.132	-0.666	-7.827	-5.413	-2.503	-1.957	3.319	-7.229	-0.712
## [4699]	-7.060	1.057	-9.450	6.100	6.738	-5.972	1.002	2.259	-7.359
## [4708]	3.946	7.555	9.008	4.678	-9.966	5.421	-3.116	-1.786	-9.937
## [4717]	-3.276	-3.512	-0.781	6.506	3.143	-2.668	5.762	-2.367	8.880
## [4726]	5.436	-3.893	-5.181	-6.902	7.763	0.803	8.130	9.975	1.104
## [4735]	-3.318	3.374	7.481	9.559	-1.921	4.688	-3.677	-0.867	-5.226
## [4744]	3.847	-1.673	3.796	8.994	-8.981	2.372	4.933	3.740	7.489
## [4753]	5.984	7.651	9.617	0.972	5.620	1.590	-8.465	-0.624	-6.986
## [4762]	1.125	-8.880	4.731	-9.956	-8.254	8.032	9.483	-3.785	2.840
## [4771]	3.071	2.246	7.626	5.630	-3.912	-8.920	-1.074	-1.480	2.739
## [4780]	4.894	1.787	-0.743	-5.613	-0.930	5.351	-4.835	4.147	5.575
## [4789]	-2.270	-3.829	-7.784	-0.402	4.735	0.354	8.466	-9.711	8.326
## [4798]	1.592	6.171	-1.329	-3.710	-1.823	-4.048	-7.597	-4.433	-4.489
## [4807]	-3.194	-9.633	5.218	-1.057	2.024	-7.902	-9.366	-6.307	-9.372
## [4816]	8.776	-0.182	6.050	1.192	-7.803	-9.790	-1.354	-5.858	6.126
## [4825]	0.220	0.774	8.562	-8.477	-9.559	1.157	-7.907	-4.738	-7.200
## [4834]	-6.940	1.392	-8.829	2.592	4.720	-2.516	4.104	-8.931	1.675
## [4843]	-4.717	-6.400	-4.249	2.077	-1.605	2.722	8.954	0.517	-5.014
## [4852]	4.861	0.608	-7.273	-5.194	-7.635	-7.851	3.016	4.752	-1.056
## [4861]	-7.066	6.810	9.938	-9.611	0.737	5.928	0.480	-5.068	-0.531
## [4870]	-4.156	9.348	3.982	4.933	-9.727	-1.982	-0.923	-7.565	-4.856
## [4879]	-7.011	-1.660	-5.675	0.454	7.756	0.333	-1.499	-4.095	-5.511
## [4888]	-8.936	5.717	9.163	8.214	-4.162	-5.127	-3.845	-3.436	2.292
## [4897]	2.701	-7.838	-0.039	-9.049	3.155	9.092	9.135	8.207	-0.482
## [4906]	-2.615	-2.211	8.816	3.726	-1.661	-0.394	4.942	4.698	-0.576
## [4915]	-1.772	-1.958	3.426	3.815	-8.821	-0.796	0.111	-6.857	-3.943
## [4924]	-5.996	-7.662	7.026	-3.204	-8.764	2.901	5.274	0.736	0.049
## [4933]	5.120	7.720	6.532	0.018	5.926	-6.143	-2.544	-1.856	-8.159
## [4942]	-9.960	-2.208	-6.829	7.579	1.810	-0.967	0.654	-8.984	5.068
## [4951]	3.768	-7.441	2.219	2.994	-3.113	3.778	1.376	-4.466	8.291
## [4960]	-9.079	8.061	6.350	-3.518	-1.649	0.085	1.042	-8.541	-3.615
## [4969]	-5.332	-8.048	9.554	-3.155	8.659	6.202	-1.813	5.951	-5.114
## [4978]	5.568	-8.283	-5.760	9.665	-9.097	0.913	3.476	-4.475	-0.141
## [4987]	7.190	-0.304	-7.285	-3.657	5.325	-9.096	8.504	3.658	-1.665
## [4996]	-7.293	8.533	-5.368	4.788	-9.838				

```
muestra_1 <- sample(x_redondeo,500); (muestra_1) #sample nos proporciona una muestra de nuestros datos, en
```

```
## [1] 6.475 4.723 8.432 -9.892 -3.993 -3.571 1.125 5.448 -3.889 5.187
## [11] 4.885 5.415 5.118 9.499 -4.750 3.088 -6.104 -4.562 6.795 -1.247
## [21] 0.780 2.466 1.643 7.324 7.528 1.655 9.189 -1.971 -8.477 2.941
## [31] 7.610 -3.533 7.756 -3.245 -0.930 7.497 3.610 -5.090 8.008 7.202
## [41] 3.460 -5.307 -8.675 -7.126 -5.078 9.136 -9.633 2.221 9.493 -2.598
## [51] 0.940 7.463 4.730 -1.030 8.632 -3.257 0.452 -9.138 -6.771 3.966
## [61] -8.421 -4.918 8.290 -8.035 -1.673 -1.660 6.756 -9.249 0.089 0.619
## [71] -0.259 5.240 4.963 4.000 8.776 5.416 8.673 -2.095 3.411 -0.770
## [81] -8.660 3.815 4.060 -4.459 -3.104 -6.716 -4.985 7.559 4.007 -4.389
## [91] -8.967 -9.612 -7.642 -2.236 -2.874 -9.543 -4.636 -3.895 2.700 9.140
## [101] 7.083 -9.325 -1.098 -7.635 7.273 -7.110 -1.644 6.246 1.374 -4.340
## [111] -7.216 -0.870 6.237 6.225 -5.019 -2.691 2.840 0.517 -4.883 2.181
## [121] 4.688 -0.380 -7.178 -2.446 4.752 -9.599 -0.952 8.048 5.208 7.329
## [131] 3.946 -2.601 -0.088 -9.030 -9.506 -9.481 -7.432 -2.283 -0.671 5.164
## [141] 4.735 -9.409 -5.128 1.949 8.214 -4.235 -9.017 -7.118 4.555 1.048
## [151] 7.087 4.334 -9.904 -6.100 5.621 4.254 1.314 -1.887 7.225 3.726
## [161] -7.208 0.222 -0.578 -0.515 -5.549 -8.159 1.324 0.425 0.889 8.395
## [171] 8.713 -6.683 4.955 -0.600 -5.744 6.887 -4.280 -4.045 -5.131 -7.824
## [181] -1.268 -8.443 4.302 -8.935 -1.584 9.948 6.330 -0.628 -1.686 -5.796
## [191] -6.483 -6.246 8.632 1.281 2.661 -7.921 2.829 7.105 8.836 1.192
## [201] 1.629 9.773 -2.827 -1.534 1.166 2.345 -6.916 -2.311 -7.259 -2.311
## [211] 0.373 2.355 1.357 -8.208 -5.025 -9.703 6.164 5.342 3.000 3.135
## [221] 9.266 6.137 6.663 -9.121 -7.033 0.848 -4.461 8.973 4.775 -9.212
## [231] 0.574 -1.932 3.500 0.129 4.922 1.002 2.947 -3.630 -7.612 8.369
## [241] 9.903 6.125 -7.136 -0.741 8.771 -4.476 -2.644 4.907 -0.951 -6.524
## [251] -7.074 -1.669 -7.412 -2.682 7.047 7.103 -7.247 1.556 6.056 2.347
## [261] 3.192 -5.086 4.276 8.514 3.403 -1.974 -1.305 5.681 7.421 4.505
## [271] 7.874 7.330 5.008 -0.457 0.197 -1.652 2.908 6.310 -3.380 7.983
## [281] 3.282 4.556 5.352 -6.848 6.454 0.847 3.488 7.996 0.788 -0.018
## [291] -6.083 -4.835 -4.846 -7.812 3.553 -3.136 -2.174 6.987 8.314 -6.549
## [301] 4.882 8.961 7.026 -8.984 -4.545 8.341 -7.016 4.916 -6.654 -6.709
## [311] -7.505 -1.660 -6.054 6.703 -7.854 3.487 -2.231 -5.541 -4.405 -7.243
## [321] 5.316 5.059 8.046 5.636 7.001 -2.642 -2.372 2.367 5.274 7.069
## [331] -0.773 2.885 7.122 4.214 2.434 -1.294 -3.629 -9.599 9.752 5.527
## [341] -0.898 -3.834 -1.847 -3.800 -7.641 -0.619 -6.435 9.125 -8.190 -7.086
## [351] 3.757 -2.880 8.587 7.026 -5.675 -0.593 4.608 -2.414 1.787 -9.504
## [361] 7.055 -4.743 7.885 -6.419 0.966 9.721 -7.963 -3.356 7.561 2.174
## [371] 3.410 5.534 5.545 -1.591 -3.280 -9.813 -9.652 -9.879 -6.186 7.042
## [381] -7.936 -1.471 5.748 6.534 -9.502 -8.277 -0.261 0.432 9.326 1.273
## [391] -5.068 8.958 3.658 -0.603 1.692 -4.378 5.728 -3.874 7.793 -4.471
## [401] 4.166 -1.958 -9.637 4.756 -2.464 -2.725 -0.870 -5.078 3.969 0.429
## [411] -2.664 1.456 -2.190 -7.224 8.030 -0.182 1.239 -3.010 -7.463 9.665
## [421] -7.189 -8.731 -6.739 -3.512 9.493 -7.300 3.320 -1.957 1.498 -1.023
## [431] -5.901 -6.530 -4.861 6.411 -5.697 -7.312 -7.618 4.827 -5.245 -8.830
## [441] 6.023 -7.842 4.593 7.868 -8.371 -9.047 -2.251 -0.314 2.459 5.412
## [451] -9.440 5.568 9.449 1.545 1.660 6.241 -4.002 -9.918 0.716 6.222
## [461] -7.693 -8.521 -0.677 5.477 7.105 -2.274 4.520 7.057 5.943 3.464
## [471] 5.995 9.132 -2.607 7.343 -8.314 7.055 -6.741 3.255 -0.381 9.975
## [481] -9.740 -6.454 8.061 4.894 -6.011 -8.779 -8.398 -4.801 -3.630 -9.201
```

```
## [491] -8.841 4.941 -9.158 -1.875 -8.501 -6.246 -4.397 3.354 9.096 6.824
```

```
nx <- runif(5000,-10,10); (nx)
```

```
## [1] 4.7710 3.0823 -3.9106 -9.5695 -2.3571 -4.9404 0.6539 3.3644
## [9] -4.1604 -0.5812 -2.4085 -8.0012 -6.4360 6.9002 -0.5471 7.5519
## [17] 6.8357 -6.5280 4.4919 -1.4714 1.7041 5.3498 8.5125 6.1226
## [25] 8.8387 -4.0302 -9.9673 -6.2035 4.8673 2.6996 0.1127 -9.4464
## [33] -8.7788 -6.5793 9.8826 5.6083 0.4467 -9.1158 -4.5963 -7.7016
## [41] 4.4545 -1.9028 7.7174 -2.3740 -1.4447 -9.2058 -1.0078 3.8951
## [49] 6.1684 -6.9296 1.0501 -1.2559 4.5737 -1.5584 -1.0095 5.1396
## [57] -5.0718 -0.5261 -5.4420 -1.0687 8.0694 9.0202 1.7211 2.9003
## [65] 8.8643 4.6707 9.0052 3.6514 -4.8318 4.3243 -0.2003 8.1836
## [73] 8.6870 -0.4554 -1.1983 1.9187 4.1306 9.5112 -0.6834 -4.0391
## [81] 6.5290 8.0330 2.2810 -4.5620 -1.3398 9.1276 0.1860 -4.9475
## [89] -1.7933 -5.8457 0.0718 6.9999 -9.5582 6.6542 -4.8947 4.1885
## [97] 3.2671 2.1529 2.3445 1.9111 3.5374 9.6010 6.7516 -8.4801
## [105] 0.5642 -6.7603 -1.3661 -7.9915 -7.8887 6.6332 9.0805 -1.9404
## [113] -2.6270 -1.0344 -1.3250 9.2546 4.8267 -4.3743 -0.1801 9.5691
## [121] 1.3493 -8.4130 -8.8140 8.1244 6.4838 2.5340 -5.7221 -3.0243
## [129] -7.1726 -1.3722 5.4677 8.1942 6.5351 9.1158 -2.4704 -9.0674
## [137] -4.2086 9.9938 -1.0250 9.9006 4.1490 -4.0064 8.1666 -0.5133
## [145] 6.3063 -4.9302 9.1619 -0.7410 -7.3980 5.4061 -7.1588 0.8232
## [153] 3.1500 1.5226 0.0998 8.6081 -5.4679 -9.7247 -4.4406 -3.7689
## [161] 8.0089 1.1483 -7.3008 -5.6643 -8.3084 7.1129 -8.8263 -3.4405
## [169] 0.0039 -3.3221 7.5411 7.6556 8.0223 5.9588 8.0296 -1.1948
## [177] -2.0522 -2.5690 0.5762 -8.5243 4.3370 -5.1438 6.8900 9.9064
## [185] -7.8998 -0.2985 3.6739 6.3902 5.3114 -1.4640 8.7194 4.5802
## [193] -3.2537 -0.4205 9.3100 8.2741 -5.4384 -3.8548 -3.0663 -9.1256
## [201] 9.2726 6.3111 3.4945 -1.8791 -4.6771 8.5664 -2.6685 0.9898
## [209] 0.4209 8.5398 0.0347 2.7383 -5.3010 -2.7293 7.8700 5.3192
## [217] 5.9089 -3.5394 -8.1498 9.0796 -0.9996 1.1654 -6.5809 -9.5763
## [225] -5.7617 -3.6319 1.6518 -8.9740 -5.0029 -9.7062 -5.6306 0.4721
## [233] -3.4027 4.8292 9.1111 -5.5360 9.5377 -8.1610 4.7329 -4.5285
## [241] -4.4708 -8.7898 -5.4391 5.1795 -3.1081 4.3437 -5.9375 -8.8935
## [249] 4.5375 -4.6742 -3.3946 7.0500 -9.8883 -1.3346 -7.6324 0.0759
## [257] 9.5171 3.2350 7.6829 4.6923 0.1055 7.7228 -0.0918 5.6022
## [265] 0.1353 -8.3166 0.9541 -2.8297 0.7644 2.3758 -5.0565 -0.8686
## [273] -7.9573 -9.7068 -8.0728 -5.5162 -7.1107 6.8968 -1.6922 1.2558
## [281] 1.7141 -8.5913 5.4105 1.7095 5.4498 -1.2610 2.6328 4.9832
## [289] 3.8956 9.6484 -4.3899 -8.7046 -1.5016 -3.8649 -1.4157 -1.1036
## [297] 2.9902 -9.6734 -5.3397 -0.2491 7.7769 -4.4693 -4.0037 9.7462
## [305] -0.7189 -9.9648 -8.7996 8.6331 -0.5456 6.8656 6.6365 5.3772
## [313] 2.0704 1.1672 0.5088 -5.2279 2.3253 7.3111 5.2873 -6.2425
## [321] -2.5527 5.1280 -8.2455 3.1279 4.3118 7.1818 -5.8035 -4.4919
## [329] 6.7071 -4.3875 6.9593 2.3741 -5.0076 2.4016 -8.8984 8.3056
## [337] 7.7135 3.7261 9.0789 1.1516 5.6645 6.0056 -4.6980 -4.1329
## [345] -5.4841 7.2559 4.4749 -6.9348 7.0602 2.6475 1.1091 -9.1215
## [353] -4.4099 1.9467 9.1347 -8.8311 -0.8506 -1.8306 0.1857 -9.2116
## [361] 7.2601 -4.7758 -0.7860 -5.6235 8.7177 4.2095 -2.0126 -4.3952
## [369] 4.0690 -5.5787 -4.1274 7.7811 -0.0138 9.0492 5.5153 7.2582
## [377] 1.5004 -8.6410 0.3584 7.9828 -9.3846 -7.8075 2.6299 -0.5736
```

##	[385]	9.8043	5.3733	5.4813	-1.6936	9.9418	-1.1955	9.3476	0.9978
##	[393]	5.0494	3.3125	6.6017	3.1918	-9.8836	-9.1572	8.5106	-1.3943
##	[401]	2.6941	2.2576	9.0655	4.4531	-5.7682	-2.4903	3.2663	-5.1773
##	[409]	1.5399	-8.2856	8.7734	-5.3388	-4.2913	-0.9672	-0.7221	-4.7471
##	[417]	-5.9464	4.4796	-1.5461	2.8675	2.5833	-8.9233	-0.4286	2.4901
##	[425]	-3.4459	4.1198	7.9937	-2.0616	-2.1444	-0.2114	-8.9399	-3.3443
##	[433]	-6.1717	-2.3077	-1.2563	-6.8608	-8.5618	3.2931	-1.6258	-7.7936
##	[441]	-0.7428	-4.4512	-8.1468	-6.3909	0.4833	-6.1620	0.3078	-2.8754
##	[449]	6.5274	-3.7250	0.4279	-5.1605	-3.4945	-2.0007	3.0048	-8.8701
##	[457]	4.1079	0.8841	-3.2165	4.7251	-4.6694	-8.0054	-2.5311	-8.4237
##	[465]	-5.4772	9.7438	-7.6325	1.8929	3.8456	0.4242	-1.9862	-1.5735
##	[473]	0.5837	3.7774	9.6753	0.2971	-0.1509	-4.1487	-6.8011	-3.3460
##	[481]	-4.1340	-2.6952	-0.4844	-2.5078	-1.4654	-5.4777	-8.8866	7.1722
##	[489]	1.2247	0.7475	2.3386	-5.6052	0.3918	4.4565	-9.4356	2.7383
##	[497]	4.8045	-1.0943	5.9477	3.1149	2.1704	-6.3526	4.8530	-1.5761
##	[505]	-7.8502	-6.8228	5.7222	-7.6351	-0.6725	-2.5966	-0.1320	1.3344
##	[513]	9.0792	-6.3876	-2.2706	-1.3897	6.9911	-9.3282	-9.5528	-3.4494
##	[521]	-3.1538	7.5827	1.2933	-7.5044	-8.7660	-7.6710	6.7174	-1.7402
##	[529]	9.3686	4.0712	-4.6235	7.8358	2.9736	9.9204	-7.7007	5.7681
##	[537]	9.5848	4.9807	5.5193	9.5743	-3.4602	-2.3653	-6.4362	-1.6895
##	[545]	-1.5436	1.9994	-1.7642	-8.4442	3.1075	-9.3356	-6.9127	9.2920
##	[553]	0.1711	-4.3471	-9.9460	-3.5296	7.6920	-6.7927	-2.3335	1.1456
##	[561]	-0.1095	4.4557	6.2589	-1.8083	-5.5030	0.1173	-3.9661	6.5860
##	[569]	-4.7073	-0.0859	-8.0663	-4.2544	6.8940	-0.4833	-0.9303	-6.0039
##	[577]	-4.8073	0.8115	4.5077	-6.3077	-6.2921	8.3666	-7.3263	-9.8710
##	[585]	-4.5487	3.1180	0.1587	-3.7447	-6.1540	9.5309	4.8793	-1.8537
##	[593]	-7.9025	3.7008	2.0511	3.0117	5.0797	1.9561	-8.2090	3.9563
##	[601]	-9.5933	9.5422	9.0319	6.4634	-5.0136	1.5572	-3.1262	9.2659
##	[609]	7.0384	4.4024	5.8205	4.8452	-3.4911	7.8688	-8.8884	1.2799
##	[617]	-3.6310	5.3185	2.4526	-2.1628	7.2903	0.9935	7.0661	0.3162
##	[625]	2.9861	-4.7765	-4.2244	9.5811	9.8506	-4.2377	2.0976	-2.7799
##	[633]	9.7692	8.1464	0.3712	-0.1645	-2.2180	7.5724	8.4496	-4.5639
##	[641]	6.9484	-0.1025	2.5307	-1.9814	-8.4450	-6.6751	-9.2466	-2.3835
##	[649]	4.2688	-4.8770	-1.8954	6.0880	5.8986	-2.2894	4.9292	-0.6597
##	[657]	-5.4911	-5.9224	-8.9486	1.7039	9.3498	-8.9302	4.3605	-9.9682
##	[665]	9.7938	9.3991	3.9926	5.5111	-1.0479	4.7439	3.5537	-9.7468
##	[673]	0.7272	5.0168	-9.5931	3.9681	7.0350	-0.6190	-4.7990	5.9065
##	[681]	-1.4968	7.3753	2.5770	-0.9723	0.4005	9.4827	3.6704	6.7702
##	[689]	-2.1042	7.7919	-9.4892	-2.4733	3.6860	-1.0432	-5.7284	-6.4378
##	[697]	3.6104	3.1626	-1.5787	2.0823	-1.9869	0.7273	6.9428	-5.0393
##	[705]	3.6355	-1.8344	0.8054	9.5287	1.9110	-6.4719	7.6733	-3.1532
##	[713]	6.9272	-8.3951	-8.7497	-7.8529	7.0815	-2.2571	7.5475	-3.0442
##	[721]	-2.5394	-4.6270	-0.4993	-5.6098	6.1549	-1.5588	4.7855	7.4321
##	[729]	7.6718	-2.7951	1.4841	-8.0058	-7.3054	-7.1495	-6.3324	-9.5497
##	[737]	-4.7041	-9.1283	-8.9250	-5.4519	-6.6092	-6.7114	1.6960	-7.4945
##	[745]	4.1300	9.5333	3.9591	9.2008	7.9842	9.8966	-5.2298	5.6254
##	[753]	3.9617	-1.2810	3.7434	-7.0317	-9.5569	-7.2900	3.8417	-3.8072
##	[761]	-3.0254	7.9964	-0.4541	4.9489	2.7208	6.2627	0.6487	-1.4843
##	[769]	-0.1344	-8.7973	2.2194	2.5588	-6.8727	-1.6993	-4.3499	-2.3884
##	[777]	5.7354	9.7105	0.1674	6.1553	9.2619	-1.8067	7.7100	-9.4302
##	[785]	3.7258	-2.2561	-4.3442	6.7802	1.5165	2.9072	-2.7513	2.1446
##	[793]	8.3725	7.0402	9.8581	-1.3421	7.5108	-2.7051	-8.6569	-2.0048

##	[801]	-0.0173	5.3736	-5.3741	-2.1234	-1.6618	5.5293	0.5628	-7.8062
##	[809]	-0.3455	-7.9629	7.9021	1.0136	8.5307	5.5297	2.8452	-3.1647
##	[817]	5.5107	-4.3635	-5.2060	-9.9548	-2.2964	2.9183	6.2420	-6.3340
##	[825]	9.6353	3.1274	6.4894	8.4648	1.9652	-3.8005	5.5798	5.1403
##	[833]	7.3997	-1.2124	6.8408	2.8758	-2.3737	-7.5224	0.8106	3.2392
##	[841]	-8.8143	8.0809	-6.1983	9.2762	-8.1498	-7.6492	-0.9783	4.0644
##	[849]	-2.4310	-3.8856	4.0766	3.9883	4.8087	-1.4164	-5.8928	-1.9589
##	[857]	-4.3552	5.0482	1.0954	5.4445	0.2416	2.1981	3.2137	3.5739
##	[865]	-7.4997	2.9491	-5.0048	2.1927	4.2017	0.7269	4.9457	6.4013
##	[873]	-1.2267	5.2223	8.6225	5.7277	9.1104	7.3169	-5.3783	4.1723
##	[881]	6.0471	-1.2160	6.8646	-7.2405	0.4039	-3.7447	1.0253	-6.1567
##	[889]	9.3511	-2.3468	6.2446	-4.0062	0.5113	-1.5593	4.3167	-6.9412
##	[897]	-2.5997	-4.6626	6.2605	2.7785	5.3319	1.6915	9.9457	-3.1384
##	[905]	-2.3751	6.8885	-2.4155	-8.9923	-9.3920	3.7073	6.9877	8.8355
##	[913]	-0.0123	-3.1661	-7.2332	3.0168	-5.8209	-4.2335	-0.7344	-0.1756
##	[921]	-3.7421	5.8527	-4.7840	-6.3929	-8.1913	-4.2098	-5.6314	4.0543
##	[929]	0.4650	3.2238	-6.7381	5.8025	-5.9965	7.8970	-8.1841	5.5861
##	[937]	-8.1861	-7.1880	-5.9889	1.2222	3.1544	-3.1017	0.5382	-0.0467
##	[945]	-3.6758	6.4105	7.7463	2.1757	-4.3123	1.2333	3.1956	-0.9561
##	[953]	2.6034	3.0865	7.3604	-7.6145	7.1096	-3.2665	6.5073	8.9403
##	[961]	-5.7365	4.6219	1.0914	-6.6385	-4.5853	-7.8076	-8.5748	-6.6261
##	[969]	2.4935	4.3819	2.7599	3.3510	-8.1531	-4.3036	9.0128	-2.8764
##	[977]	-6.8548	6.7204	6.9006	-2.4121	3.6360	-0.4658	-6.9750	2.7556
##	[985]	6.4976	-1.4467	-1.0881	-5.3768	-8.4381	-3.2035	-5.4984	-7.2030
##	[993]	1.3714	3.6115	-3.4513	8.1715	-5.1129	1.9297	3.3737	5.8363
##	[1001]	-1.1595	-4.8504	1.2280	-1.1158	1.2769	-1.0827	8.9049	-8.4747
##	[1009]	-3.4506	0.7083	3.4842	3.6823	-9.8920	5.0565	-4.6703	8.1171
##	[1017]	-0.3448	-8.5915	-2.9264	4.6319	-0.2349	5.2667	-4.5694	-1.6635
##	[1025]	4.7580	4.5313	-5.8123	-8.4279	0.5140	1.6976	-8.2963	-9.9825
##	[1033]	-6.5586	2.4500	4.3844	8.9075	-0.4644	8.7692	-2.1214	-9.9218
##	[1041]	-9.6202	4.0737	9.1999	-2.7345	-2.4182	-0.1995	8.5692	-9.7332
##	[1049]	-7.8173	2.9979	-5.4744	-4.0161	7.2281	2.9151	-3.0096	-1.5165
##	[1057]	-0.2531	-6.9484	0.8690	2.6101	-1.6910	4.4820	8.5129	0.4332
##	[1065]	-1.2767	4.2935	-4.6684	-3.5736	-7.4449	8.0114	9.3384	-2.7964
##	[1073]	0.3565	-4.3785	-6.7268	7.0240	-1.5446	7.3510	2.4295	5.0391
##	[1081]	-9.4500	1.7452	-9.1050	8.8926	-1.0150	-6.8781	1.3105	7.4478
##	[1089]	-8.7385	9.1354	7.2279	0.4355	6.7552	4.4194	9.9548	4.5794
##	[1097]	0.0089	7.6736	-6.1768	-6.0955	-8.7109	6.2959	4.0132	-9.1254
##	[1105]	8.3905	9.6219	6.3199	-4.3223	-0.9817	-2.4814	-8.7917	2.0428
##	[1113]	-3.4445	-7.8881	-9.9827	-2.0737	-3.3869	0.4903	-2.3348	-2.6612
##	[1121]	-9.6292	9.5727	0.9908	-9.2602	0.2797	-0.8393	0.2039	7.6154
##	[1129]	-9.0237	-5.9694	-8.9566	-1.1051	-7.5299	9.4802	-7.5436	-9.3815
##	[1137]	-0.1237	-5.7837	-1.4089	-7.6924	-4.3638	4.7099	-4.3993	-6.1358
##	[1145]	-8.1680	3.6008	-4.9133	-4.8160	-7.0202	-1.2123	3.3074	6.0812
##	[1153]	8.5352	1.0897	3.2432	-9.4771	8.1007	-4.0796	-3.6729	5.2452
##	[1161]	-8.2059	9.3714	6.6859	-7.3508	-7.5832	-1.8084	-7.7809	-0.6503
##	[1169]	5.5712	-5.3618	-2.2590	-8.3322	-6.2975	9.3405	3.6598	8.6068
##	[1177]	1.8209	-8.4051	6.9004	-9.4043	6.2932	-2.1850	-7.1319	3.3312
##	[1185]	2.3717	-1.0247	6.1320	7.4121	9.7237	-1.9473	-2.9574	-7.5620
##	[1193]	9.8123	8.7146	8.7806	-1.8504	5.4436	-2.2357	-9.3474	-2.0757
##	[1201]	9.1785	0.4466	3.4759	-9.7959	0.1119	5.0558	-0.9659	5.2923
##	[1209]	1.2516	4.7081	4.6066	-3.1902	3.9498	0.7911	7.5784	-8.4869

## [1217]	-2.4673	2.8099	-9.4935	-5.8610	2.6761	8.1213	-0.0821	-6.4546
## [1225]	5.0937	-9.6051	2.3969	1.3708	-6.9726	-8.9702	-9.1619	-6.4651
## [1233]	-5.1025	5.7862	-6.5275	9.3884	-0.1753	-2.0720	0.1972	6.6878
## [1241]	-8.8780	-8.3160	8.6530	9.1646	-2.0664	-0.0406	-1.9908	-5.6454
## [1249]	4.9729	3.0903	0.4091	-5.8163	4.7271	0.4014	-6.6598	0.8106
## [1257]	-2.4447	7.5765	7.3945	-4.2901	-3.9968	-2.9259	-0.7376	0.9868
## [1265]	-0.4252	6.3048	-6.3611	9.4199	2.7409	8.8639	0.4083	6.6307
## [1273]	9.4213	-9.4297	-6.4860	6.5653	2.4812	5.4880	5.3336	-5.1470
## [1281]	-5.2167	-4.2788	-8.4704	-8.2905	1.3391	-0.1001	-4.4923	0.7221
## [1289]	0.2801	-9.2049	-1.4486	-8.8620	-2.2641	1.3852	-7.3269	-6.9552
## [1297]	-3.9867	-2.3523	-6.0106	6.8641	-3.7903	-3.2983	3.6664	1.0111
## [1305]	-7.5838	4.6195	-1.9130	-7.3927	6.2580	7.8601	8.6693	6.4087
## [1313]	-0.8390	7.9697	-3.6082	-3.3004	4.0070	-8.5689	0.3977	3.9076
## [1321]	4.3960	4.6378	9.0675	9.8893	-2.4023	9.1448	1.6833	-3.5798
## [1329]	-8.0972	-6.8596	-7.3003	4.6539	-4.5238	9.5377	-2.5510	-7.6324
## [1337]	-7.3156	-5.6490	7.7659	9.1948	-6.3620	9.3382	-5.0634	1.5426
## [1345]	-7.6810	-8.9482	-1.0547	-4.1257	-2.0183	7.5696	4.3658	9.6874
## [1353]	-9.9618	-0.1267	-2.7822	-4.7225	-4.2724	2.3985	7.1043	1.1519
## [1361]	-6.5144	-6.3683	-6.7280	4.4607	9.2065	8.3561	-1.4976	0.1047
## [1369]	5.6518	4.6852	-0.7947	1.9902	9.1526	3.5561	-1.7498	-1.6916
## [1377]	-7.9329	7.6903	9.7432	1.8331	8.0925	-3.8492	3.5194	4.7160
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## [1393]	0.0525	-5.4712	7.2440	-2.7302	-0.2003	8.9447	1.1519	-9.4231
## [1401]	8.1697	-7.3151	-4.0530	-5.4641	-6.8985	-8.1078	1.3482	7.3010
## [1409]	-9.3356	7.9395	6.9339	3.8354	3.1621	-3.3963	-6.9757	-3.2820
## [1417]	0.8111	-4.3001	-1.4040	-4.0021	-1.4980	3.7078	-9.7530	6.3679
## [1425]	-7.7509	-3.5203	3.0942	2.5027	6.6004	-9.8296	7.8906	0.6045
## [1433]	-3.4605	-6.2030	2.5473	7.2803	3.1499	7.6431	-8.3643	-9.7194
## [1441]	-5.8886	6.4364	-3.5892	-4.2842	-7.0985	-1.3022	7.5768	6.3108
## [1449]	0.8686	5.0928	6.0008	-5.8723	-4.5127	2.5108	-5.3934	-3.5955
## [1457]	-6.6801	7.8938	5.5693	-8.8393	2.5303	5.8764	-0.6116	8.3580
## [1465]	-1.1707	-9.9093	5.4253	4.8285	-1.1977	-2.8338	-4.0033	0.8777
## [1473]	-4.6037	-2.1290	-3.7450	5.1112	-8.9704	6.4674	8.2446	4.4691
## [1481]	-7.1155	-0.8465	-4.5005	-3.2853	-6.5289	-1.1159	4.9229	-1.7980
## [1489]	2.7112	1.6248	3.6142	-5.9040	6.6515	4.4117	-5.7605	-6.8187
## [1497]	0.8449	-8.6227	-2.9247	2.0342	-6.6785	3.2207	-0.5526	6.9483
## [1505]	-4.1617	5.3221	8.7870	1.7396	9.5158	5.1081	9.5734	-6.7316
## [1513]	5.7695	-7.4597	-4.8437	-7.7582	4.9361	0.3568	5.7254	4.9169
## [1521]	7.4067	6.3527	6.0802	8.6211	0.4559	2.5800	-6.2942	-1.0200
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## [1537]	-4.8730	-5.0161	-3.5950	-6.8552	-2.4824	-6.7437	9.1202	8.5812
## [1545]	-0.8172	1.9789	-1.3786	-3.3286	3.3993	-8.8571	-8.3168	1.5293
## [1553]	-7.8129	-3.8102	-9.1043	-0.6845	2.0818	-1.6411	4.5918	5.6228
## [1561]	0.4015	-0.5472	5.3056	-1.4128	4.2018	9.8235	1.7602	7.2305
## [1569]	7.7127	1.0345	1.7268	-5.3534	0.7803	7.6165	-1.2334	-6.4900
## [1577]	-6.0840	-0.3833	-3.4064	6.0611	-8.5946	-3.1916	-2.0836	-2.8654
## [1585]	3.7568	3.4029	-6.2473	1.1092	-1.6119	-1.9683	1.5542	-1.6584
## [1593]	-8.9526	1.9811	-4.2730	3.5931	-2.7332	8.0442	7.0825	-3.2495
## [1601]	8.8400	8.1205	-9.4645	-8.5714	4.1051	-4.8113	-6.3461	-8.2391
## [1609]	5.2370	2.4301	-2.7679	3.2614	4.5759	-8.2807	-7.7473	0.3818
## [1617]	8.8844	-6.3921	1.3221	3.7858	-9.9835	-8.8003	2.6465	6.1281
## [1625]	9.8563	-0.3542	9.0403	-6.9977	9.1573	-0.6609	-6.7375	-2.0950

## [1633]	6.2997	-2.8449	9.6334	-8.6570	3.4915	-4.4159	-4.2890	-1.3761
## [1641]	5.6444	-8.6199	6.9801	2.3762	-5.3161	8.3028	5.5756	5.6584
## [1649]	-5.2511	-5.7544	7.8960	-4.8873	3.7948	-3.5856	3.7328	4.5541
## [1657]	1.6680	-0.3152	-8.7286	-6.5537	6.7642	-8.6946	9.2649	4.2404
## [1665]	-1.7053	-7.0350	0.6123	3.8082	-0.3979	0.5825	-1.6931	5.2956
## [1673]	-4.7023	7.2612	7.8595	-8.4535	4.0266	6.9336	3.3682	5.0843
## [1681]	5.5770	-2.4150	-2.9114	-0.1576	-8.8532	3.7473	9.1572	4.9107
## [1689]	-6.7204	-9.1348	0.1714	8.5540	8.3581	-8.4084	-1.9511	-9.6258
## [1697]	9.3514	6.6004	-9.8518	-8.6936	-0.0441	-1.7735	6.8699	-8.2187
## [1705]	3.9930	-9.2728	1.2871	-3.4159	6.4252	4.8244	3.8134	2.4792
## [1713]	1.2130	-4.7943	-9.2769	-0.1257	6.6770	-0.7862	4.1476	-0.1877
## [1721]	-0.9115	-3.4443	-9.2499	2.8186	1.8492	-7.2106	-3.0121	2.9421
## [1729]	2.9287	-2.6570	-3.0115	-2.6283	8.3948	5.1969	-7.1157	-3.3104
## [1737]	-5.1125	-6.7555	9.4586	-7.7104	-8.5870	4.9653	-7.3119	-9.8037
## [1745]	7.1251	-3.2326	-0.1858	1.4399	-7.7468	-8.3086	7.3420	0.0556
## [1753]	-0.1422	4.9730	-6.5402	-9.3191	1.7116	-6.2504	-6.1504	9.2198
## [1761]	-6.7548	-4.8729	1.7453	-6.6600	2.8845	-6.0895	0.6410	-3.3594
## [1769]	1.5329	1.4749	-4.4496	-5.1836	8.0732	1.5986	-0.2965	5.1729
## [1777]	-9.1208	2.6327	2.4910	5.4664	-2.0205	-5.7680	2.8565	-0.9769
## [1785]	1.1782	4.9349	-7.2411	0.4383	-7.4748	1.6844	-5.8477	0.4056
## [1793]	7.1314	1.4514	3.5257	-0.1968	9.1275	-7.2840	-1.6979	1.3813
## [1801]	2.5837	-3.5115	6.8625	8.4893	7.9952	-4.0360	0.8390	1.2093
## [1809]	6.5331	-9.5153	1.4637	-4.9003	3.8176	-5.9259	-6.4302	7.8194
## [1817]	-0.7650	-1.2009	1.7478	1.1525	2.4620	2.3605	-7.0594	-4.0466
## [1825]	5.2965	2.3637	3.0740	8.6809	2.1786	4.6276	-6.6900	5.2502
## [1833]	5.6534	-5.9601	-1.8307	1.4210	-0.2539	2.3074	-4.5877	-4.8760
## [1841]	5.6692	0.9347	-6.6704	9.5156	-0.2307	-6.7021	-5.7294	4.1970
## [1849]	-7.9755	-7.1710	-1.8099	2.7516	-9.5202	-7.0251	-4.8499	4.4504
## [1857]	-1.3719	-7.9536	-2.3255	0.1634	0.3916	8.2018	-6.5497	-1.7606
## [1865]	-1.6763	-0.8473	-9.2793	8.3974	5.0568	9.5828	7.7058	9.8698
## [1873]	-2.4937	-3.0614	5.5625	-7.0864	0.0617	4.8505	-5.9720	3.3687
## [1881]	-1.1243	8.0128	-4.1317	1.1940	-9.4288	1.2776	1.4341	8.6991
## [1889]	3.4765	1.2869	-6.6029	5.0491	-3.9182	3.7198	-7.5557	-0.3547
## [1897]	-8.2637	-4.0722	-7.4100	-2.6430	-0.5775	-9.4865	-1.2654	7.9891
## [1905]	-5.4771	-8.8931	-3.7010	-4.4528	-0.5331	-2.8886	3.4126	7.7200
## [1913]	-1.6857	0.4613	-3.2365	6.5495	-6.7210	-2.3142	3.5726	9.0535
## [1921]	-5.3891	3.7374	-8.2808	4.1582	8.5274	6.1980	-6.4989	-8.6172
## [1929]	1.0643	-0.0840	6.5331	1.6978	3.5315	-5.5676	-3.5563	0.0971
## [1937]	8.8684	-5.2930	-4.5992	1.7393	-8.5718	-9.2571	4.8336	2.2390
## [1945]	5.0608	-4.7540	6.1057	6.5244	-6.3021	-9.5152	-3.2072	0.3391
## [1953]	9.9001	-1.7484	-1.9068	6.5305	6.7611	-8.8590	-0.8269	-4.1681
## [1961]	-3.7977	5.2345	-9.7111	4.6004	-7.4382	2.7841	-9.1019	2.2709
## [1969]	-7.5758	-9.0479	4.4052	6.0527	5.3228	1.7084	-0.4245	-9.3043
## [1977]	-5.1961	-1.3770	3.6698	6.5980	-6.8378	5.1843	-1.3539	-4.7215
## [1985]	-1.5823	6.3591	9.1451	-3.0861	6.3832	2.9751	-5.9412	6.4687
## [1993]	4.7543	3.0389	-5.5400	-4.8605	-2.0567	5.8777	4.5750	8.6044
## [2001]	-9.8284	-6.4721	2.3776	-1.4457	-4.0509	6.8446	-4.9780	-6.0578
## [2009]	-9.3020	-9.1445	0.9823	-3.7919	-6.4930	7.1380	9.0494	-8.2675
## [2017]	2.2934	-1.9610	-6.6067	6.6392	3.3580	0.3743	0.5341	4.2923
## [2025]	7.3282	8.9562	7.9088	6.3229	-6.9581	4.4561	2.8840	3.7656
## [2033]	-6.3015	-7.5331	1.3435	5.9864	6.8448	4.1225	7.7392	3.1351
## [2041]	-7.0148	0.4561	-4.4742	-7.9629	-9.3818	1.2207	5.0990	7.9051

## [2049]	5.2380	-4.5927	4.6183	-9.2827	2.2659	9.3394	2.4436	-4.2743
## [2057]	-3.2978	7.4586	-5.4276	4.6741	-2.5814	9.0263	9.7942	-0.3357
## [2065]	2.2505	-2.3616	-2.5767	9.1677	2.4079	9.6896	-3.4047	2.4991
## [2073]	8.0417	1.3487	6.0661	5.4212	-9.5070	-4.9876	-4.4371	3.8139
## [2081]	-8.4771	-6.7764	0.0137	-2.3390	-0.1427	5.7332	5.8496	-5.5451
## [2089]	-7.5731	6.0515	-7.9339	3.8446	-2.1588	8.4835	-8.9241	9.6966
## [2097]	6.0576	-4.3496	-3.3335	3.0394	-8.2888	8.8092	-2.8779	1.5777
## [2105]	5.0744	-4.8677	7.8985	-6.8151	8.2398	8.3862	3.0611	9.9066
## [2113]	3.2466	-7.0504	4.4330	0.8802	9.5752	-8.8313	-7.1663	0.5554
## [2121]	3.6907	-2.4129	-2.8166	-0.0962	7.8013	9.9621	9.2229	-8.3809
## [2129]	-9.3119	3.3434	-9.0173	9.3315	-4.1437	-4.5882	-0.2816	-5.8741
## [2137]	-5.4222	0.3853	-2.6809	8.0659	-5.1671	-2.6850	9.6176	-6.9501
## [2145]	-6.0959	0.5624	-1.8277	-9.5639	-1.4918	-8.5991	7.7846	7.2273
## [2153]	-4.0885	3.5363	-7.9668	-9.7029	7.4199	4.6317	8.2420	-9.7882
## [2161]	-9.1997	-3.4510	-2.2710	2.3543	-0.7924	-8.4684	-0.3276	7.9502
## [2169]	-3.3039	7.7545	-6.2045	-5.0023	7.1643	7.3772	-2.5095	1.5582
## [2177]	-0.2231	4.8438	-5.4878	4.9778	6.4974	-8.0091	-4.4826	-9.5725
## [2185]	1.9249	9.7257	6.9075	1.8299	-3.1329	2.9282	-9.2565	-7.6712
## [2193]	0.3315	7.4260	-7.9486	9.3415	8.9798	8.7346	5.1720	-9.5699
## [2201]	-1.4684	-7.0190	-1.3870	0.7790	4.8194	5.1410	8.2512	-4.1348
## [2209]	1.1146	-6.0941	5.5281	1.8174	0.8647	-6.5414	8.1243	9.5900
## [2217]	9.7911	-9.3207	-7.0689	3.8910	-7.7904	-9.5410	-9.7159	-3.7892
## [2225]	2.2911	8.6228	-5.2757	7.3804	7.3166	-8.3844	-9.5325	-8.6008
## [2233]	1.5748	0.4634	3.1723	-6.5940	-8.5306	3.1509	-9.6965	-6.7485
## [2241]	2.1526	-8.9263	-8.7100	-3.3293	-7.3165	-7.7913	-4.1117	7.2280
## [2249]	4.5354	-2.6223	-8.3910	-0.3019	-6.4945	8.3260	-8.6820	9.3681
## [2257]	-2.5920	-0.4223	-6.0170	2.6112	8.6398	8.0950	1.1793	-1.3850
## [2265]	9.4385	-7.0334	-2.8056	6.7987	5.4714	-8.7265	-1.0838	-1.9374
## [2273]	-4.1933	-3.1243	8.6449	-3.9885	8.2182	-9.9411	-4.0517	-4.4273
## [2281]	-8.0989	7.9460	1.4780	-3.6084	9.2139	-4.6411	7.2833	-4.9468
## [2289]	-9.6490	-6.1828	-9.6622	1.1610	1.4035	4.4667	7.0778	8.5258
## [2297]	-1.2595	-8.8267	-4.4286	3.0742	8.9079	-1.2179	9.4514	2.2457
## [2305]	1.0519	-6.9780	-1.5389	0.3630	2.2882	-7.5134	-2.6055	4.9056
## [2313]	-6.1914	-6.5595	-8.8540	0.5067	-3.4432	-9.6660	4.9422	2.5631
## [2321]	9.4271	7.3202	-0.5795	5.5642	5.2642	7.0797	5.1850	-8.2488
## [2329]	3.6752	-1.3444	0.0080	-0.5557	0.3523	5.2860	3.1683	-1.3325
## [2337]	-1.6648	-1.6108	-7.0666	9.0668	-5.1664	7.2470	7.3039	-9.2099
## [2345]	-3.4709	-2.9947	-1.8924	-6.7888	8.8774	9.9704	4.0673	-3.7893
## [2353]	-4.8719	7.4905	8.2456	0.7669	1.9551	0.8666	2.3242	3.7957
## [2361]	5.8398	-1.1343	-0.0570	5.9512	-0.3832	4.2440	2.4255	-9.0803
## [2369]	-2.6820	-4.2152	-3.1636	-3.9927	6.5464	-3.8722	5.2889	7.1325
## [2377]	0.5000	8.6299	6.8736	-6.4337	-9.0142	4.1179	-8.8354	-7.5791
## [2385]	-1.1006	-4.2253	-1.5855	2.3465	6.8771	-1.4985	-3.1795	-5.0378
## [2393]	-8.6304	-7.9141	1.7458	-2.4478	-6.3860	0.4187	6.4112	4.2604
## [2401]	-8.5509	-6.1887	-9.5627	4.4263	-1.5384	-7.5551	-5.3534	-4.1501
## [2409]	9.4740	3.4756	-6.4596	-5.1139	-1.2637	-5.6437	2.4430	-0.3775
## [2417]	5.3853	0.2827	-3.6665	6.4994	8.5669	-1.8454	-6.9054	-9.1102
## [2425]	6.1360	-4.8807	3.2865	-4.7161	-7.2480	-6.4883	-0.4617	2.5247
## [2433]	8.5018	9.8764	-4.8438	5.7290	-4.9064	-1.2305	7.1420	3.8550
## [2441]	-3.2721	8.5091	-1.2677	-2.4288	6.1260	4.9362	1.3634	4.2582
## [2449]	-3.6335	-0.9760	8.6479	0.4509	-3.3956	4.8553	5.8326	-0.2093
## [2457]	4.7157	-2.6577	-3.3188	-4.4748	9.8950	2.2892	-2.1652	-0.4388

## [2465]	-9.1432	-2.8991	-1.4164	1.4017	2.5622	-5.6052	6.9192	-5.7769
## [2473]	5.2205	8.3042	-4.2571	-6.2187	9.6626	2.4784	9.0768	-6.6715
## [2481]	-4.0337	-1.8238	-9.0028	3.0373	-5.5417	-6.8728	4.4199	4.4818
## [2489]	-4.7241	1.7328	5.7222	-2.4846	-6.8624	1.9390	-7.8729	0.5982
## [2497]	4.6399	-5.9890	0.8119	-8.3761	-2.6529	0.8452	-1.0748	-6.8497
## [2505]	-3.0216	-0.2810	-9.4559	-9.0066	-8.3756	-6.4713	5.2728	2.1446
## [2513]	4.7622	7.0413	9.9358	1.3442	-8.6398	-6.7999	-3.0274	-3.2211
## [2521]	-0.3075	7.2589	4.2721	3.9602	-7.0877	5.6874	1.7266	6.8109
## [2529]	-5.7261	2.9417	0.0199	7.3273	-8.8754	-6.6955	-6.3437	2.3555
## [2537]	-5.5438	-2.6575	3.0990	1.5988	7.5474	8.1496	-6.2623	3.8559
## [2545]	-6.3547	-4.7107	-0.4393	-2.7378	-2.8159	-6.5857	0.3279	-0.7157
## [2553]	-4.4710	-3.8264	-3.8130	5.7139	-3.9093	-3.7779	-1.2802	5.2249
## [2561]	0.2605	5.2892	4.8269	-3.3668	1.6020	8.4627	3.3468	0.2148
## [2569]	-6.5701	-4.8613	-0.5247	0.5931	1.1969	-6.7830	-7.7279	4.8495
## [2577]	-5.4941	-6.2930	-3.7865	8.0399	0.2079	-9.6194	-9.7549	-7.3406
## [2585]	8.1775	-2.6490	2.5189	-5.1651	4.5303	8.2200	3.6873	-0.4818
## [2593]	1.7024	2.7726	-9.8876	5.2066	0.3698	1.2151	-0.4063	6.6345
## [2601]	-9.2978	3.3986	5.6958	5.1496	-8.5992	-8.6272	-2.7181	-7.3997
## [2609]	-8.8891	-9.7401	-3.8963	-1.9994	6.9379	8.4779	-0.3229	-6.7212
## [2617]	-2.5461	2.4729	8.1068	3.4076	8.3856	1.8853	-2.4908	-2.2376
## [2625]	-5.4246	-4.6352	3.7426	-4.7572	3.6475	0.1392	7.6607	8.0628
## [2633]	-7.2886	-5.7252	-0.0960	-0.9752	3.5634	0.7907	-5.9201	5.7130
## [2641]	-8.0621	-1.8294	-9.4181	4.7634	-0.3706	-8.6554	-0.3168	5.2749
## [2649]	-3.4714	0.7677	-0.6072	-7.2677	-8.7568	-9.8021	5.6986	-2.8514
## [2657]	1.3288	4.9642	-4.5411	2.5032	-3.2273	-2.8427	5.7338	0.3993
## [2665]	-2.9177	1.2396	9.6529	-5.4820	6.9184	0.4351	-5.4302	0.8939
## [2673]	-7.1710	-5.3022	4.8163	1.5711	-0.0175	2.4387	4.9868	4.0134
## [2681]	-6.6967	5.2294	0.7391	3.1842	-0.5140	2.3689	6.7245	-0.7603
## [2689]	1.9896	3.4640	-2.6943	-7.9565	-2.4350	4.8706	1.5221	-4.0729
## [2697]	-4.6862	9.8059	1.8507	-4.4905	-8.3812	-9.9612	-3.2499	8.2996
## [2705]	6.4442	-4.5060	-0.5361	4.4679	7.7441	1.7860	0.1336	7.2706
## [2713]	6.7648	-6.3362	2.5190	-7.6558	-4.2693	7.1311	6.1134	-6.6222
## [2721]	6.4286	-8.7490	-6.7005	5.6341	1.9258	-0.5203	8.6428	9.7729
## [2729]	1.3249	2.6886	5.0675	-2.7598	8.2409	-4.6946	1.5816	9.0794
## [2737]	-9.1197	-7.7561	-1.5021	4.6274	5.9005	4.2567	-6.8873	-4.3322
## [2745]	-2.5091	8.7539	-3.4348	-5.4533	-9.8056	-0.1554	5.8047	3.0171
## [2753]	-6.7709	9.7983	-8.2251	4.6155	-5.2733	9.1665	-4.0528	-5.3373
## [2761]	2.2483	-9.1909	-8.8208	6.2619	-7.3680	-8.9657	-4.1473	-8.2047
## [2769]	0.3107	4.5865	7.0689	-0.4448	-4.0236	6.0302	2.4419	8.4819
## [2777]	7.4301	6.3862	-9.9428	2.9778	-1.8047	4.1906	5.8147	1.1515
## [2785]	-1.4706	9.1244	5.4648	0.4178	-1.4942	5.8042	6.0789	-6.5284
## [2793]	1.2848	0.3130	-2.7918	-9.5424	0.7578	1.1034	-3.4479	-8.6332
## [2801]	8.3687	-1.0571	7.8462	-1.8976	3.6746	6.6896	-2.9443	-4.7928
## [2809]	-4.2447	-4.9342	-6.1983	-9.5151	-4.6332	4.6499	-1.9675	7.2370
## [2817]	2.9369	-1.7886	9.6279	8.6119	5.5178	-3.1802	-4.8390	-6.8381
## [2825]	7.4192	1.8124	-8.2898	5.0747	-6.8843	-8.6470	-7.5928	-8.9808
## [2833]	-0.1772	0.3702	-9.3339	3.6908	8.4614	-9.8095	5.3549	-9.1203
## [2841]	7.2631	2.4174	-0.2898	6.2585	7.0858	-2.9675	2.5878	7.7002
## [2849]	6.9055	-4.6064	-5.0912	-1.3239	-4.7451	8.2290	3.8023	-0.6500
## [2857]	-7.2603	0.2064	9.4595	-6.3181	1.8587	0.5343	5.1024	-4.8060
## [2865]	2.3511	-9.2805	9.1817	-6.6820	-3.8222	-1.8862	-1.0073	1.2998
## [2873]	1.0714	7.1450	-1.4665	0.3074	0.5986	-1.0138	-3.4697	7.3133

## [2881]	6.2017	9.8427	-1.3423	-0.6809	-0.4108	7.6120	0.3065	3.7316
## [2889]	-6.4620	-0.6062	3.7258	-9.4525	-2.9450	-6.6673	1.7319	-4.2230
## [2897]	0.7508	-7.3309	2.8559	-7.5446	3.2121	-3.2034	2.3439	6.5885
## [2905]	5.9839	-6.4252	-4.6951	0.9407	-3.2908	-1.6086	3.2130	5.7228
## [2913]	-6.5049	0.3066	7.5906	-1.3354	-1.6584	-4.4091	-0.6333	6.2755
## [2921]	-6.2433	7.4544	7.0622	0.1551	3.0430	9.9893	2.8119	-0.0239
## [2929]	-3.7017	9.9654	-5.3575	-1.0650	-4.7012	-8.8375	9.1171	-8.4578
## [2937]	-0.6783	-6.1650	9.6236	8.6070	0.3864	9.1149	4.1291	3.0908
## [2945]	-5.5701	2.4229	7.5520	-4.0870	-2.8635	-5.0139	3.8997	-5.2718
## [2953]	-8.0368	-6.1600	4.7689	-4.1075	3.3644	-4.6651	-9.7732	4.0287
## [2961]	-5.8302	5.5919	-6.9529	8.9111	6.3415	-9.8566	4.2673	3.7948
## [2969]	-5.8131	-2.4987	-1.6800	-2.1006	5.6502	-8.9935	-8.9131	-0.2623
## [2977]	0.2387	-5.6226	2.2188	2.7798	3.3093	-7.3421	-7.1483	9.9943
## [2985]	-4.9142	-5.6584	3.4209	-2.2221	1.4193	-7.8084	5.9125	-2.5495
## [2993]	6.6952	9.6718	1.1459	-3.0116	9.2697	4.5827	1.2201	-6.5267
## [3001]	2.8364	8.7604	3.3367	3.6047	-7.4567	-5.2640	1.1169	1.8225
## [3009]	-2.5142	-3.0114	1.5816	3.8904	-0.1271	-5.8171	-3.1648	3.5679
## [3017]	-7.5850	8.8662	-6.6762	7.9333	-9.6620	2.3963	2.1899	2.6138
## [3025]	-1.2417	3.2661	4.7863	6.3102	2.1924	3.0045	-4.9260	4.5555
## [3033]	-4.4006	4.7640	-8.0655	1.8019	-5.6052	5.2875	9.8555	0.4612
## [3041]	4.9856	6.8812	-7.8402	-3.3153	-9.7307	1.5364	-4.5917	0.2590
## [3049]	3.7346	-7.1725	2.2429	7.6940	-4.6603	-7.5494	-5.0522	8.0815
## [3057]	2.2085	-7.1469	4.1366	-6.7394	9.9641	6.1070	-1.5789	-0.6515
## [3065]	-2.2015	-9.3499	3.2072	8.6727	-4.8387	-9.8717	6.4424	-5.1139
## [3073]	-8.0328	-8.7157	1.1126	-4.2994	-9.4724	0.7066	4.2911	-2.6127
## [3081]	-1.7725	-4.3731	8.7584	-1.3011	-7.1462	2.1220	-7.7312	-6.0787
## [3089]	8.1626	-9.6200	0.9582	-6.4547	-9.2111	-9.4314	5.1830	4.7479
## [3097]	-5.9275	9.3854	3.1272	-7.4798	7.0573	-6.2408	-0.2178	-6.1872
## [3105]	4.5158	5.6953	8.9221	0.2501	-4.3859	3.7512	-6.2305	5.5957
## [3113]	4.6205	1.9574	1.4428	3.4573	1.7292	-7.5499	3.3266	-0.4927
## [3121]	1.6859	-4.6217	4.0417	-8.0805	-7.8014	-7.5759	3.3681	6.5155
## [3129]	9.5148	-8.0659	-0.0671	-3.1403	-3.2379	6.7418	4.0635	2.5804
## [3137]	8.0165	4.8582	-0.3398	2.5964	4.0486	-1.1835	-3.5120	-9.7217
## [3145]	-4.8811	-4.5737	-3.6423	-8.3189	8.1713	-2.9463	8.7768	9.6003
## [3153]	-7.4223	1.0994	1.4760	-5.2795	3.4200	6.2583	-8.4355	3.7684
## [3161]	1.9853	-0.6404	-3.2651	-1.8007	-9.2822	-1.1567	5.3070	-0.5694
## [3169]	-4.0135	4.6443	-8.6660	1.0158	9.6661	-4.6959	5.7788	5.9593
## [3177]	7.5827	-5.1508	1.1906	8.0955	-9.8360	-6.7300	-4.8034	5.4553
## [3185]	-8.5783	7.9153	-3.7008	7.7836	2.1752	-0.8930	4.0315	5.1119
## [3193]	0.5562	-9.6745	-8.4655	3.4313	4.3757	-0.8703	-5.5309	4.5138
## [3201]	-9.6487	0.8661	-8.3734	8.0461	5.7106	-1.3550	2.2810	5.3689
## [3209]	-5.6981	-0.8438	-9.9981	4.8802	4.5192	-7.8178	4.2969	4.5586
## [3217]	-1.4605	-3.0316	-5.4938	3.0702	-8.4561	-6.1860	-4.3478	4.1193
## [3225]	-2.9100	9.9722	-6.5009	8.7657	-7.0992	6.7310	7.1406	-4.6409
## [3233]	-4.0157	2.5471	2.6934	-1.2581	2.0460	-9.5287	3.3940	-1.8980
## [3241]	-3.7702	-2.4841	4.3352	-0.7142	1.1419	-4.0528	7.7136	4.8549
## [3249]	1.1372	9.4496	-5.1958	7.5218	8.9621	4.9086	-2.1643	0.4392
## [3257]	1.4693	0.6037	8.9463	7.5294	2.4623	8.0003	5.0938	-4.4611
## [3265]	-9.8456	6.5779	-7.8071	-7.6323	-6.8633	-7.6700	2.1138	8.8946
## [3273]	4.7266	0.2947	-8.3723	0.2695	5.8035	6.7243	9.9945	-1.5590
## [3281]	6.9485	-6.0282	-6.3202	1.1004	-7.6895	-9.1248	-4.7595	-3.6341
## [3289]	8.8925	-1.2455	-0.9221	-8.5086	-2.5296	1.4659	7.1656	-5.3628

## [3297]	9.1453	7.3475	-5.5322	-2.2132	2.3094	-7.2473	-4.5721	1.8907
## [3305]	-6.7841	-0.4244	6.9881	0.3216	-2.5459	1.7059	4.5261	-3.0643
## [3313]	-6.7780	-2.5602	2.0101	4.8302	-9.6647	8.9621	-7.3870	-9.4586
## [3321]	6.8847	-1.0793	-4.9688	6.7276	-0.4951	-7.6169	-7.5794	7.5671
## [3329]	-8.9756	-2.9548	-1.7087	5.2090	3.9060	8.5946	-6.7139	6.3112
## [3337]	1.0830	7.7537	8.7751	-9.7274	-1.6838	-7.9711	-0.7051	4.3907
## [3345]	-0.7596	-6.3759	1.5578	-4.1897	4.6542	8.3371	-4.8222	0.9274
## [3353]	5.2995	0.3615	-5.6262	-1.9400	1.2008	-8.8993	5.3205	-9.3997
## [3361]	-7.3770	-1.7239	2.2385	2.7721	2.8748	-8.1281	-1.5817	-9.2936
## [3369]	-3.6694	7.9787	-5.2152	3.8784	7.8495	3.7026	-7.8180	6.4091
## [3377]	4.0002	4.2093	-0.9672	-7.0880	-7.7265	7.8876	-6.1096	-7.9852
## [3385]	-6.5126	-3.6777	3.4836	2.6132	-2.1735	2.2129	7.7189	4.4436
## [3393]	-0.1298	6.1929	-3.8805	-5.8187	-4.1033	-8.0205	-6.2076	-7.5709
## [3401]	-0.4927	4.3436	-5.9916	-3.5646	-0.4277	3.2033	3.1786	7.6318
## [3409]	-9.2684	-4.9825	-3.3996	6.4488	2.4423	-4.8100	-0.2324	4.6264
## [3417]	-7.7396	0.2718	3.1316	-6.0492	-0.2616	6.5679	7.1846	5.3008
## [3425]	2.9566	-6.1494	-6.1398	0.3007	5.1069	9.8834	-2.9636	-5.4787
## [3433]	4.2864	-1.3240	8.4432	8.3935	-2.4473	-7.3459	5.1731	9.8271
## [3441]	-9.1379	-4.9542	-9.9240	-3.9793	-9.2673	-0.3075	0.9559	7.1748
## [3449]	-9.9012	9.9879	-9.9150	3.7113	9.7365	4.6449	2.5436	-8.3041
## [3457]	2.1264	1.1232	-0.4819	-9.8171	8.0195	-2.0380	-1.8857	8.5610
## [3465]	5.5029	-1.5701	0.8258	-0.4597	-4.4078	0.3115	-6.2450	-0.9847
## [3473]	-9.2385	0.7970	-0.5950	-8.6927	-8.8652	-3.6939	-3.6657	-2.6734
## [3481]	-7.6281	4.3117	0.7923	1.6428	8.0458	-5.5190	6.6817	6.0413
## [3489]	-6.1454	7.8002	-9.6652	-7.9925	-1.6275	1.4600	-7.3156	-5.4616
## [3497]	-4.1694	-4.5691	2.5021	3.2541	-0.6777	6.0013	9.7664	2.9617
## [3505]	0.1822	-4.0724	7.1444	7.4140	-6.9191	6.9788	-1.7493	-6.7752
## [3513]	5.7142	2.4585	5.0155	-7.5381	-8.2086	8.5755	-4.6201	2.1289
## [3521]	-0.0588	7.1409	-1.3297	3.0209	-4.1366	-0.8622	-7.7037	5.4704
## [3529]	8.6302	-7.3011	5.6037	-8.4805	7.3041	-4.7206	0.2868	2.0793
## [3537]	7.2275	-3.2090	3.3085	-3.5815	-0.8698	6.4287	-5.0974	-3.8447
## [3545]	-7.8572	6.7017	2.3857	-5.6044	6.7513	-2.0242	-1.9511	-5.9137
## [3553]	-6.3521	-3.3004	5.9835	-7.5464	-7.1454	3.4512	-1.5851	7.6292
## [3561]	-0.5663	2.2310	6.3545	9.1315	-9.2038	-8.9902	4.3594	-5.4768
## [3569]	-6.7095	4.0017	-2.6581	4.7351	0.1100	2.5633	-0.0612	9.1913
## [3577]	-1.1167	9.0614	-6.7343	-8.5552	2.7804	-3.2362	2.4865	-8.4844
## [3585]	-3.7493	-6.9735	4.3756	1.9606	8.6325	2.7472	2.6142	-5.6487
## [3593]	5.8385	2.3704	3.4837	-5.3329	5.9124	1.9696	3.5812	0.9008
## [3601]	-2.0517	-2.2360	2.6954	-6.6299	5.3781	-3.5830	-5.9297	3.7668
## [3609]	1.7834	-3.7882	7.6761	-2.5520	-4.8206	-7.1527	-0.1254	-3.3579
## [3617]	-1.0584	-0.1983	8.1307	-5.3406	-5.9542	-2.9807	1.0163	-7.2897
## [3625]	-2.0641	5.4624	-4.7511	9.7288	9.7035	-3.2933	2.6235	3.5926
## [3633]	7.1602	8.8634	4.0985	-2.9935	6.4584	-3.0285	9.5943	4.8255
## [3641]	0.4907	-6.5908	6.4423	-1.3746	8.1582	-4.1985	8.2243	-6.6687
## [3649]	9.9471	3.2602	2.2608	-1.1040	0.4765	9.6996	2.0191	-9.9553
## [3657]	7.1538	4.8126	6.7541	-5.8854	2.4400	-2.2530	3.5981	-7.2376
## [3665]	6.7114	4.3701	-8.8042	8.2472	8.0991	3.7524	-6.4600	-6.3816
## [3673]	1.7679	8.9358	9.1852	-1.3005	1.8425	1.4186	-5.0115	-0.3927
## [3681]	-3.1947	-1.7086	-0.4206	0.6454	-6.8874	-6.9725	-0.1971	7.5118
## [3689]	-0.6517	4.8193	-1.7305	0.6283	3.0184	6.5555	-0.5613	-7.7835
## [3697]	-3.6192	7.2214	-3.6593	1.6789	-4.2149	3.8519	-4.2325	-3.5140
## [3705]	-3.9573	-5.6596	-5.0514	-4.1507	-6.2783	6.9992	2.0894	7.2463

## [3713]	0.5351	8.8002	8.2013	-0.4943	-6.7772	-5.5038	5.0442	8.5910
## [3721]	-4.8041	-8.7932	2.7196	-1.2624	-9.3555	4.5366	-5.8544	-7.5488
## [3729]	7.9561	-8.8126	6.2511	3.7529	0.5773	8.9499	-0.7205	-5.2523
## [3737]	-4.0057	-5.0772	-0.7426	-6.5360	-5.1713	8.6238	-4.5176	-5.2101
## [3745]	9.2174	-5.9766	5.8590	-5.7862	9.9137	-4.4340	6.3677	-4.4997
## [3753]	-1.4984	4.5578	2.8476	2.3714	1.8709	3.6503	7.9597	-4.7144
## [3761]	-2.6260	5.3125	9.5239	-9.9766	-5.2784	7.4055	-1.2333	1.5840
## [3769]	-0.9996	-3.1460	-1.7345	3.6559	9.2460	-7.7872	-4.8475	-4.7308
## [3777]	4.3267	0.8693	-9.0136	-8.8184	-7.2580	-0.3210	5.8896	-0.2898
## [3785]	-6.9333	-2.1970	0.2185	-8.0403	-6.2458	7.5671	-2.9054	-8.6256
## [3793]	-6.9625	-8.6036	-0.4423	-4.3986	0.8027	8.7869	-2.9505	0.8512
## [3801]	-9.5141	2.0480	-3.3186	-1.7100	-9.6110	2.1349	-5.6356	4.7339
## [3809]	3.1820	5.2073	-4.3922	8.2953	-7.6199	-5.6346	5.5210	-1.2563
## [3817]	-8.5407	3.2079	-6.8277	5.9067	2.6750	-2.6088	-9.3868	4.8481
## [3825]	4.4496	-4.1662	2.2658	-2.6726	5.4283	6.8853	6.8487	-9.4027
## [3833]	-8.0681	1.7533	-1.1427	6.4552	-9.3776	-2.6974	7.3959	4.6793
## [3841]	8.7369	-5.2012	8.2112	1.3594	-0.2238	0.0780	-1.3779	-0.2281
## [3849]	-2.0591	-7.9797	6.8197	-1.2916	-4.6374	-5.8433	-6.0737	-2.8086
## [3857]	3.3257	6.9428	-6.0872	-4.2901	-1.3397	-0.2611	5.1490	0.8206
## [3865]	9.3208	-1.5027	-6.5948	5.7981	-6.3092	-0.3386	-4.5969	1.2310
## [3873]	9.4641	-6.2669	4.5392	-2.9773	3.2067	-4.8237	-5.9847	3.6470
## [3881]	-8.2149	7.5049	4.2227	-3.5494	9.4743	-9.7326	-4.5249	-6.0218
## [3889]	-4.4971	-6.4984	7.6036	7.6230	9.3678	1.1370	-9.7483	8.8236
## [3897]	0.0596	3.1151	0.3175	-8.7827	4.6351	3.7934	4.1906	-5.4545
## [3905]	-2.8575	0.0048	9.0582	-0.8097	-8.5910	-1.1455	-0.8264	-2.5963
## [3913]	-1.6395	9.4529	-9.6268	8.6892	9.9961	-6.2615	7.4719	1.5749
## [3921]	6.9836	7.5042	0.5614	7.2977	4.1844	-6.2942	-5.4377	8.4140
## [3929]	-8.0446	-5.1894	8.1444	1.0078	5.1262	1.3401	1.1209	-9.4514
## [3937]	0.9216	-1.0605	-2.2745	-9.6431	5.5860	-9.5074	1.0846	2.5142
## [3945]	-2.0689	2.6708	4.3892	6.7011	-3.4122	2.9572	1.8325	-4.9954
## [3953]	5.0558	-5.7671	1.5456	5.7892	-8.8235	8.5323	-0.5634	5.7559
## [3961]	-0.1259	1.7173	-3.3691	-0.5501	2.4995	8.9656	5.2210	-9.8729
## [3969]	7.2532	6.7925	-2.4592	-1.3318	-2.7681	-7.4030	-6.7782	5.9722
## [3977]	6.5957	-6.7305	2.2354	2.6916	-5.9867	1.4719	5.9540	1.1229
## [3985]	-5.5408	-9.3599	3.8185	1.2902	8.6340	-3.7811	-0.8250	-5.1843
## [3993]	6.2907	-7.6344	-3.9735	-7.4517	6.3170	-6.0223	1.8898	7.8760
## [4001]	-6.1161	5.1305	-1.5968	2.1656	7.4595	9.1393	-5.5966	8.9306
## [4009]	-1.7625	6.3345	-5.5836	-7.6337	2.9234	8.4817	3.6723	-1.6018
## [4017]	0.4958	-1.9174	-2.8223	1.8776	-9.4213	0.5018	-1.8847	-3.7861
## [4025]	7.5006	5.0657	-5.3313	-3.7022	9.4788	-4.2779	4.7162	-2.0073
## [4033]	-1.8797	-5.5256	8.7452	1.2214	8.0413	-0.7747	-2.3391	-2.3269
## [4041]	4.6202	3.5720	7.4679	-9.3197	-6.1358	1.9930	2.4732	6.3254
## [4049]	-0.4747	6.2138	-6.3690	9.7437	9.5195	-4.7772	0.8596	-6.7297
## [4057]	-3.0563	2.5565	5.1722	-9.0806	7.8642	7.7399	5.8870	-7.0495
## [4065]	7.4438	3.3901	-9.9697	-2.6449	0.8313	-9.8266	-1.8326	2.0463
## [4073]	-5.3755	-7.2997	6.6386	6.4363	-3.4103	-9.8577	-5.5593	-1.0423
## [4081]	0.5115	-7.0784	1.0180	7.4033	9.0291	-5.3329	-5.8031	6.5039
## [4089]	7.9114	-3.6042	2.6607	1.0329	-5.6834	8.5636	4.8795	-5.0487
## [4097]	-1.2478	9.3103	4.9904	6.7594	7.8628	8.0107	7.1298	-9.7532
## [4105]	6.6499	-1.3382	-7.1682	-8.5752	-8.8606	5.5231	3.8289	7.7859
## [4113]	-7.9605	-4.6312	-2.1755	-3.9119	5.4931	-8.9511	9.1220	5.7102
## [4121]	3.6315	8.5613	-1.6624	-9.1086	7.0539	-9.3009	-8.6714	7.4743

## [4129]	-4.1703	8.2252	-6.4301	2.8978	0.9255	1.9141	-4.0560	-4.6557
## [4137]	-6.9374	2.6542	-6.5099	-3.4883	-3.5206	9.5897	-1.3621	-3.4681
## [4145]	5.8328	-9.9830	-2.3373	3.8290	5.6532	-7.9130	8.6875	5.8936
## [4153]	-1.1282	-8.7792	4.4843	-2.0420	0.9221	3.0376	3.6234	-1.5638
## [4161]	6.6269	-9.4846	9.7234	-8.4660	-3.4611	-6.0491	7.0865	8.9572
## [4169]	-7.6288	-1.0077	-0.6730	7.9790	-7.7293	1.4245	8.0117	-3.7882
## [4177]	-3.5096	7.4051	-3.4265	-7.4860	-2.8756	8.6140	7.5036	6.4043
## [4185]	-9.5646	2.6247	5.6183	4.3449	-6.2843	1.6135	-1.7242	-3.4515
## [4193]	-1.5835	-2.6880	7.4141	-4.5405	-2.1336	7.8453	6.4612	4.5116
## [4201]	-1.2034	-0.8659	-0.3913	-7.4524	-4.2783	2.5092	3.7659	-6.8585
## [4209]	6.0018	-6.0557	0.7860	9.5853	-1.6595	-8.3728	-6.6036	-6.2276
## [4217]	-9.4201	-4.3863	-2.7206	-4.4866	-7.6131	-7.3963	7.4423	-1.4891
## [4225]	-7.9952	-7.0075	7.8159	8.2055	6.8466	0.9830	-4.7025	6.0919
## [4233]	-8.6714	-8.4011	-4.4340	-0.1153	7.1207	4.1464	-1.6843	0.7816
## [4241]	-7.1347	0.1404	-7.7493	-6.9899	7.9459	-4.7047	5.0076	-3.1445
## [4249]	2.7131	-5.7766	4.4423	5.5502	9.1940	9.2153	-0.6784	8.1155
## [4257]	3.4855	3.2759	3.0221	-0.9418	0.2826	4.3539	-1.9329	-5.6966
## [4265]	-1.7540	2.0893	-0.6346	1.3221	-1.9589	-9.8097	-6.0554	9.4212
## [4273]	4.0716	-8.1700	3.5577	-9.9136	-4.8184	-0.0027	-4.2815	0.8554
## [4281]	2.8025	-0.0639	5.0142	1.1179	7.3644	-7.6395	6.2983	4.4648
## [4289]	9.3529	8.3262	-7.6912	-5.9838	3.1126	2.4030	9.6618	-2.1353
## [4297]	9.3487	-0.4847	-3.8043	6.7079	1.9172	2.1881	2.8466	9.4662
## [4305]	0.2702	-2.6033	-6.4321	8.6164	-3.2751	-3.5136	7.5671	-1.8294
## [4313]	9.5880	-2.5642	2.3223	2.2658	-7.8633	3.5917	-9.9791	1.1542
## [4321]	0.7890	4.5104	-3.2410	-5.6575	-5.9014	-1.8880	-9.0924	4.8569
## [4329]	8.0651	7.0077	-4.2096	2.6489	-6.4681	5.2192	6.4790	5.9913
## [4337]	5.3926	4.5768	2.6816	-1.7702	-1.7248	-1.3054	-2.0122	4.6605
## [4345]	2.6793	-3.5117	-3.2201	-1.0500	-4.0954	-0.7315	8.5880	-8.8744
## [4353]	-7.1382	2.6176	-6.8382	4.3133	-7.0331	-0.4872	-4.4003	9.2720
## [4361]	4.3242	-1.3954	-2.3718	4.3728	-5.0071	4.0569	-2.6545	5.6736
## [4369]	3.4294	3.2384	-4.1828	-1.5169	7.3329	4.2681	-3.8908	2.0014
## [4377]	-2.4847	1.9240	2.5587	5.2822	-6.0025	5.6032	-5.6800	-0.0680
## [4385]	3.5844	-3.0278	-9.2001	8.1590	-8.1347	0.4228	6.5123	-2.9328
## [4393]	0.9718	-6.5597	-4.6794	2.1283	2.6490	3.0136	8.2949	-7.4490
## [4401]	5.7555	-8.2246	2.1178	1.4614	2.6822	-5.8716	-0.5036	8.2254
## [4409]	-6.7384	-7.3179	5.0187	-8.7859	-1.0182	-0.5569	-8.4011	3.6022
## [4417]	-4.5195	1.2466	-2.9975	-0.9620	5.2703	2.9516	7.4031	-5.5774
## [4425]	-7.1843	1.7957	4.5085	-5.9554	1.9223	-5.7883	-5.0404	0.8499
## [4433]	-5.3321	3.1927	-7.7786	-6.1522	6.2139	-8.7909	3.6425	-6.9527
## [4441]	-7.0071	-9.5452	-2.8052	1.0266	-6.9897	4.4130	0.8492	9.2844
## [4449]	8.9233	-6.2907	-0.2911	-3.6346	6.5255	-0.8641	9.3936	1.8780
## [4457]	9.7765	-5.7533	-8.1039	6.4728	-4.3438	-2.9582	3.2954	6.7546
## [4465]	3.8887	-8.9673	-0.1311	-1.2888	8.0825	2.7049	-4.6014	-3.7742
## [4473]	-0.7168	-3.6784	4.7081	-4.7635	6.2488	0.8237	9.2603	7.9316
## [4481]	-0.5608	-6.9431	-5.0117	-0.7886	1.6784	-3.7060	-5.6973	8.4240
## [4489]	6.7465	9.4611	-7.2746	3.8344	9.7370	0.5287	6.3242	8.2814
## [4497]	-3.1212	-3.5564	-6.1297	-0.4531	6.3483	7.0639	8.2147	0.8419
## [4505]	-9.7489	-3.6551	9.4291	1.4676	-4.0584	-3.4558	-7.2595	9.3850
## [4513]	0.7841	-2.9845	-3.0420	6.5317	8.0300	0.0091	4.9173	8.1660
## [4521]	-5.0433	6.4800	5.3482	-4.8350	-2.0379	2.2210	-0.3621	-1.1593
## [4529]	0.9345	3.8488	-0.6264	-0.3698	-7.1562	3.6188	9.1204	-0.9312
## [4537]	-5.1966	6.6075	-6.2670	2.4585	-8.4196	-8.1178	3.5672	-0.1654

## [4545]	-6.3251	-2.6363	-3.5534	7.3576	1.0733	-7.5054	-2.9667	-9.5914
## [4553]	-1.7824	3.5998	1.0538	6.5379	5.5198	3.4939	9.5173	8.2387
## [4561]	-6.7278	6.3292	7.1934	-6.4779	5.9492	6.4790	-9.9593	-4.4399
## [4569]	-0.4719	7.3088	4.1192	6.7923	9.0263	-0.3971	-8.3876	7.1445
## [4577]	-1.6378	-2.1122	-9.6361	6.8799	3.7342	-9.7891	7.7460	-7.0692
## [4585]	0.8151	1.1731	5.2237	-3.9041	-6.4317	-9.9237	7.2725	-5.9641
## [4593]	-0.3513	8.9346	1.5507	-7.7325	0.5280	1.3258	3.5864	-3.7862
## [4601]	3.6699	0.1461	5.9634	-1.4078	5.8376	-1.0811	8.8079	-0.9750
## [4609]	5.3572	-9.7052	-0.2055	0.0045	-9.4407	1.0213	0.8302	3.1991
## [4617]	-8.7228	8.2845	7.1256	0.2932	-7.1267	2.1343	-3.8434	-8.4661
## [4625]	6.5782	0.7932	-6.5949	-8.8016	4.3140	9.5434	-7.8251	-2.7893
## [4633]	-8.3804	2.7528	5.1238	3.4103	3.9548	-8.0895	0.6811	3.5941
## [4641]	-6.2163	-7.1668	2.7042	3.5341	7.4173	-4.6396	-1.2447	-5.0346
## [4649]	5.6443	3.2128	6.0328	2.8595	-0.1988	-1.9657	1.7472	-5.3847
## [4657]	-3.2570	1.5383	-4.9685	-0.3888	-2.2334	6.6269	-9.6550	-4.7267
## [4665]	1.7396	-0.9795	7.7542	9.7684	4.0250	-8.0083	8.5658	2.4895
## [4673]	-0.0892	3.1867	-9.4963	-3.4649	-3.9125	-5.2209	-4.3103	-8.5494
## [4681]	6.6880	-8.4369	4.6587	-2.0824	9.0822	7.8972	-2.2938	0.4283
## [4689]	5.9012	8.7783	-2.6721	-4.5298	-4.3096	4.3714	3.1797	0.9900
## [4697]	-5.9673	2.8313	7.1561	1.6274	2.7774	5.6327	-1.1369	-0.0617
## [4705]	-5.3499	6.6721	-6.8711	3.7600	-9.8685	-0.0761	4.3653	-0.8982
## [4713]	-2.7000	-7.9828	-4.5734	-4.8340	6.8039	-2.0769	9.8085	6.5054
## [4721]	-0.2393	5.7124	1.5351	-5.7992	6.1141	-3.6616	8.9416	9.1764
## [4729]	1.8639	-8.4220	-0.5419	1.4234	0.7707	2.1872	1.4292	-3.8412
## [4737]	2.7761	-8.3009	-5.2183	-3.0606	-2.0180	9.9282	-9.1495	-0.8741
## [4745]	-1.6844	-1.7656	-3.9009	1.4235	5.8469	-7.5752	-0.4020	5.4747
## [4753]	-6.5701	5.2810	-4.0564	-2.7227	-2.6623	0.4731	-7.2199	-3.8647
## [4761]	6.7834	5.0974	1.8780	-9.0284	3.4474	-1.7588	-8.8443	4.6699
## [4769]	-7.8009	9.0846	-4.1436	-6.3639	7.9104	-7.3670	-2.8989	-0.8617
## [4777]	7.4804	-0.4587	3.8238	5.5157	-1.4198	-0.9107	-5.1848	-4.3404
## [4785]	8.2537	9.4123	4.5117	-2.5796	-0.1935	6.2162	1.1813	5.4338
## [4793]	-4.6500	-3.3988	9.3788	9.0357	4.4300	-7.5224	6.3543	-6.5162
## [4801]	-6.8076	7.5684	-8.7964	-8.8064	9.5085	4.7611	0.5624	1.5652
## [4809]	-4.1525	5.4038	-0.5625	5.6545	-3.6938	-4.9430	6.9587	-5.0629
## [4817]	5.2173	5.8452	9.7776	6.2567	-8.2202	-7.3458	-2.6481	-7.8201
## [4825]	3.7854	-7.1747	9.0320	-0.1179	-6.9128	-0.1680	0.1623	9.8007
## [4833]	-3.0646	4.0913	-8.8441	-1.1997	-7.1717	-4.0246	-7.5972	-4.0739
## [4841]	-9.2730	-4.6466	1.8595	-7.2611	1.2935	1.3874	5.3468	-9.9857
## [4849]	7.9106	-7.5778	-9.4804	6.7859	-3.2318	1.1860	-7.3271	4.2085
## [4857]	1.2422	-4.4460	6.7465	8.4133	2.9399	-2.3387	-0.6554	4.4303
## [4865]	-2.8149	0.1091	0.3740	3.6813	6.0654	-3.4112	6.3636	7.8650
## [4873]	3.0639	-3.9529	7.3597	9.9576	0.4033	-1.2848	8.5802	1.0037
## [4881]	-6.8357	-4.6484	2.1560	5.3646	-9.3030	1.4981	3.6017	4.7479
## [4889]	-9.8245	5.2426	-2.2446	-5.5804	0.6269	5.9112	3.8657	-7.6591
## [4897]	-4.4815	1.2189	-4.2680	6.3856	0.3847	-8.8849	-4.3238	-3.4108
## [4905]	3.6622	-3.1302	4.1349	-9.6082	-7.7987	-1.6052	-2.5347	8.4075
## [4913]	7.8507	-8.4069	-3.9106	-7.2843	-4.6828	-3.0574	6.6721	-0.4742
## [4921]	-7.8680	0.8356	-3.1387	1.4990	-4.8888	-2.1914	-8.6762	9.8169
## [4929]	-2.0470	-3.4558	1.6462	-8.1824	-4.9832	-8.8237	-0.7948	-1.5811
## [4937]	3.4189	-4.1058	1.6717	7.0608	-4.1942	-1.4849	-5.9005	0.6313
## [4945]	3.0218	-8.9426	6.5287	-1.8962	-1.4221	-2.1429	-7.0347	-8.3939
## [4953]	-9.8078	-6.3873	-7.2119	-6.1744	-5.0862	5.9187	7.7334	4.2317

```
## [4961]  1.4307  5.1258  2.4805  8.4022  0.3796 -2.5209 -5.6144  7.6321
## [4969]  3.8603 -2.0956 -6.0216  7.5239  4.9906 -7.8981 -6.5136 -4.5429
## [4977]  5.6459  6.3898  8.8213  9.0257 -7.7891 -0.3101 -7.1573 -2.5958
## [4985]  6.4785  4.6873  9.2827 -0.9603 -4.0389  0.1087 -6.6382  1.9912
## [4993]  3.7553  7.2407  8.2021 -7.1338 -4.3160 -3.4097  0.9835  4.5171
```

```
head(nx,10)
```

```
## [1]  4.77  3.08 -3.91 -9.57 -2.36 -4.94  0.65  3.36 -4.16 -0.58
```

```
tail(nx,10)
```

```
## [1] -6.64  1.99  3.76  7.24  8.20 -7.13 -4.32 -3.41  0.98  4.52
```

```
nx_red <- round(nx,3)
```

```
head(nx,15)
```

```
## [1]  4.77  3.08 -3.91 -9.57 -2.36 -4.94  0.65  3.36 -4.16 -0.58 -2.41 -8.00
```

```
## [13] -6.44  6.90 -0.55
```

```
tail(nx,15)
```

```
## [1]  4.69  9.28 -0.96 -4.04  0.11 -6.64  1.99  3.76  7.24  8.20 -7.13 -4.32
```

```
## [13] -3.41  0.98  4.52
```

```
muestra1 <- sample(nx_red,500);
```

```
head(muestra1,10)
```

```
## [1] -5.47  1.16  9.13 -3.36 -1.69 -0.46  0.36  4.34  4.32 -2.60
```

```
tail(muestra1,10)
```

```
## [1] -6.33  0.54 -8.39  7.59  1.88  5.95  2.94 -6.44 -2.82  3.02
```

Convertimos todos los números a positivos

```
muestra2 <- abs(muestra1)
```

```
head(muestra2,15)
```

```
## [1]  5.47  1.16  9.13  3.36  1.69  0.46  0.36  4.34  4.32  2.60  1.01  7.18  0.24  4.70  8.32
```

```
tail(muestra2,15)
```

```
## [1]  6.50  1.52  4.45  9.27  9.07  6.33  0.54  8.39  7.59  1.88  5.95  2.94  6.44  2.82  3.02
```

```
nx_red <- round(muestra1,0)
```

```
muestra3 <- abs(nx_red)
```

```
head(muestra3,15)
```

```
## [1]  5  1  9  3  2  0  0  4  4  3  1  7  0  5  8
```

```
tail(muestra3,15)
```

```
## [1]  6  2  4  9  9  6  1  8  8  2  6  3  6  3  3
```

2.6.4. Sesión 4

```
soluciones <- function(a,b,c){
```

```
  d<- (b^2)-(4*a*c)
```

```
  if(d<0){
```

```

    resp<- "No hay soluciones"
  }
  else{
    resp<-"Sí hay soluciones"
  }
  respuesta<- cat(resp,"",d)
  return(respuesta)
}
intento1<- soluciones(5,2,3)

## No hay soluciones -56
valores<- c(15,3,-2,13,27,10,8,-4,11)
A<- matrix(valores,nrow = 3)
valoresb<- c(-3,2,-1)
b<- matrix(valoresb,nrow=3);(b)

##      [,1]
## [1,]  -3
## [2,]   2
## [3,]  -1

Ab<- cbind(A,b);(Ab)

##      [,1] [,2] [,3] [,4]
## [1,]  15  13   8  -3
## [2,]   3  27  -4   2
## [3,]  -2  10  11  -1

bA<- cbind(b,A);(bA)

##      [,1] [,2] [,3] [,4]
## [1,]  -3  15  13   8
## [2,]   2   3  27  -4
## [3,]  -1  -2  10  11

Abt<- rbind(A,t(b))
btA<- rbind(t(b),A)
(Abt)

##      [,1] [,2] [,3]
## [1,]  15  13   8
## [2,]   3  27  -4
## [3,]  -2  10  11
## [4,]  -3   2  -1

(btA)

##      [,1] [,2] [,3]
## [1,]  -3   2  -1
## [2,]  15  13   8
## [3,]   3  27  -4
## [4,]  -2  10  11

valores2<- c(15,13,8,3,27,-4,-2,10,4)
tA<- matrix(t(valores),nrow=3)

```



```
A2<- matrix(valores2,nrow = 3)
(tA)
```

```
##      [,1] [,2] [,3]
## [1,]  15  13   8
## [2,]   3  27  -4
## [3,]  -2  10  11
```

```
(A2)
```

```
##      [,1] [,2] [,3]
## [1,]  15   3  -2
## [2,]  13  27  10
## [3,]   8  -4   4
```

```
r1<- Ab[1,]
r2<- Ab[2,]
r3<- Ab[3,]
c1<- Ab[,1]
c2<- Ab[,2]
c3<- Ab[,3]
a11<-Ab[1,1]
a22<-Ab[2,2]
a32<-Ab[3,2]
a14<-Ab[1,2]
a24<-Ab[2,4]
```

```
m1<- -Ab[2,1]/Ab[1,1]
(m1)
```

```
## [1] -0.2
```

```
m2<- -Ab[3,1]/Ab[1,1]
(m2)
```

```
## [1] 0.13
```

```
A1<- Ab[1,]
(A1)
```

```
## [1] 15 13 8 -3
```

```
A2<- m1*Ab[1,]+Ab[2,]
(A2)
```

```
## [1] 0.0 24.4 -5.6 2.6
```

```
A3<-m2*Ab[1,]+Ab[3,]
(A3)
```

```
## [1] 0.0 11.7 12.1 -1.4
```

```
NuevaA<- rbind(A1,A2,A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1    15   13  8.0 -3.0
## A2     0   24 -5.6 2.6
```

```
## A3      0    12 12.1 -1.4
```

```
Ab<- NuevaA
m3<--Ab[3,2]/Ab[2,2]
(m3)
```

```
##      A3
## -0.48
A3<- m3*Ab[2,]+Ab[3,]
(A3)
```

```
## [1]  0.0  0.0 14.8 -2.7
```

```
NuevaA<- rbind(A1,A2,A3)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4]
## A1      15   13  8.0 -3.0
## A2       0   24 -5.6  2.6
## A3       0    0 14.8 -2.7
```

2.6.5. Sesión 5 y Tareita 1

Eliminación Gaussiana con pivoteo parcial

El sistema a resolver es:

$$5x+2y-3z+4w=-1$$

$$-x+y+z-3w=2$$

$$6x+6y-4z+w=0$$

$$x-y-z-w=2$$

```
valores<- c(5,-1,6,1,2,1,6,-1,-3,1,-4,-1,4,-3,1,-1,-1,2,0,2)
A<- matrix(valores,nrow = 4)
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    5    2   -3    4   -1
## [2,]   -1    1    1   -3    2
## [3,]    6    6   -4    1    0
## [4,]    1   -1   -1   -1    2
```

Primera eliminación

Cambiar columnas

```
pivoteo<- A[c(3,2,1,4), ]
(pivoteo)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    6    6   -4    1    0
## [2,]   -1    1    1   -3    2
## [3,]    5    2   -3    4   -1
## [4,]    1   -1   -1   -1    2
```

Volver ceros

```

m1<- -A[2,1]/A[1,1]
m2<- -A[3,1]/A[1,1]
m3<- -A[4,1]/A[1,1]
R1<- pivoteo[1,]
R2<- m1*A[1,]+A[2,]
R3<- m2*A[1,]+A[3,]
R4<- m3*A[1,]+A[4,]
NuevaA<- rbind(R1,R2,R3,R4)
(NuevaA)

```

```

##      [,1] [,2] [,3] [,4] [,5]
## R1      6  6.0 -4.0  1.0  0.0
## R2      0  1.4  0.4 -2.2  1.8
## R3      0  3.6 -0.4 -3.8  1.2
## R4      0 -1.4 -0.4 -1.8  2.2

```

Segunda eliminación

Cambiar columnas Segunda vez

```

pivoteo<- NuevaA[c(1,3,2,4), ]
(pivoteo)

```

```

##      [,1] [,2] [,3] [,4] [,5]
## R1      6  6.0 -4.0  1.0  0.0
## R3      0  3.6 -0.4 -3.8  1.2
## R2      0  1.4  0.4 -2.2  1.8
## R4      0 -1.4 -0.4 -1.8  2.2

```

Volver ceros

```

m4<- -pivoteo[3,2]/pivoteo[2,2]
m5<- -pivoteo[4,2]/pivoteo[2,2]
R1<- pivoteo[1,]
R2<- pivoteo[2,]
R3<- m4*pivoteo[2,]+pivoteo[3,]
R4<- m5*pivoteo[2,]+pivoteo[4,]
NuevaA<- rbind(R1,R2,R3,R4)
(NuevaA)

```

```

##      [,1] [,2] [,3] [,4] [,5]
## R1      6  6.0 -4.00  1.00  0.0
## R2      0  3.6 -0.40 -3.80  1.2
## R3      0  0.0  0.56 -0.72  1.3
## R4      0  0.0 -0.56 -3.28  2.7

```

Tercera eliminación

Cambiar columnas tercera vez

No es necesario cambiar las columnas porque el máximo valor absoluto de las columnas, ya se encuentra acomodado

Volver ceros

```

m6<- -NuevaA[4,3]/NuevaA[3,3]
R1<- NuevaA[1,]
R2<- NuevaA[2,]

```

```
R3<- NuevaA[3,]
R4<- m6*NuevaA[3,]+NuevaA[4,]
NuevaA<- rbind(R1,R2,R3,R4)
(NuevaA)

##      [,1] [,2] [,3] [,4] [,5]
## R1      6  6.0 -4.00  1.00  0.0
## R2      0  3.6 -0.40 -3.80  1.2
## R3      0  0.0  0.56 -0.72  1.3
## R4      0  0.0  0.00 -4.00  4.0
```

Cuarta eliminación (Aquí comienza la eliminación hacia atrás)

Volver ceros

```
m7<- -NuevaA[3,4]/NuevaA[4,4]
m8<- -NuevaA[2,4]/NuevaA[4,4]
m9<- -NuevaA[1,4]/NuevaA[4,4]

R1<- m9 * NuevaA[4,] + NuevaA[1,]
R2<- m8 * NuevaA[4,] + NuevaA[2,]
R3<- m7 * NuevaA[4,] + NuevaA[3,]
R4<- NuevaA[4,]

NuevaA<- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## R1      6  6.0 -4.00  0  1.00
## R2      0  3.6 -0.40  0 -2.60
## R3      0  0.0  0.56  0  0.61
## R4      0  0.0  0.00 -4  4.00
```

Quinta eliminación

```
m10<- -NuevaA[2,3]/NuevaA[3,3]
m11<- -NuevaA[1,3]/NuevaA[3,3]

R4<- NuevaA[4,]
R3<- NuevaA[3,]
R2<- m10 * NuevaA[3,] + NuevaA[2,]
R1<- m11 * NuevaA[3,] + NuevaA[1,]

NuevaA<- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## R1      6  6.0 0.00  0  5.40
## R2      0  3.6 0.00  0 -2.16
## R3      0  0.0 0.56  0  0.61
## R4      0  0.0 0.00 -4  4.00
```

Sexta eliminación

```
m12<- -NuevaA[1,2]/NuevaA[2,2]
```

```
R4<- NuevaA[4,]
R3<- NuevaA[3,]
R2<- NuevaA[2,]
R1<- m12 * NuevaA[2,] + NuevaA[1,]
```

```
NuevaA<- rbind(R1, R2, R3, R4)
(NuevaA)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## R1      6  0.0 0.00  0  9.00
## R2      0  3.6 0.00  0 -2.16
## R3      0  0.0 0.56  0  0.61
## R4      0  0.0 0.00 -4  4.00
```

Soluciones

```
x<- NuevaA[1,5] / NuevaA[1,1]
y<- NuevaA[2,5] / NuevaA[2,2]
z<- NuevaA[3,5] / NuevaA[3,3]
w<- NuevaA[4,5] / NuevaA[4,4]
```

```
cat("Soluciones del sistema:\n")
```

```
## Soluciones del sistema:
```

```
cat("x =", x, "\n")
```

```
## x = 1.5
```

```
cat("y =", y, "\n")
```

```
## y = -0.6
```

```
cat("z =", z, "\n")
```

```
## z = 1.1
```

```
cat("w =", w, "\n")
```

```
## w = -1
```

2.6.6. Sesión 6 y Tareita 2

Inversa de una matriz

```
A=matrix(c(1,3,-2,1,
           1,4,-1,2,
           0,1,-1,4,
           2,6,1,2),nrow = 4,byrow = TRUE)
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    3   -2    1
## [2,]    1    4   -1    2
## [3,]    0    1   -1    4
## [4,]    2    6    1    2
```

```
Aorig<- A
```

1)

```
I = diag(4);(I)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

2)

```
AInv <- cbind(A,I);
(AInv)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    1    4   -1    2    0    1    0    0
## [3,]    0    1   -1    4    0    0    1    0
## [4,]    2    6    1    2    0    0    0    1
```

3)

```
A <- AInv
l21 <- A[2,1]/A[1,1];(l21)
```

```
## [1] 1
```

```
l31 <- A[3,1]/A[1,1];(l31)
```

```
## [1] 0
```

```
l41 <- A[4,1]/A[1,1];(l41)
```

```
## [1] 2
```

4)

```
A[2,] <- A[2,]-l21*A[1,];
A[3,] <- A[3,]-l31*A[1,];
A[4,] <- A[4,]-l41*A[1,];
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    0    1    1    1   -1    1    0    0
## [3,]    0    1   -1    4    0    0    1    0
## [4,]    0    0    5    0   -2    0    0    1
```

6)

```
l32 <- A[3,2]/A[2,2];(l32)
```

```
## [1] 1
```

```
l42 <- A[4,2]/A[2,2];(l42)
```

```
## [1] 0
```

7)

```
A[3,] <- A[3,]-132*A[2,];
A[4,] <- A[4,]-142*A[2,];
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    0    1    1    1   -1    1    0    0
## [3,]    0    0   -2    3    1   -1    1    0
## [4,]    0    0    5    0   -2    0    0    1
```

8)

```
esc <- 1/A[3,3]
A[3,] <- esc*A[3,]
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2  1.0  1.0  0.0  0.0    0
## [2,]    0    1    1  1.0 -1.0  1.0  0.0    0
## [3,]    0    0    1 -1.5 -0.5  0.5 -0.5    0
## [4,]    0    0    5  0.0 -2.0  0.0  0.0    1
```

9)

```
143 <- A[4,3]/A[3,3]; (143)
```

```
## [1] 5
```

10)

```
A[4,] <- A[4,]-143*A[3,];
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2  1.0  1.0  0.0  0.0    0
## [2,]    0    1    1  1.0 -1.0  1.0  0.0    0
## [3,]    0    0    1 -1.5 -0.5  0.5 -0.5    0
## [4,]    0    0    0  7.5  0.5 -2.5  2.5    1
```

11)

```
esc <- 1/A[4,4]
A[4,] <- esc*A[4,]
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2  1.0  1.000  0.00  0.00  0.00
## [2,]    0    1    1  1.0 -1.000  1.00  0.00  0.00
## [3,]    0    0    1 -1.5 -0.500  0.50 -0.50  0.00
## [4,]    0    0    0  1.0  0.067 -0.33  0.33  0.13
```

12)

```
134 <- A[3,4]/A[4,4]; (134); A[3,] <- A[3,]-134*A[4,]
```

```
## [1] -1.5
```

```
124 <- A[2,4]/A[4,4];(124); A[2,] <- A[2,]-124*A[4,]
```

```
## [1] 1
```

```
114 <- A[1,4]/A[4,4];(114); A[1,] <- A[1,]-114*A[4,]
```

```
## [1] 1
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    0 0.933 0.33 -0.33 -0.13
## [2,]    0    1    1    0 -1.067 1.33 -0.33 -0.13
## [3,]    0    0    1    0 -0.400 0.00 0.00 0.20
## [4,]    0    0    0    1 0.067 -0.33 0.33 0.13
```

13)

```
123 <- A[2,3]/A[3,3];(123); A[2,] <- A[2,]-123*A[3,]
```

```
## [1] 1
```

```
113 <- A[1,3]/A[3,3];(113); A[1,] <- A[1,]-113*A[3,]
```

```
## [1] -2
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3    0    0 0.133 0.33 -0.33 0.27
## [2,]    0    1    0    0 -0.667 1.33 -0.33 -0.33
## [3,]    0    0    1    0 -0.400 0.00 0.00 0.20
## [4,]    0    0    0    1 0.067 -0.33 0.33 0.13
```

14)

```
112 <- A[1,2]/A[2,2];(112); A[1,] <- A[1,]-112*A[2,]
```

```
## [1] 3
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    0    0    0 2.133 -3.67 0.67 1.27
## [2,]    0    1    0    0 -0.667 1.33 -0.33 -0.33
## [3,]    0    0    1    0 -0.400 0.00 0.00 0.20
## [4,]    0    0    0    1 0.067 -0.33 0.33 0.13
```

15)

```
AInv <- A[,5:8]; (AInv)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 2.133 -3.67 0.67 1.27
## [2,] -0.667 1.33 -0.33 -0.33
## [3,] -0.400 0.00 0.00 0.20
## [4,] 0.067 -0.33 0.33 0.13
```

16)


```
IdCalc <- Aorig%*%AInv;
(IdCalc)
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 1.0e+00 -5.6e-17 1.1e-16 -1.9e-16
## [2,] 1.4e-16 1.0e+00 1.1e-16 -2.2e-16
## [3,] 5.6e-17 0.0e+00 1.0e+00 -1.1e-16
## [4,] 1.4e-16 -1.1e-16 2.2e-16 1.0e+00
```

Factorización LU Ejemplo 1

```
A=matrix(c(1,1,1,1,
           2,3,1,5,
          -1,1,-5,3,
           3,1,7,-2),byrow = TRUE,nrow = 4)
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    2    3    1    5
## [3,]   -1    1   -5    3
## [4,]    3    1    7   -2
```

```
b=matrix(c(10,31,-2,18),nrow = 4)
(b)
```

```
##      [,1]
## [1,]   10
## [2,]   31
## [3,]   -2
## [4,]   18
```

```
Ab <- cbind(A,b); (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    2    3    1    5   31
## [3,]   -1    1   -5    3   -2
## [4,]    3    1    7   -2   18
```

Los pivotes se definen como $a_{kk}^{(k)}$, luego construimos los multiplicadores: $l_{i,k} = a_{ik}^{(k)} / a_{kk}^{(k)}$, $l_{21} = a_{21} / a_{11}$ y $l_{31} = a_{31} / a_{11}$ y $l_{41} = a_{41} / a_{11}$

```
A <- Ab;
l21 <- A[2,1]/A[1,1]; (l21)
```

```
## [1] 2
```

```
l31 <- A[3,1]/A[1,1]; (l31)
```

```
## [1] -1
```

```
l41 <- A[4,1]/A[1,1]; (l41)
```

```
## [1] 3
```

Construimos la matriz U , donde las entradas $a_{ij}^{(k+1)} = a_{ik}^{(k)} - l_{ik} \times a_{kj}^{(k)}$

```
A[2,] <- A[2,]-121*A[1,]; (A[2,])
```

```
## [1] 0 1 -1 3 11
```

```
A[3,] <- A[3,]-131*A[1,]; (A[3,])
```

```
## [1] 0 2 -4 4 8
```

```
A[4,] <- A[4,]-141*A[1,]; (A[4,])
```

```
## [1] 0 -2 4 -5 -12
```

Es decir la matriz resultante es:

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,] 1    1    1    1    10
## [2,] 0    1   -1    3    11
## [3,] 0    2   -4    4    8
## [4,] 0   -2    4   -5   -12
```

entonces el pivote es $A_{22} = 1$, calculemos los l_{32} y l_{42}

```
l32 <- A[3,2]/A[2,2]; (l32)
```

```
## [1] 2
```

```
l42 <- A[4,2]/A[2,2]; (l42)
```

```
## [1] -2
```

Haciendo cero debajo del pivote

```
A[3,] <- A[3,]-l32*A[2,]; (A[3,])
```

```
## [1] 0 0 -2 -2 -14
```

```
A[4,] <- A[4,]-l42*A[2,]; (A[4,])
```

```
## [1] 0 0 2 1 10
```

La matriz A queda de la forma:

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,] 1    1    1    1    10
## [2,] 0    1   -1    3    11
## [3,] 0    0   -2   -2   -14
## [4,] 0    0    2    1    10
```

por tanto el pivote es $A_{33} = -2$, y resta calcular l_{43}

```
l43 <- A[4,3]/A[3,3]; (l43)
```

```
## [1] -1
```

haciendo cero debajo del pivote

```
A[4,] <- A[4,]-l43*A[3,]; (A[4,])
```

```
## [1] 0 0 0 -1 -4
```

Por lo tanto la matriz A resultante es

(A)

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    0   -2   -2  -14
## [4,]    0    0    0   -1   -4
```

entonces podemos construir la matriz L con los valores l_{ij} calculados en los pasos anteriores

```
L=diag(4); (L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

```
L[2,1] <- 121
L[3,1] <- 131
L[4,1] <- 141
L[3,2] <- 132
L[4,2] <- 142
L[4,3] <- 143
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    2    1    0    0
## [3,]   -1    2    1    0
## [4,]    3   -2   -1    1
```

```
U <- A[,1:4]; (U)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    0    1   -1    3
## [3,]    0    0   -2   -2
## [4,]    0    0    0   -1
```

Verifiquemos que efectivamente $A = LU$

```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    2    3    1    5
## [3,]   -1    1   -5    3
## [4,]    3    1    7   -2
```

Ejemplo 2 Apliquemos lo desarrollado para la siguiente matriz A

```
A=matrix(c(4,0,1,1,
           3,1,3,1,
           0,1,2,0,
```

```
3,2,4,1),nrow = 4,byrow = TRUE);
```

(A)

```
##      [,1] [,2] [,3] [,4]
## [1,]    4    0    1    1
## [2,]    3    1    3    1
## [3,]    0    1    2    0
## [4,]    3    2    4    1
```

```
b=matrix(c(5,6,13,1),nrow = 4);(b)
```

```
##      [,1]
## [1,]    5
## [2,]    6
## [3,]   13
## [4,]    1
```

1)

```
Ab <- cbind(A,b)
A <- Ab;
```

2)

```
l21 <- A[2,1]/A[1,1];(l21)
```

```
## [1] 0.75
```

```
l31 <- A[3,1]/A[1,1];(l31)
```

```
## [1] 0
```

```
l41 <- A[4,1]/A[1,1];(l41)
```

```
## [1] 0.75
```

3)

```
A[2,] <- A[2,]-l21*A[1,]; (A[2,])
```

```
## [1] 0.00 1.00 2.25 0.25 2.25
```

```
A[3,] <- A[3,]-l31*A[1,]; (A[3,])
```

```
## [1] 0 1 2 0 13
```

```
A[4,] <- A[4,]-l41*A[1,]; (A[4,])
```

```
## [1] 0.00 2.00 3.25 0.25 -2.75
```

4)

```
l32 <- A[3,2]/A[2,2];(l32)
```

```
## [1] 1
```

```
l42 <- A[4,2]/A[2,2];(l42)
```

```
## [1] 2
```

5)

```
A[3,] <- A[3,]-132*A[2,];(A[3,])
```

```
## [1] 0.00 0.00 -0.25 -0.25 10.75
```

```
A[4,] <- A[4,]-142*A[2,];(A[4,])
```

```
## [1] 0.00 0.00 -1.25 -0.25 -7.25
```

6)

```
143 <- A[4,3]/A[3,3];(143)
```

```
## [1] 5
```

7)

```
A[4,] <- A[4,]-143*A[3,];(A[4,])
```

```
## [1] 0 0 0 1 -61
```

8)

```
L=diag(4);(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 1    0    0    0
## [2,] 0    1    0    0
## [3,] 0    0    1    0
## [4,] 0    0    0    1
```

9)

```
L[2,1] <- 121
L[3,1] <- 131
L[4,1] <- 141
L[3,2] <- 132
L[4,2] <- 142
L[4,3] <- 143;
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 1.00 0    0    0
## [2,] 0.75 1    0    0
## [3,] 0.00 1    1    0
## [4,] 0.75 2    5    1
```

10)

```
U <- A[,1:4];(U)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 4    0 1.00 1.00
## [2,] 0    1 2.25 0.25
## [3,] 0    0 -0.25 -0.25
## [4,] 0    0 0.00 1.00
```

11)

```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    4    0    1    1
## [2,]    3    1    3    1
## [3,]    0    1    2    0
## [4,]    3    2    4    1
```

Ejercicio Aplicar la factorización LU a la matriz A definida por

```
A=matrix(c(2,3,2,4,
           4,10,-4,0,
           -3,-2,-5,-2,
           -2,4,4,-7),nrow = 4,byrow = TRUE)
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    2    3    2    4
## [2,]    4   10   -4    0
## [3,]   -3   -2   -5   -2
## [4,]   -2    4    4   -7
```

```
Aorig<- A
```

Factorización LU

Ejercicio

```
A=matrix(c(2,3,2,4,
           4,10,-4,0,
           -3,-2,-5,-2,
           -2,4,4,-7),nrow = 4,byrow = TRUE)
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    2    3    2    4
## [2,]    4   10   -4    0
## [3,]   -3   -2   -5   -2
## [4,]   -2    4    4   -7
```

```
b<- matrix(c(10,31,-2,18),nrow = 4)
(b)
```

```
##      [,1]
## [1,]   10
## [2,]   31
## [3,]   -2
## [4,]   18
```

```
Ab <- cbind(A,b); (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2    3    2    4   10
## [2,]    4   10   -4    0   31
## [3,]   -3   -2   -5   -2   -2
## [4,]   -2    4    4   -7   18
```

Los pivotes se definen como $a_{kk}^{(k)}$, luego construimos los multiplicadores: $l_{i,k} = a_{ik}^{(k)} / a_{kk}^{(k)}$, $l_{21} = a_{21} / a_{11}$ y $l_{31} = a_{31} / a_{11}$ y $l_{41} = a_{41} / a_{11}$

```
A <- Ab;
121 <- A[2,1]/A[1,1]; (121)
```

```
## [1] 2
```

```
131 <- A[3,1]/A[1,1]; (131)
```

```
## [1] -1.5
```

```
141 <- A[4,1]/A[1,1]; (141)
```

```
## [1] -1
```

Construimos la matriz U , donde las entradas $a_{ij}^{(k+1)} = a_{ik}^{(k)} - l_{ik} \times a_{kj}^{(k)}$

```
A[2,] <- A[2,]-121*A[1,]; (A[2,])
```

```
## [1] 0 4 -8 -8 11
```

```
A[3,] <- A[3,]-131*A[1,]; (A[3,])
```

```
## [1] 0.0 2.5 -2.0 4.0 13.0
```

```
A[4,] <- A[4,]-141*A[1,]; (A[4,])
```

```
## [1] 0 7 6 -3 28
```

Es decir la matriz resultante es:

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2 3.0    2    4    10
## [2,]    0 4.0   -8   -8    11
## [3,]    0 2.5   -2    4    13
## [4,]    0 7.0    6   -3    28
```

entonces el pivote es $A_{22} = 1$, calculemos los l_{32} y l_{42}

```
132 <- A[3,2]/A[2,2]; (132)
```

```
## [1] 0.62
```

```
142 <- A[4,2]/A[2,2]; (142)
```

```
## [1] 1.8
```

Haciendo cero debajo del pivote

```
A[3,] <- A[3,]-132*A[2,]; (A[3,])
```

```
## [1] 0.0 0.0 3.0 9.0 6.1
```

```
A[4,] <- A[4,]-142*A[2,]; (A[4,])
```

```
## [1] 0.0 0.0 20.0 11.0 8.8
```

La matriz A queda de la forma:

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2    3    2    4 10.0
```

```
## [2,]    0    4   -8   -8 11.0
## [3,]    0    0    3    9  6.1
## [4,]    0    0   20   11  8.8
```

por tanto el pivote es $A_{33} = -2$, y resta calcular l_{43}

```
143 <- A[4,3]/A[3,3];(143)
```

```
## [1] 6.7
```

haciendo cero debajo del pivote

```
A[4,] <- A[4,]-143*A[3,];(A[4,])
```

```
## [1]    0    0    0 -49 -32
```

Por lo tanto la matriz A resultante es

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    2    3    2    4 10.0
## [2,]    0    4   -8   -8 11.0
## [3,]    0    0    3    9  6.1
## [4,]    0    0    0  -49 -32.1
```

entonces podemos construir la matriz L con los valores l_{ij} calculados en los pasos anteriores

```
L=diag(4);(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

```
L[2,1] <- 121
L[3,1] <- 131
L[4,1] <- 141
L[3,2] <- 132
L[4,2] <- 142
L[4,3] <- 143
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]  1.0 0.00 0.0   0
## [2,]  2.0 1.00 0.0   0
## [3,] -1.5 0.62 1.0   0
## [4,] -1.0 1.75 6.7   1
```

```
U <- A[,1:4];(U)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    2    3    2    4
## [2,]    0    4   -8   -8
## [3,]    0    0    3    9
## [4,]    0    0    0  -49
```

Verifiquemos que efectivamente $A = LU$


```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    2    3    2    4
## [2,]    4   10   -4    0
## [3,]   -3   -2   -5   -2
## [4,]   -2    4    4   -7
```

Notas importantes **Nota 1** Si se fija $l_{ii} = 1$ se le denomina factorización *Doolittle*. **Nota 2** Una matriz A tiene factorización de Doolittle si y sólo si se le puede aplicar el método de eliminación de Gauss sin pivoteo. **Nota 3** Si $U = L^T$ entonces la factorización se le denomina *factorización de Cholesky*. **Nota 4** Si la matriz es simétrica y definida positiva, entonces $A = LL^T$.

2.6.7. Sesión 7

Gauss-Seidel

```
A <- matrix(c(4, -1, 0, 0, 0,
              -1, 4, -1, 0, 0,
              0, -1, 4, -1, 0,
              0, 0, -1, 4, -1,
              0, 0, 0, -1, 4),
            nrow = 5, byrow = TRUE)
b <- c(1, 2, 3, 4, 5)
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    4   -1    0    0    0
## [2,]   -1    4   -1    0    0
## [3,]    0   -1    4   -1    0
## [4,]    0    0   -1    4   -1
## [5,]    0    0    0   -1    4
```

(b)

```
## [1] 1 2 3 4 5
```

```
x <- rep(0, 5) # x^(0)=(0,0,0,0,0)
cat("x^(0)=", paste(sprintf("%.8f", x), collapse=" "), "\n")
```

```
## x^(0)= 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.00000000
```

Iteración 1

```
x_old <- x
cat("\nIteración 1\n")
```

```
##
```

```
## Iteración 1
```

```
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(1) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))
```

```
## x1^(1) = (1 + 0)/4 = 0.25000000
```

```
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(1) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))
```

```

##  $x_2^{(1)} = (2 + 0.25000000 + 0)/4 = 0.56250000$ 
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(1) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

##  $x_3^{(1)} = (3 + 0.56250000 + 0)/4 = 0.89062500$ 
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(1) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

##  $x_4^{(1)} = (4 + 0.89062500 + 0)/4 = 1.22265625$ 
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(1) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

##  $x_5^{(1)} = (5 + 1.22265625)/4 = 1.55566406$ 
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(1) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(1)} = 0.25000000, 0.56250000, 0.89062500, 1.22265625, 1.55566406$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.55566406e+00 | ||r||_inf = 1.55566406e+00

Iteración 2
x_old <- x
cat("\nIteración 2\n")

##
## Iteración 2
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(2) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

##  $x_1^{(2)} = (1 + 0.5625)/4 = 0.39062500$ 
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(2) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

##  $x_2^{(2)} = (2 + 0.39062500 + 0.890625)/4 = 0.82031250$ 
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(2) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

##  $x_3^{(2)} = (3 + 0.82031250 + 1.22266)/4 = 1.26074219$ 
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(2) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

##  $x_4^{(2)} = (4 + 1.26074219 + 1.55566)/4 = 1.70410156$ 
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(2) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

##  $x_5^{(2)} = (5 + 1.70410156)/4 = 1.67602539$ 

```

```

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(2) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(2) = 0.39062500, 0.82031250, 1.26074219, 1.70410156, 1.67602539
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.81445312e-01 | ||r||_inf = 4.81445312e-01

Iteración 3
x_old <- x
cat("\nIteración 3\n")

##
## Iteración 3
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(3) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(3) = (1 + 0.820312)/4 = 0.45507812
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(3) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(3) = (2 + 0.45507812 + 1.26074)/4 = 0.92895508
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(3) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(3) = (3 + 0.92895508 + 1.7041)/4 = 1.40826416
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(3) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(3) = (4 + 1.40826416 + 1.67603)/4 = 1.77107239
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(3) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(3) = (5 + 1.77107239)/4 = 1.69276810
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(3) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(3) = 0.45507812, 0.92895508, 1.40826416, 1.77107239, 1.69276810
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.47521973e-01 | ||r||_inf = 1.47521973e-01

Iteración 4
x_old <- x
cat("\nIteración 4\n")

##
## Iteración 4

```

```

x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(4) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(4) = (1 + 0.928955)/4 = 0.48223877
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(4) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(4) = (2 + 0.48223877 + 1.40826)/4 = 0.97262573
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(4) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(4) = (3 + 0.97262573 + 1.77107)/4 = 1.43592453
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(4) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(4) = (4 + 1.43592453 + 1.69277)/4 = 1.78217316
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(4) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(4) = (5 + 1.78217316)/4 = 1.69554329
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(4) = %s\n", paste(sprintf("%.8f", x), collapse=", ")))

## x^(4) = 0.48223877, 0.97262573, 1.43592453, 1.78217316, 1.69554329
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.36706543e-02 | ||r||_inf = 4.36706543e-02

```

Iteración 5

```

x_old <- x
cat("\nIteración 5\n")

##
## Iteración 5
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(5) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(5) = (1 + 0.972626)/4 = 0.49315643
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(5) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(5) = (2 + 0.49315643 + 1.43592)/4 = 0.98227024
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(5) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(5) = (3 + 0.98227024 + 1.78217)/4 = 1.44111085
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(5) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

```

```

##  $x_4^{(5)} = (4 + 1.44111085 + 1.69554)/4 = 1.78416353$ 
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(5) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

##  $x_5^{(5)} = (5 + 1.78416353)/4 = 1.69604088$ 
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(5) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(5)} = 0.49315643, 0.98227024, 1.44111085, 1.78416353, 1.69604088$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.09176636e-02 | ||r||_inf = 9.64450836e-03
Iteración 6
x_old <- x
cat("\nIteración 6\n")

##
## Iteración 6
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(6) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

##  $x_1^{(6)} = (1 + 0.98227)/4 = 0.49556756$ 
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(6) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

##  $x_2^{(6)} = (2 + 0.49556756 + 1.44111)/4 = 0.98416960$ 
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(6) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

##  $x_3^{(6)} = (3 + 0.98416960 + 1.78416)/4 = 1.44208328$ 
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(6) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

##  $x_4^{(6)} = (4 + 1.44208328 + 1.69604)/4 = 1.78453104$ 
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(6) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

##  $x_5^{(6)} = (5 + 1.78453104)/4 = 1.69613276$ 
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(6) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(6)} = 0.49556756, 0.98416960, 1.44208328, 1.78453104, 1.69613276$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.41112709e-03 | ||r||_inf = 1.89936161e-03
Iteración 7

```

```

x_old <- x
cat("\nIteración 7\n")

##
## Iteración 7
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(7) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(7) = (1 + 0.98417)/4 = 0.49604240
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(7) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(7) = (2 + 0.49604240 + 1.44208)/4 = 0.98453142
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(7) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(7) = (3 + 0.98453142 + 1.78453)/4 = 1.44226562
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(7) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(7) = (4 + 1.44226562 + 1.69613)/4 = 1.78459959
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(7) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(7) = (5 + 1.78459959)/4 = 1.69614990
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(7) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(7) = 0.49604240, 0.98453142, 1.44226562, 1.78459959, 1.69614990
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.74840403e-04 | ||r||_inf = 3.61818820e-04

Iteración 8
x_old <- x
cat("\nIteración 8\n")

##
## Iteración 8
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(8) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(8) = (1 + 0.984531)/4 = 0.49613286
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(8) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(8) = (2 + 0.49613286 + 1.44227)/4 = 0.98459962

```

```

x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(8) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(8) = (3 + 0.98459962 + 1.7846)/4 = 1.44229980
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(8) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(8) = (4 + 1.44229980 + 1.69615)/4 = 1.78461243
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(8) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(8) = (5 + 1.78461243)/4 = 1.69615311
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(8) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(8) = 0.49613286, 0.98459962, 1.44229980, 1.78461243, 1.69615311
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 9.04547051e-05 | ||r||_inf = 6.81965612e-05

Iteración 9
x_old <- x
cat("\nIteración 9\n")

##
## Iteración 9
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(9) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(9) = (1 + 0.9846)/4 = 0.49614990
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(9) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(9) = (2 + 0.49614990 + 1.4423)/4 = 0.98461243
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(9) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(9) = (3 + 0.98461243 + 1.78461)/4 = 1.44230621
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(9) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(9) = (4 + 1.44230621 + 1.69615)/4 = 1.78461483
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(9) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(9) = (5 + 1.78461483)/4 = 1.69615371

```

```

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(9) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(9) = 0.49614990, 0.98461243, 1.44230621, 1.78461483, 1.69615371
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.70491403e-05 | ||r||_inf = 1.28090760e-05

Iteración 10
x_old <- x
cat("\nIteración 10\n")

##
## Iteración 10
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(10) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(10) = (1 + 0.984612)/4 = 0.49615311
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(10) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(10) = (2 + 0.49615311 + 1.44231)/4 = 0.98461483
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(10) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(10) = (3 + 0.98461483 + 1.78461)/4 = 1.44230741
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(10) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(10) = (4 + 1.44230741 + 1.69615)/4 = 1.78461528
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(10) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(10) = (5 + 1.78461528)/4 = 1.69615382
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(10) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(10) = 0.49615311, 0.98461483, 1.44230741, 1.78461528, 1.69615382
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.20226900e-06 | ||r||_inf = 2.40309055e-06

Iteración 11
x_old <- x
cat("\nIteración 11\n")

##
## Iteración 11

```



```

x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(11) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(11) = (1 + 0.984615)/4 = 0.49615371
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(11) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(11) = (2 + 0.49615371 + 1.44231)/4 = 0.98461528
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(11) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(11) = (3 + 0.98461528 + 1.78462)/4 = 1.44230764
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(11) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(11) = (4 + 1.44230764 + 1.69615)/4 = 1.78461537
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(11) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(11) = (5 + 1.78461537)/4 = 1.69615384
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(11) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(11) = 0.49615371, 0.98461528, 1.44230764, 1.78461537, 1.69615384
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 6.00772637e-07 | ||r||_inf = 4.50666278e-07

Iteración 12
x_old <- x
cat("\nIteración 12\n")

##
## Iteración 12
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(12) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(12) = (1 + 0.984615)/4 = 0.49615382
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(12) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(12) = (2 + 0.49615382 + 1.44231)/4 = 0.98461537
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(12) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(12) = (3 + 0.98461537 + 1.78462)/4 = 1.44230768
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(12) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

```

```
## x4^(12) = (4 + 1.44230768 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(12) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(12) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(12) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(12) = 0.49615382, 0.98461537, 1.44230768, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.12666569e-07 | ||r||_inf = 8.45053520e-08
```

Iteración 13

```
x_old <- x
cat("\nIteración 13\n")

##
## Iteración 13
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(13) = (1 + %.8g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(13) = (1 + 0.984615)/4 = 0.49615384
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(13) = (2 + %.8f + %.8g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(13) = (2 + 0.49615384 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(13) = (3 + %.8f + %.8g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(13) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(13) = (4 + %.8f + %.8g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(13) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(13) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(13) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(13) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(13) = 0.49615384, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.11263380e-08 | ||r||_inf = 1.58450926e-08
```

Iteración 14

```

x_old <- x
cat("\nIteración 14\n")

##
## Iteración 14
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(14) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(14) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(14) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(14) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(14) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(14) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(14) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(14) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(14) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(14) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(14) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(14) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.96127314e-09 | ||r||_inf = 2.97097602e-09

Iteración 15
x_old <- x
cat("\nIteración 15\n")

##
## Iteración 15
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(15) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(15) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(15) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(15) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538

```

```

x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(15) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(15) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(15) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(15) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(15) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(15) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(15) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(15) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 7.42744033e-10 | ||r||_inf = 5.57059288e-10

Iteración 16
x_old <- x
cat("\nIteración 16\n")

##
## Iteración 16
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(16) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(16) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(16) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(16) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(16) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(16) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(16) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(16) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(16) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(16) = (5 + 1.78461538)/4 = 1.69615385

```

```

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(16) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(16) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.39264822e-10 | ||r||_inf = 1.04448672e-10

Iteración 17
x_old <- x
cat("\nIteración 17\n")

##
## Iteración 17
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(17) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(17) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(17) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(17) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(17) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(17) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(17) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(17) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(17) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(17) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(17) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(17) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.61121680e-11 | ||r||_inf = 1.95841121e-11

Iteración 18
x_old <- x
cat("\nIteración 18\n")

##
## Iteración 18

```

```

x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(18) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(18) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(18) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(18) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(18) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(18) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(18) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(18) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(18) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(18) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(18) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(18) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 4.89602803e-12 | ||r||_inf = 3.67206265e-12

```

Iteración 19

```

x_old <- x
cat("\nIteración 19\n")

##
## Iteración 19
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(19) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(19) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(19) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(19) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(19) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(19) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(19) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

```

```
##  $x_4^{(19)} = (4 + 1.44230769 + 1.69615)/4 = 1.78461538$ 
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(19) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

##  $x_5^{(19)} = (5 + 1.78461538)/4 = 1.69615385$ 
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(19) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(19)} = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 9.18043419e-13 | ||r||_inf = 6.88338275e-13
```

Iteración 20

```
x_old <- x
cat("\nIteración 20\n")

##
## Iteración 20
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(20) = (1 + %.8g)/4 = %.8f\n", x_old[2], x[1]))

##  $x_1^{(20)} = (1 + 0.984615)/4 = 0.49615385$ 
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(20) = (2 + %.8f + %.8g)/4 = %.8f\n", x[1], x_old[3], x[2]))

##  $x_2^{(20)} = (2 + 0.49615385 + 1.44231)/4 = 0.98461538$ 
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(20) = (3 + %.8f + %.8g)/4 = %.8f\n", x[2], x_old[4], x[3]))

##  $x_3^{(20)} = (3 + 0.98461538 + 1.78462)/4 = 1.44230769$ 
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(20) = (4 + %.8f + %.8g)/4 = %.8f\n", x[3], x_old[5], x[4]))

##  $x_4^{(20)} = (4 + 1.44230769 + 1.69615)/4 = 1.78461538$ 
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(20) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

##  $x_5^{(20)} = (5 + 1.78461538)/4 = 1.69615385$ 
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(20) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

##  $x^{(20)} = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385$ 
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.72084569e-13 | ||r||_inf = 1.29007915e-13
```

Iteración 21

```

x_old <- x
cat("\nIteración 21\n")

##
## Iteración 21
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(21) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(21) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(21) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(21) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(21) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(21) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(21) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(21) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(21) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(21) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(21) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(21) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 3.22519789e-14 | ||r||_inf = 2.43138842e-14

Iteración 22
x_old <- x
cat("\nIteración 22\n")

##
## Iteración 22
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(22) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(22) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(22) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(22) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538

```



```

x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(22) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(22) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(22) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(22) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(22) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(22) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(22) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(22) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 6.05071548e-15 | ||r||_inf = 4.66293670e-15

Iteración 23
x_old <- x
cat("\nIteración 23\n")

##
## Iteración 23
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(23) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(23) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(23) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(23) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(23) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(23) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(23) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(23) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(23) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(23) = (5 + 1.78461538)/4 = 1.69615385

```

```

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(23) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(23) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 1.16573418e-15 | ||r||_inf = 8.88178420e-16

Iteración 24
x_old <- x
cat("\nIteración 24\n")

##
## Iteración 24
x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(24) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(24) = (1 + 0.984615)/4 = 0.49615385
x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(24) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(24) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538
x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(24) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(24) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769
x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(24) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(24) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538
x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(24) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(24) = (5 + 1.78461538)/4 = 1.69615385
r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(24) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(24) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385
cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 2.22044605e-16 | ||r||_inf = 2.22044605e-16

Iteración 25
x_old <- x
cat("\nIteración 25\n")

##
## Iteración 25

```

```

x[1] <- (b[1] + x_old[2]) / A[1,1]
cat(sprintf("x1^(25) = (1 + %g)/4 = %.8f\n", x_old[2], x[1]))

## x1^(25) = (1 + 0.984615)/4 = 0.49615385

x[2] <- (b[2] + x[1] + x_old[3]) / A[2,2]
cat(sprintf("x2^(25) = (2 + %.8f + %g)/4 = %.8f\n", x[1], x_old[3], x[2]))

## x2^(25) = (2 + 0.49615385 + 1.44231)/4 = 0.98461538

x[3] <- (b[3] + x[2] + x_old[4]) / A[3,3]
cat(sprintf("x3^(25) = (3 + %.8f + %g)/4 = %.8f\n", x[2], x_old[4], x[3]))

## x3^(25) = (3 + 0.98461538 + 1.78462)/4 = 1.44230769

x[4] <- (b[4] + x[3] + x_old[5]) / A[4,4]
cat(sprintf("x4^(25) = (4 + %.8f + %g)/4 = %.8f\n", x[3], x_old[5], x[4]))

## x4^(25) = (4 + 1.44230769 + 1.69615)/4 = 1.78461538

x[5] <- (b[5] + x[4]) / A[5,5]
cat(sprintf("x5^(25) = (5 + %.8f)/4 = %.8f\n", x[4], x[5]))

## x5^(25) = (5 + 1.78461538)/4 = 1.69615385

r <- b - A %*% x
dx_inf <- max(abs(x - x_old))
cat(sprintf("x^(25) = %s\n", paste(sprintf("%.8f", x), collapse=" ")))

## x^(25) = 0.49615385, 0.98461538, 1.44230769, 1.78461538, 1.69615385

cat(sprintf("||dx||_inf = %.8e | ||r||_inf = %.8e\n", dx_inf, max(abs(r))))

## ||dx||_inf = 5.55111512e-17 | ||r||_inf = 0.00000000e+00

```

2.6.8. Sesión 8

```

N <- 1000
a <- -10
b <- 10
paso <- 0.01
x <- seq(a,b,paso); (x)

##      [1] -10.00 -9.99 -9.98 -9.97 -9.96 -9.95 -9.94 -9.93 -9.92 -9.91
##     [11] -9.90 -9.89 -9.88 -9.87 -9.86 -9.85 -9.84 -9.83 -9.82 -9.81
##     [21] -9.80 -9.79 -9.78 -9.77 -9.76 -9.75 -9.74 -9.73 -9.72 -9.71
##     [31] -9.70 -9.69 -9.68 -9.67 -9.66 -9.65 -9.64 -9.63 -9.62 -9.61
##     [41] -9.60 -9.59 -9.58 -9.57 -9.56 -9.55 -9.54 -9.53 -9.52 -9.51
##     [51] -9.50 -9.49 -9.48 -9.47 -9.46 -9.45 -9.44 -9.43 -9.42 -9.41
##     [61] -9.40 -9.39 -9.38 -9.37 -9.36 -9.35 -9.34 -9.33 -9.32 -9.31
##     [71] -9.30 -9.29 -9.28 -9.27 -9.26 -9.25 -9.24 -9.23 -9.22 -9.21
##     [81] -9.20 -9.19 -9.18 -9.17 -9.16 -9.15 -9.14 -9.13 -9.12 -9.11
##     [91] -9.10 -9.09 -9.08 -9.07 -9.06 -9.05 -9.04 -9.03 -9.02 -9.01
##    [101] -9.00 -8.99 -8.98 -8.97 -8.96 -8.95 -8.94 -8.93 -8.92 -8.91
##    [111] -8.90 -8.89 -8.88 -8.87 -8.86 -8.85 -8.84 -8.83 -8.82 -8.81
##    [121] -8.80 -8.79 -8.78 -8.77 -8.76 -8.75 -8.74 -8.73 -8.72 -8.71

```

##	[131]	-8.70	-8.69	-8.68	-8.67	-8.66	-8.65	-8.64	-8.63	-8.62	-8.61
##	[141]	-8.60	-8.59	-8.58	-8.57	-8.56	-8.55	-8.54	-8.53	-8.52	-8.51
##	[151]	-8.50	-8.49	-8.48	-8.47	-8.46	-8.45	-8.44	-8.43	-8.42	-8.41
##	[161]	-8.40	-8.39	-8.38	-8.37	-8.36	-8.35	-8.34	-8.33	-8.32	-8.31
##	[171]	-8.30	-8.29	-8.28	-8.27	-8.26	-8.25	-8.24	-8.23	-8.22	-8.21
##	[181]	-8.20	-8.19	-8.18	-8.17	-8.16	-8.15	-8.14	-8.13	-8.12	-8.11
##	[191]	-8.10	-8.09	-8.08	-8.07	-8.06	-8.05	-8.04	-8.03	-8.02	-8.01
##	[201]	-8.00	-7.99	-7.98	-7.97	-7.96	-7.95	-7.94	-7.93	-7.92	-7.91
##	[211]	-7.90	-7.89	-7.88	-7.87	-7.86	-7.85	-7.84	-7.83	-7.82	-7.81
##	[221]	-7.80	-7.79	-7.78	-7.77	-7.76	-7.75	-7.74	-7.73	-7.72	-7.71
##	[231]	-7.70	-7.69	-7.68	-7.67	-7.66	-7.65	-7.64	-7.63	-7.62	-7.61
##	[241]	-7.60	-7.59	-7.58	-7.57	-7.56	-7.55	-7.54	-7.53	-7.52	-7.51
##	[251]	-7.50	-7.49	-7.48	-7.47	-7.46	-7.45	-7.44	-7.43	-7.42	-7.41
##	[261]	-7.40	-7.39	-7.38	-7.37	-7.36	-7.35	-7.34	-7.33	-7.32	-7.31
##	[271]	-7.30	-7.29	-7.28	-7.27	-7.26	-7.25	-7.24	-7.23	-7.22	-7.21
##	[281]	-7.20	-7.19	-7.18	-7.17	-7.16	-7.15	-7.14	-7.13	-7.12	-7.11
##	[291]	-7.10	-7.09	-7.08	-7.07	-7.06	-7.05	-7.04	-7.03	-7.02	-7.01
##	[301]	-7.00	-6.99	-6.98	-6.97	-6.96	-6.95	-6.94	-6.93	-6.92	-6.91
##	[311]	-6.90	-6.89	-6.88	-6.87	-6.86	-6.85	-6.84	-6.83	-6.82	-6.81
##	[321]	-6.80	-6.79	-6.78	-6.77	-6.76	-6.75	-6.74	-6.73	-6.72	-6.71
##	[331]	-6.70	-6.69	-6.68	-6.67	-6.66	-6.65	-6.64	-6.63	-6.62	-6.61
##	[341]	-6.60	-6.59	-6.58	-6.57	-6.56	-6.55	-6.54	-6.53	-6.52	-6.51
##	[351]	-6.50	-6.49	-6.48	-6.47	-6.46	-6.45	-6.44	-6.43	-6.42	-6.41
##	[361]	-6.40	-6.39	-6.38	-6.37	-6.36	-6.35	-6.34	-6.33	-6.32	-6.31
##	[371]	-6.30	-6.29	-6.28	-6.27	-6.26	-6.25	-6.24	-6.23	-6.22	-6.21
##	[381]	-6.20	-6.19	-6.18	-6.17	-6.16	-6.15	-6.14	-6.13	-6.12	-6.11
##	[391]	-6.10	-6.09	-6.08	-6.07	-6.06	-6.05	-6.04	-6.03	-6.02	-6.01
##	[401]	-6.00	-5.99	-5.98	-5.97	-5.96	-5.95	-5.94	-5.93	-5.92	-5.91
##	[411]	-5.90	-5.89	-5.88	-5.87	-5.86	-5.85	-5.84	-5.83	-5.82	-5.81
##	[421]	-5.80	-5.79	-5.78	-5.77	-5.76	-5.75	-5.74	-5.73	-5.72	-5.71
##	[431]	-5.70	-5.69	-5.68	-5.67	-5.66	-5.65	-5.64	-5.63	-5.62	-5.61
##	[441]	-5.60	-5.59	-5.58	-5.57	-5.56	-5.55	-5.54	-5.53	-5.52	-5.51
##	[451]	-5.50	-5.49	-5.48	-5.47	-5.46	-5.45	-5.44	-5.43	-5.42	-5.41
##	[461]	-5.40	-5.39	-5.38	-5.37	-5.36	-5.35	-5.34	-5.33	-5.32	-5.31
##	[471]	-5.30	-5.29	-5.28	-5.27	-5.26	-5.25	-5.24	-5.23	-5.22	-5.21
##	[481]	-5.20	-5.19	-5.18	-5.17	-5.16	-5.15	-5.14	-5.13	-5.12	-5.11
##	[491]	-5.10	-5.09	-5.08	-5.07	-5.06	-5.05	-5.04	-5.03	-5.02	-5.01
##	[501]	-5.00	-4.99	-4.98	-4.97	-4.96	-4.95	-4.94	-4.93	-4.92	-4.91
##	[511]	-4.90	-4.89	-4.88	-4.87	-4.86	-4.85	-4.84	-4.83	-4.82	-4.81
##	[521]	-4.80	-4.79	-4.78	-4.77	-4.76	-4.75	-4.74	-4.73	-4.72	-4.71
##	[531]	-4.70	-4.69	-4.68	-4.67	-4.66	-4.65	-4.64	-4.63	-4.62	-4.61
##	[541]	-4.60	-4.59	-4.58	-4.57	-4.56	-4.55	-4.54	-4.53	-4.52	-4.51
##	[551]	-4.50	-4.49	-4.48	-4.47	-4.46	-4.45	-4.44	-4.43	-4.42	-4.41
##	[561]	-4.40	-4.39	-4.38	-4.37	-4.36	-4.35	-4.34	-4.33	-4.32	-4.31
##	[571]	-4.30	-4.29	-4.28	-4.27	-4.26	-4.25	-4.24	-4.23	-4.22	-4.21
##	[581]	-4.20	-4.19	-4.18	-4.17	-4.16	-4.15	-4.14	-4.13	-4.12	-4.11
##	[591]	-4.10	-4.09	-4.08	-4.07	-4.06	-4.05	-4.04	-4.03	-4.02	-4.01
##	[601]	-4.00	-3.99	-3.98	-3.97	-3.96	-3.95	-3.94	-3.93	-3.92	-3.91
##	[611]	-3.90	-3.89	-3.88	-3.87	-3.86	-3.85	-3.84	-3.83	-3.82	-3.81
##	[621]	-3.80	-3.79	-3.78	-3.77	-3.76	-3.75	-3.74	-3.73	-3.72	-3.71
##	[631]	-3.70	-3.69	-3.68	-3.67	-3.66	-3.65	-3.64	-3.63	-3.62	-3.61
##	[641]	-3.60	-3.59	-3.58	-3.57	-3.56	-3.55	-3.54	-3.53	-3.52	-3.51

##	[651]	-3.50	-3.49	-3.48	-3.47	-3.46	-3.45	-3.44	-3.43	-3.42	-3.41
##	[661]	-3.40	-3.39	-3.38	-3.37	-3.36	-3.35	-3.34	-3.33	-3.32	-3.31
##	[671]	-3.30	-3.29	-3.28	-3.27	-3.26	-3.25	-3.24	-3.23	-3.22	-3.21
##	[681]	-3.20	-3.19	-3.18	-3.17	-3.16	-3.15	-3.14	-3.13	-3.12	-3.11
##	[691]	-3.10	-3.09	-3.08	-3.07	-3.06	-3.05	-3.04	-3.03	-3.02	-3.01
##	[701]	-3.00	-2.99	-2.98	-2.97	-2.96	-2.95	-2.94	-2.93	-2.92	-2.91
##	[711]	-2.90	-2.89	-2.88	-2.87	-2.86	-2.85	-2.84	-2.83	-2.82	-2.81
##	[721]	-2.80	-2.79	-2.78	-2.77	-2.76	-2.75	-2.74	-2.73	-2.72	-2.71
##	[731]	-2.70	-2.69	-2.68	-2.67	-2.66	-2.65	-2.64	-2.63	-2.62	-2.61
##	[741]	-2.60	-2.59	-2.58	-2.57	-2.56	-2.55	-2.54	-2.53	-2.52	-2.51
##	[751]	-2.50	-2.49	-2.48	-2.47	-2.46	-2.45	-2.44	-2.43	-2.42	-2.41
##	[761]	-2.40	-2.39	-2.38	-2.37	-2.36	-2.35	-2.34	-2.33	-2.32	-2.31
##	[771]	-2.30	-2.29	-2.28	-2.27	-2.26	-2.25	-2.24	-2.23	-2.22	-2.21
##	[781]	-2.20	-2.19	-2.18	-2.17	-2.16	-2.15	-2.14	-2.13	-2.12	-2.11
##	[791]	-2.10	-2.09	-2.08	-2.07	-2.06	-2.05	-2.04	-2.03	-2.02	-2.01
##	[801]	-2.00	-1.99	-1.98	-1.97	-1.96	-1.95	-1.94	-1.93	-1.92	-1.91
##	[811]	-1.90	-1.89	-1.88	-1.87	-1.86	-1.85	-1.84	-1.83	-1.82	-1.81
##	[821]	-1.80	-1.79	-1.78	-1.77	-1.76	-1.75	-1.74	-1.73	-1.72	-1.71
##	[831]	-1.70	-1.69	-1.68	-1.67	-1.66	-1.65	-1.64	-1.63	-1.62	-1.61
##	[841]	-1.60	-1.59	-1.58	-1.57	-1.56	-1.55	-1.54	-1.53	-1.52	-1.51
##	[851]	-1.50	-1.49	-1.48	-1.47	-1.46	-1.45	-1.44	-1.43	-1.42	-1.41
##	[861]	-1.40	-1.39	-1.38	-1.37	-1.36	-1.35	-1.34	-1.33	-1.32	-1.31
##	[871]	-1.30	-1.29	-1.28	-1.27	-1.26	-1.25	-1.24	-1.23	-1.22	-1.21
##	[881]	-1.20	-1.19	-1.18	-1.17	-1.16	-1.15	-1.14	-1.13	-1.12	-1.11
##	[891]	-1.10	-1.09	-1.08	-1.07	-1.06	-1.05	-1.04	-1.03	-1.02	-1.01
##	[901]	-1.00	-0.99	-0.98	-0.97	-0.96	-0.95	-0.94	-0.93	-0.92	-0.91
##	[911]	-0.90	-0.89	-0.88	-0.87	-0.86	-0.85	-0.84	-0.83	-0.82	-0.81
##	[921]	-0.80	-0.79	-0.78	-0.77	-0.76	-0.75	-0.74	-0.73	-0.72	-0.71
##	[931]	-0.70	-0.69	-0.68	-0.67	-0.66	-0.65	-0.64	-0.63	-0.62	-0.61
##	[941]	-0.60	-0.59	-0.58	-0.57	-0.56	-0.55	-0.54	-0.53	-0.52	-0.51
##	[951]	-0.50	-0.49	-0.48	-0.47	-0.46	-0.45	-0.44	-0.43	-0.42	-0.41
##	[961]	-0.40	-0.39	-0.38	-0.37	-0.36	-0.35	-0.34	-0.33	-0.32	-0.31
##	[971]	-0.30	-0.29	-0.28	-0.27	-0.26	-0.25	-0.24	-0.23	-0.22	-0.21
##	[981]	-0.20	-0.19	-0.18	-0.17	-0.16	-0.15	-0.14	-0.13	-0.12	-0.11
##	[991]	-0.10	-0.09	-0.08	-0.07	-0.06	-0.05	-0.04	-0.03	-0.02	-0.01
##	[1001]	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
##	[1011]	0.10	0.11	0.12	0.13	0.14	0.15	0.16	0.17	0.18	0.19
##	[1021]	0.20	0.21	0.22	0.23	0.24	0.25	0.26	0.27	0.28	0.29
##	[1031]	0.30	0.31	0.32	0.33	0.34	0.35	0.36	0.37	0.38	0.39
##	[1041]	0.40	0.41	0.42	0.43	0.44	0.45	0.46	0.47	0.48	0.49
##	[1051]	0.50	0.51	0.52	0.53	0.54	0.55	0.56	0.57	0.58	0.59
##	[1061]	0.60	0.61	0.62	0.63	0.64	0.65	0.66	0.67	0.68	0.69
##	[1071]	0.70	0.71	0.72	0.73	0.74	0.75	0.76	0.77	0.78	0.79
##	[1081]	0.80	0.81	0.82	0.83	0.84	0.85	0.86	0.87	0.88	0.89
##	[1091]	0.90	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99
##	[1101]	1.00	1.01	1.02	1.03	1.04	1.05	1.06	1.07	1.08	1.09
##	[1111]	1.10	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19
##	[1121]	1.20	1.21	1.22	1.23	1.24	1.25	1.26	1.27	1.28	1.29
##	[1131]	1.30	1.31	1.32	1.33	1.34	1.35	1.36	1.37	1.38	1.39
##	[1141]	1.40	1.41	1.42	1.43	1.44	1.45	1.46	1.47	1.48	1.49
##	[1151]	1.50	1.51	1.52	1.53	1.54	1.55	1.56	1.57	1.58	1.59
##	[1161]	1.60	1.61	1.62	1.63	1.64	1.65	1.66	1.67	1.68	1.69

## [1171]	1.70	1.71	1.72	1.73	1.74	1.75	1.76	1.77	1.78	1.79
## [1181]	1.80	1.81	1.82	1.83	1.84	1.85	1.86	1.87	1.88	1.89
## [1191]	1.90	1.91	1.92	1.93	1.94	1.95	1.96	1.97	1.98	1.99
## [1201]	2.00	2.01	2.02	2.03	2.04	2.05	2.06	2.07	2.08	2.09
## [1211]	2.10	2.11	2.12	2.13	2.14	2.15	2.16	2.17	2.18	2.19
## [1221]	2.20	2.21	2.22	2.23	2.24	2.25	2.26	2.27	2.28	2.29
## [1231]	2.30	2.31	2.32	2.33	2.34	2.35	2.36	2.37	2.38	2.39
## [1241]	2.40	2.41	2.42	2.43	2.44	2.45	2.46	2.47	2.48	2.49
## [1251]	2.50	2.51	2.52	2.53	2.54	2.55	2.56	2.57	2.58	2.59
## [1261]	2.60	2.61	2.62	2.63	2.64	2.65	2.66	2.67	2.68	2.69
## [1271]	2.70	2.71	2.72	2.73	2.74	2.75	2.76	2.77	2.78	2.79
## [1281]	2.80	2.81	2.82	2.83	2.84	2.85	2.86	2.87	2.88	2.89
## [1291]	2.90	2.91	2.92	2.93	2.94	2.95	2.96	2.97	2.98	2.99
## [1301]	3.00	3.01	3.02	3.03	3.04	3.05	3.06	3.07	3.08	3.09
## [1311]	3.10	3.11	3.12	3.13	3.14	3.15	3.16	3.17	3.18	3.19
## [1321]	3.20	3.21	3.22	3.23	3.24	3.25	3.26	3.27	3.28	3.29
## [1331]	3.30	3.31	3.32	3.33	3.34	3.35	3.36	3.37	3.38	3.39
## [1341]	3.40	3.41	3.42	3.43	3.44	3.45	3.46	3.47	3.48	3.49
## [1351]	3.50	3.51	3.52	3.53	3.54	3.55	3.56	3.57	3.58	3.59
## [1361]	3.60	3.61	3.62	3.63	3.64	3.65	3.66	3.67	3.68	3.69
## [1371]	3.70	3.71	3.72	3.73	3.74	3.75	3.76	3.77	3.78	3.79
## [1381]	3.80	3.81	3.82	3.83	3.84	3.85	3.86	3.87	3.88	3.89
## [1391]	3.90	3.91	3.92	3.93	3.94	3.95	3.96	3.97	3.98	3.99
## [1401]	4.00	4.01	4.02	4.03	4.04	4.05	4.06	4.07	4.08	4.09
## [1411]	4.10	4.11	4.12	4.13	4.14	4.15	4.16	4.17	4.18	4.19
## [1421]	4.20	4.21	4.22	4.23	4.24	4.25	4.26	4.27	4.28	4.29
## [1431]	4.30	4.31	4.32	4.33	4.34	4.35	4.36	4.37	4.38	4.39
## [1441]	4.40	4.41	4.42	4.43	4.44	4.45	4.46	4.47	4.48	4.49
## [1451]	4.50	4.51	4.52	4.53	4.54	4.55	4.56	4.57	4.58	4.59
## [1461]	4.60	4.61	4.62	4.63	4.64	4.65	4.66	4.67	4.68	4.69
## [1471]	4.70	4.71	4.72	4.73	4.74	4.75	4.76	4.77	4.78	4.79
## [1481]	4.80	4.81	4.82	4.83	4.84	4.85	4.86	4.87	4.88	4.89
## [1491]	4.90	4.91	4.92	4.93	4.94	4.95	4.96	4.97	4.98	4.99
## [1501]	5.00	5.01	5.02	5.03	5.04	5.05	5.06	5.07	5.08	5.09
## [1511]	5.10	5.11	5.12	5.13	5.14	5.15	5.16	5.17	5.18	5.19
## [1521]	5.20	5.21	5.22	5.23	5.24	5.25	5.26	5.27	5.28	5.29
## [1531]	5.30	5.31	5.32	5.33	5.34	5.35	5.36	5.37	5.38	5.39
## [1541]	5.40	5.41	5.42	5.43	5.44	5.45	5.46	5.47	5.48	5.49
## [1551]	5.50	5.51	5.52	5.53	5.54	5.55	5.56	5.57	5.58	5.59
## [1561]	5.60	5.61	5.62	5.63	5.64	5.65	5.66	5.67	5.68	5.69
## [1571]	5.70	5.71	5.72	5.73	5.74	5.75	5.76	5.77	5.78	5.79
## [1581]	5.80	5.81	5.82	5.83	5.84	5.85	5.86	5.87	5.88	5.89
## [1591]	5.90	5.91	5.92	5.93	5.94	5.95	5.96	5.97	5.98	5.99
## [1601]	6.00	6.01	6.02	6.03	6.04	6.05	6.06	6.07	6.08	6.09
## [1611]	6.10	6.11	6.12	6.13	6.14	6.15	6.16	6.17	6.18	6.19
## [1621]	6.20	6.21	6.22	6.23	6.24	6.25	6.26	6.27	6.28	6.29
## [1631]	6.30	6.31	6.32	6.33	6.34	6.35	6.36	6.37	6.38	6.39
## [1641]	6.40	6.41	6.42	6.43	6.44	6.45	6.46	6.47	6.48	6.49
## [1651]	6.50	6.51	6.52	6.53	6.54	6.55	6.56	6.57	6.58	6.59
## [1661]	6.60	6.61	6.62	6.63	6.64	6.65	6.66	6.67	6.68	6.69
## [1671]	6.70	6.71	6.72	6.73	6.74	6.75	6.76	6.77	6.78	6.79
## [1681]	6.80	6.81	6.82	6.83	6.84	6.85	6.86	6.87	6.88	6.89

```
## [1691] 6.90 6.91 6.92 6.93 6.94 6.95 6.96 6.97 6.98 6.99
## [1701] 7.00 7.01 7.02 7.03 7.04 7.05 7.06 7.07 7.08 7.09
## [1711] 7.10 7.11 7.12 7.13 7.14 7.15 7.16 7.17 7.18 7.19
## [1721] 7.20 7.21 7.22 7.23 7.24 7.25 7.26 7.27 7.28 7.29
## [1731] 7.30 7.31 7.32 7.33 7.34 7.35 7.36 7.37 7.38 7.39
## [1741] 7.40 7.41 7.42 7.43 7.44 7.45 7.46 7.47 7.48 7.49
## [1751] 7.50 7.51 7.52 7.53 7.54 7.55 7.56 7.57 7.58 7.59
## [1761] 7.60 7.61 7.62 7.63 7.64 7.65 7.66 7.67 7.68 7.69
## [1771] 7.70 7.71 7.72 7.73 7.74 7.75 7.76 7.77 7.78 7.79
## [1781] 7.80 7.81 7.82 7.83 7.84 7.85 7.86 7.87 7.88 7.89
## [1791] 7.90 7.91 7.92 7.93 7.94 7.95 7.96 7.97 7.98 7.99
## [1801] 8.00 8.01 8.02 8.03 8.04 8.05 8.06 8.07 8.08 8.09
## [1811] 8.10 8.11 8.12 8.13 8.14 8.15 8.16 8.17 8.18 8.19
## [1821] 8.20 8.21 8.22 8.23 8.24 8.25 8.26 8.27 8.28 8.29
## [1831] 8.30 8.31 8.32 8.33 8.34 8.35 8.36 8.37 8.38 8.39
## [1841] 8.40 8.41 8.42 8.43 8.44 8.45 8.46 8.47 8.48 8.49
## [1851] 8.50 8.51 8.52 8.53 8.54 8.55 8.56 8.57 8.58 8.59
## [1861] 8.60 8.61 8.62 8.63 8.64 8.65 8.66 8.67 8.68 8.69
## [1871] 8.70 8.71 8.72 8.73 8.74 8.75 8.76 8.77 8.78 8.79
## [1881] 8.80 8.81 8.82 8.83 8.84 8.85 8.86 8.87 8.88 8.89
## [1891] 8.90 8.91 8.92 8.93 8.94 8.95 8.96 8.97 8.98 8.99
## [1901] 9.00 9.01 9.02 9.03 9.04 9.05 9.06 9.07 9.08 9.09
## [1911] 9.10 9.11 9.12 9.13 9.14 9.15 9.16 9.17 9.18 9.19
## [1921] 9.20 9.21 9.22 9.23 9.24 9.25 9.26 9.27 9.28 9.29
## [1931] 9.30 9.31 9.32 9.33 9.34 9.35 9.36 9.37 9.38 9.39
## [1941] 9.40 9.41 9.42 9.43 9.44 9.45 9.46 9.47 9.48 9.49
## [1951] 9.50 9.51 9.52 9.53 9.54 9.55 9.56 9.57 9.58 9.59
## [1961] 9.60 9.61 9.62 9.63 9.64 9.65 9.66 9.67 9.68 9.69
## [1971] 9.70 9.71 9.72 9.73 9.74 9.75 9.76 9.77 9.78 9.79
## [1981] 9.80 9.81 9.82 9.83 9.84 9.85 9.86 9.87 9.88 9.89
## [1991] 9.90 9.91 9.92 9.93 9.94 9.95 9.96 9.97 9.98 9.99
## [2001] 10.00
```

```
head(x,10)
```

```
## [1] -10.0 -10.0 -10.0 -10.0 -10.0 -9.9 -9.9 -9.9 -9.9 -9.9
```

Graficamos la función cúbica de $f(x) = -x^3 - 6x^2 + 11x - 6$

```
A <- -1
```

```
B <- -6
```

```
C <- 11
```

```
D <- -6
```

```
f_x <- A*(x^3)+B*(x^2)+C*x+D
```

```
head(f_x,10)
```

```
## [1] 284 282 281 279 277 276 274 272 271 269
```

Graficamos la función

```
plot(x,f_x, col = "red",
     main = "Gráfica de una función cúbica",
     type = "l",
     xlab = "x",
```